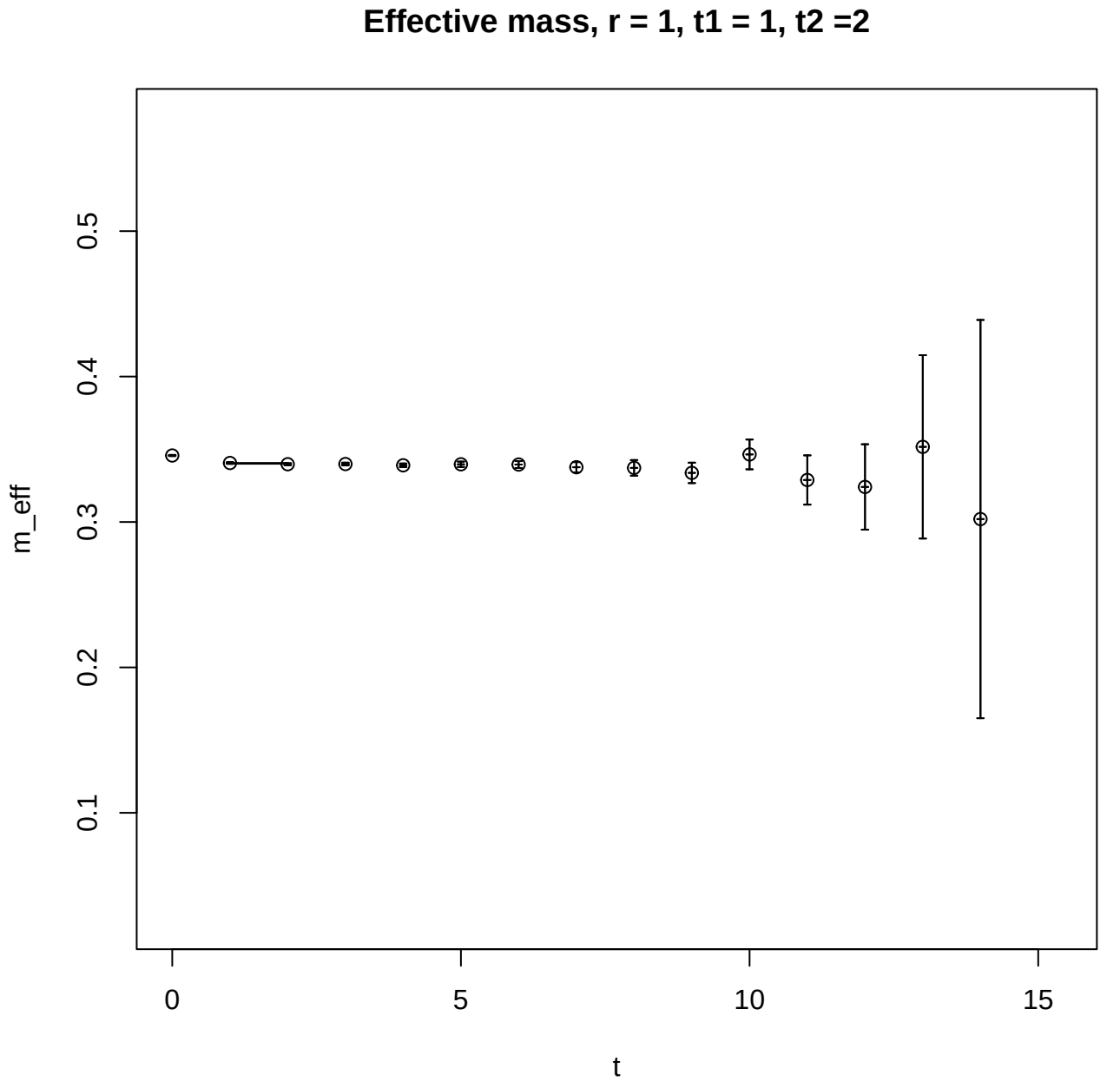
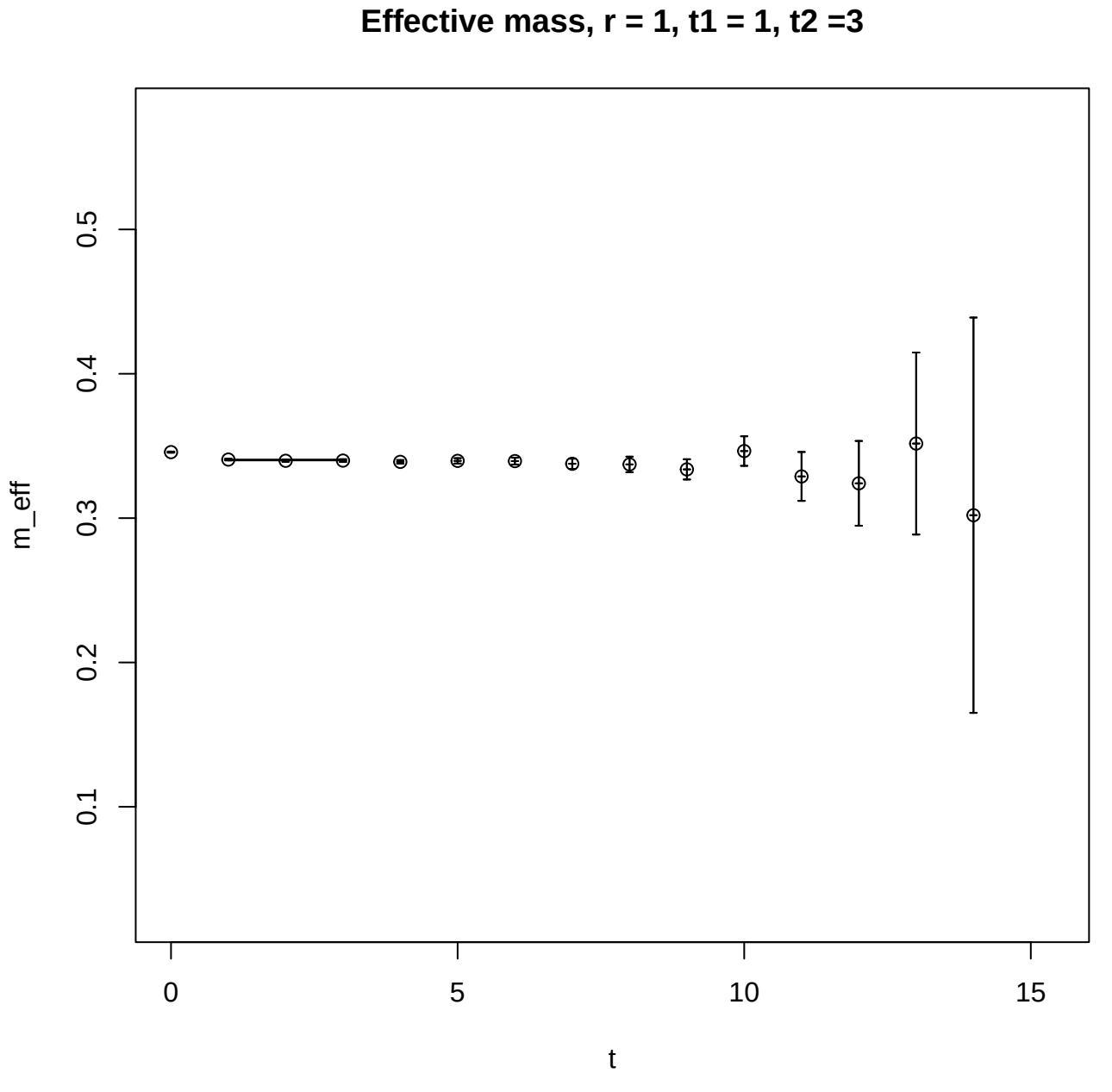
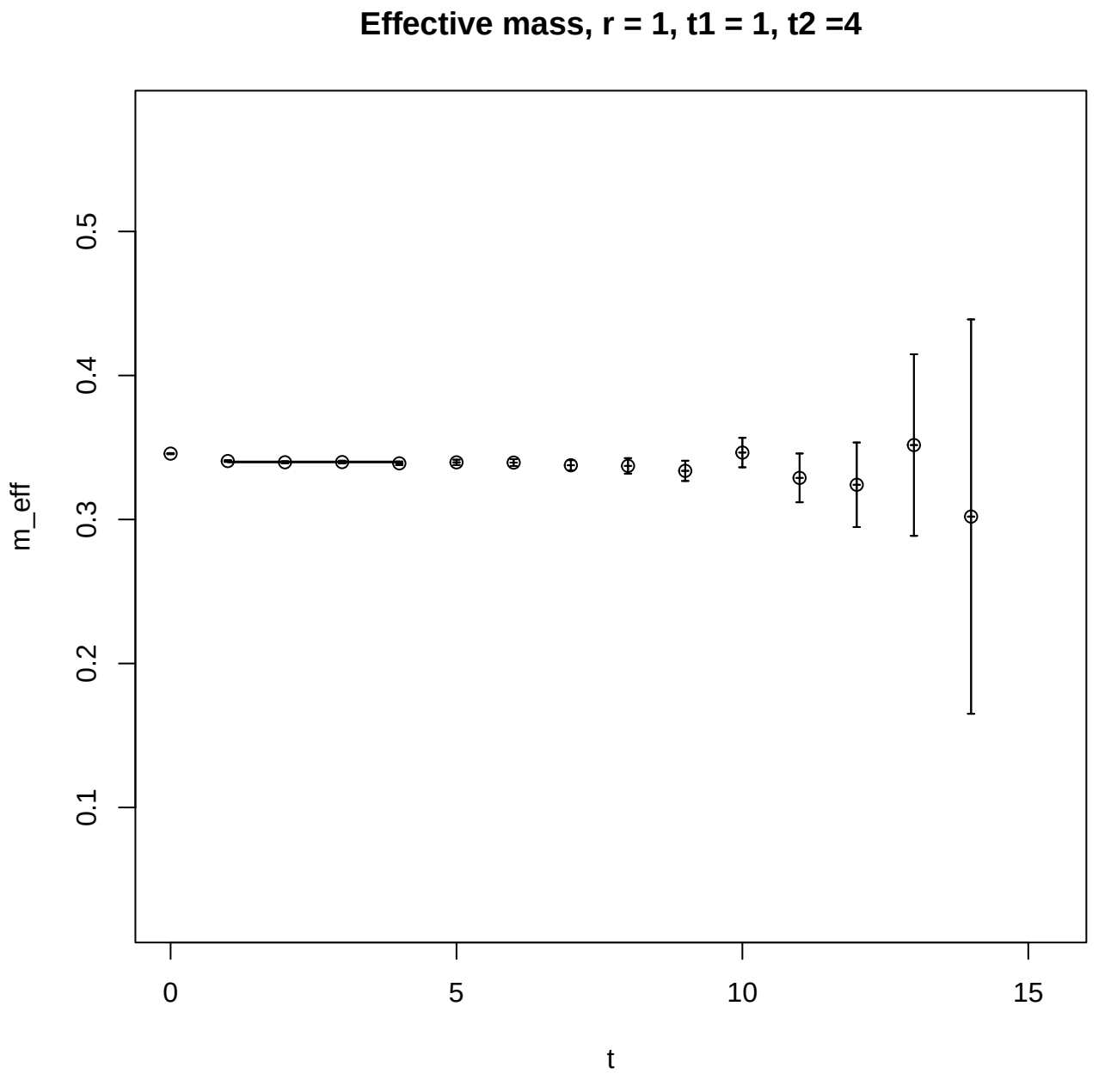
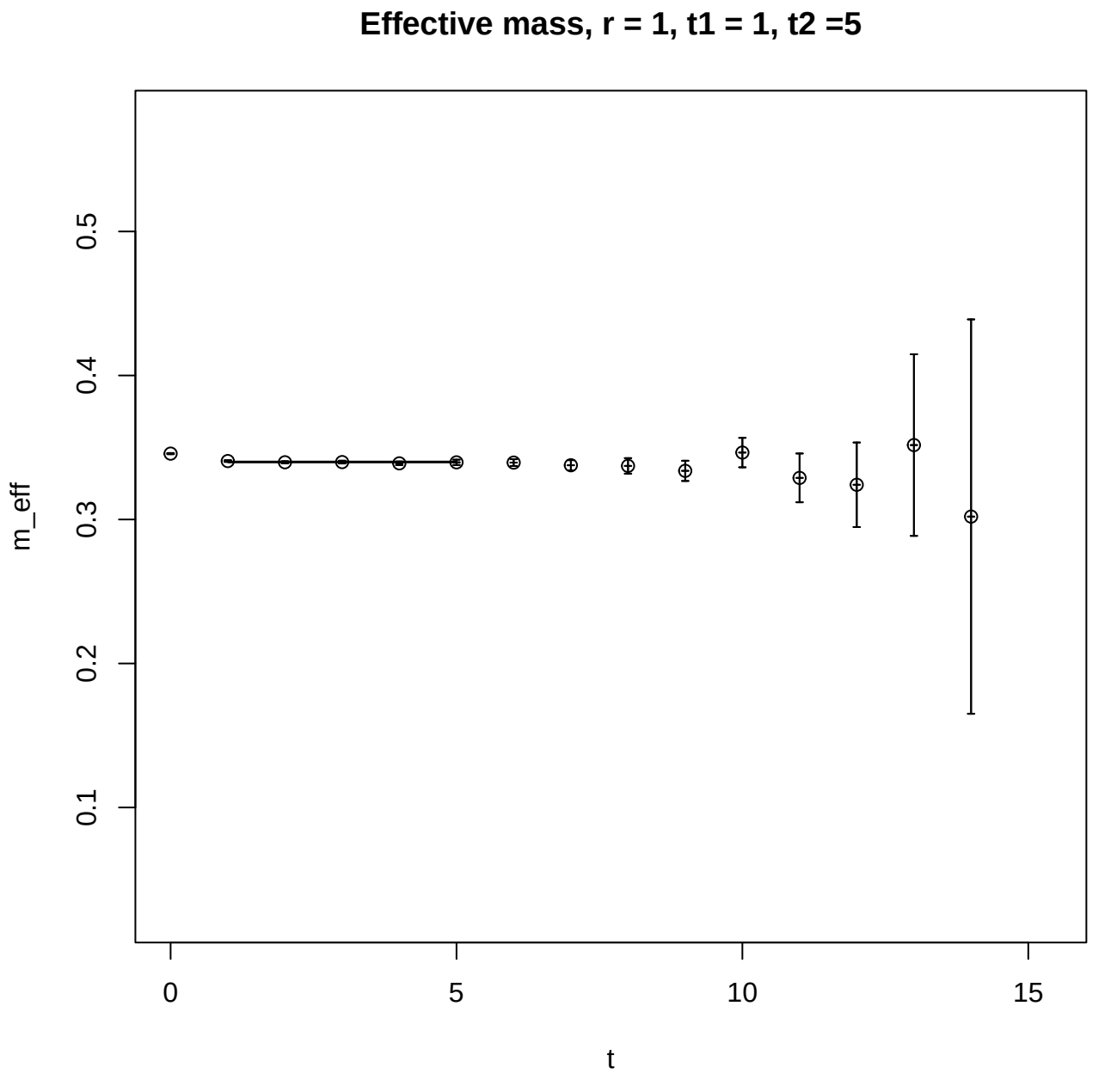
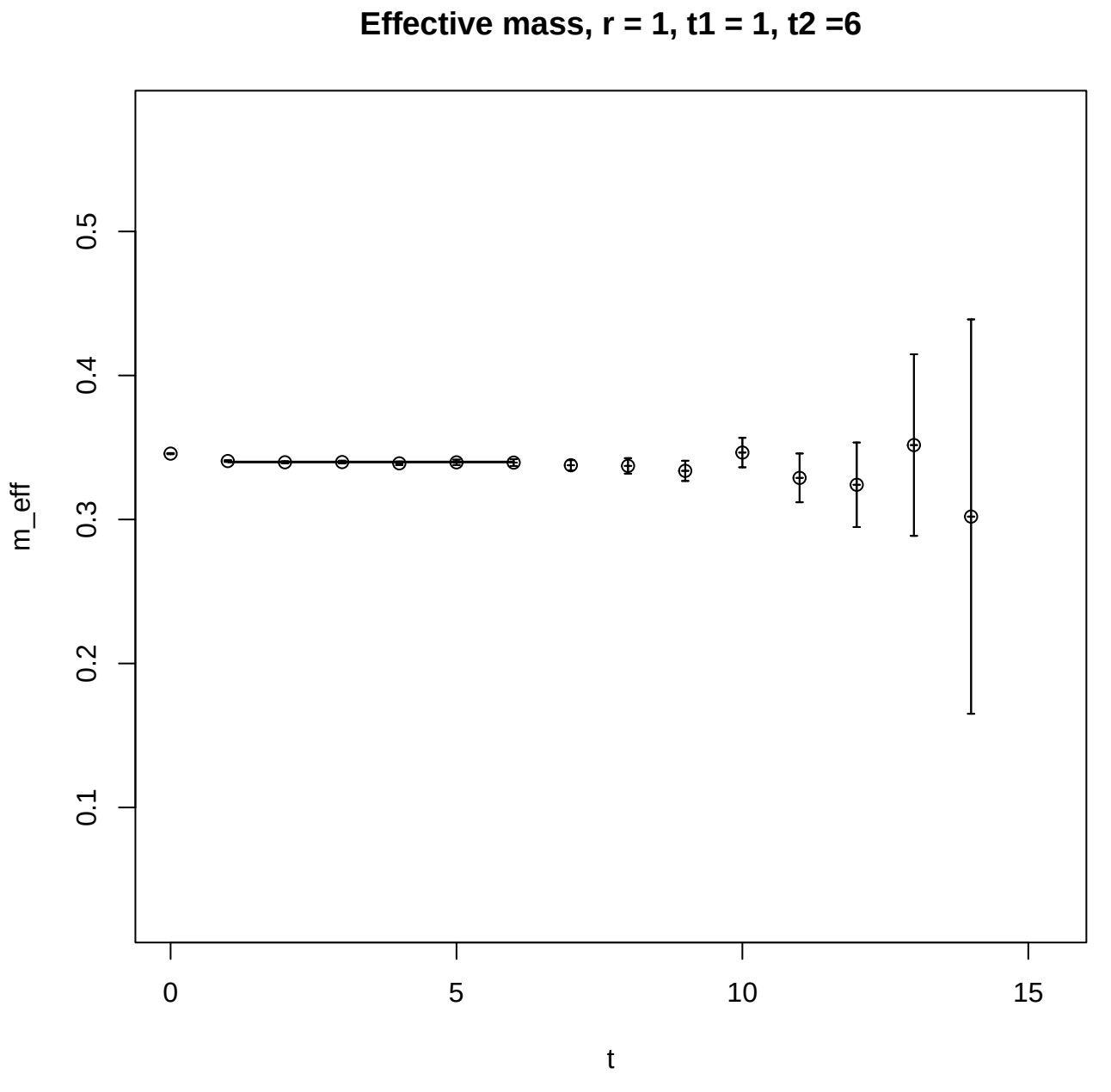
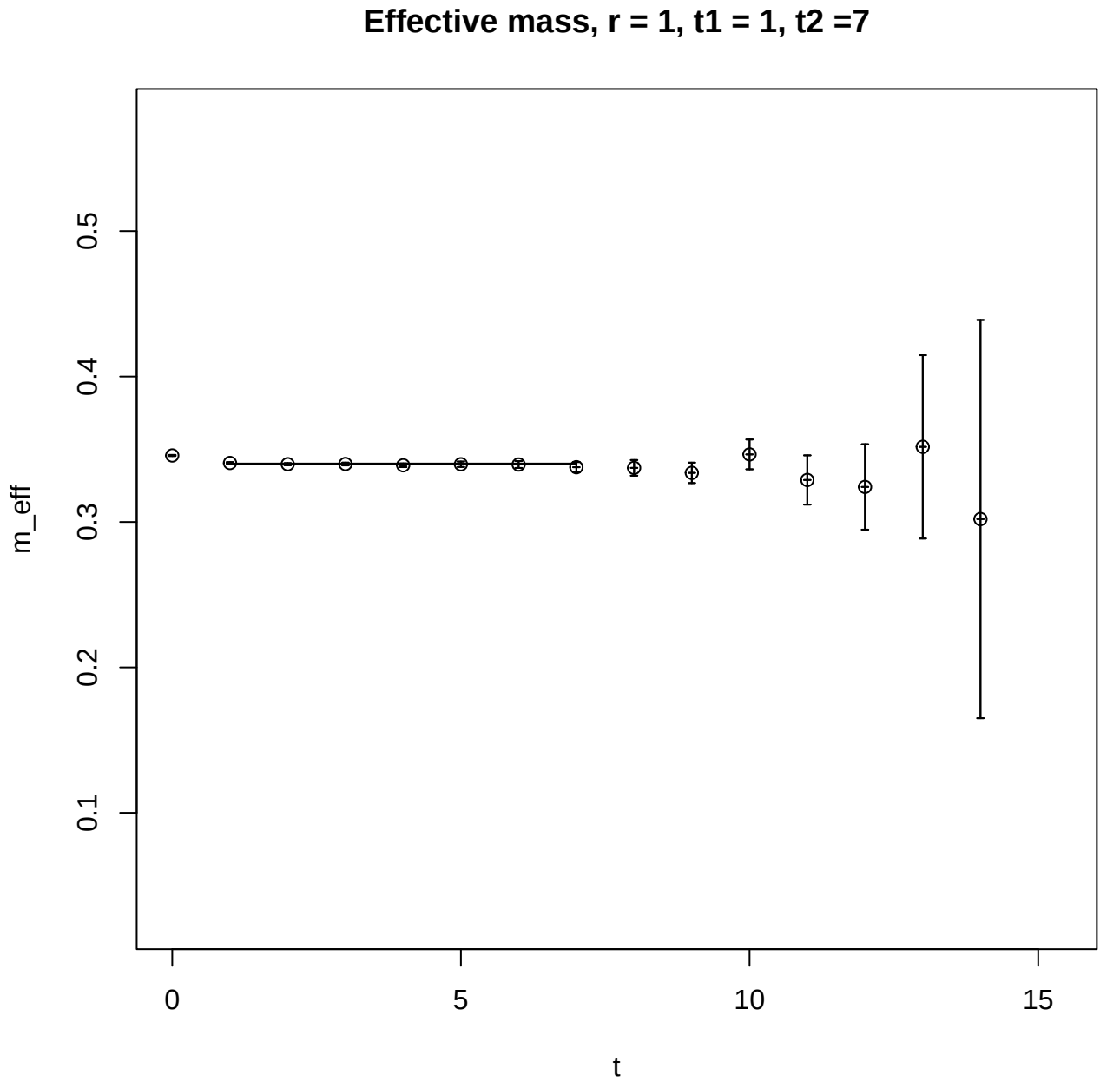

Effective mass, r = 1, t1 = 1, t2 =2
m_eff
0.1
0.2
0.3
0.4
0.5
0
5
10
15
t

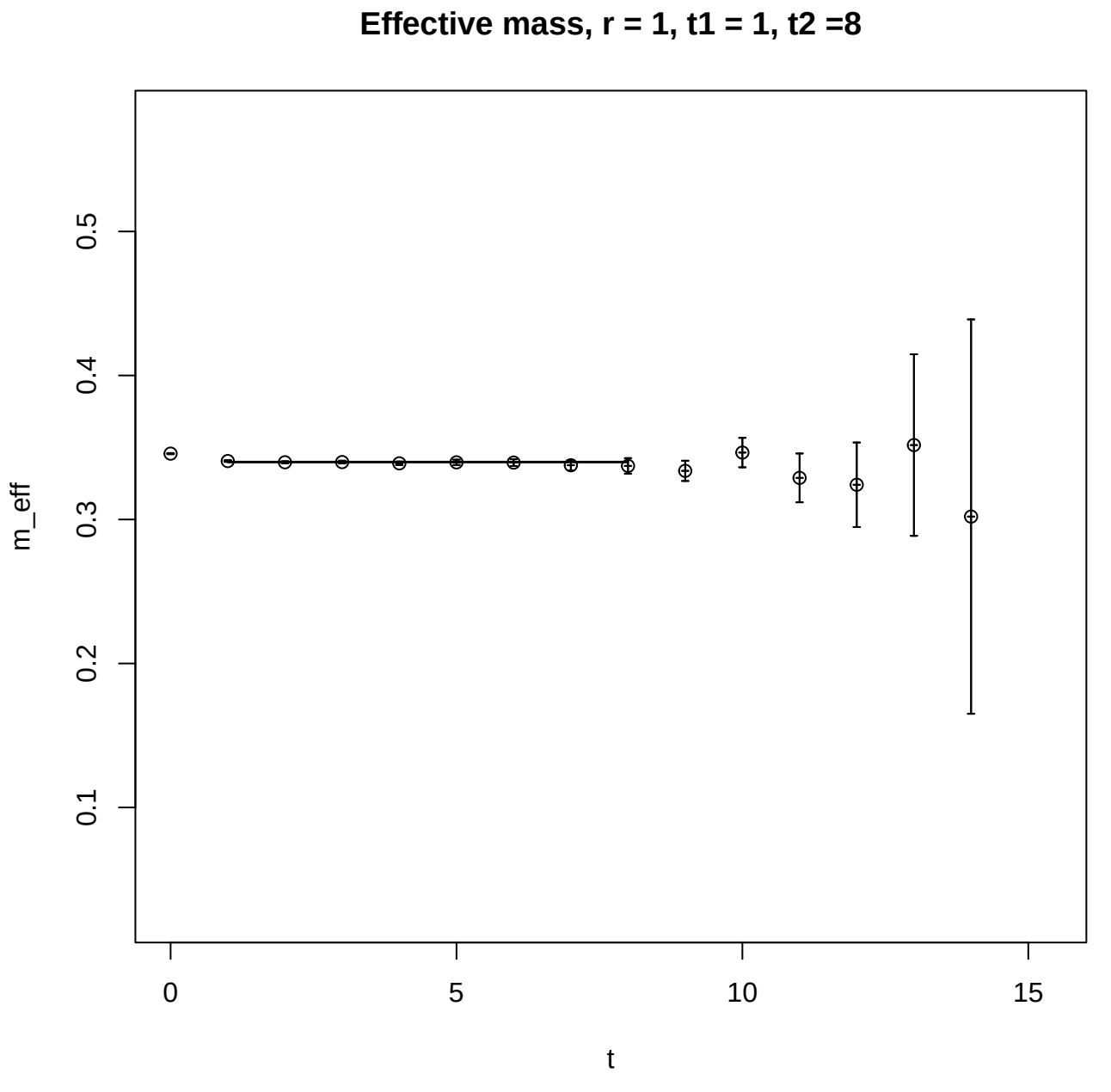

Effective mass, r = 1, t1 = 1, t2 =3
m_eff
0.1
0.2
0.3
0.4
0.5
0
5
10
15
t

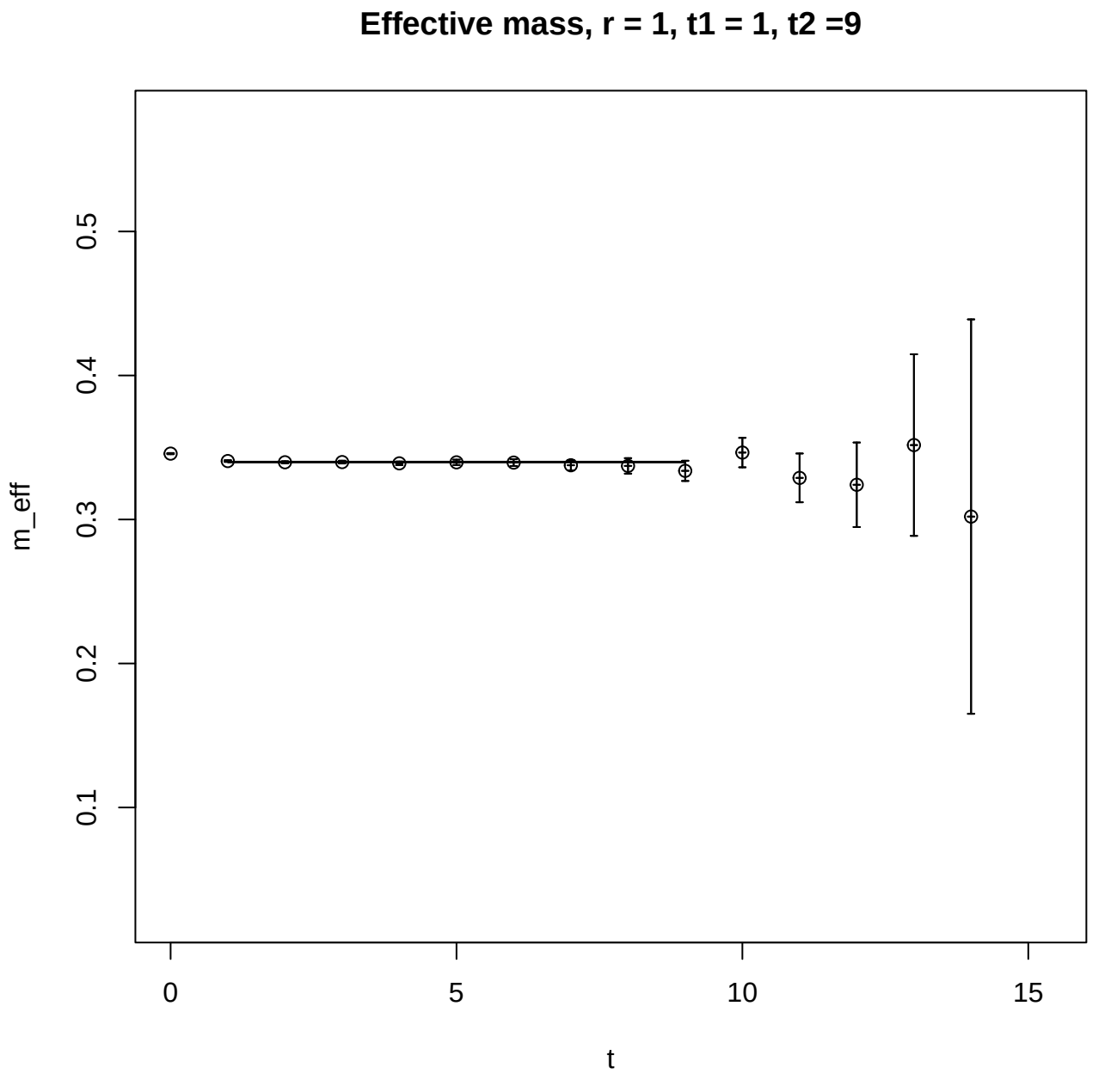

Effective mass, r = 1, t1 = 1, t2 =4
m_eff
0.1
0.2
0.3
0.4
0.5
0
5
10
15
t

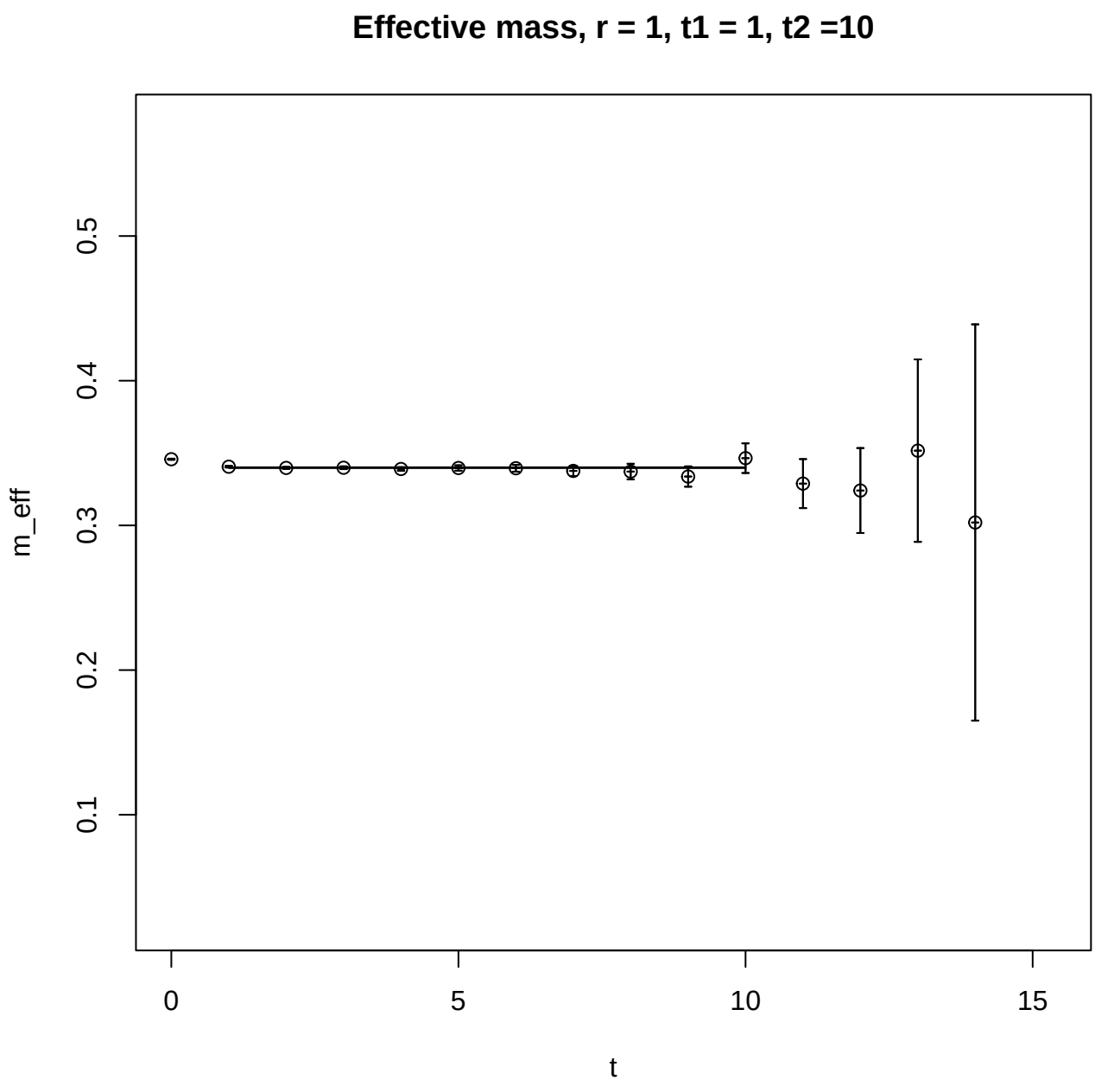

Effective mass, r = 1, t1 = 1, t2 =5
m_eff
0.1
0.2
0.3
0.4
0.5
0
5
10
15
t

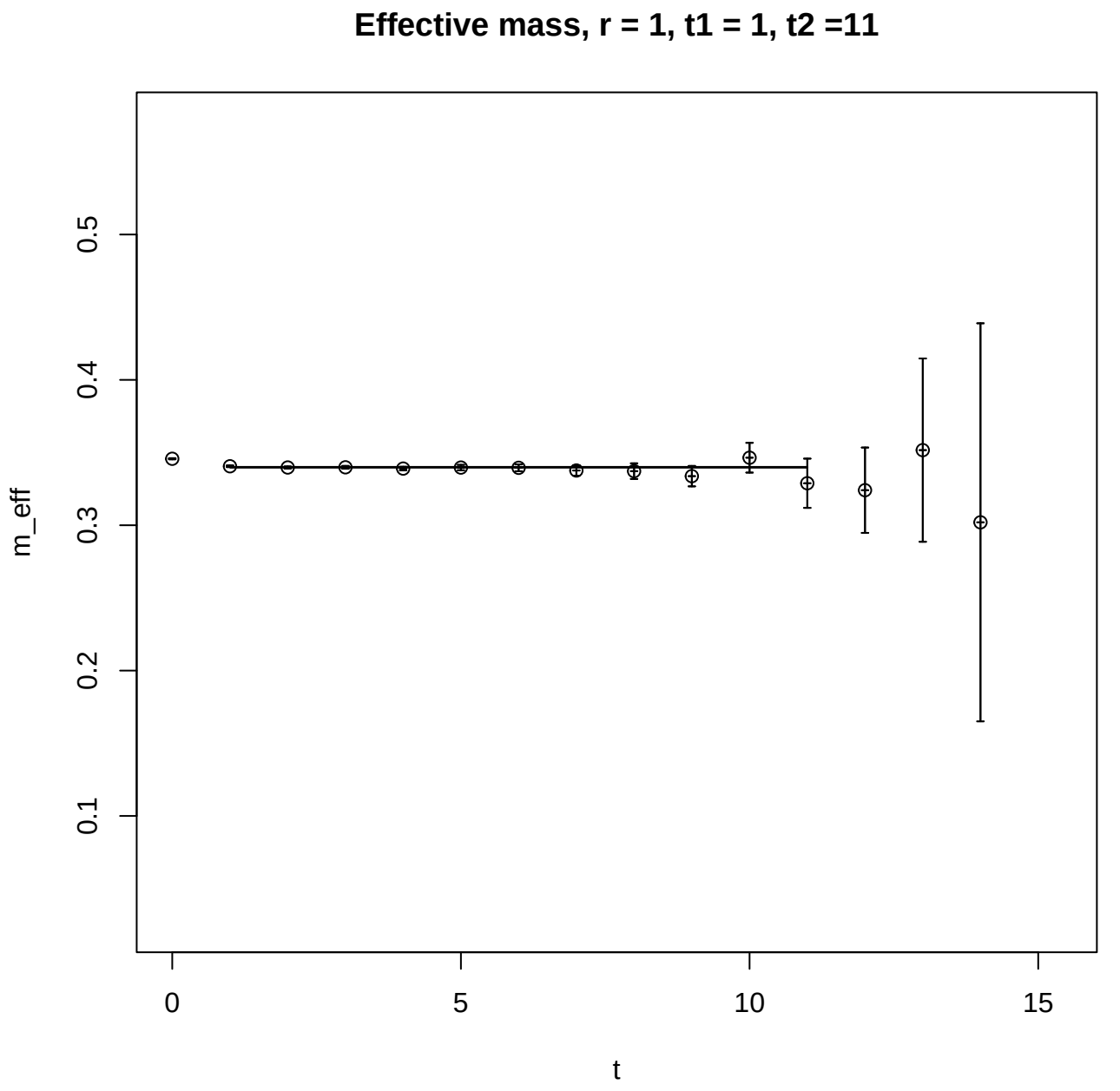

Effective mass, r = 1, t1 = 1, t2 =6
m_eff
0.1
0.2
0.3
0.4
0.5
0
5
10
15
t

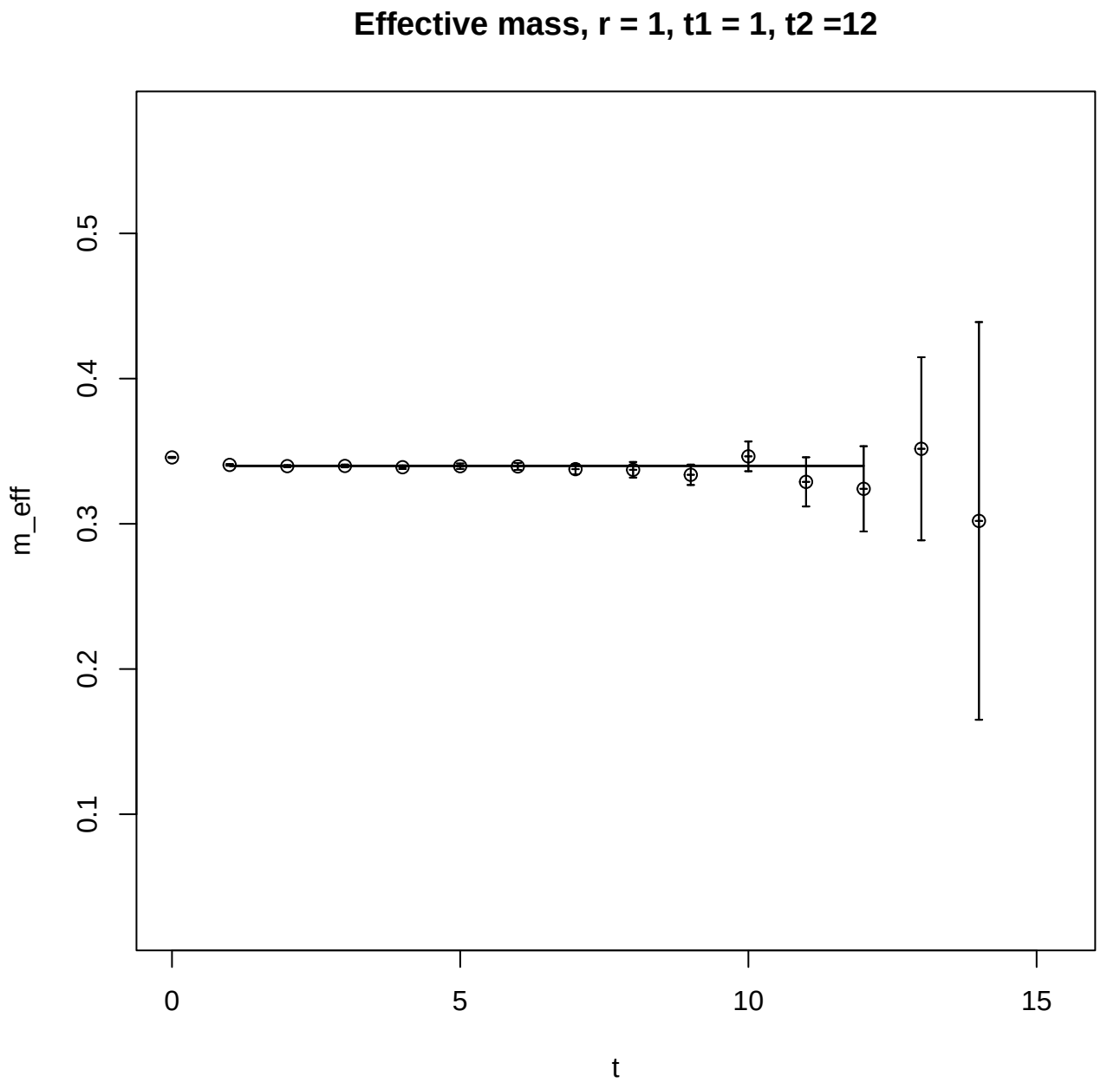

Effective mass, r = 1, t1 = 1, t2 =7
m_eff
0.1
0.2
0.3
0.4
0.5
0
5
10
15
t

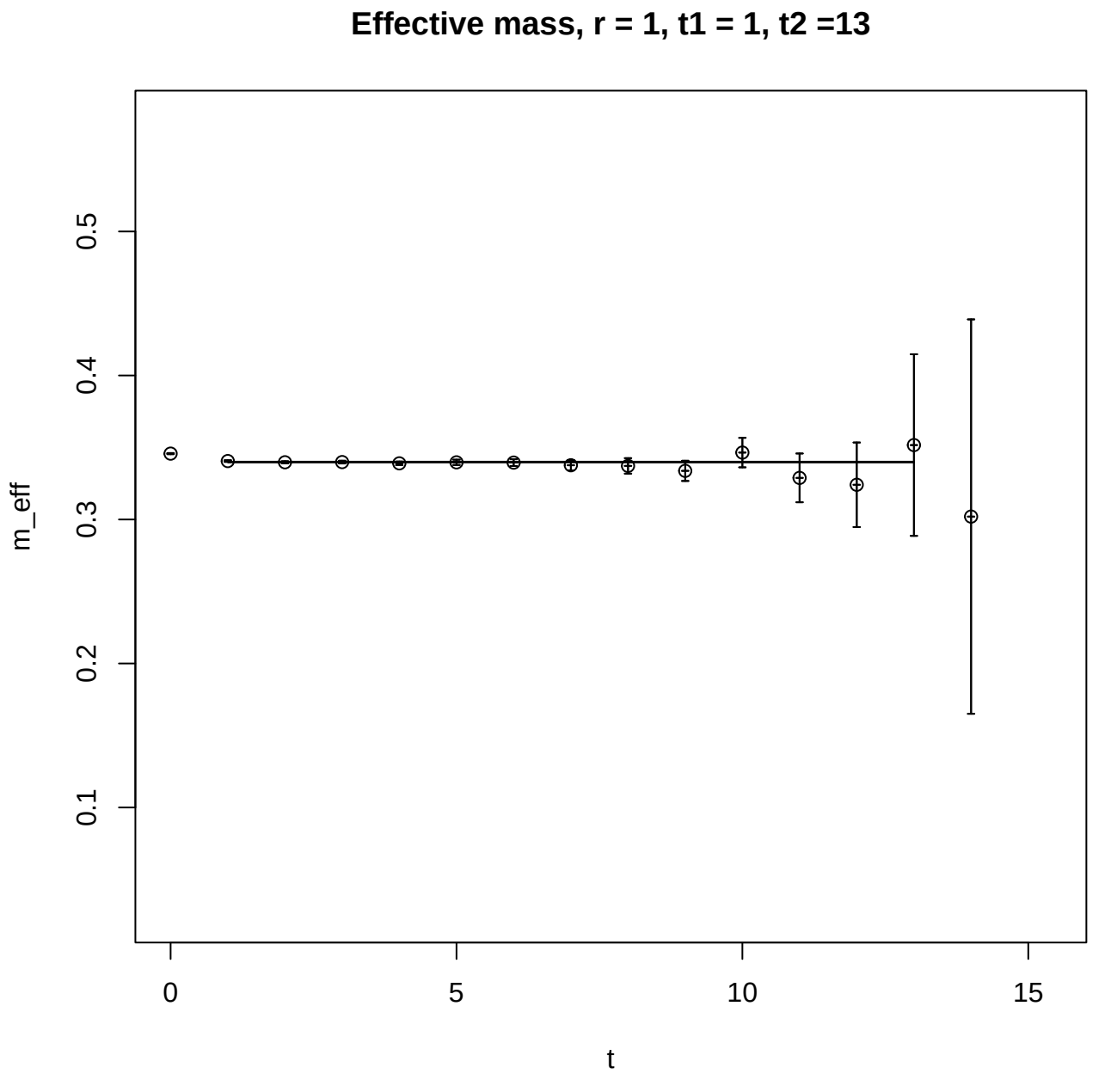

Effective mass, r = 1, t1 = 1, t2 =8
m_eff
0.1
0.2
0.3
0.4
0.5
0
5
10
15
t

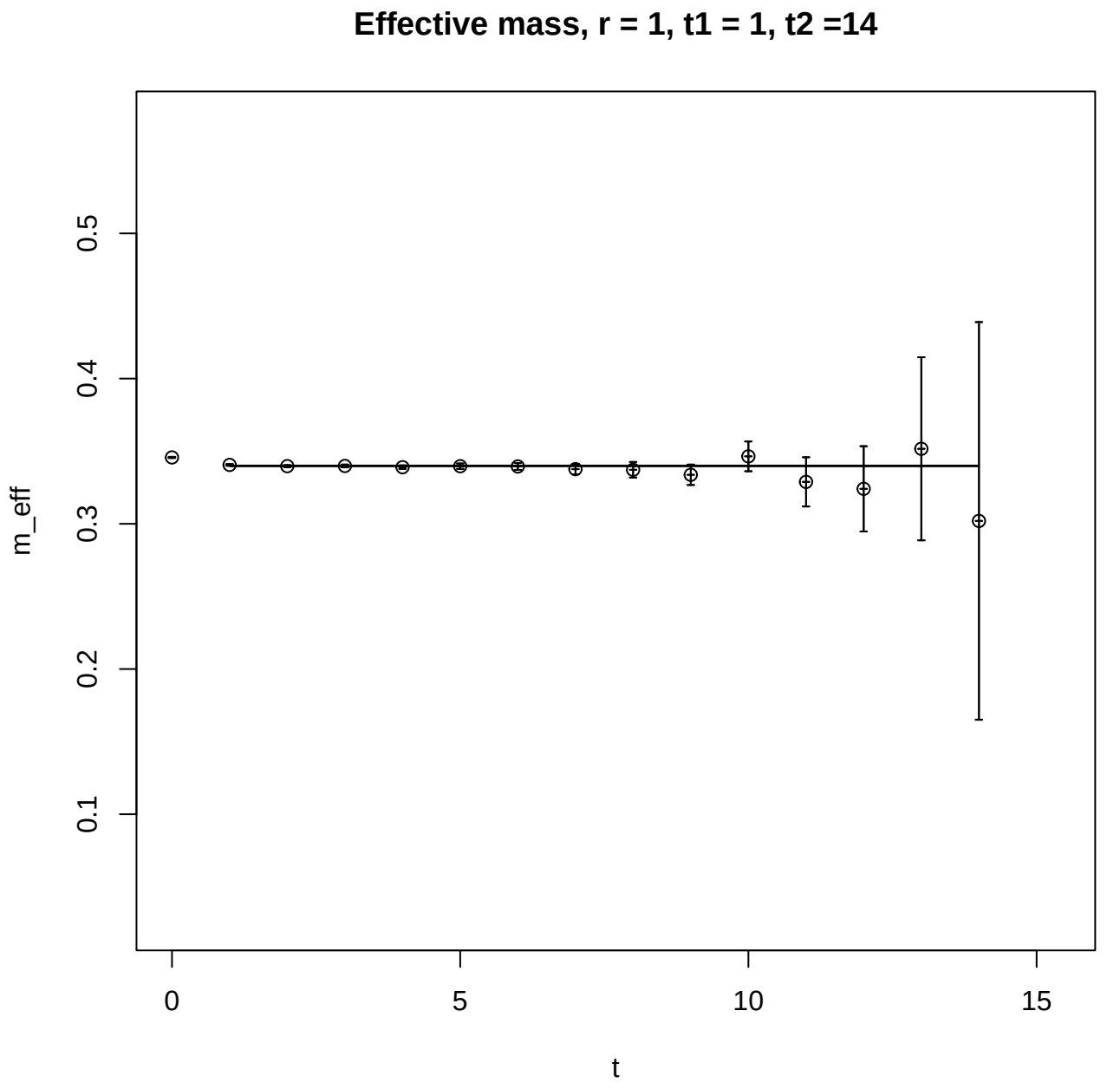

Effective mass, r = 1, t1 = 1, t2 =9
m_eff
0.1
0.2
0.3
0.4
0.5
0
5
10
15
t

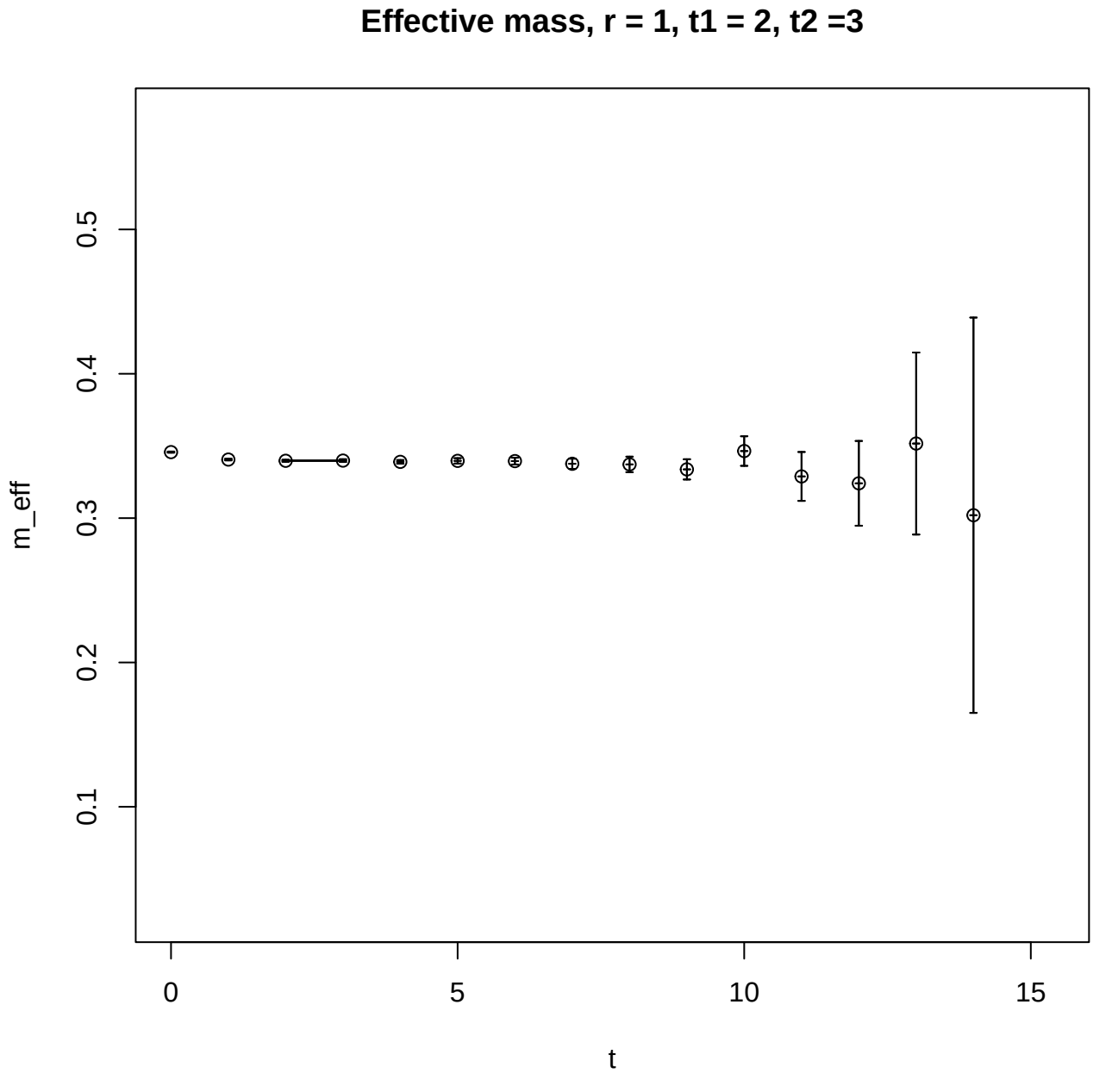

Effective mass, r = 1, t1 = 1, t2 =10
m_eff
0.1
0.2
0.3
0.4
0.5
0
5
10
15
t

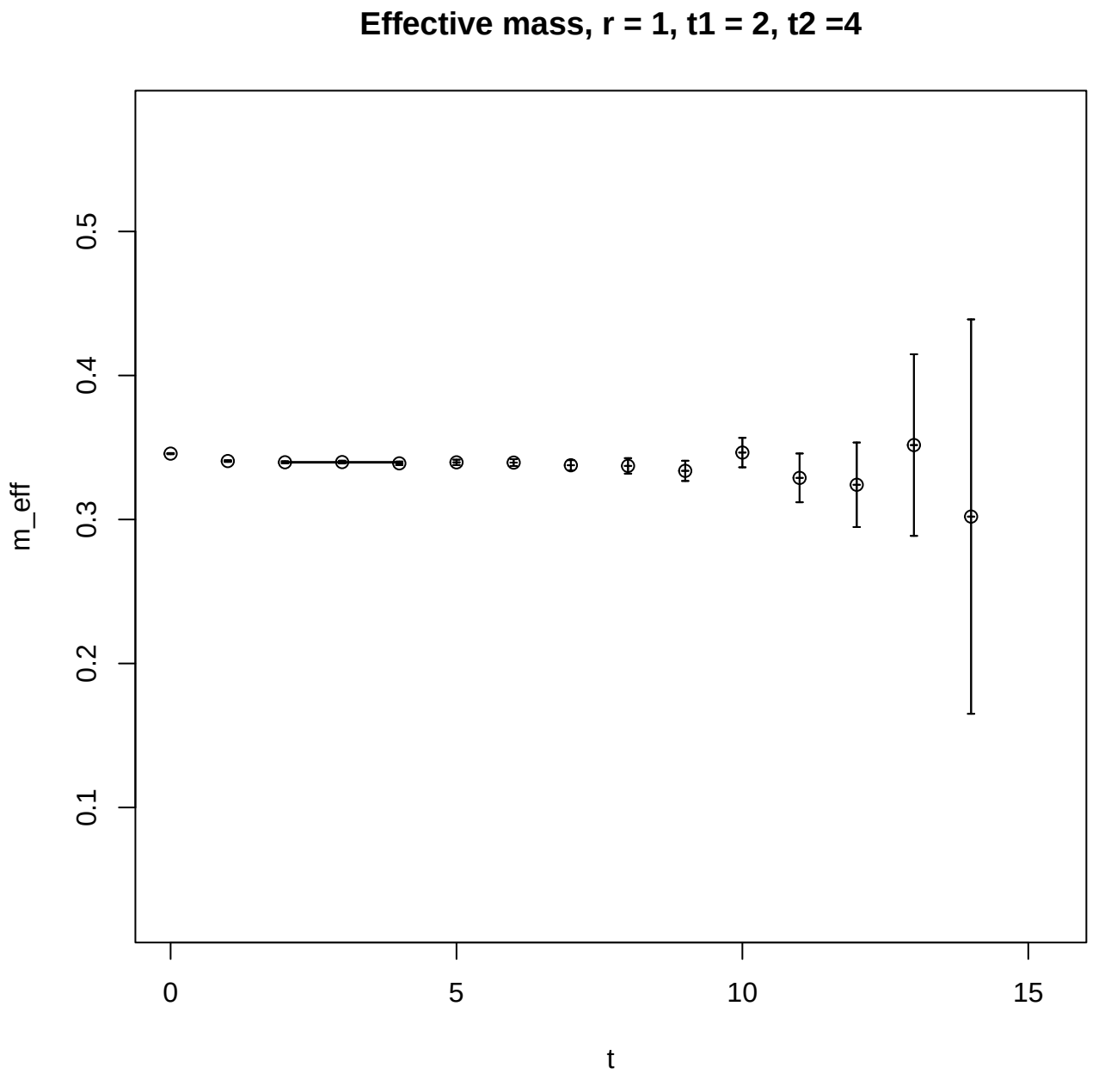

Effective mass, r = 1, t1 = 1, t2 =11
m_eff
0.1
0.2
0.3
0.4
0.5
0
5
10
15
t

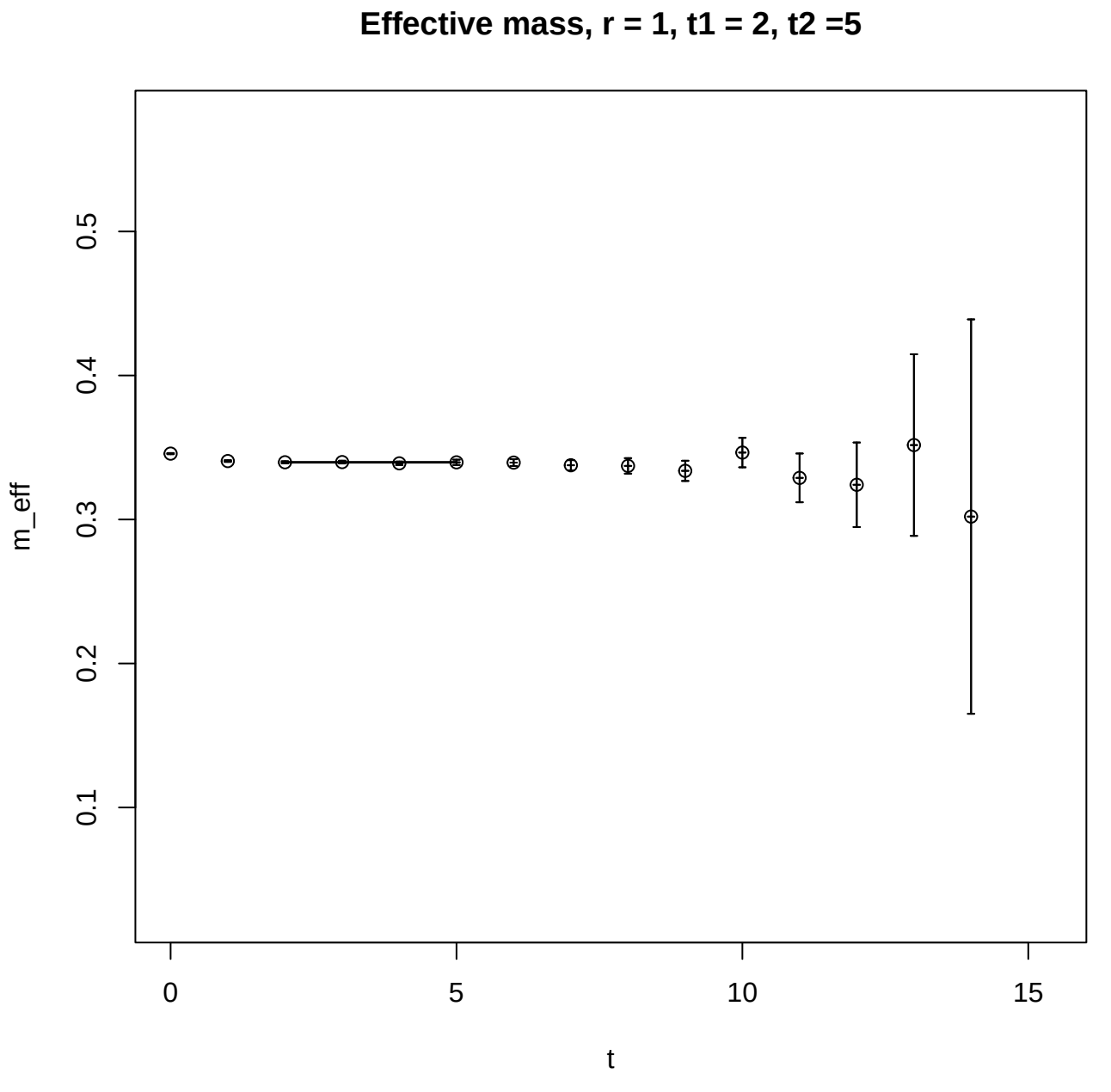

Effective mass, r = 1, t1 = 1, t2 =12
m_eff
0.1
0.2
0.3
0.4
0.5
0
5
10
15
t

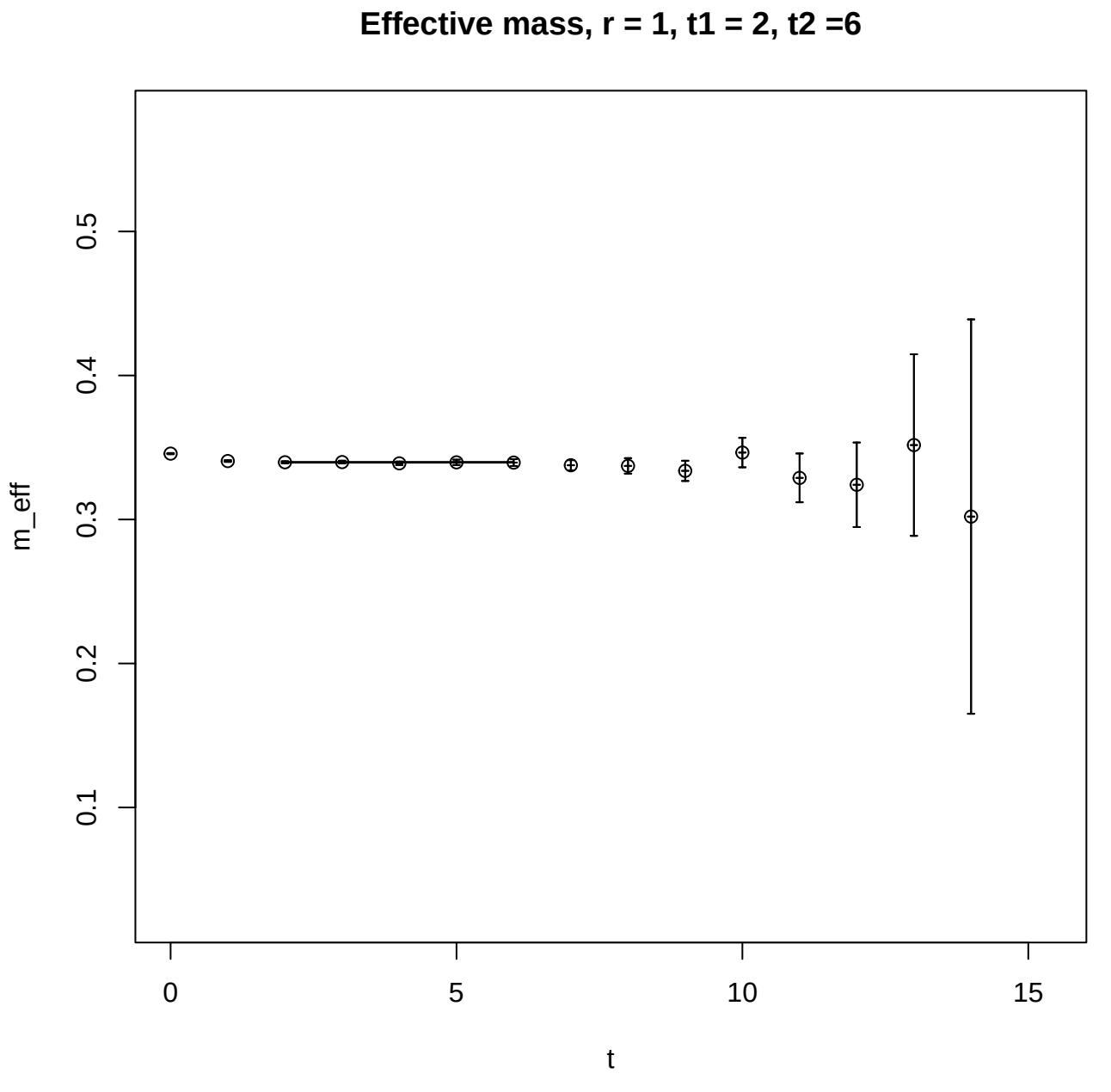

Effective mass, r = 1, t1 = 1, t2 =13
m_eff
0.1
0.2
0.3
0.4
0.5
0
5
10
15
t

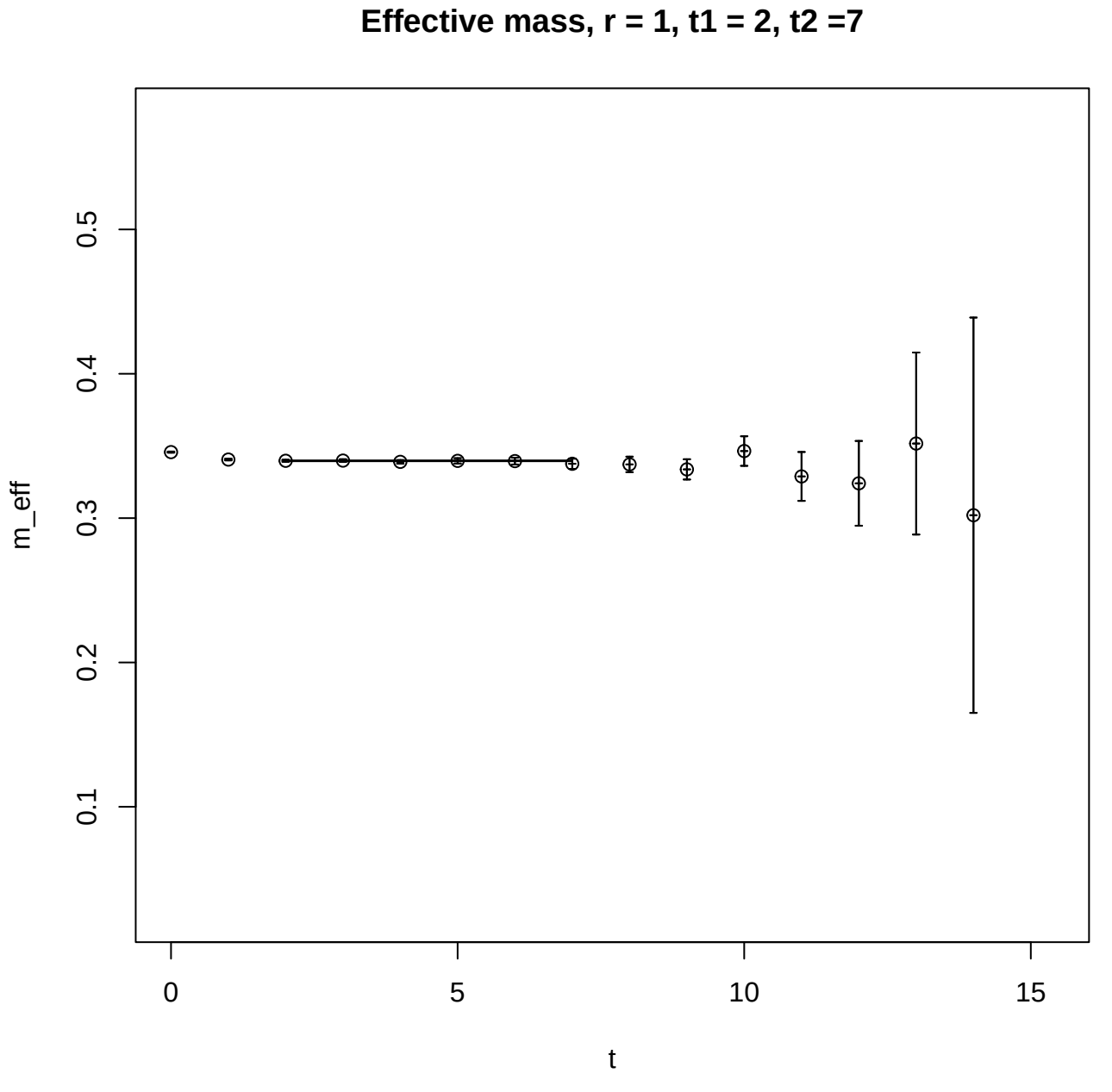

Effective mass, r = 1, t1 = 1, t2 =14
m_eff
0.1
0.2
0.3
0.4
0.5
0
5
10
15
t

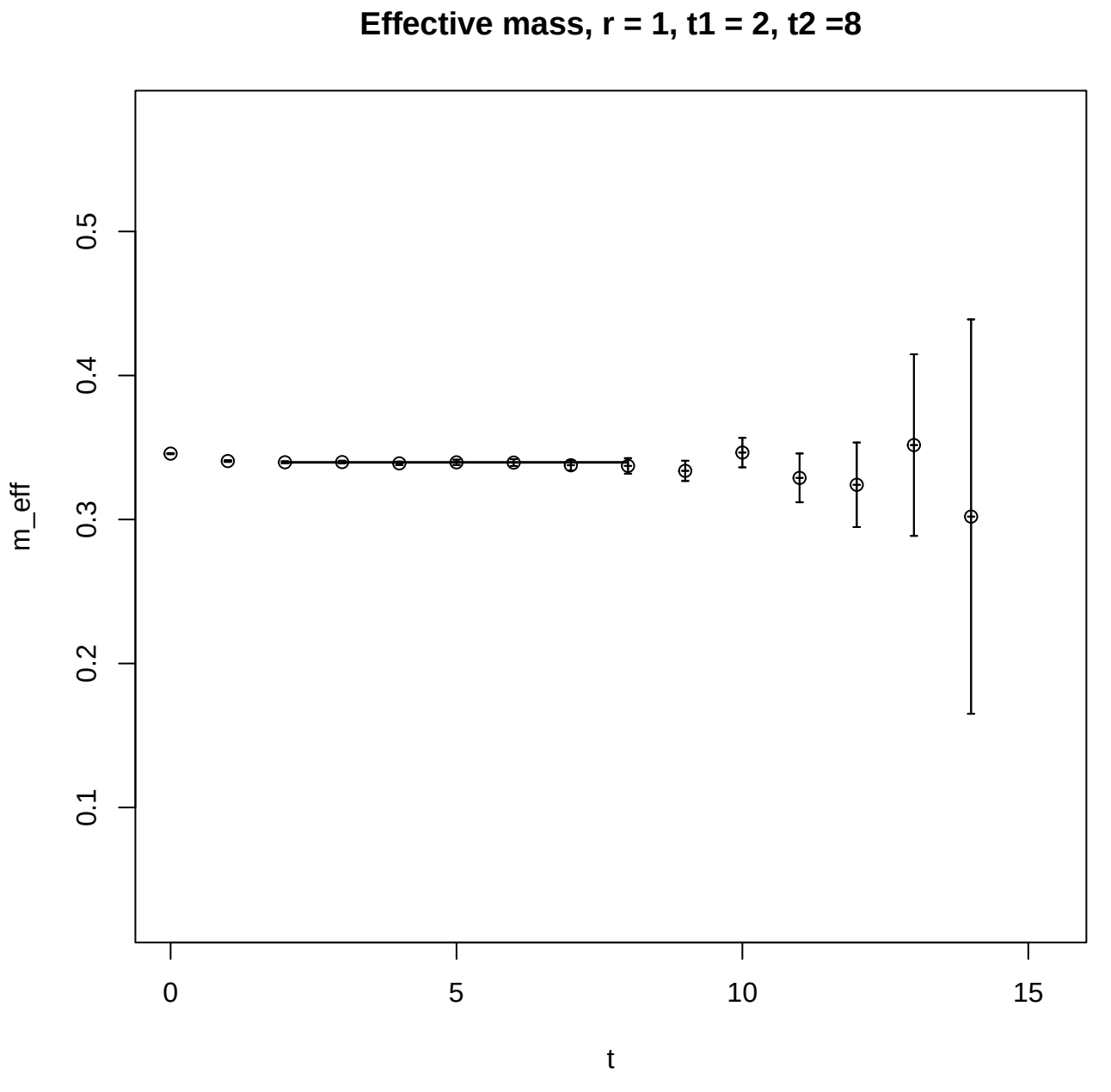

Effective mass, r = 1, t1 = 2, t2 =3
m_eff
0.1
0.2
0.3
0.4
0.5
0
5
10
15
t

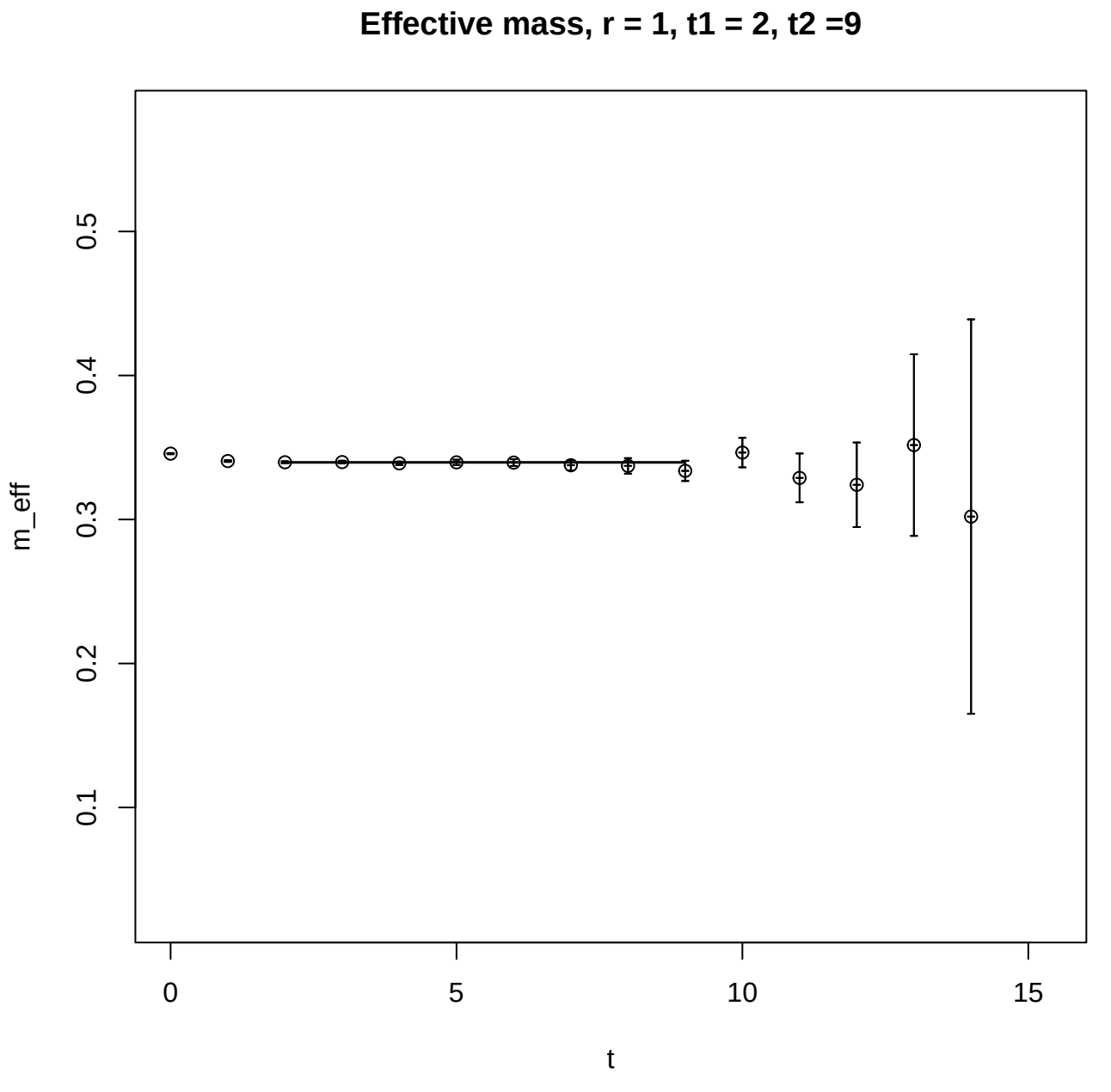

Effective mass, r = 1, t1 = 2, t2 =4
m_eff
0.1
0.2
0.3
0.4
0.5
0
5
10
15
t

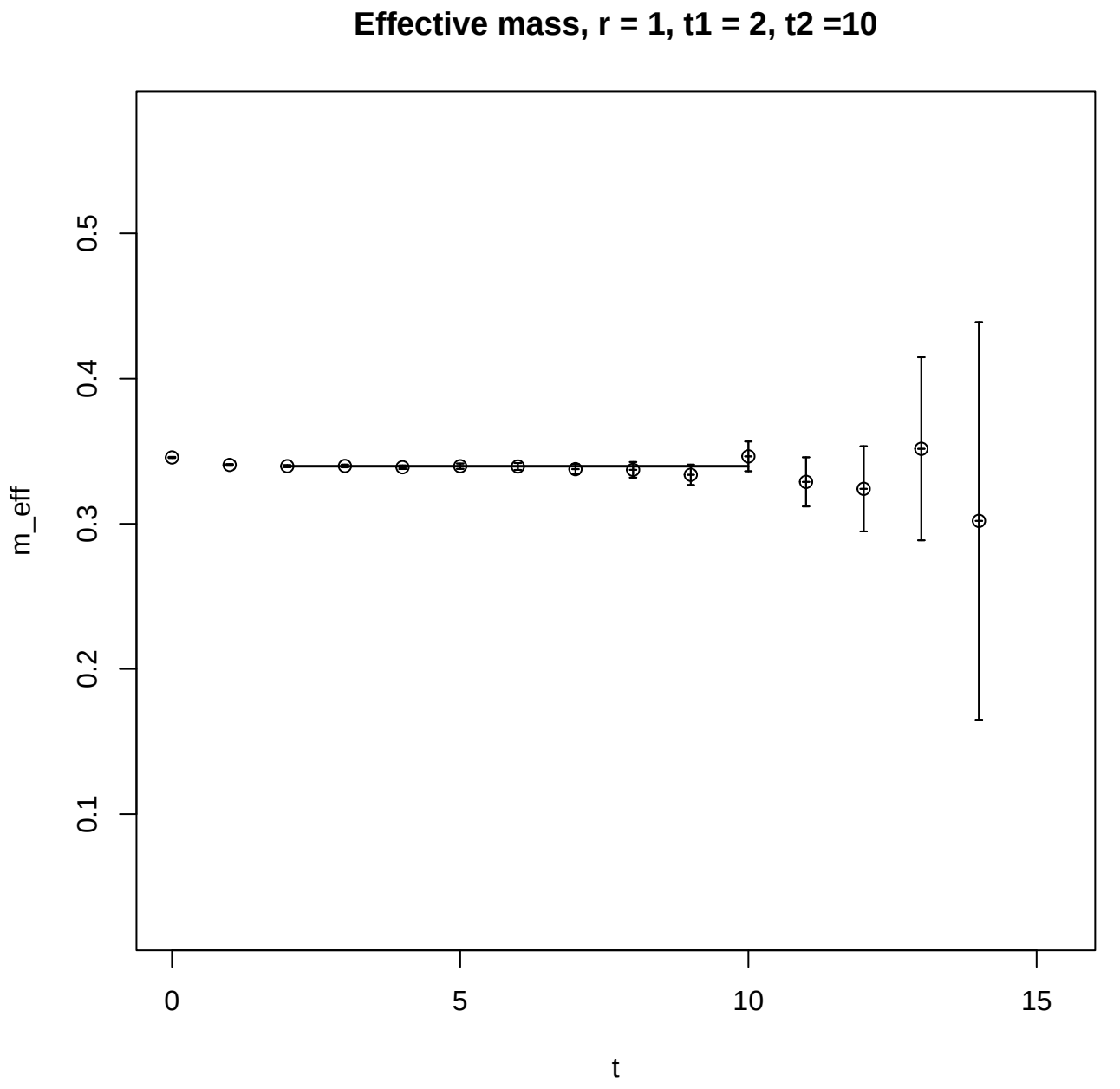

Effective mass, r = 1, t1 = 2, t2 =5
m_eff
0.1
0.2
0.3
0.4
0.5
0
5
10
15
t

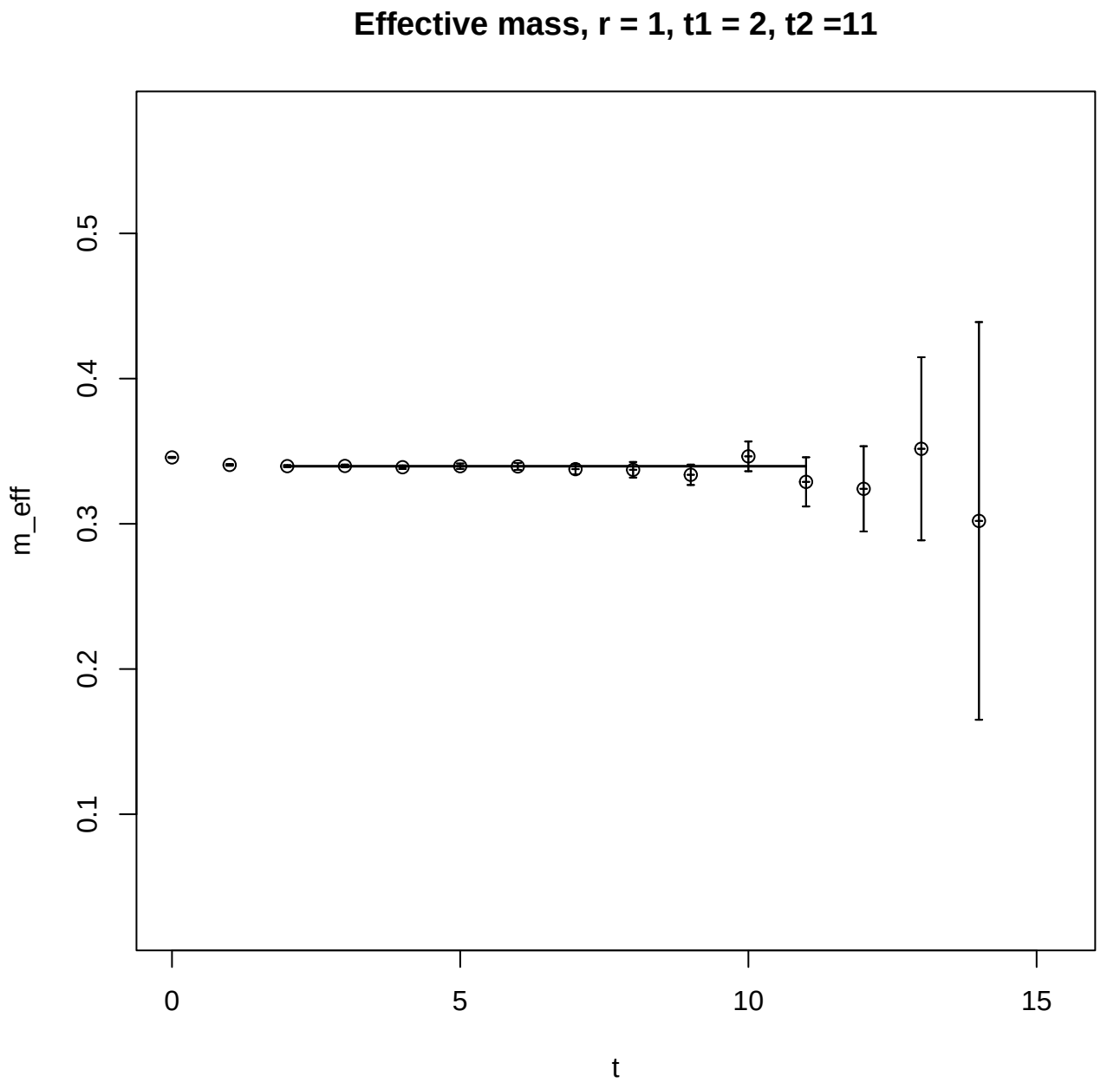

Effective mass, r = 1, t1 = 2, t2 =6
m_eff
0.1
0.2
0.3
0.4
0.5
0
5
10
15
t

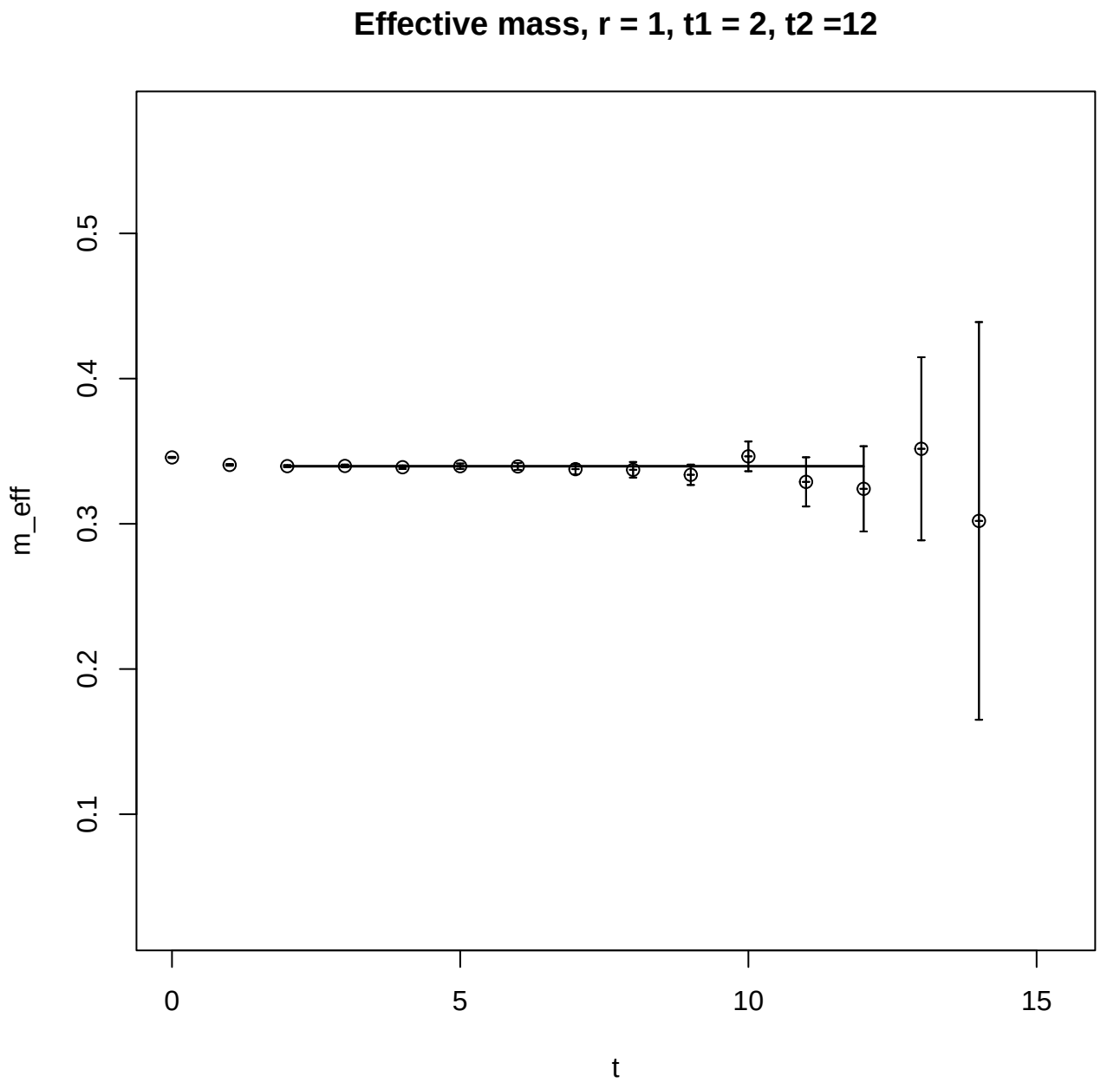

Effective mass, r = 1, t1 = 2, t2 =7
m_eff
0.1
0.2
0.3
0.4
0.5
0
5
10
15
t

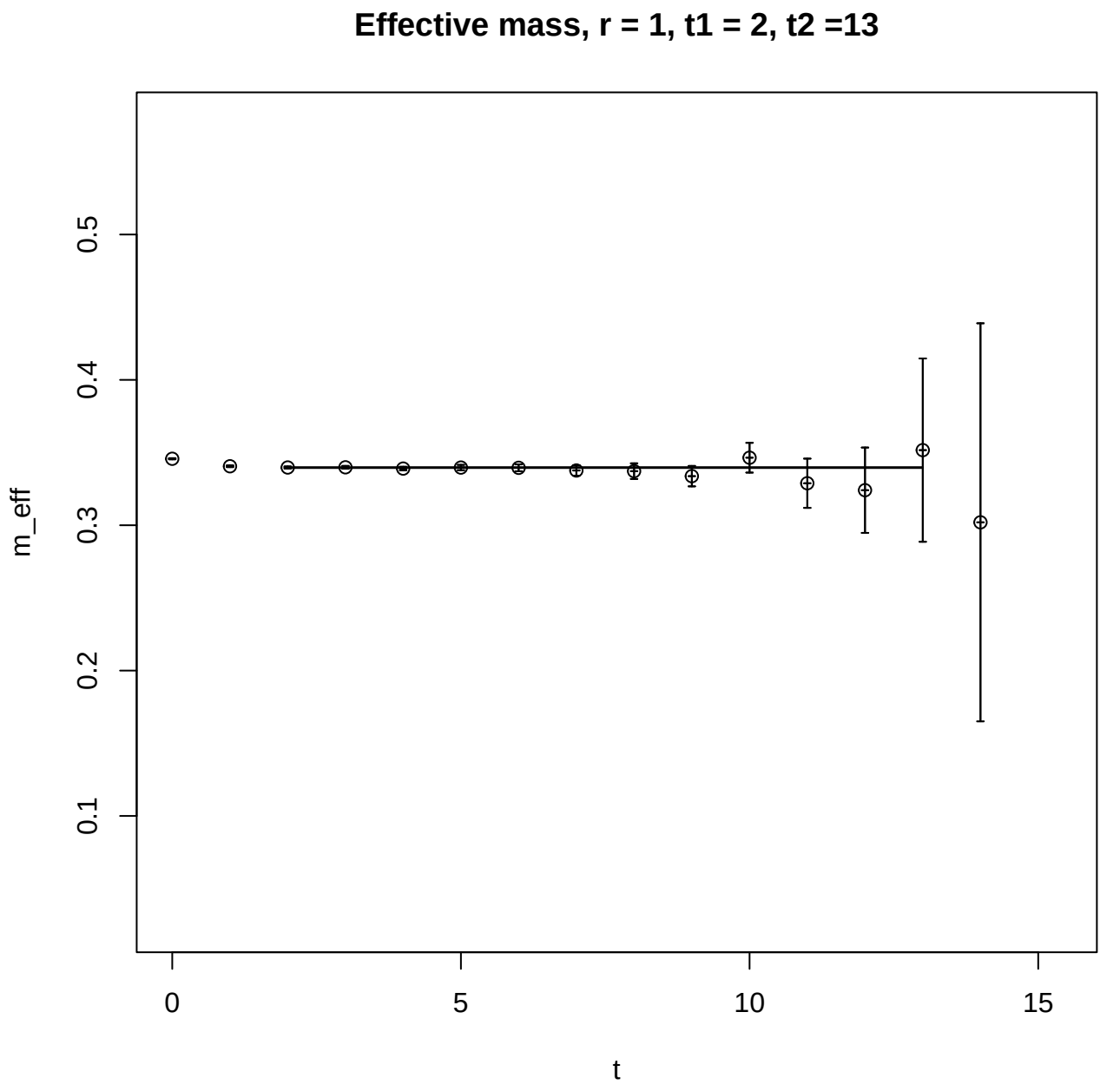

Effective mass, r = 1, t1 = 2, t2 =8
m_eff
0.1
0.2
0.3
0.4
0.5
0
5
10
15
t

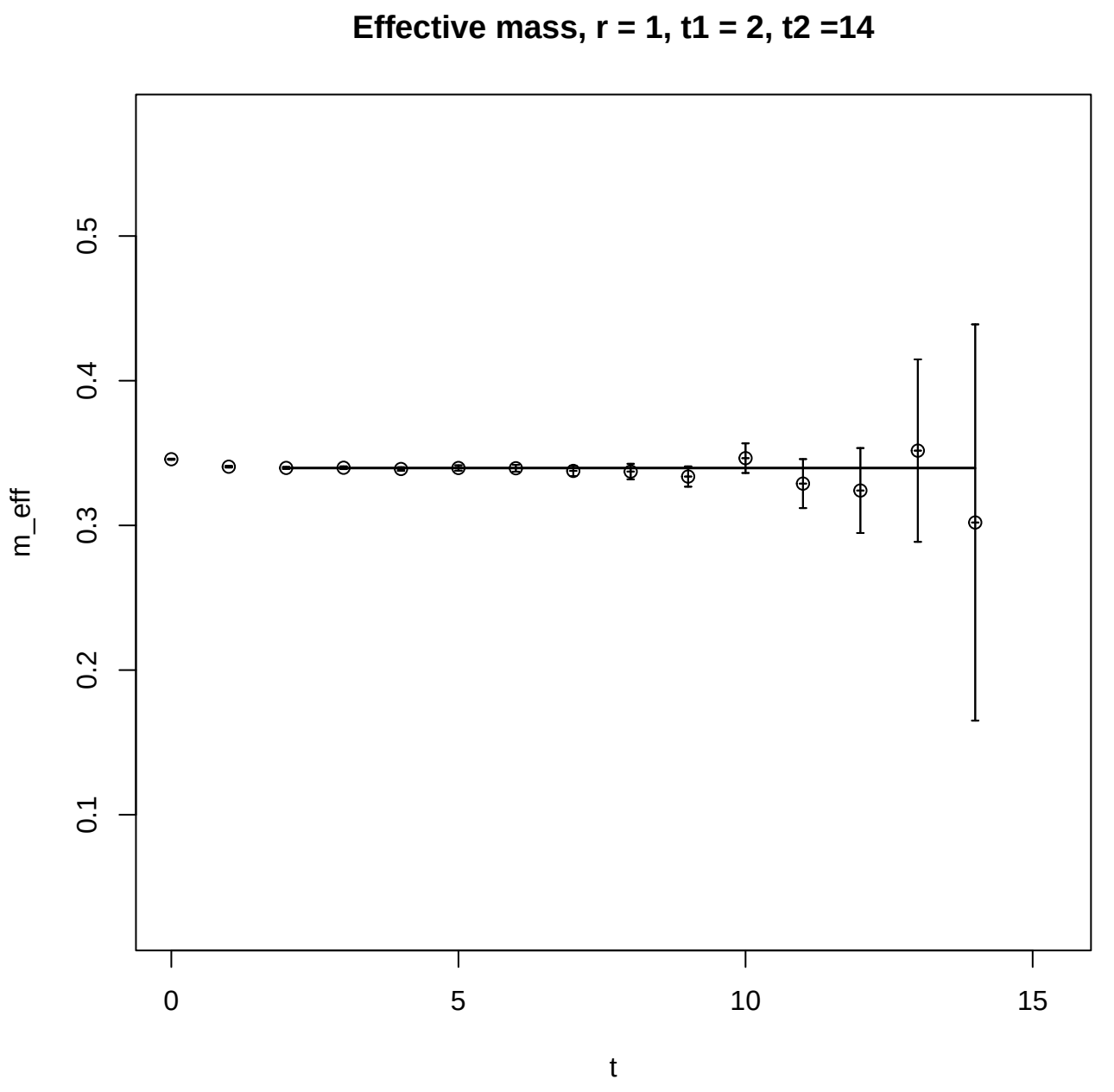

Effective mass, r = 1, t1 = 2, t2 =9
m_eff
0.1
0.2
0.3
0.4
0.5
0
5
10
15
t

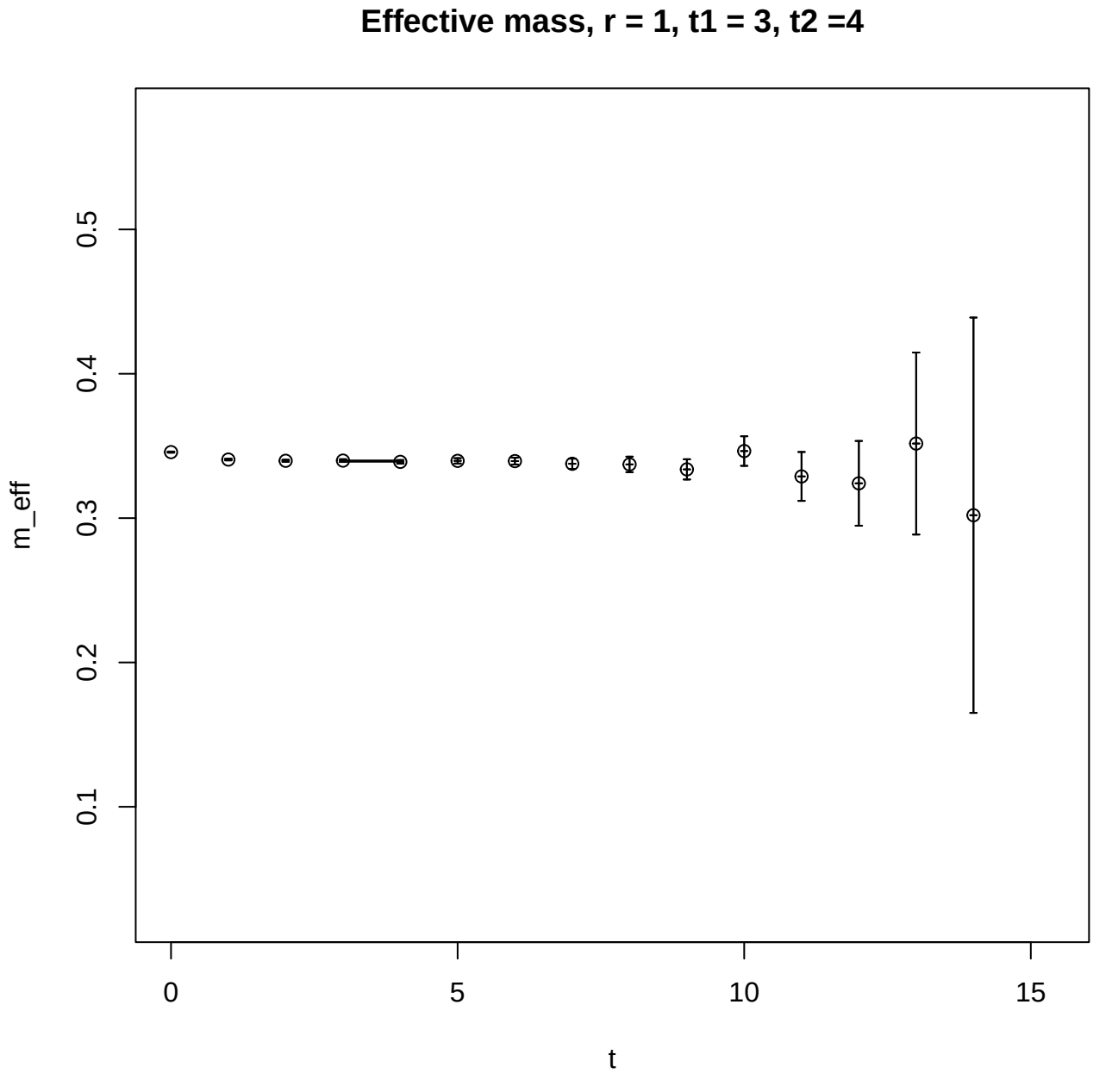

Effective mass, r = 1, t1 = 2, t2 =10
m_eff
0.1
0.2
0.3
0.4
0.5
0
5
10
15
t

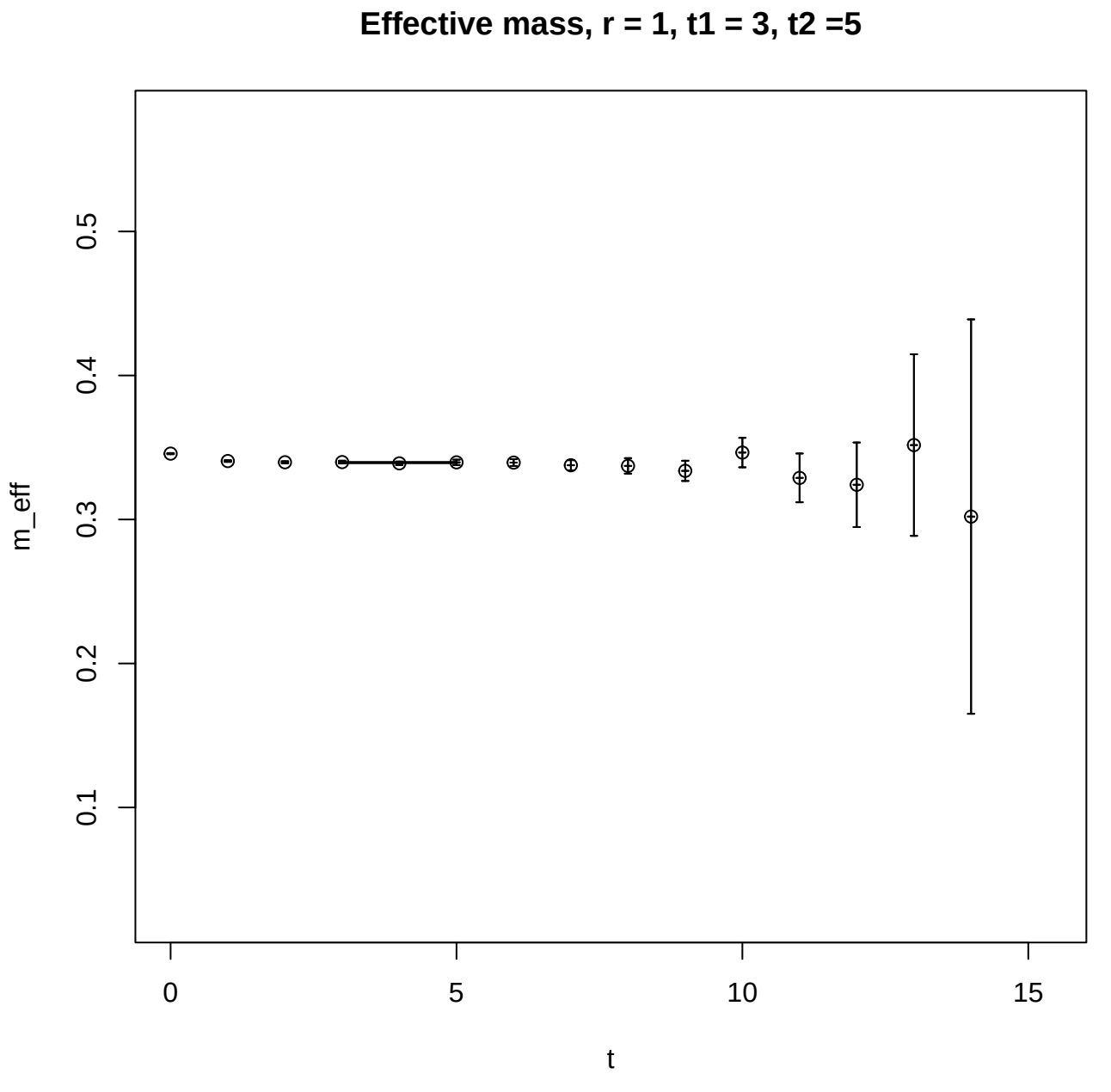

Effective mass, r = 1, t1 = 2, t2 =11
m_eff
0.1
0.2
0.3
0.4
0.5
0
5
10
15
t

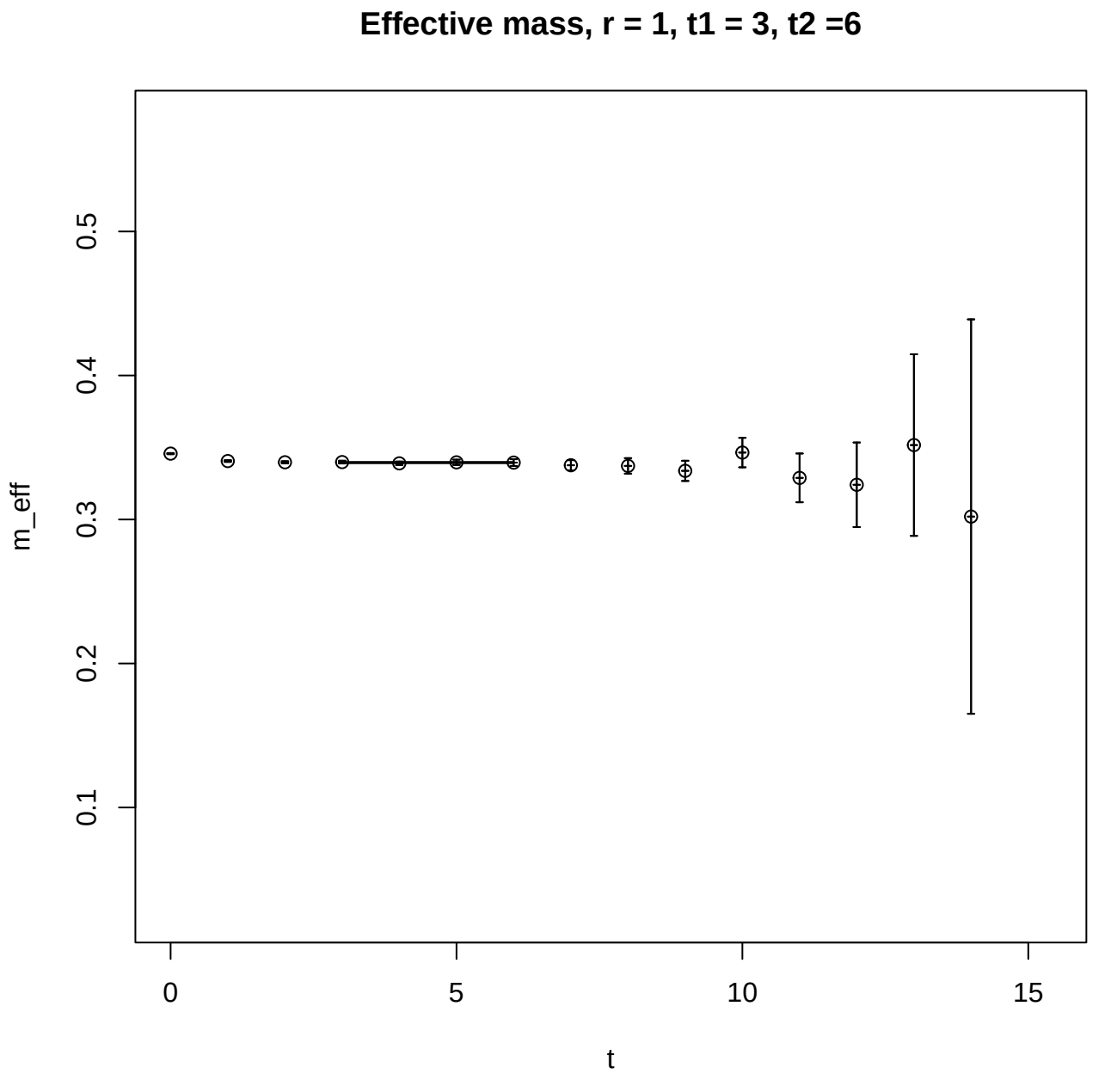

Effective mass, r = 1, t1 = 2, t2 =12
m_eff
0.1
0.2
0.3
0.4
0.5
0
5
10
15
t

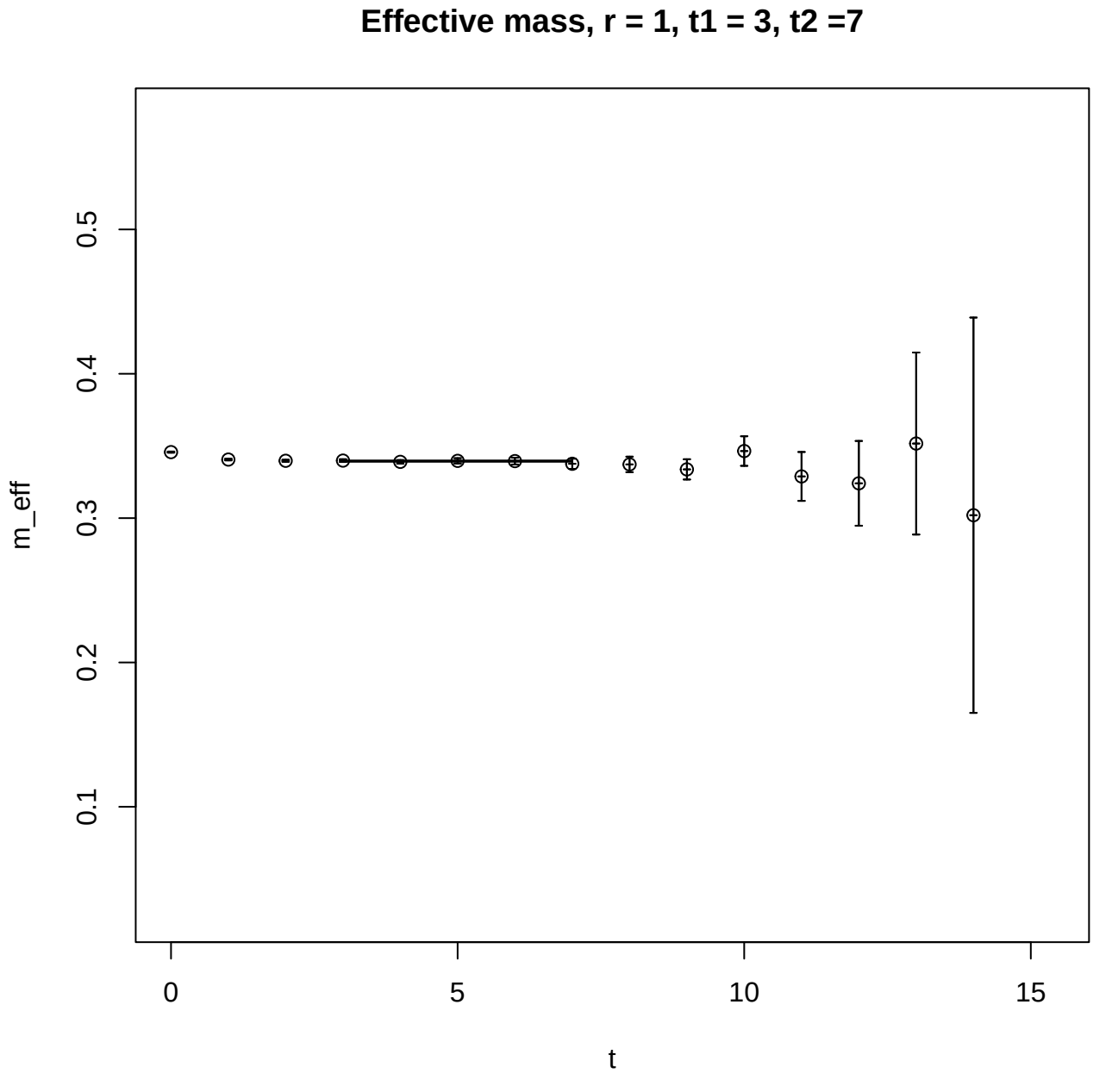

Effective mass, r = 1, t1 = 2, t2 =13
m_eff
0.1
0.2
0.3
0.4
0.5
0
5
10
15
t

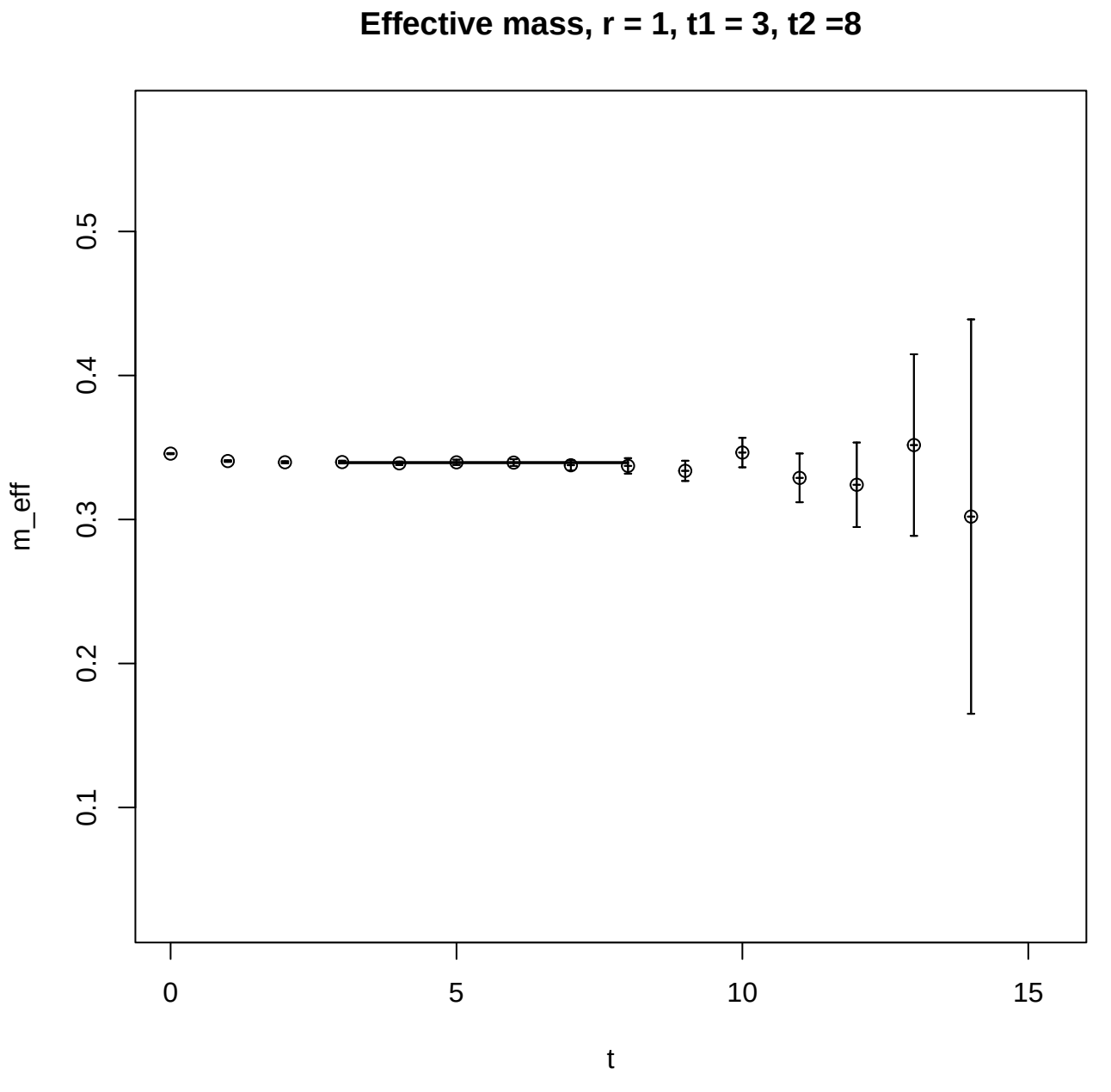

Effective mass, r = 1, t1 = 2, t2 =14
m_eff
0.1
0.2
0.3
0.4
0.5
0
5
10
15
t


Effective mass, r = 1, t1 = 3, t2 =4
m_eff
0.1
0.2
0.3
0.4
0.5
0
5
10
15
t


Effective mass, r = 1, t1 = 3, t2 =5
m_eff
0.1
0.2
0.3
0.4
0.5
0
5
10
15
t


Effective mass, r = 1, t1 = 3, t2 =6
m_eff
0.1
0.2
0.3
0.4
0.5
0
5
10
15
t


Effective mass, r = 1, t1 = 3, t2 =7
m_eff
0.1
0.2
0.3
0.4
0.5
0
5
10
15
t


Effective mass, r = 1, t1 = 3, t2 =8
m_eff
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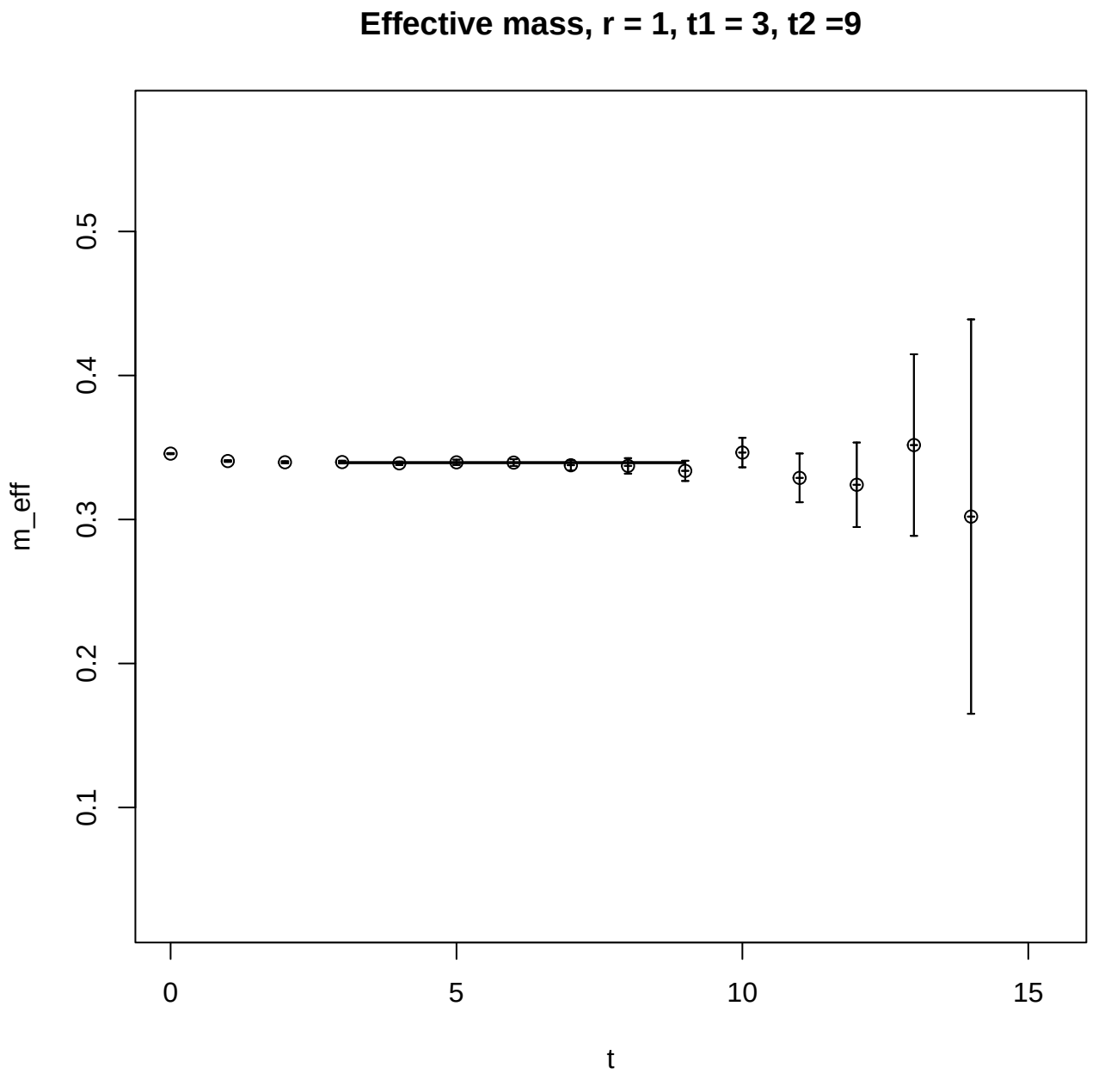
0.2
0.3
0.4
0.5
0
5
10
15
t


Effective mass, r = 1, t1 = 3, t2 =9
m_eff
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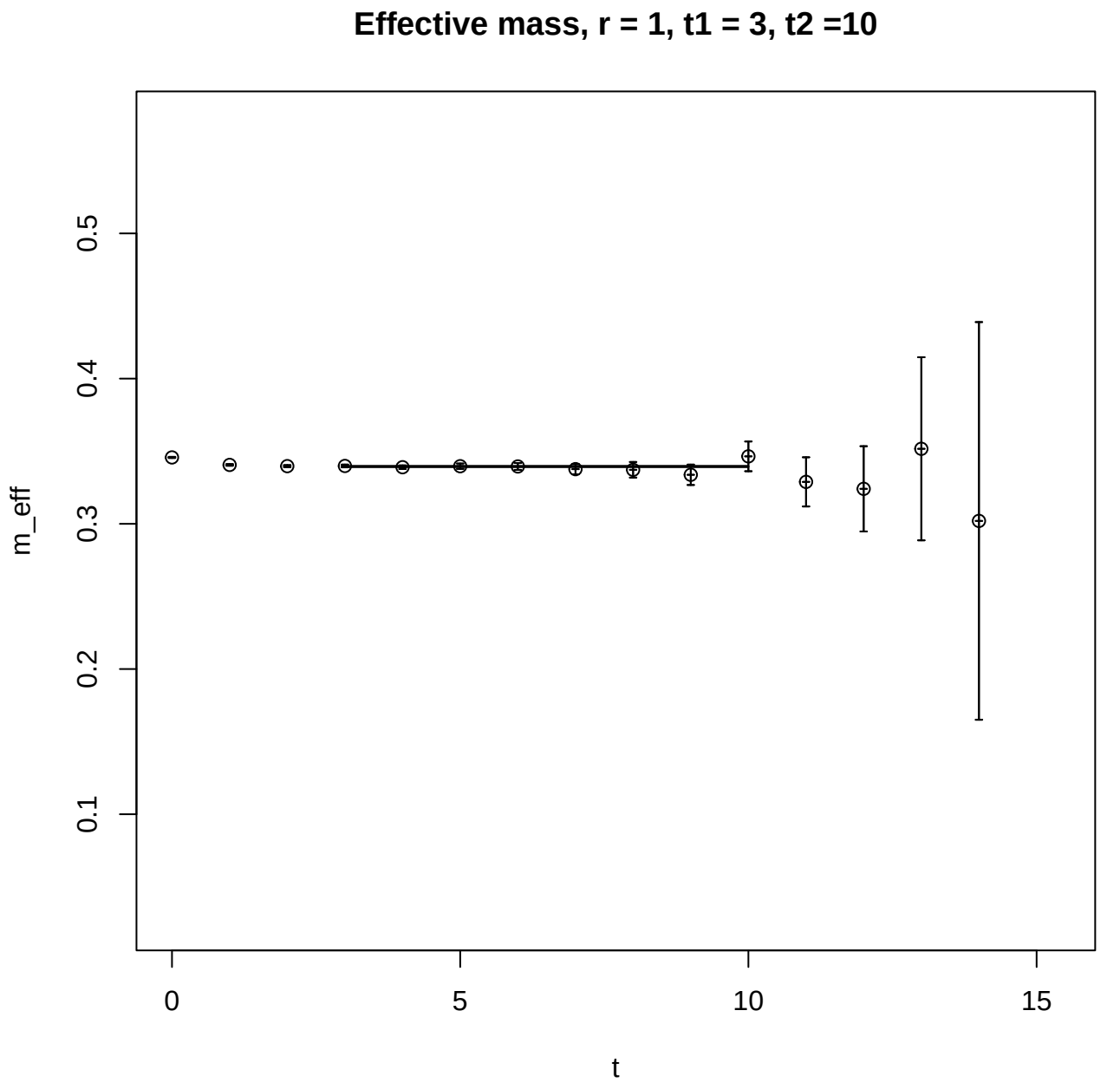
0.2
0.3
0.4
0.5
0
5
10
15
t


Effective mass, r = 1, t1 = 3, t2 =10
m_eff
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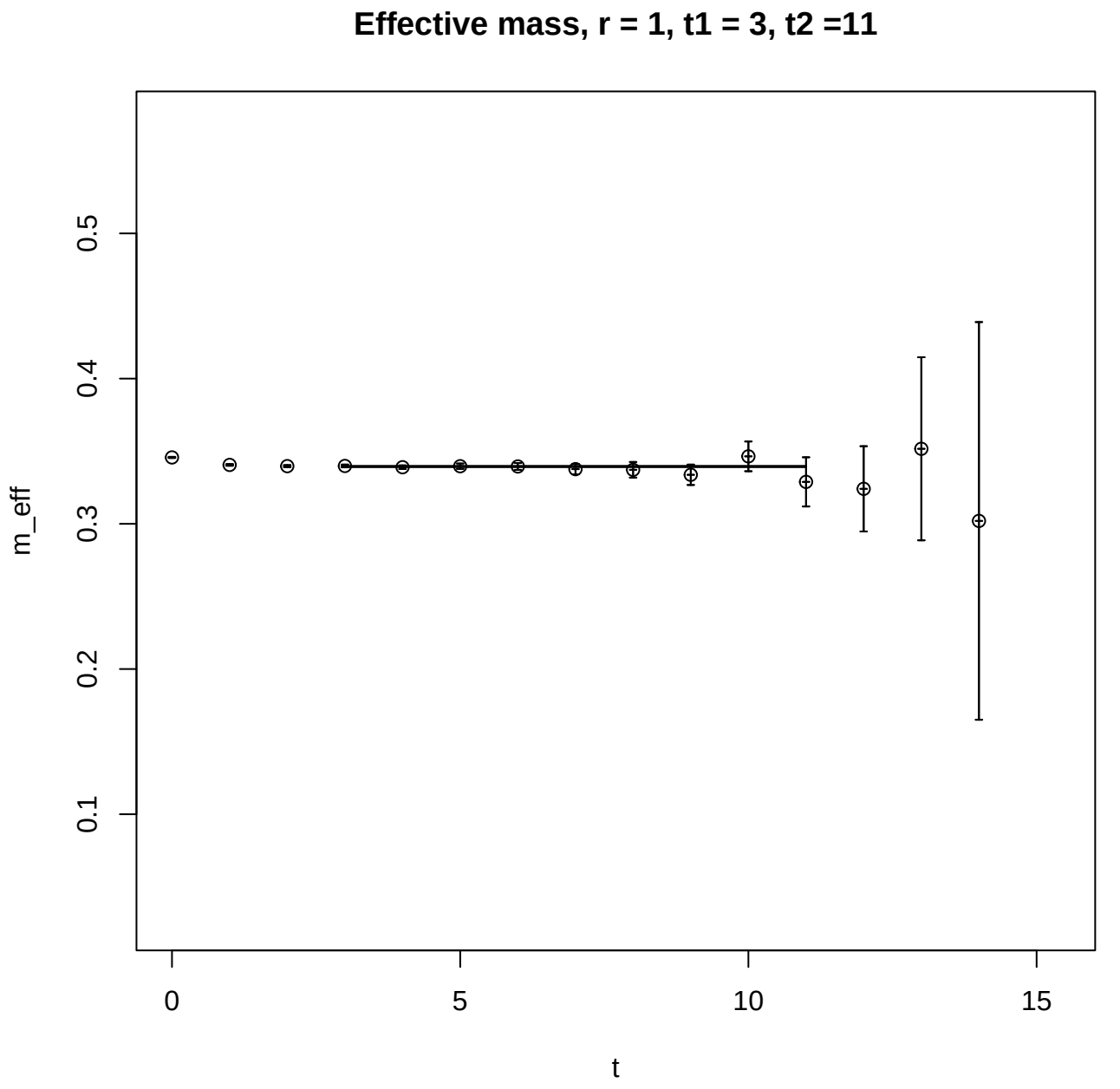
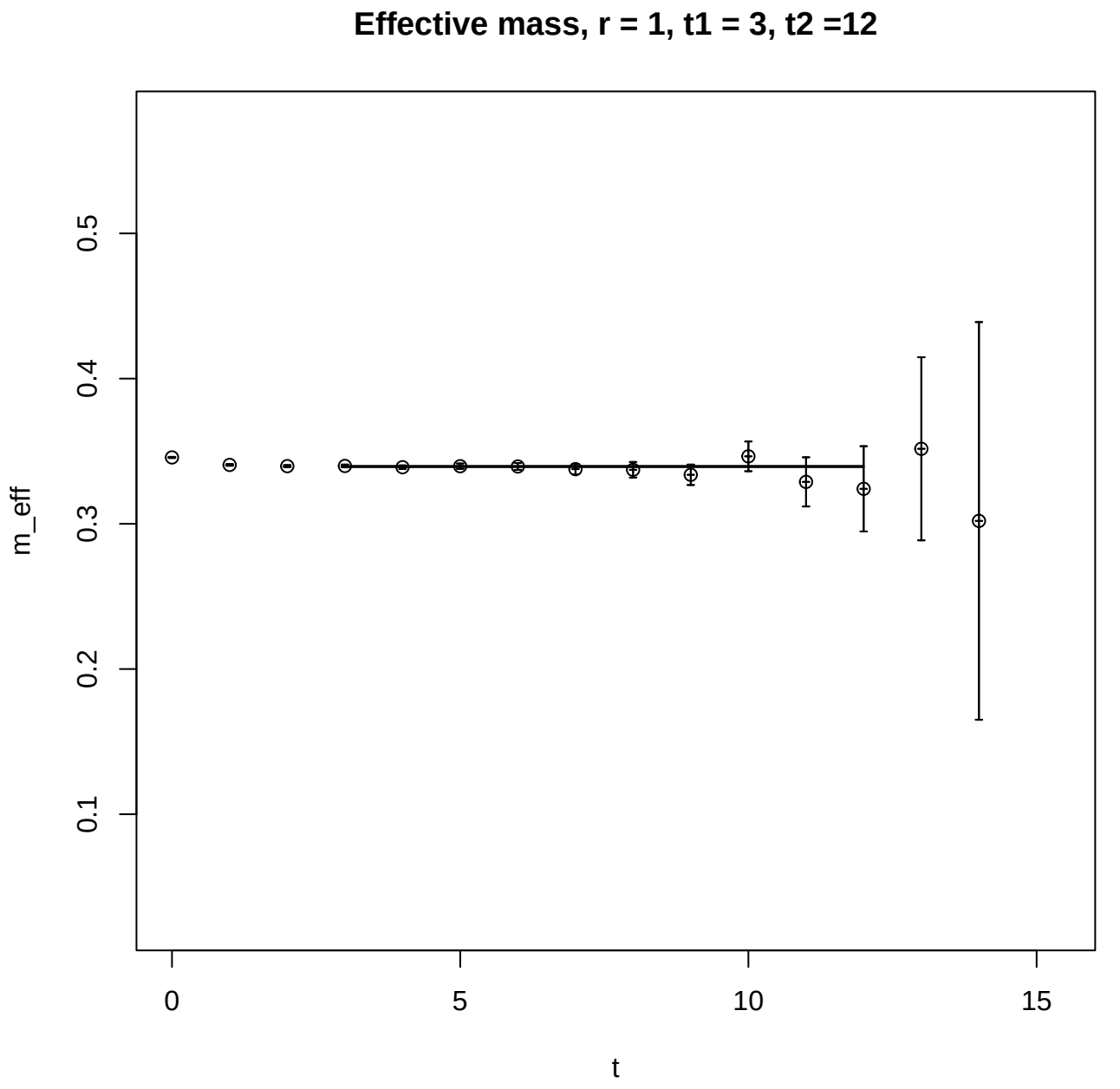
0.2
0.3
0.4
0.5
0
5
10
15
t


Effective mass, r = 1, t1 = 3, t2 =11
m_eff
0.1
0.2
0.3
0.4
0.5
0
5
10
15
t


Effective mass, r = 1, t1 = 3, t2 =12
m_eff
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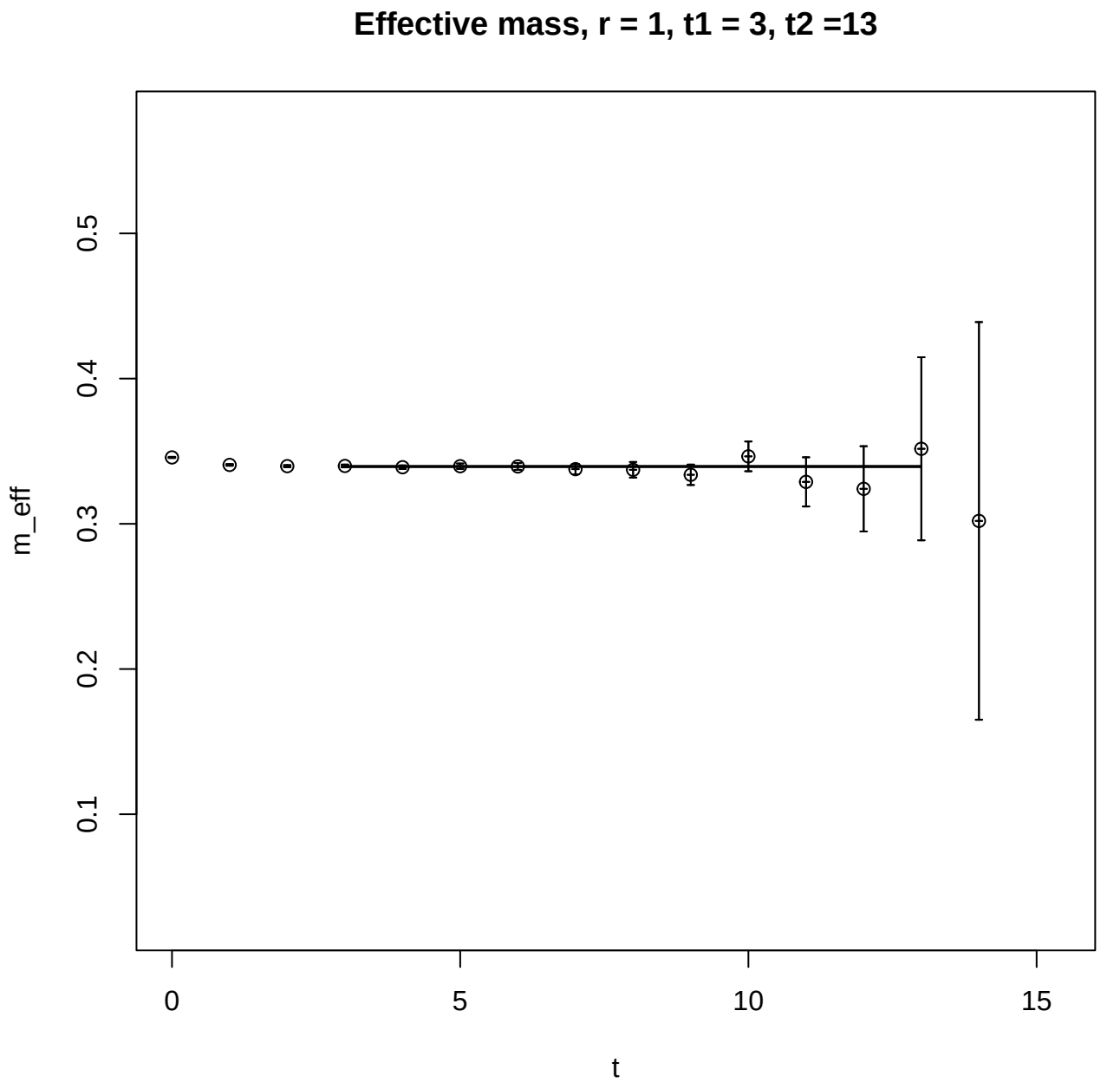
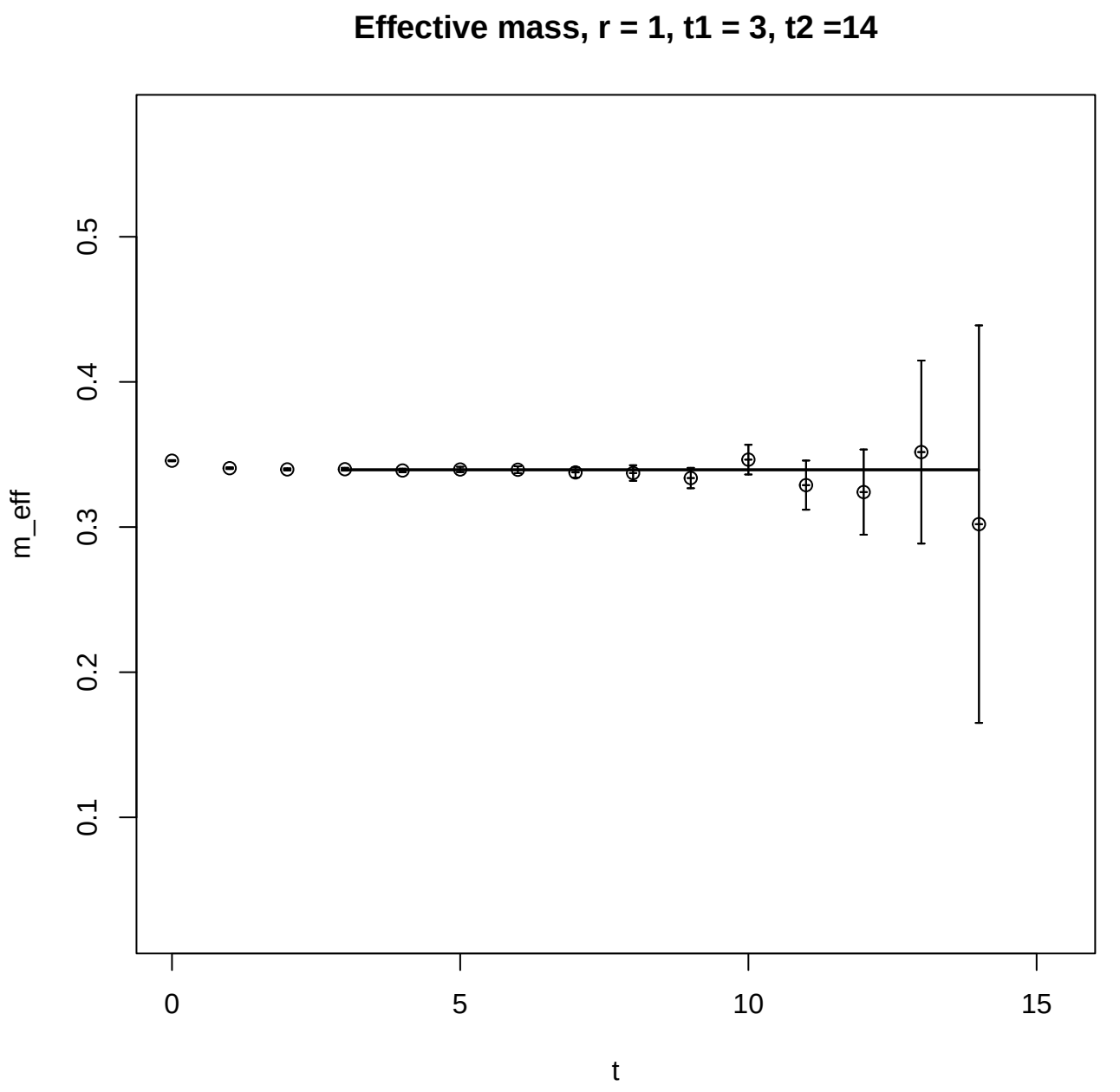
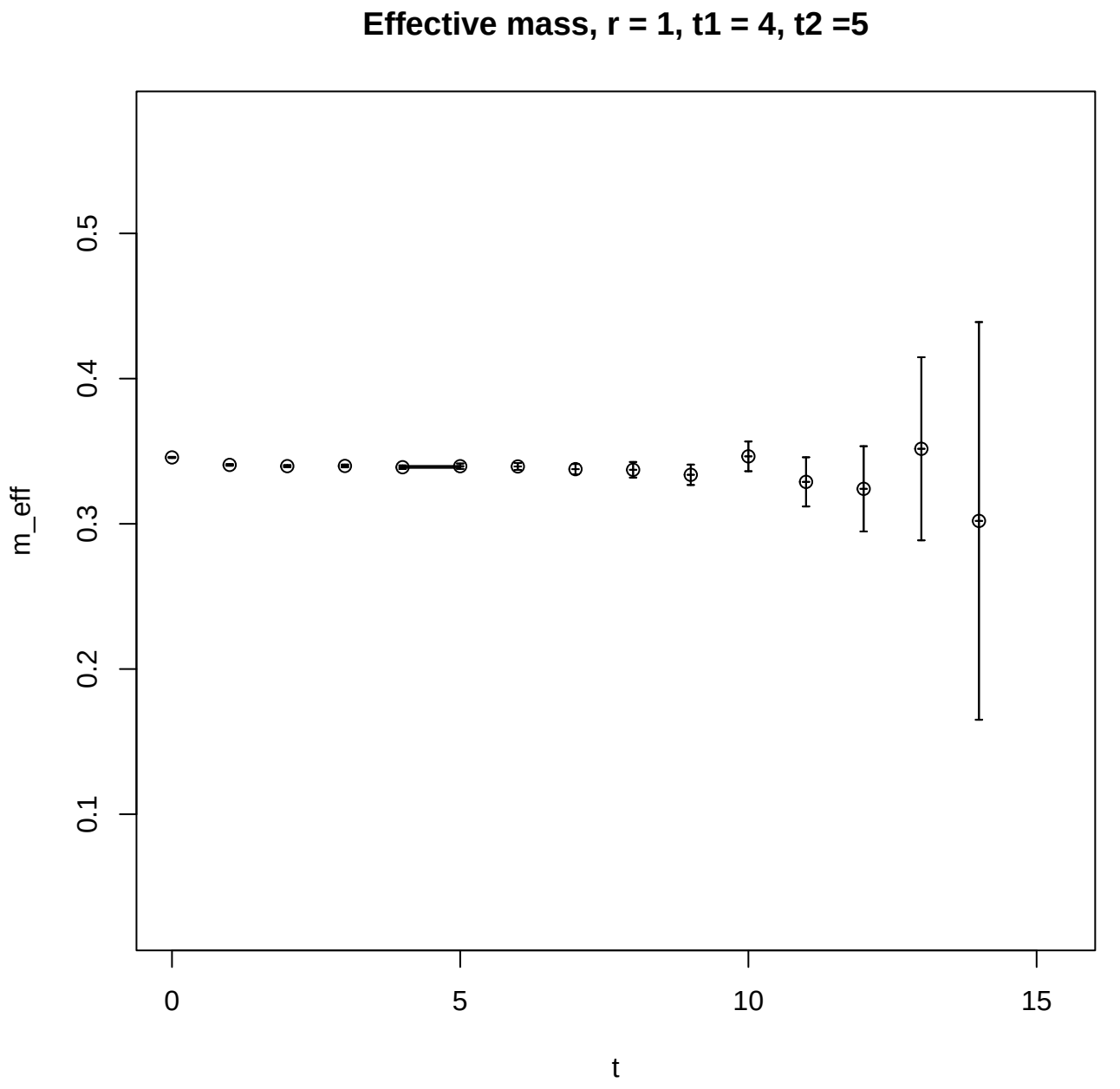
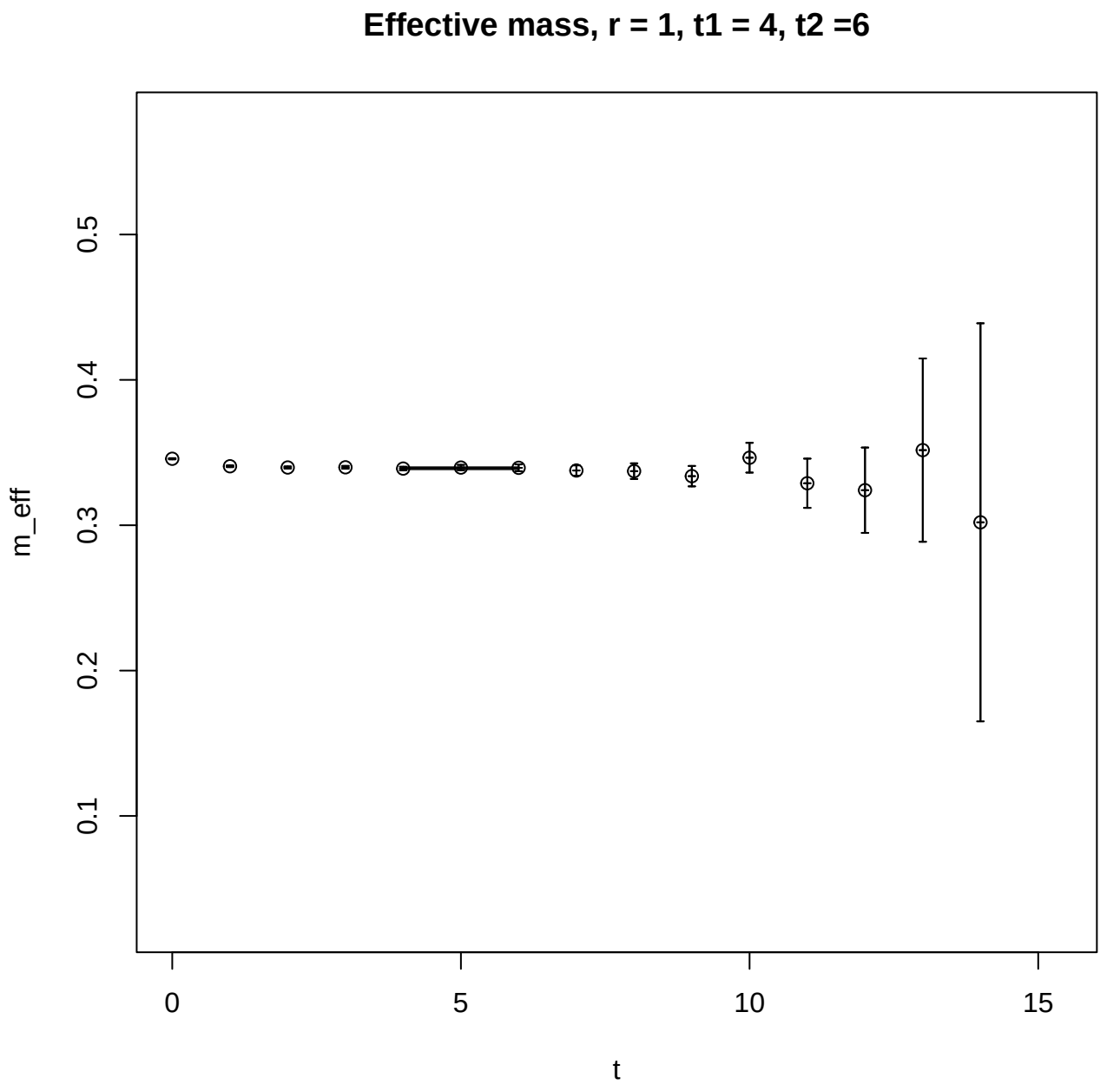
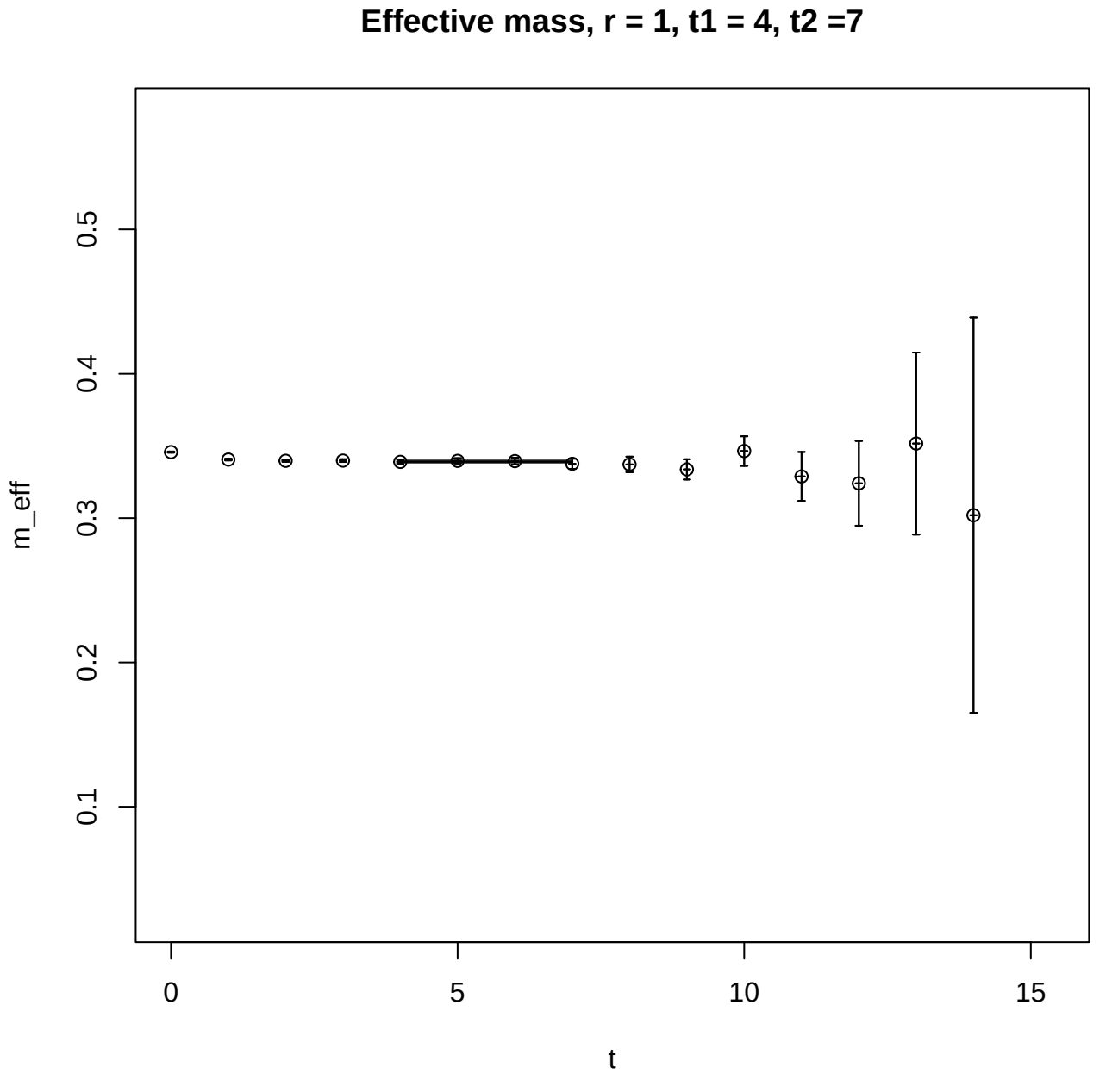
0.2
0.3
0.4
0.5
0
5
10
15
t

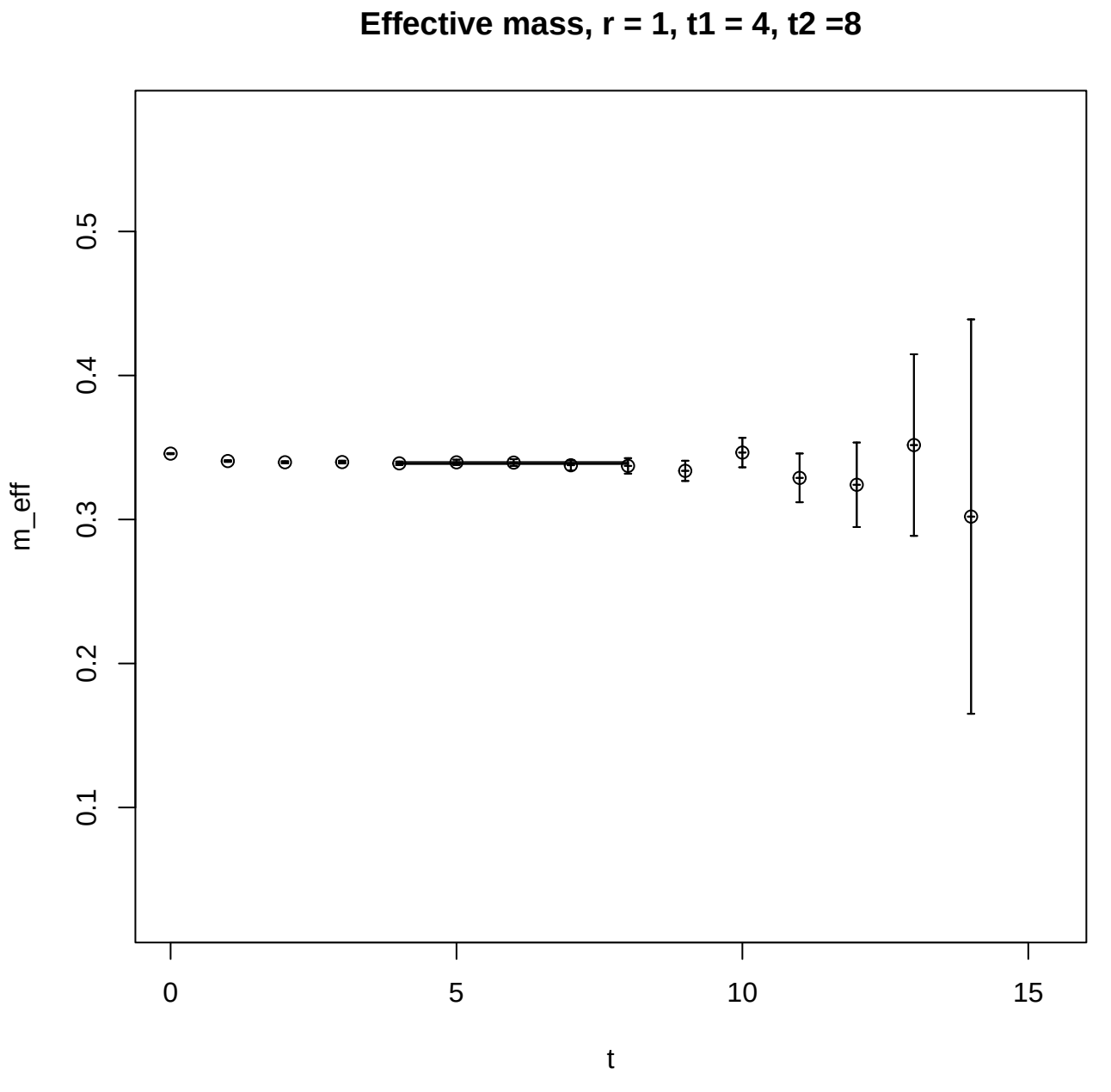

Effective mass, r = 1, t1 = 3, t2 =13
m_eff
0.1
0.2
0.3
0.4
0.5
0
5
10
15
t

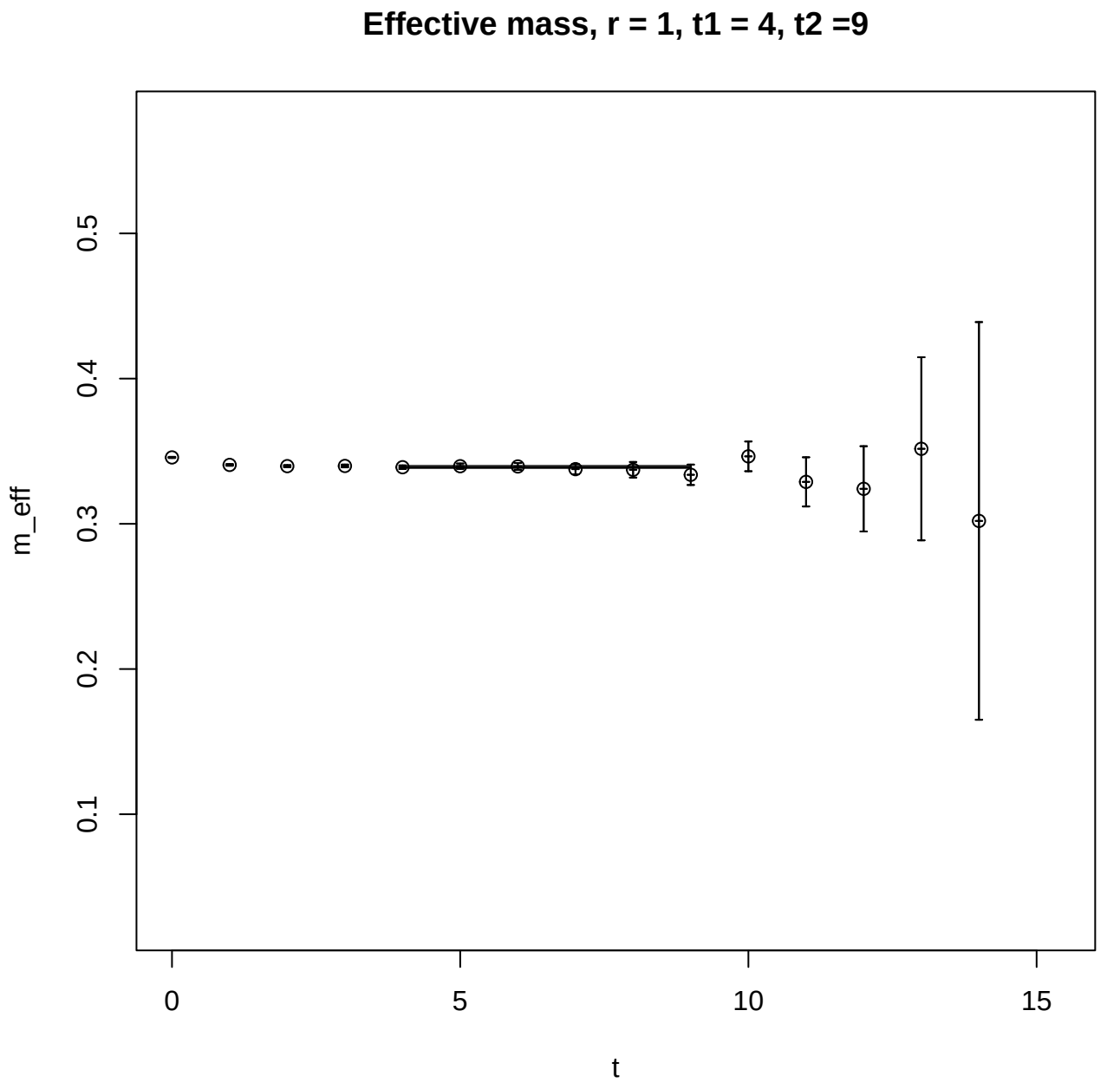

Effective mass, r = 1, t1 = 3, t2 =14
m_eff
0.1
0.2
0.3
0.4
0.5
0
5
10
15
t


Effective mass, r = 1, t1 = 4, t2 =5
m_eff
0.1
0.2
0.3
0.4
0.5
0
5
10
15
t


Effective mass, r = 1, t1 = 4, t2 =6
m_eff
0.1
0.2
0.3
0.4
0.5
0
5
10
15
t


Effective mass, r = 1, t1 = 4, t2 =7
m_eff
0.1
0.2
0.3
0.4
0.5
0
5
10
15
t


Effective mass, r = 1, t1 = 4, t2 =8
m_eff
0.1
0.2
0.3
0.4
0.5
0
5
10
15
t


Effective mass, r = 1, t1 = 4, t2 =9
m_eff
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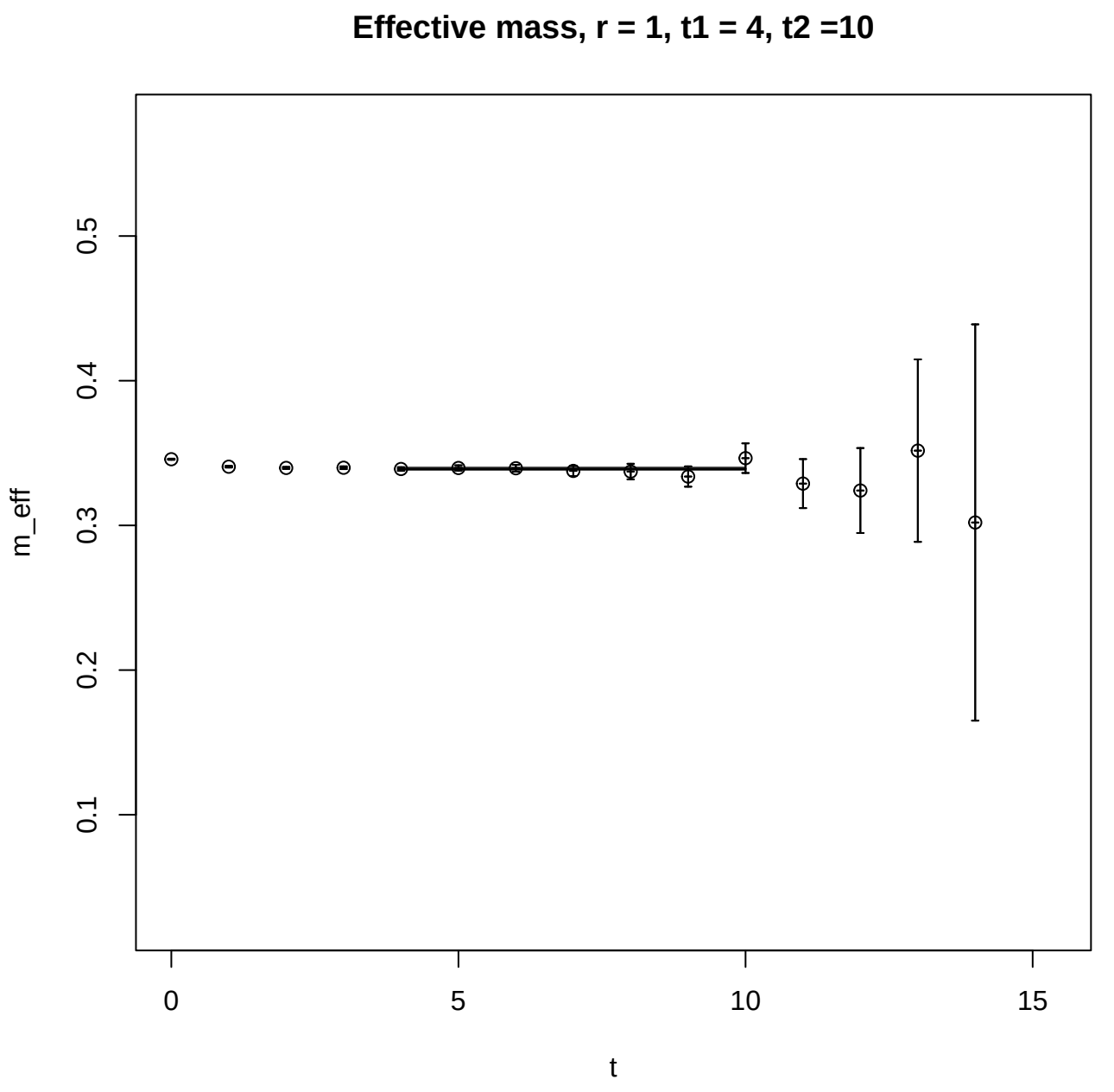
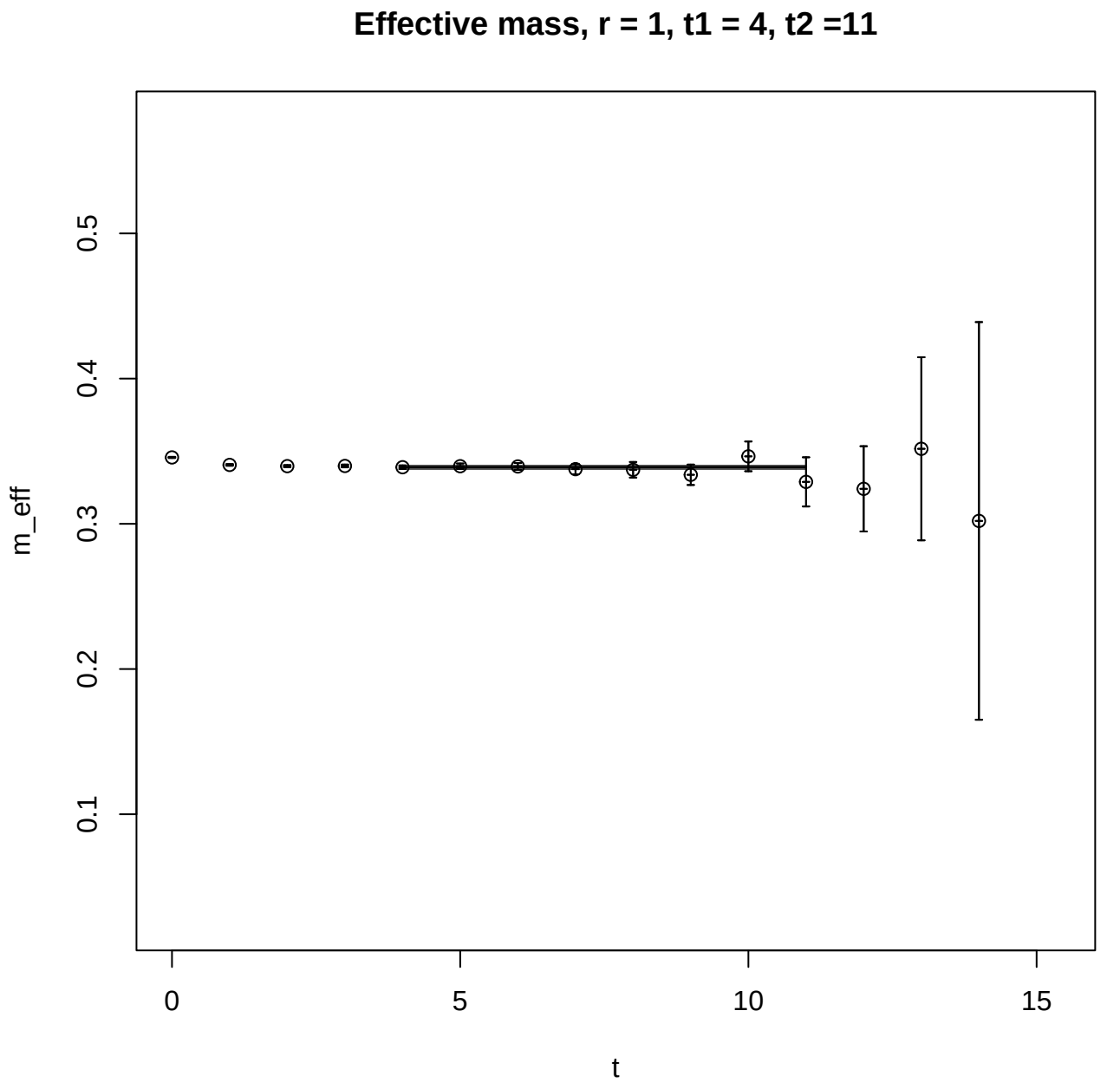
0.2
0.3
0.4
0.5
0
5
10
15
t


Effective mass, r = 1, t1 = 4, t2 =10
m_eff
0.1
0.2
0.3
0.4
0.5
0
5
10
15
t


Effective mass, r = 1, t1 = 4, t2 =11
m_eff
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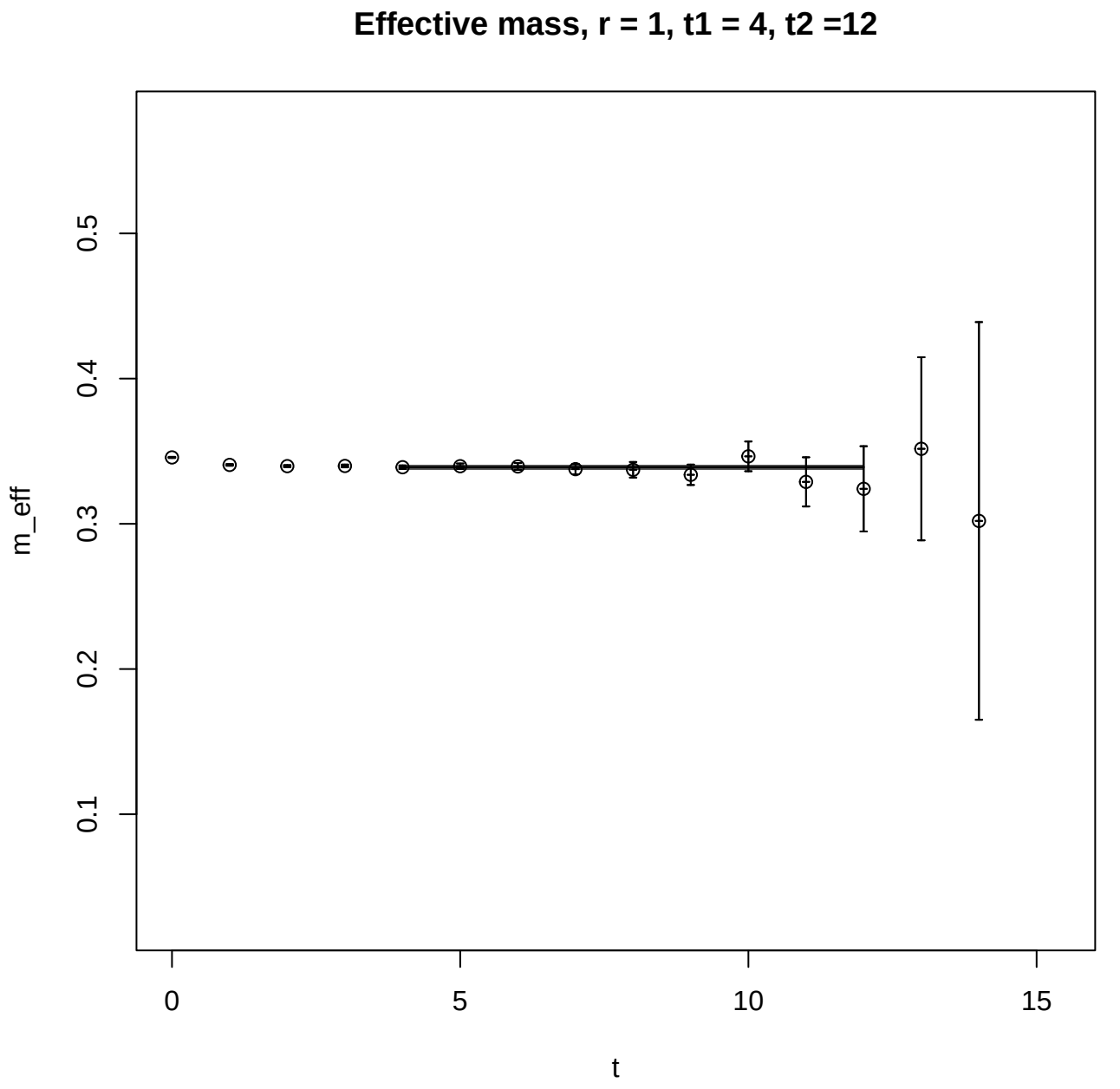
0.2
0.3
0.4
0.5
0
5
10
15
t


Effective mass, r = 1, t1 = 4, t2 =12
m_eff
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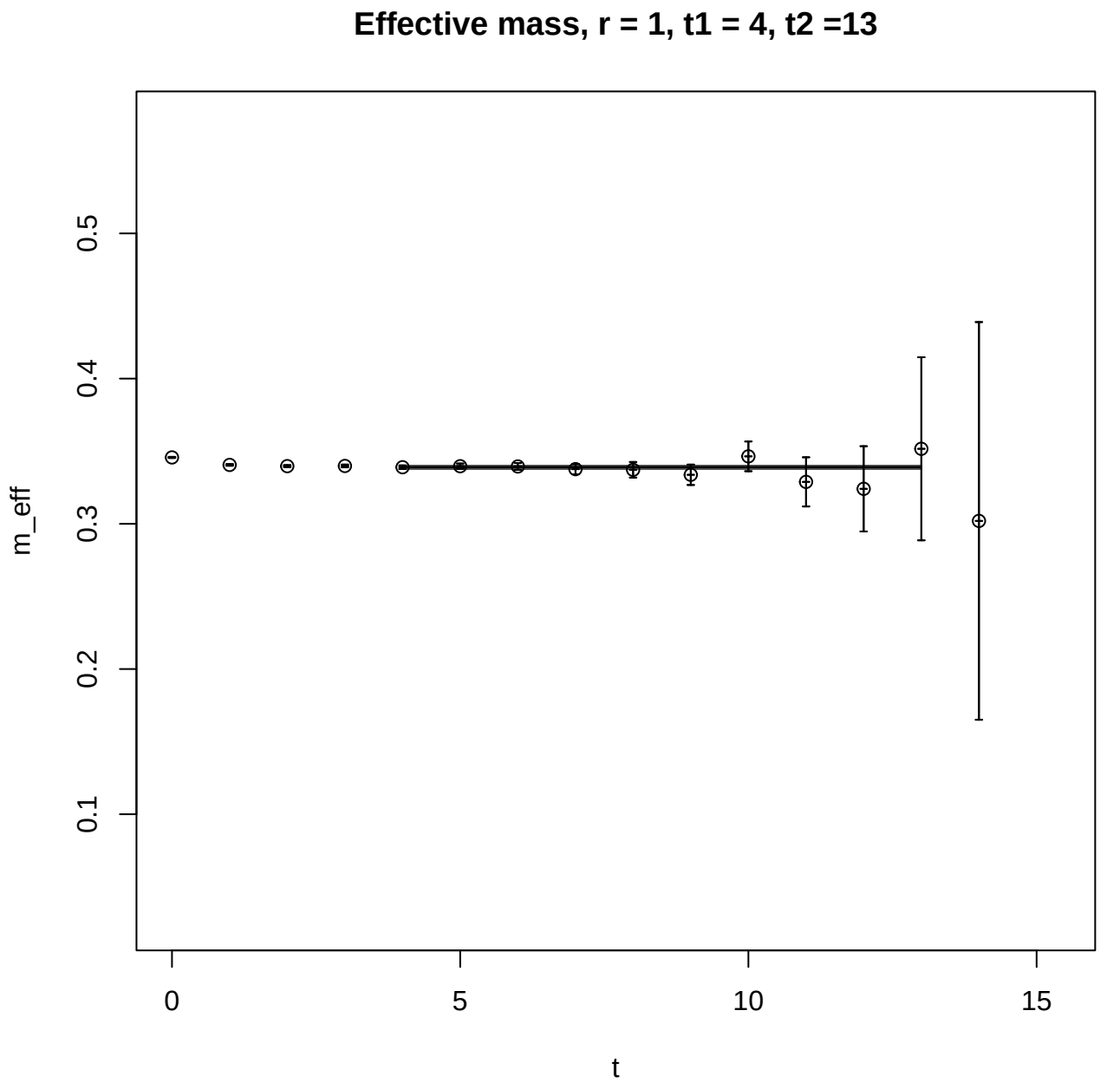
0.2
0.3
0.4
0.5
0
5
10
15
t


Effective mass, r = 1, t1 = 4, t2 =13
m_eff
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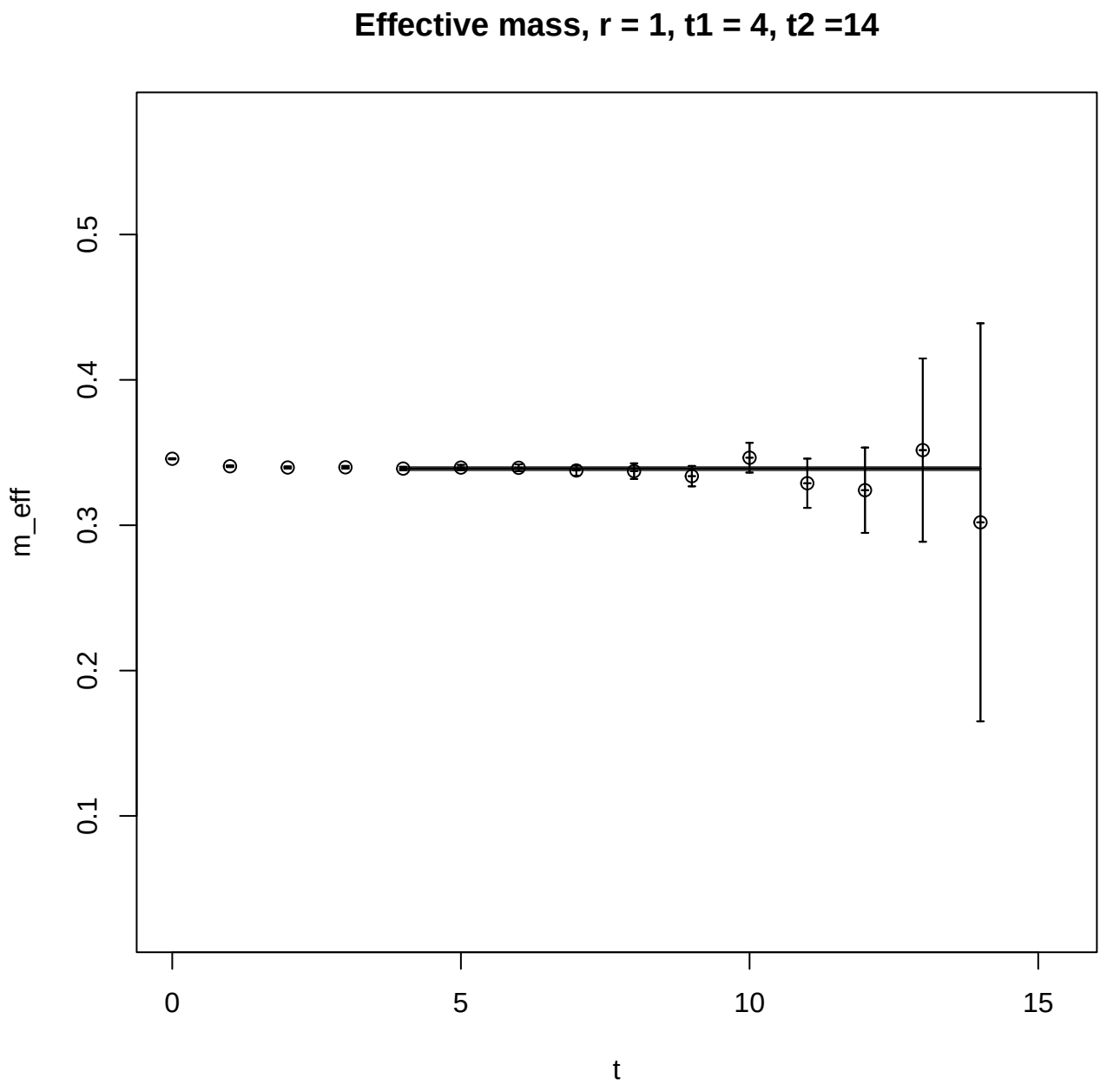
0.2
0.3
0.4
0.5
0
5
10
15
t


Effective mass, r = 1, t1 = 4, t2 =14
m_eff
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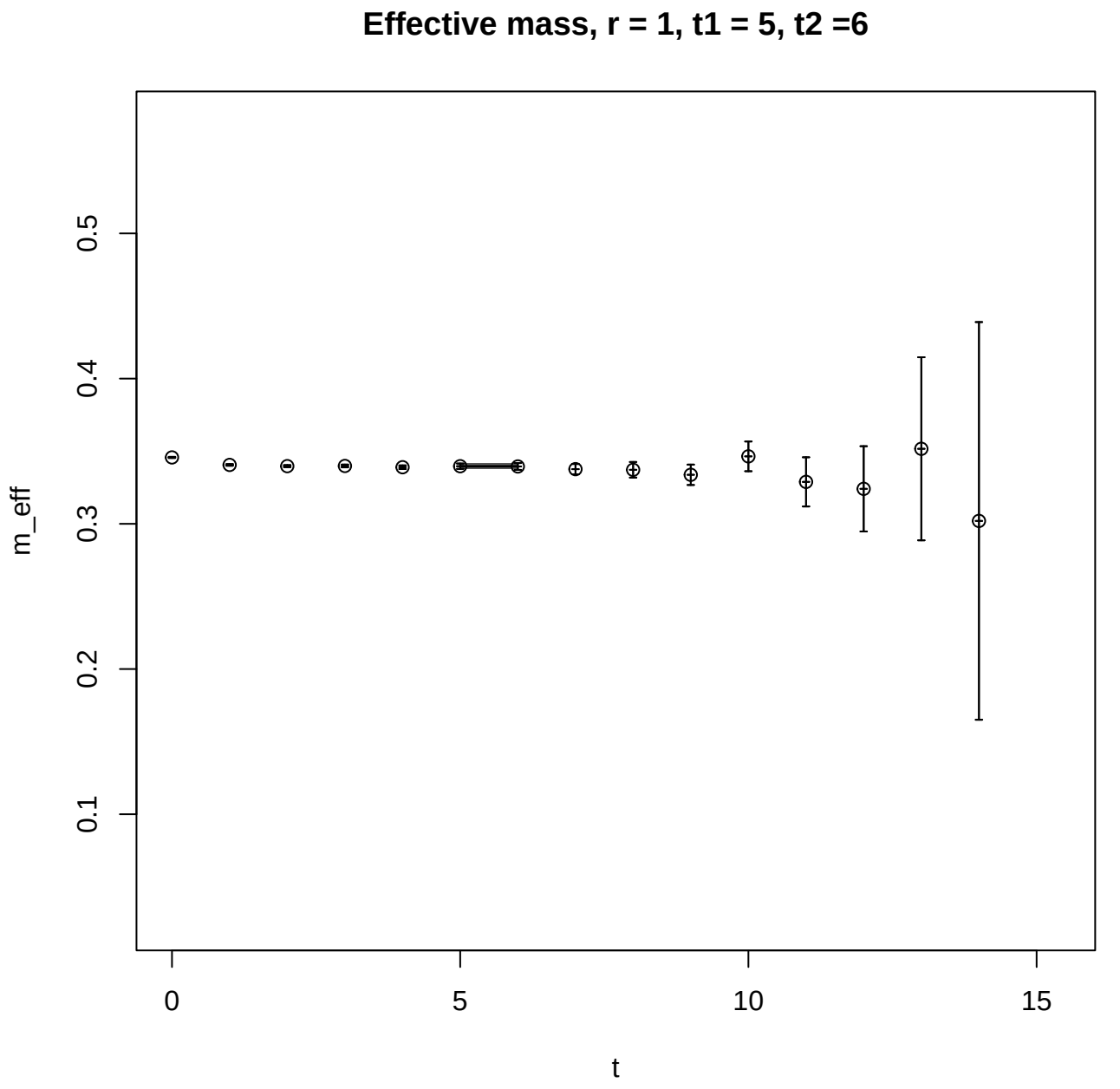
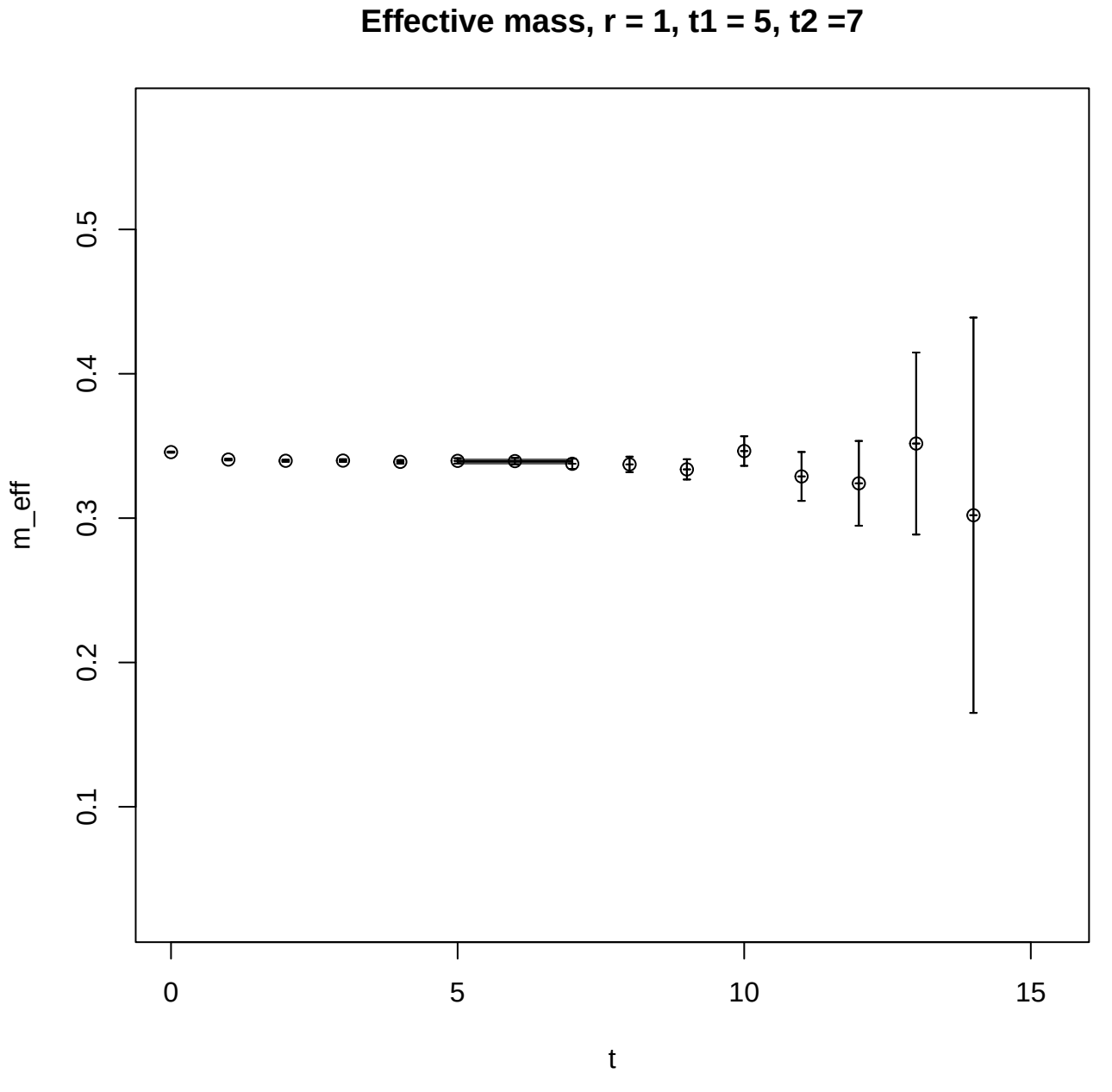
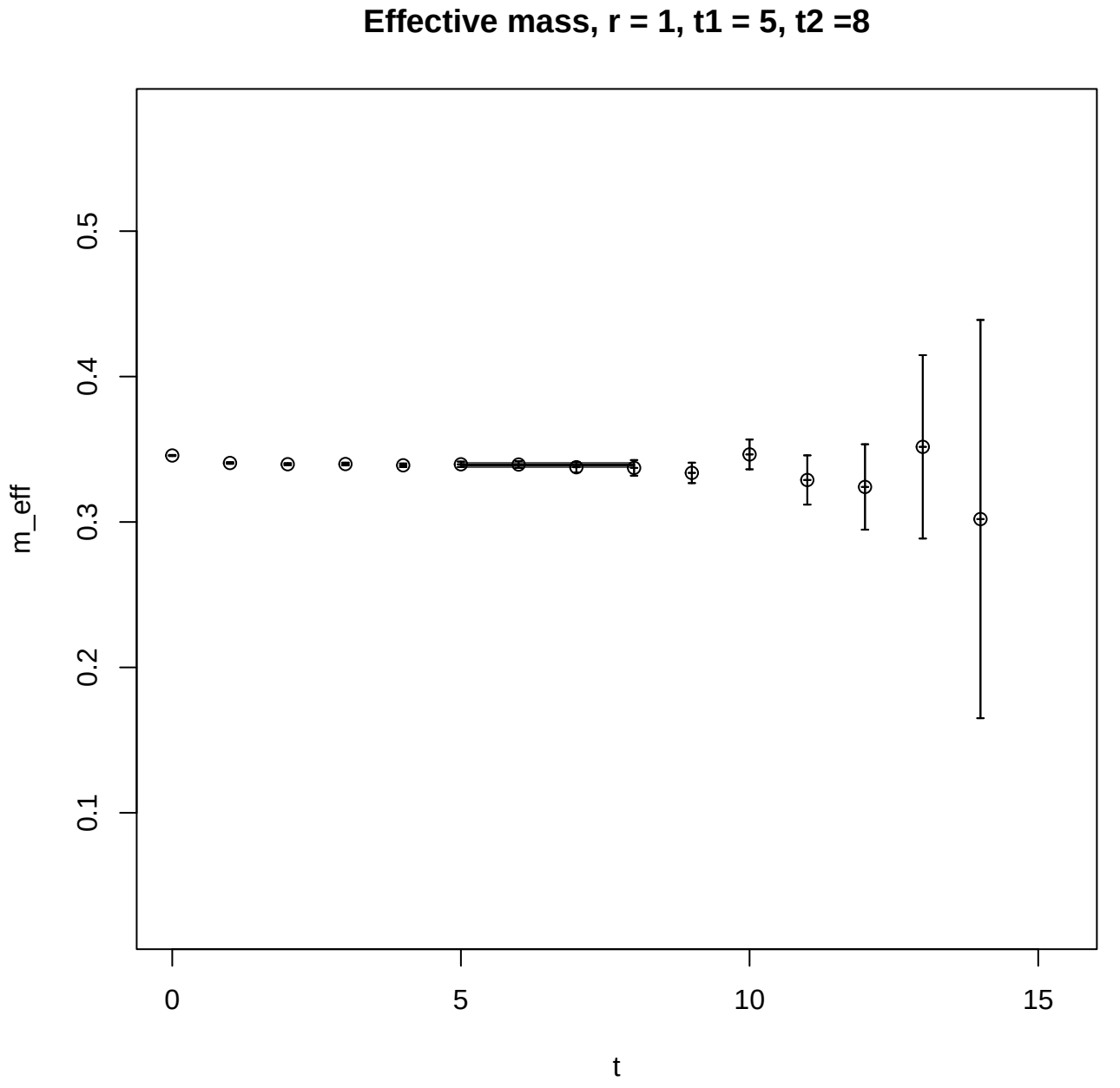
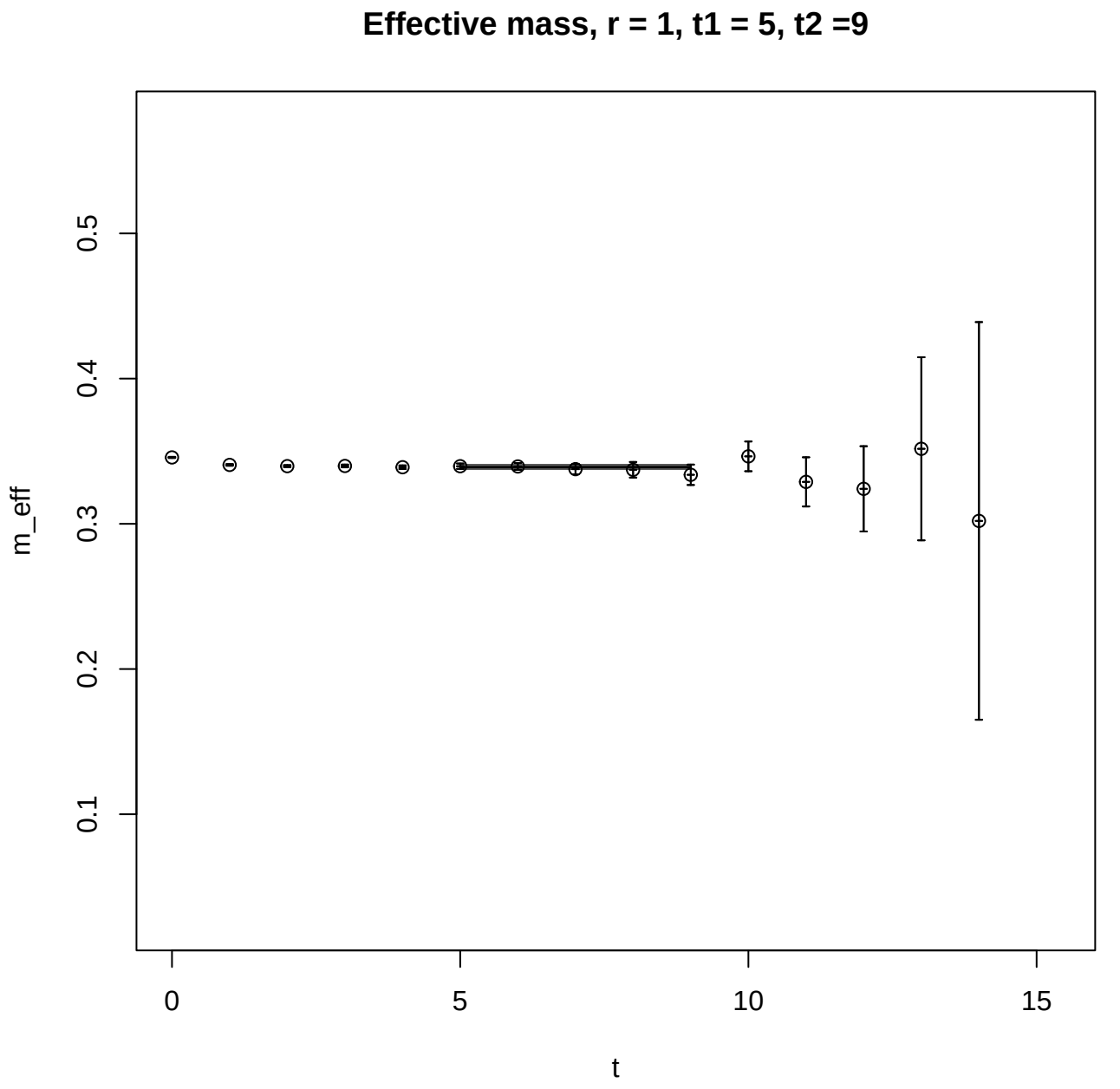
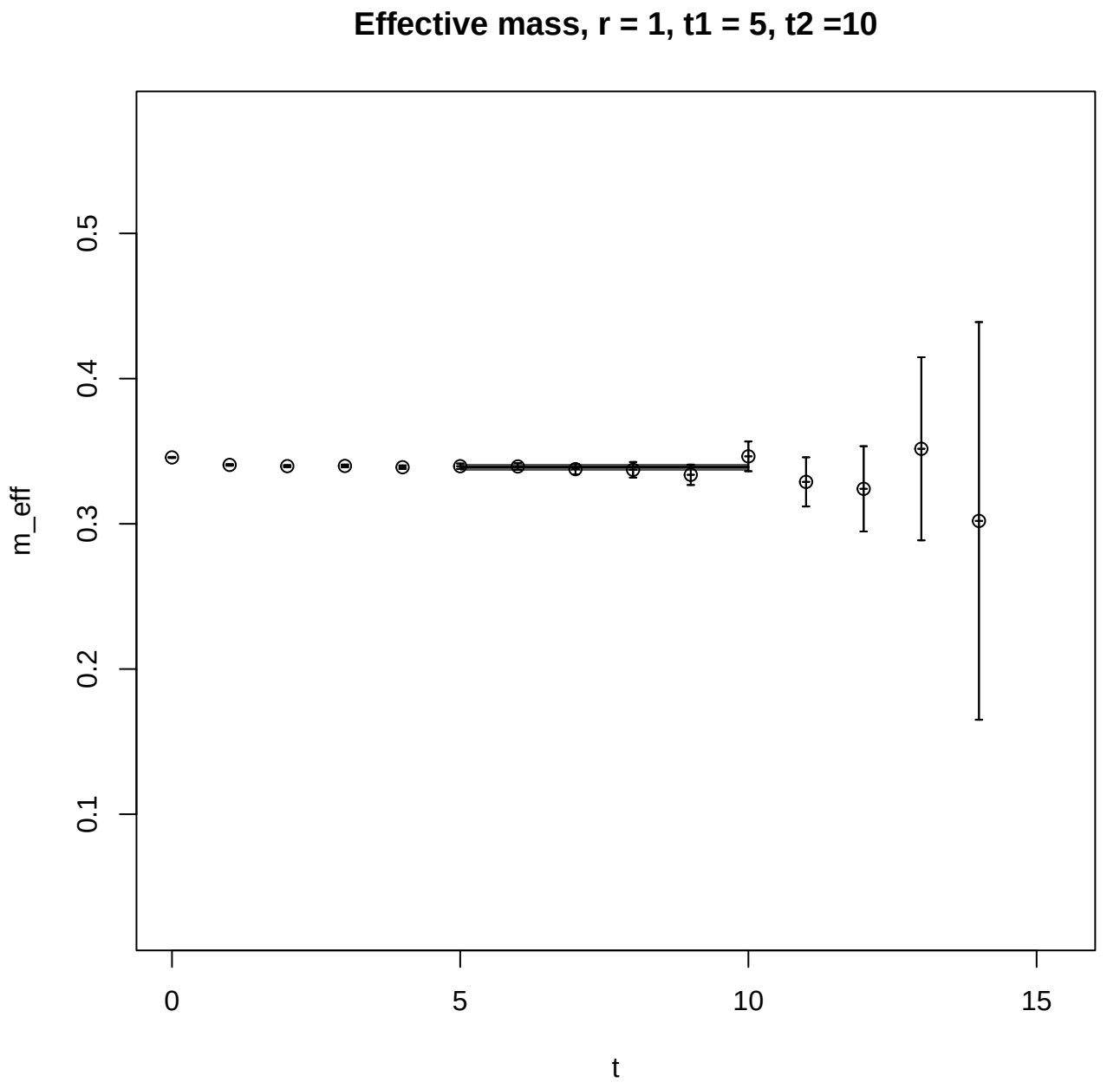
0.2
0.3
0.4
0.5
0
5
10
15
t

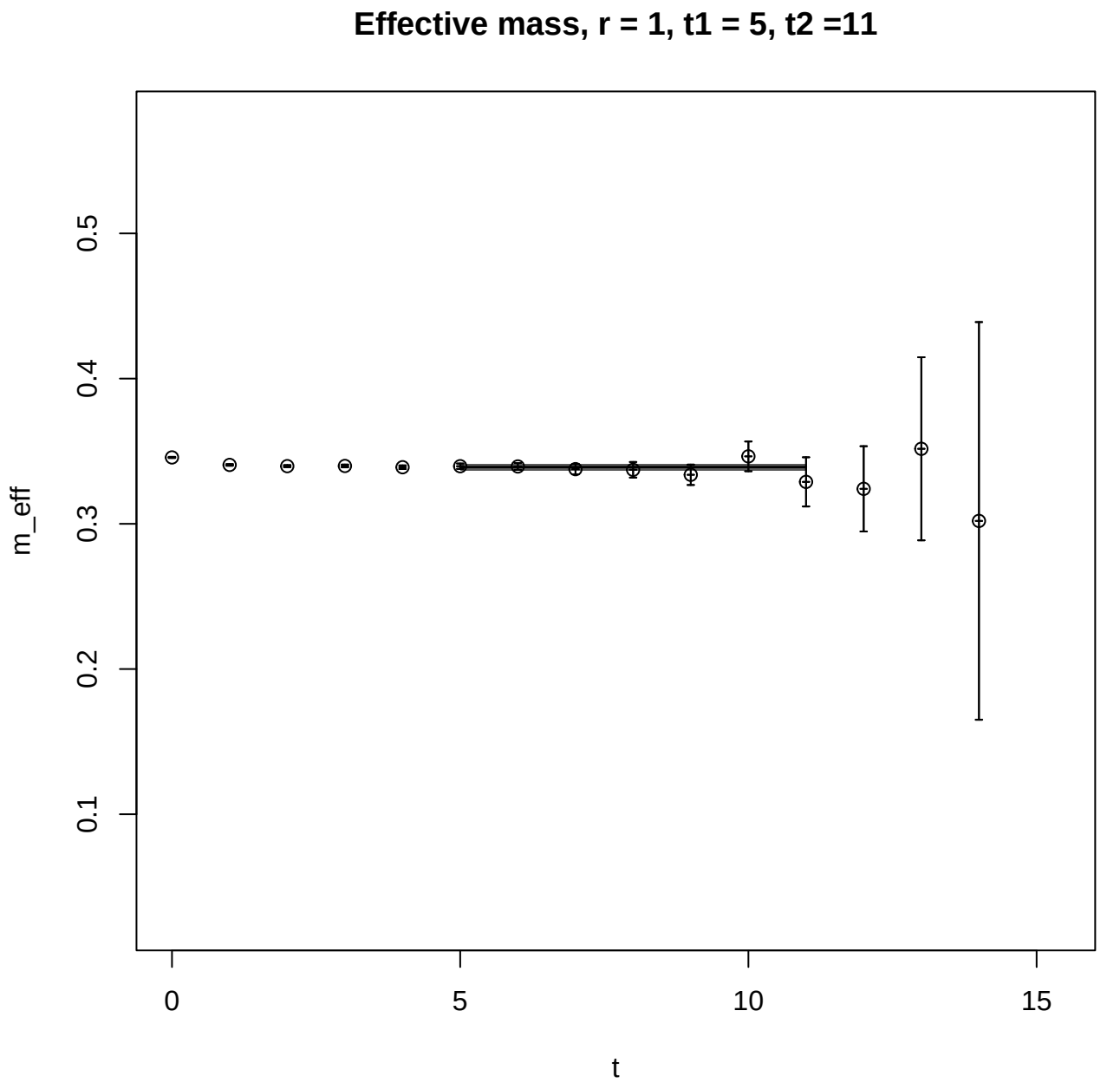

Effective mass, r = 1, t1 = 5, t2 =6
m_eff
0.1
0.2
0.3
0.4
0.5
0
5
10
15
t

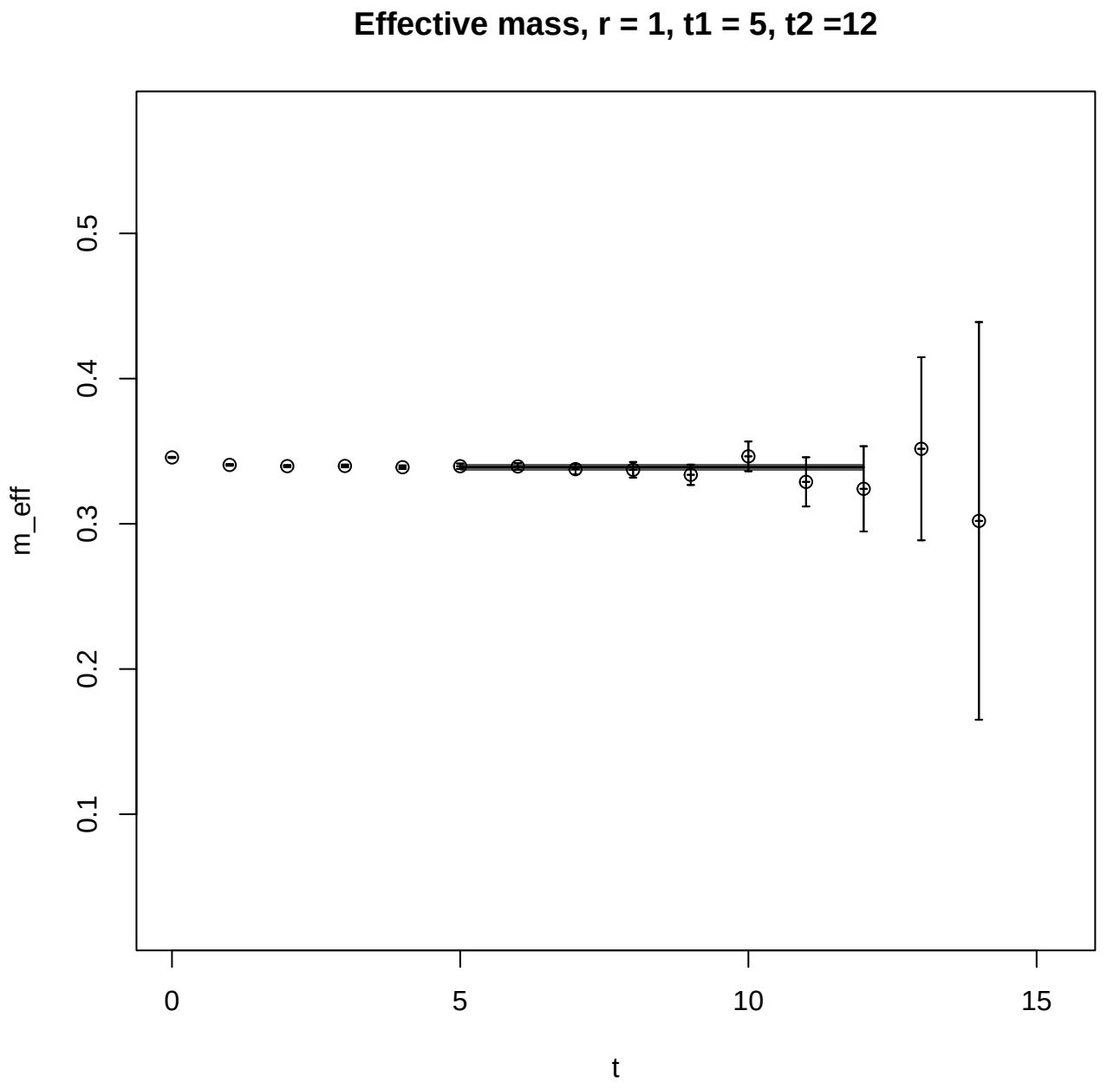

Effective mass, r = 1, t1 = 5, t2 =7
m_eff
0.1
0.2
0.3
0.4
0.5
0
5
10
15
t

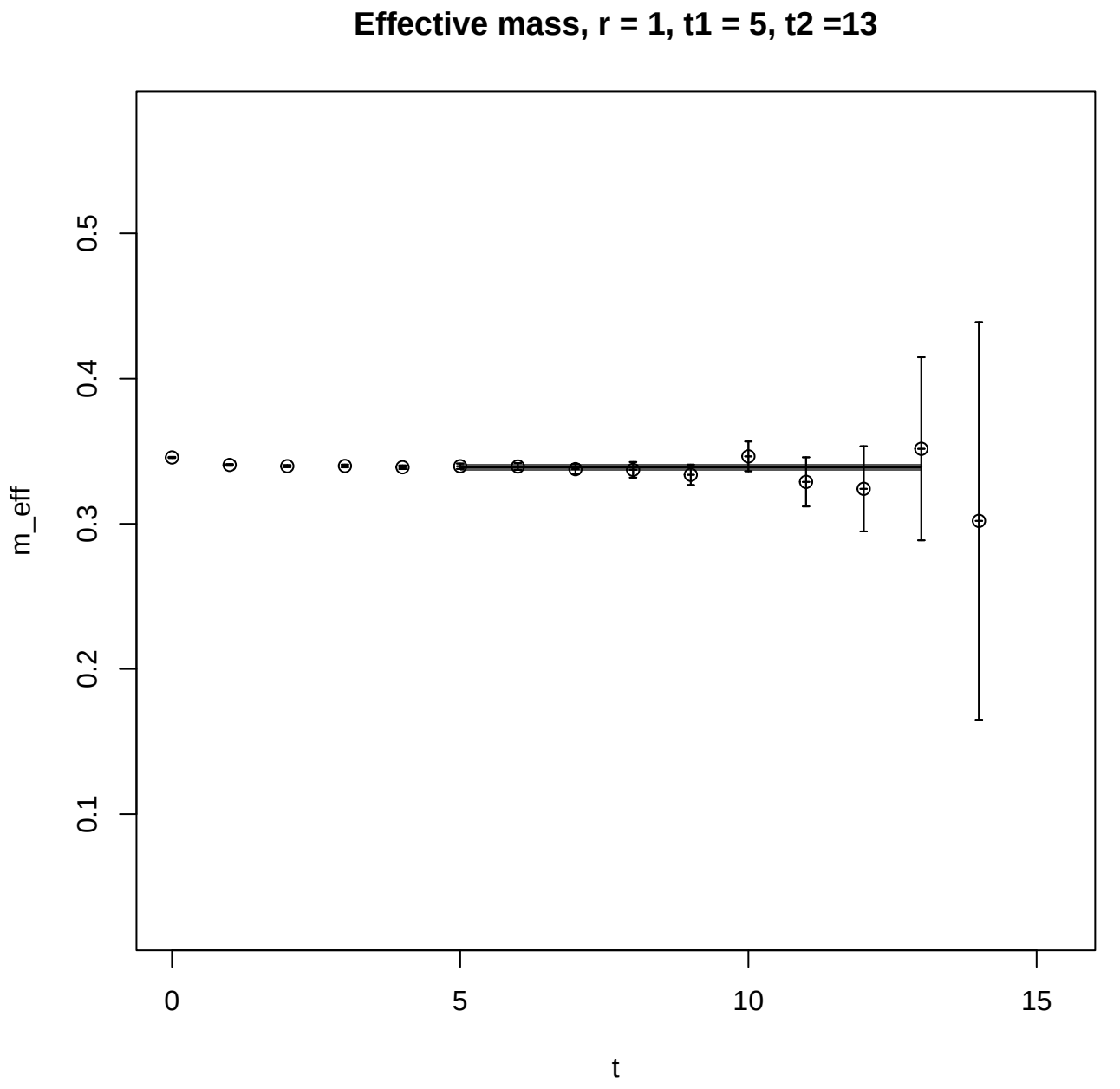

Effective mass, r = 1, t1 = 5, t2 =8
m_eff
0.1
0.2
0.3
0.4
0.5
0
5
10
15
t

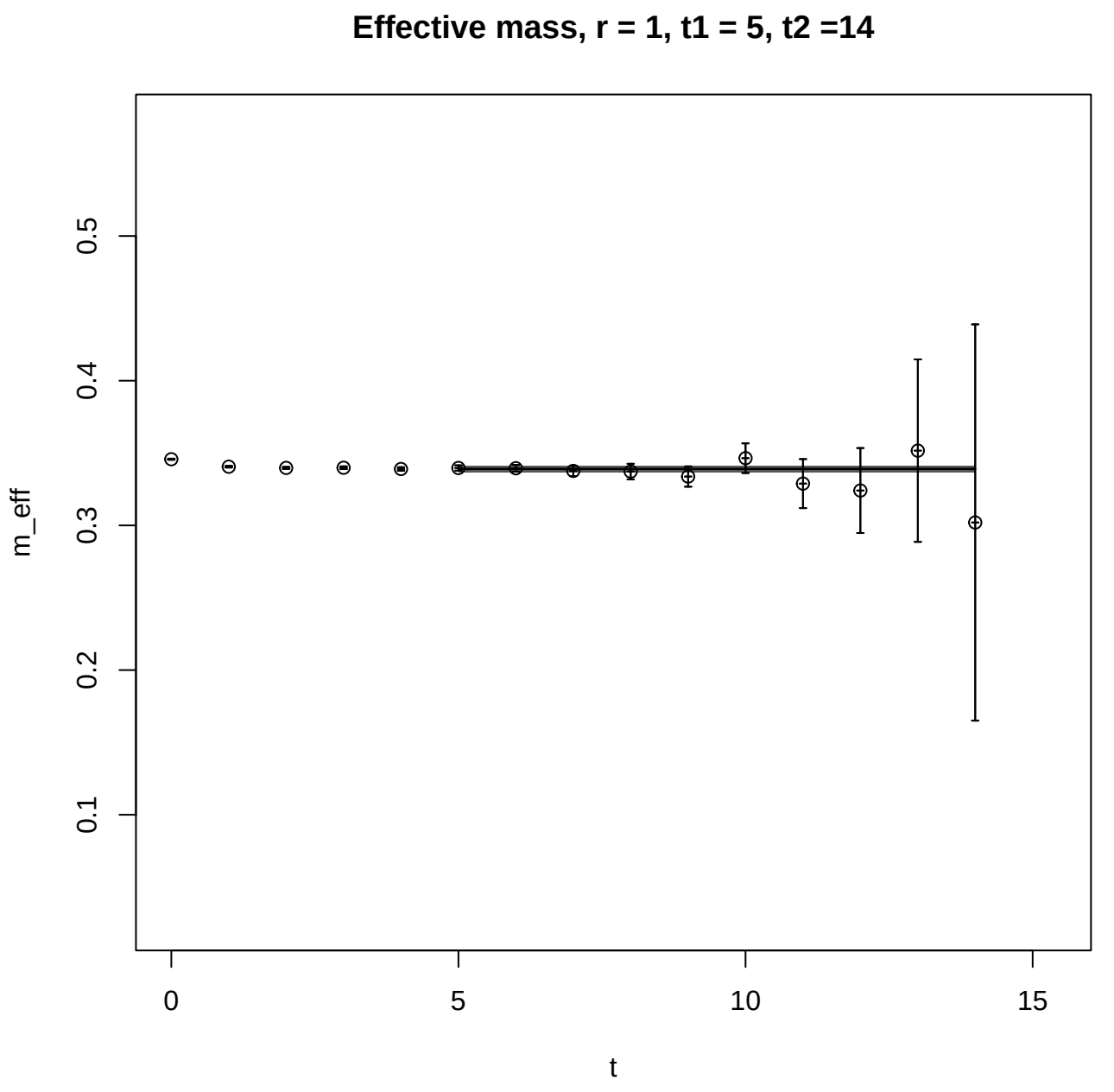

Effective mass, r = 1, t1 = 5, t2 =9
m_eff
0.1
0.2
0.3
0.4
0.5
0
5
10
15
t

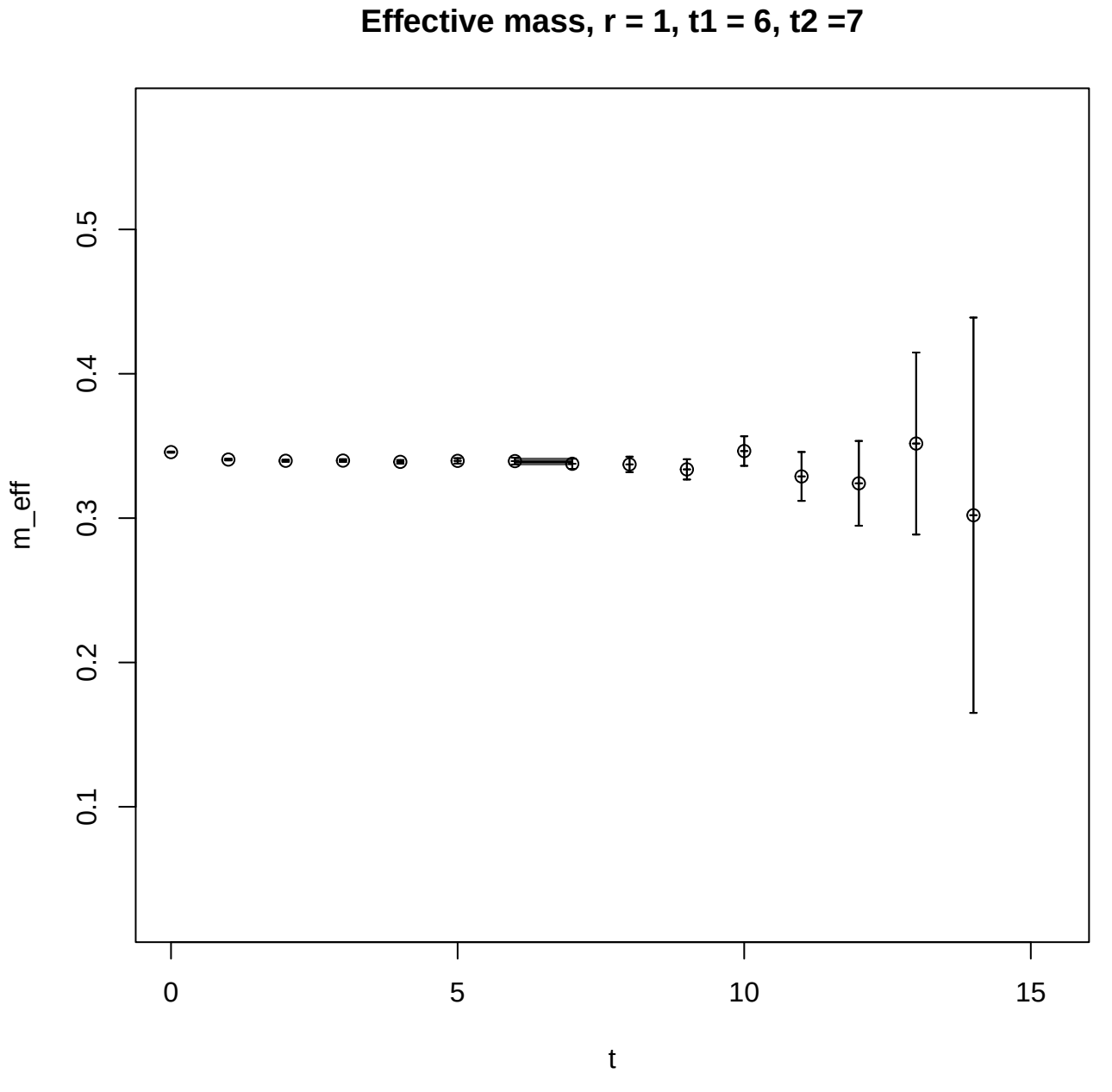

Effective mass, r = 1, t1 = 5, t2 =10
m_eff
0.1
0.2
0.3
0.4
0.5
0
5
10
15
t

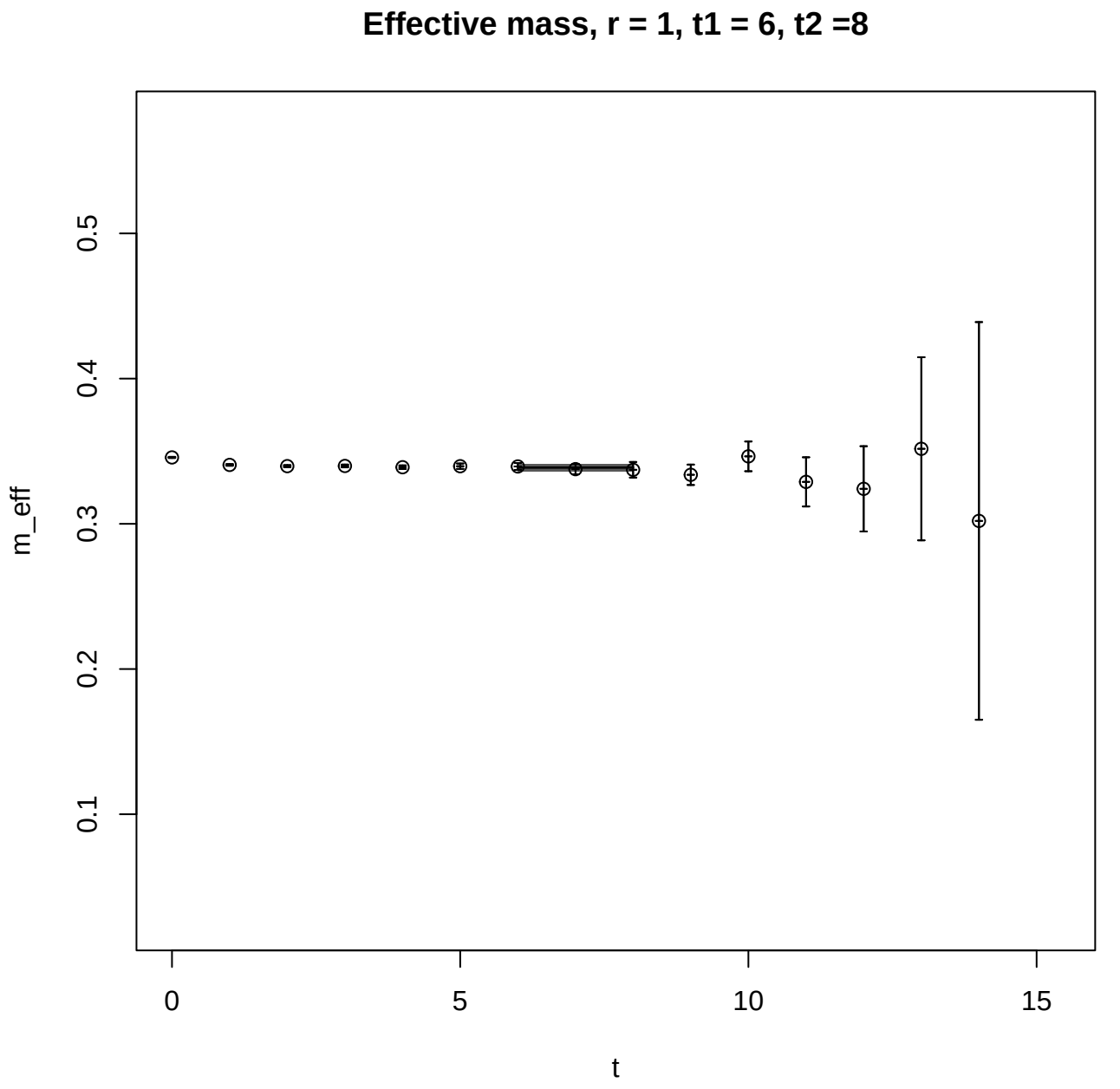

Effective mass, r = 1, t1 = 5, t2 =11
m_eff
0.1
0.2
0.3
0.4
0.5
0
5
10
15
t

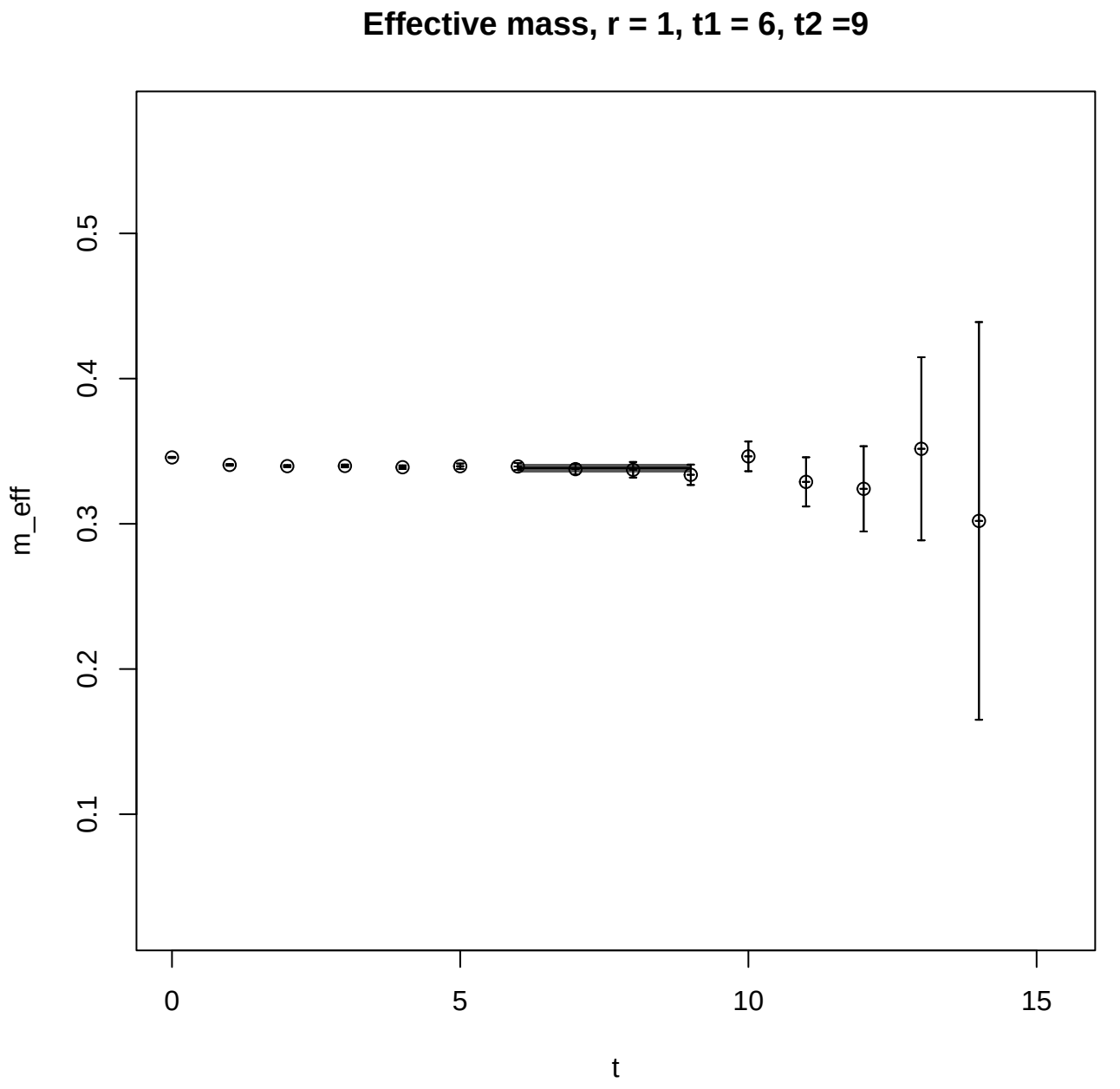

Effective mass, r = 1, t1 = 5, t2 =12
m_eff
0.1
0.2
0.3
0.4
0.5
0
5
10
15
t

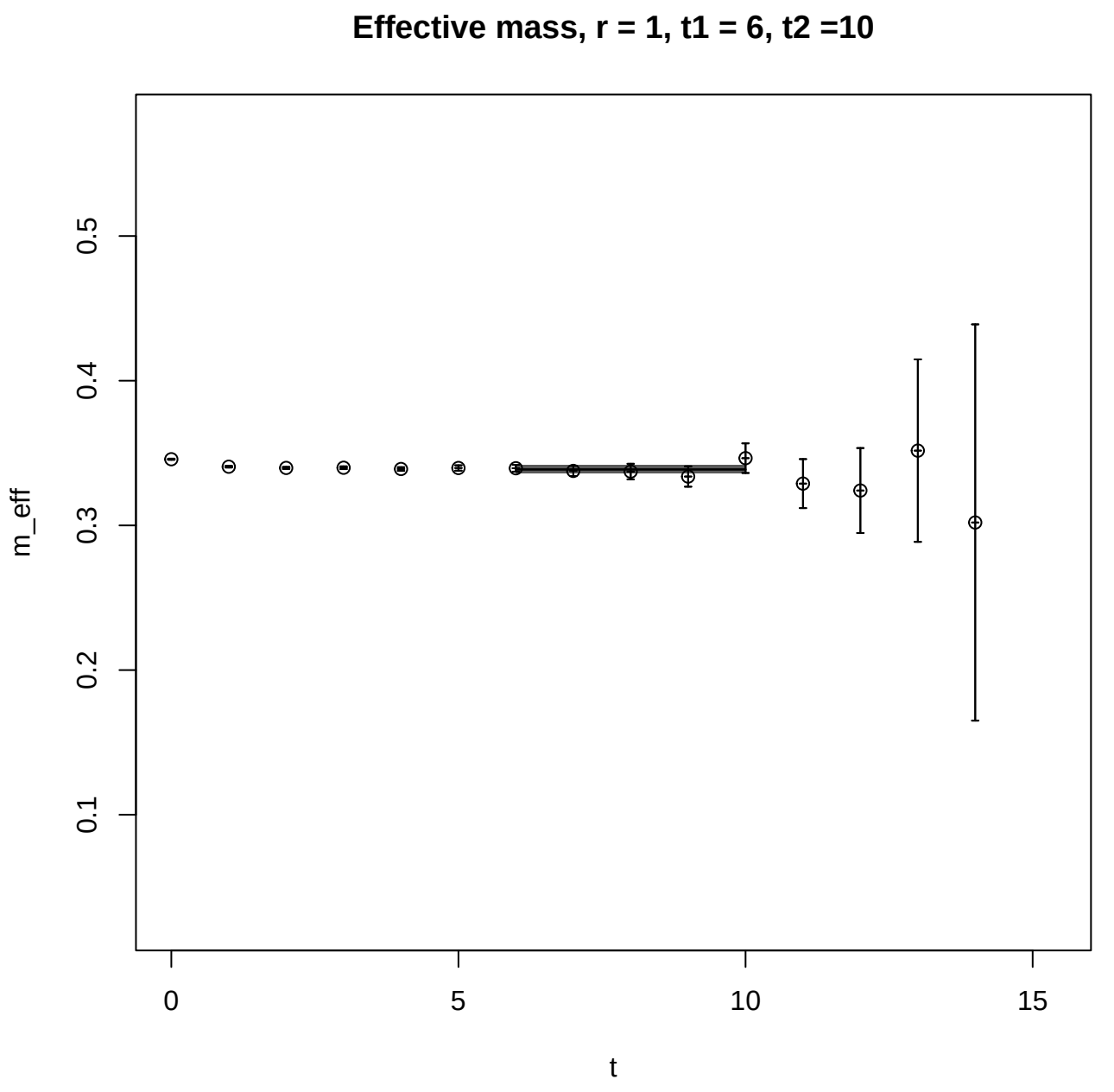

Effective mass, r = 1, t1 = 5, t2 =13
m_eff
0.1
0.2
0.3
0.4
0.5
0
5
10
15
t

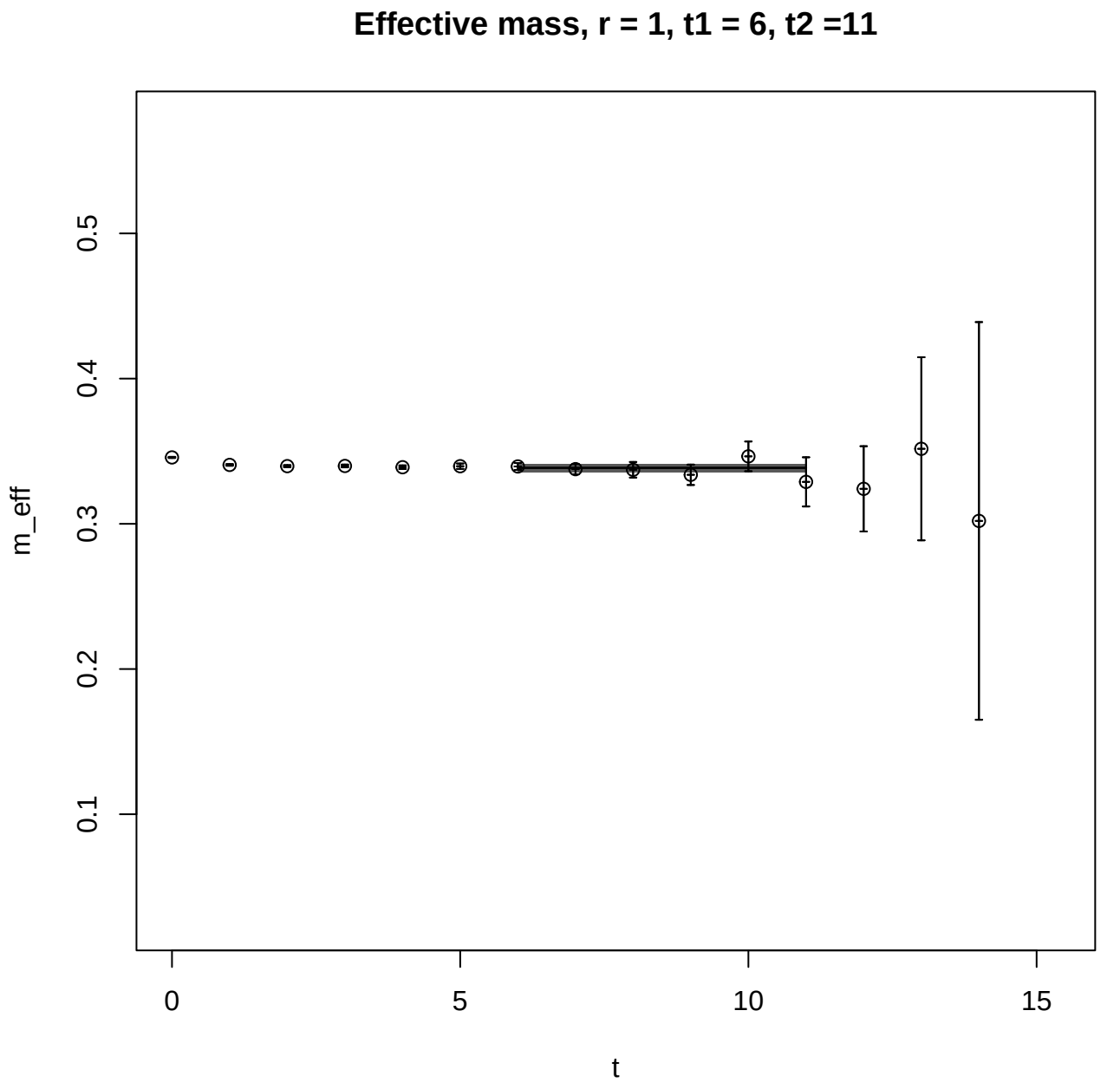

Effective mass, r = 1, t1 = 5, t2 =14
m_eff
0.1
0.2
0.3
0.4
0.5
0
5
10
15
t

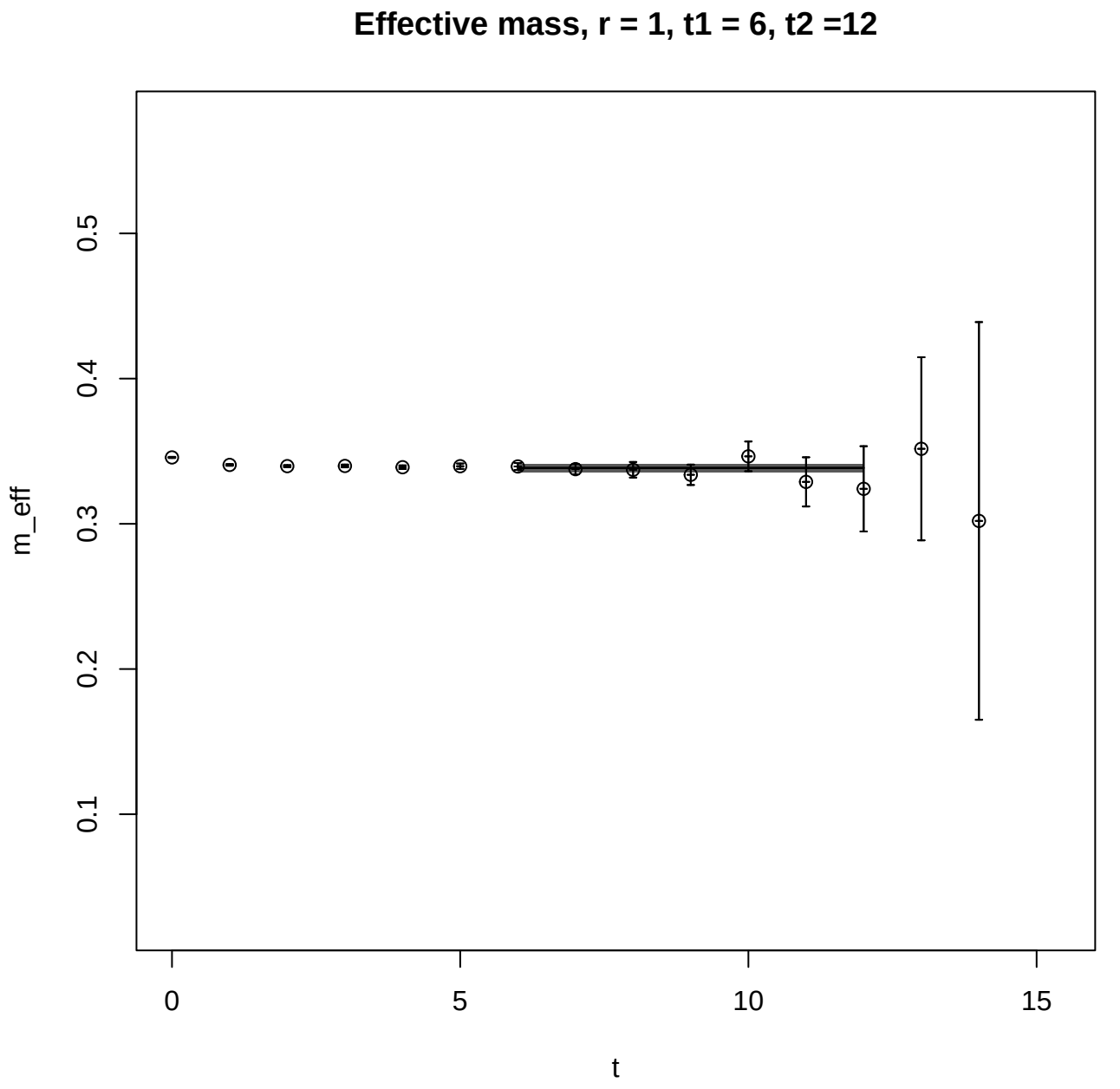

Effective mass, r = 1, t1 = 6, t2 =7
m_eff
0.1
0.2
0.3
0.4
0.5
0
5
10
15
t

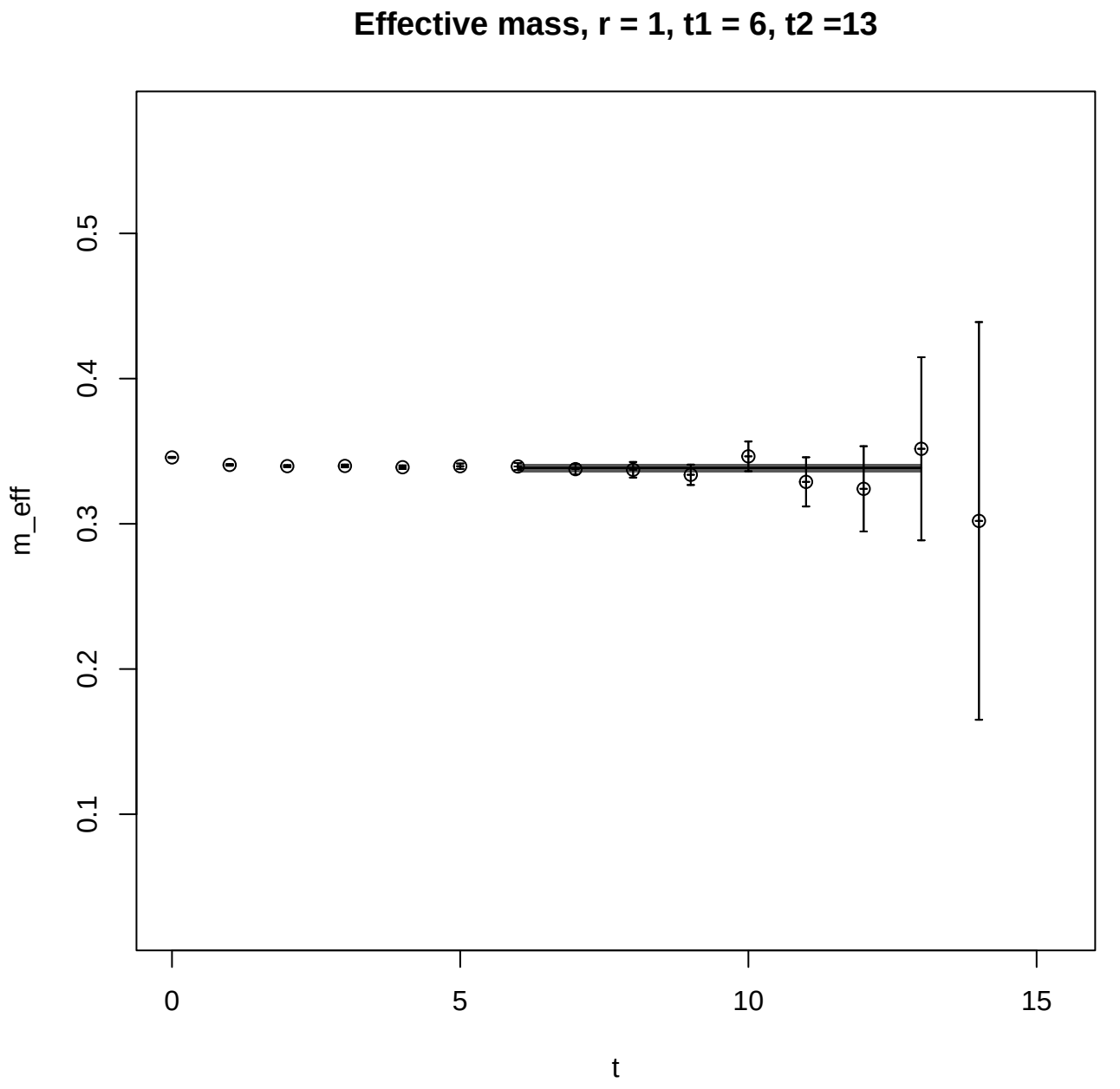

Effective mass, r = 1, t1 = 6, t2 =8
m_eff
0.1
0.2
0.3
0.4
0.5
0
5
10
15
t

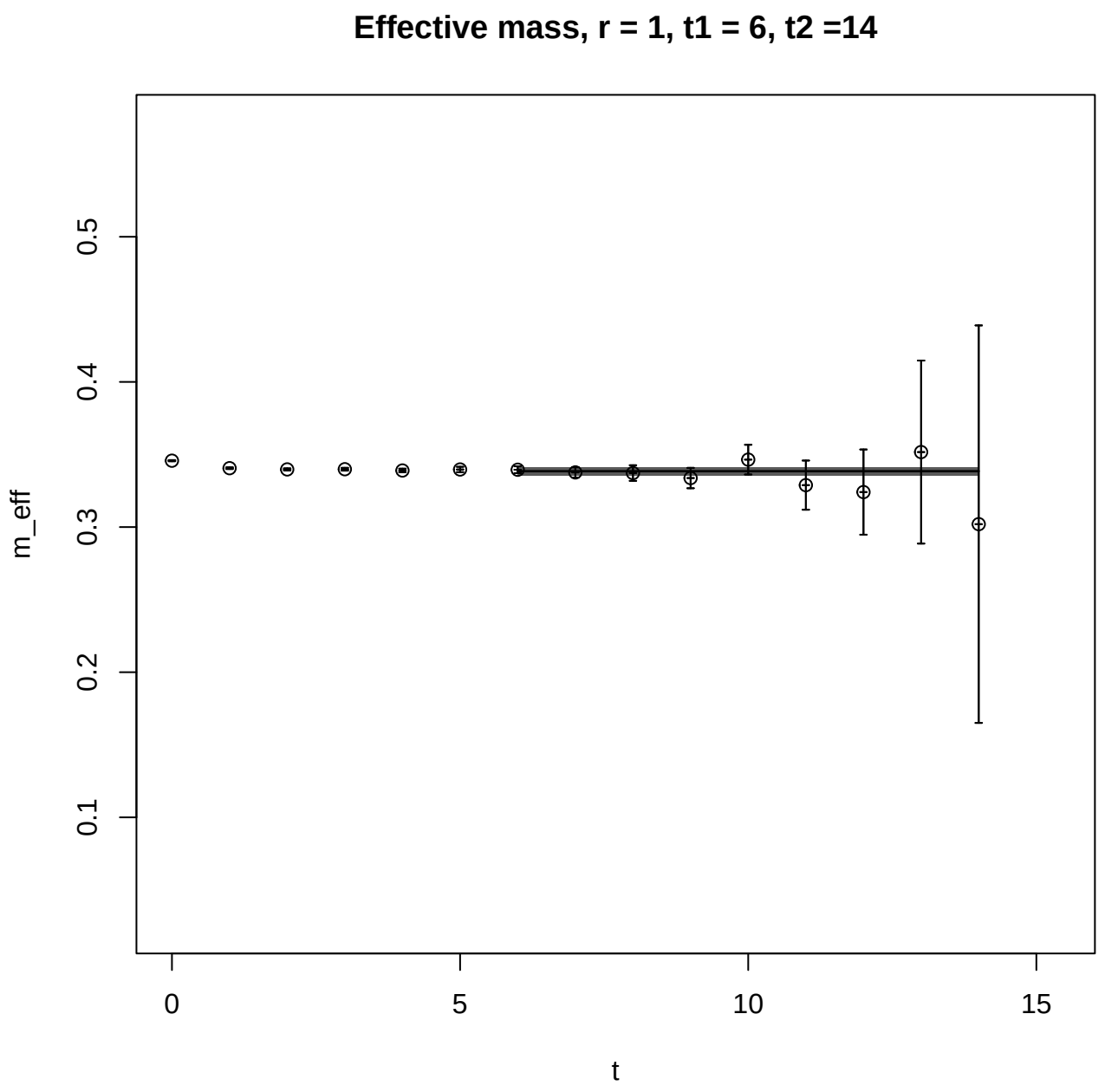

Effective mass, r = 1, t1 = 6, t2 =9
m_eff
0.1
0.2
0.3
0.4
0.5
0
5
10
15
t


Effective mass, r = 1, t1 = 6, t2 =10
m_eff
0.1
0.2
0.3
0.4
0.5
0
5
10
15
t


Effective mass, r = 1, t1 = 6, t2 =11
m_eff
0.1
0.2
0.3
0.4
0.5
0
5
10
15
t


Effective mass, r = 1, t1 = 6, t2 =12
m_eff
0.1
0.2
0.3
0.4
0.5
0
5
10
15
t


Effective mass, r = 1, t1 = 6, t2 =13
m_eff
0.1
0.2
0.3
0.4
0.5
0
5
10
15
t


Effective mass, r = 1, t1 = 6, t2 =14
m_eff
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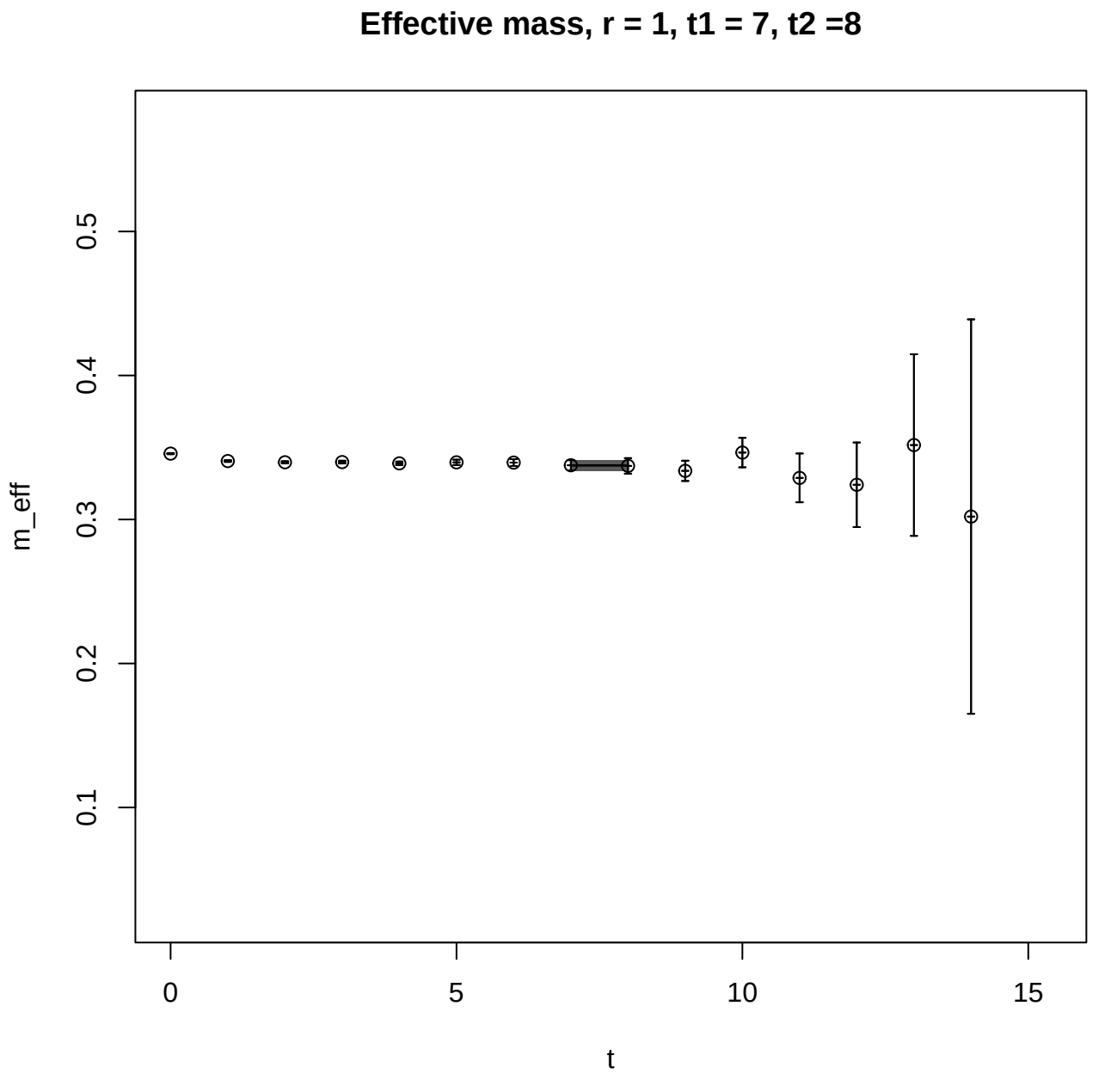
0.2
0.3
0.4
0.5
0
5
10
15
t


Effective mass, r = 1, t1 = 7, t2 =8
m_eff
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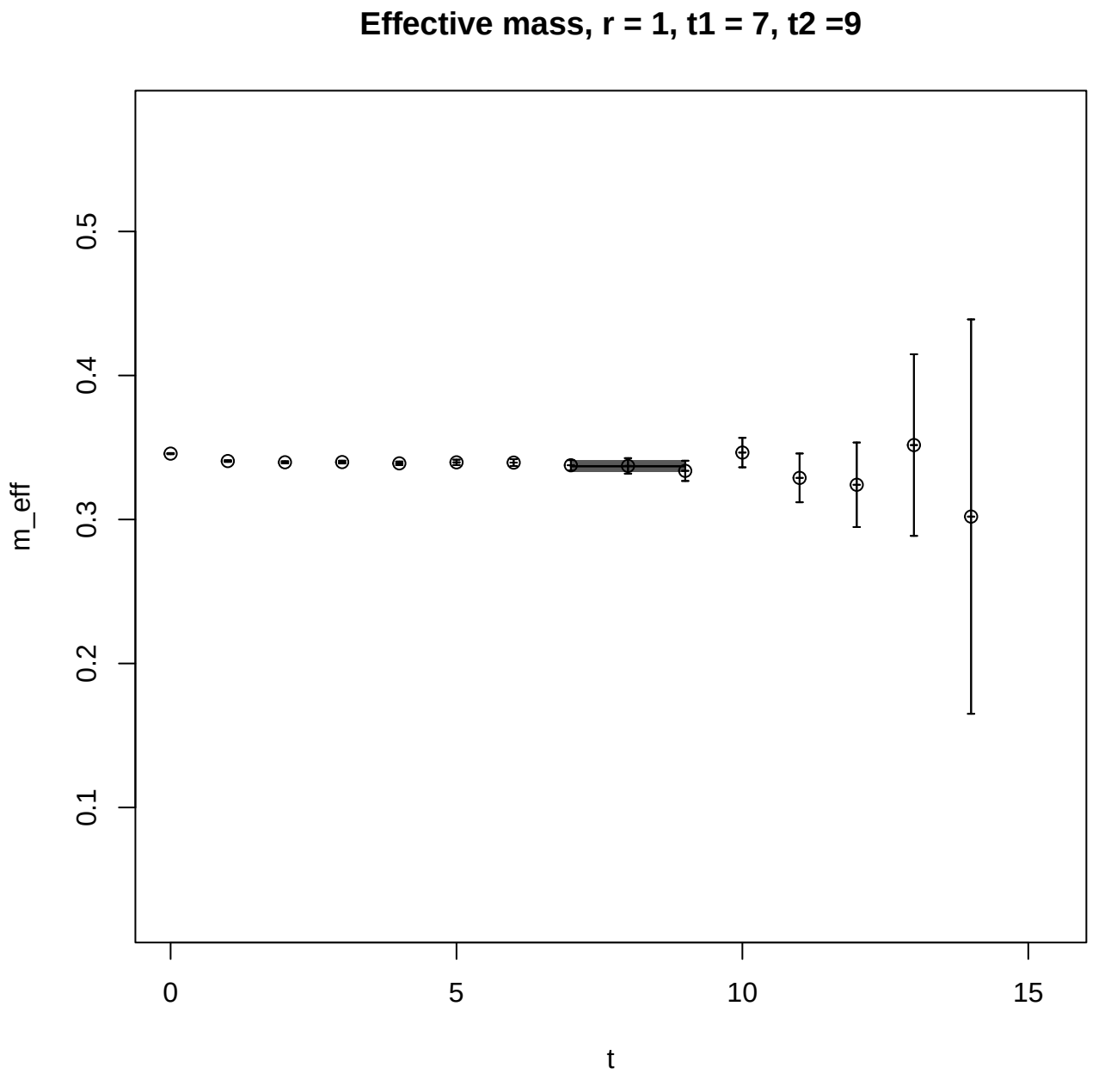
0.2
0.3
0.4
0.5
0
5
10
15
t


Effective mass, r = 1, t1 = 7, t2 =9
m_eff
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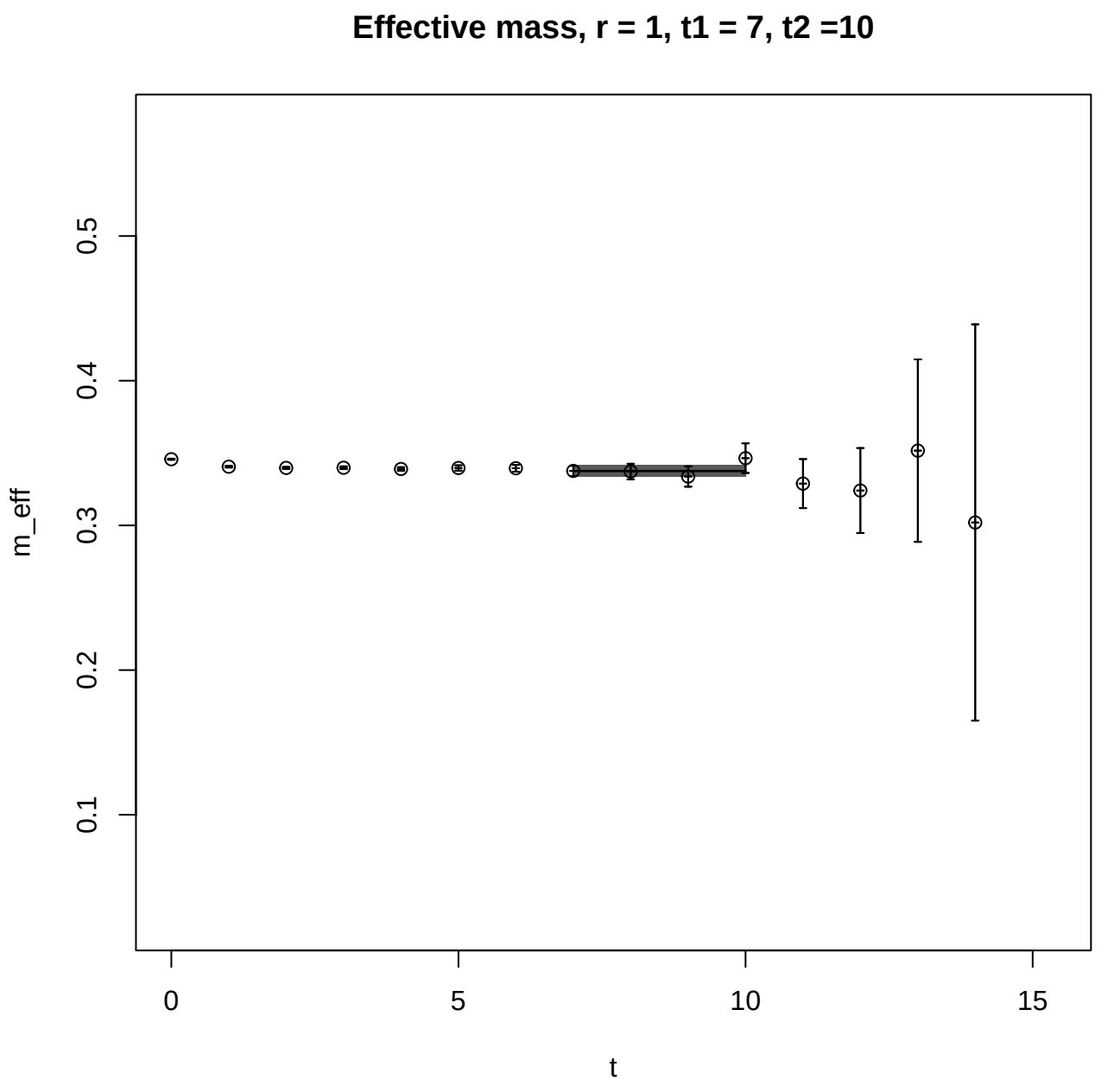
0.2
0.3
0.4
0.5
0
5
10
15
t


Effective mass, r = 1, t1 = 7, t2 =10
m_eff
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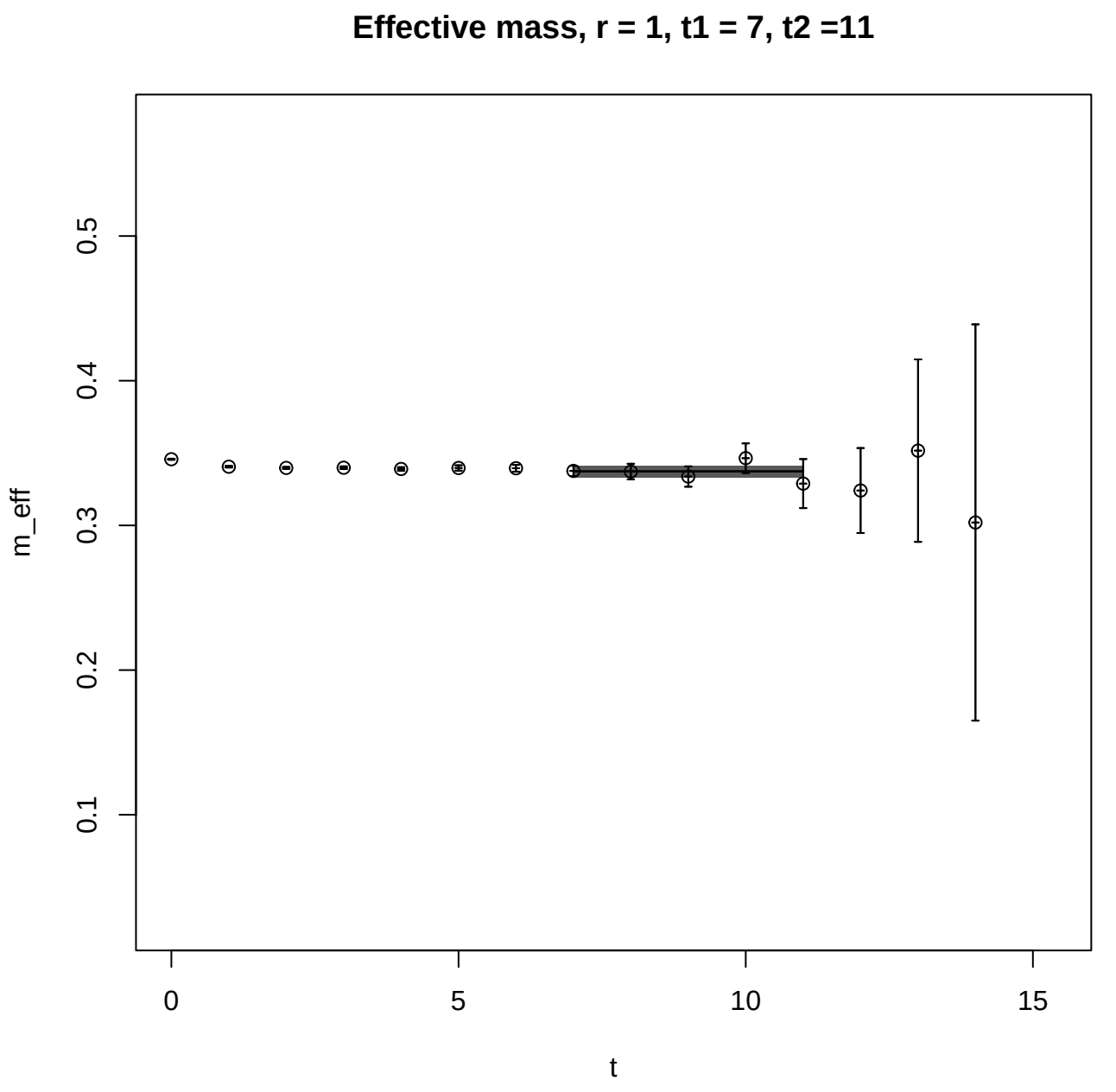
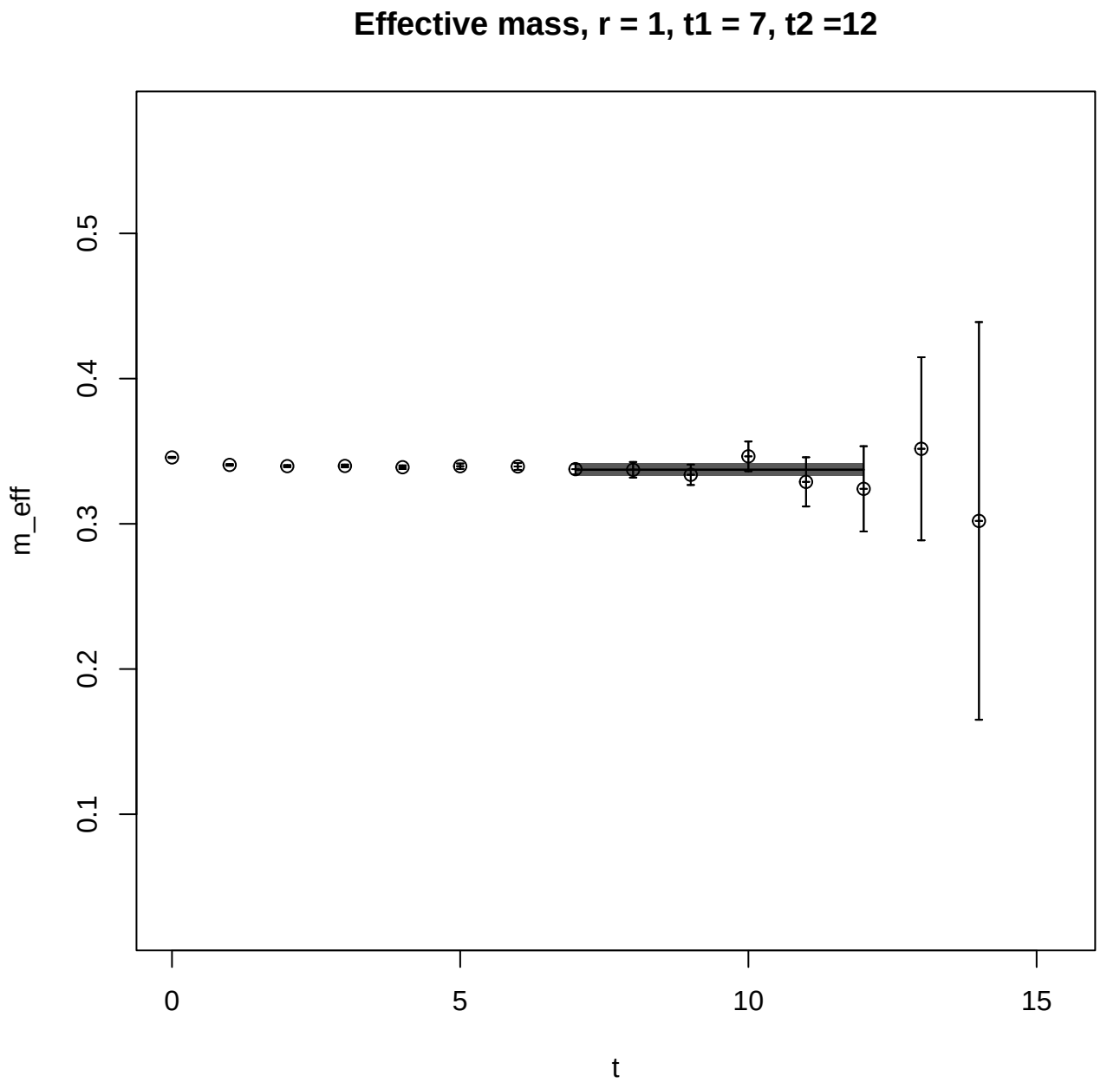
0.2
0.3
0.4
0.5
0
5
10
15
t


Effective mass, r = 1, t1 = 7, t2 =11
m_eff
0.1
0.2
0.3
0.4
0.5
0
5
10
15
t


Effective mass, r = 1, t1 = 7, t2 =12
m_eff
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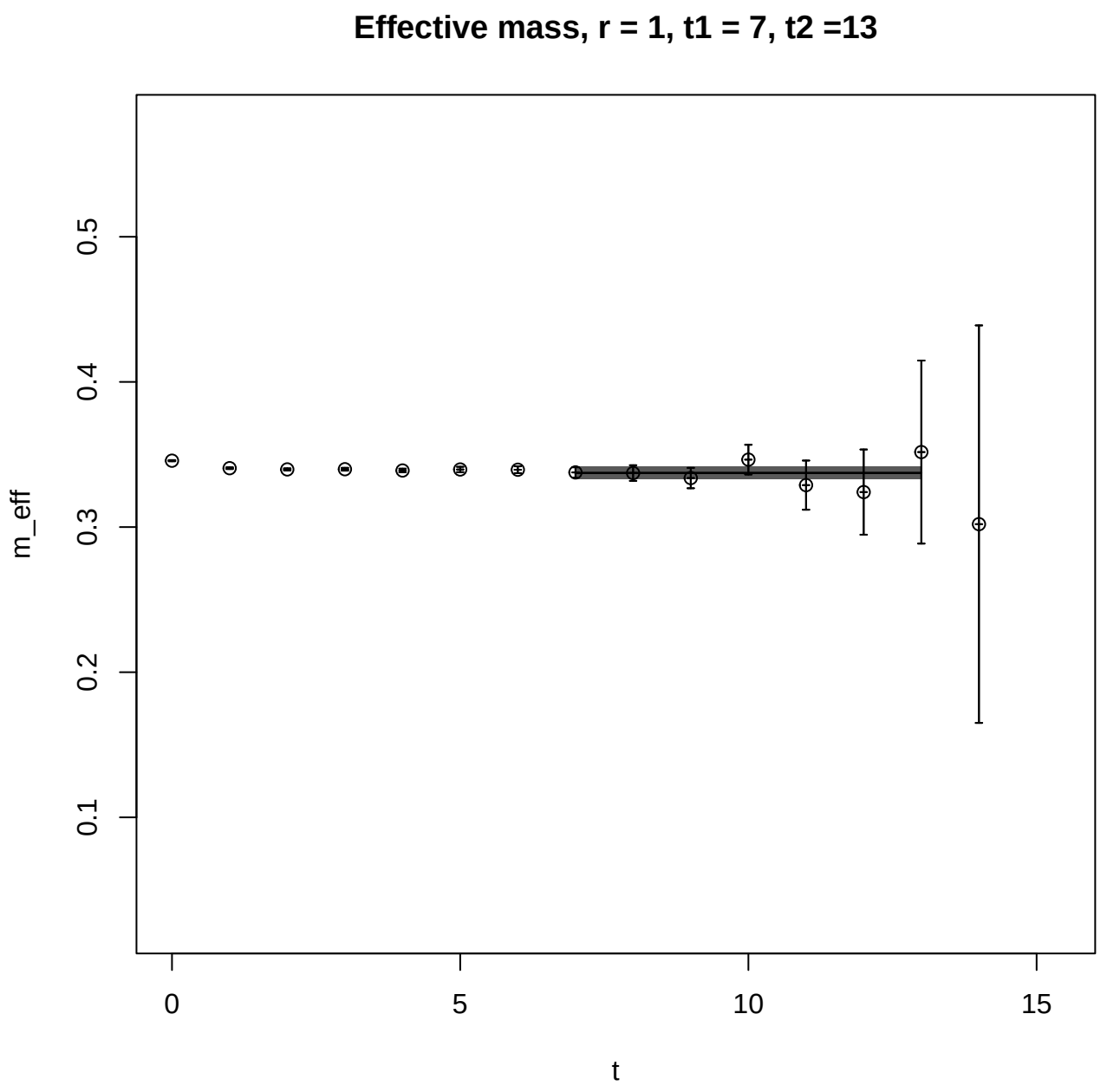
0.2
0.3
0.4
0.5
0
5
10
15
t


Effective mass, r = 1, t1 = 7, t2 =13
m_eff
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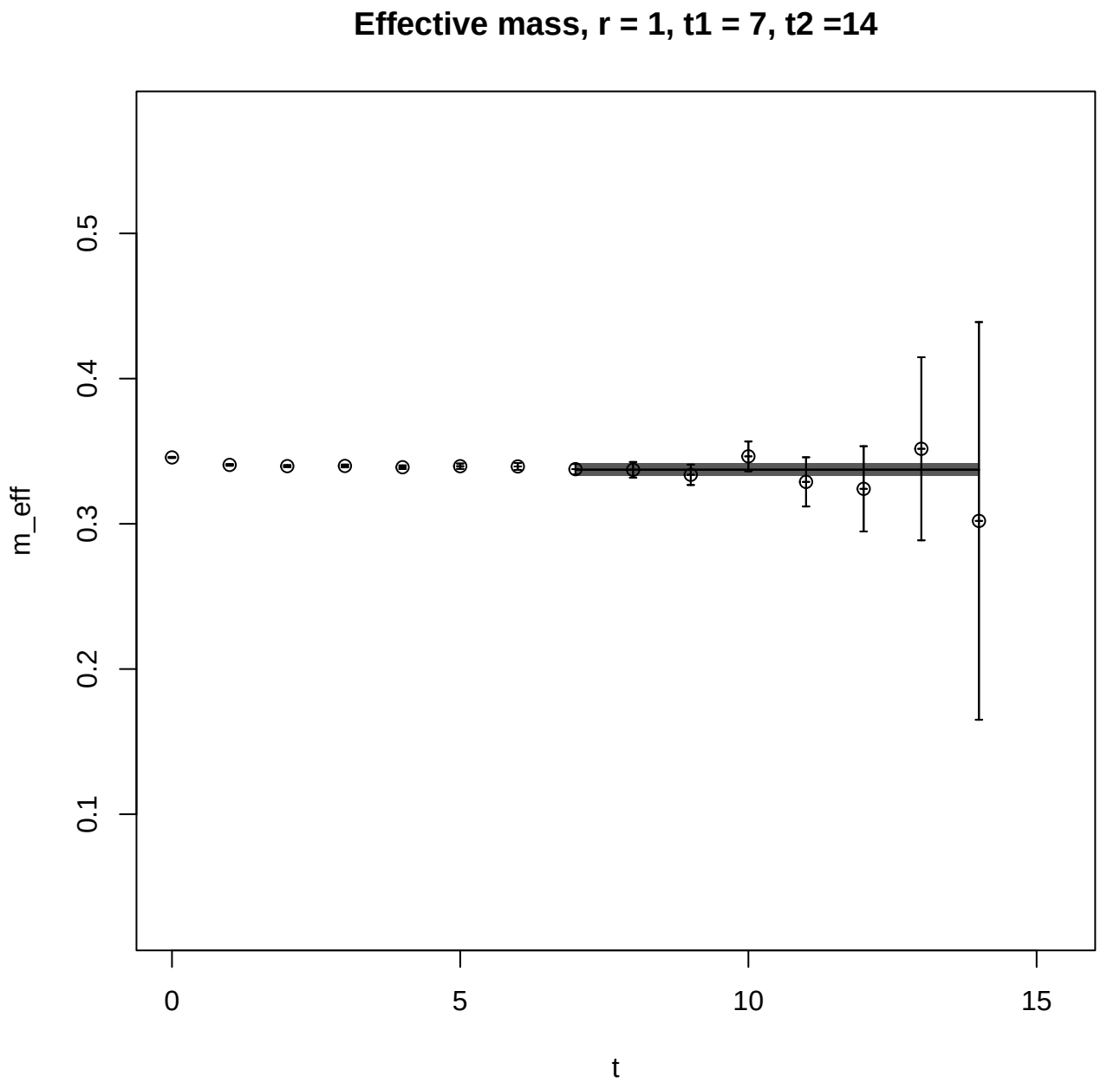
0.2
0.3
0.4
0.5
0
5
10
15
t


Effective mass, r = 1, t1 = 7, t2 =14
m_eff
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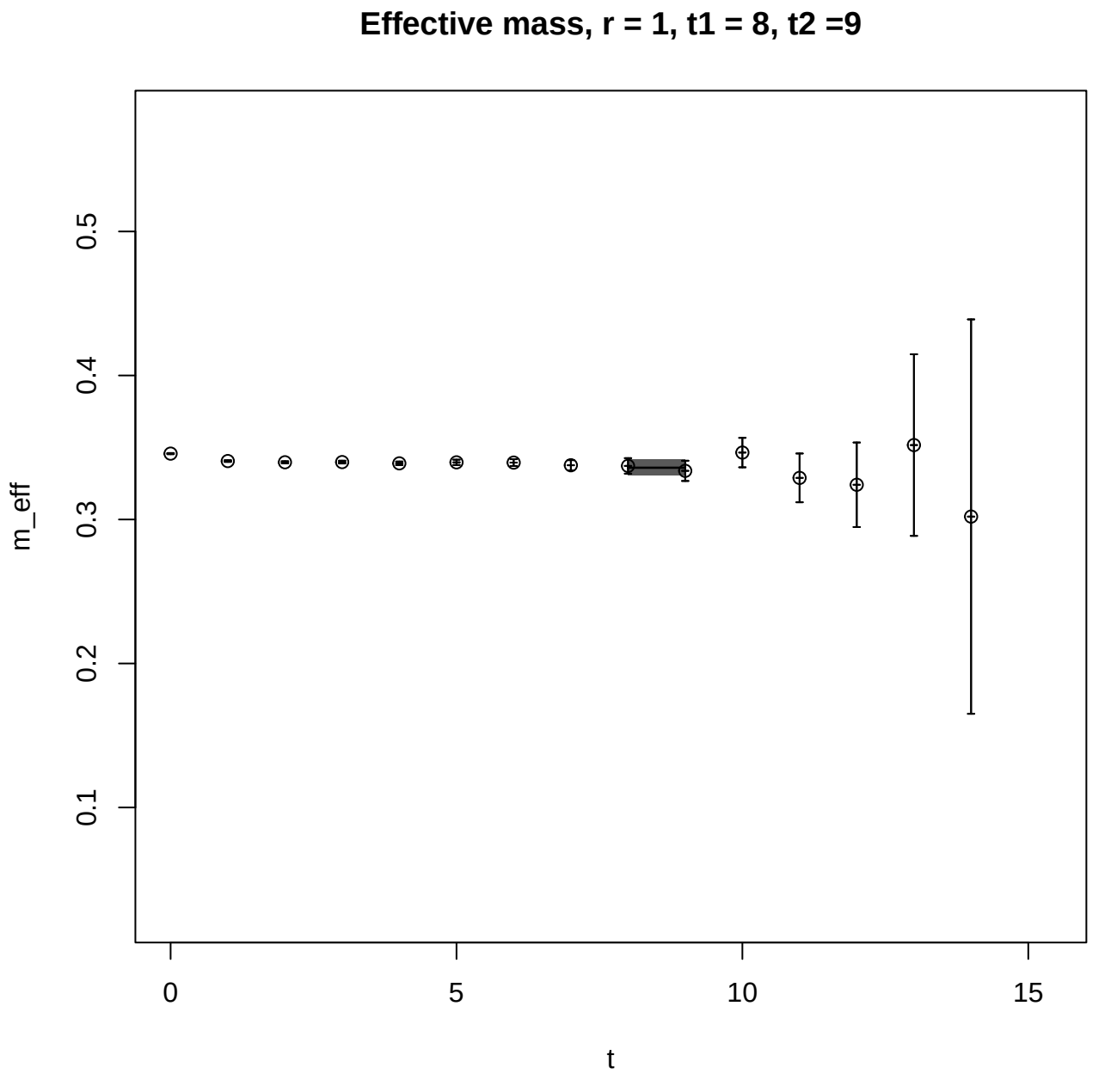
0.2
0.3
0.4
0.5
0
5
10
15
t


Effective mass, r = 1, t1 = 8, t2 =9
m_eff
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0.4
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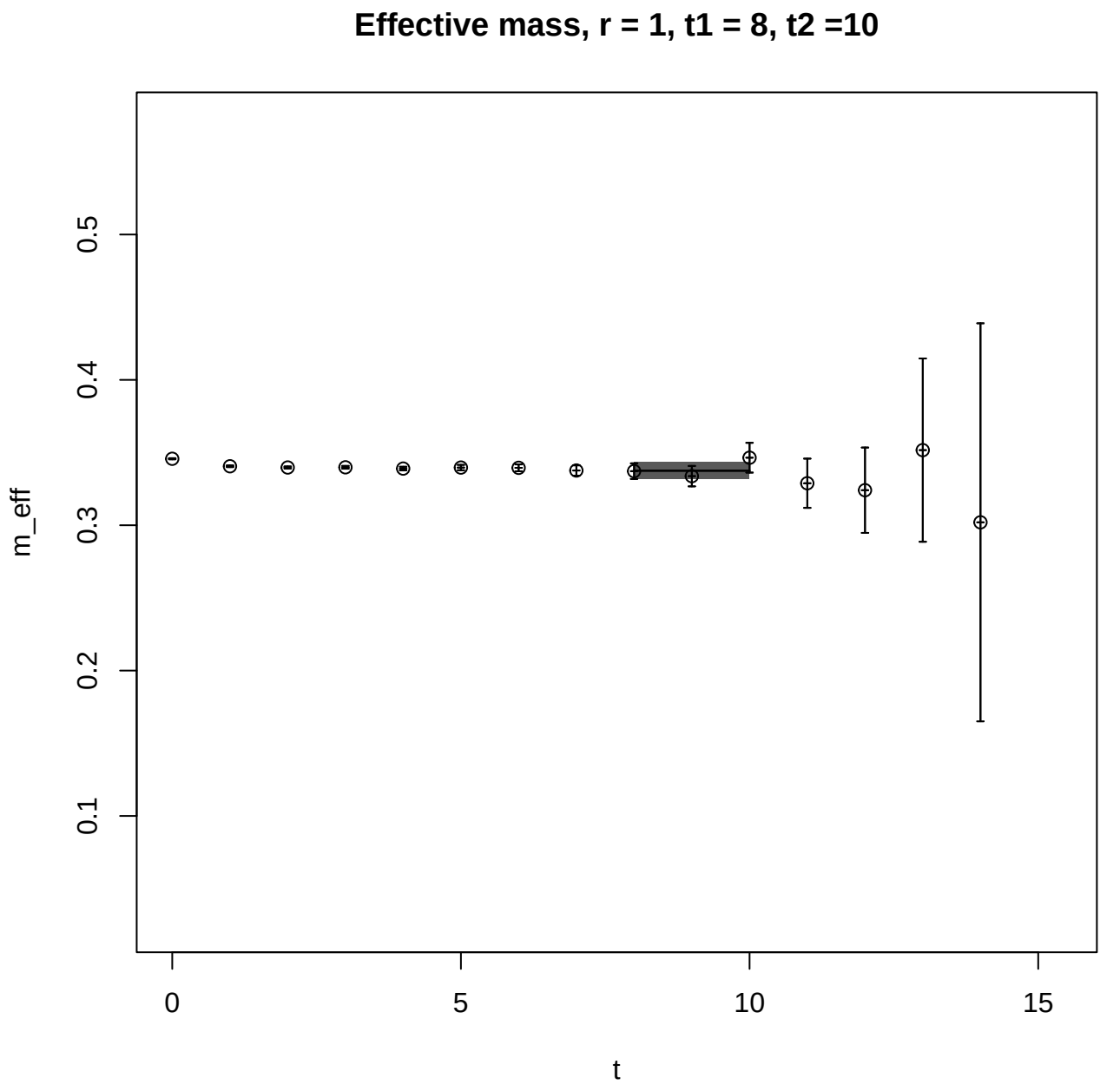
0
5
10
15
t


Effective mass, r = 1, t1 = 8, t2 =10
m_eff
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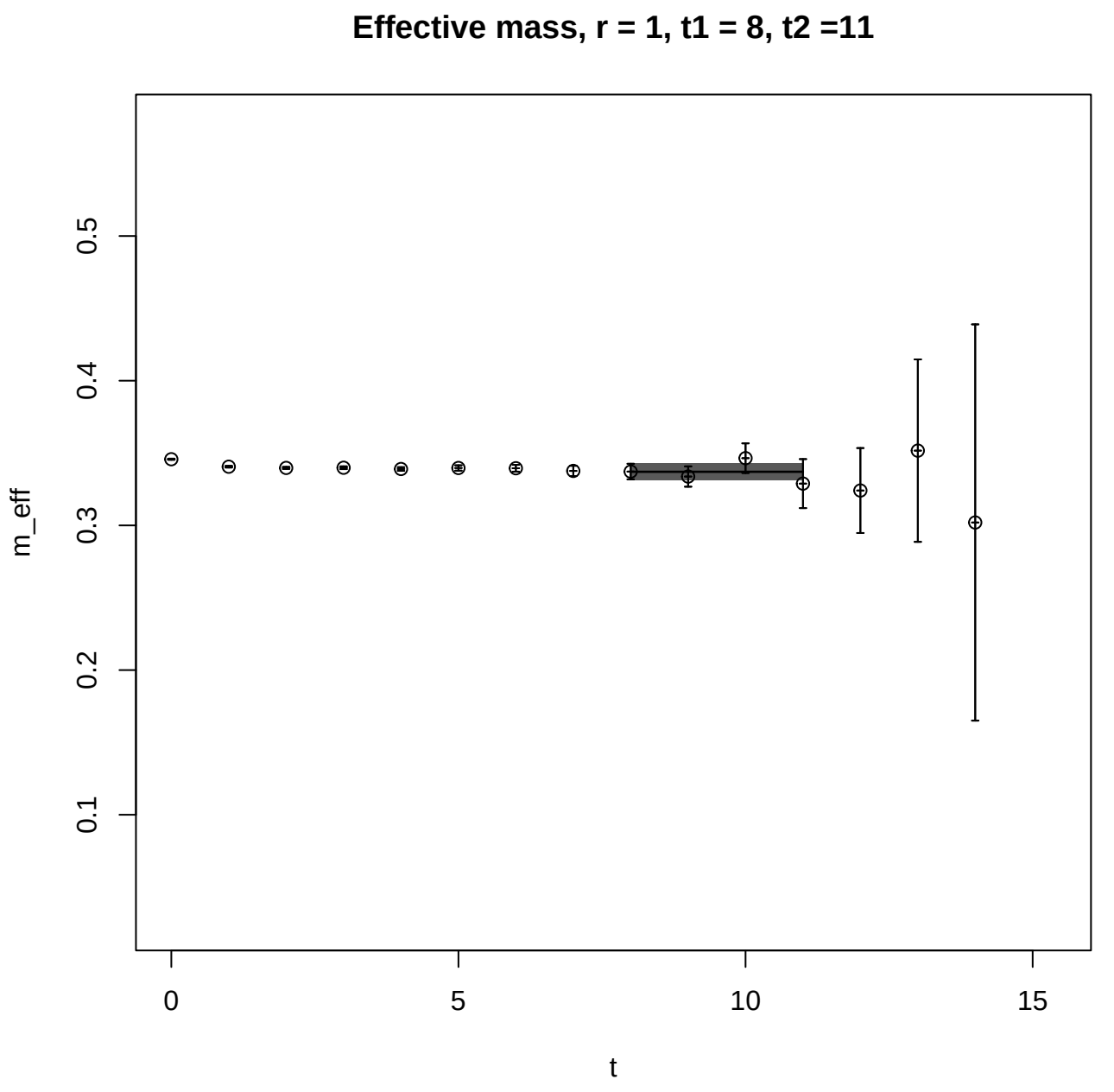
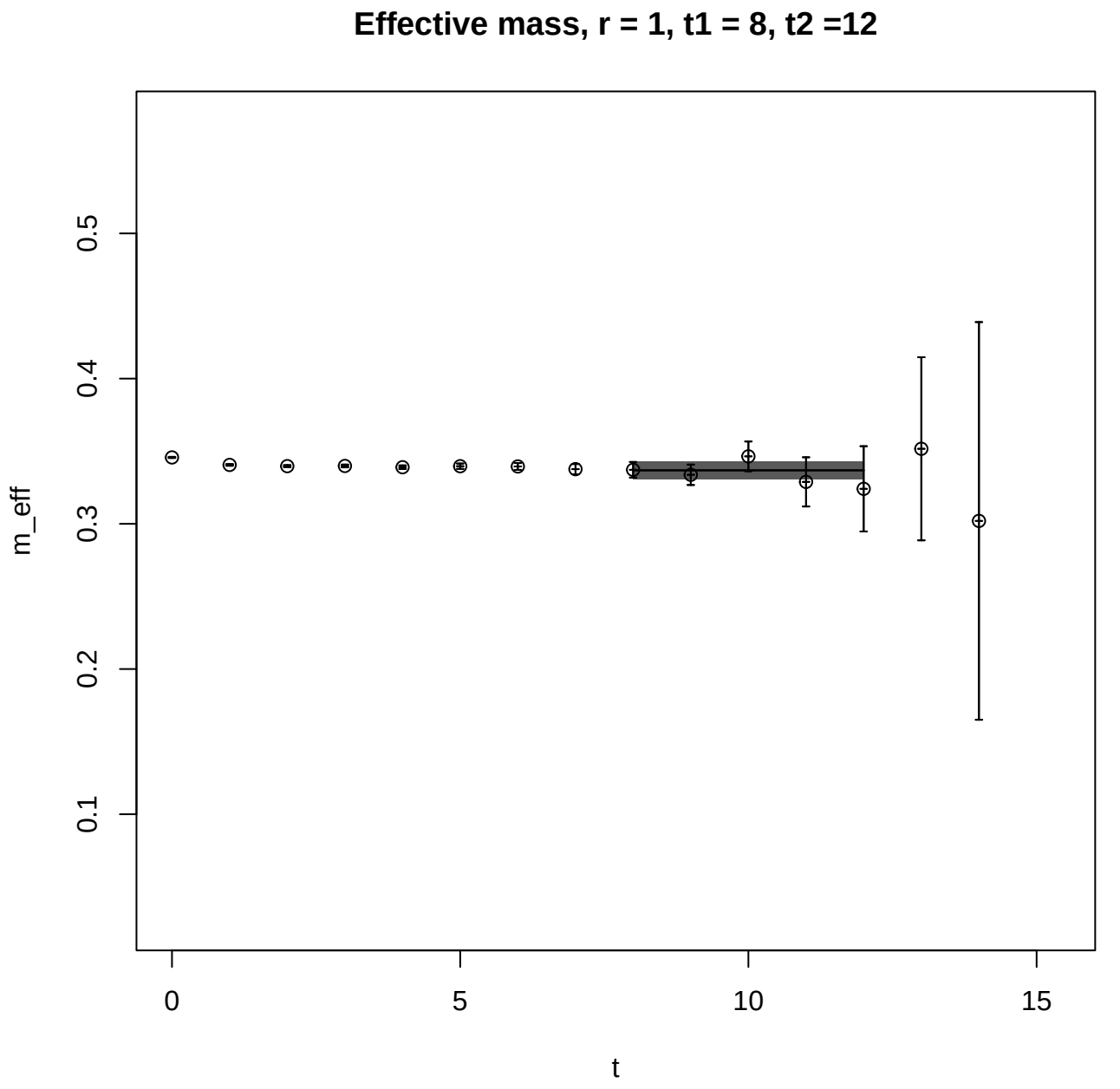
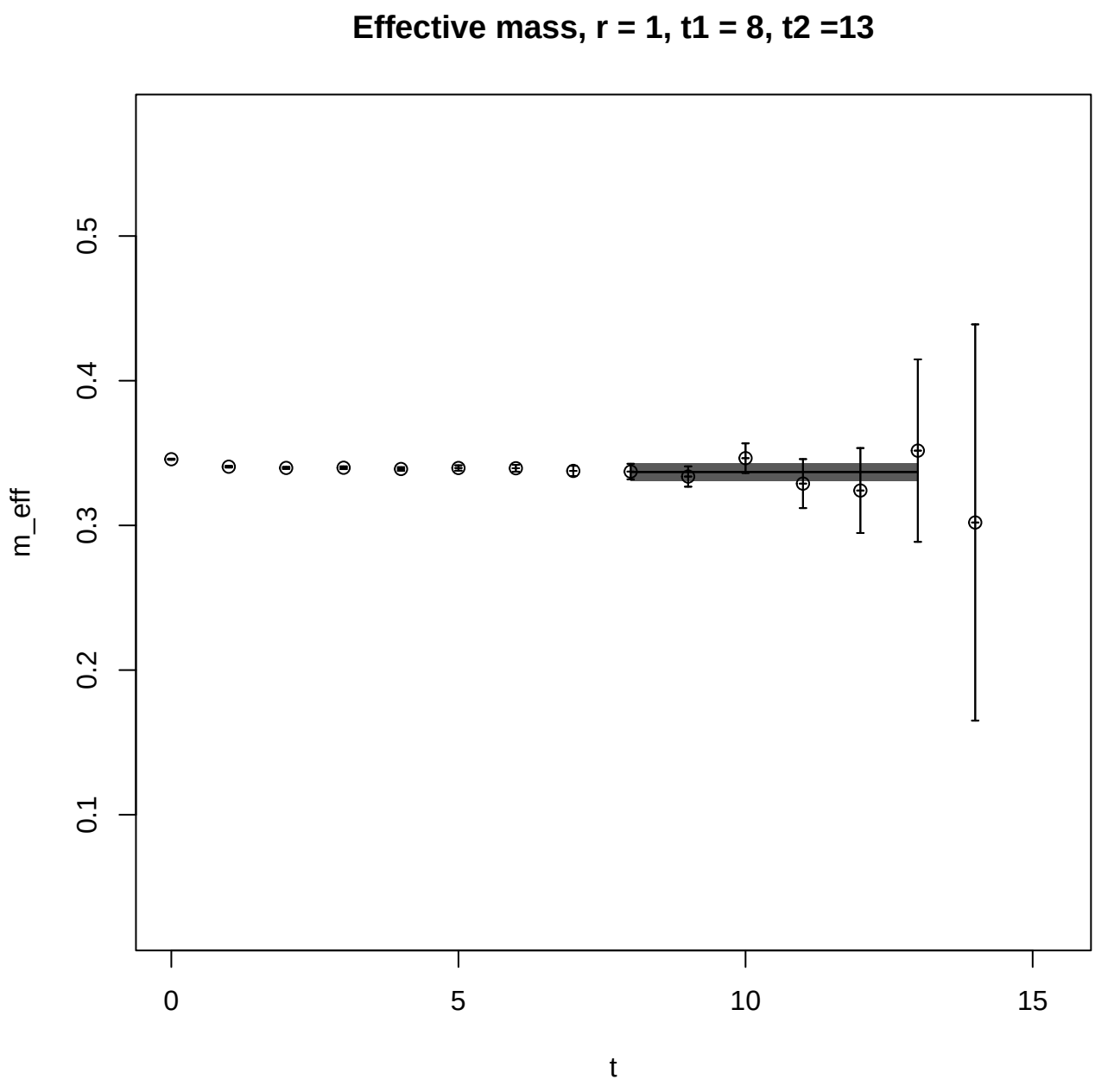
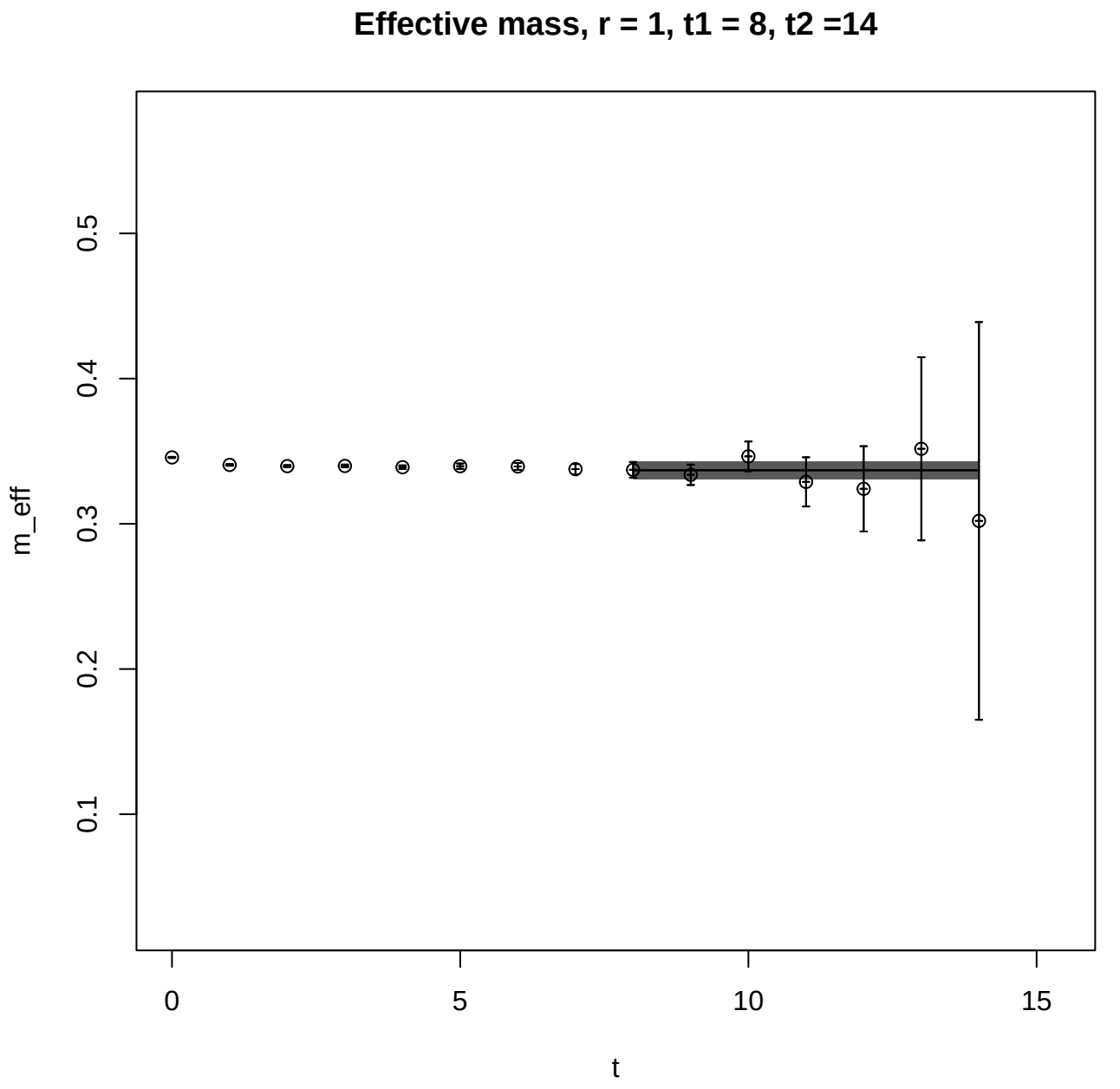
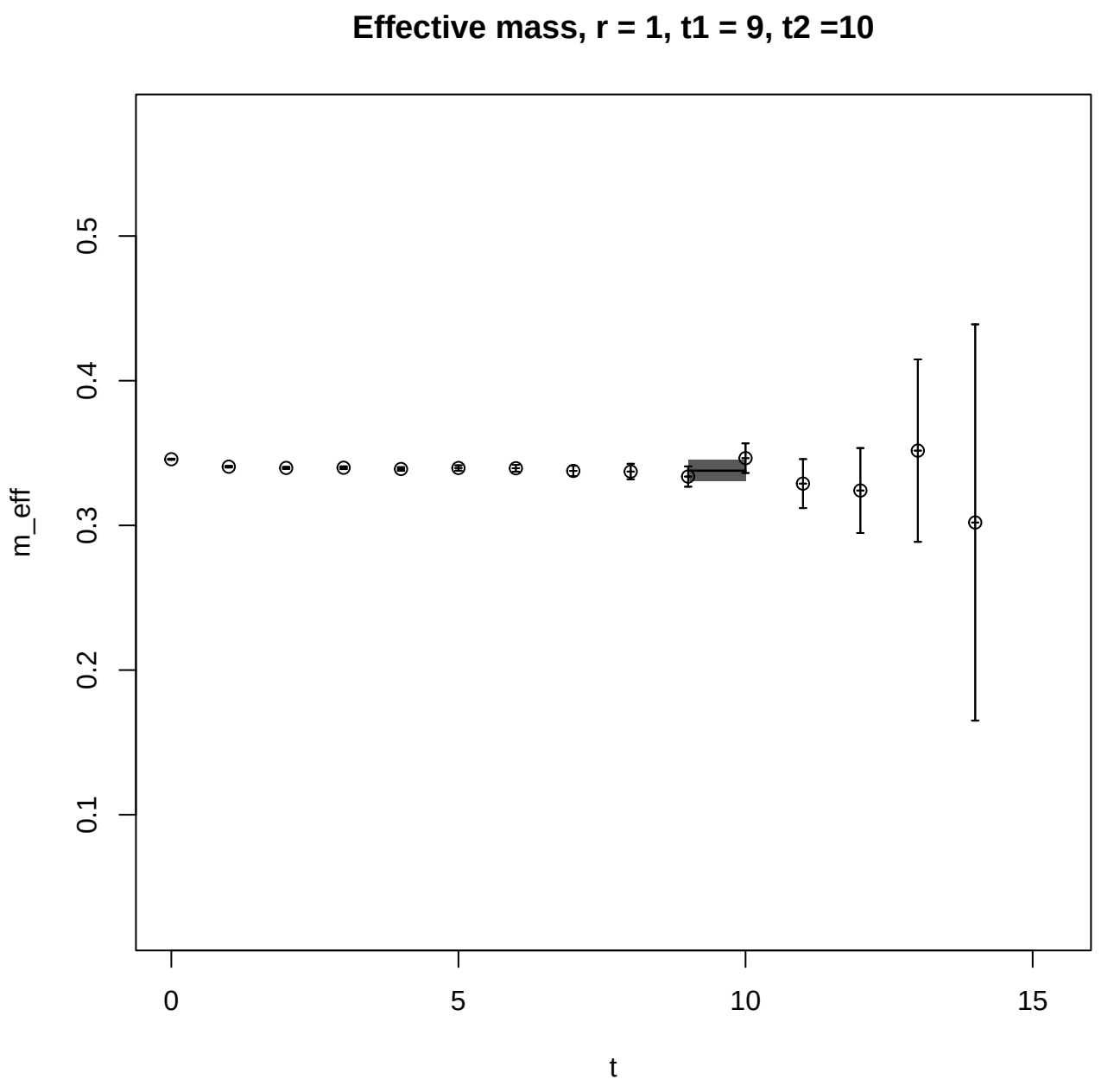
0.2
0.3
0.4
0.5
0
5
10
15
t

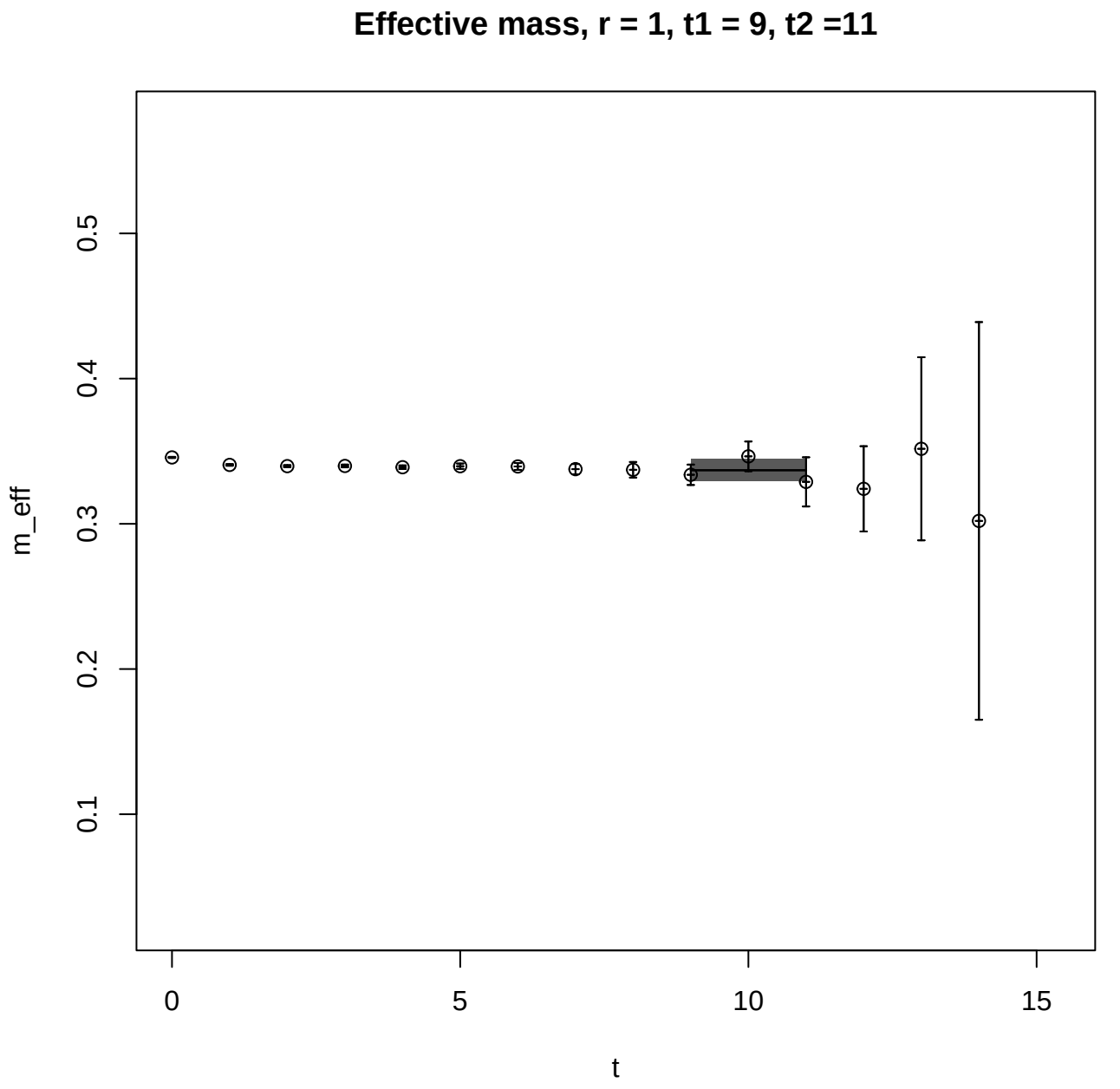

Effective mass, r = 1, t1 = 8, t2 =11
m_eff
0.1
0.2
0.3
0.4
0.5
0
5
10
15
t

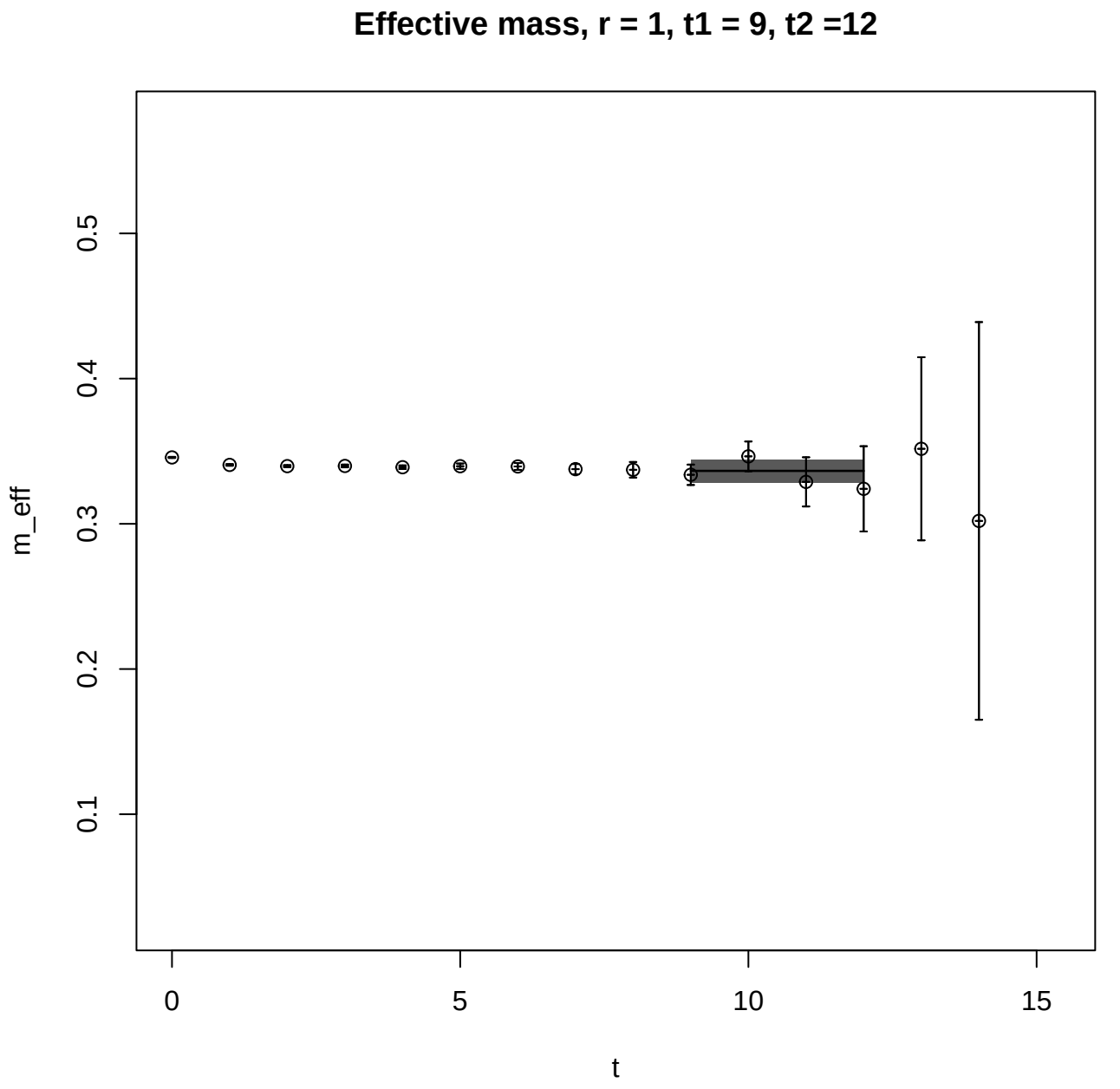

Effective mass, r = 1, t1 = 8, t2 =12
m_eff
0.1
0.2
0.3
0.4
0.5
0
5
10
15
t

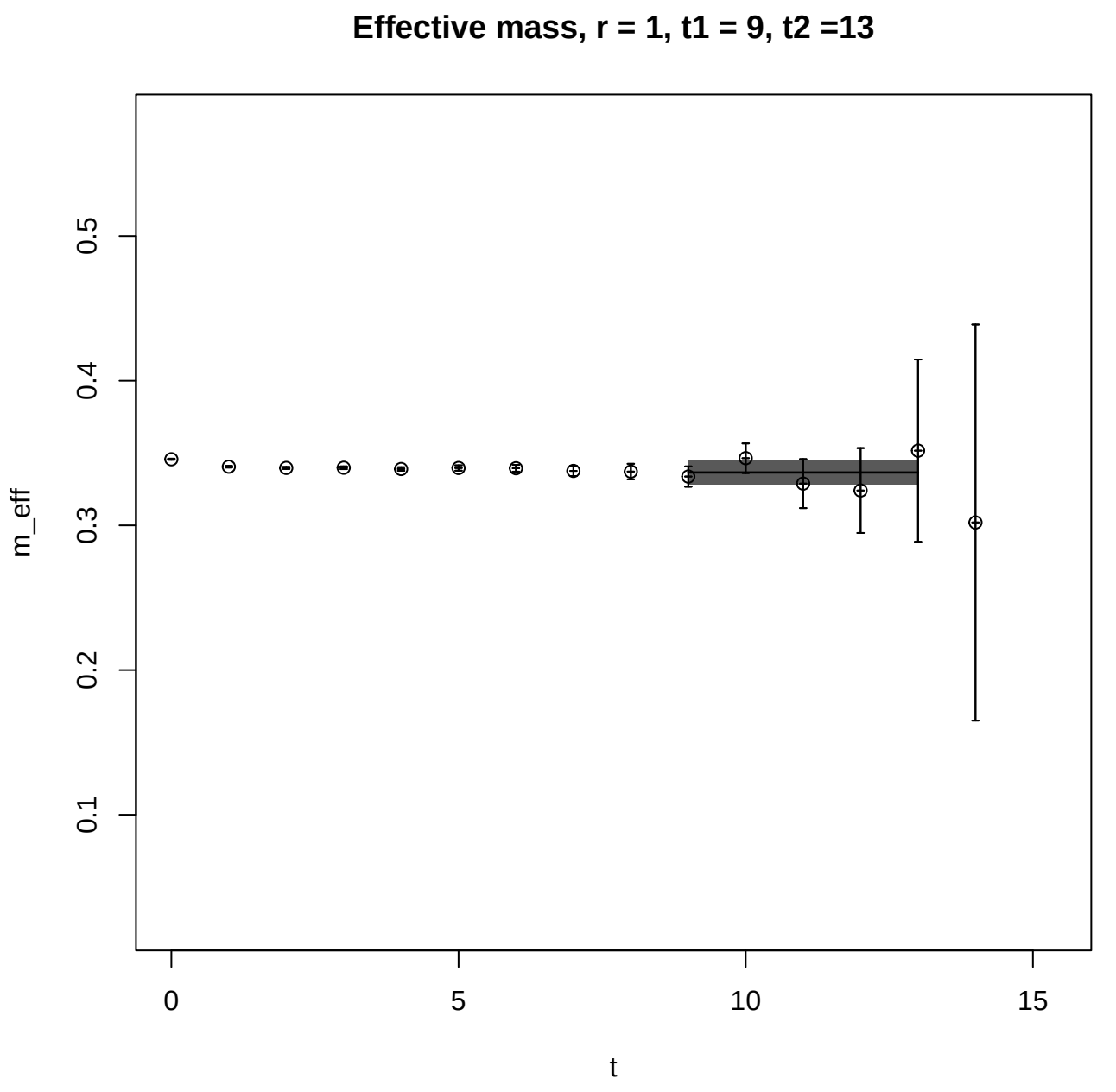

Effective mass, r = 1, t1 = 8, t2 =13
m_eff
0.1
0.2
0.3
0.4
0.5
0
5
10
15
t

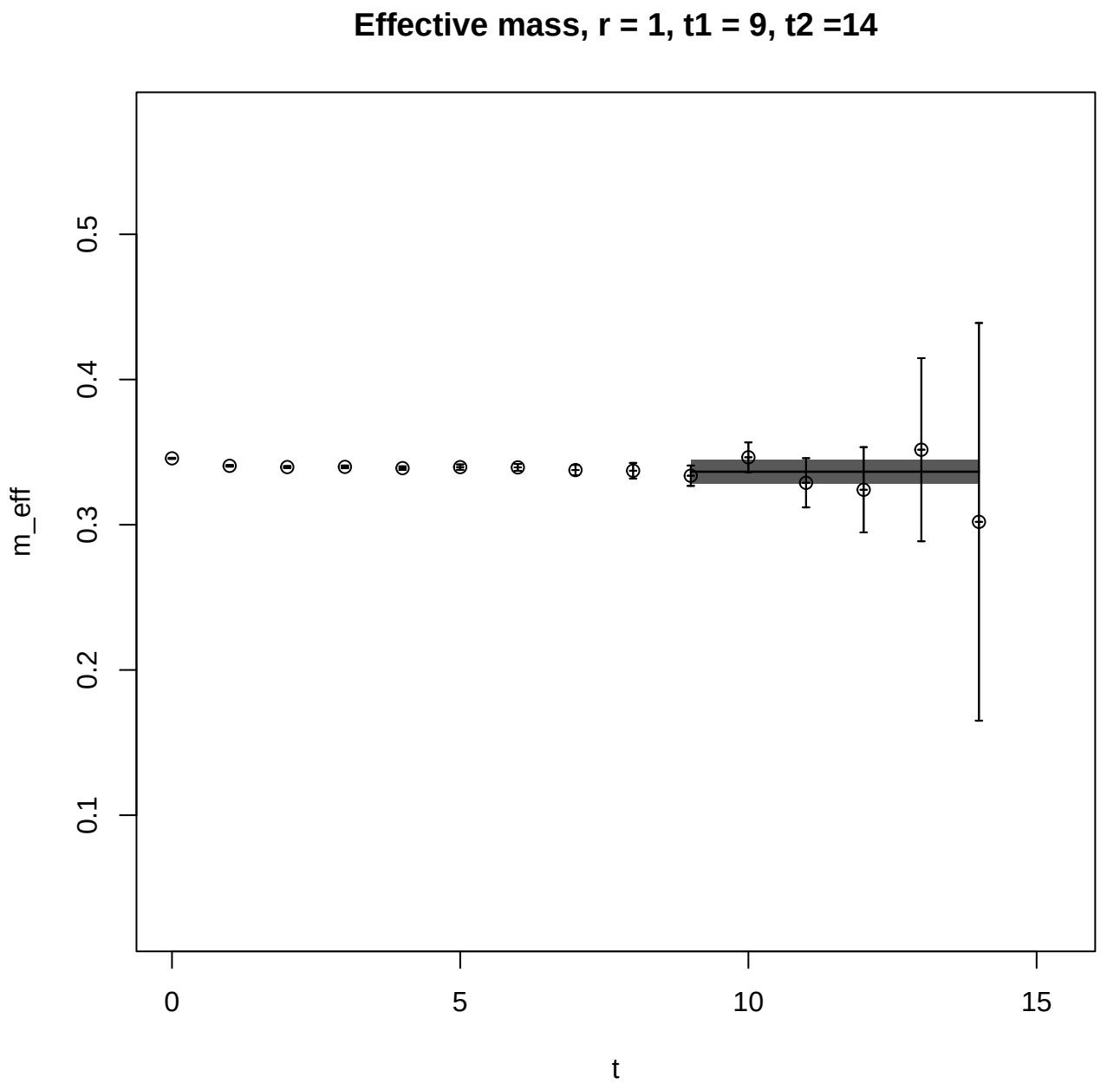

Effective mass, r = 1, t1 = 8, t2 =14
m_eff
0.1
0.2
0.3
0.4
0.5
0
5
10
15
t


Effective mass, r = 1, t1 = 9, t2 =10
m_eff
0.1
0.2
0.3
0.4
0.5
0
5
10
15
t


Effective mass, r = 1, t1 = 9, t2 =11
m_eff
0.1
0.2
0.3
0.4
0.5
0
5
10
15
t


Effective mass, r = 1, t1 = 9, t2 =12
m_eff
0.1
0.2
0.3
0.4
0.5
0
5
10
15
t


Effective mass, r = 1, t1 = 9, t2 =13
m_eff
0.1
0.2
0.3
0.4
0.5
0
5
10
15
t


Effective mass, r = 1, t1 = 9, t2 =14
m_eff
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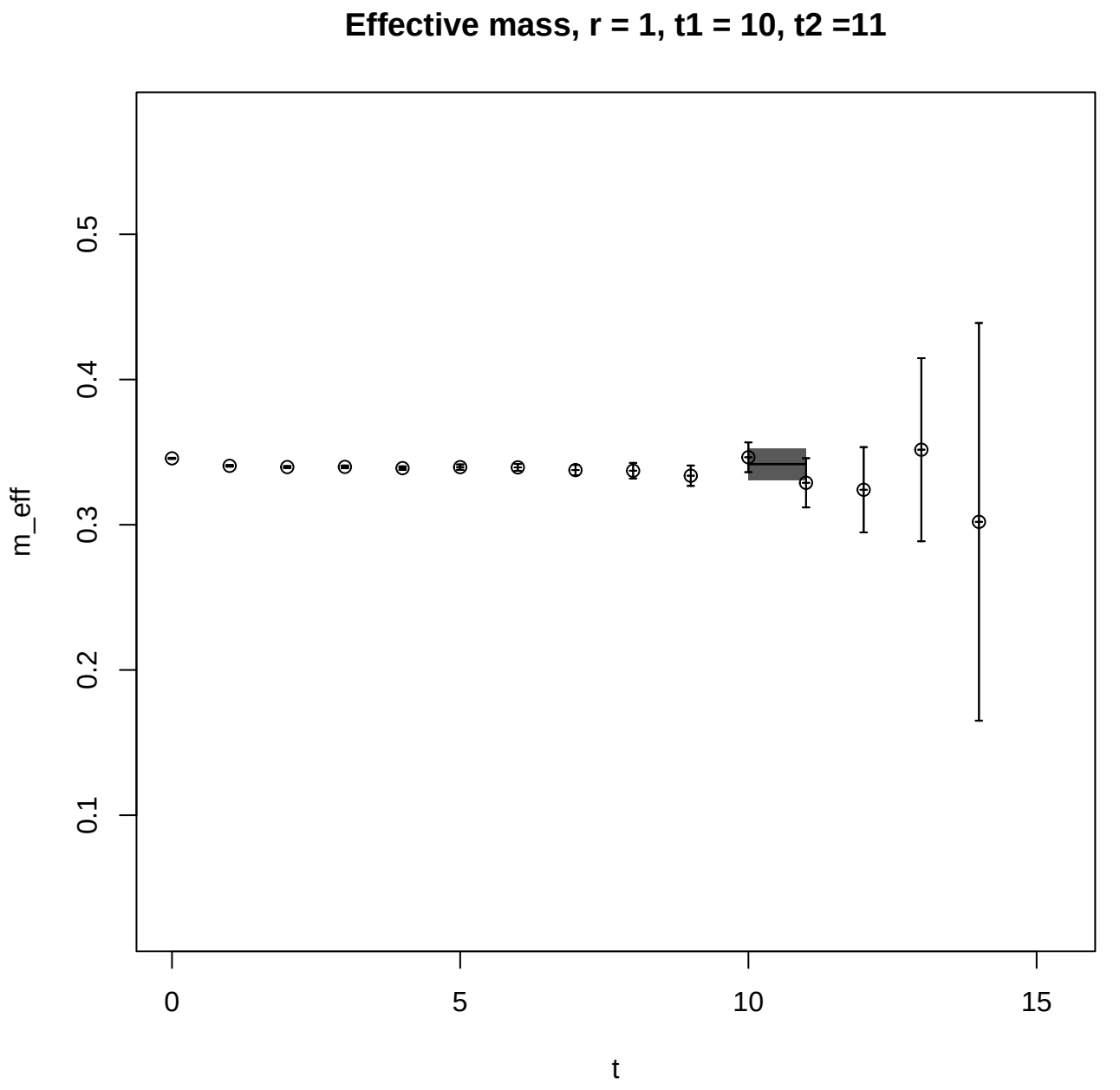
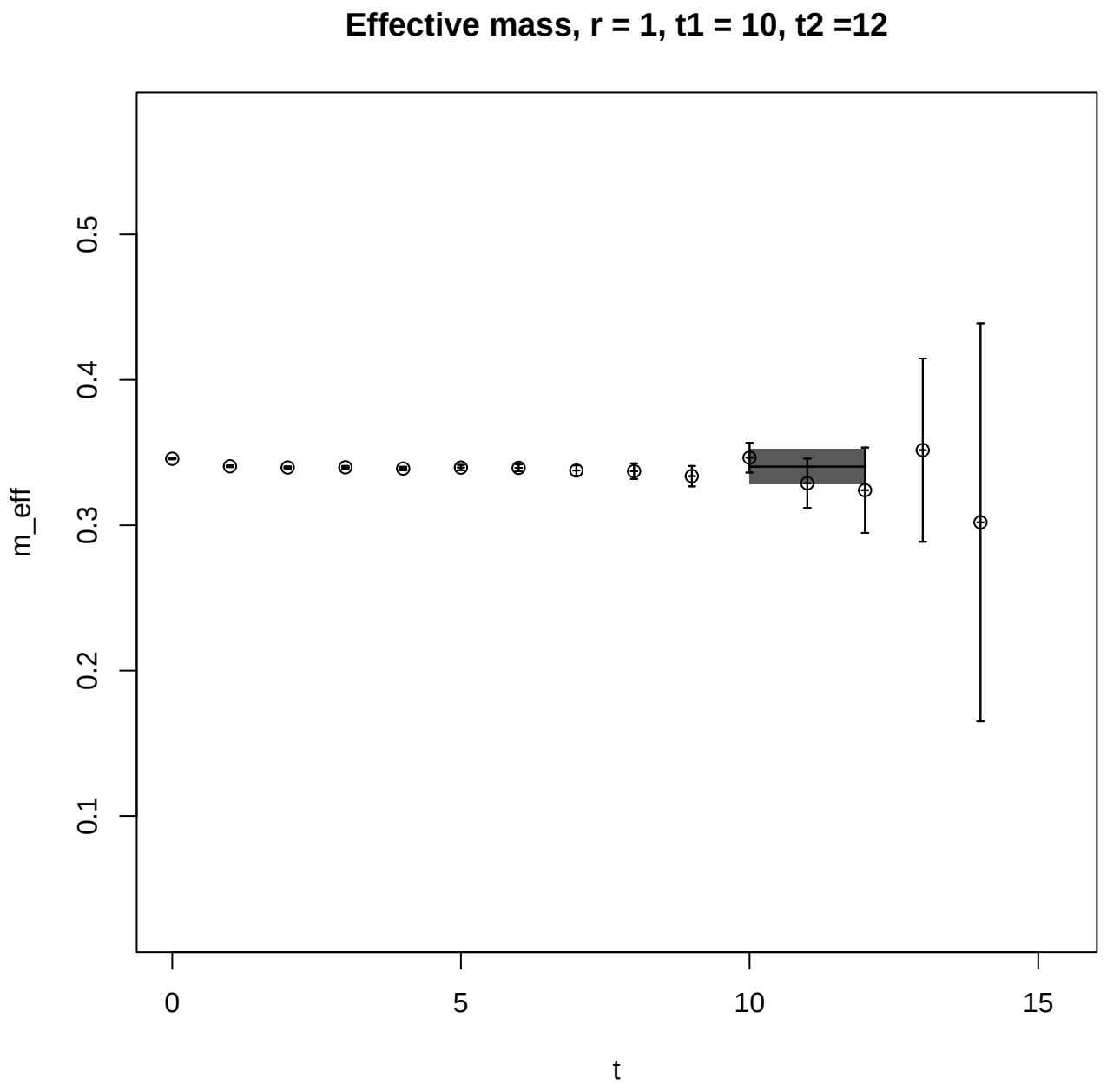
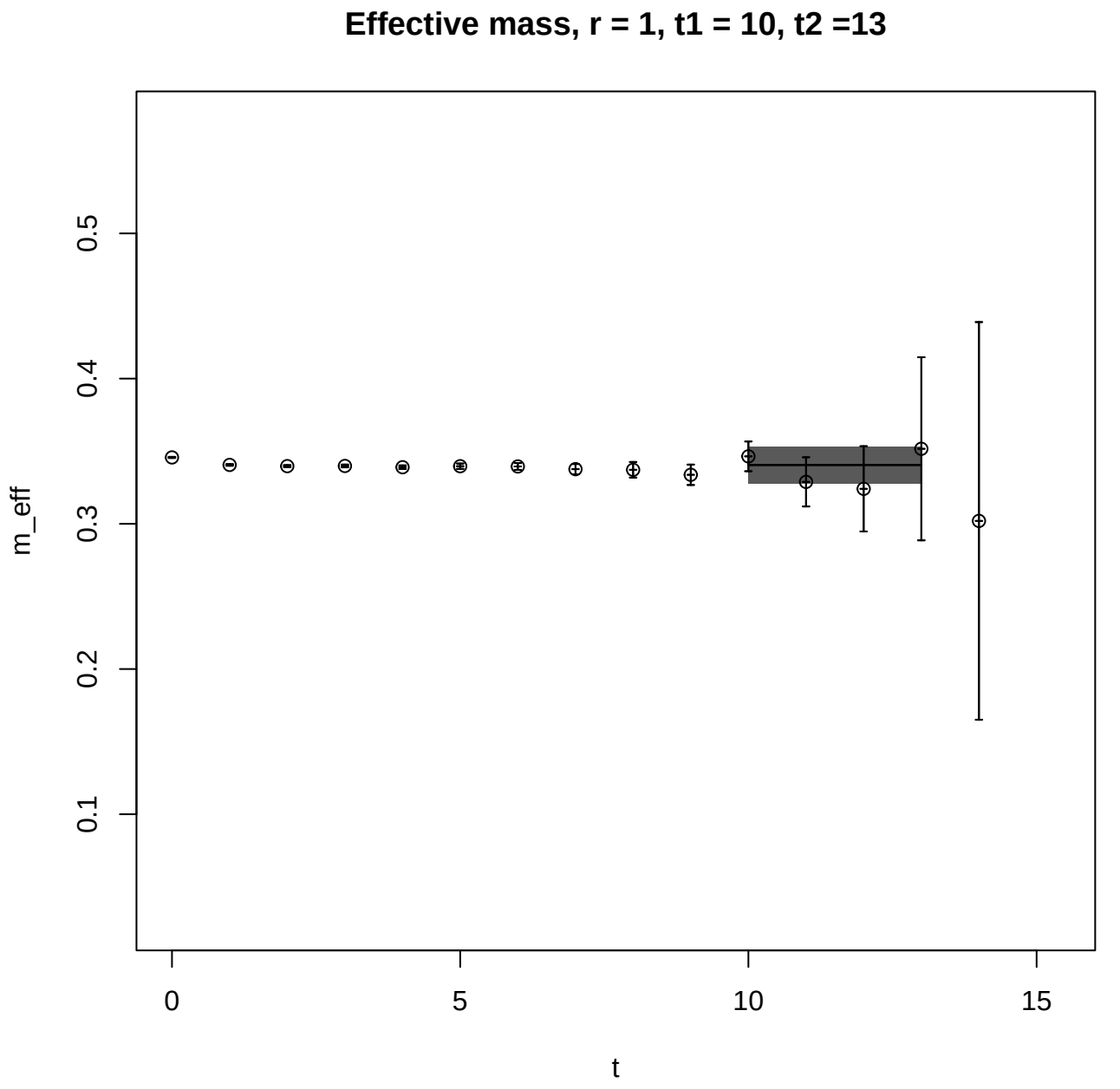
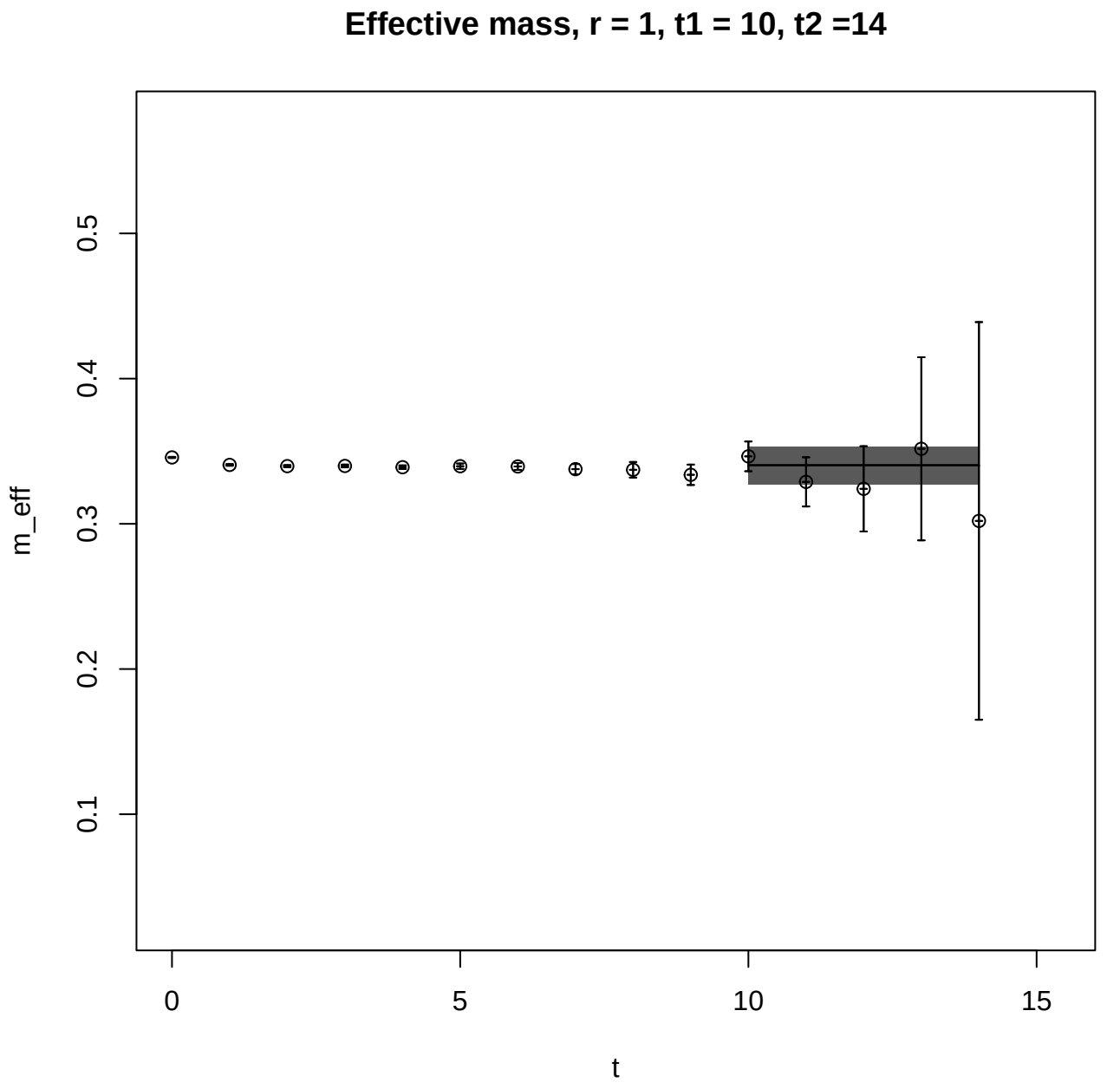
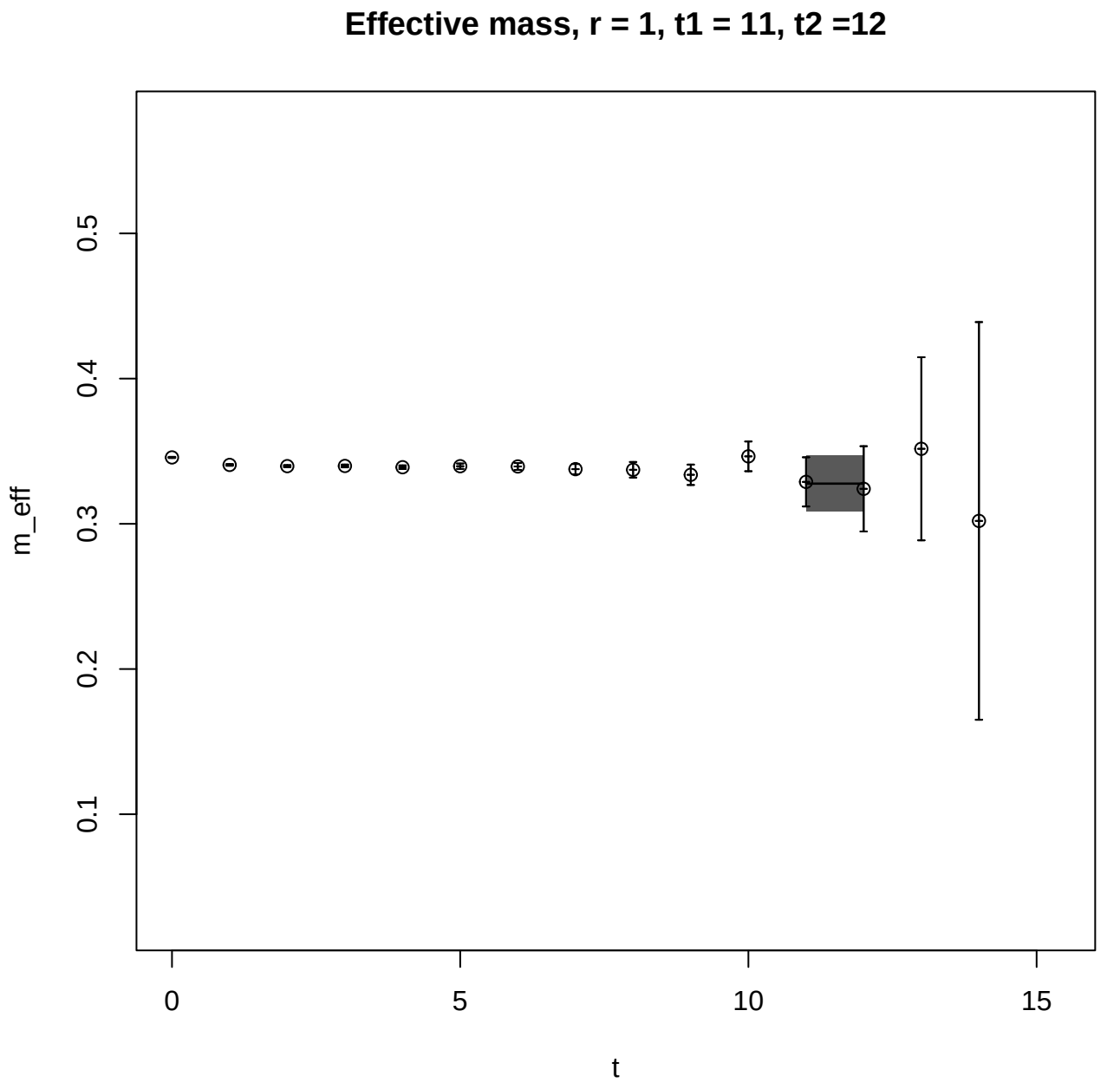
0.2
0.3
0.4
0.5
0
5
10
15
t

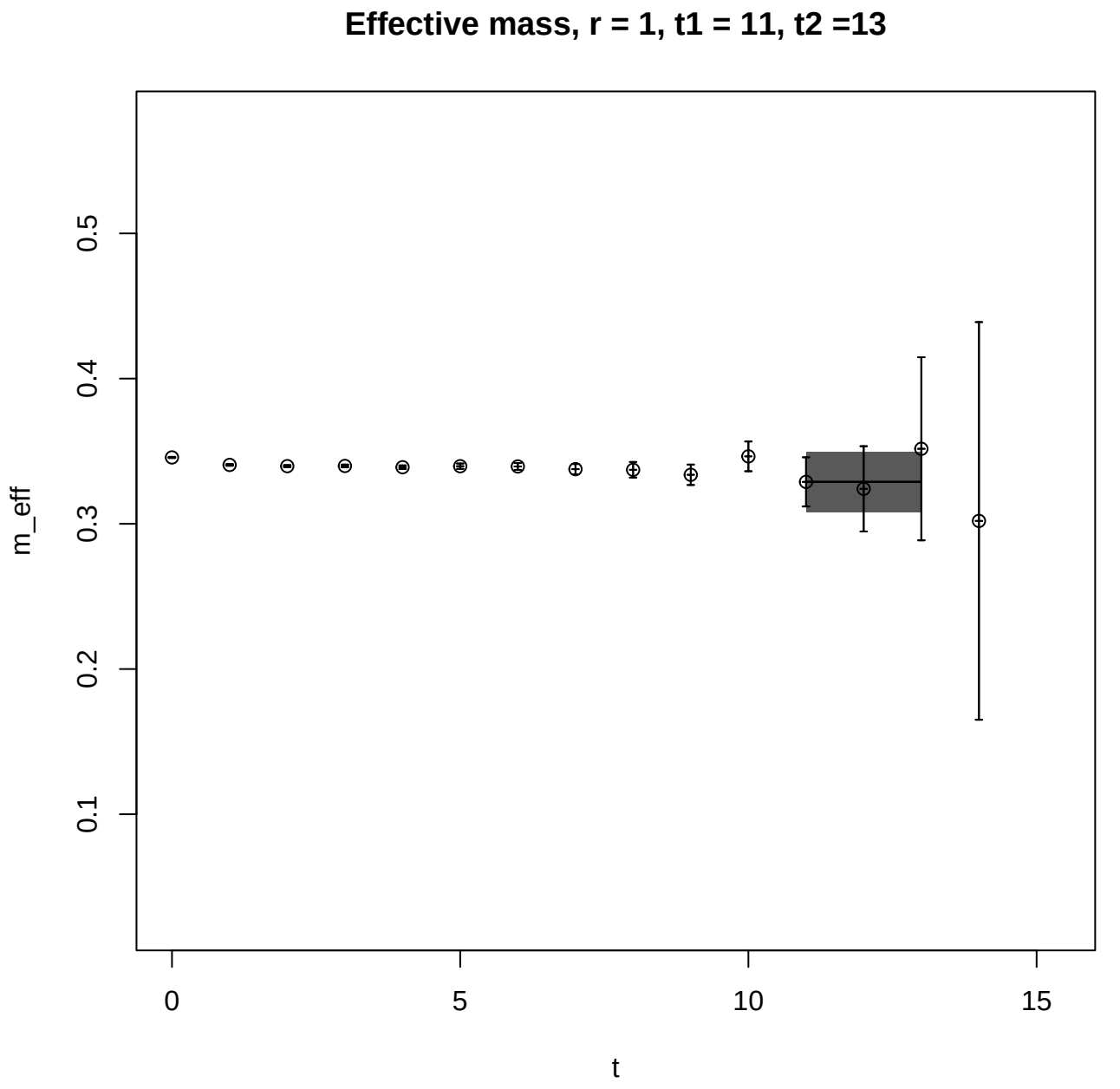

Effective mass, r = 1, t1 = 10, t2 =11
m_eff
0.1
0.2
0.3
0.4
0.5
0
5
10
15
t

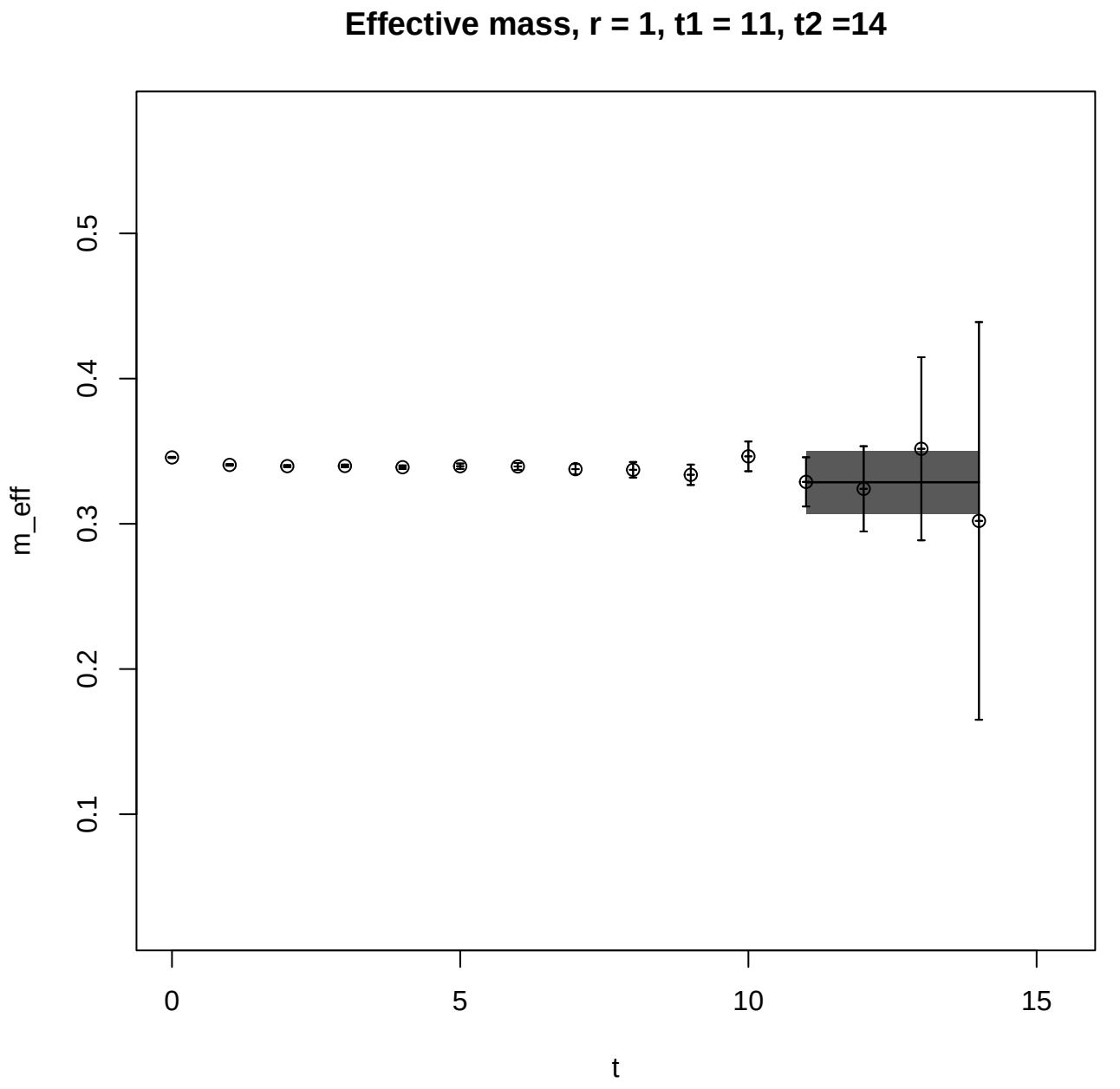

Effective mass, r = 1, t1 = 10, t2 =12
m_eff
0.1
0.2
0.3
0.4
0.5
0
5
10
15
t

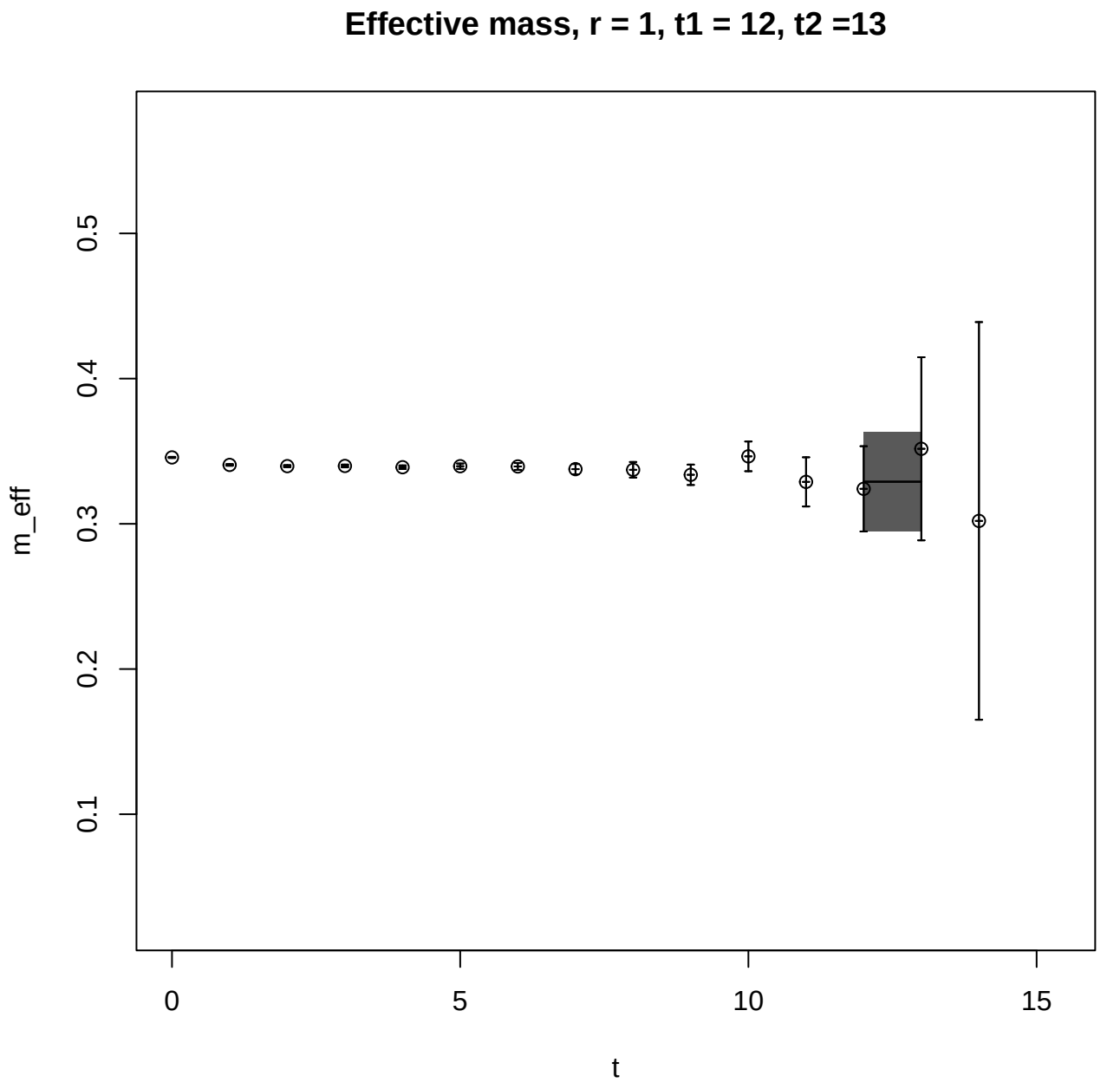

Effective mass, r = 1, t1 = 10, t2 =13
m_eff
0.1
0.2
0.3
0.4
0.5
0
5
10
15
t

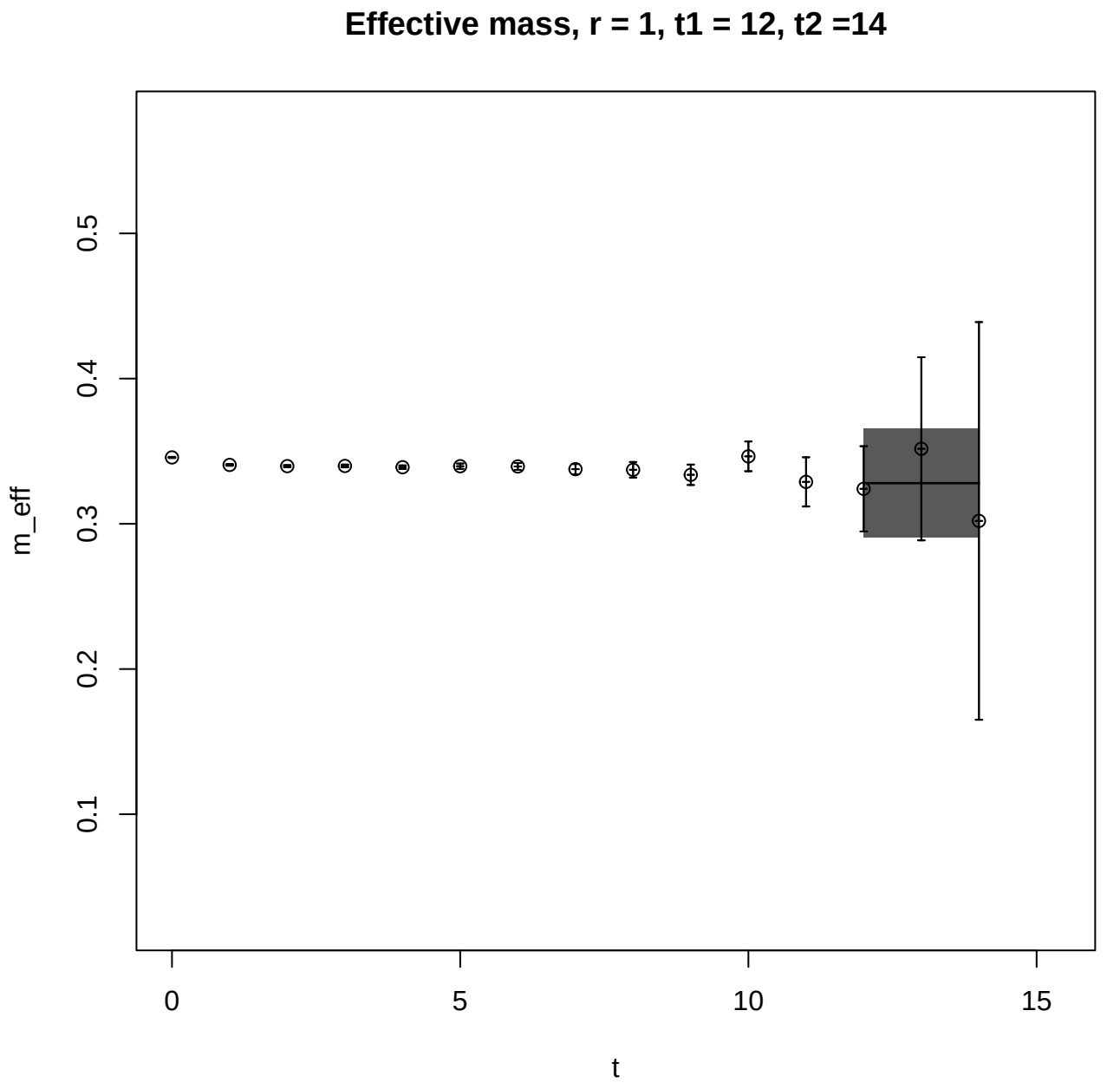

Effective mass, r = 1, t1 = 10, t2 =14
m_eff
0.1
0.2
0.3
0.4
0.5
0
5
10
15
t

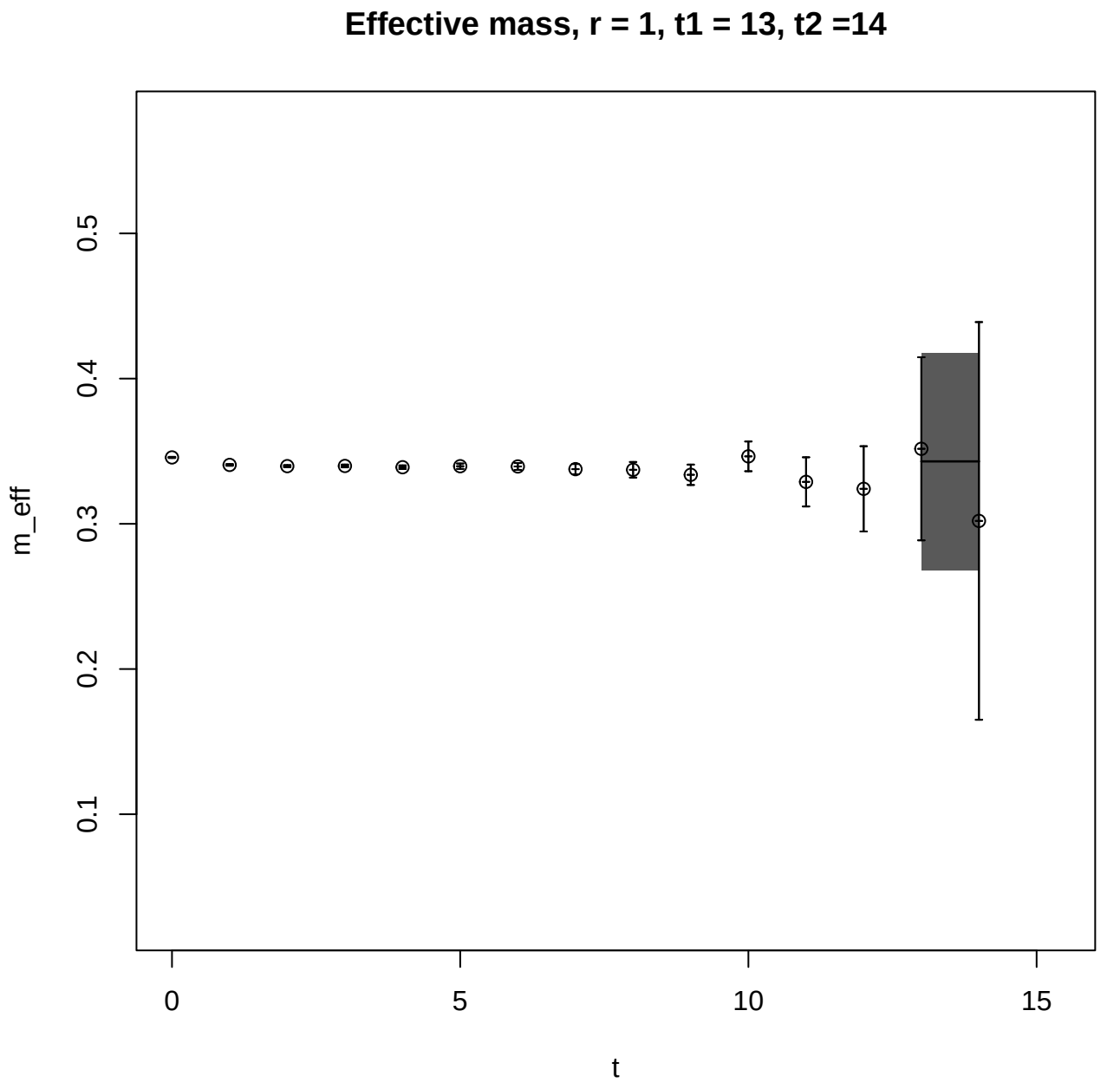

Effective mass, r = 1, t1 = 11, t2 =12
m_eff
0.1
0.2
0.3
0.4
0.5
0
5
10
15
t

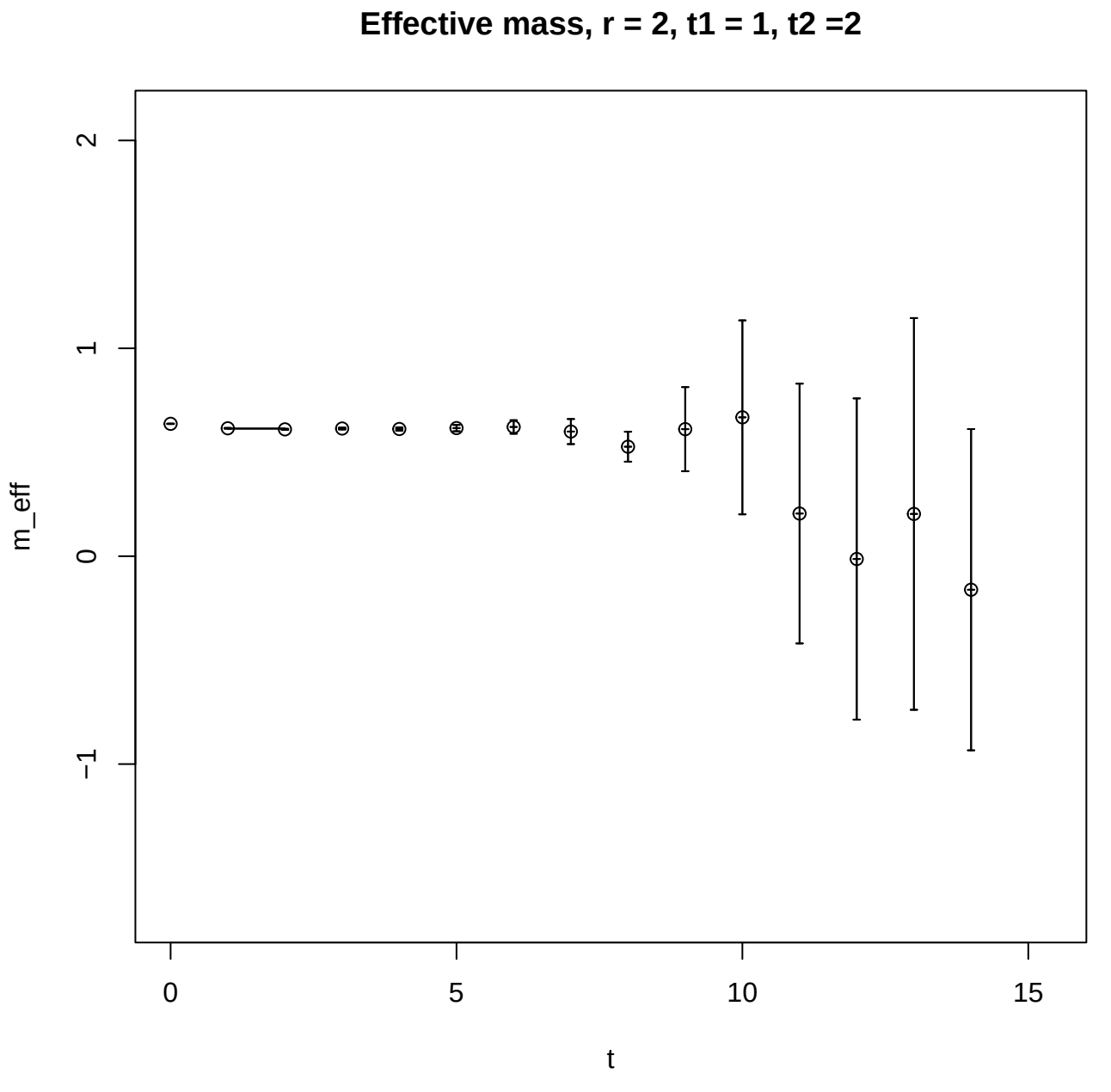

Effective mass, r = 1, t1 = 11, t2 =13
m_eff
0.1
0.2
0.3
0.4
0.5
0
5
10
15
t

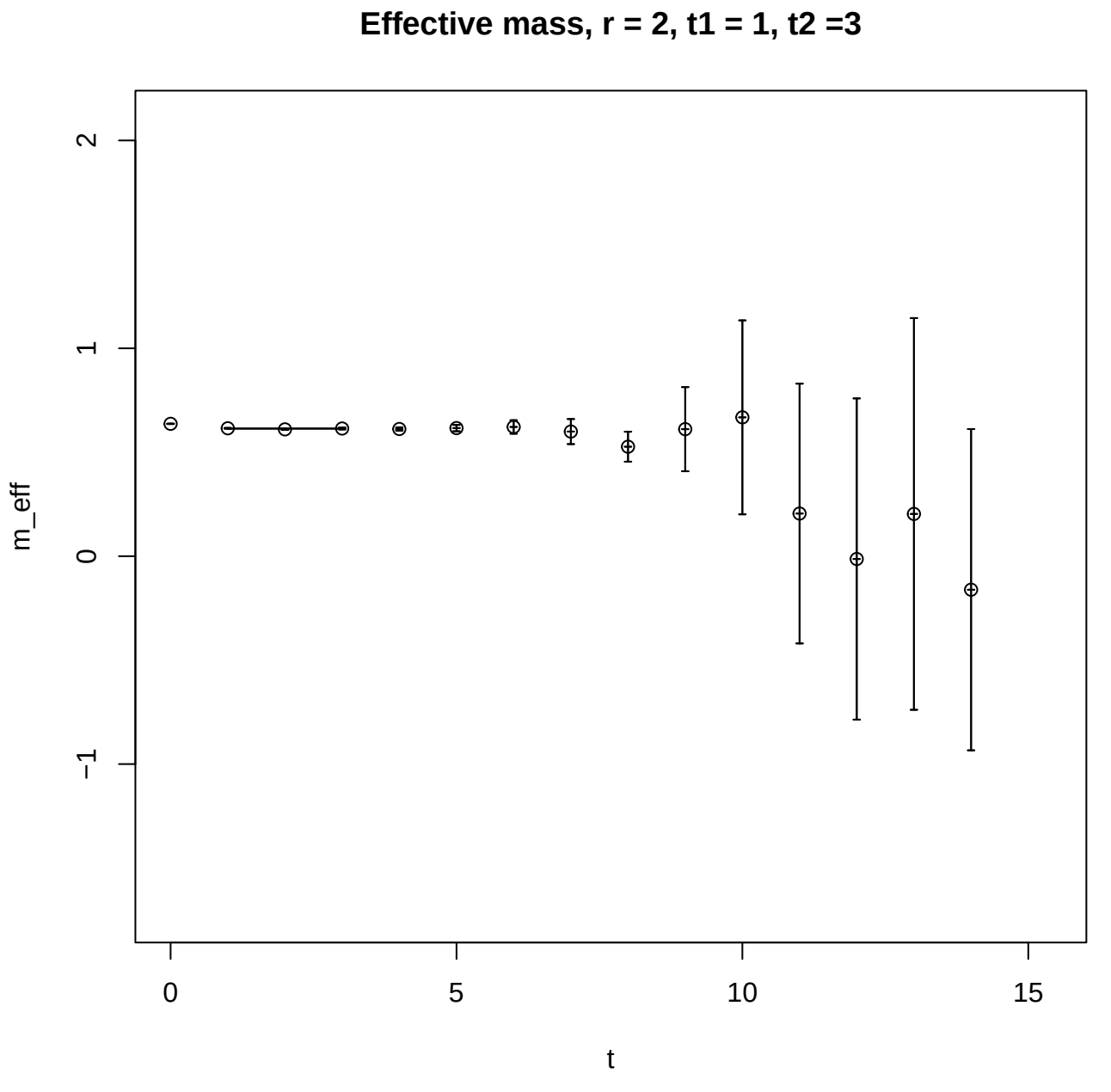

Effective mass, r = 1, t1 = 11, t2 =14
m_eff
0.1
0.2
0.3
0.4
0.5
0
5
10
15
t

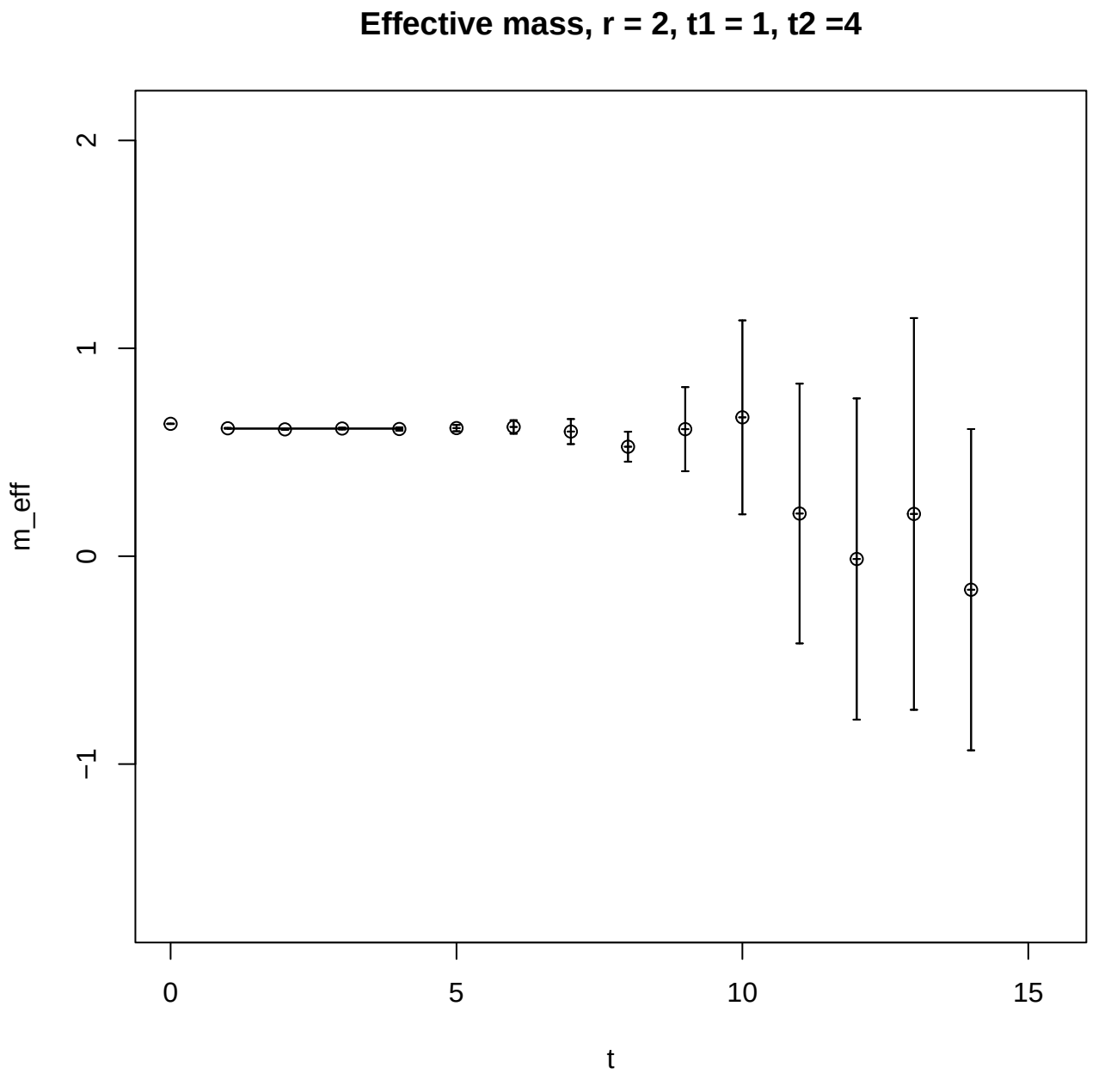

Effective mass, r = 1, t1 = 12, t2 =13
m_eff
0.1
0.2
0.3
0.4
0.5
0
5
10
15
t

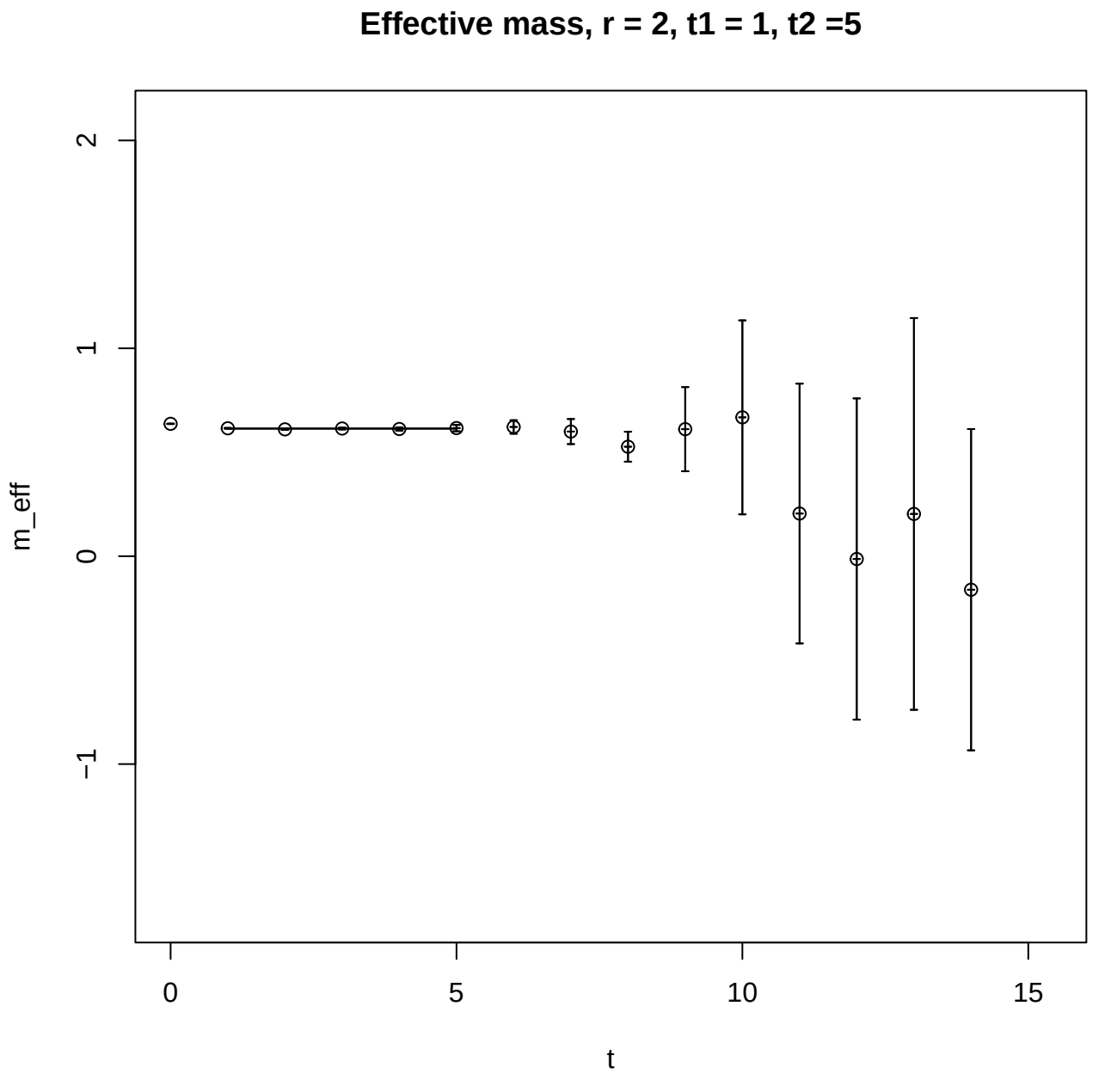

Effective mass, r = 1, t1 = 12, t2 =14
m_eff
0.1
0.2
0.3
0.4
0.5
0
5
10
15
t

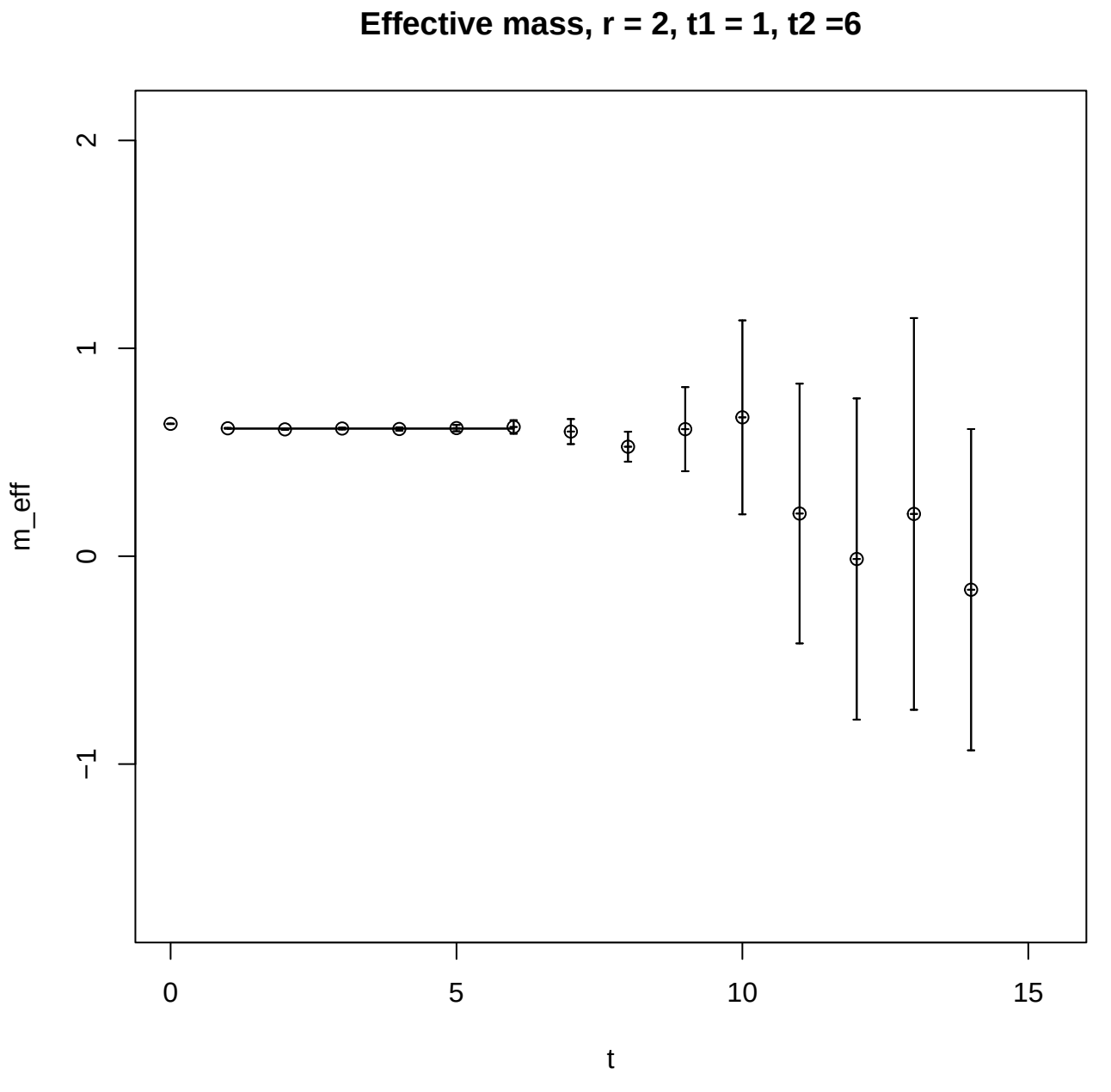

Effective mass, r = 1, t1 = 13, t2 =14
m_eff
0.1
0.2
0.3
0.4
0.5
0
5
10
15
t

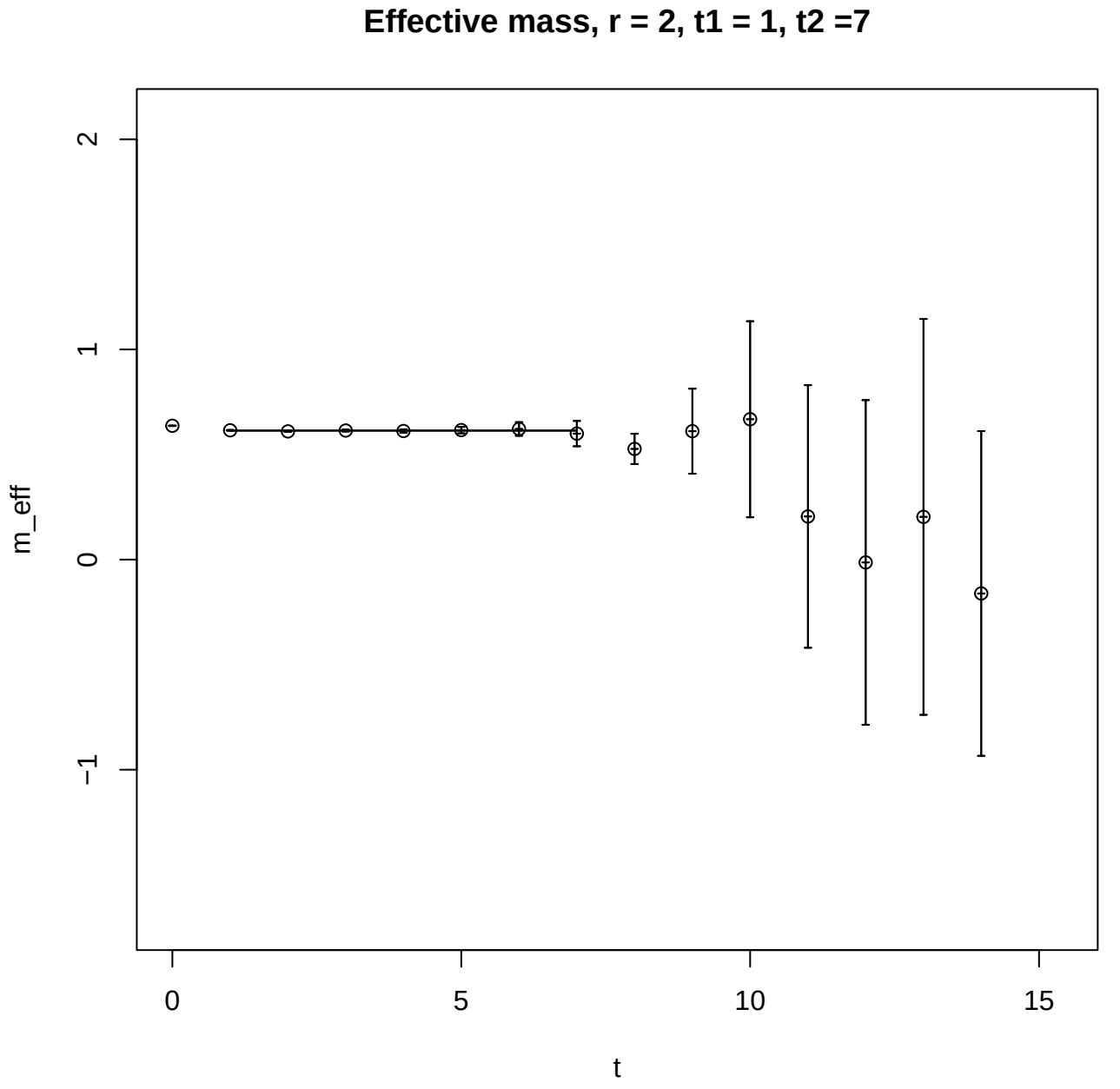

Effective mass, r = 2, t1 = 1, t2 =2
m_eff
2
1
0
−1
0
5
10
15
t

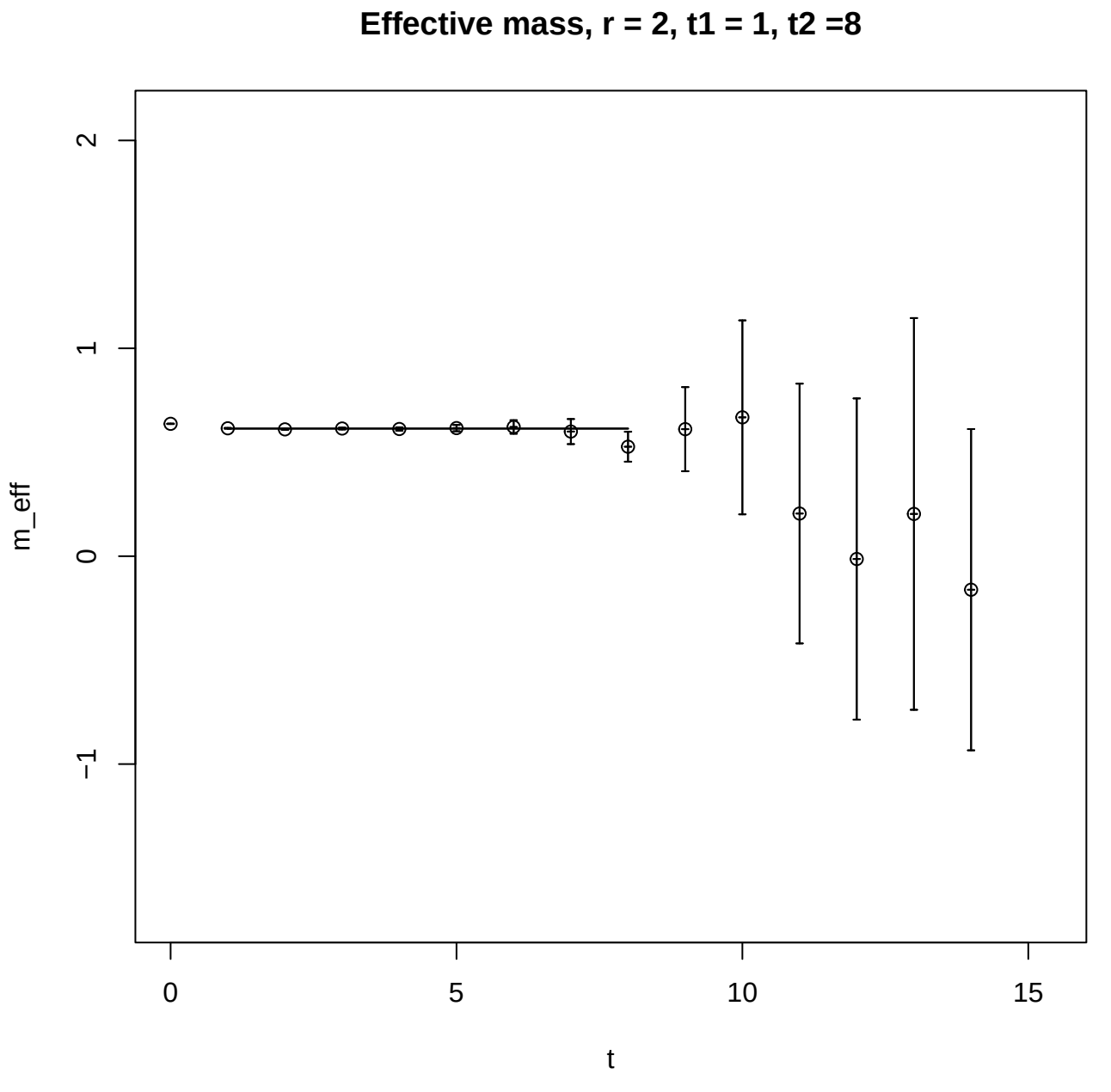

Effective mass, r = 2, t1 = 1, t2 =3
m_eff
2
1
0
−1
0
5
10
15
t

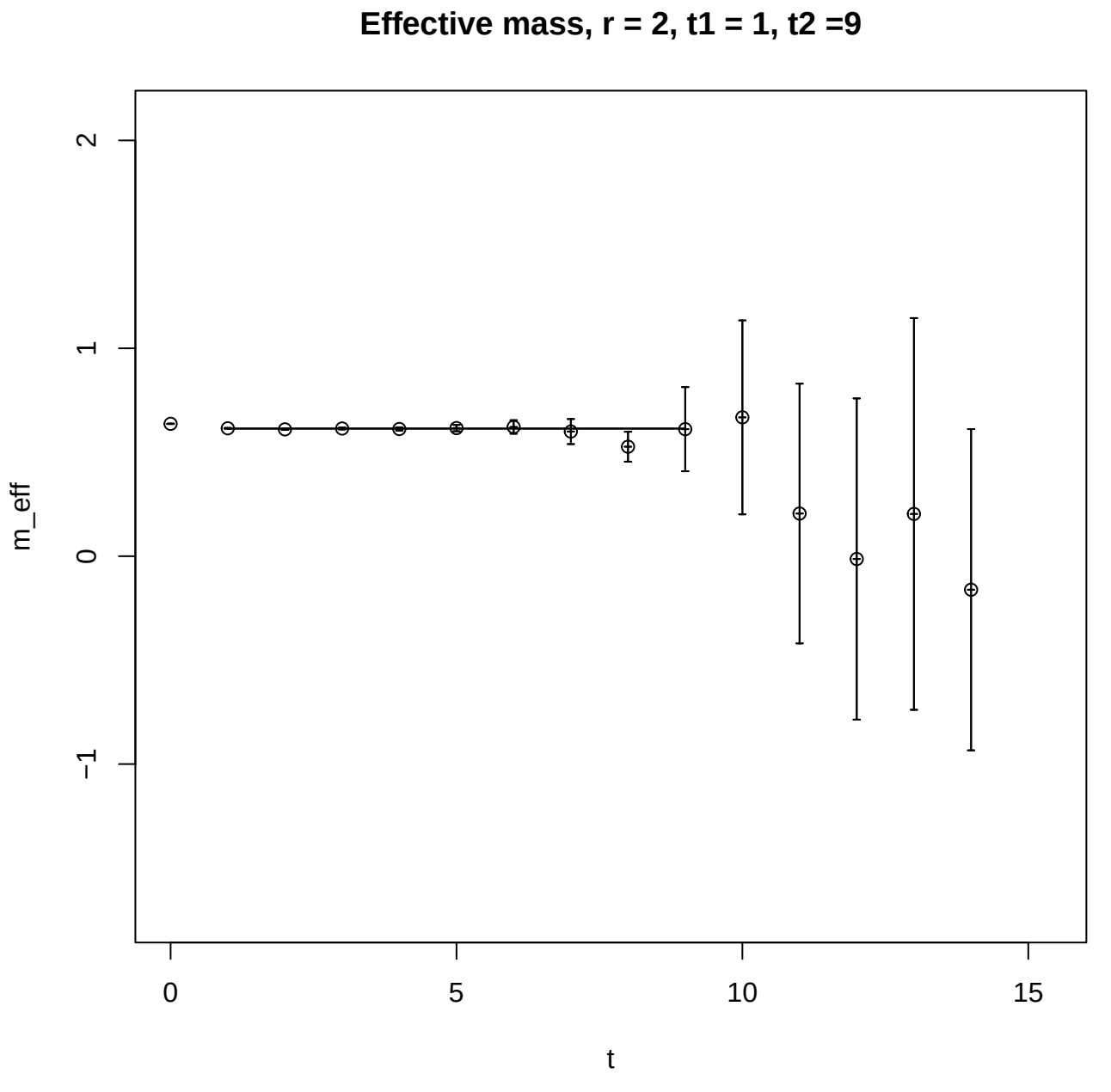

Effective mass, r = 2, t1 = 1, t2 =4
m_eff
2
1
0
−1
0
5
10
15
t

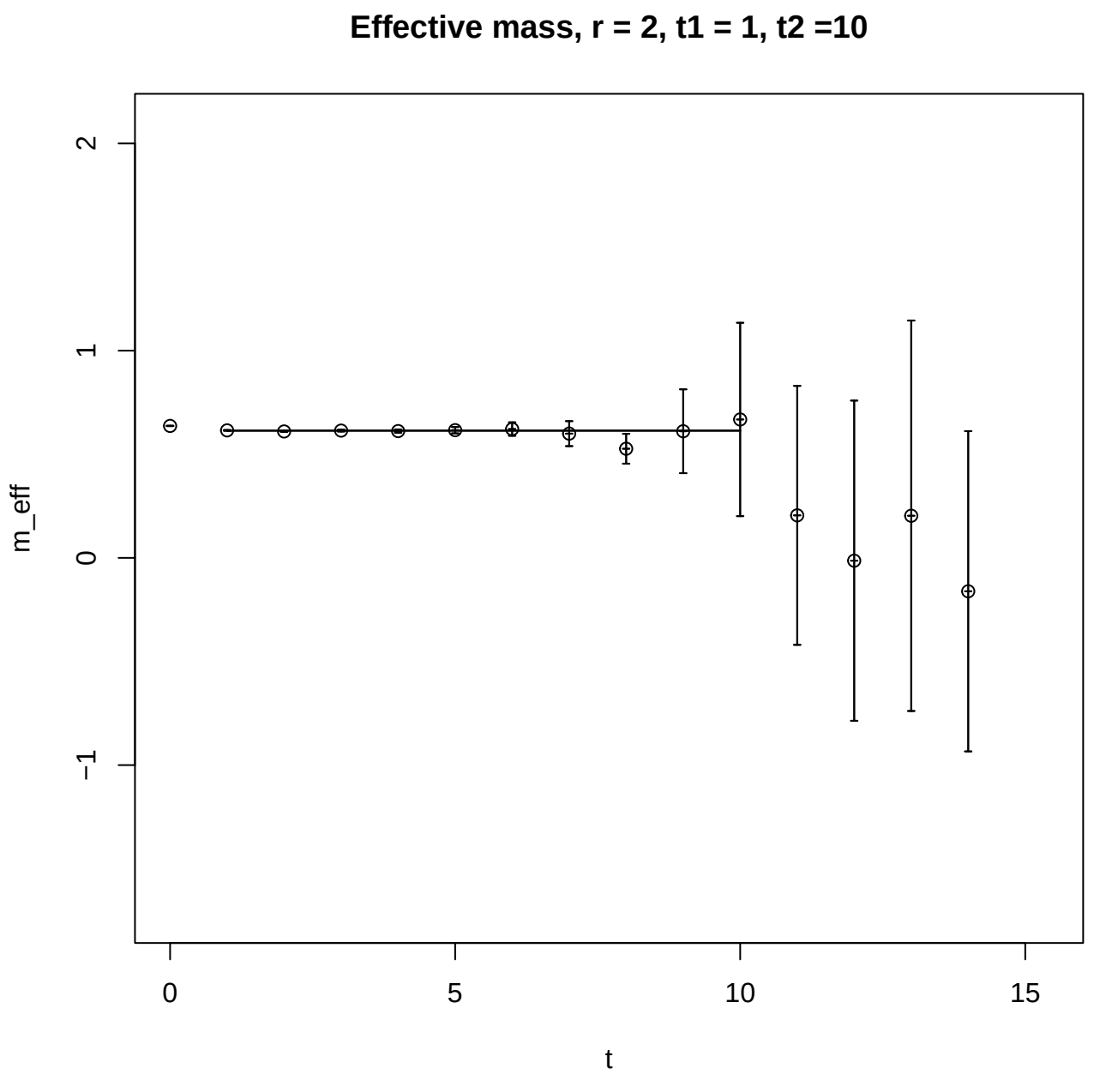

Effective mass, r = 2, t1 = 1, t2 =5
m_eff
2
1
0
−1
0
5
10
15
t

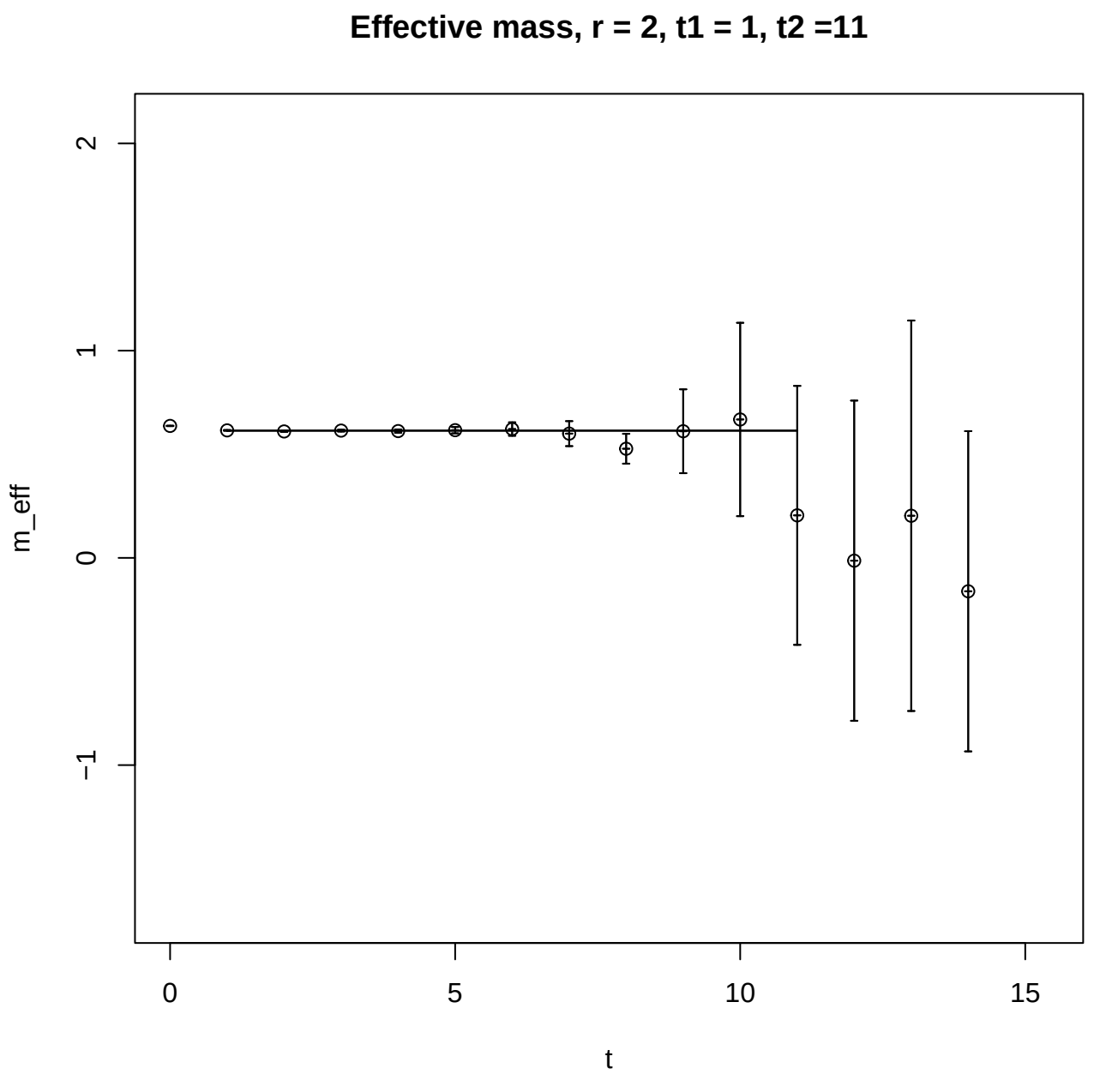

Effective mass, r = 2, t1 = 1, t2 =6
m_eff
2
1
0
−1
0
5
10
15
t

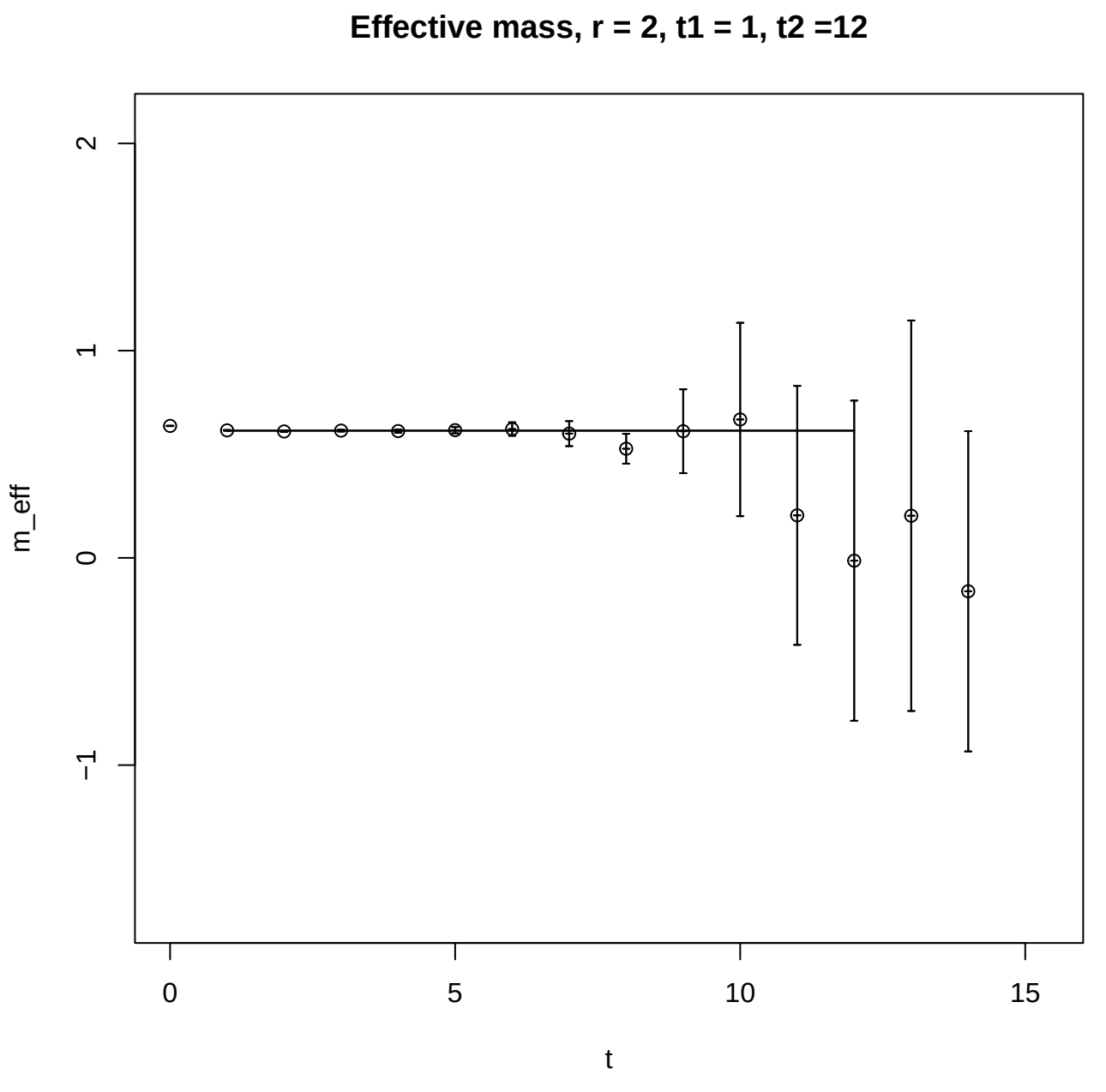

Effective mass, r = 2, t1 = 1, t2 =7
m_eff
2
1
0
−1
0
5
10
15
t

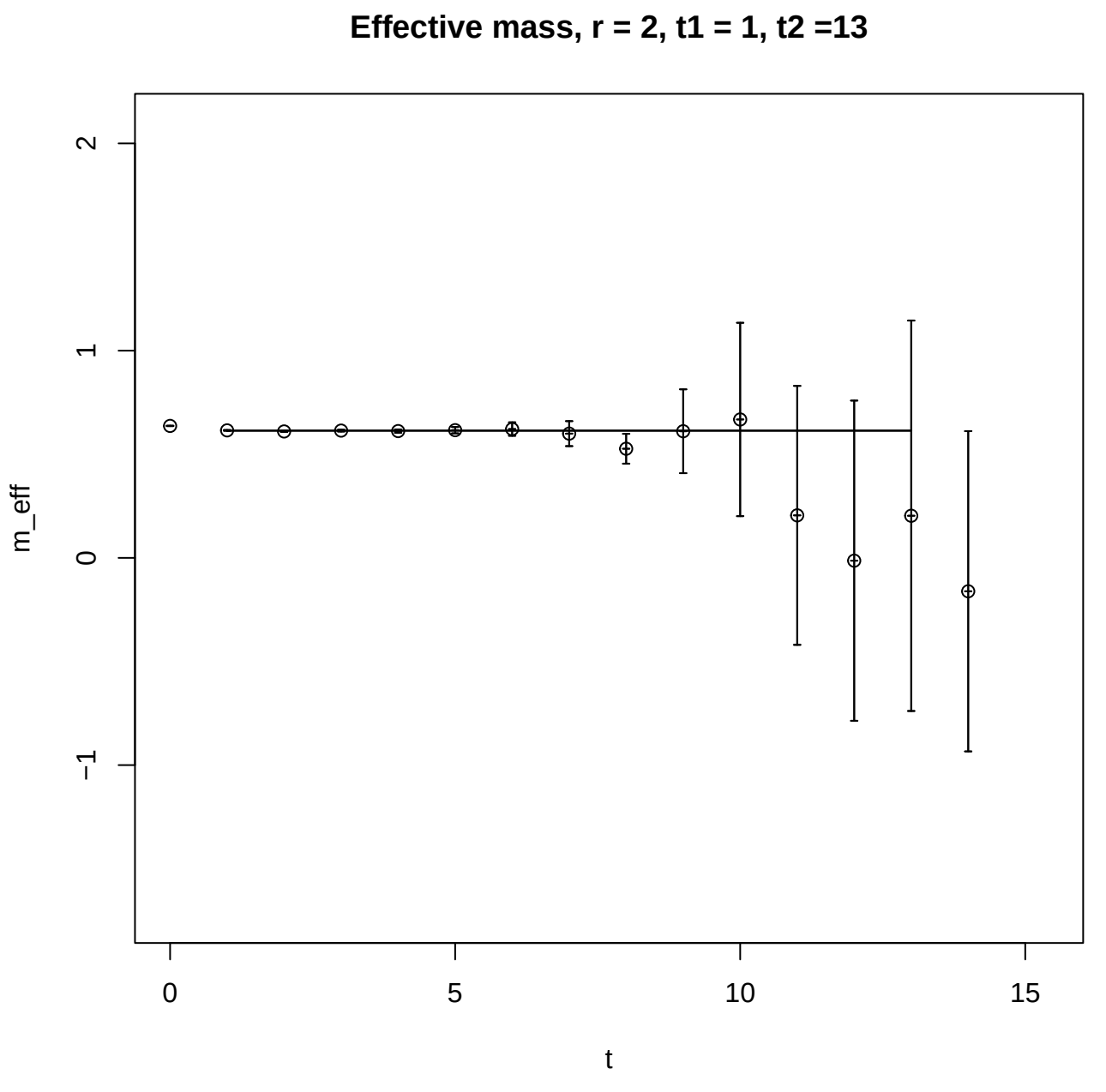

Effective mass, r = 2, t1 = 1, t2 =8
m_eff
2
1
0
−1
0
5
10
15
t

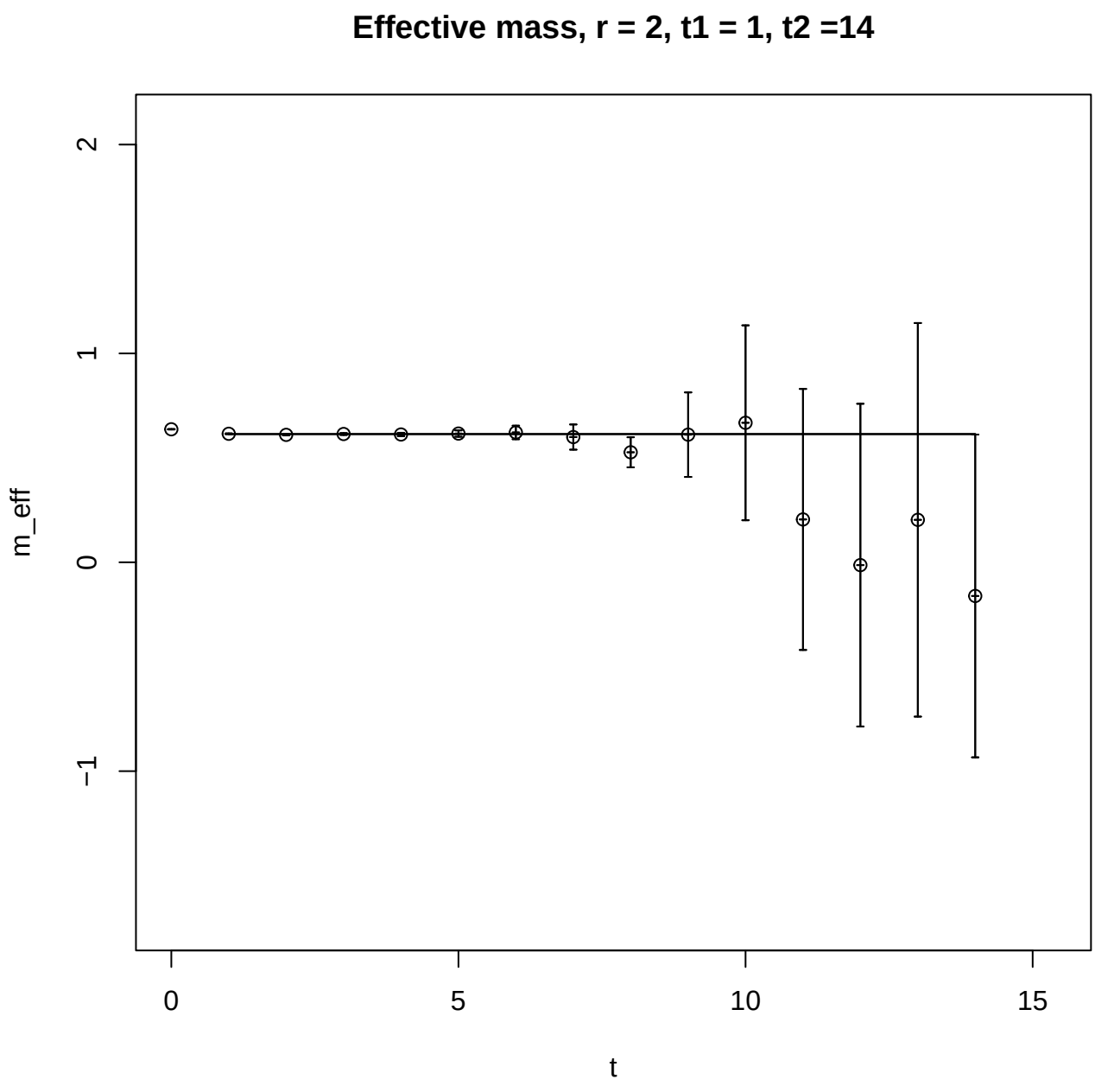

Effective mass, r = 2, t1 = 1, t2 =9
m_eff
2
1
0
−1
0
5
10
15
t

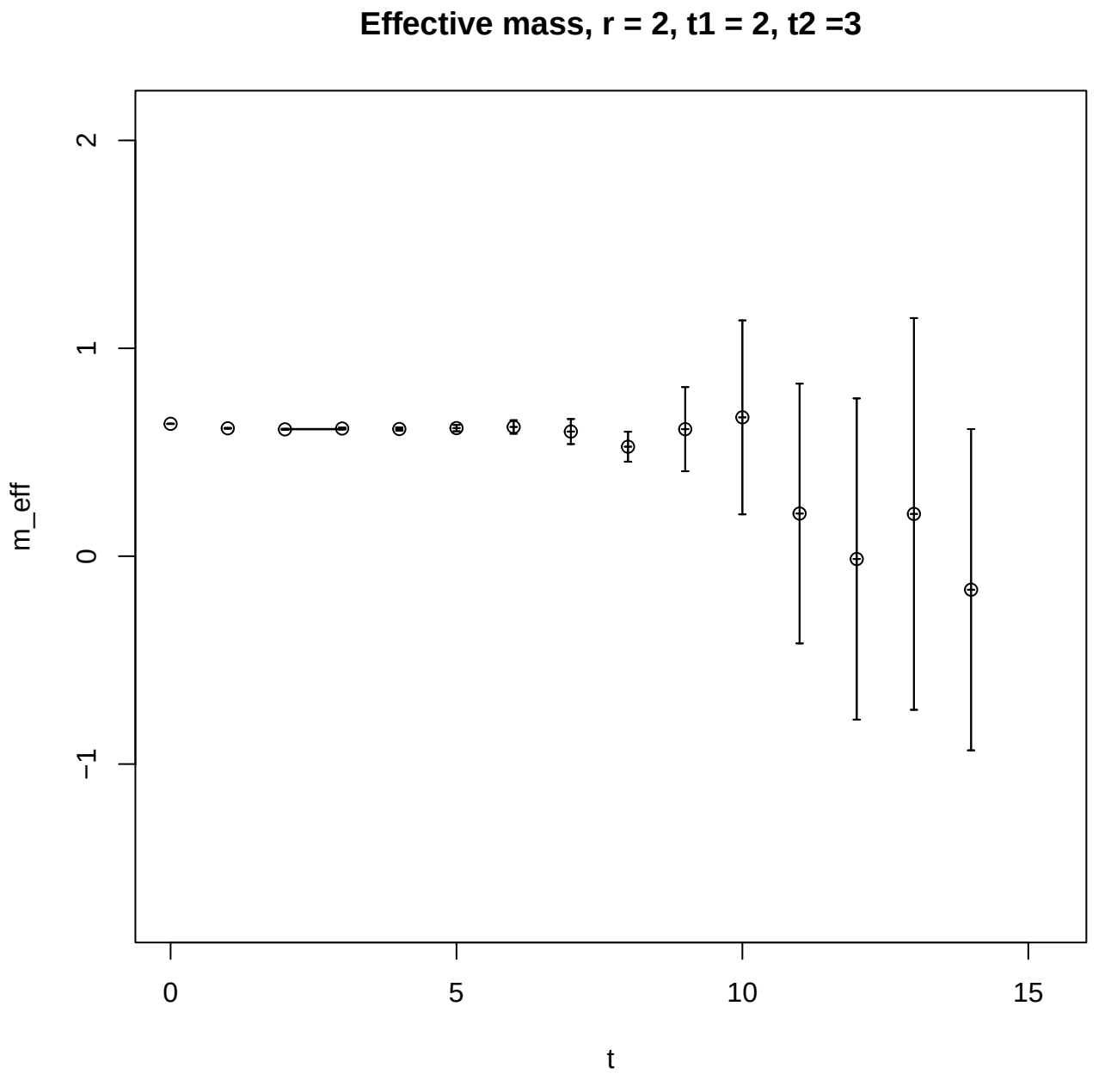

Effective mass, r = 2, t1 = 1, t2 =10
m_eff
2
1
0
−1
0
5
10
15
t

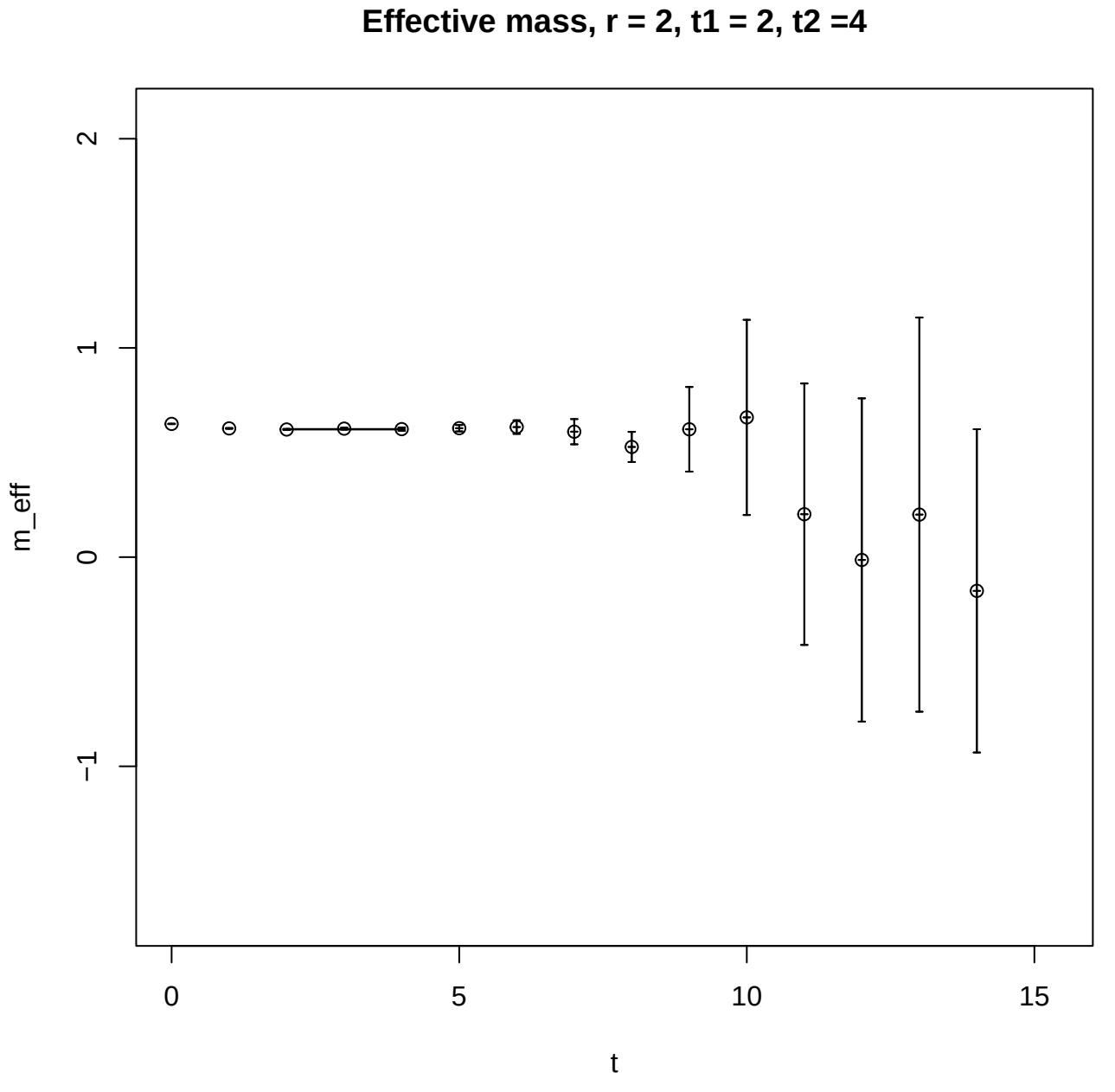

Effective mass, r = 2, t1 = 1, t2 =11
m_eff
2
1
0
−1
0
5
10
15
t

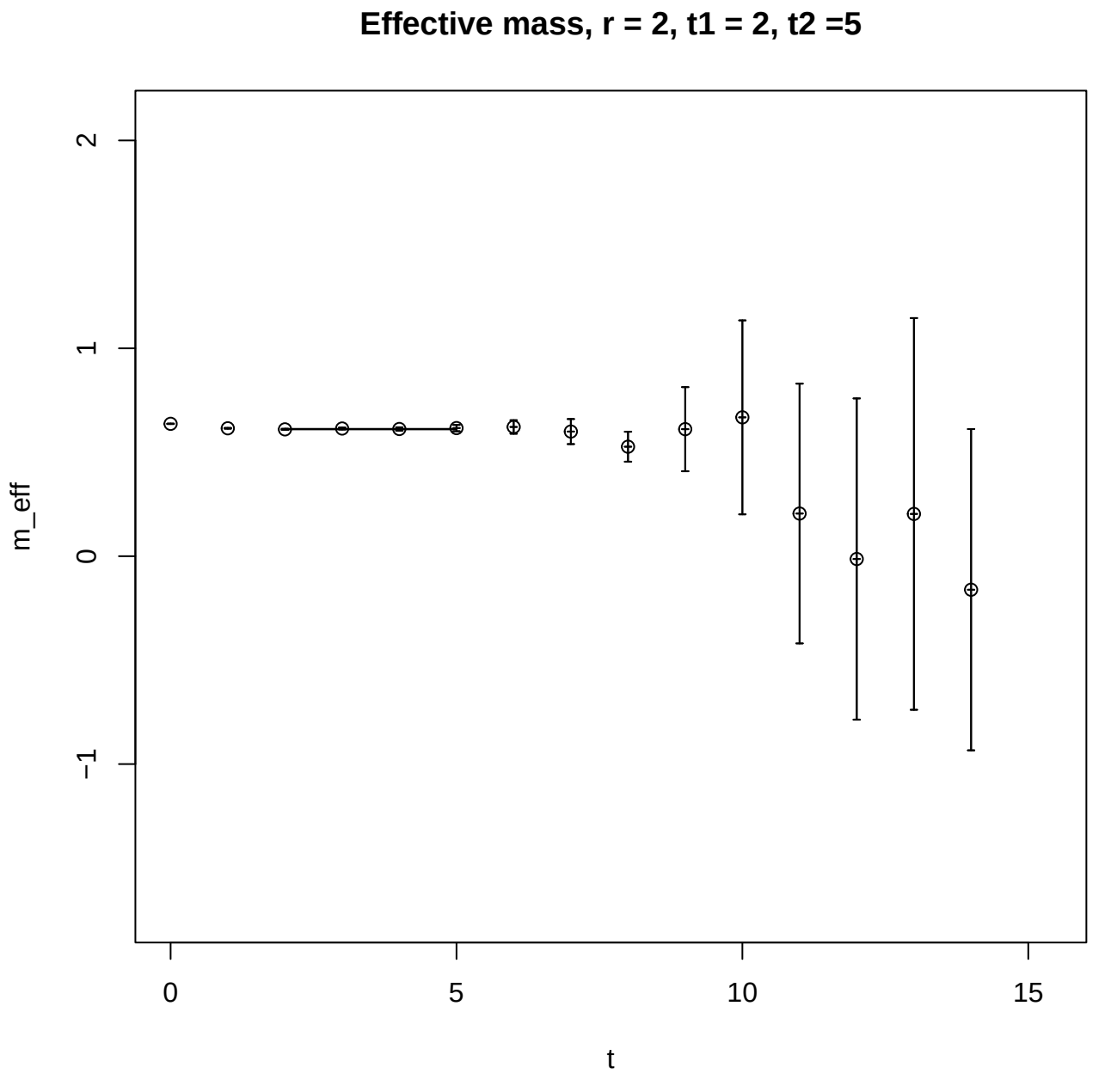

Effective mass, r = 2, t1 = 1, t2 =12
m_eff
2
1
0
−1
0
5
10
15
t

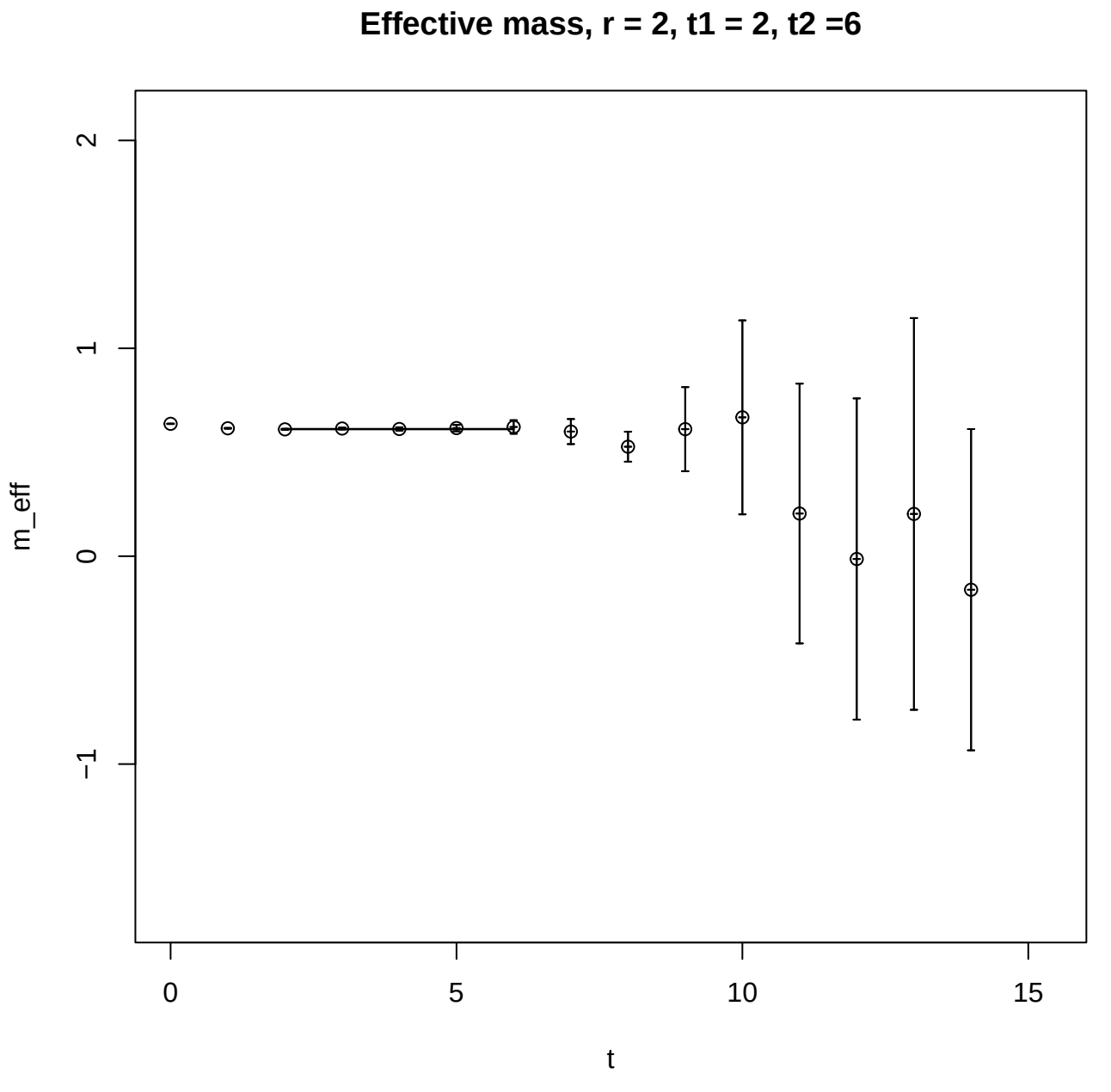

Effective mass, r = 2, t1 = 1, t2 =13
m_eff
2
1
0
−1
0
5
10
15
t

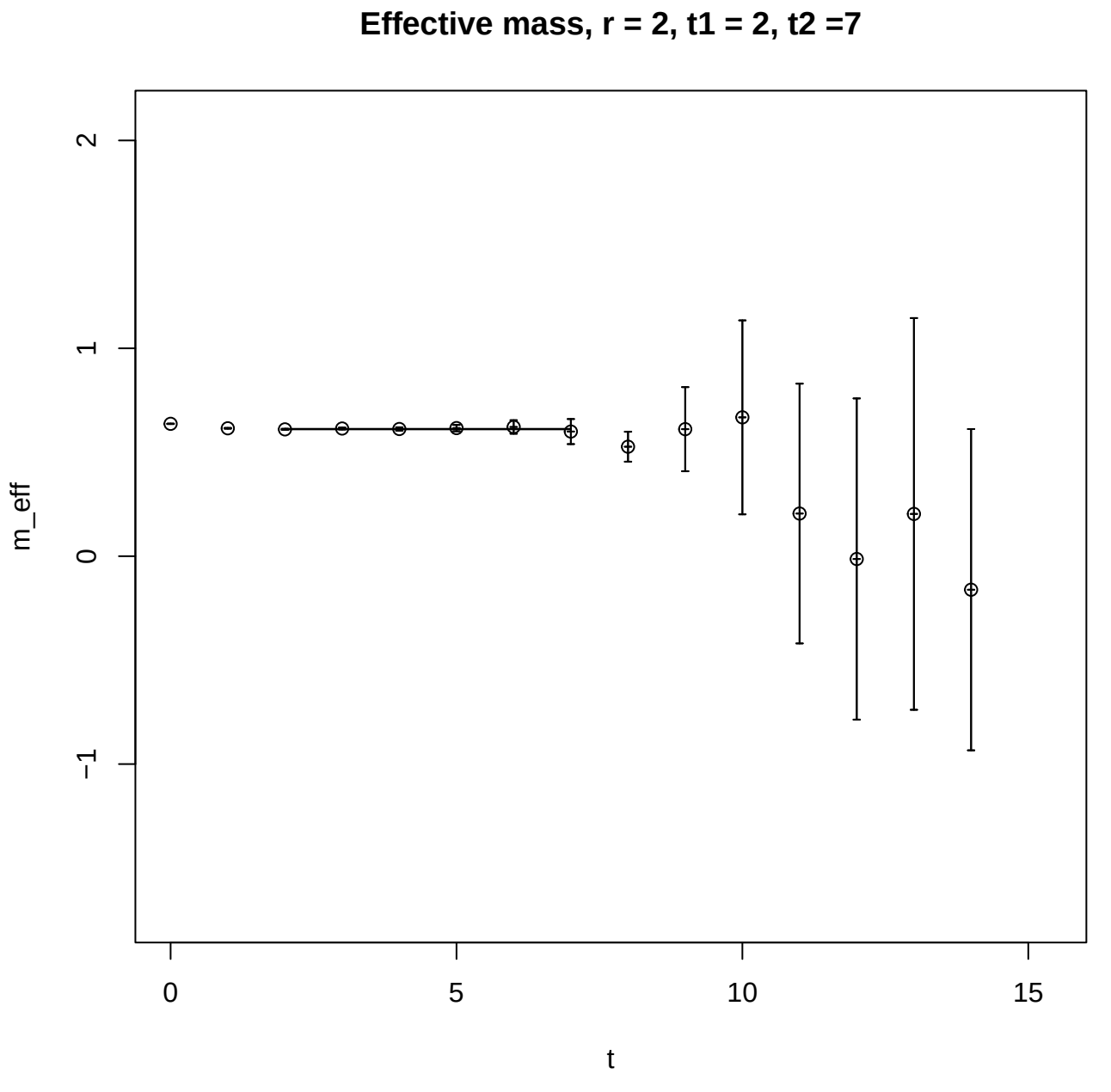

Effective mass, r = 2, t1 = 1, t2 =14
m_eff
t
2
1
0
−1
0
5
10
15

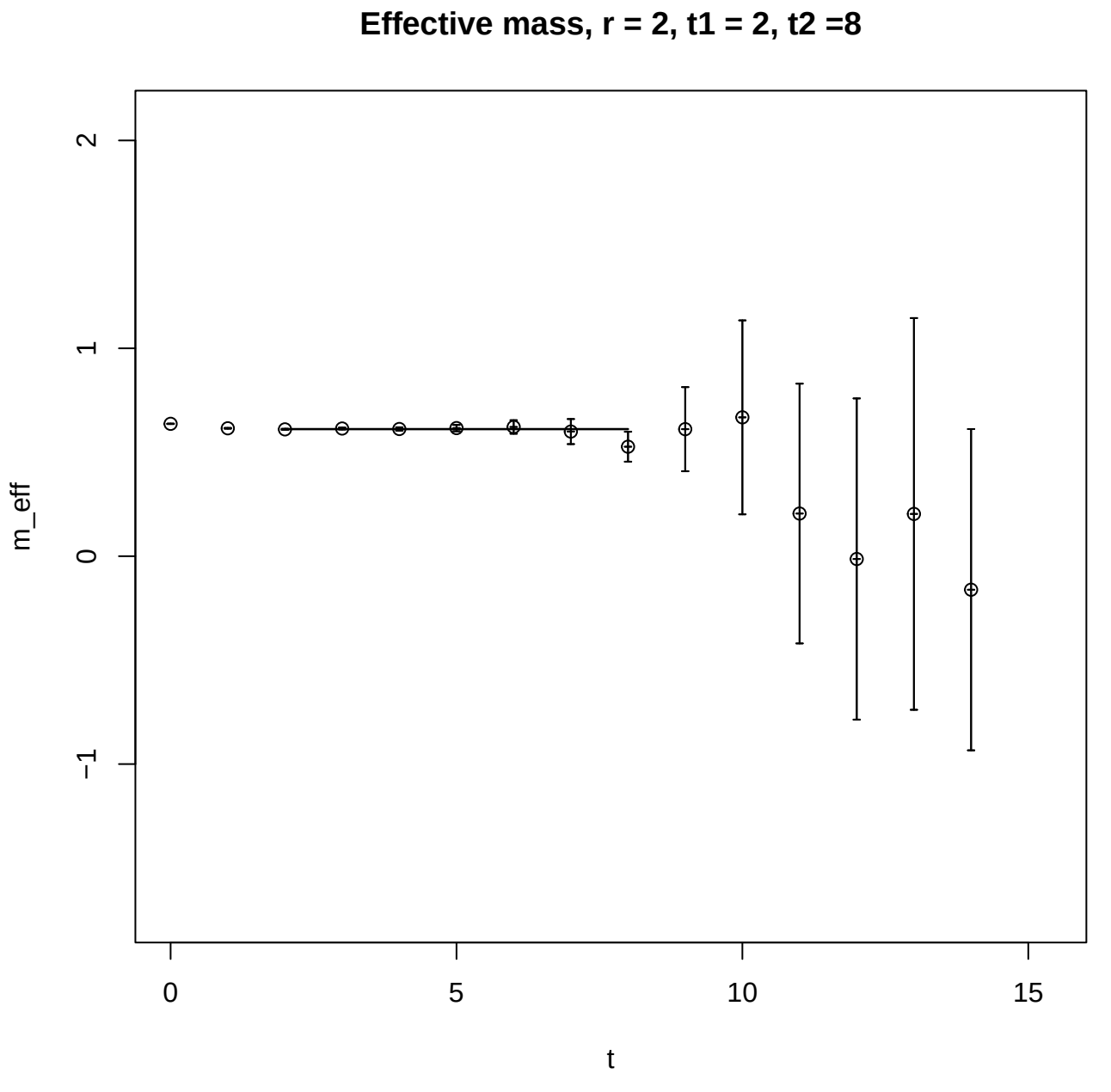

Effective mass, r = 2, t1 = 2, t2 =3
m_eff
2
1
0
−1
0
5
10
15
t

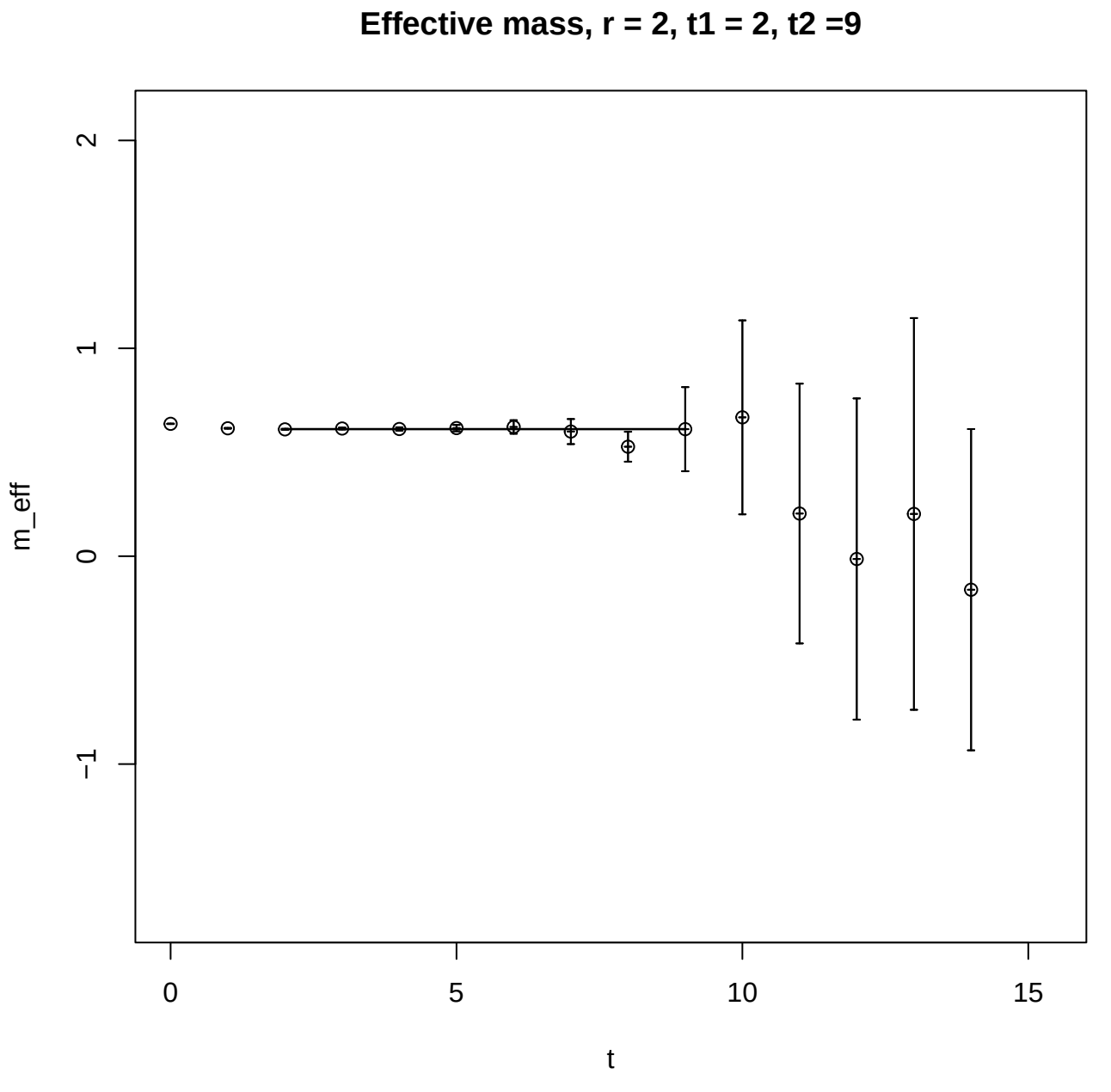

Effective mass, r = 2, t1 = 2, t2 =4
m_eff
2
1
0
−1
0
5
10
15
t


Effective mass, r = 2, t1 = 2, t2 =5
m_eff
2
1
0
−1
0
5
10
15
t


Effective mass, r = 2, t1 = 2, t2 =6
m_eff
2
1
0
−1
0
5
10
15
t


Effective mass, r = 2, t1 = 2, t2 =7
m_eff
2
1
0
−1
0
5
10
15
t


Effective mass, r = 2, t1 = 2, t2 =8
m_eff
2
1
0
−1
0
5
10
15
t


Effective mass, r = 2, t1 = 2, t2 =9
m_eff
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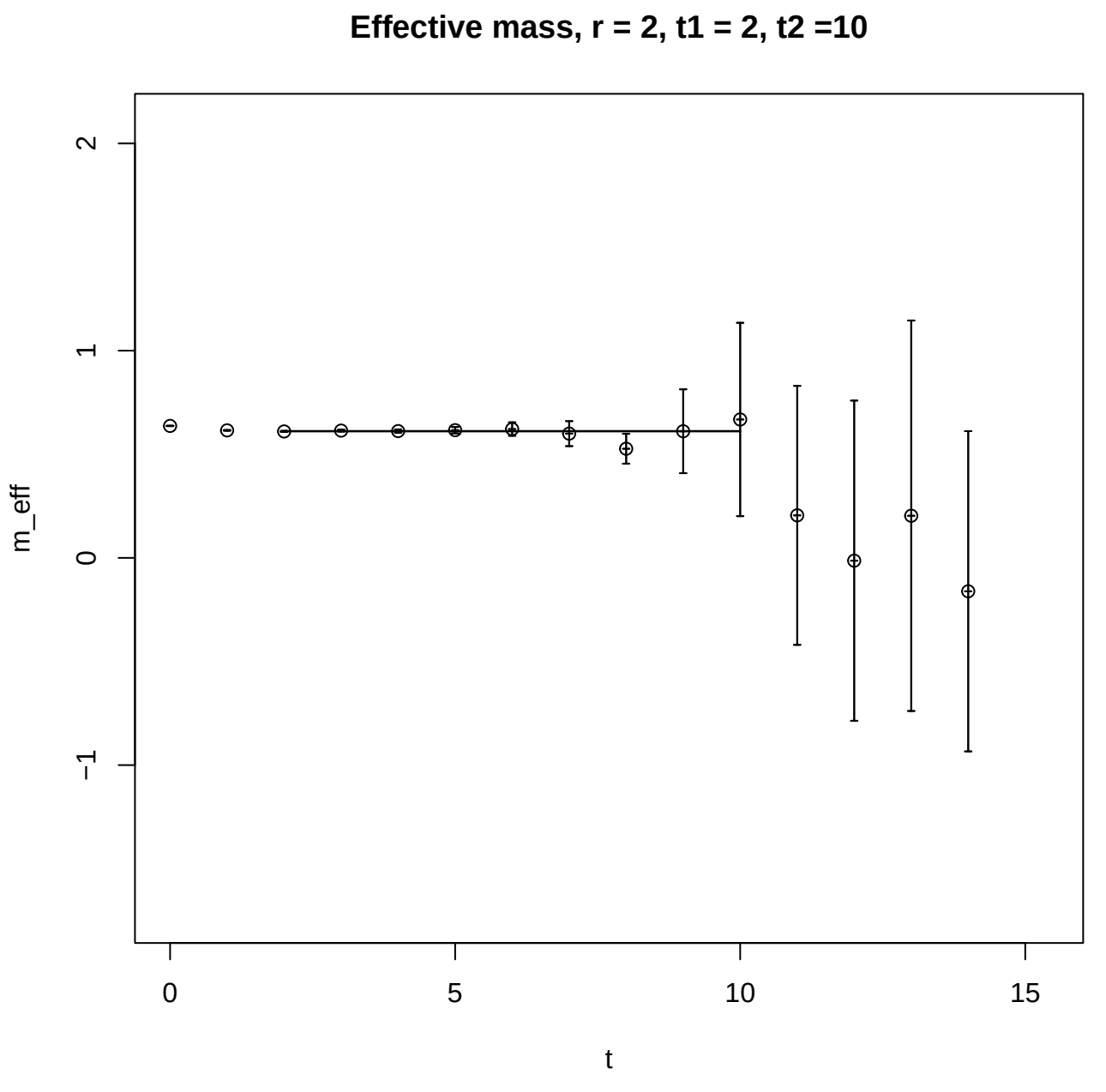
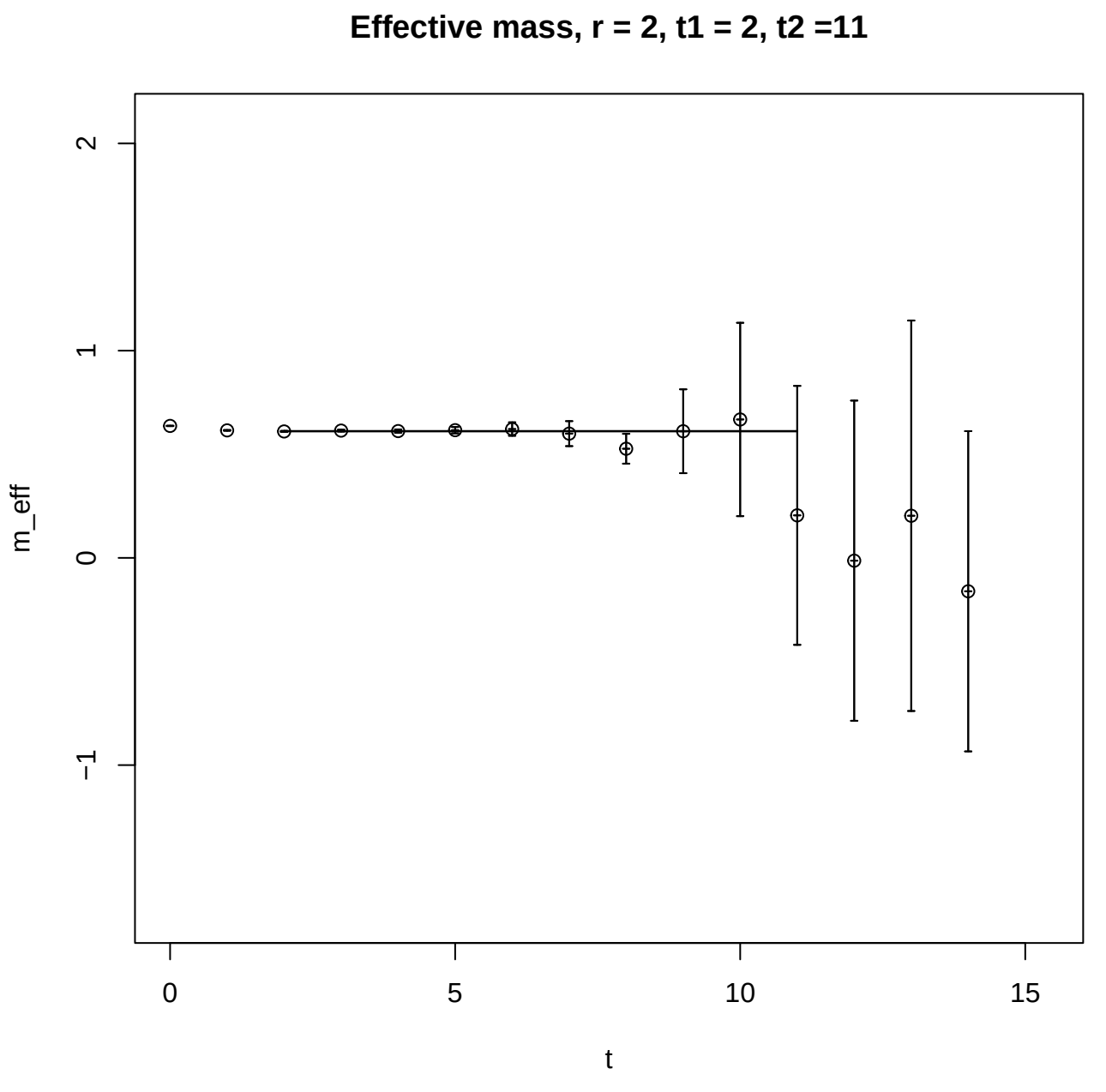
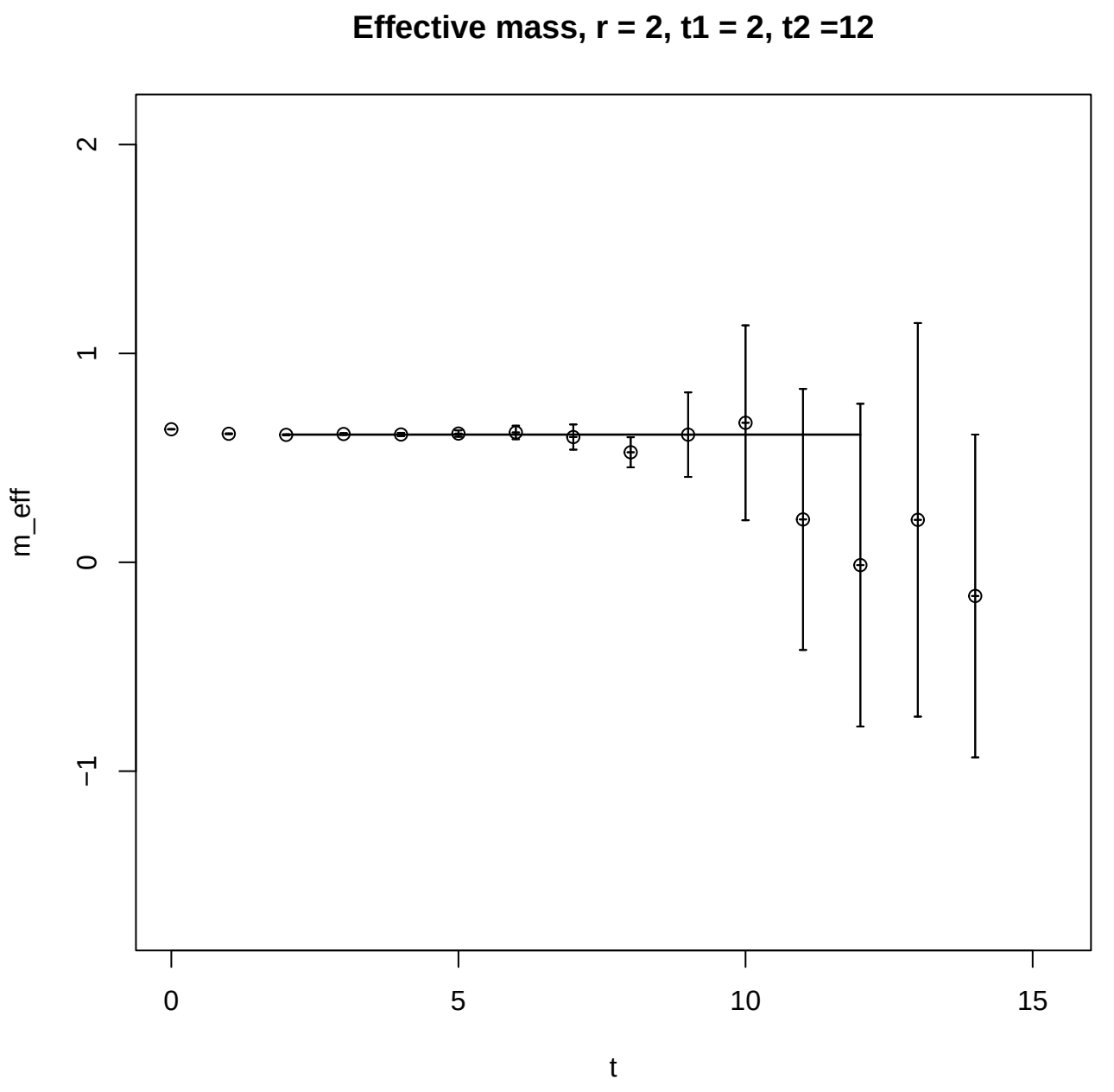
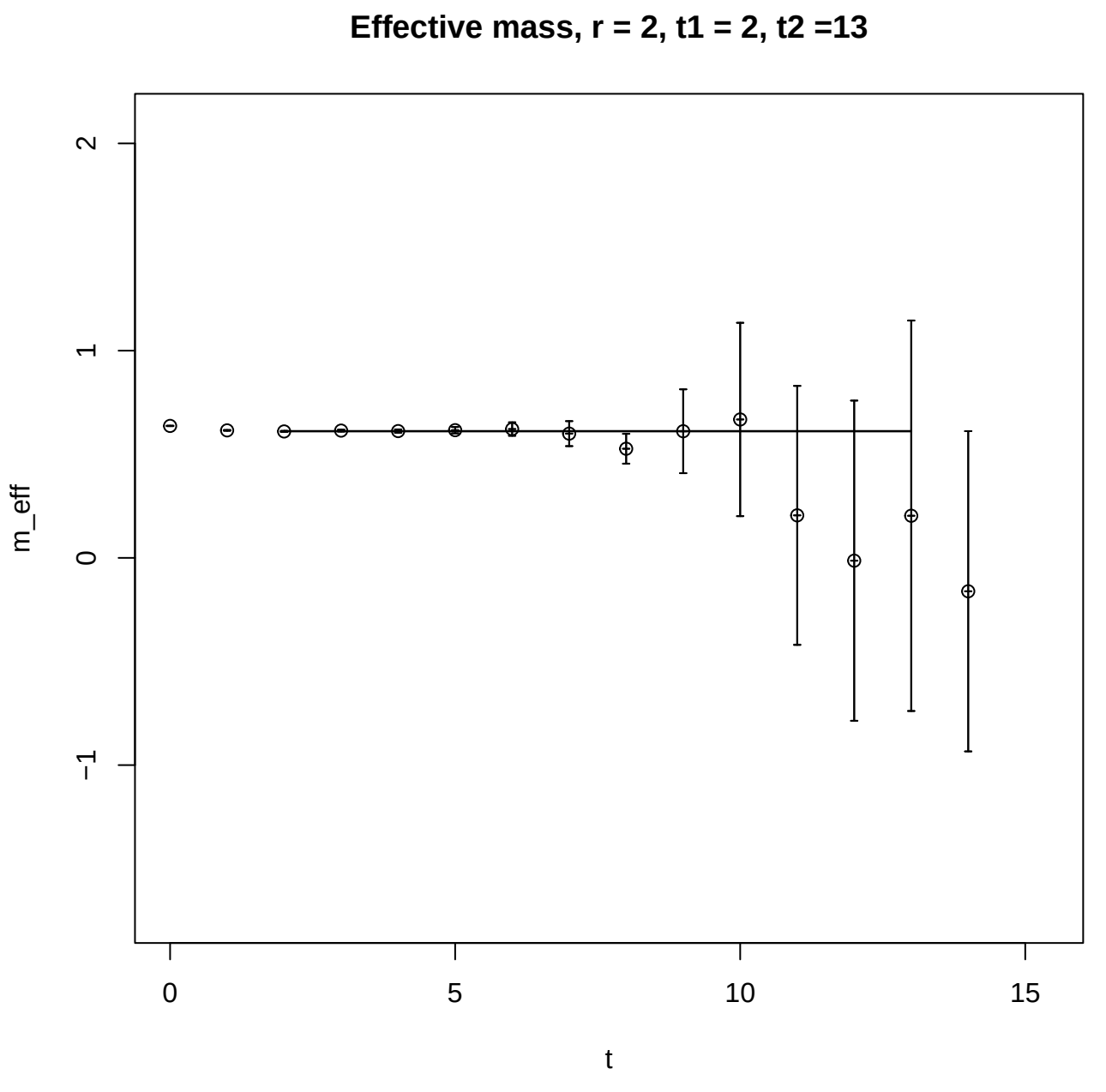
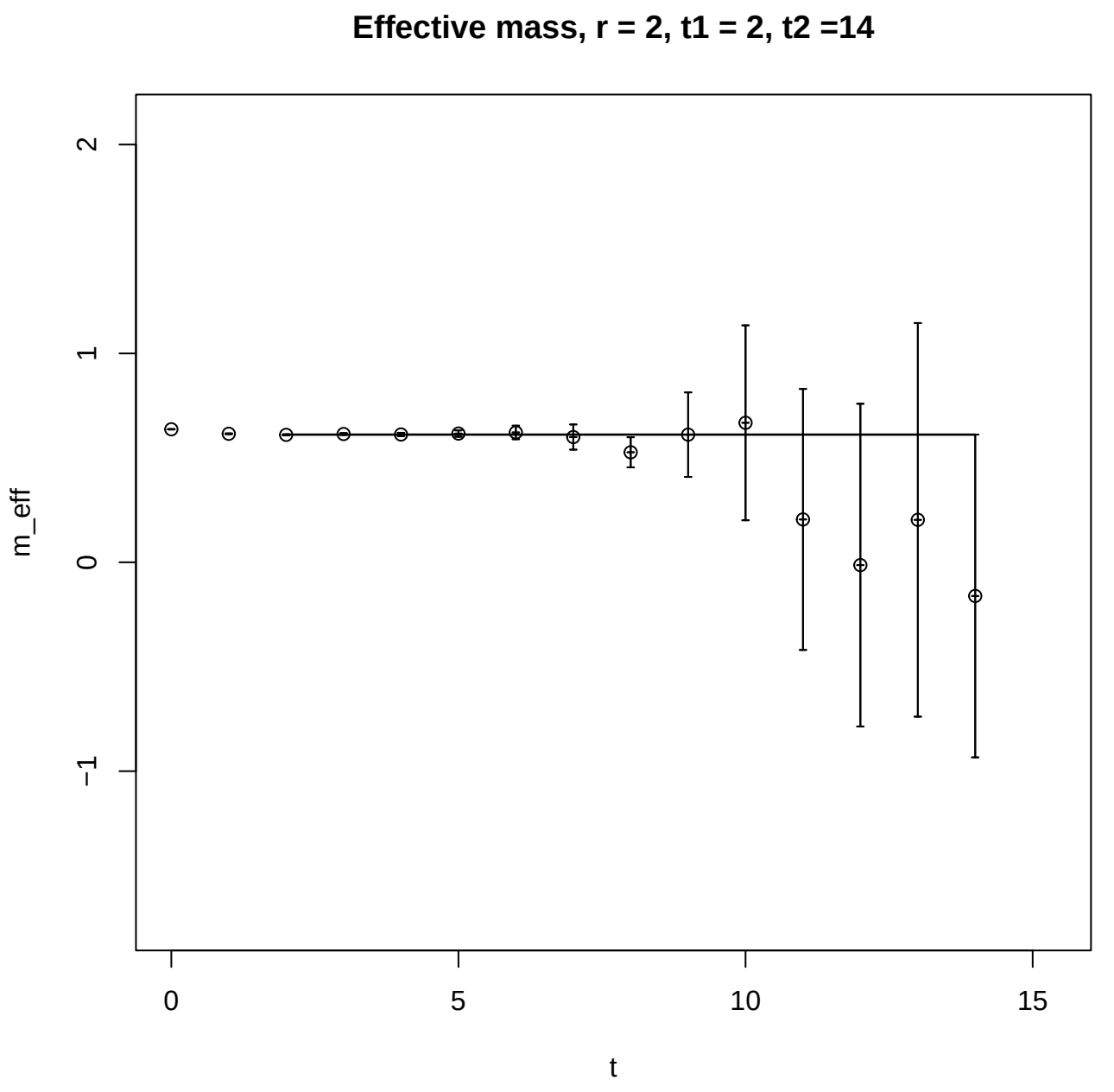
1
0
−1
0
5
10
15
t

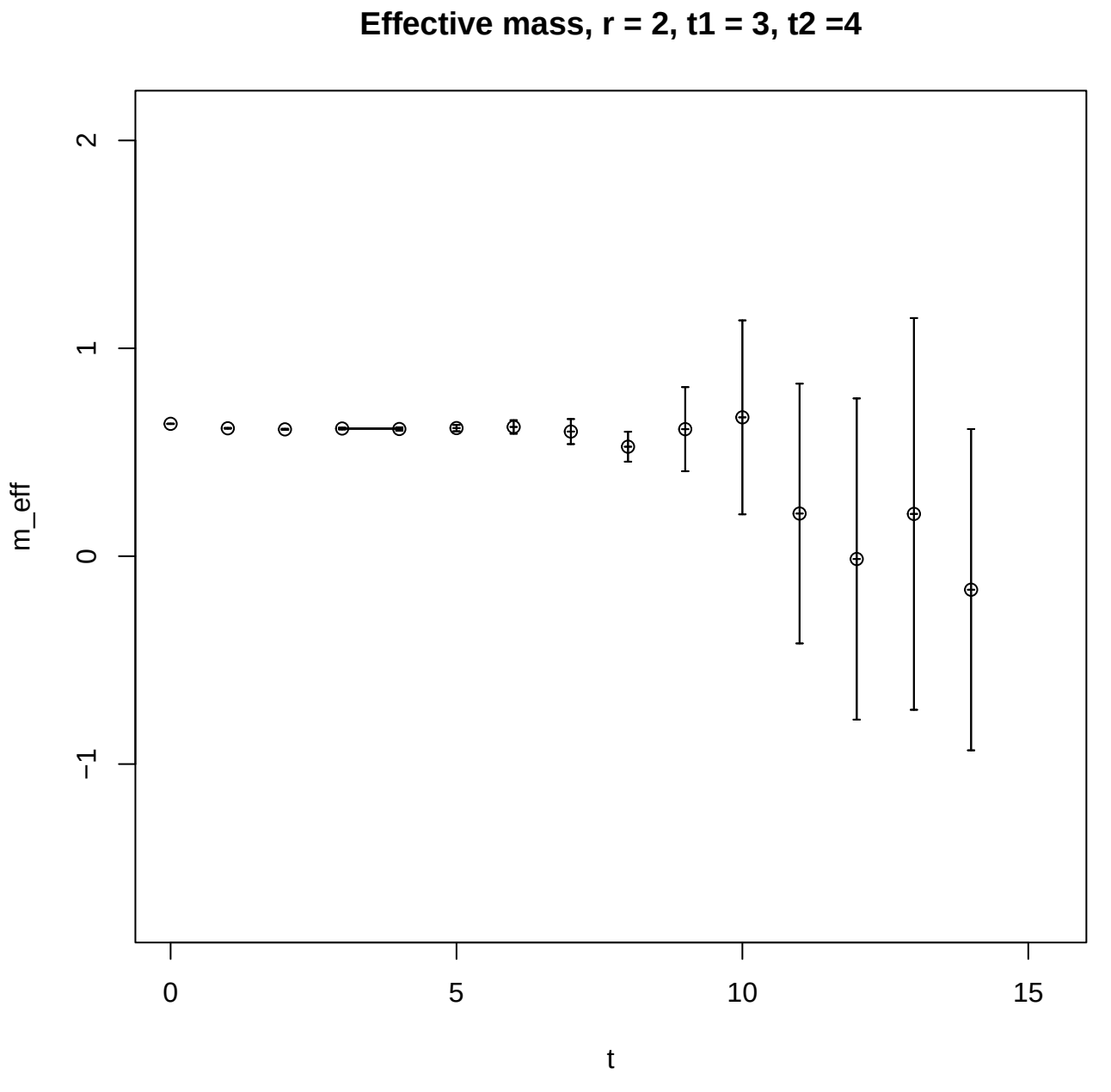

Effective mass, r = 2, t1 = 2, t2 =10
m_eff
2
1
0
−1
0
5
10
15
t

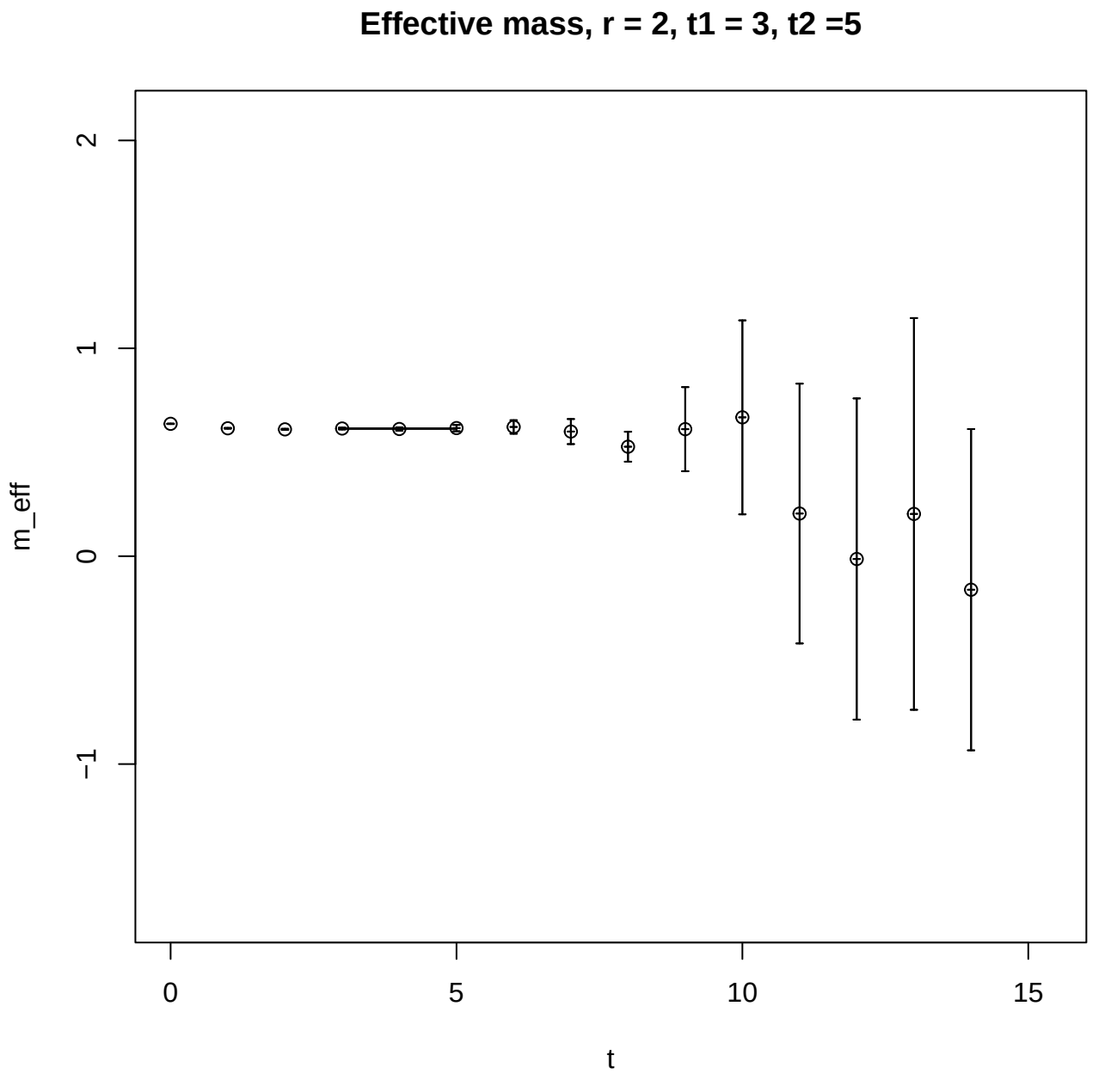

Effective mass, r = 2, t1 = 2, t2 =11
m_eff
2
1
0
−1
0
5
10
15
t


Effective mass, r = 2, t1 = 2, t2 =12
m_eff
2
1
0
−1
0
5
10
15
t


Effective mass, r = 2, t1 = 2, t2 =13
m_eff
2
1
0
−1
0
5
10
15
t


Effective mass, r = 2, t1 = 2, t2 =14
m_eff
2
1
0
−1
0
5
10
15
t


Effective mass, r = 2, t1 = 3, t2 =4
m_eff
2
1
0
−1
0
5
10
15
t


Effective mass, r = 2, t1 = 3, t2 =5
m_eff
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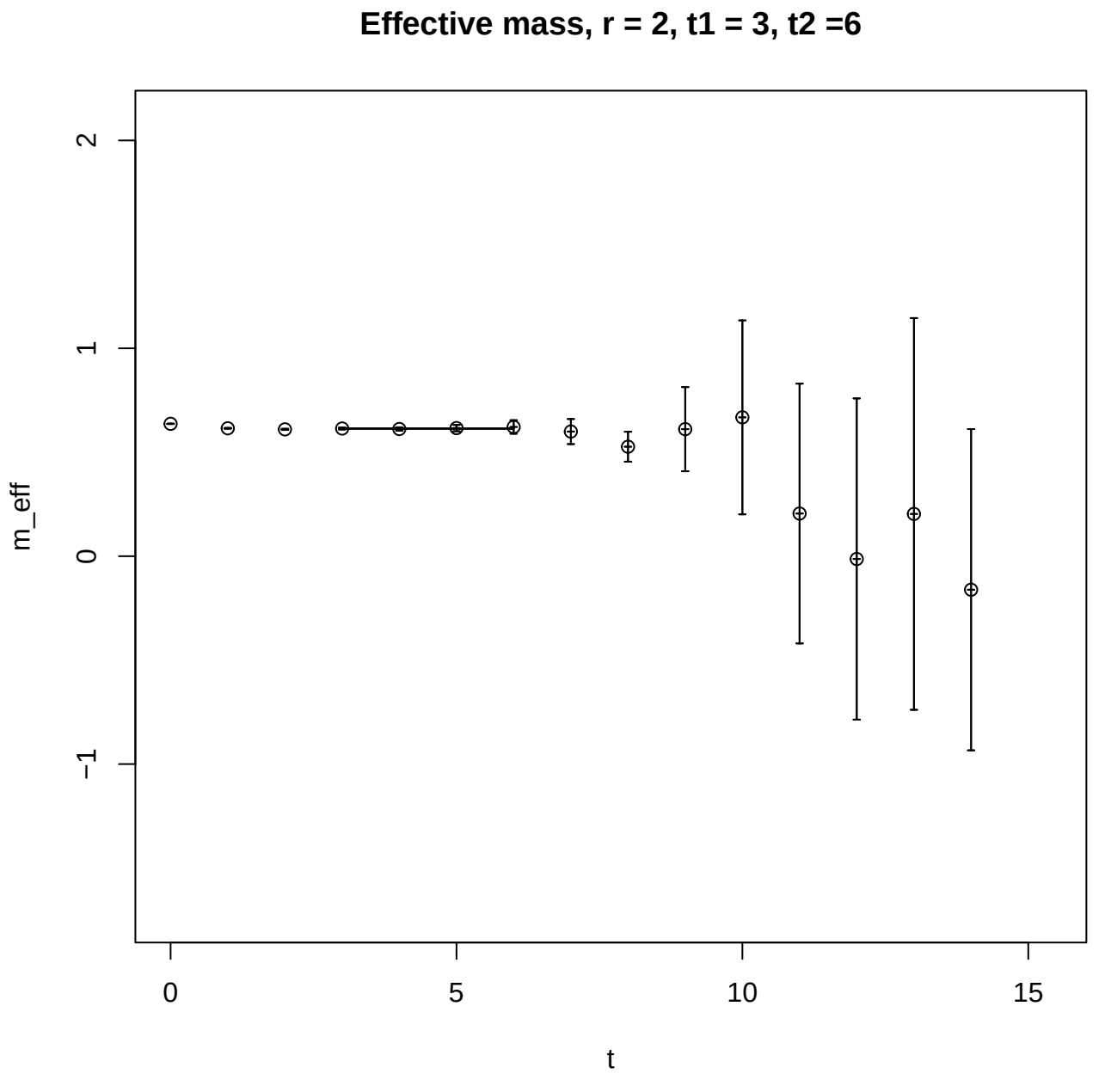
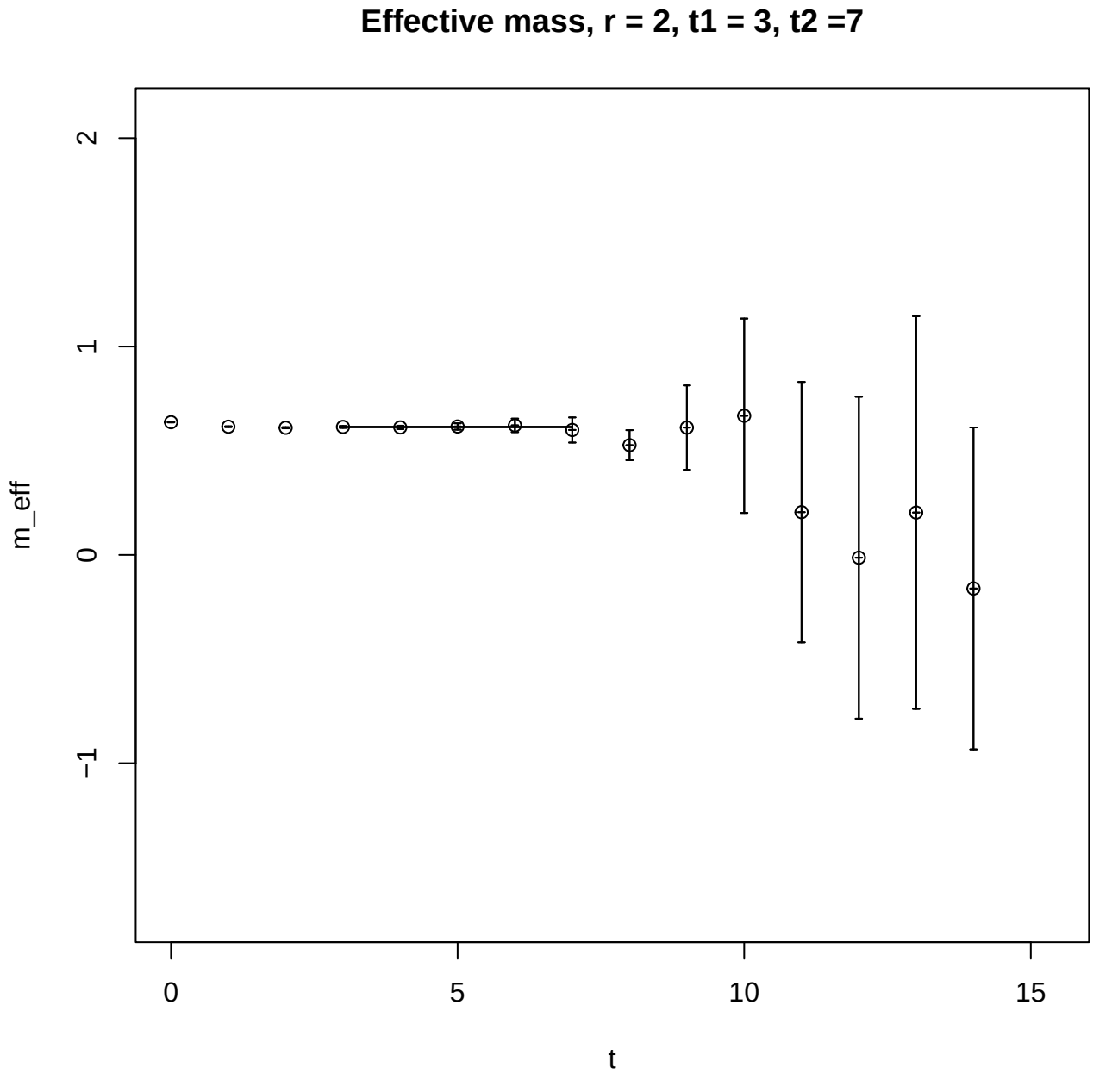
1
0
−1
0
5
10
15
t


Effective mass, r = 2, t1 = 3, t2 =6
m_eff
2
1
0
−1
0
5
10
15
t


Effective mass, r = 2, t1 = 3, t2 =7
m_eff
2
1
0
−1
0
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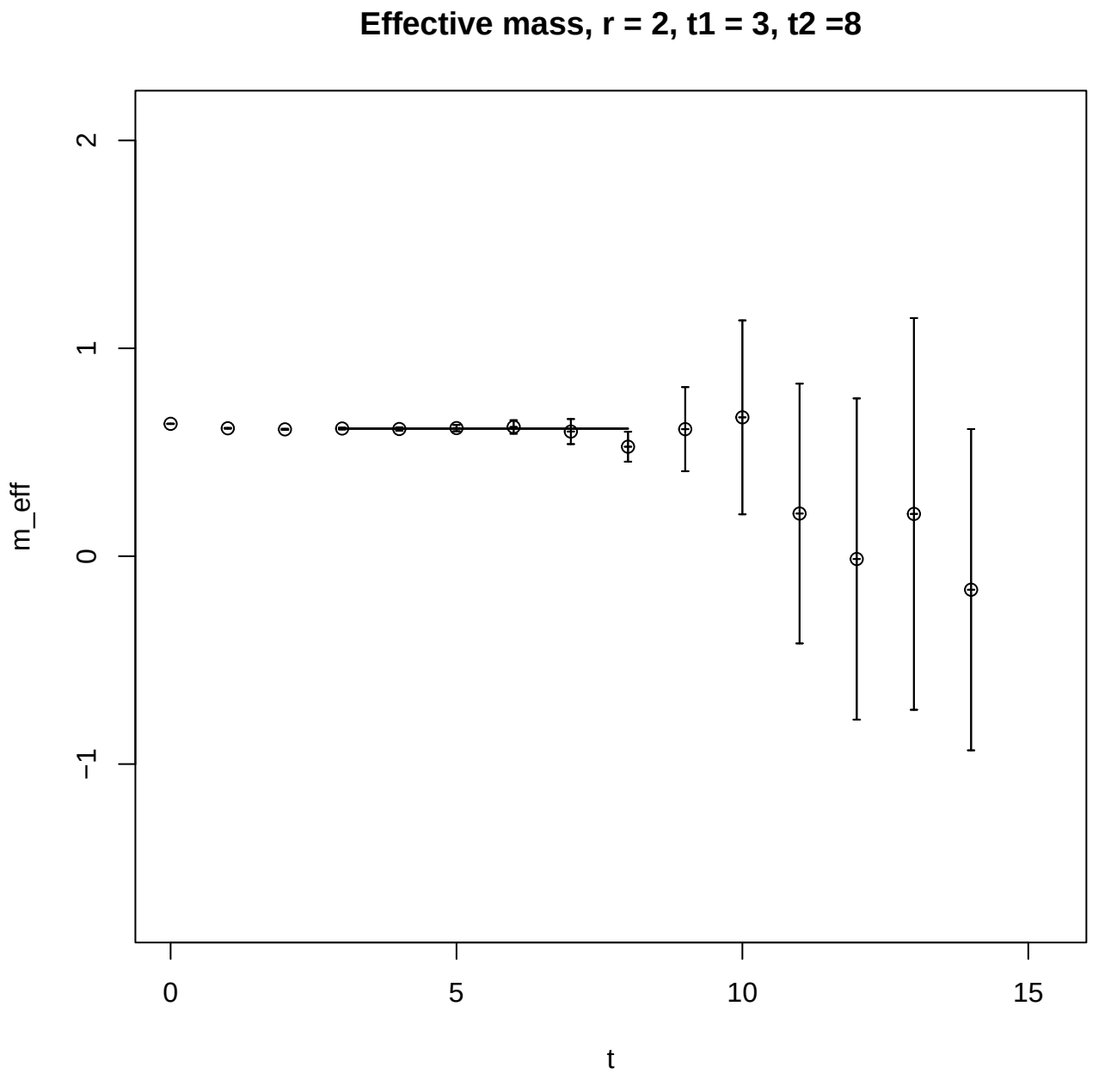
15
t


Effective mass, r = 2, t1 = 3, t2 =8
m_eff
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1
0
−1
0
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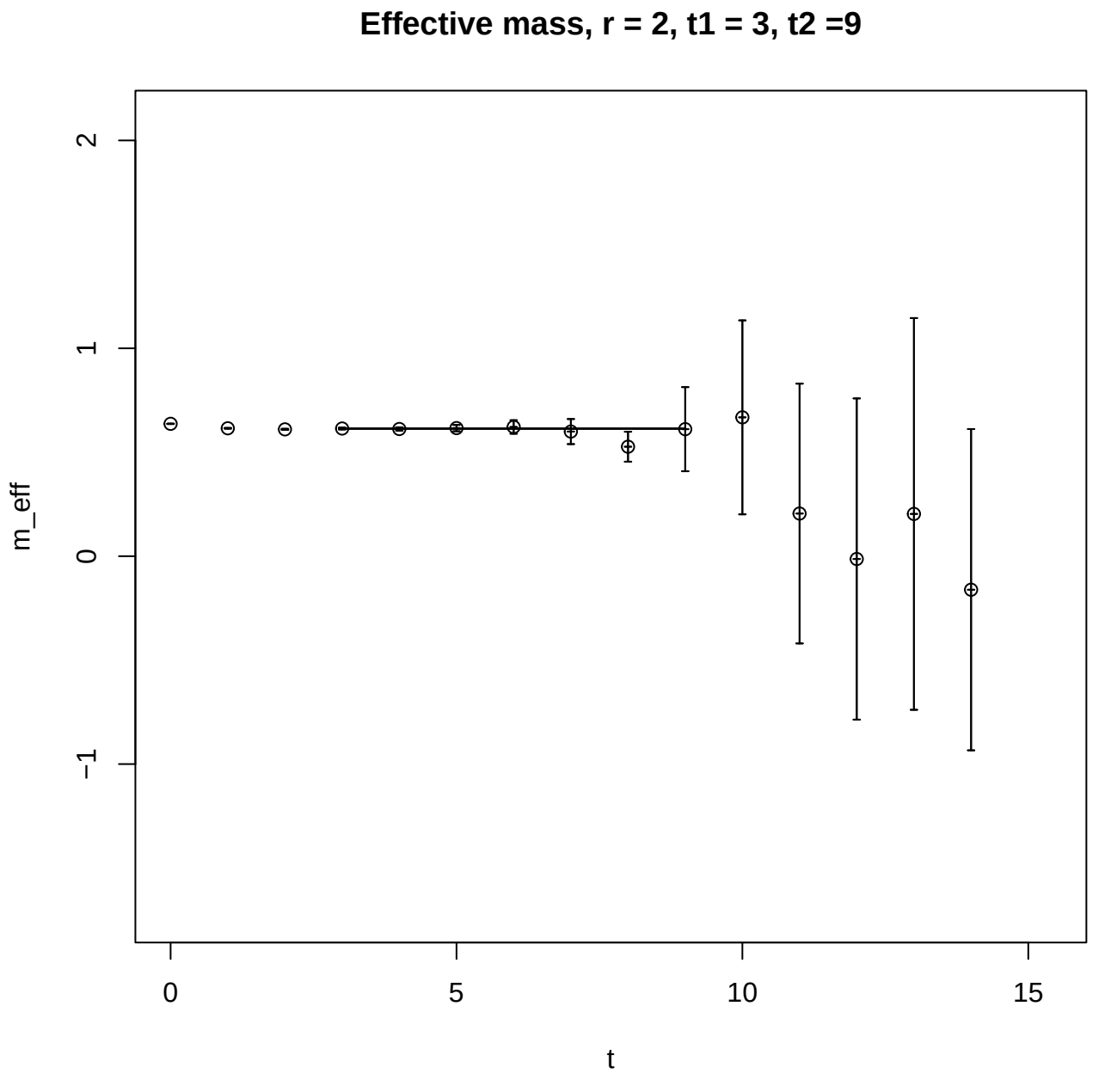
15
t


Effective mass, r = 2, t1 = 3, t2 =9
m_eff
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1
0
−1
0
5
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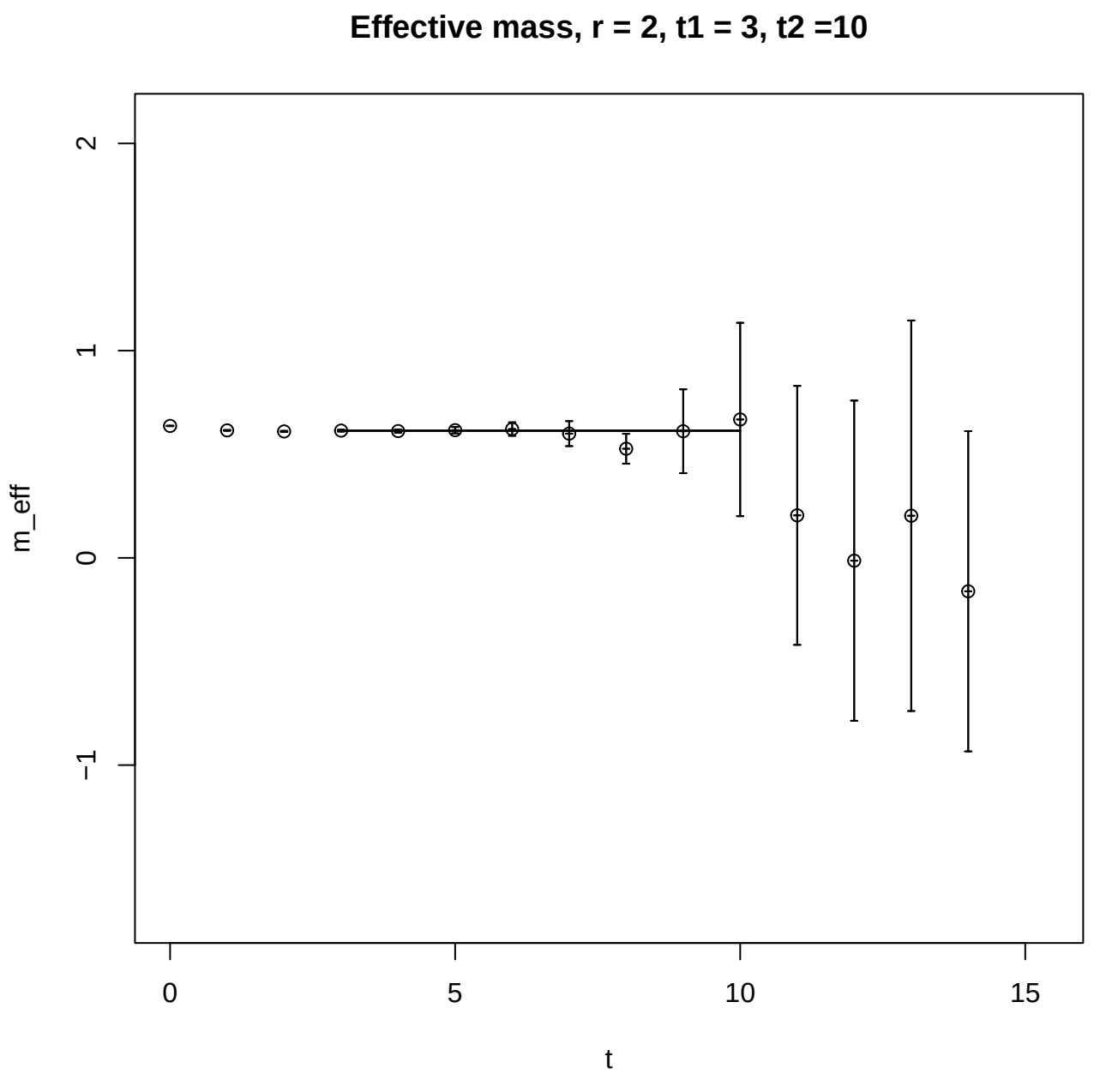
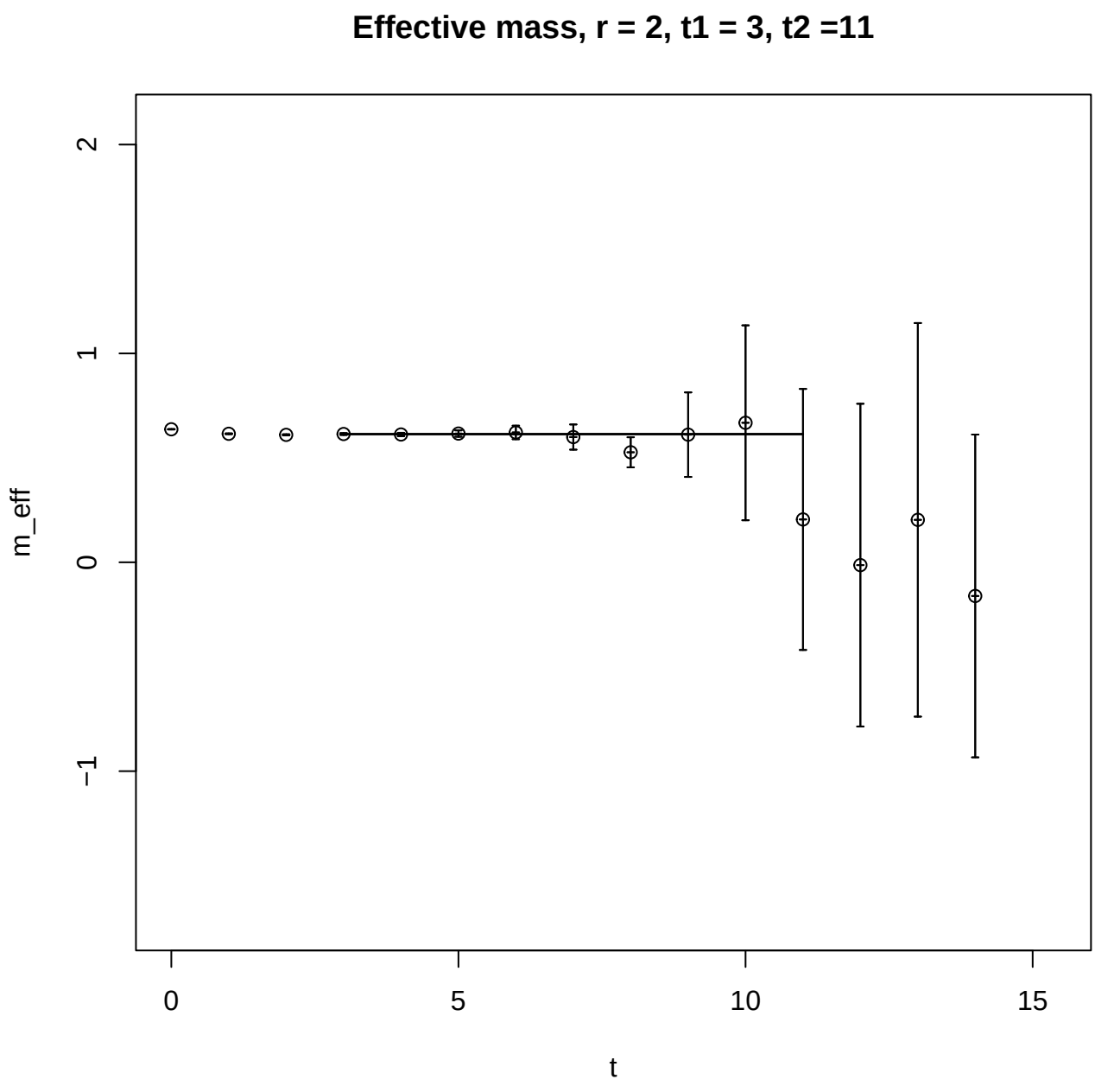
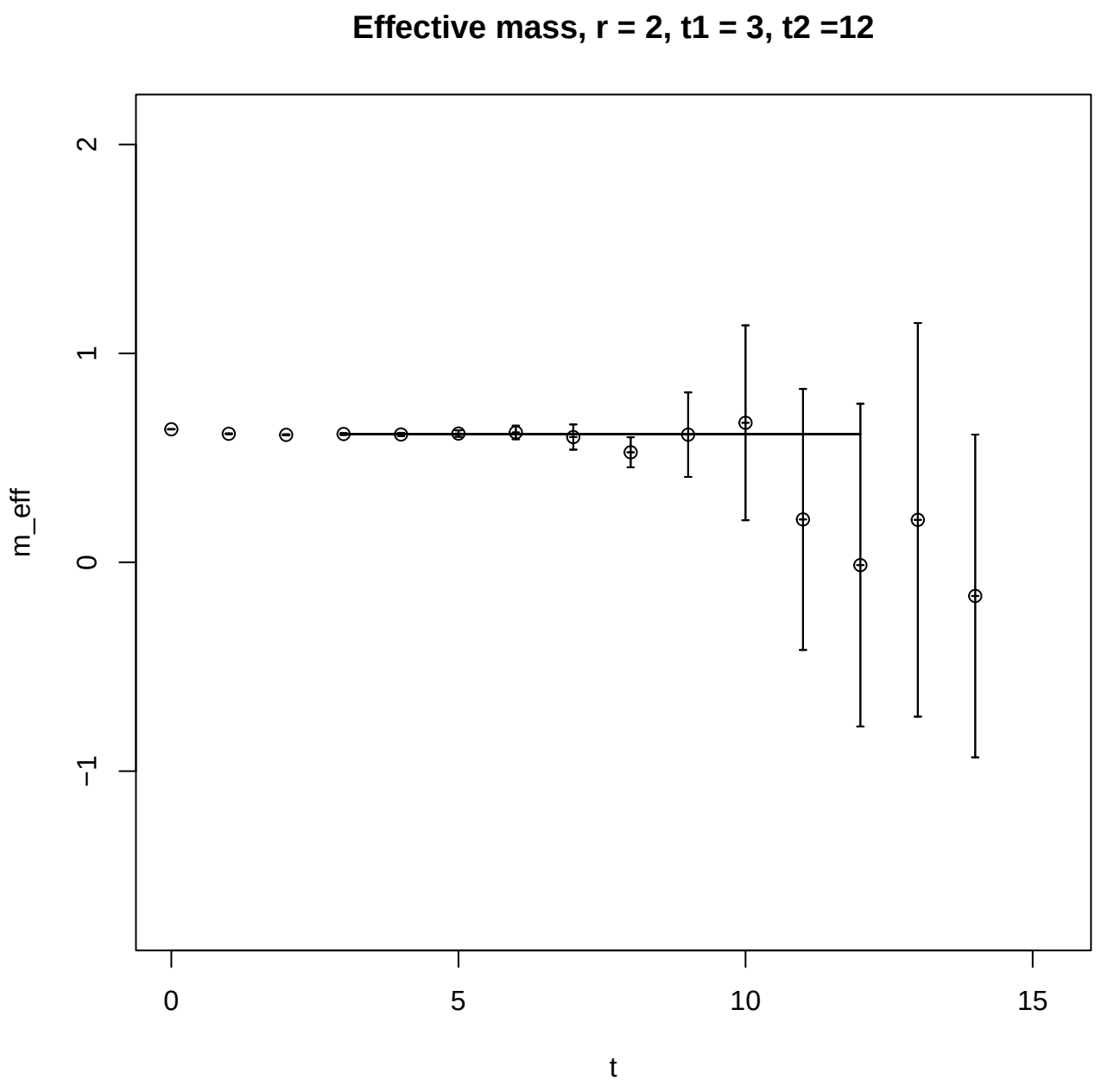
15
t


Effective mass, r = 2, t1 = 3, t2 =10
m_eff
2
1
0
−1
0
5
10
15
t


Effective mass, r = 2, t1 = 3, t2 =11
m_eff
2
1
0
−1
0
5
10
15
t


Effective mass, r = 2, t1 = 3, t2 =12
m_eff
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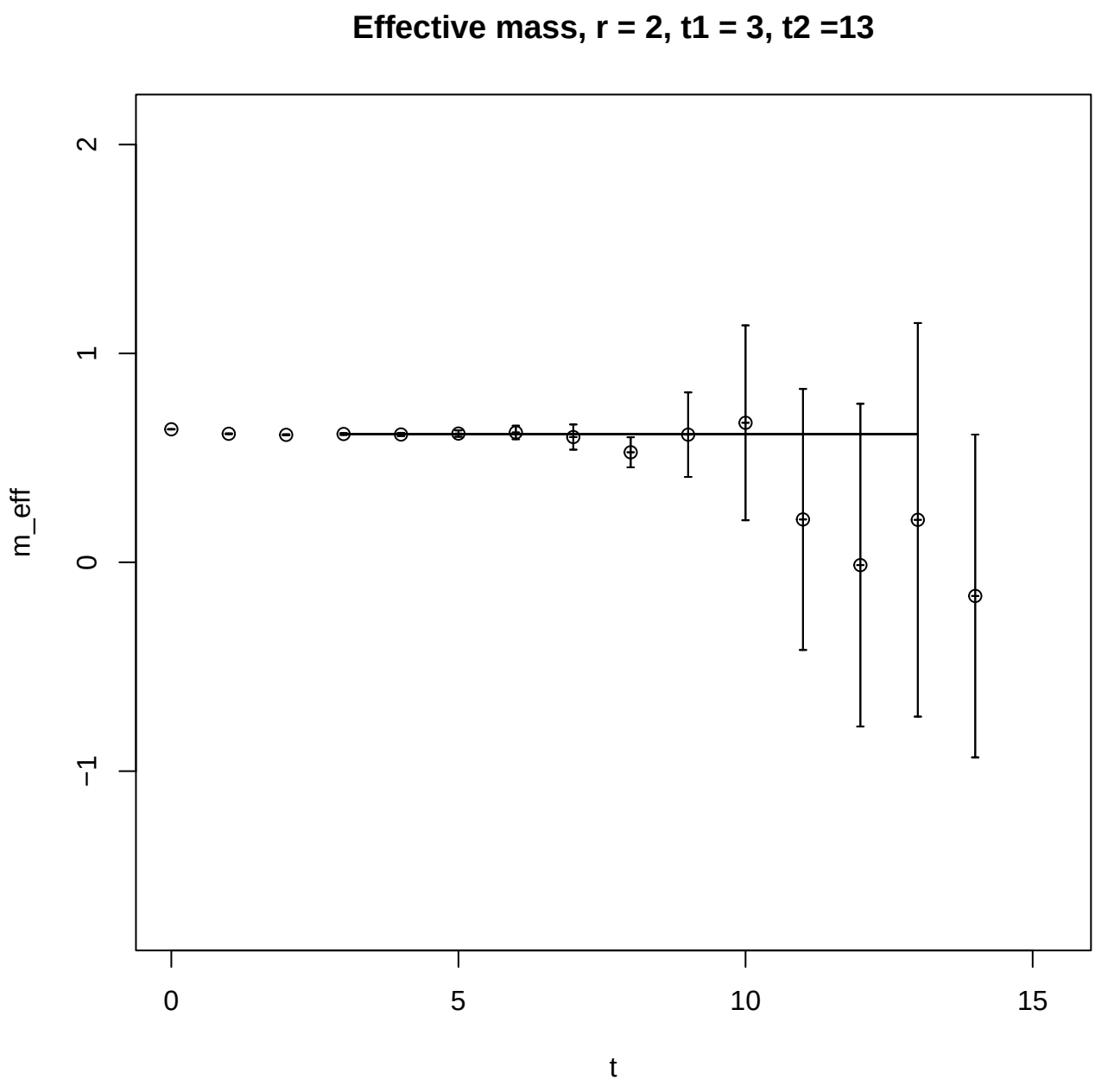
1
0
−1
0
5
10
15
t


Effective mass, r = 2, t1 = 3, t2 =13
m_eff
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1
0
−1
0
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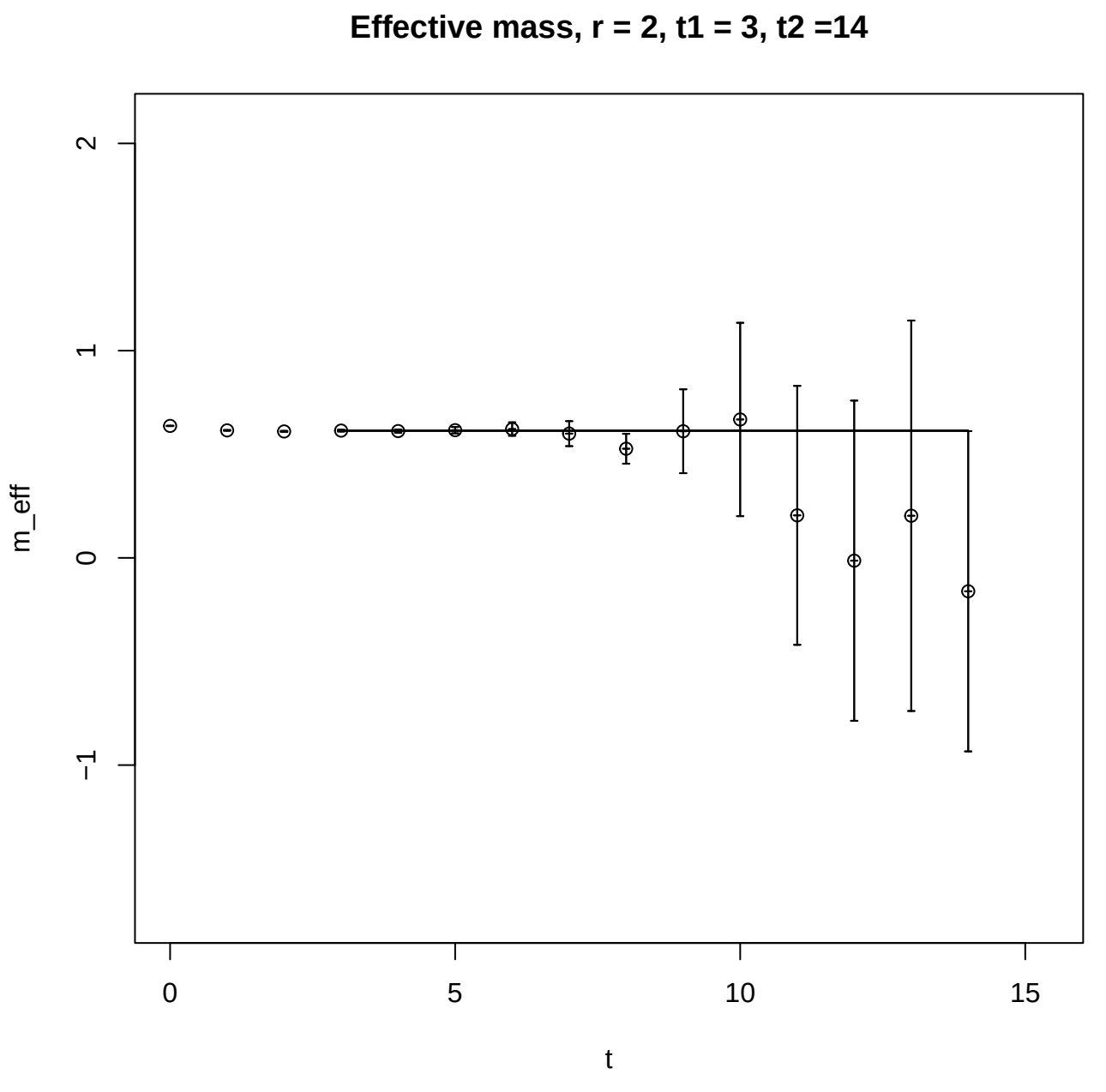
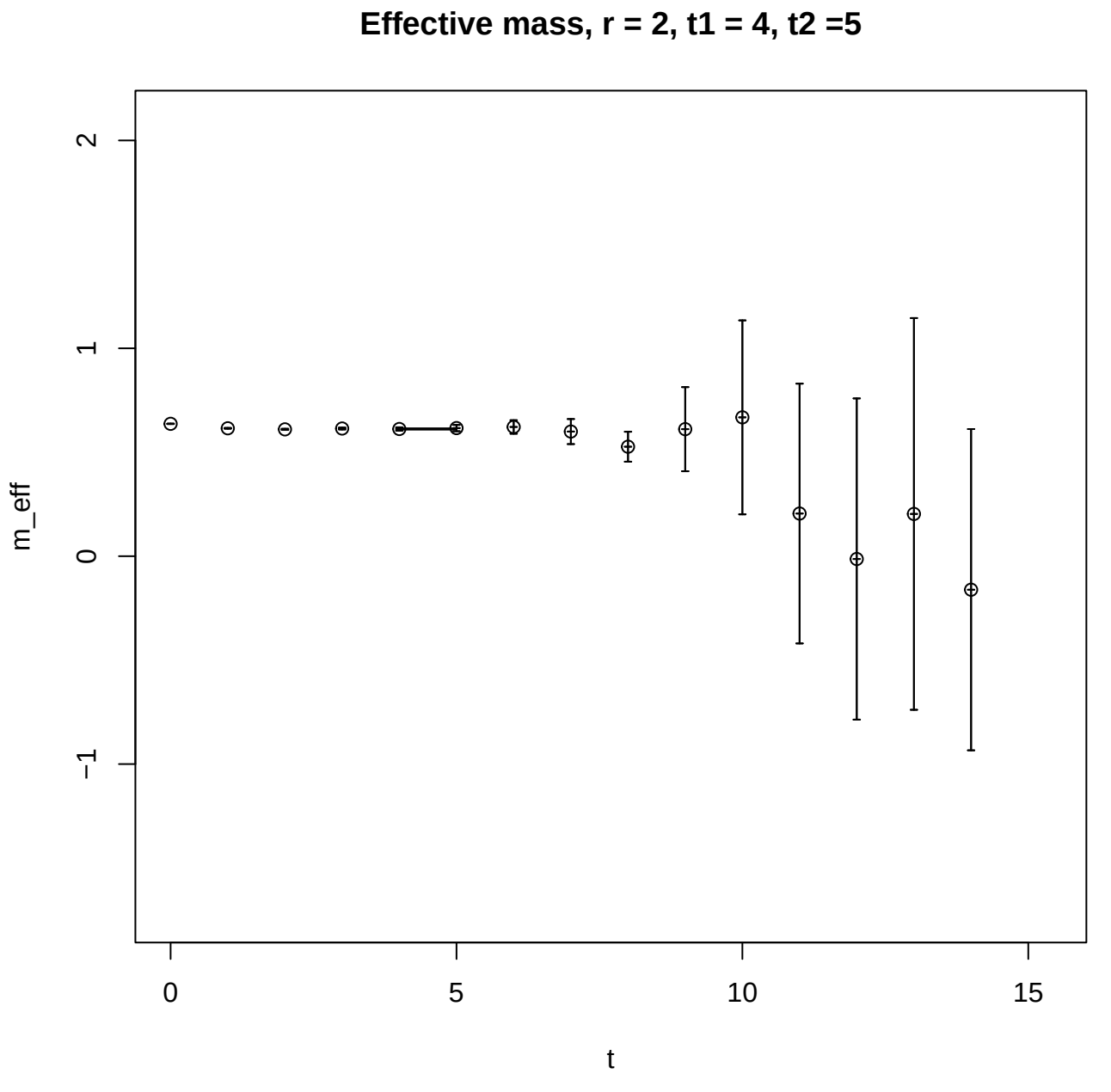
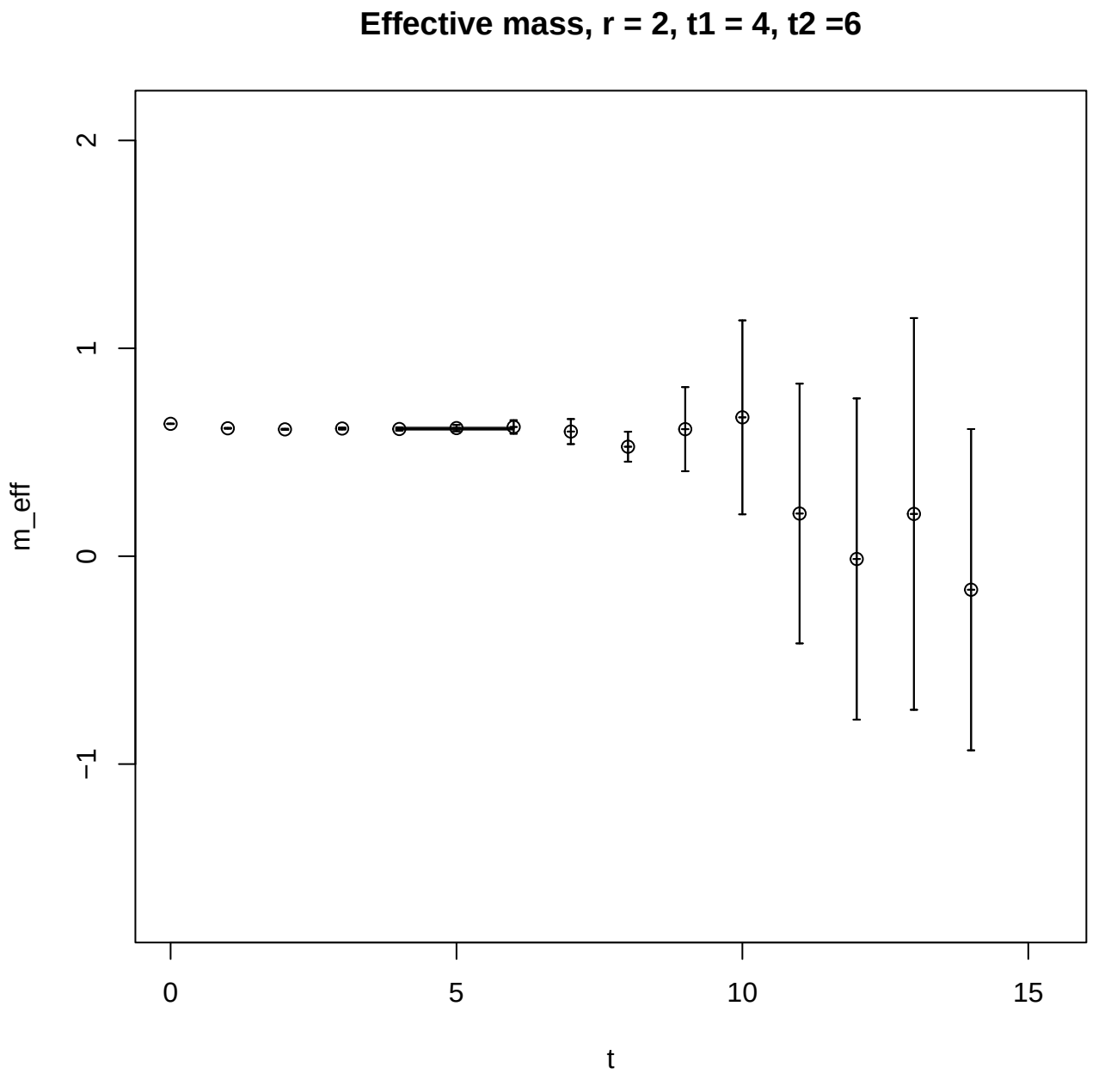
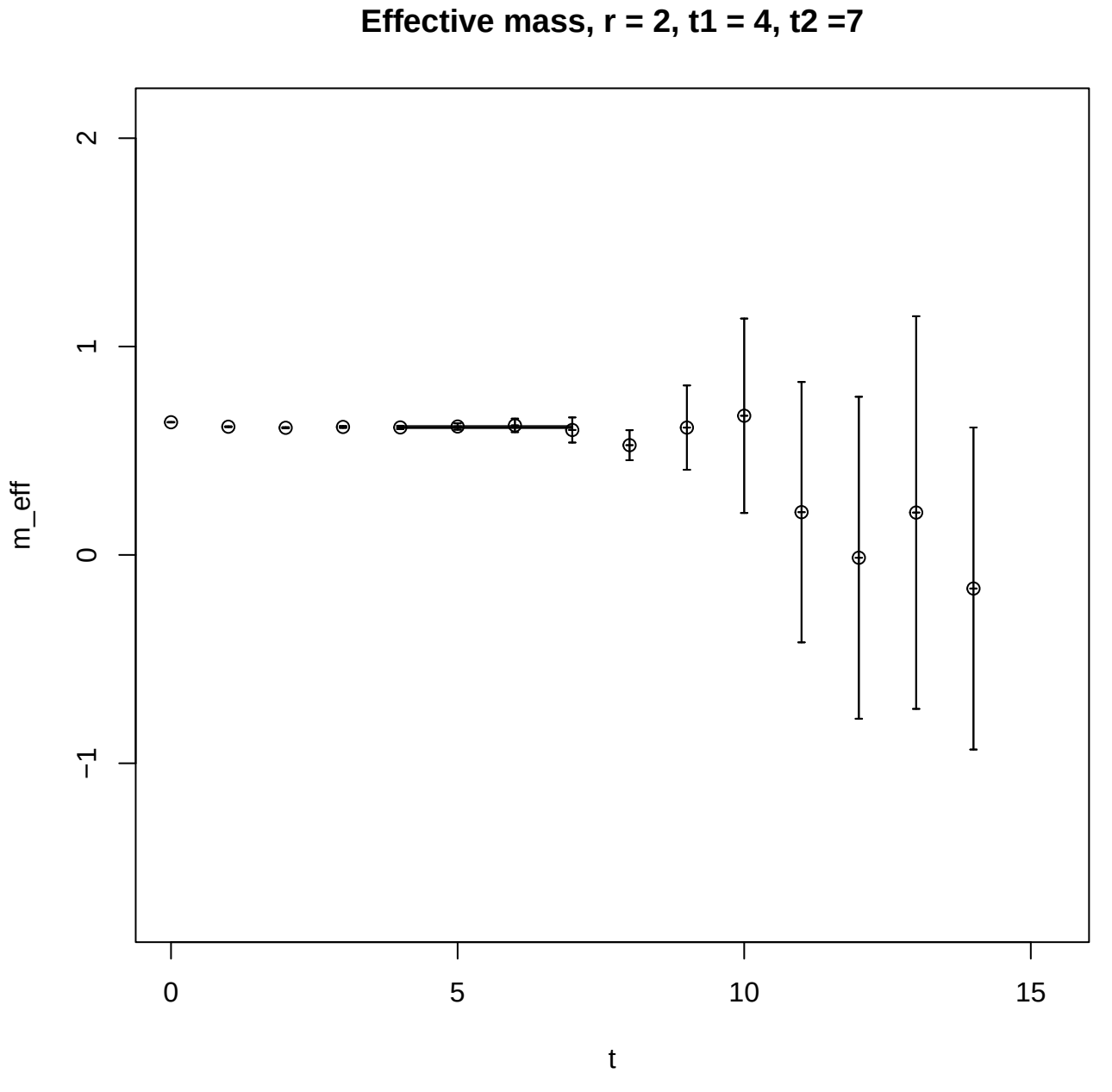
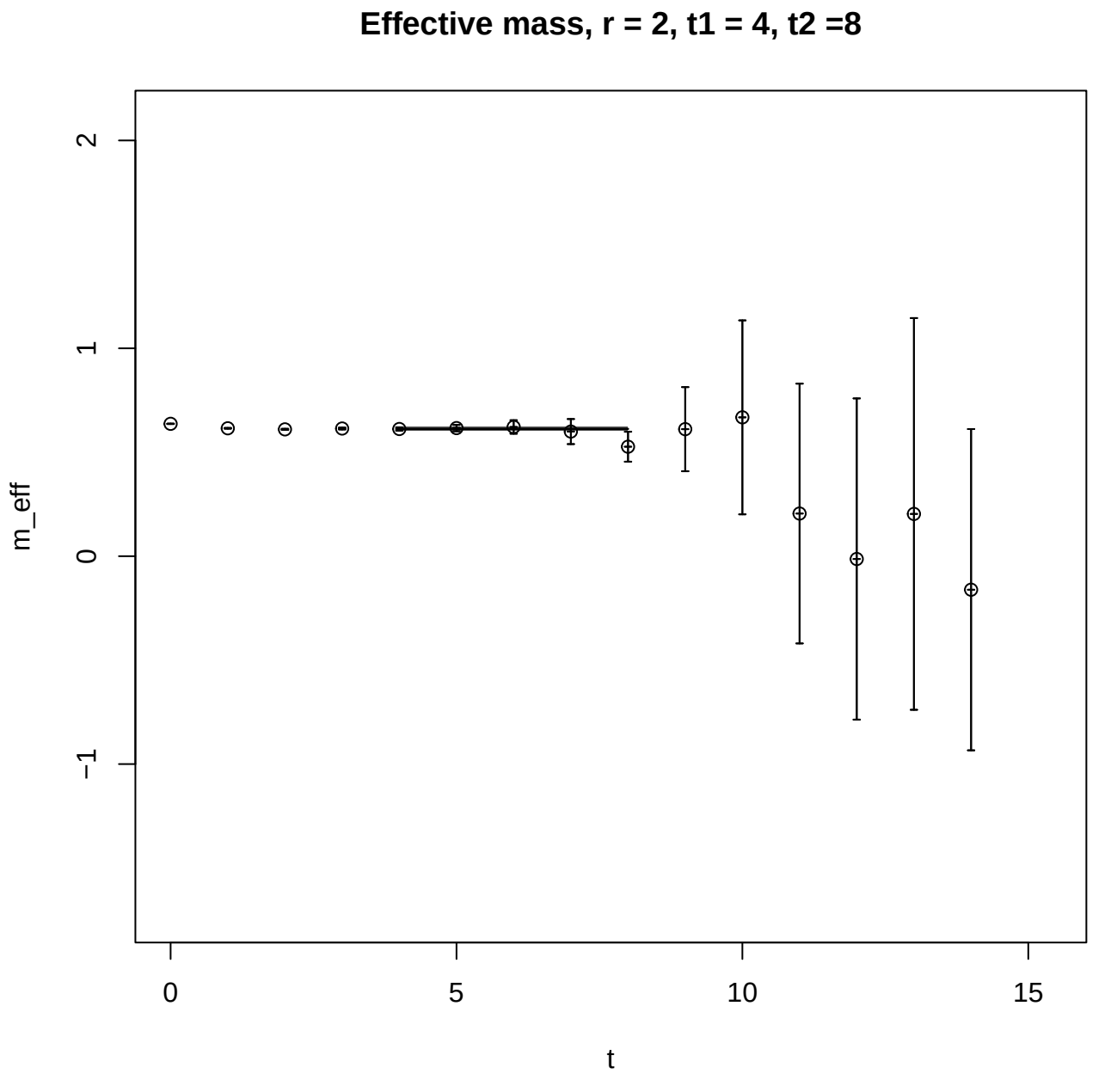
15
t

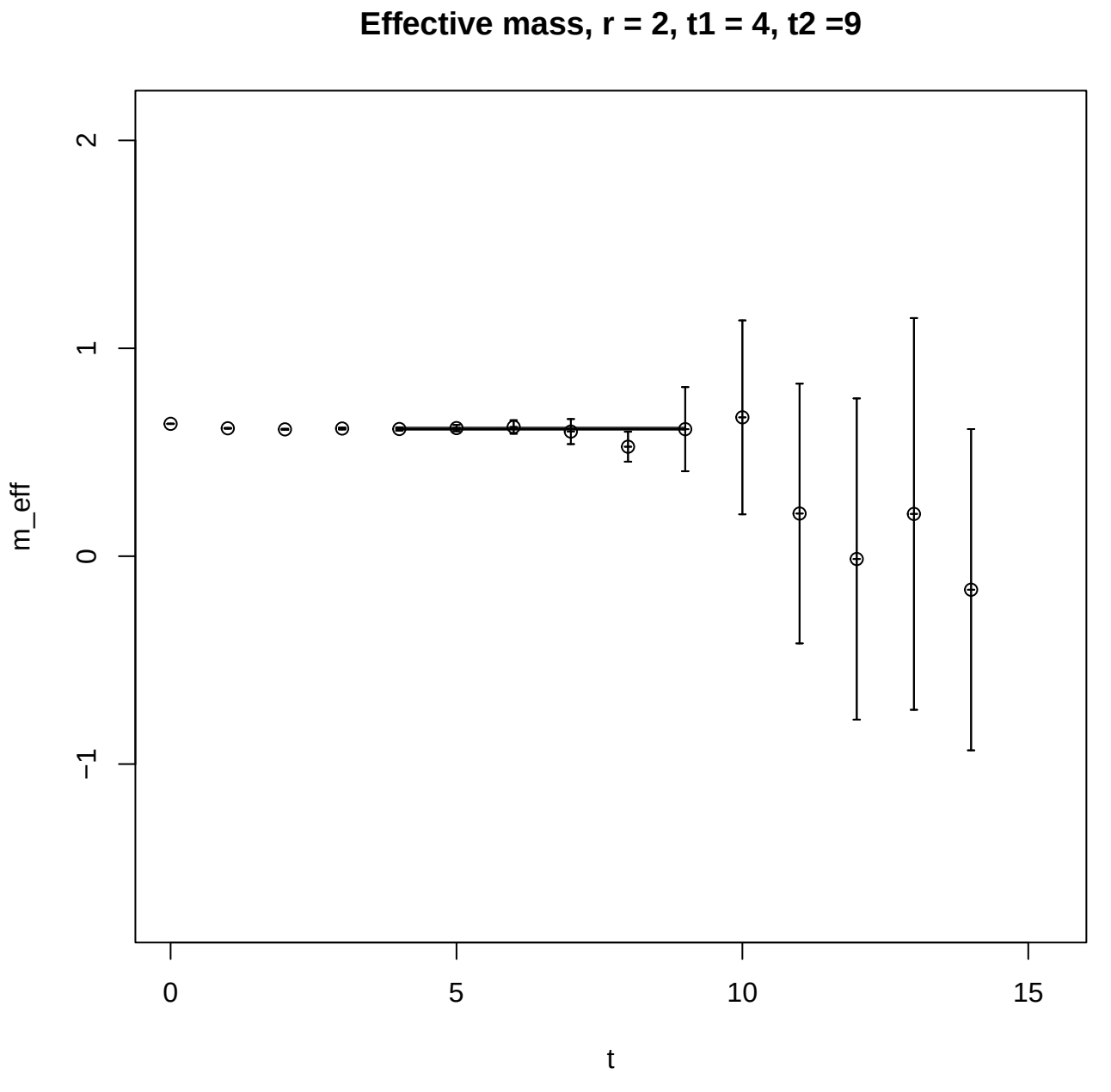

Effective mass, r = 2, t1 = 3, t2 =14
m_eff
2
1
0
−1
0
5
10
15
t

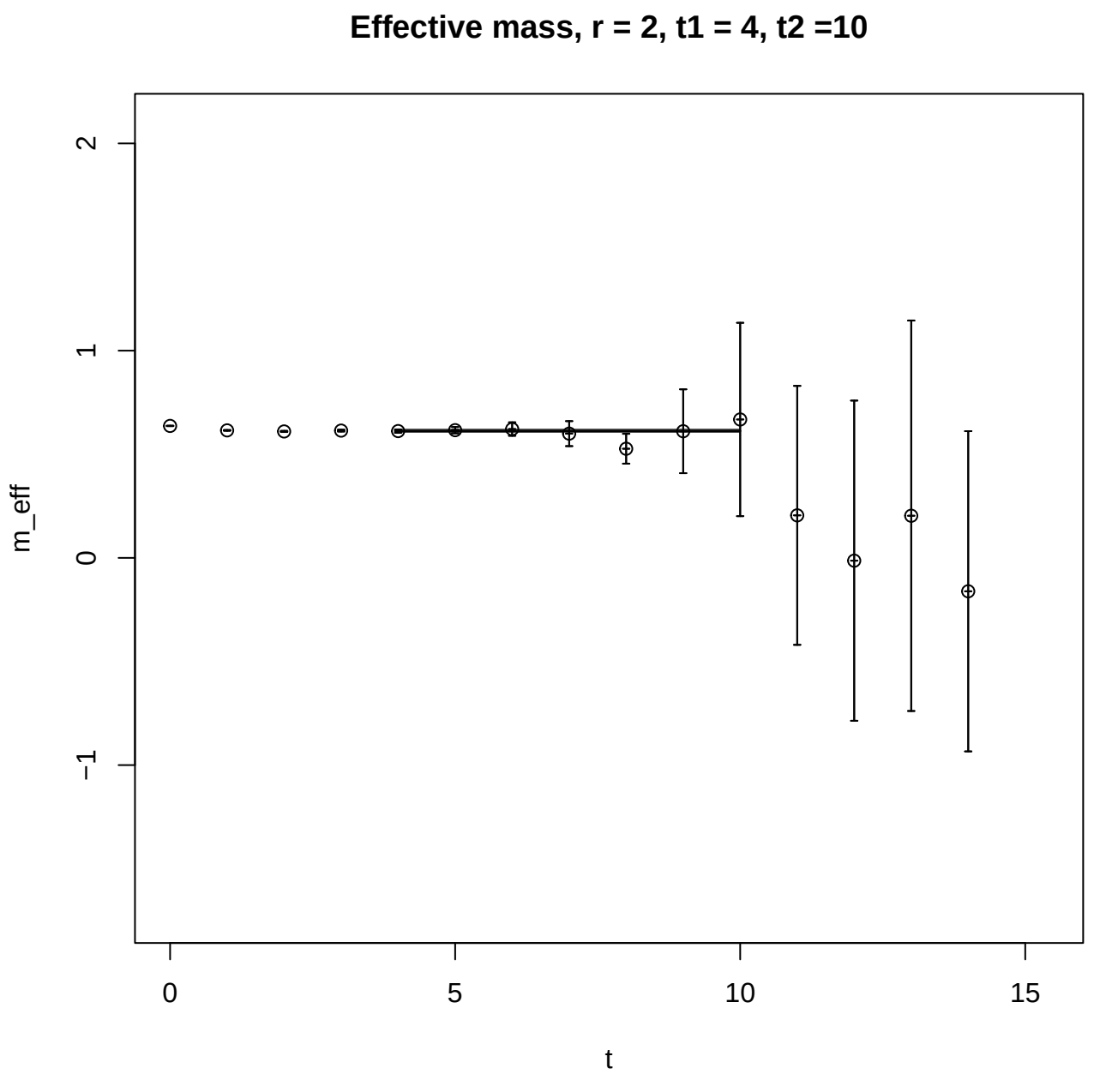

Effective mass, r = 2, t1 = 4, t2 =5
m_eff
2
1
0
−1
0
5
10
15
t

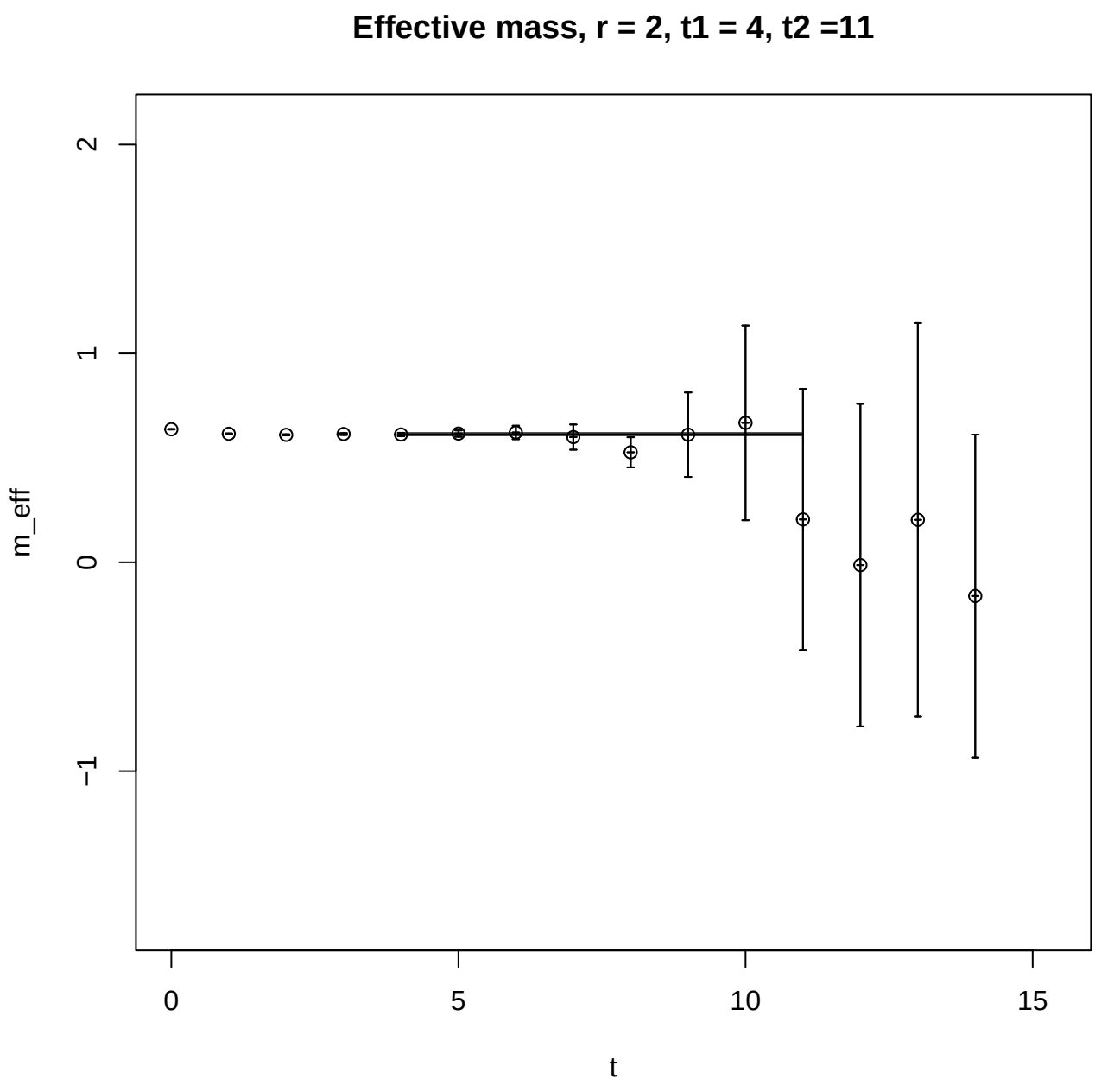

Effective mass, r = 2, t1 = 4, t2 =6
m_eff
2
1
0
−1
0
5
10
15
t

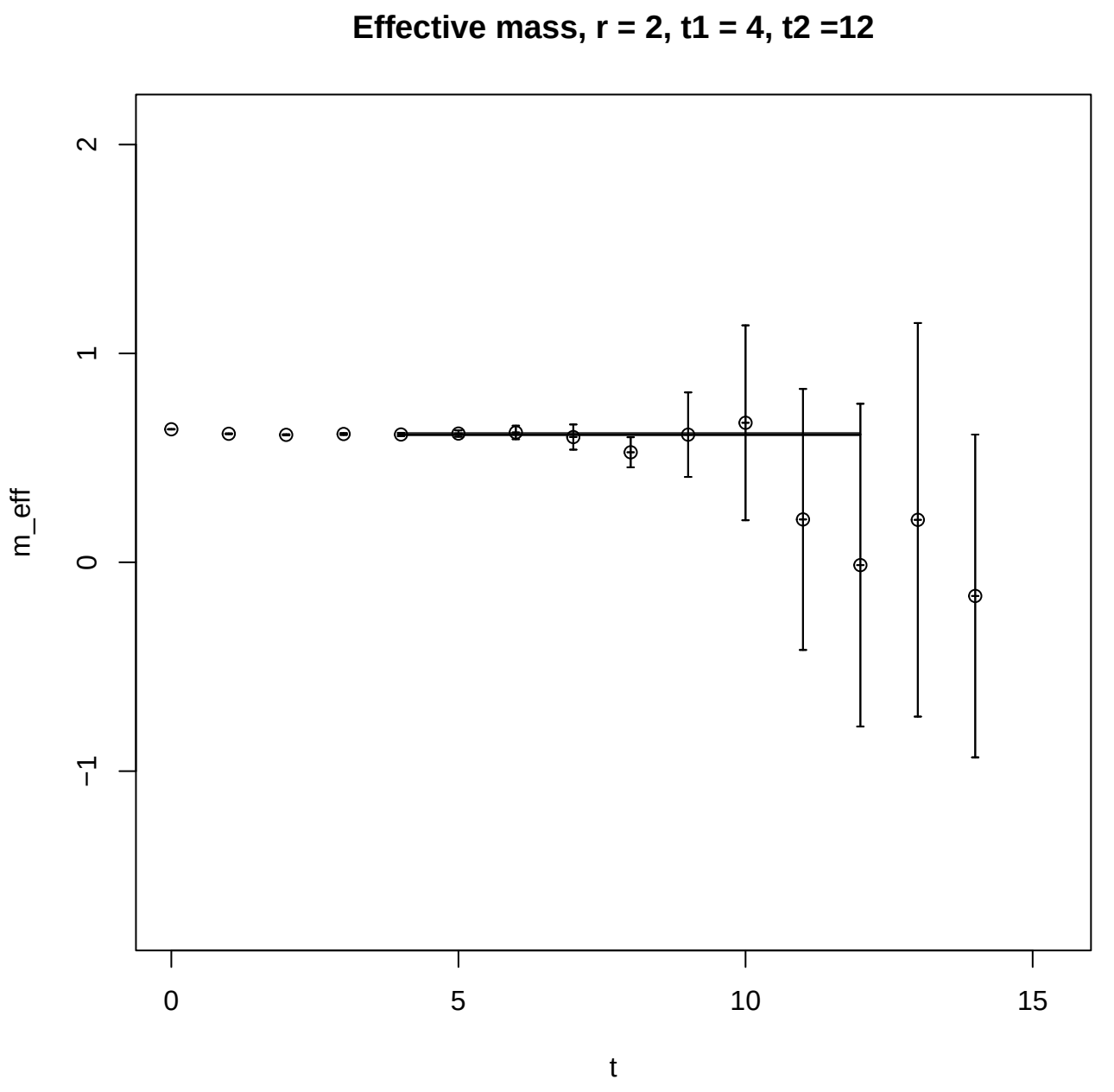

Effective mass, r = 2, t1 = 4, t2 =7
m_eff
2
1
0
−1
0
5
10
15
t

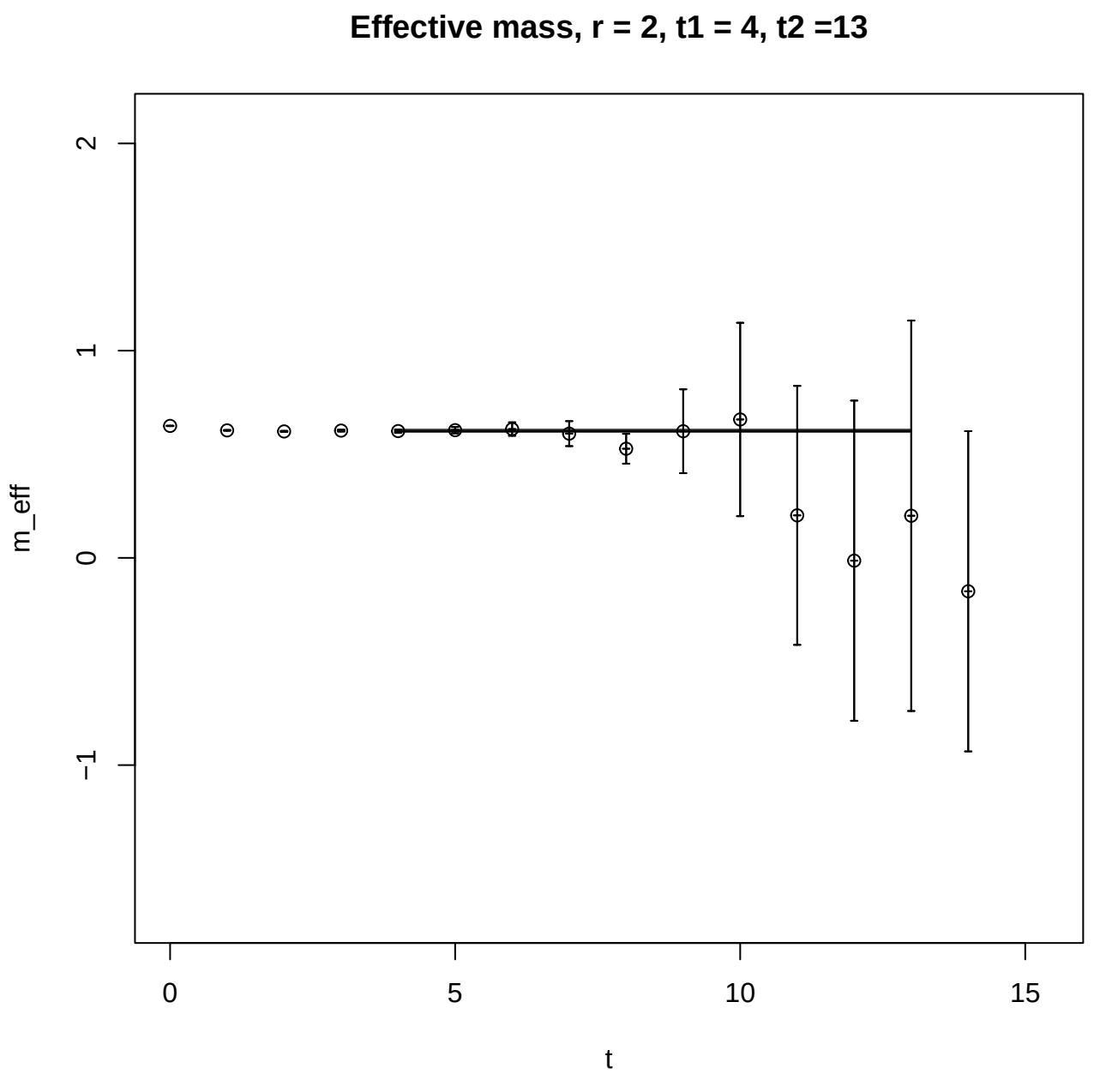

Effective mass, r = 2, t1 = 4, t2 =8
m_eff
2
1
0
−1
0
5
10
15
t

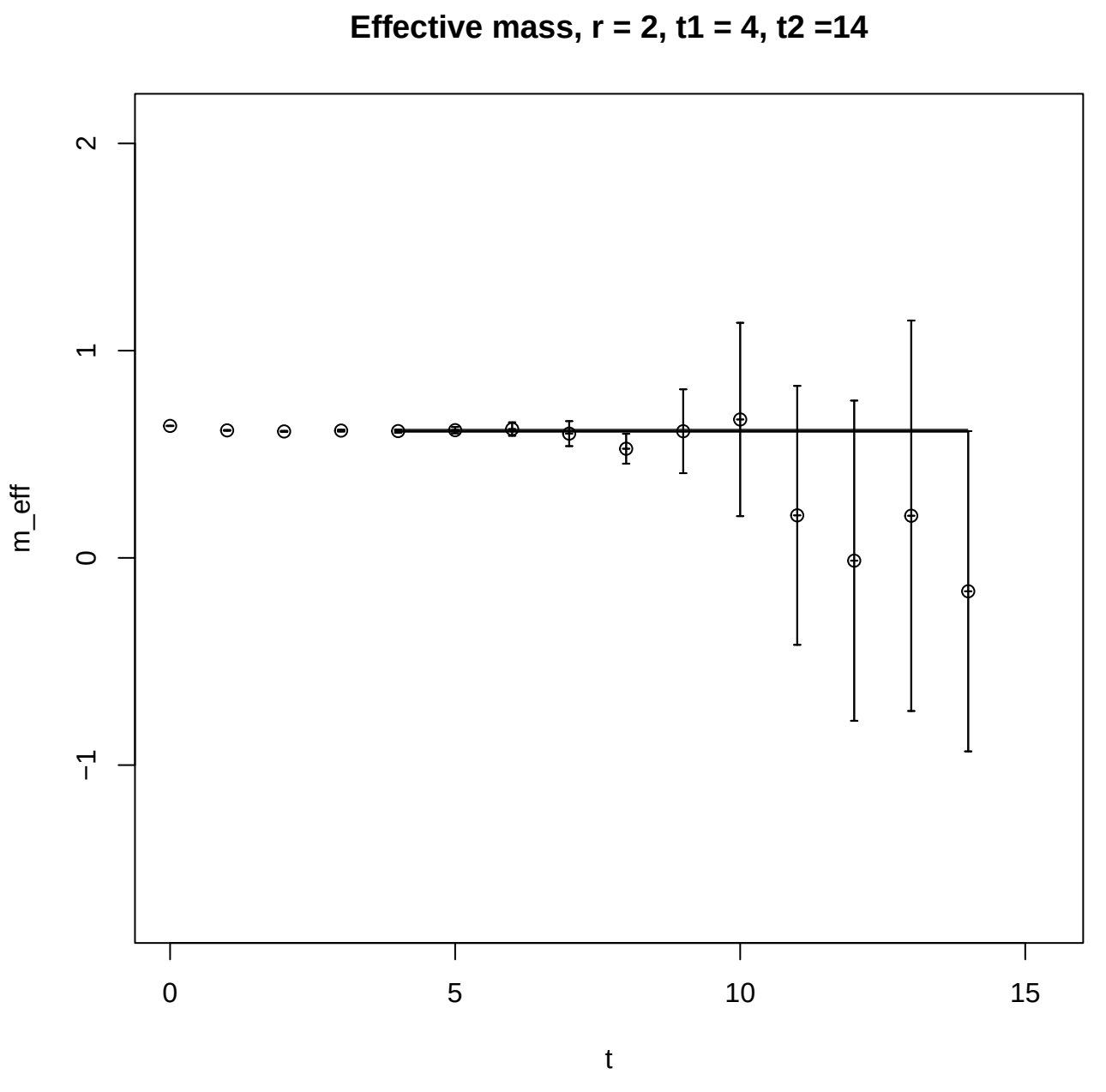

Effective mass, r = 2, t1 = 4, t2 =9
m_eff
2
1
0
−1
0
5
10
15
t

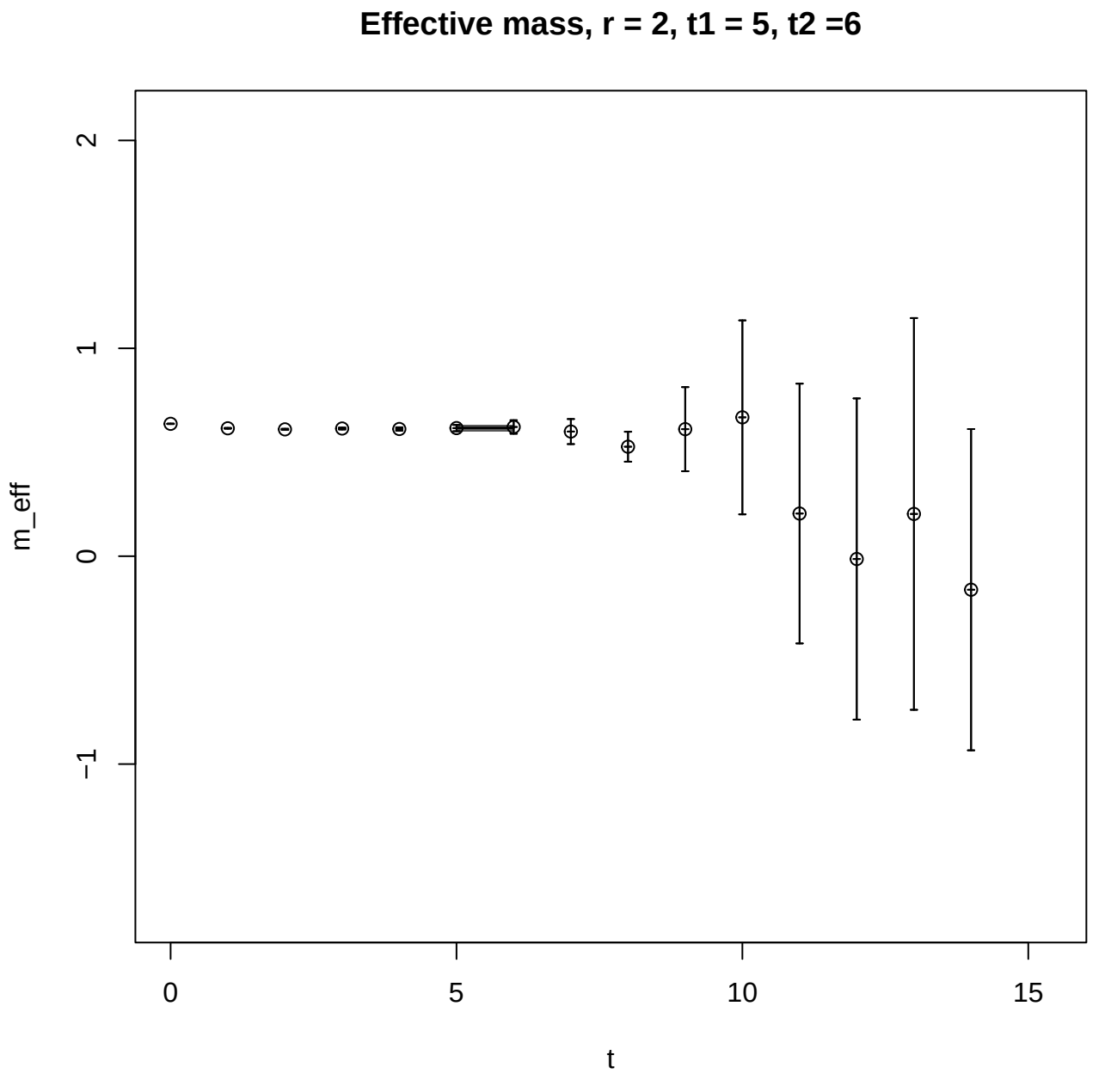

Effective mass, r = 2, t1 = 4, t2 =10
m_eff
2
1
0
−1
0
5
10
15
t

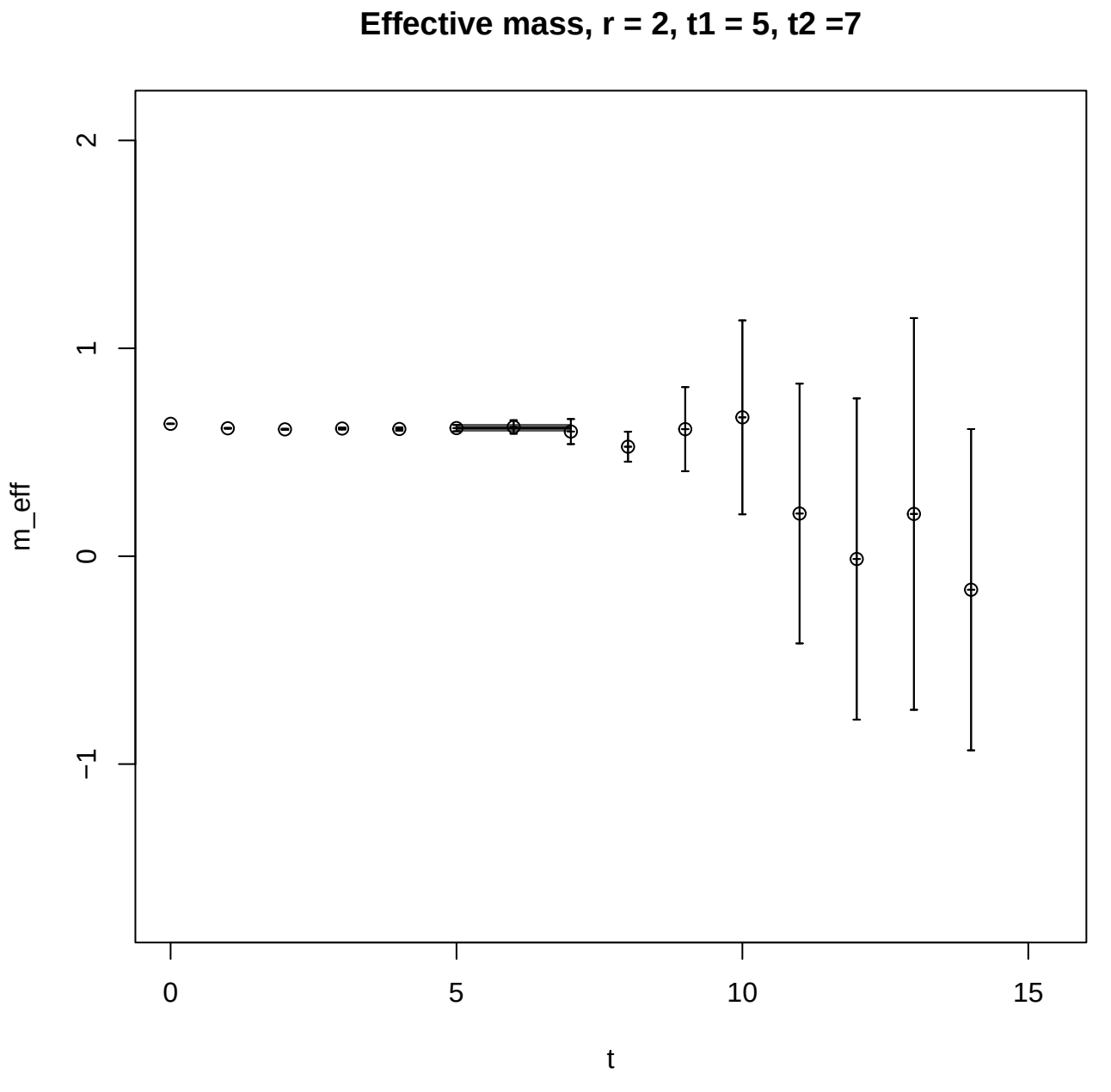

Effective mass, r = 2, t1 = 4, t2 =11
m_eff
2
1
0
−1
0
5
10
15
t

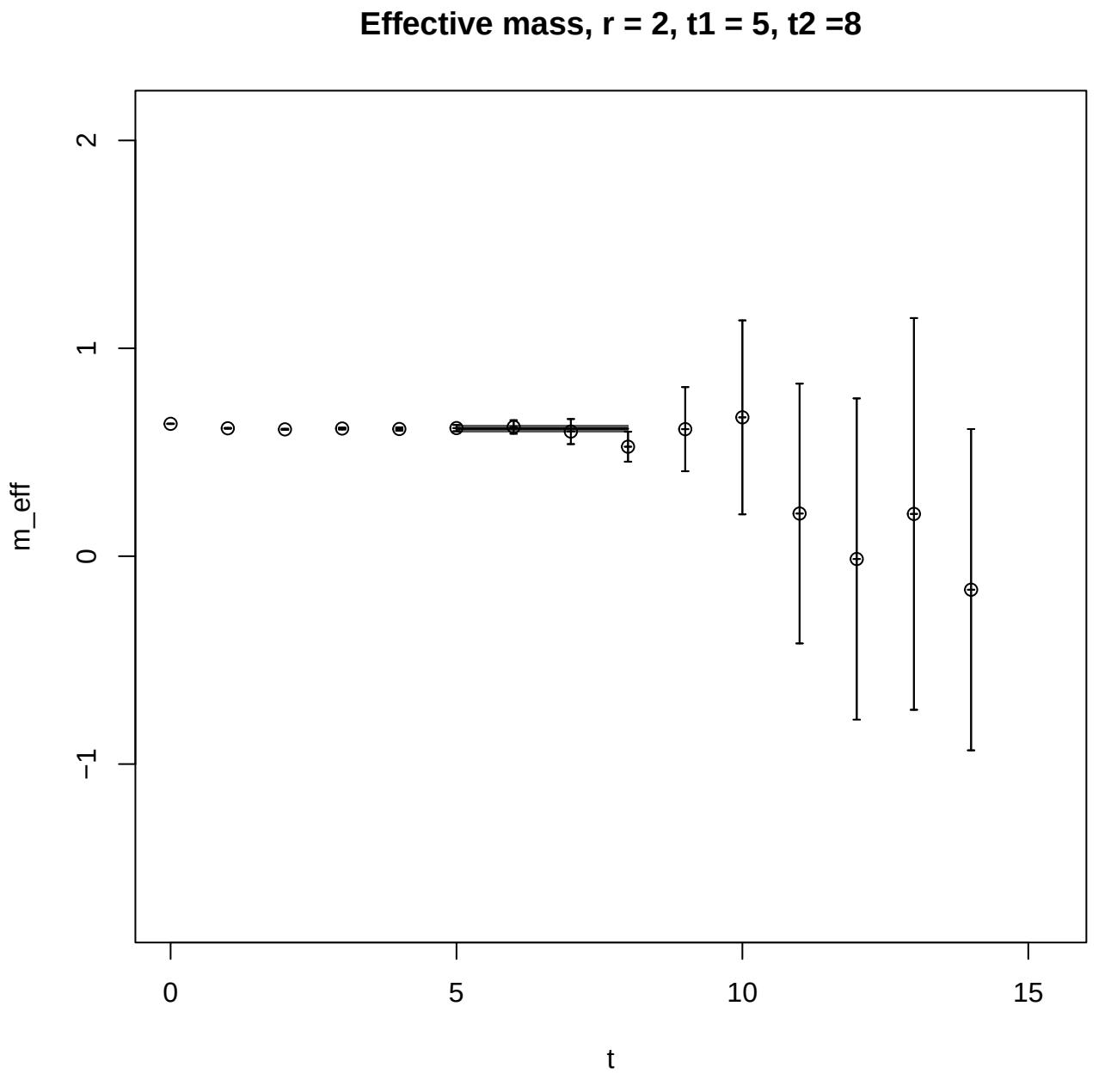

Effective mass, r = 2, t1 = 4, t2 =12
m_eff
2
1
0
−1
0
5
10
15
t


Effective mass, r = 2, t1 = 4, t2 =13
m_eff
2
1
0
−1
0
5
10
15
t


Effective mass, r = 2, t1 = 4, t2 =14
m_eff
t
2
1
0
−1
0
5
10
15


Effective mass, r = 2, t1 = 5, t2 =6
m_eff
2
1
0
−1
0
5
10
15
t


Effective mass, r = 2, t1 = 5, t2 =7
m_eff
2
1
0
−1
0
5
10
15
t


Effective mass, r = 2, t1 = 5, t2 =8
m_eff
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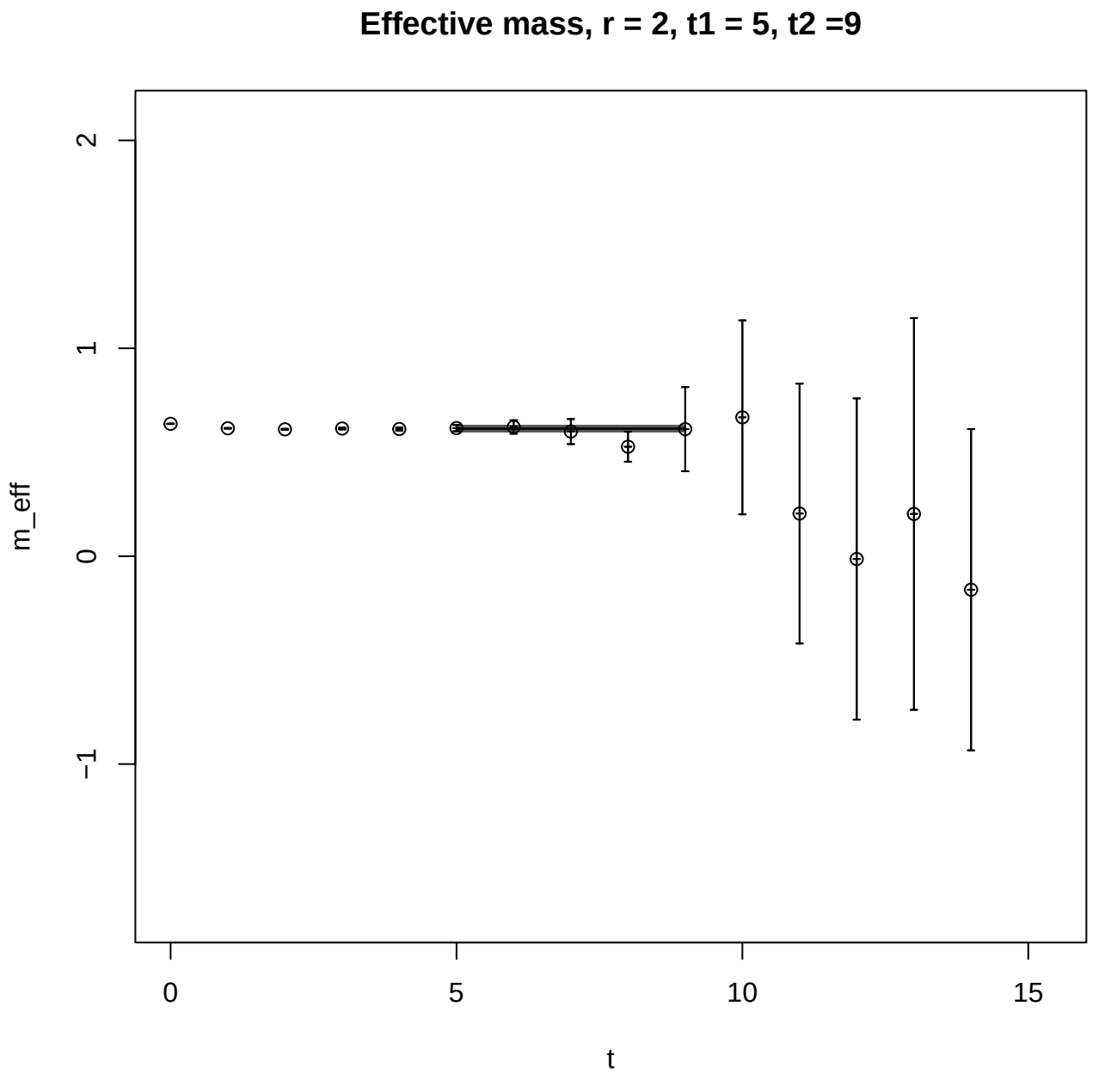
1
0
−1
0
5
10
15
t


Effective mass, r = 2, t1 = 5, t2 =9
m_eff
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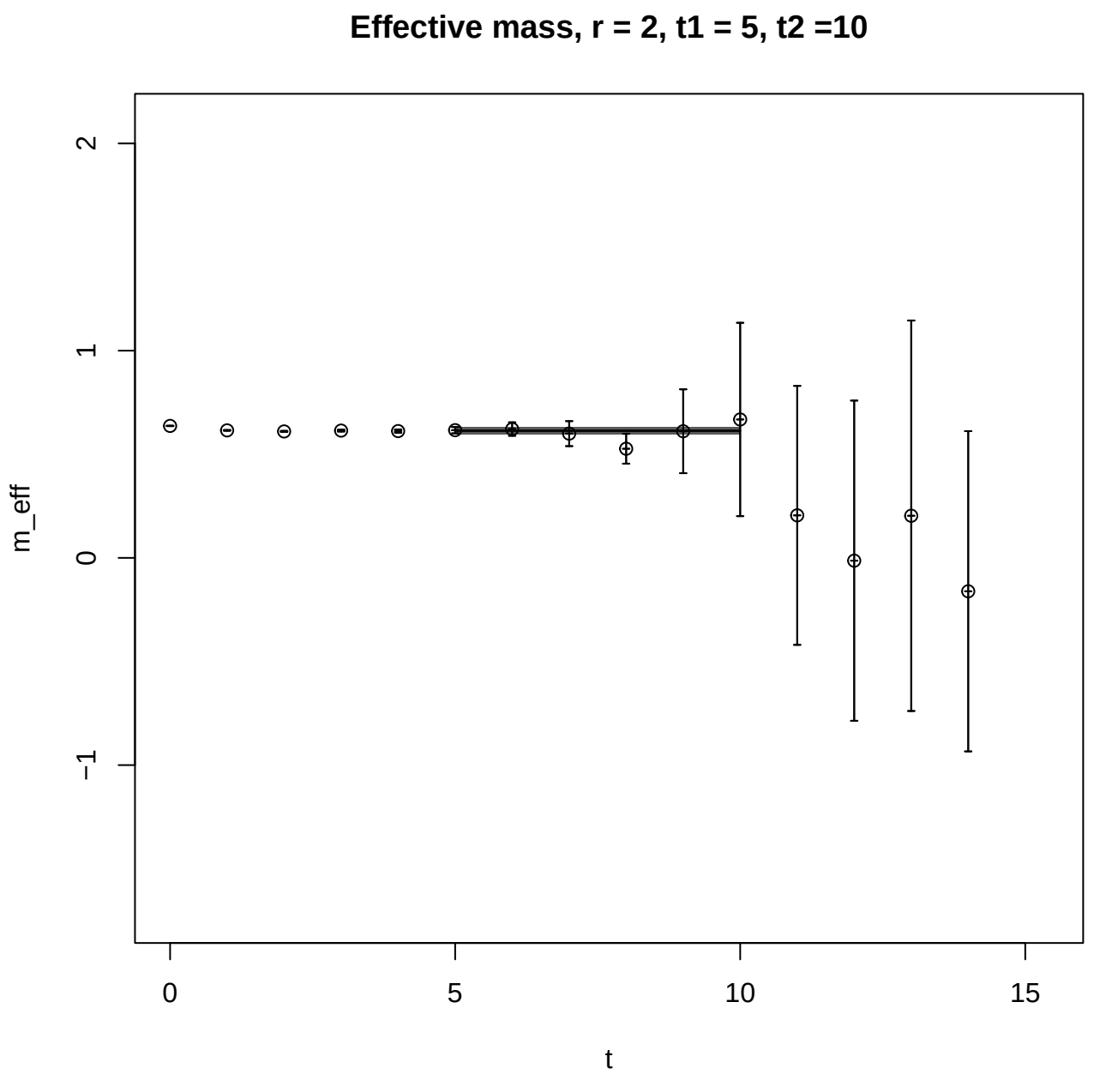
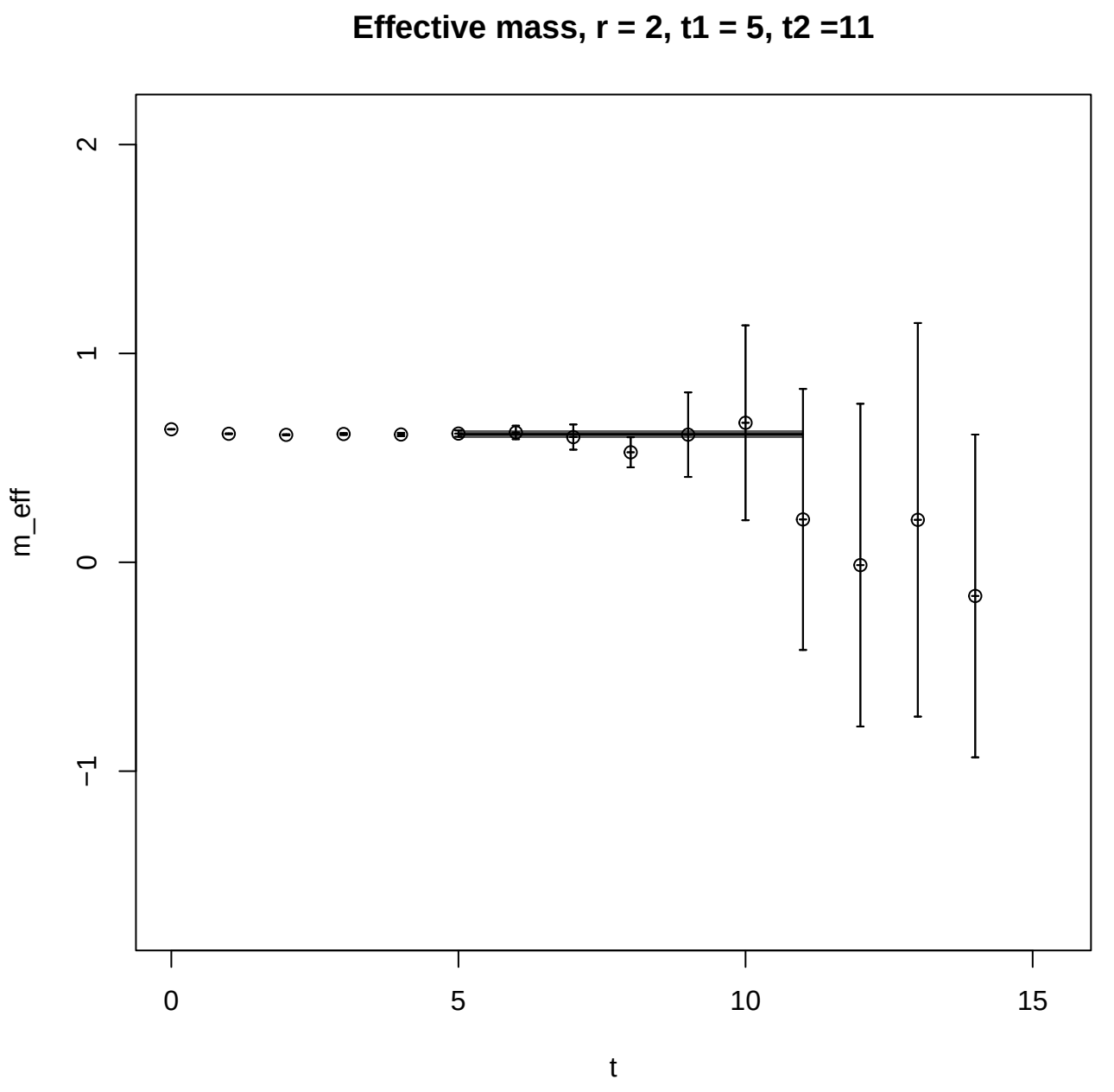
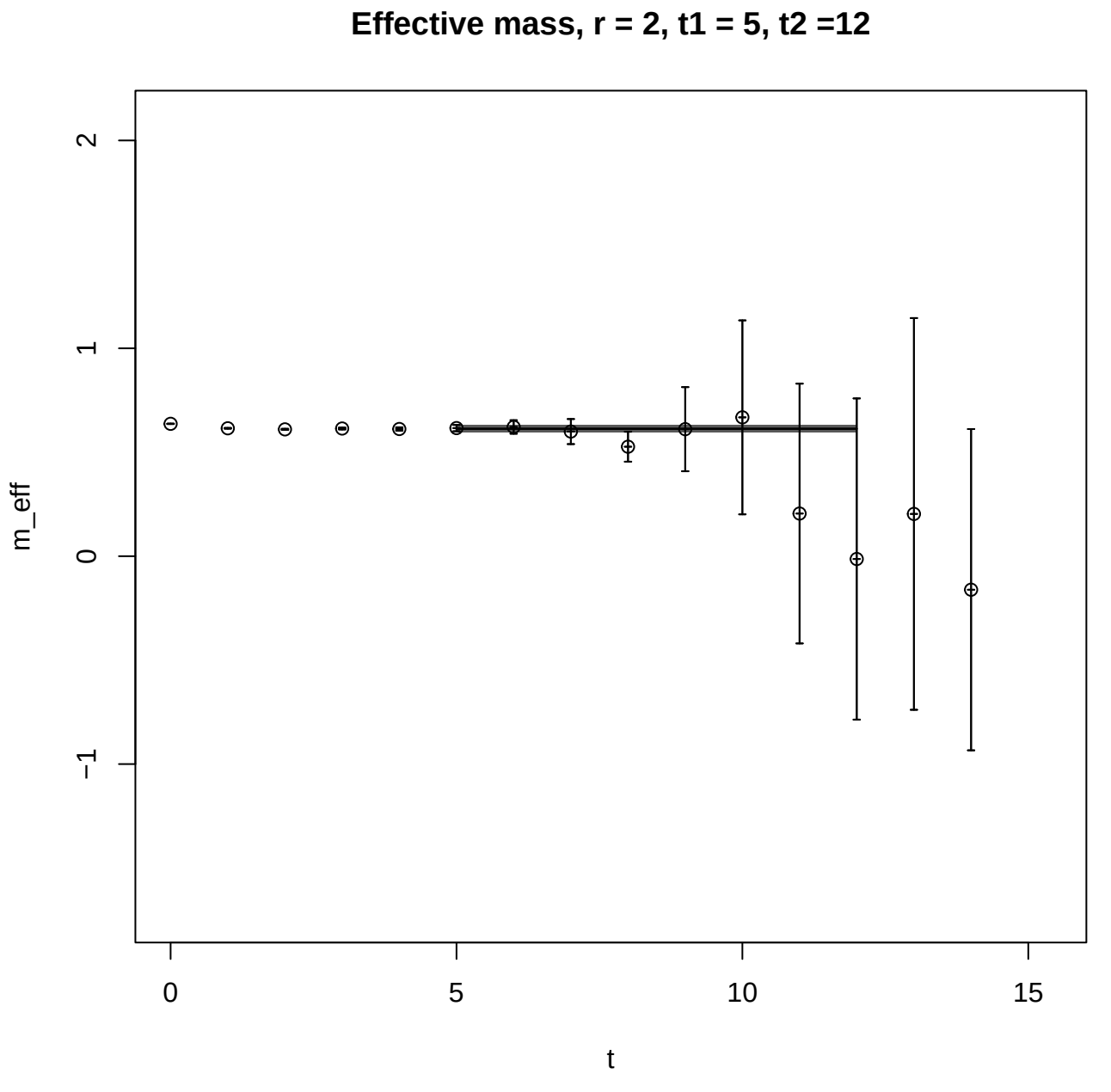
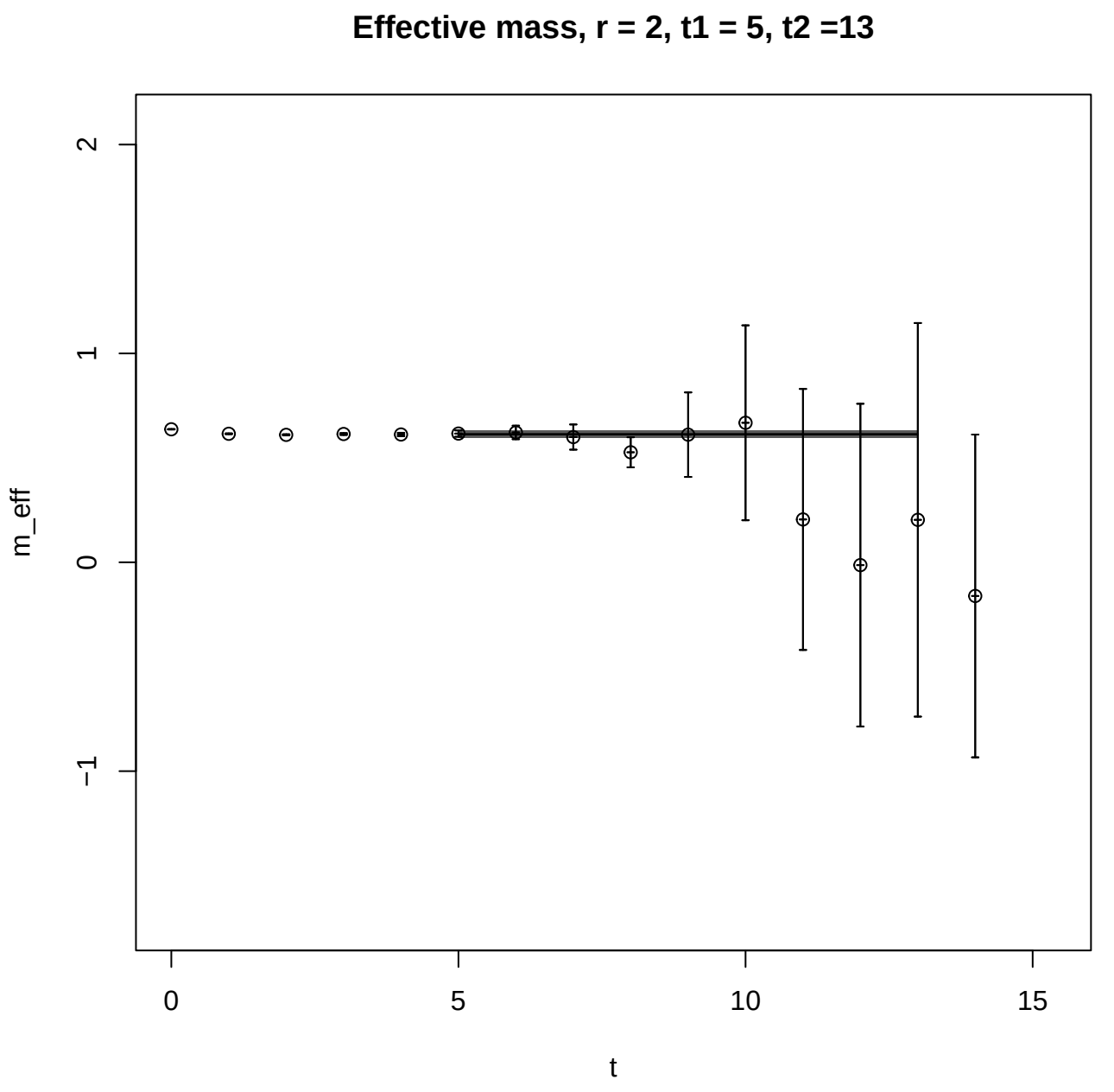
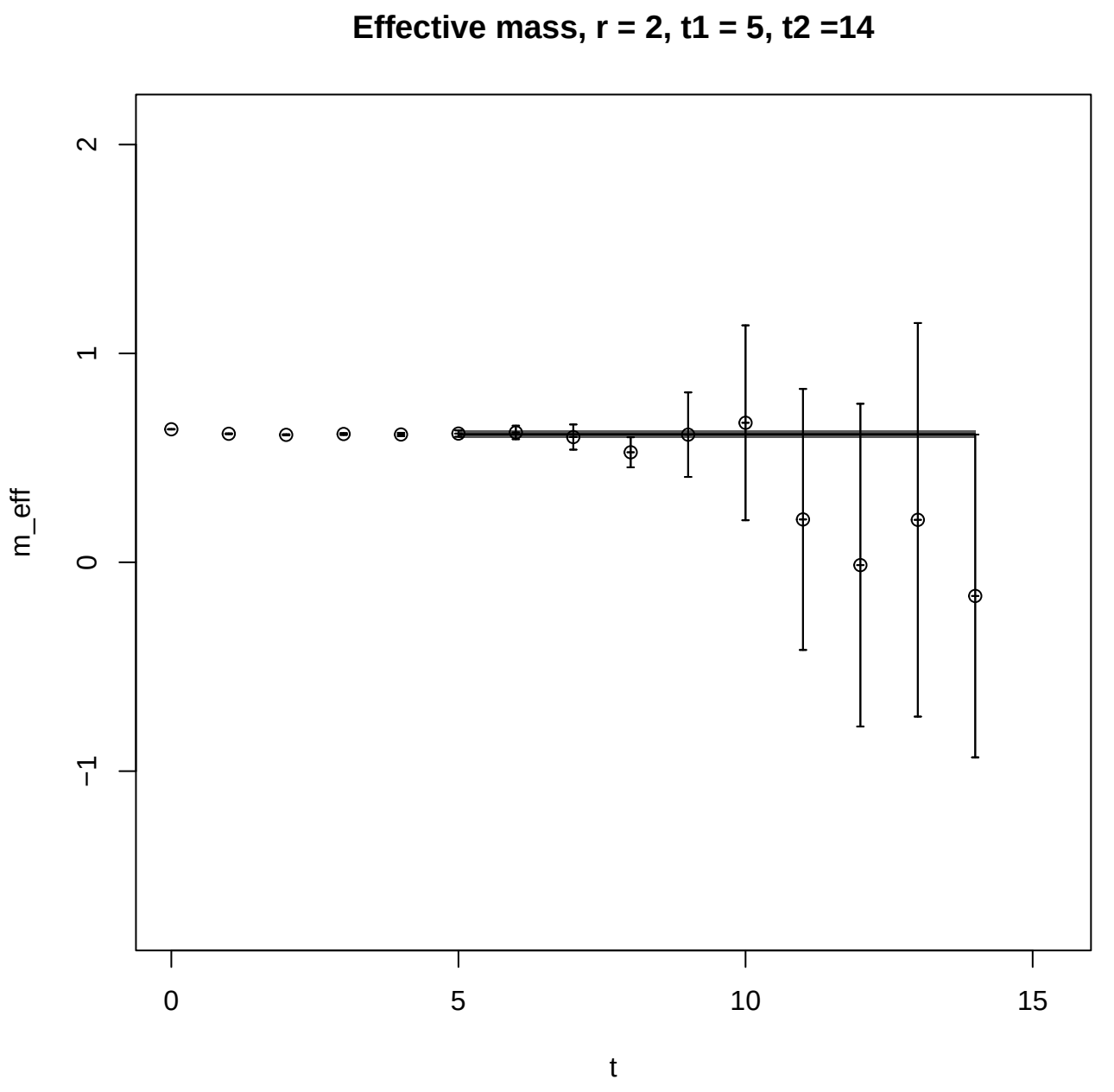
1
0
−1
0
5
10
15
t

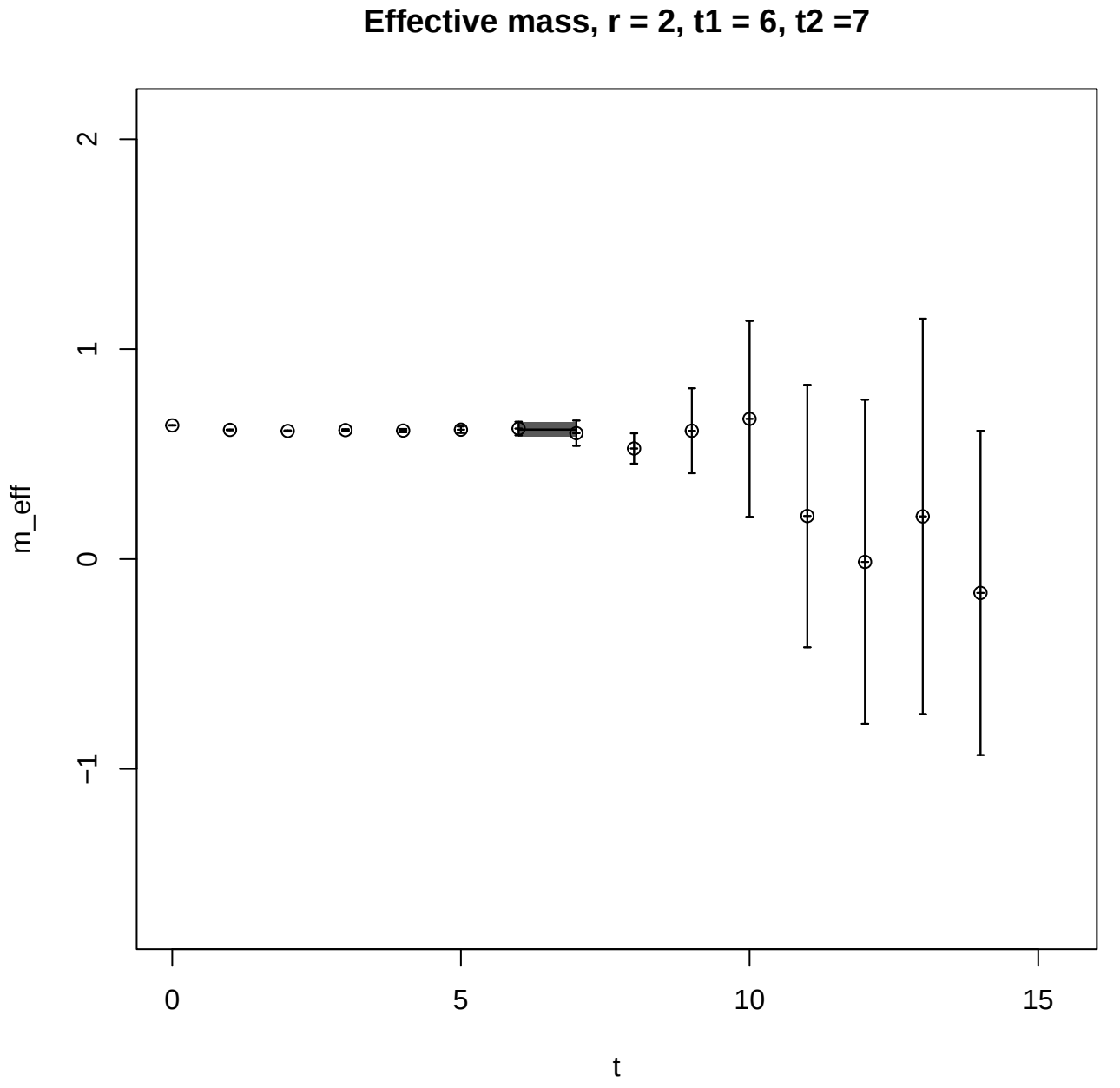

Effective mass, r = 2, t1 = 5, t2 =10
m_eff
2
1
0
−1
0
5
10
15
t

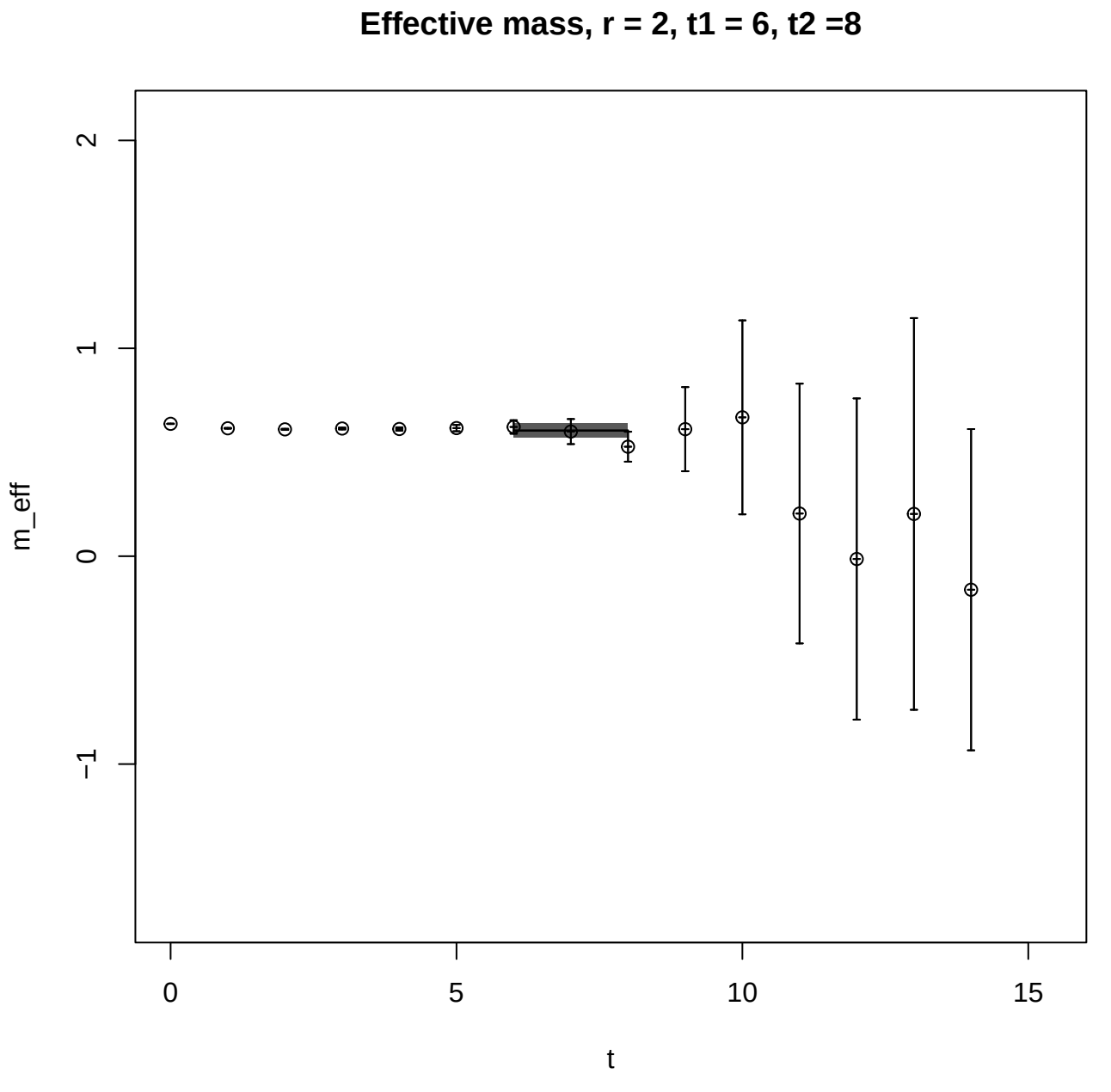

Effective mass, r = 2, t1 = 5, t2 =11
m_eff
t
2
1
0
−1
0
5
10
15

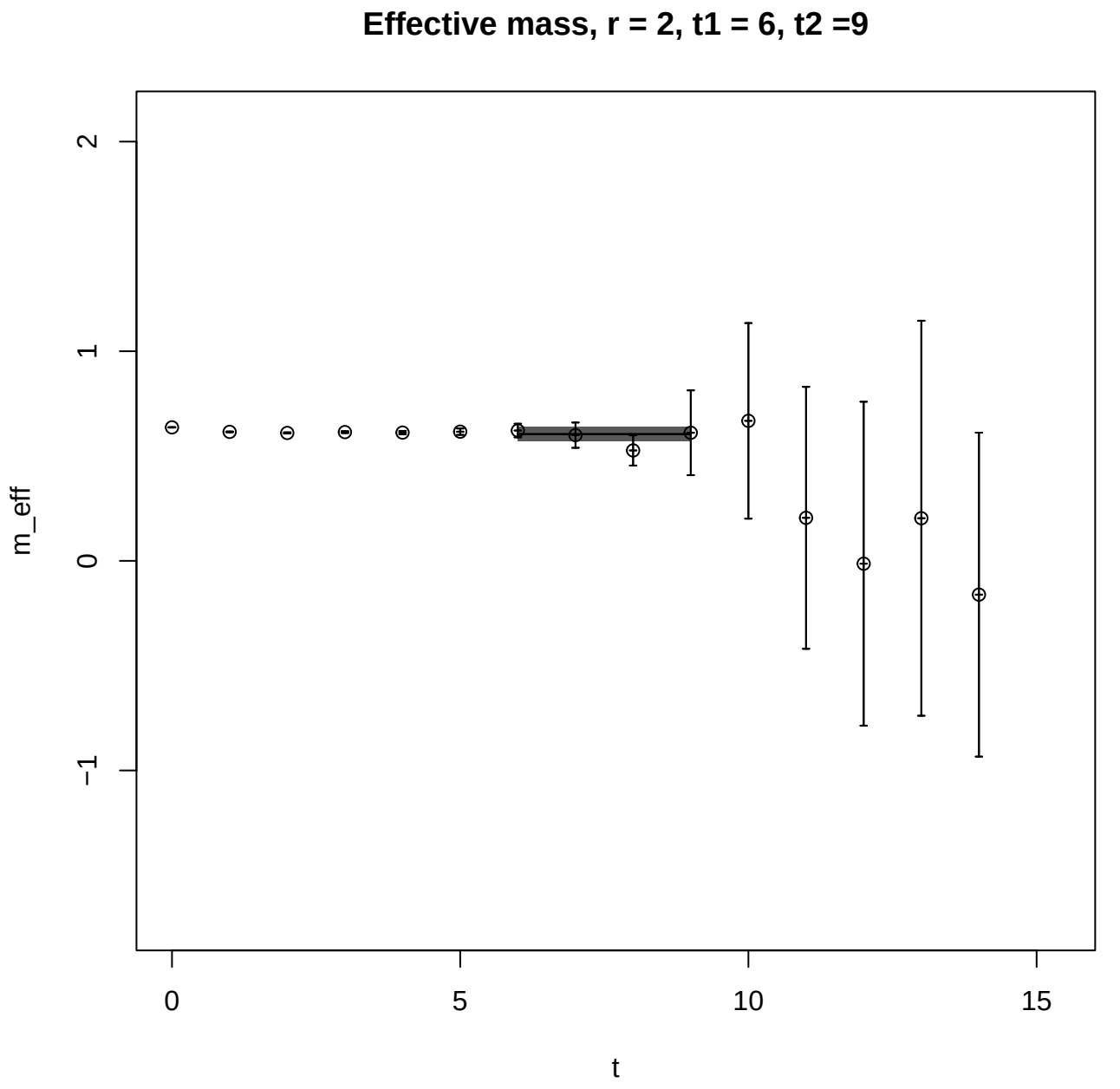

Effective mass, r = 2, t1 = 5, t2 =12
m_eff
t
2
1
0
−1
0
5
10
15

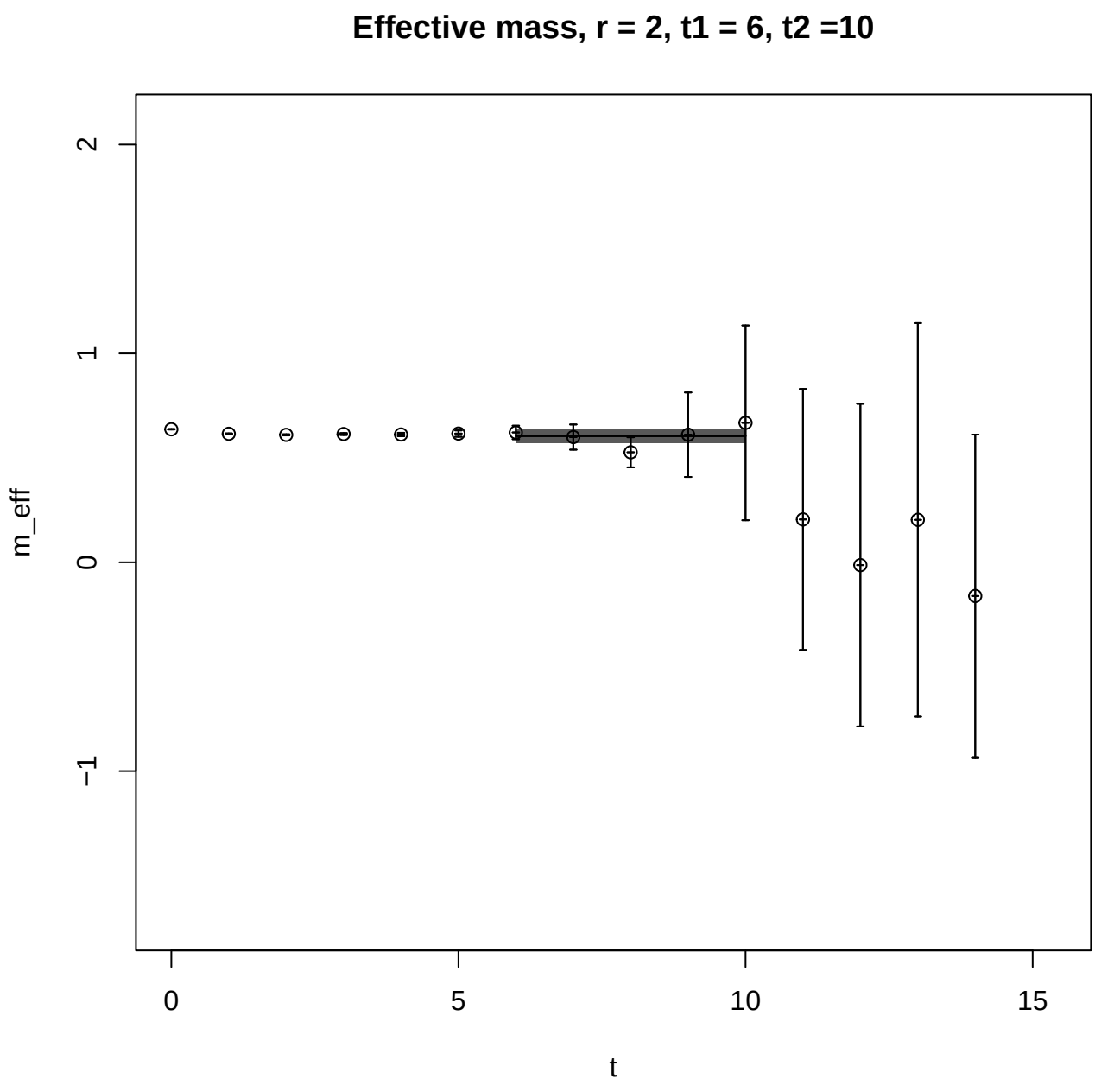

Effective mass, r = 2, t1 = 5, t2 =13
m_eff
2
1
0
−1
0
5
10
15
t

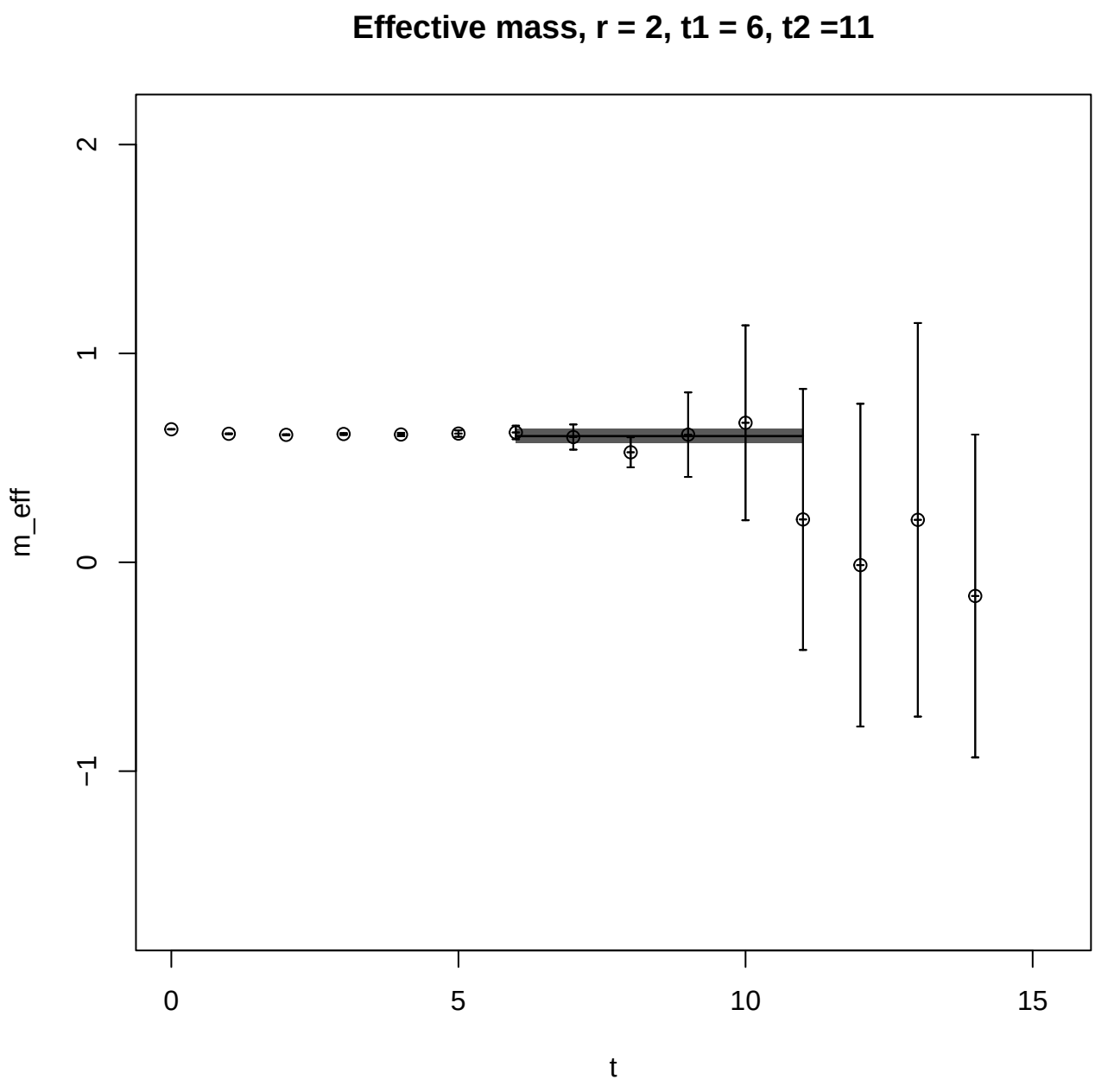

Effective mass, r = 2, t1 = 5, t2 =14
m_eff
t
2
1
0
−1
0
5
10
15

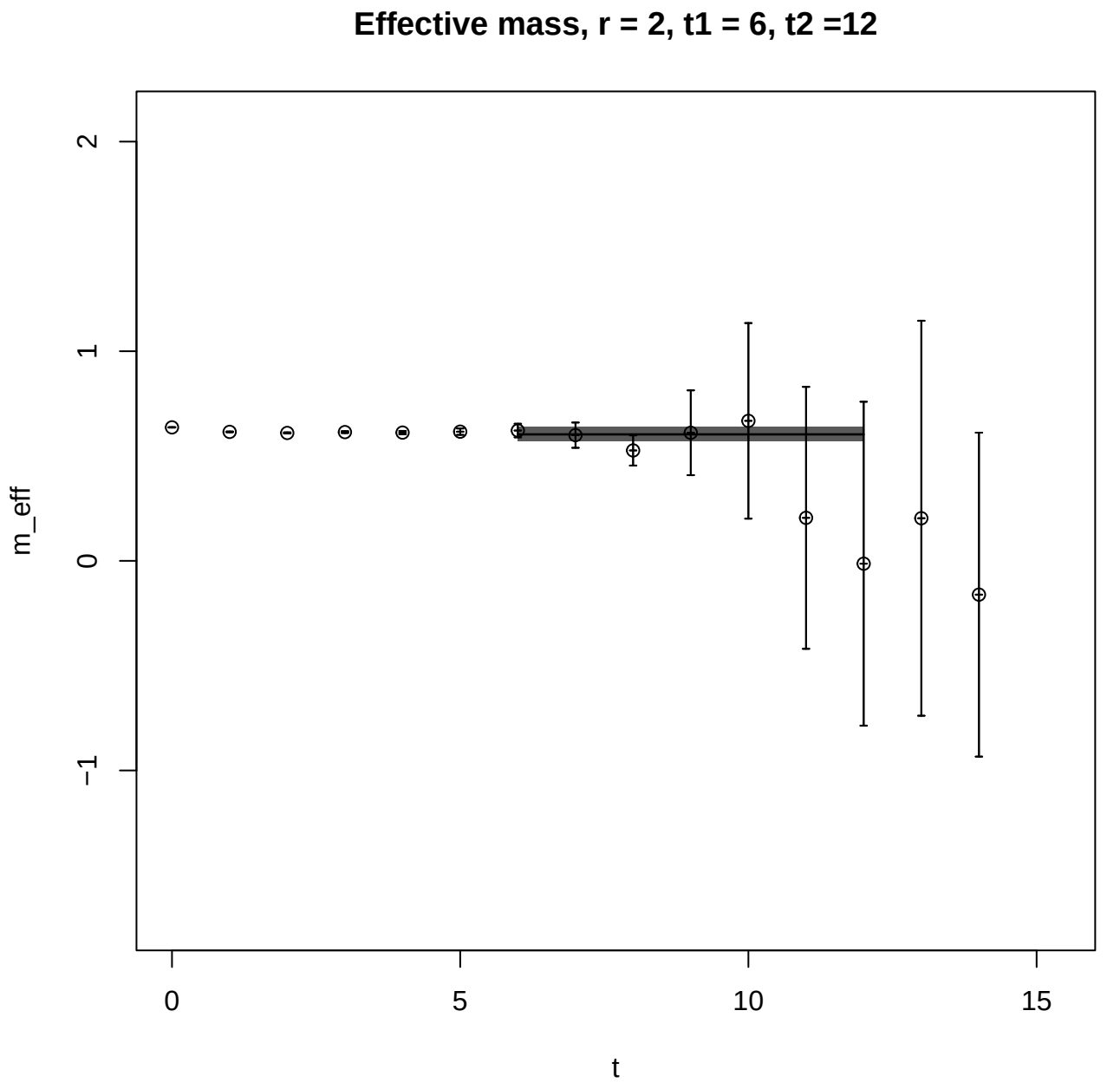

Effective mass, r = 2, t1 = 6, t2 =7
m_eff
2
1
0
−1
0
5
10
15
t

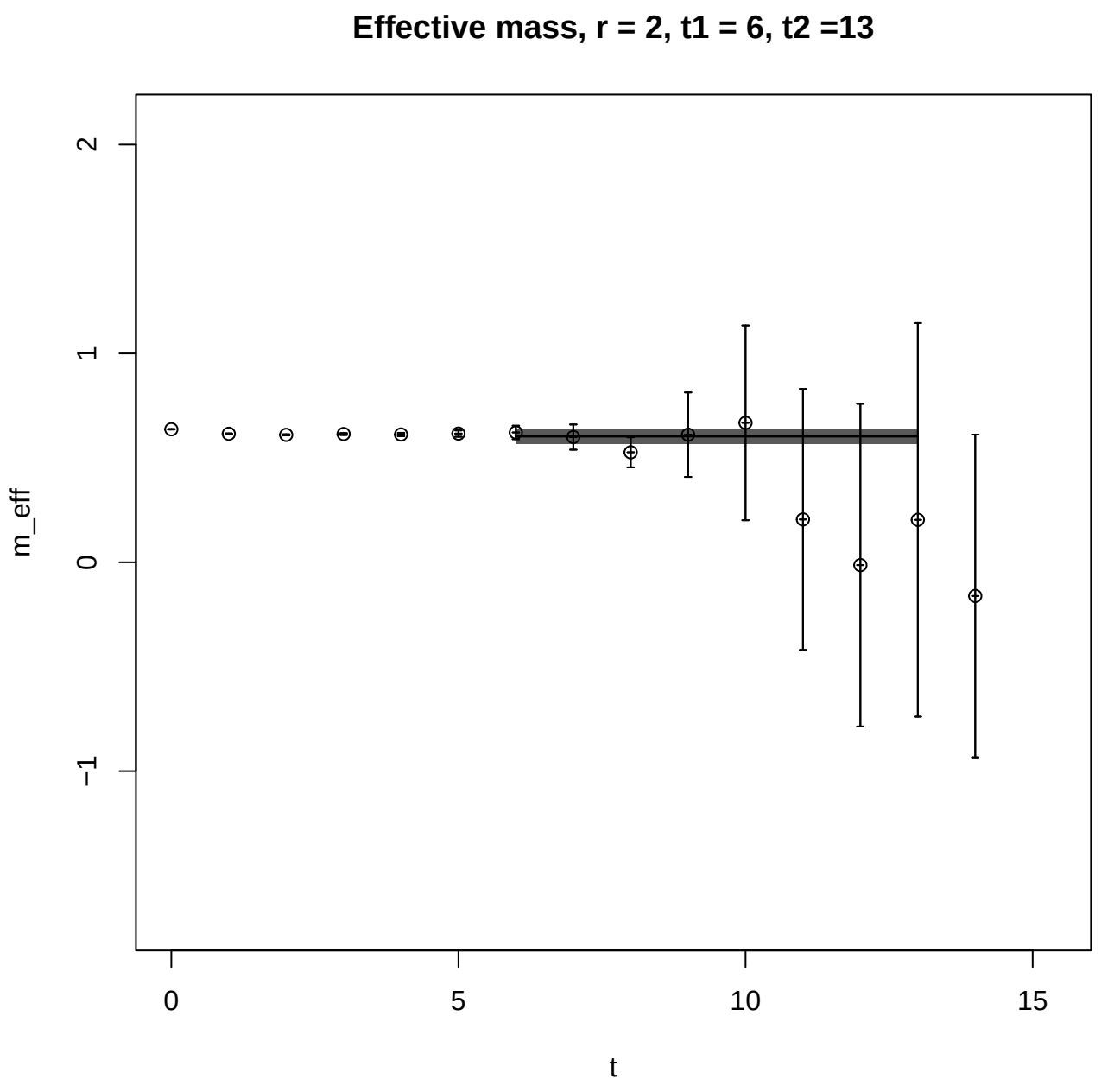

Effective mass, r = 2, t1 = 6, t2 =8
m_eff
2
1
0
−1
0
5
10
15
t

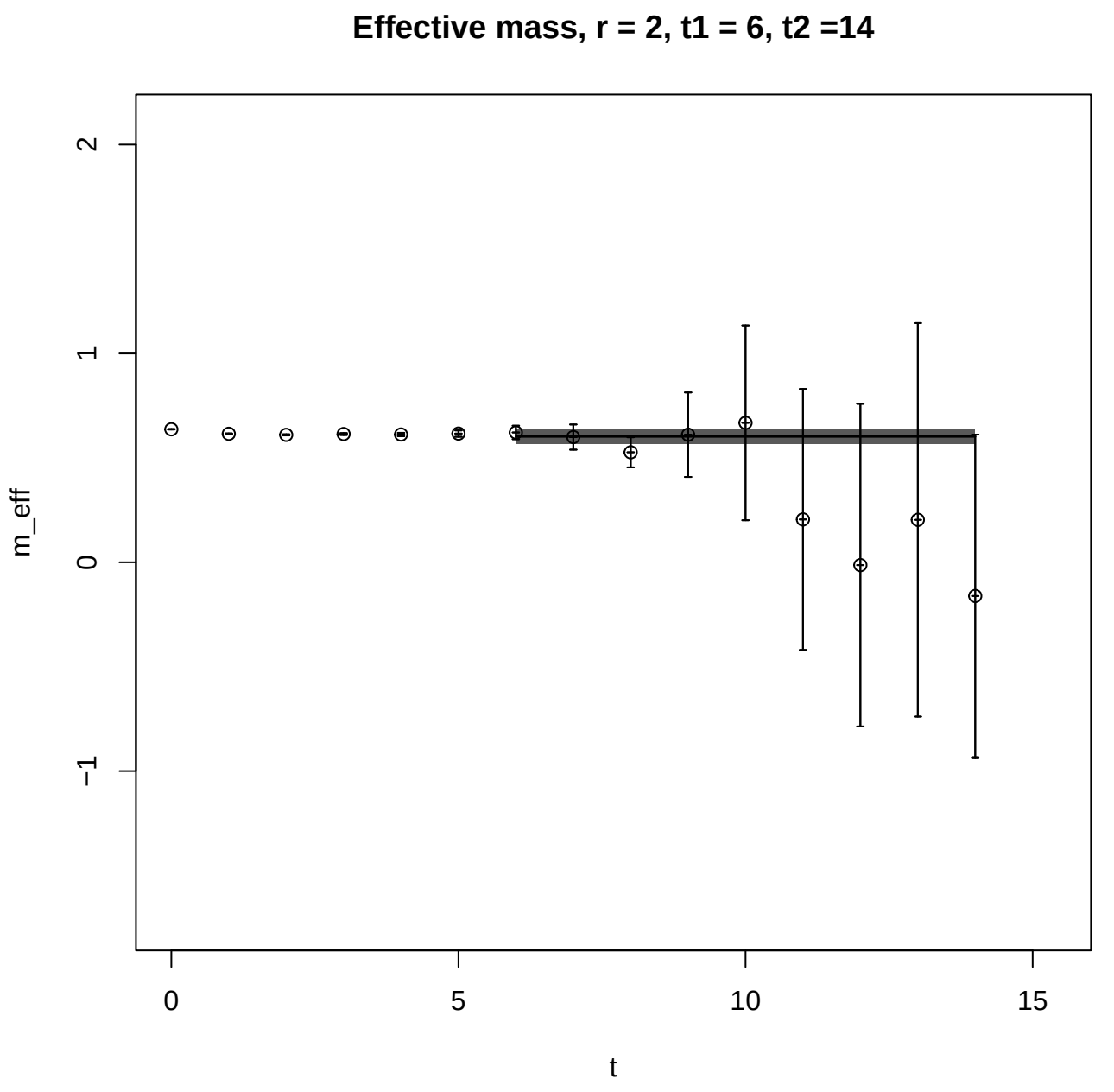

Effective mass, r = 2, t1 = 6, t2 =9
m_eff
2
1
0
−1
0
5
10
15
t


Effective mass, r = 2, t1 = 6, t2 =10
m_eff
t
2
1
0
−1
0
5
10
15


Effective mass, r = 2, t1 = 6, t2 =11
m_eff
t
2
1
0
−1
0
5
10
15


Effective mass, r = 2, t1 = 6, t2 =12
m_eff
2
1
0
−1
0
5
10
15
t


Effective mass, r = 2, t1 = 6, t2 =13
m_eff
2
1
0
−1
0
5
10
15
t


Effective mass, r = 2, t1 = 6, t2 =14
m_eff
t
2
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0
−1
0
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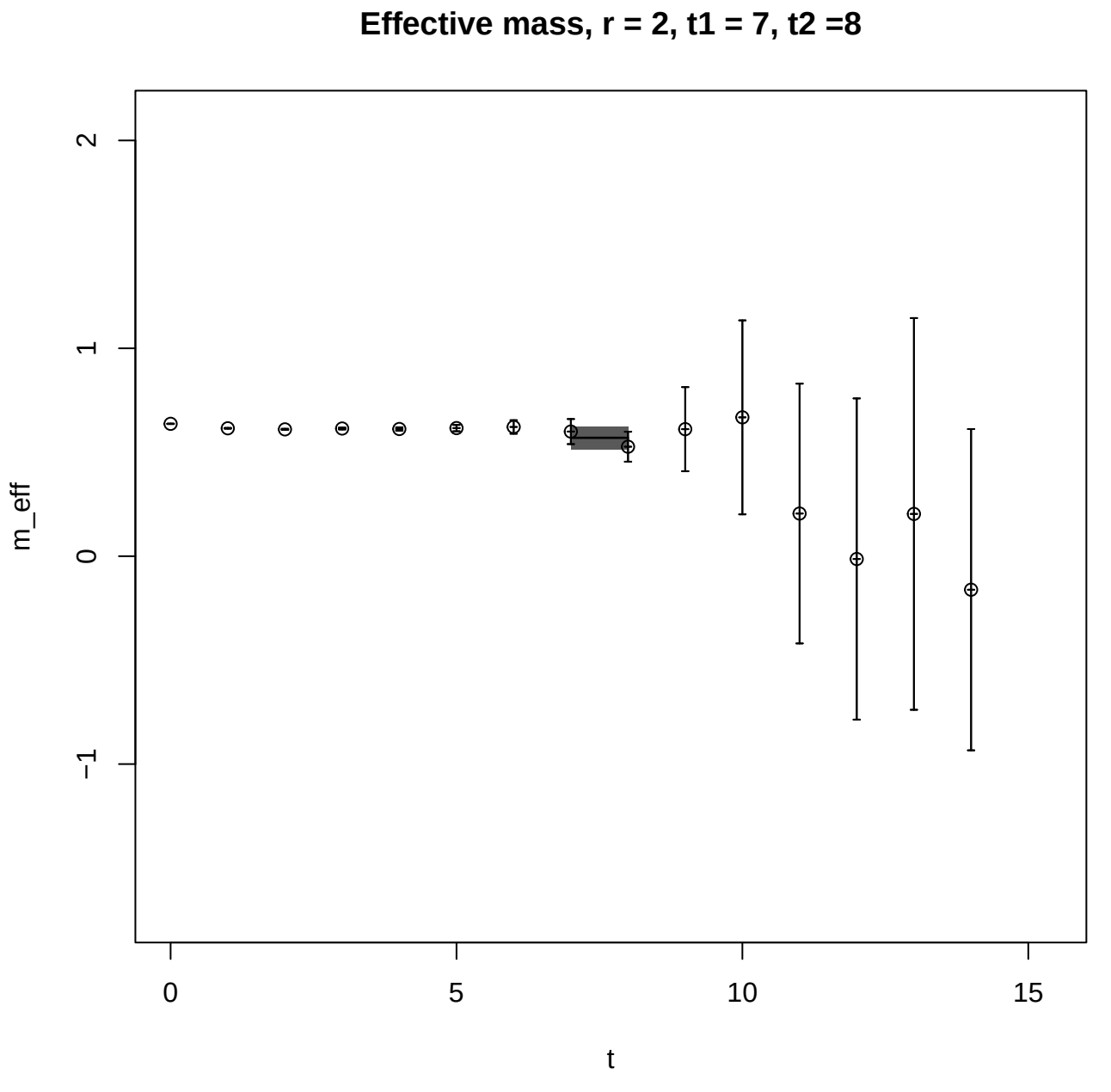
10
15


Effective mass, r = 2, t1 = 7, t2 =8
m_eff
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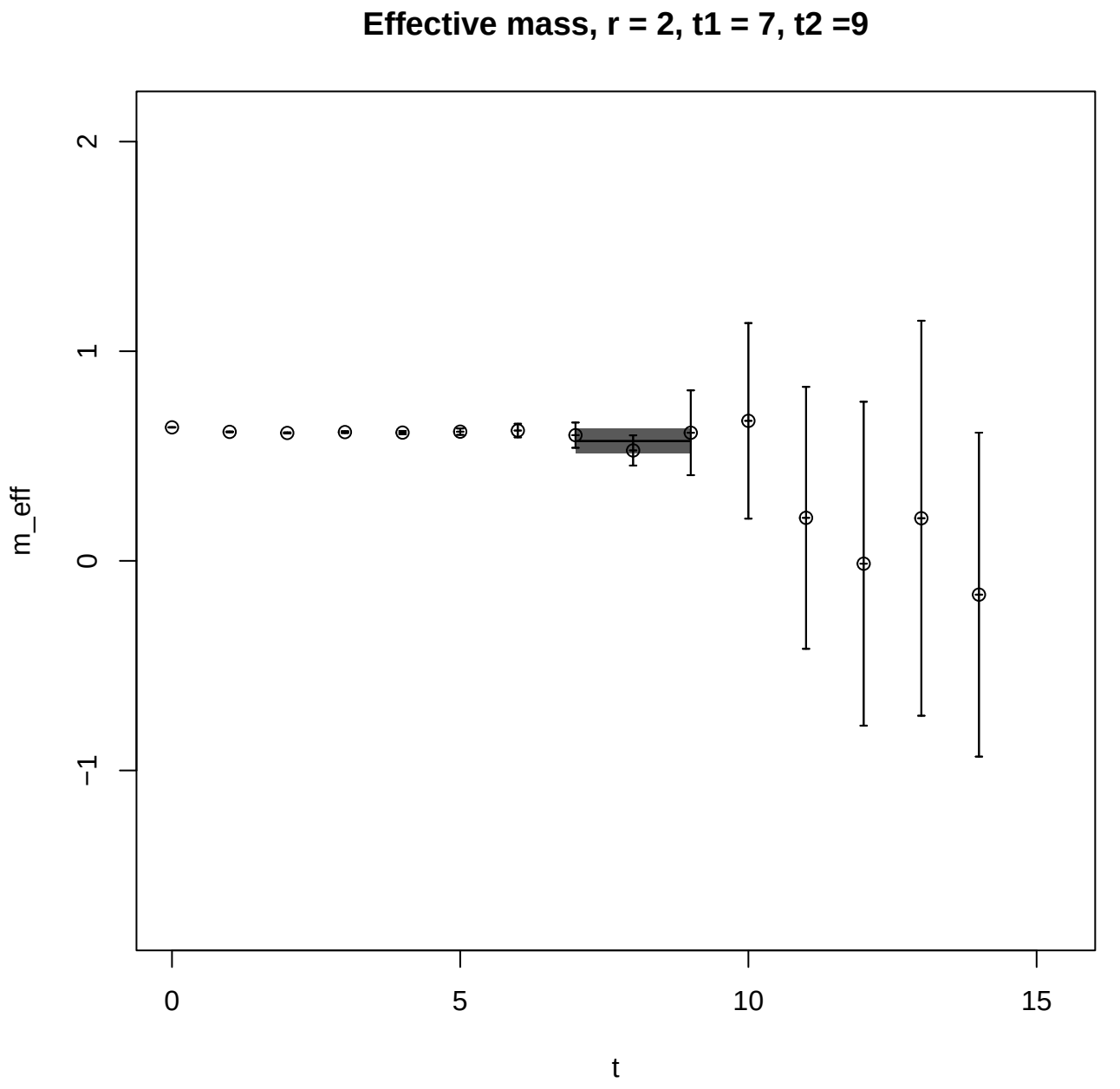
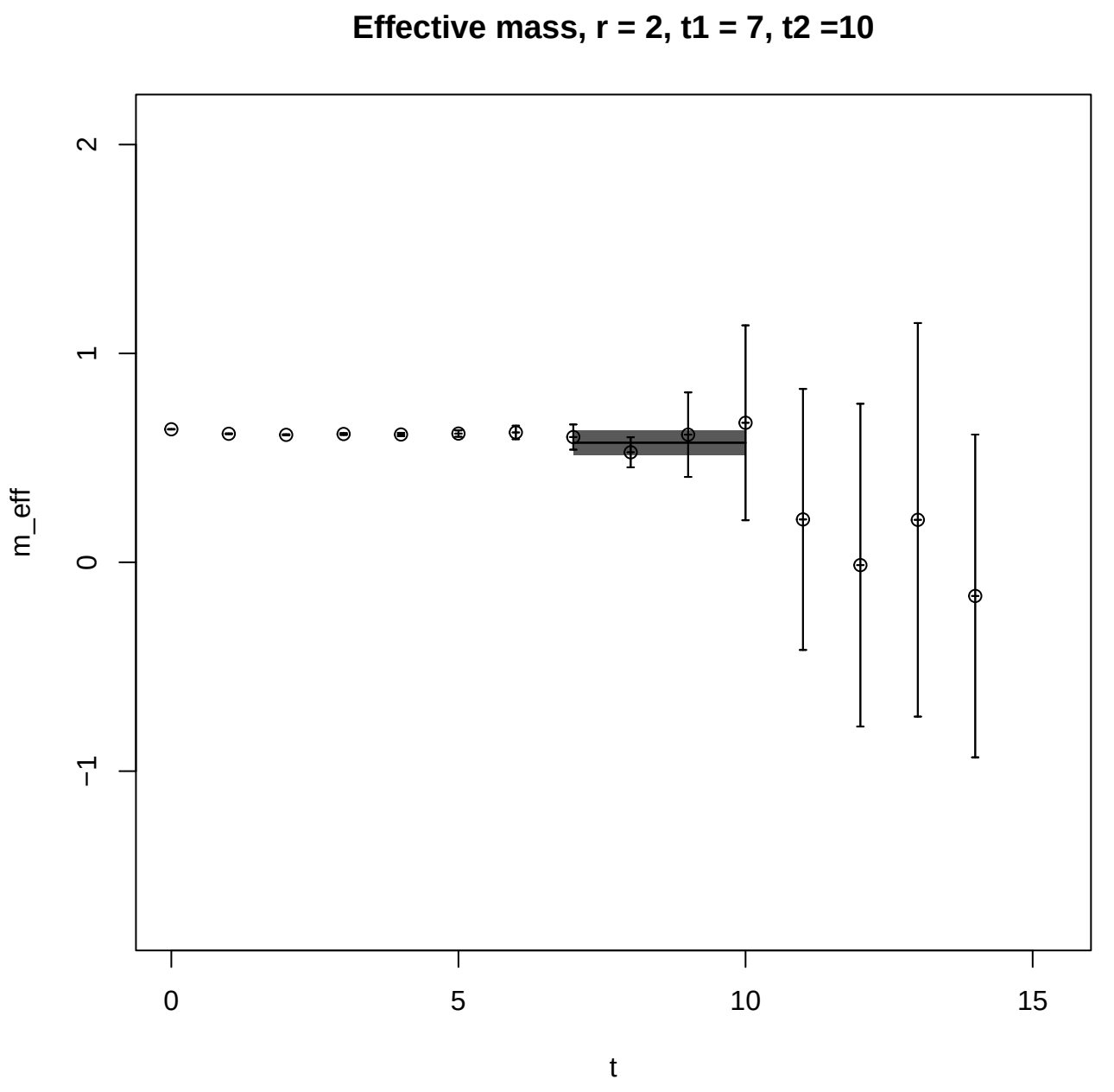
1
0
−1
0
5
10
15
t


Effective mass, r = 2, t1 = 7, t2 =9
m_eff
2
1
0
−1
0
5
10
15
t


Effective mass, r = 2, t1 = 7, t2 =10
m_eff
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1
0
−1
0
5
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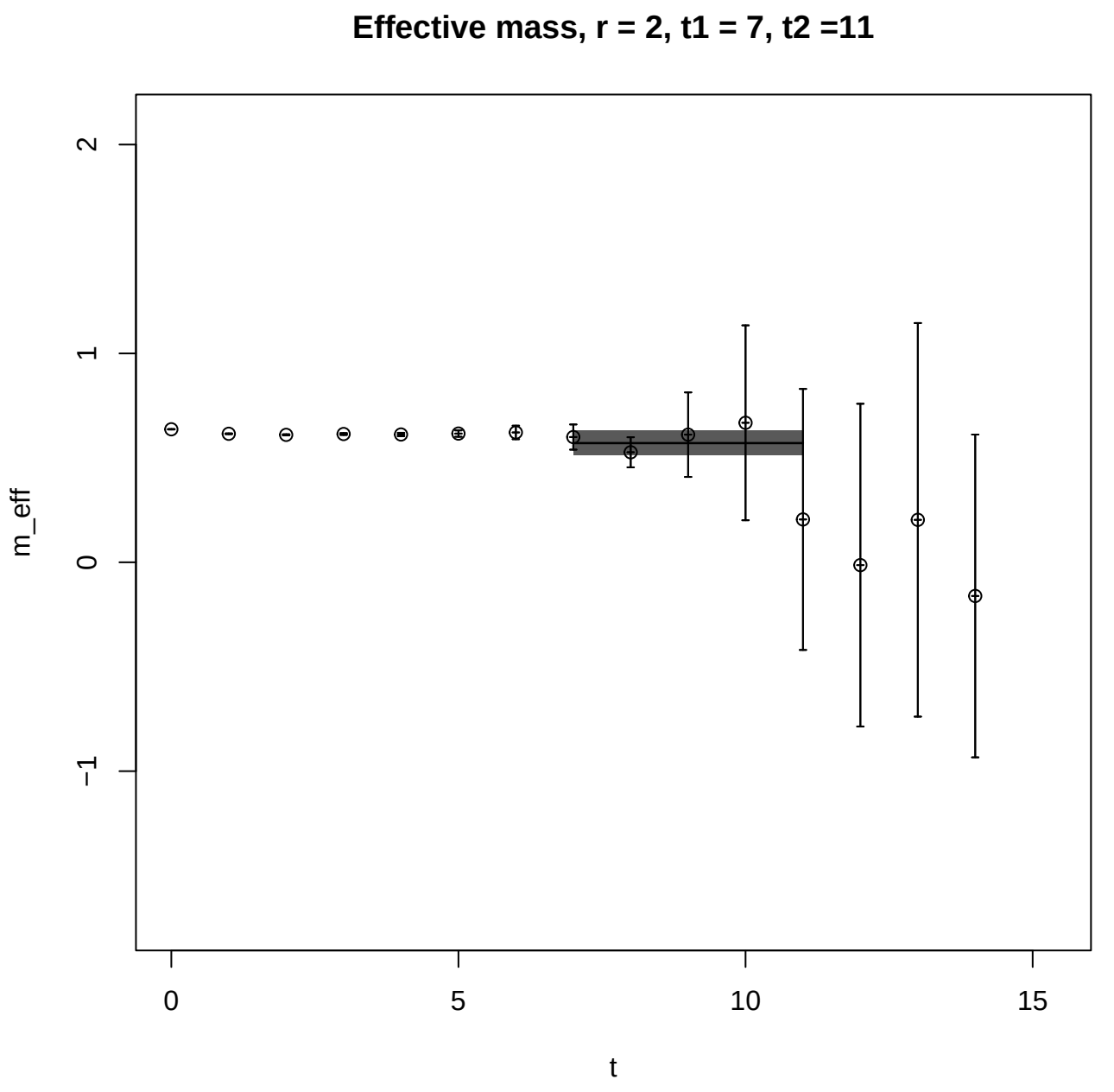
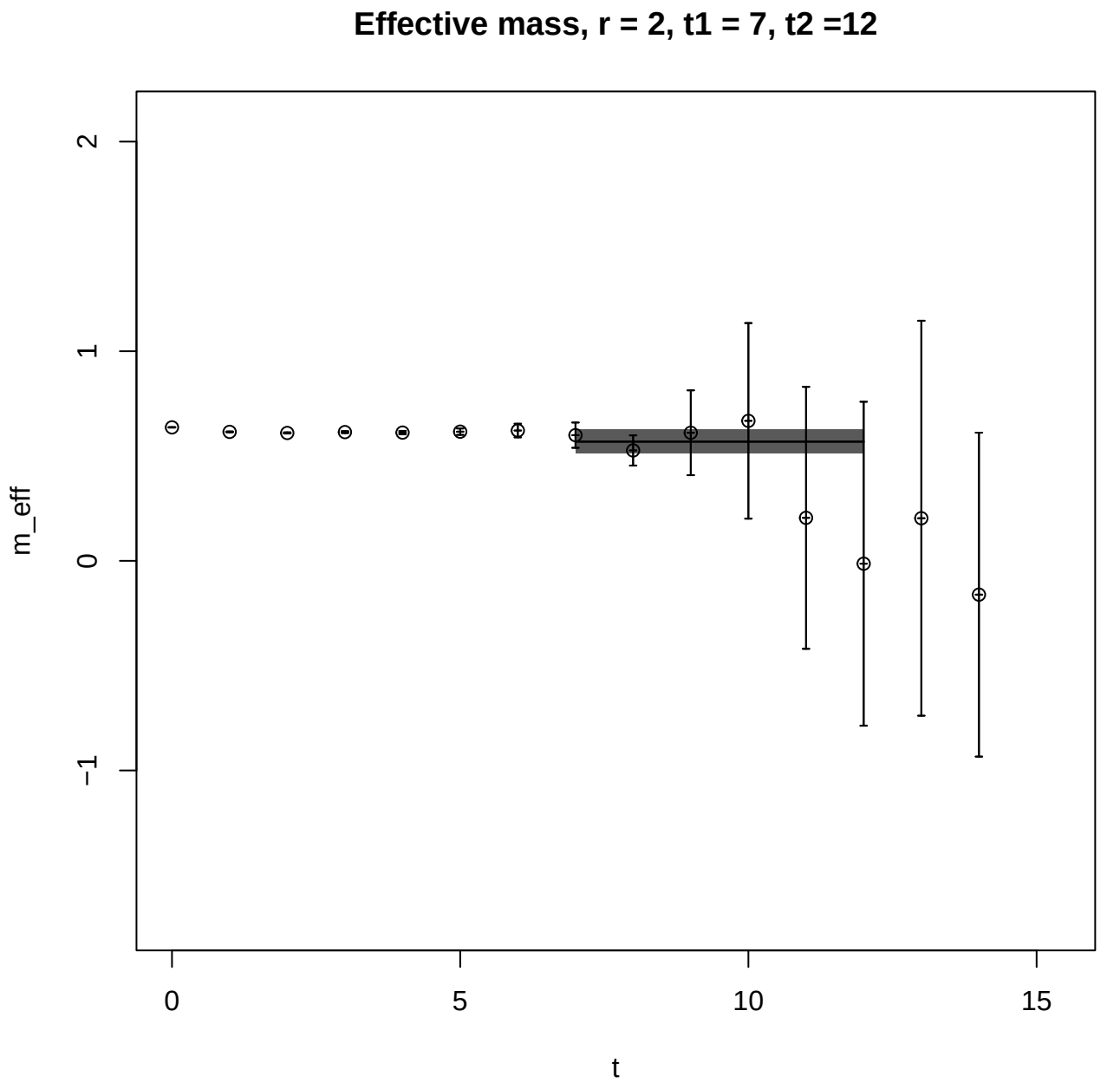
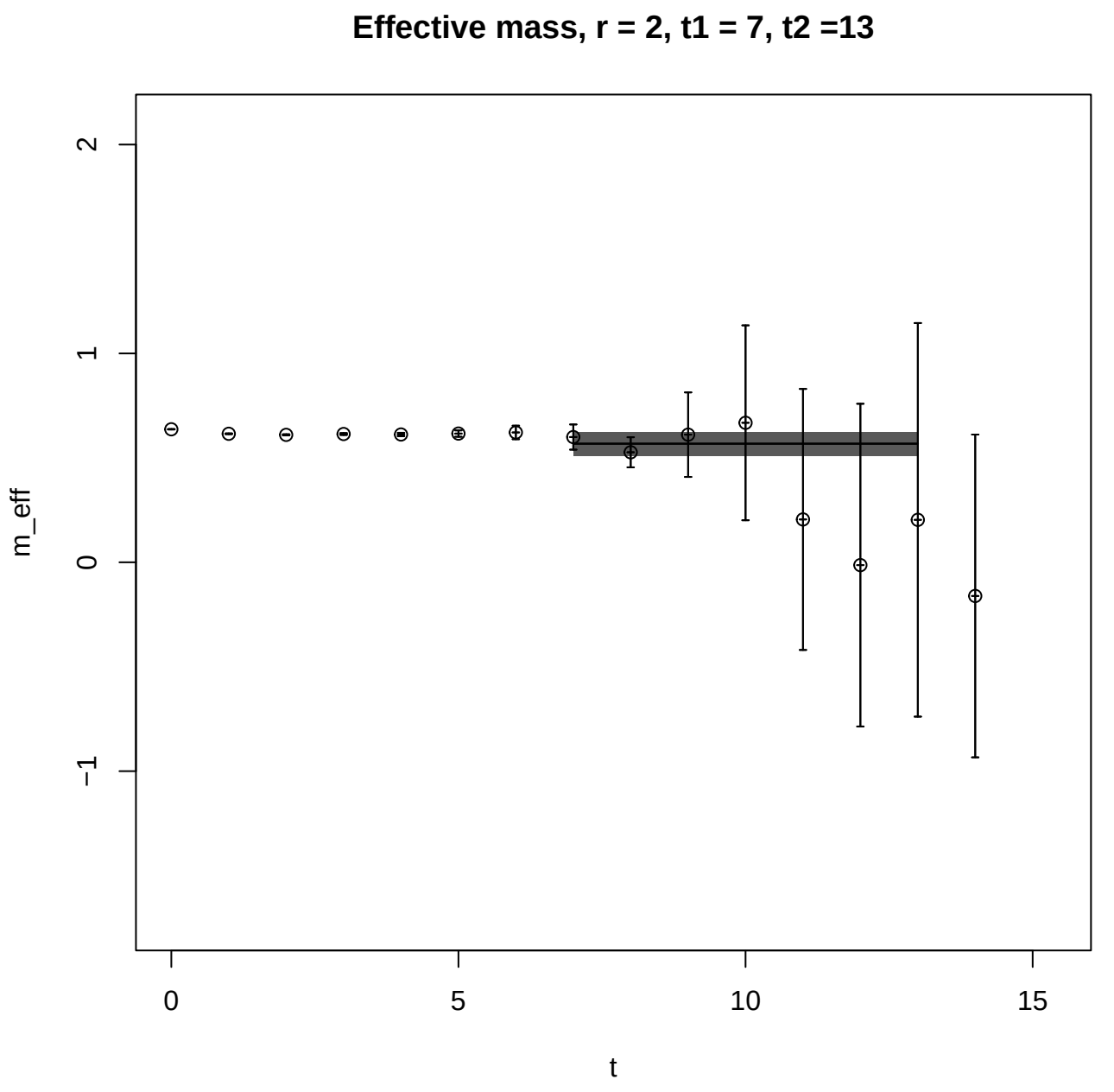
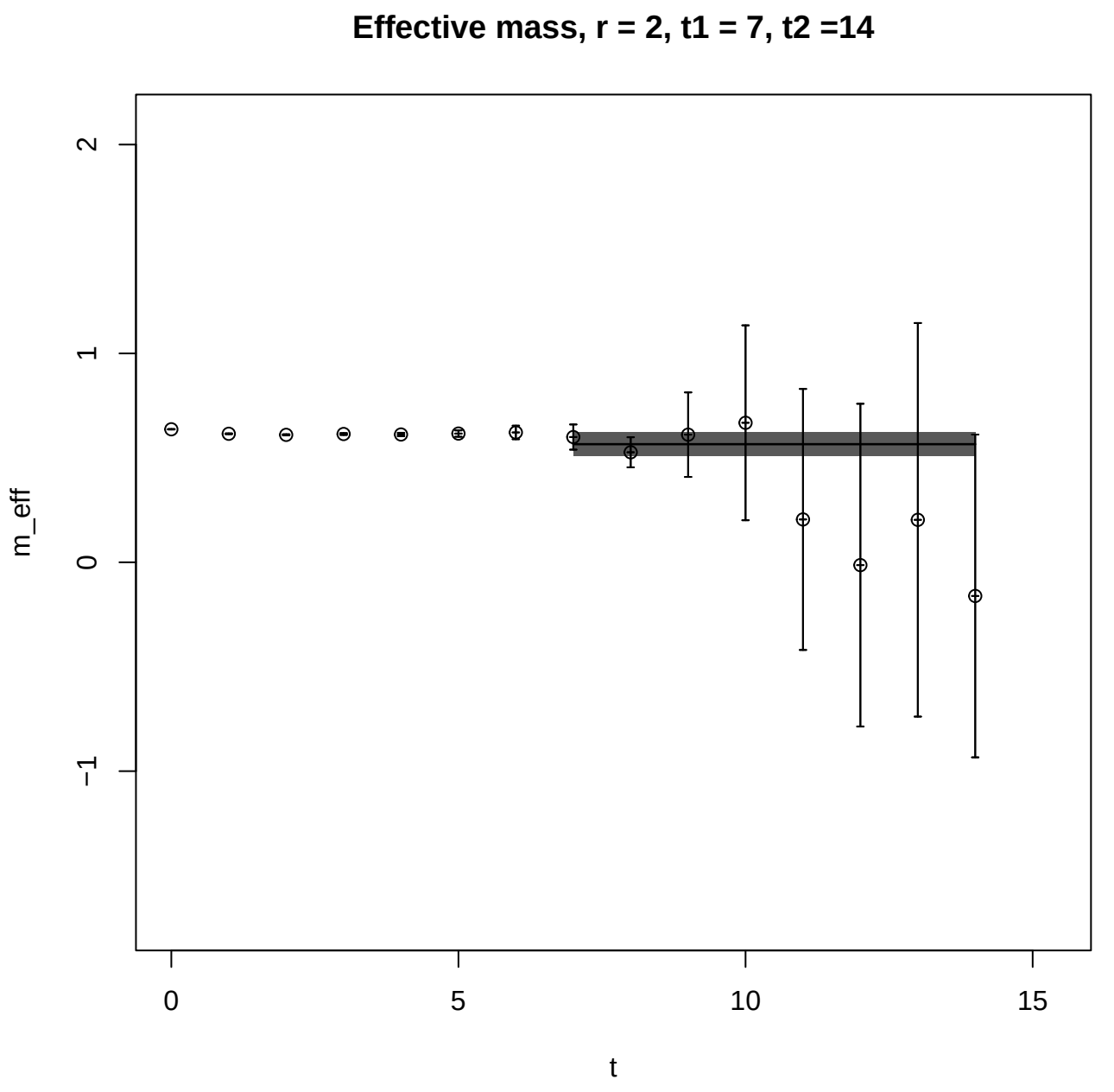
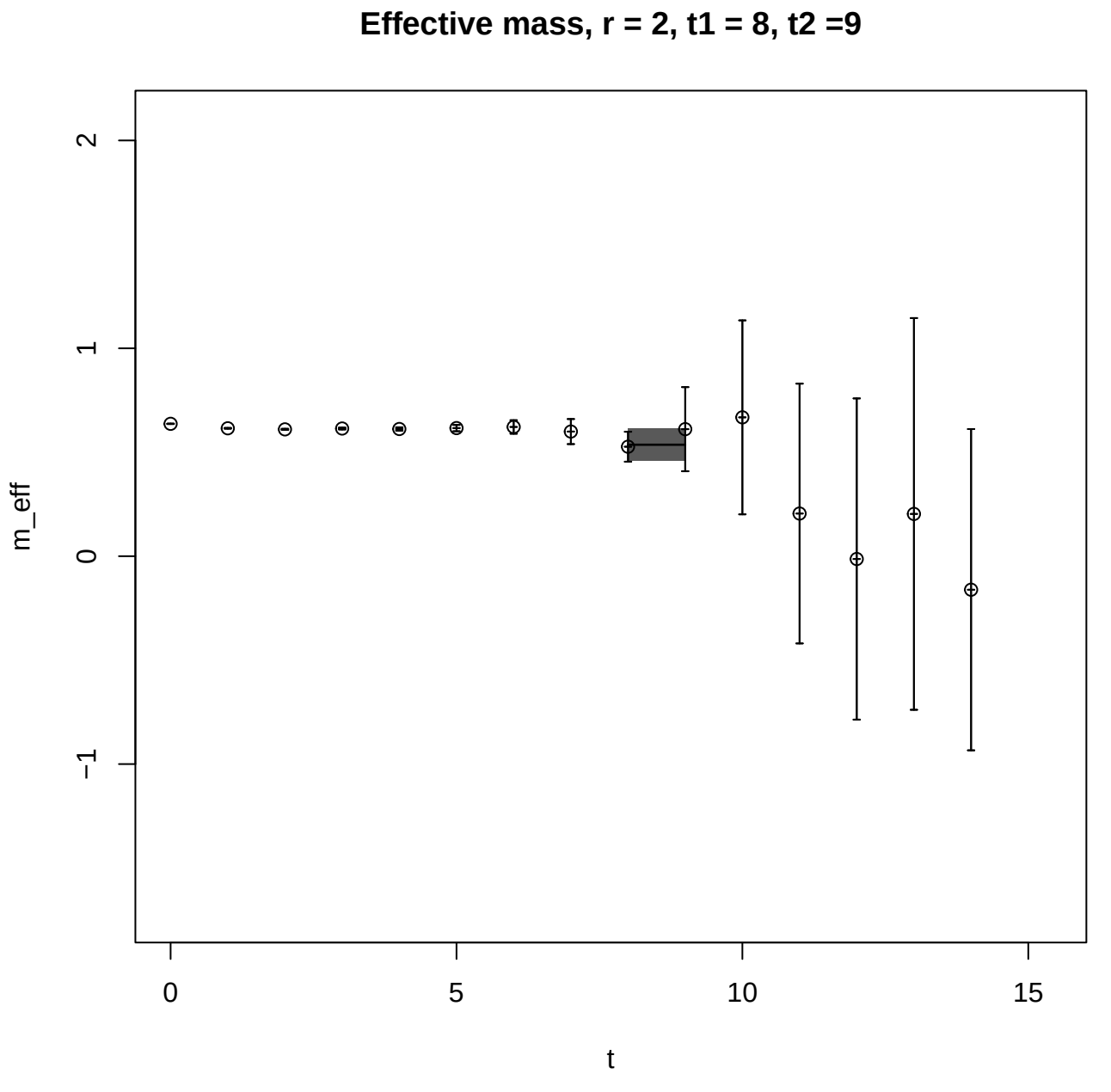
15
t

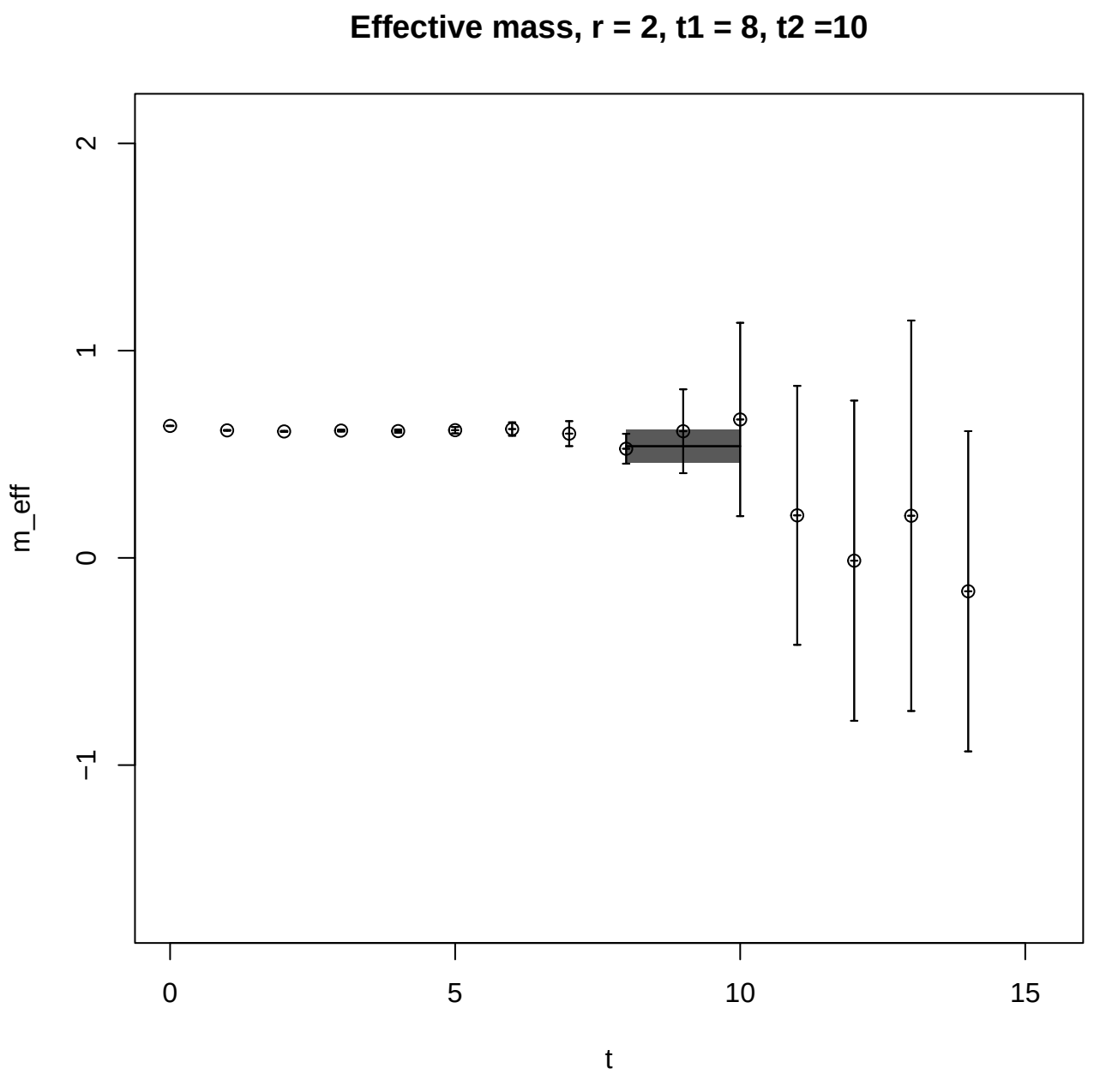

Effective mass, r = 2, t1 = 7, t2 =11
m_eff
2
1
0
−1
0
5
10
15
t

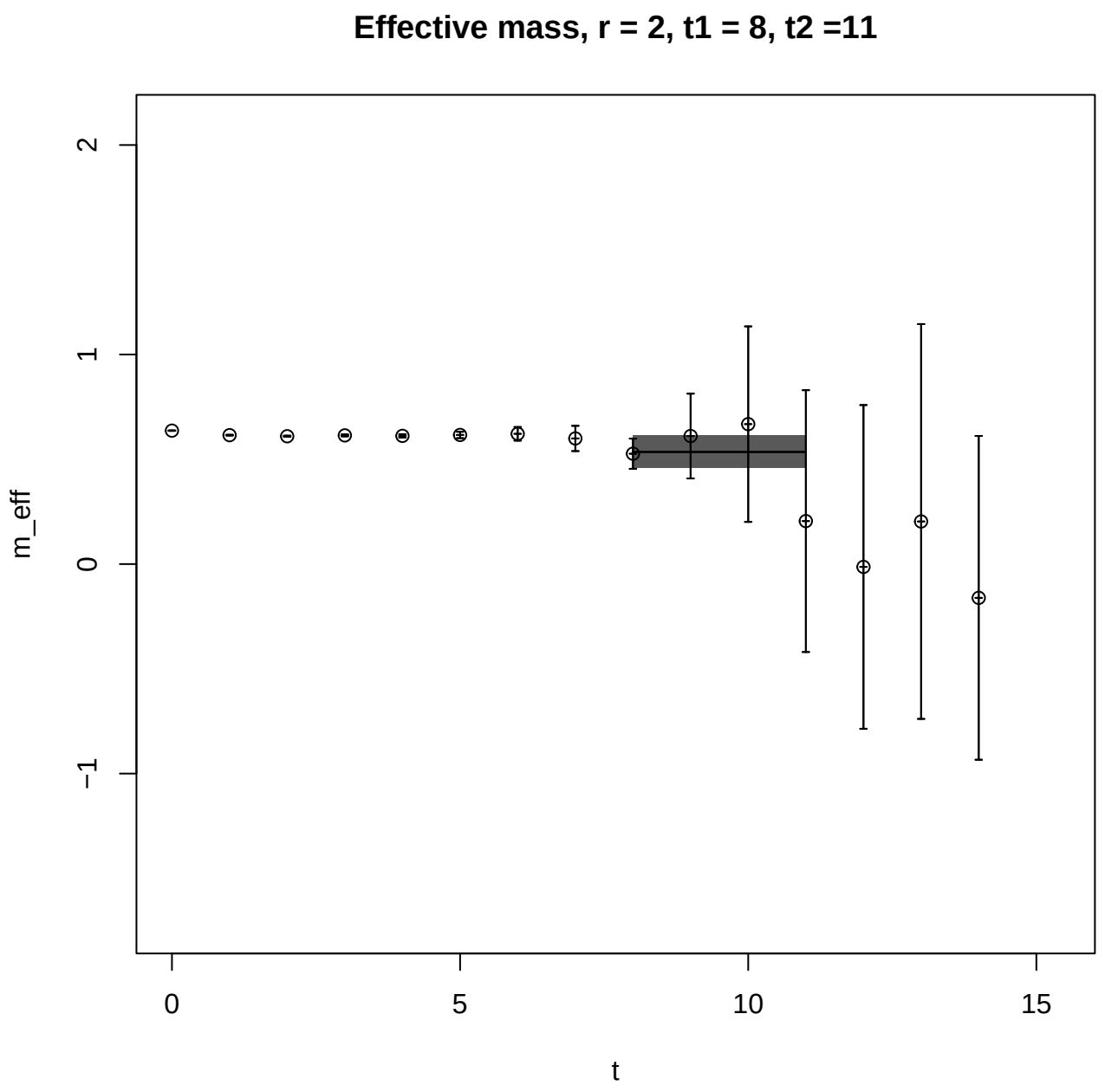

Effective mass, r = 2, t1 = 7, t2 =12
m_eff
t
2
1
0
−1
0
5
10
15

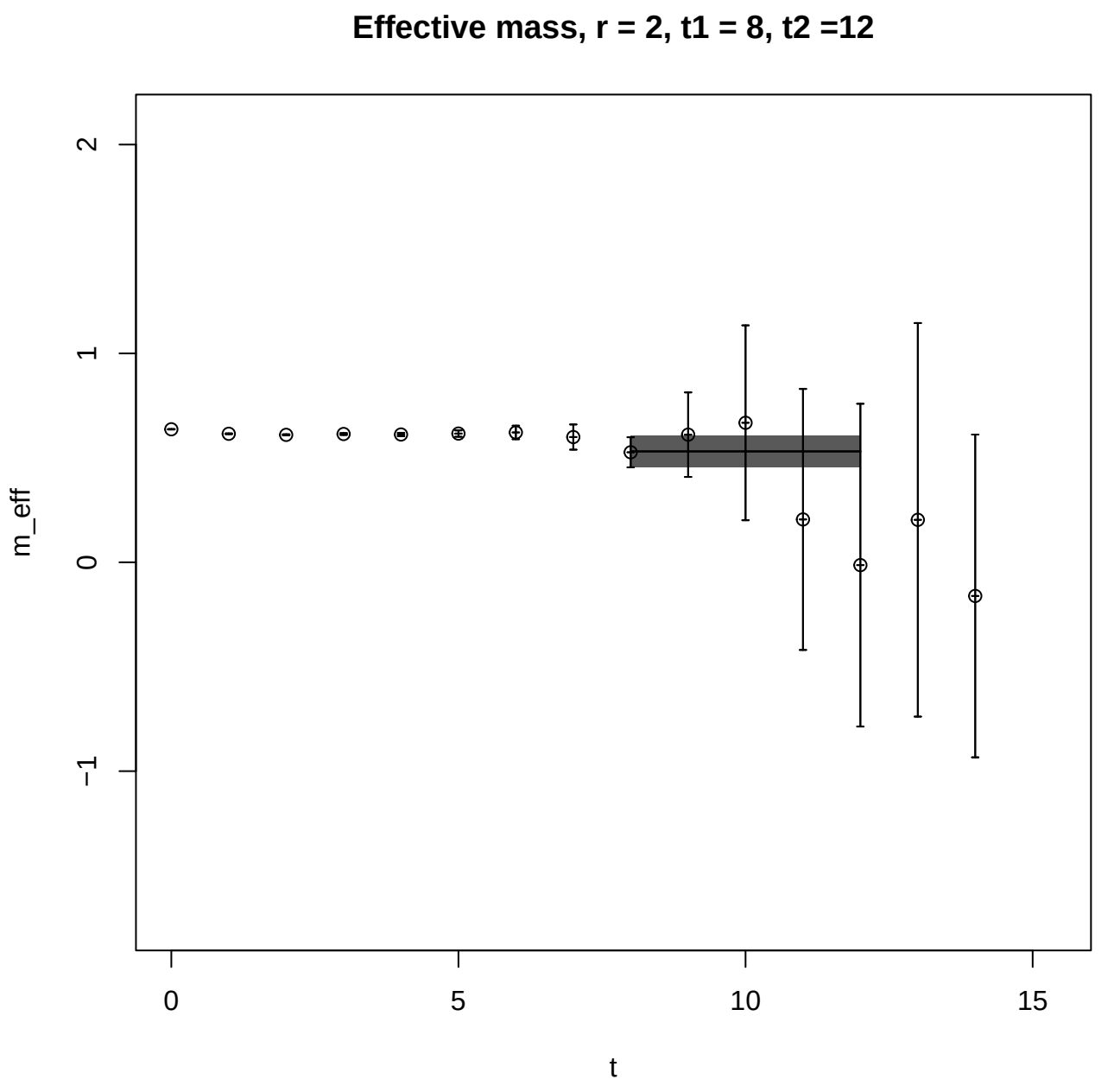

Effective mass, r = 2, t1 = 7, t2 =13
m_eff
t
2
1
0
−1
0
5
10
15

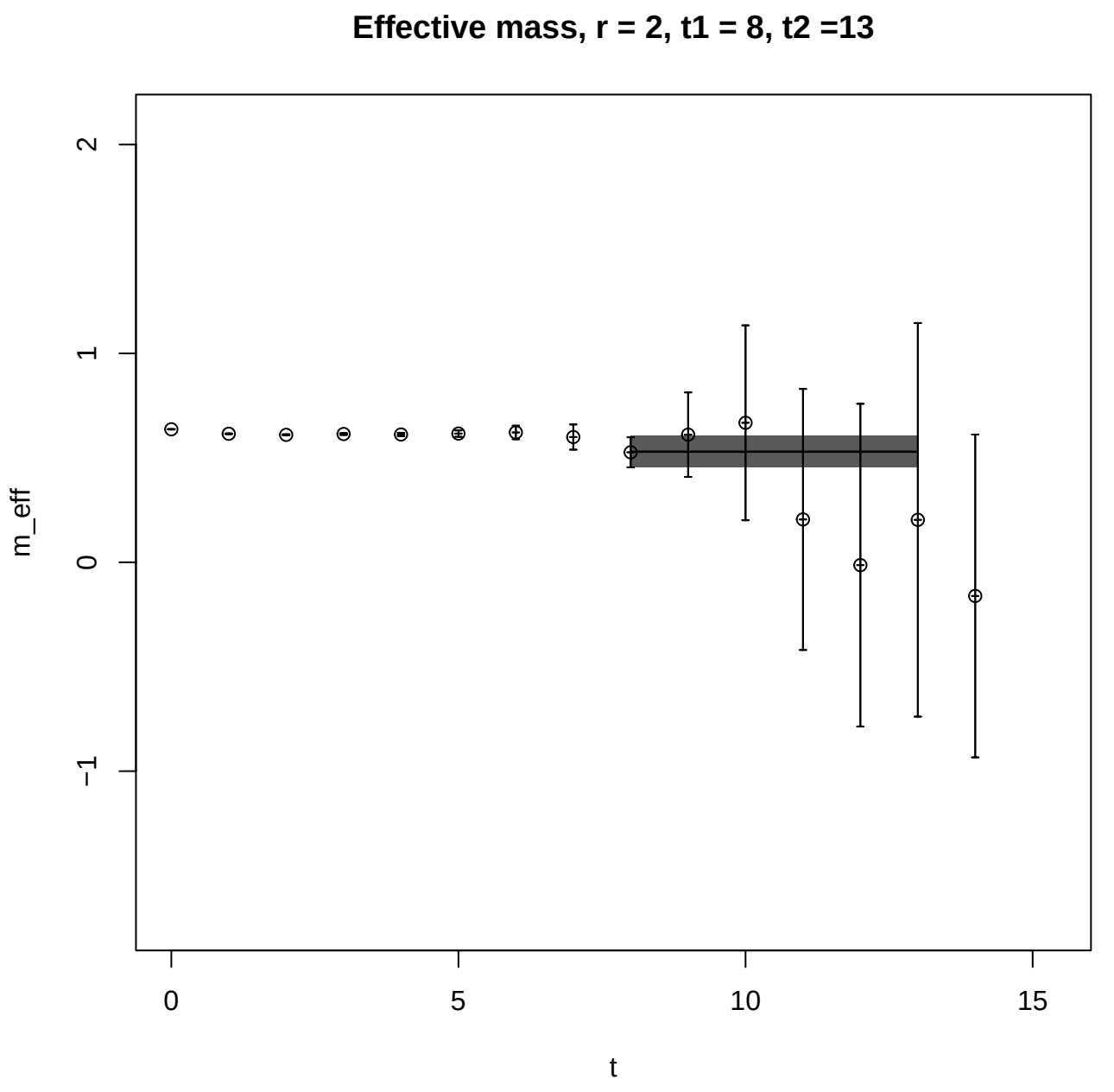

Effective mass, r = 2, t1 = 7, t2 =14
m_eff
t
2
1
0
−1
0
5
10
15

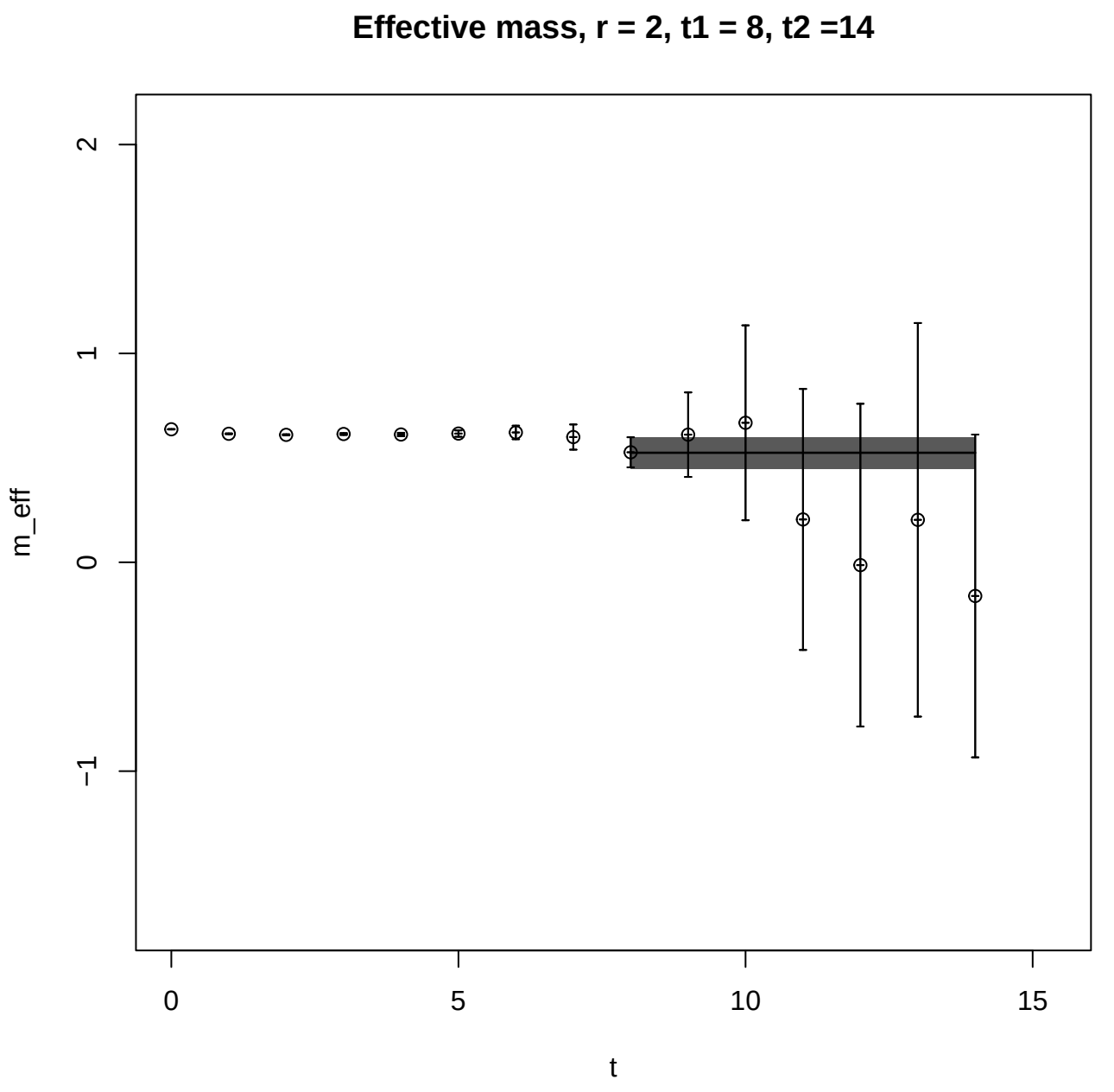

Effective mass, r = 2, t1 = 8, t2 =9
m_eff
2
1
0
−1
0
5
10
15
t

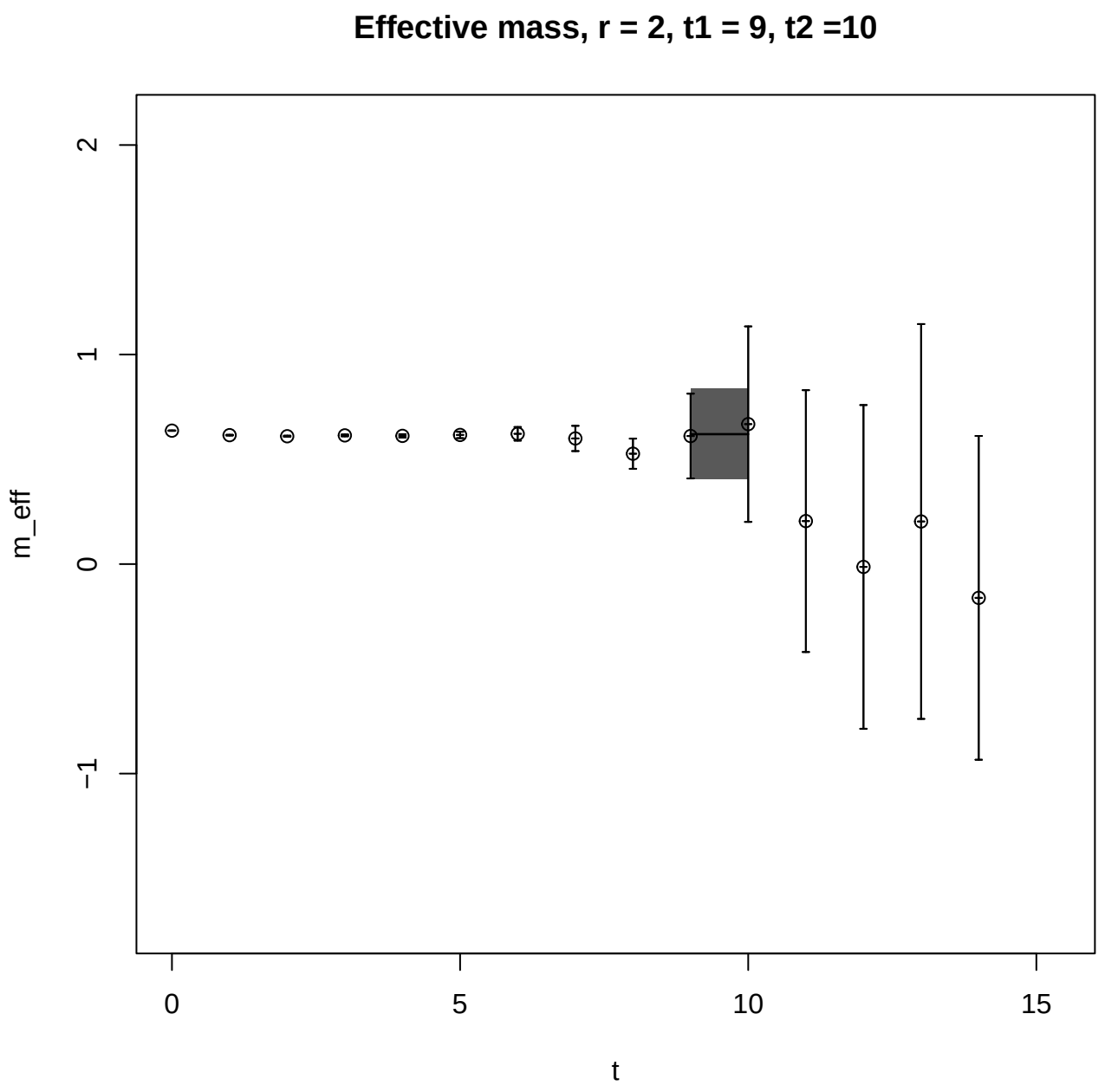

Effective mass, r = 2, t1 = 8, t2 =10
m_eff
2
1
0
−1
0
5
10
15
t

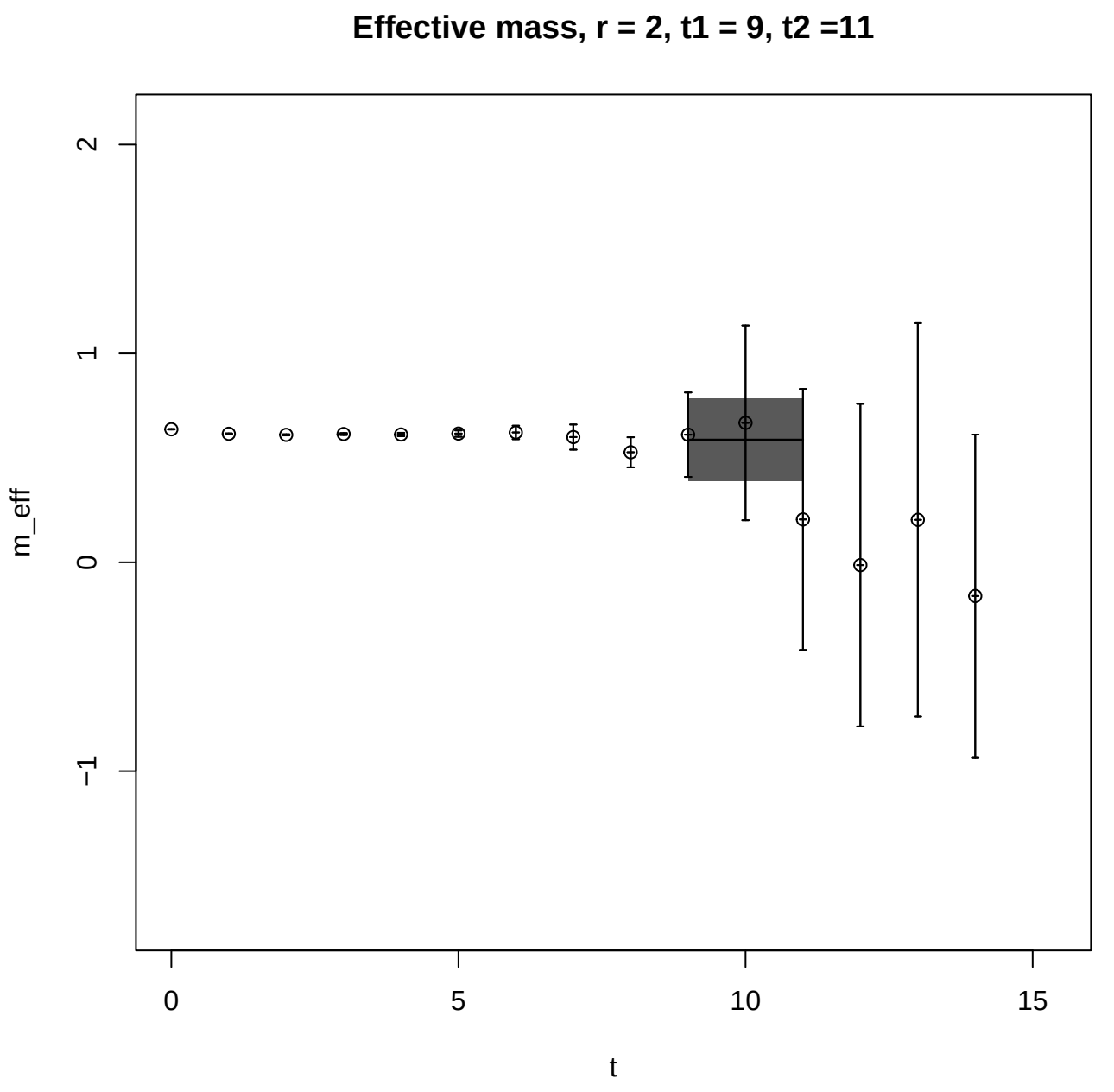

Effective mass, r = 2, t1 = 8, t2 =11
m_eff
2
1
0
−1
0
5
10
15
t

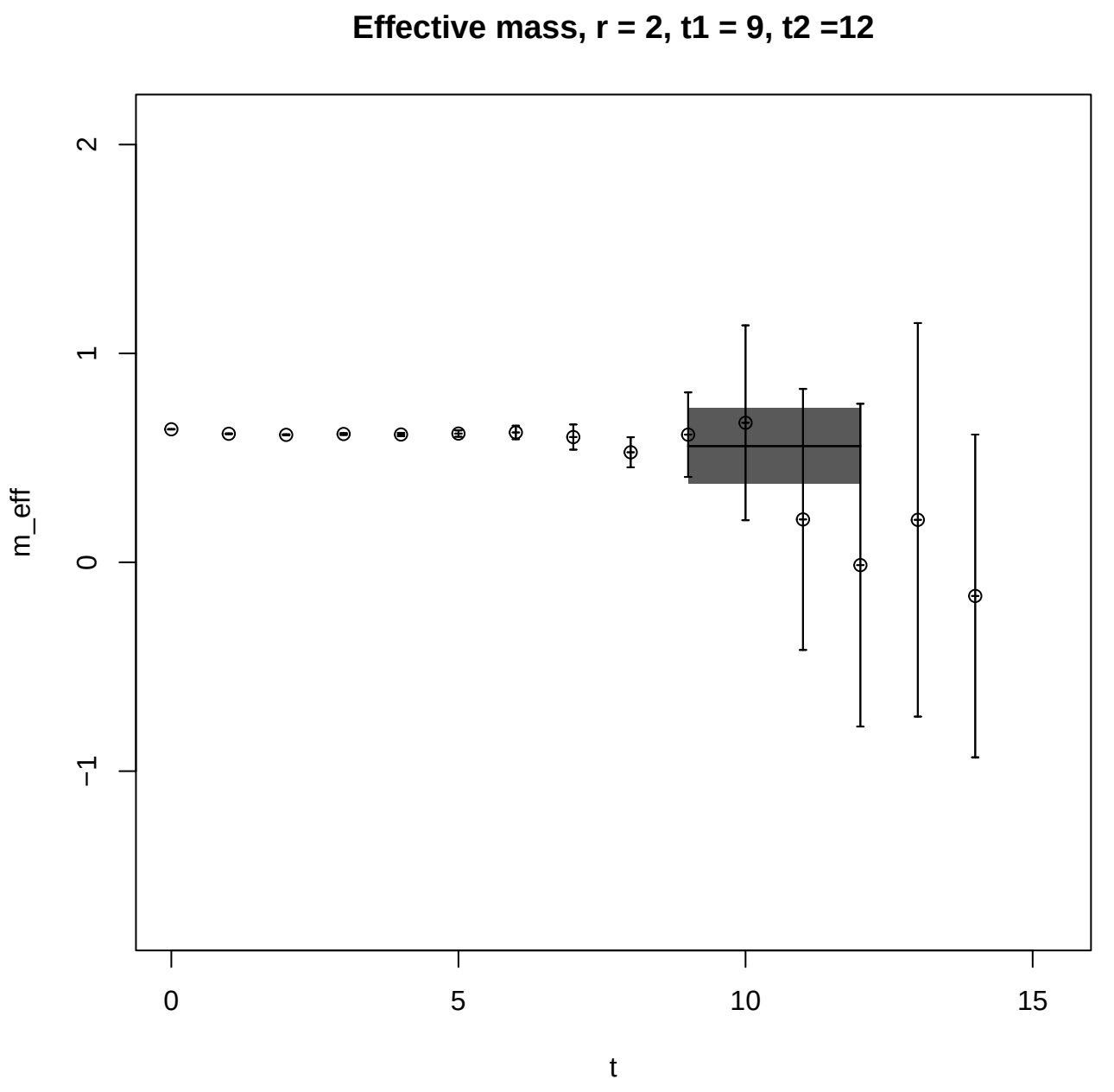

Effective mass, r = 2, t1 = 8, t2 =12
m_eff
2
1
0
−1
0
5
10
15
t

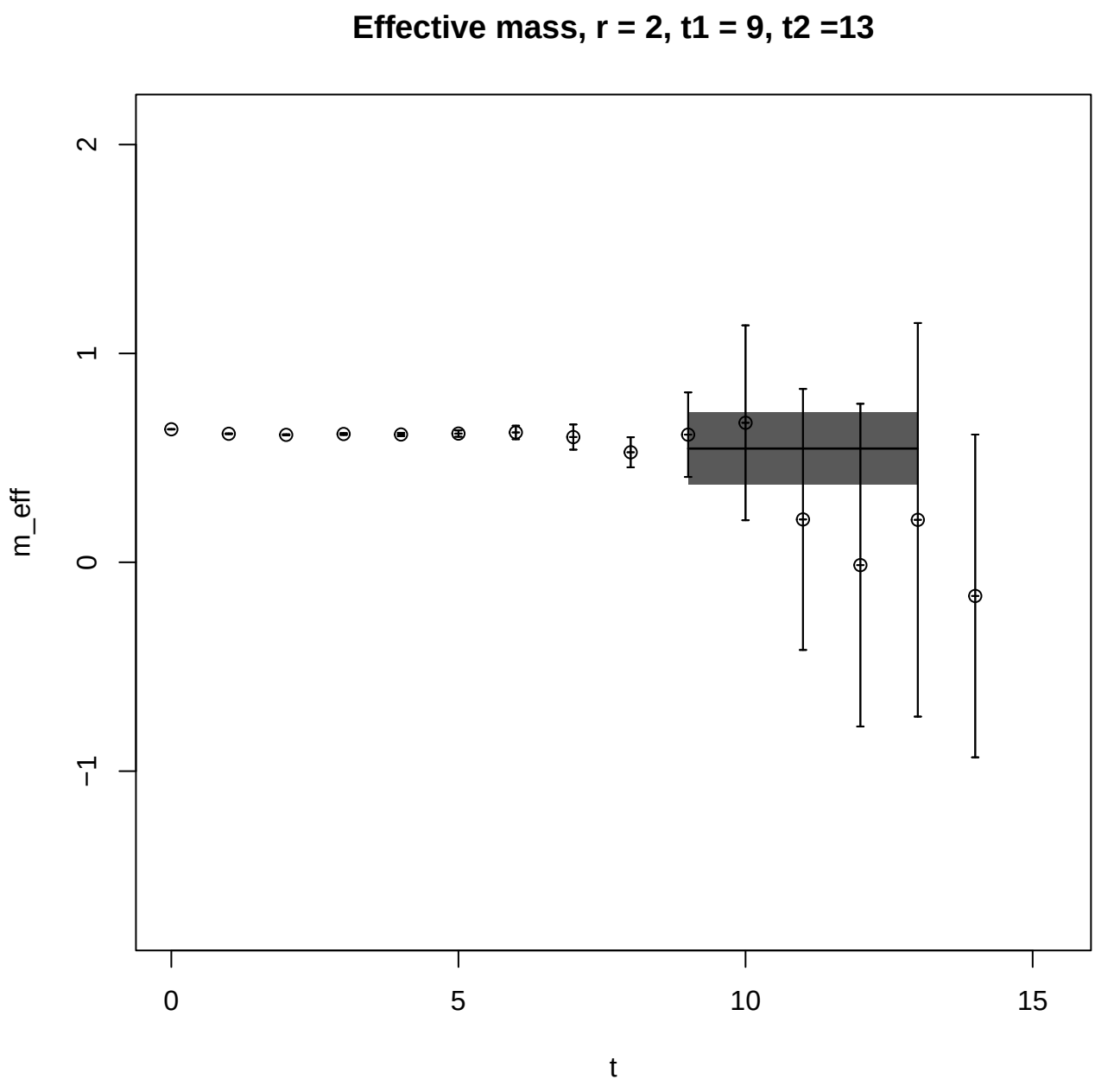

Effective mass, r = 2, t1 = 8, t2 =13
m_eff
2
1
0
−1
0
5
10
15
t

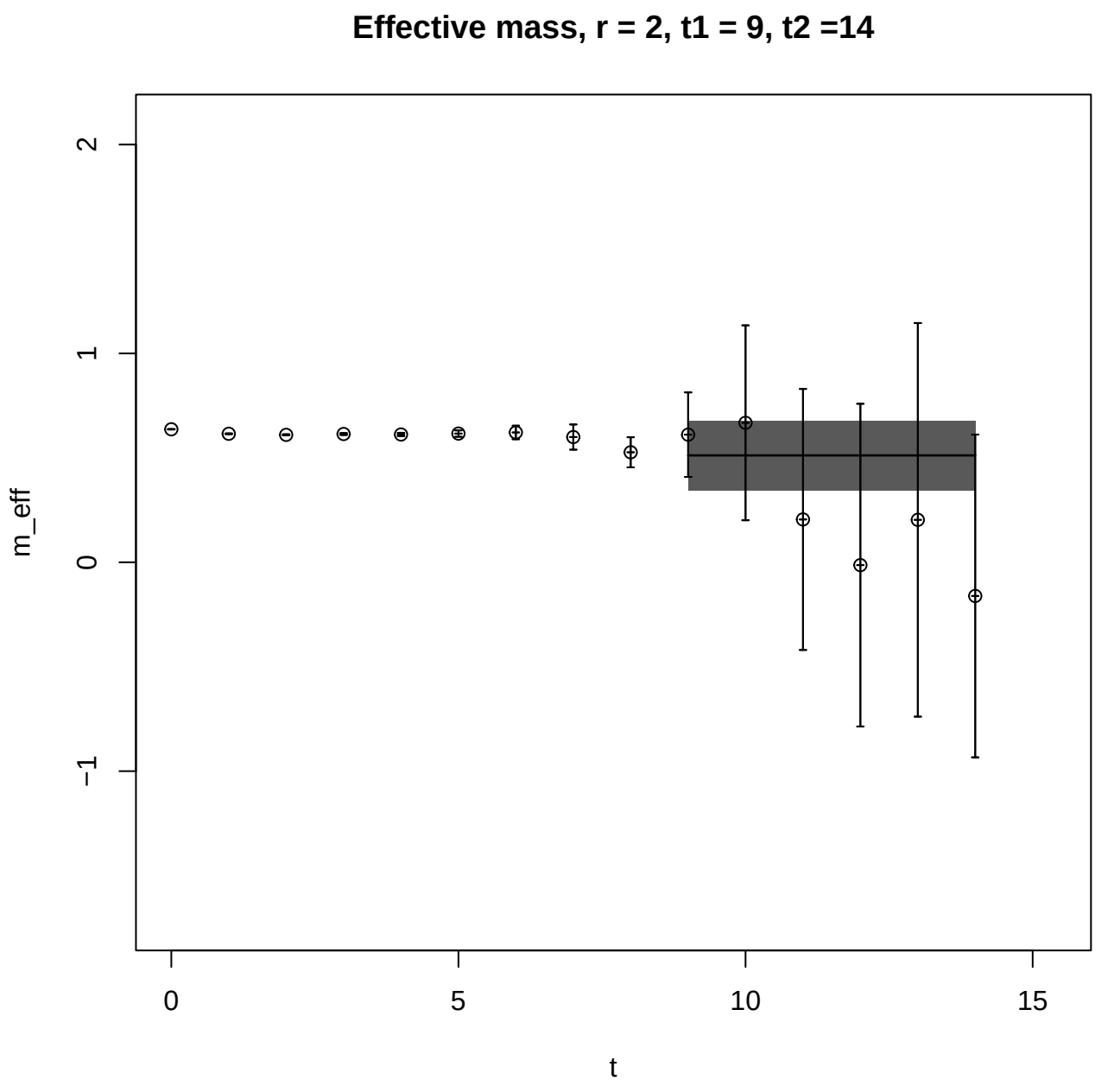

Effective mass, r = 2, t1 = 8, t2 =14
m_eff
t
2
1
0
−1
0
5
10
15

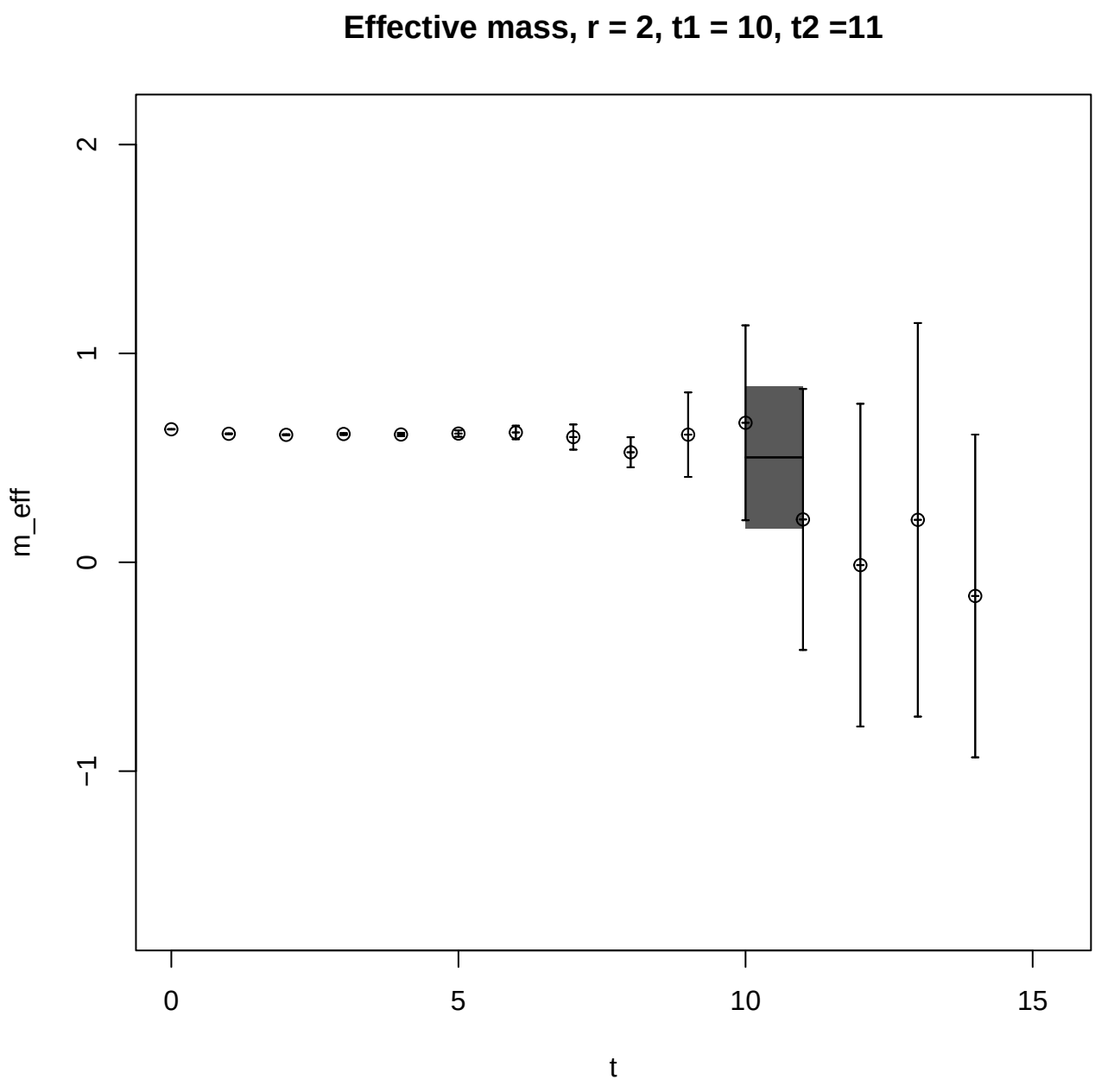

Effective mass, r = 2, t1 = 9, t2 =10
m_eff
2
1
0
−1
0
5
10
15
t

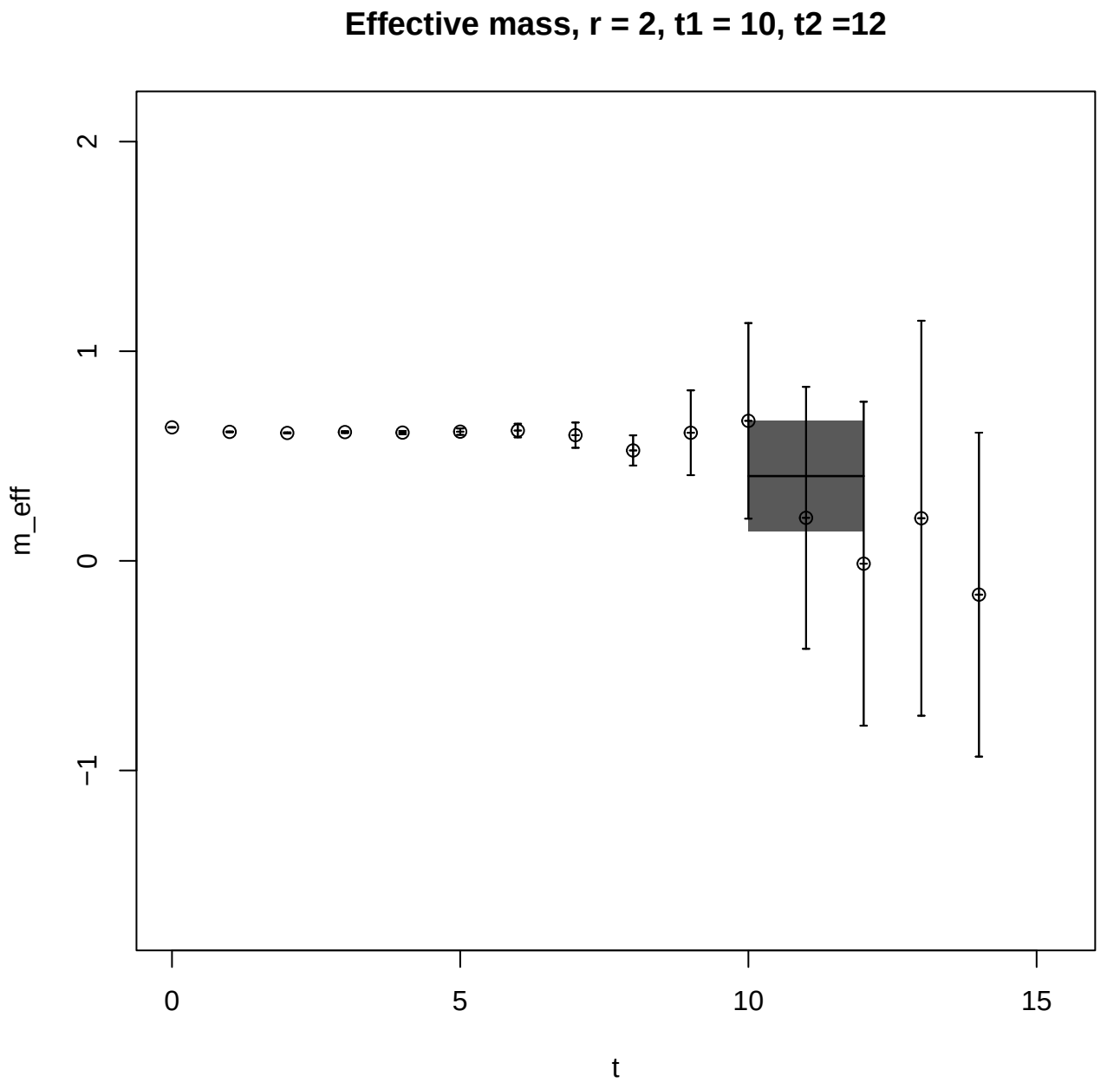

Effective mass, r = 2, t1 = 9, t2 =11
m_eff
2
1
0
−1
0
5
10
15
t

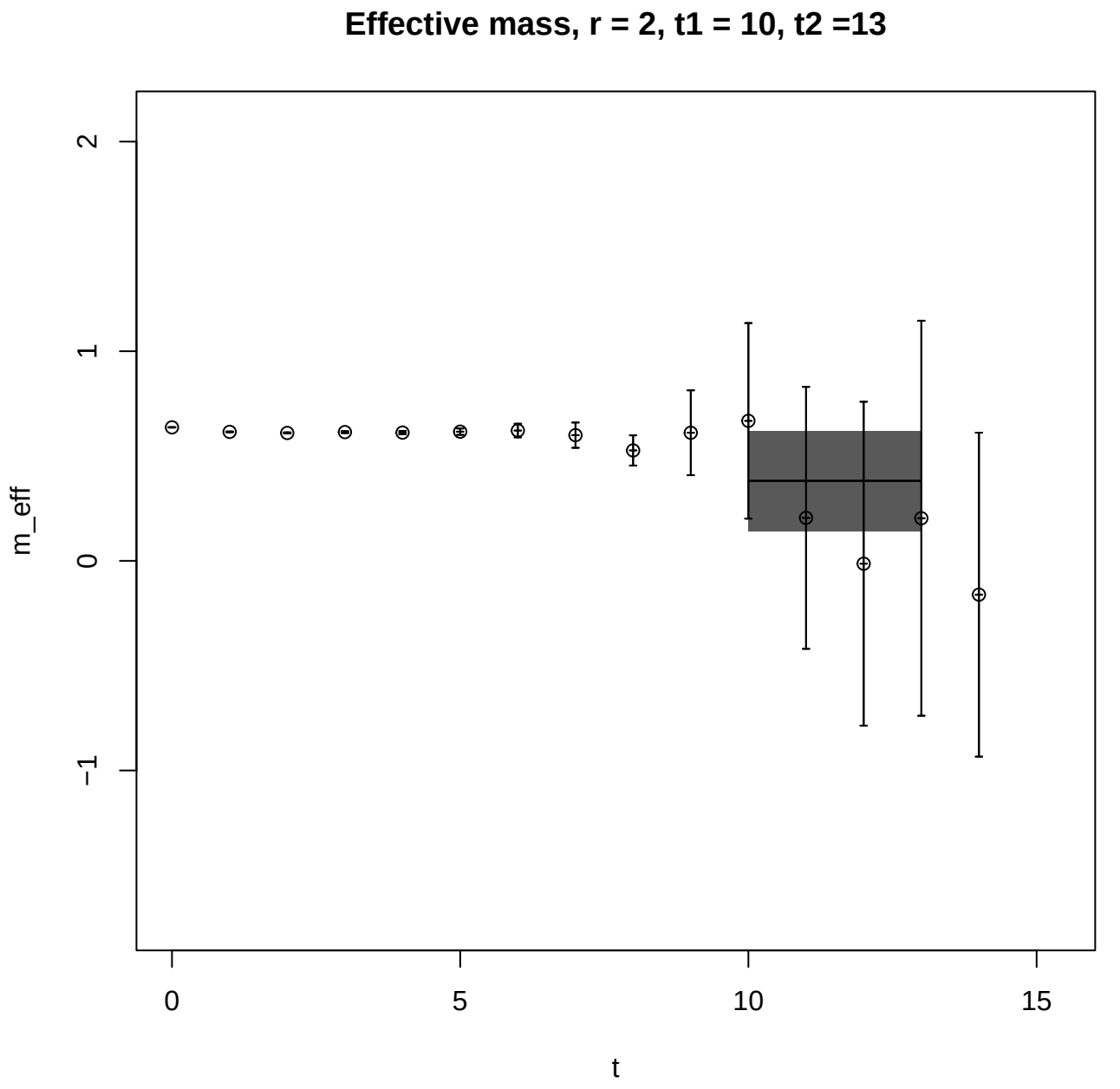

Effective mass, r = 2, t1 = 9, t2 =12
m_eff
2
1
0
−1
0
5
10
15
t

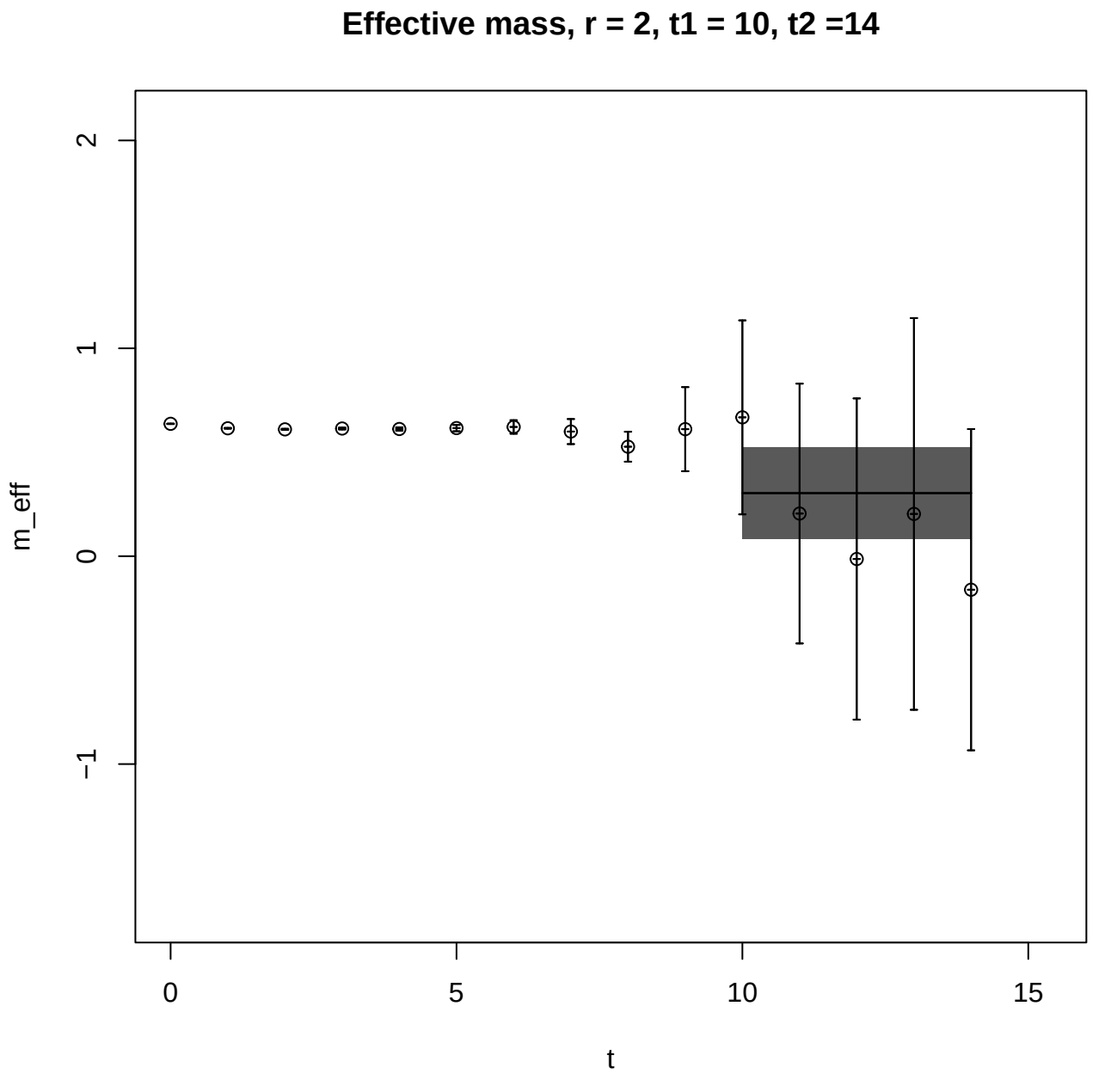

Effective mass, r = 2, t1 = 9, t2 =13
m_eff
2
1
0
−1
0
5
10
15
t

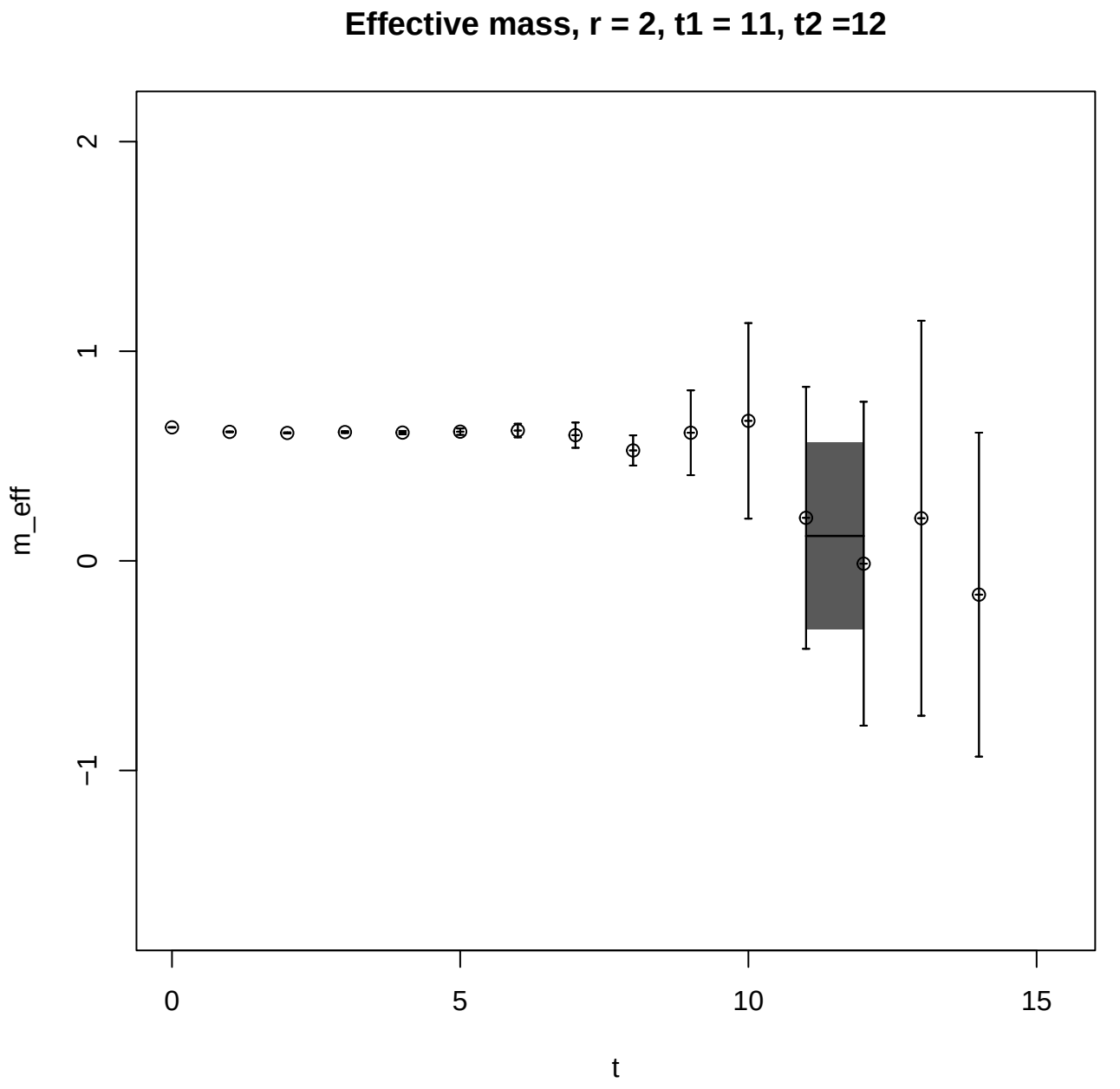

Effective mass, r = 2, t1 = 9, t2 =14
m_eff
t
2
1
0
−1
0
5
10
15

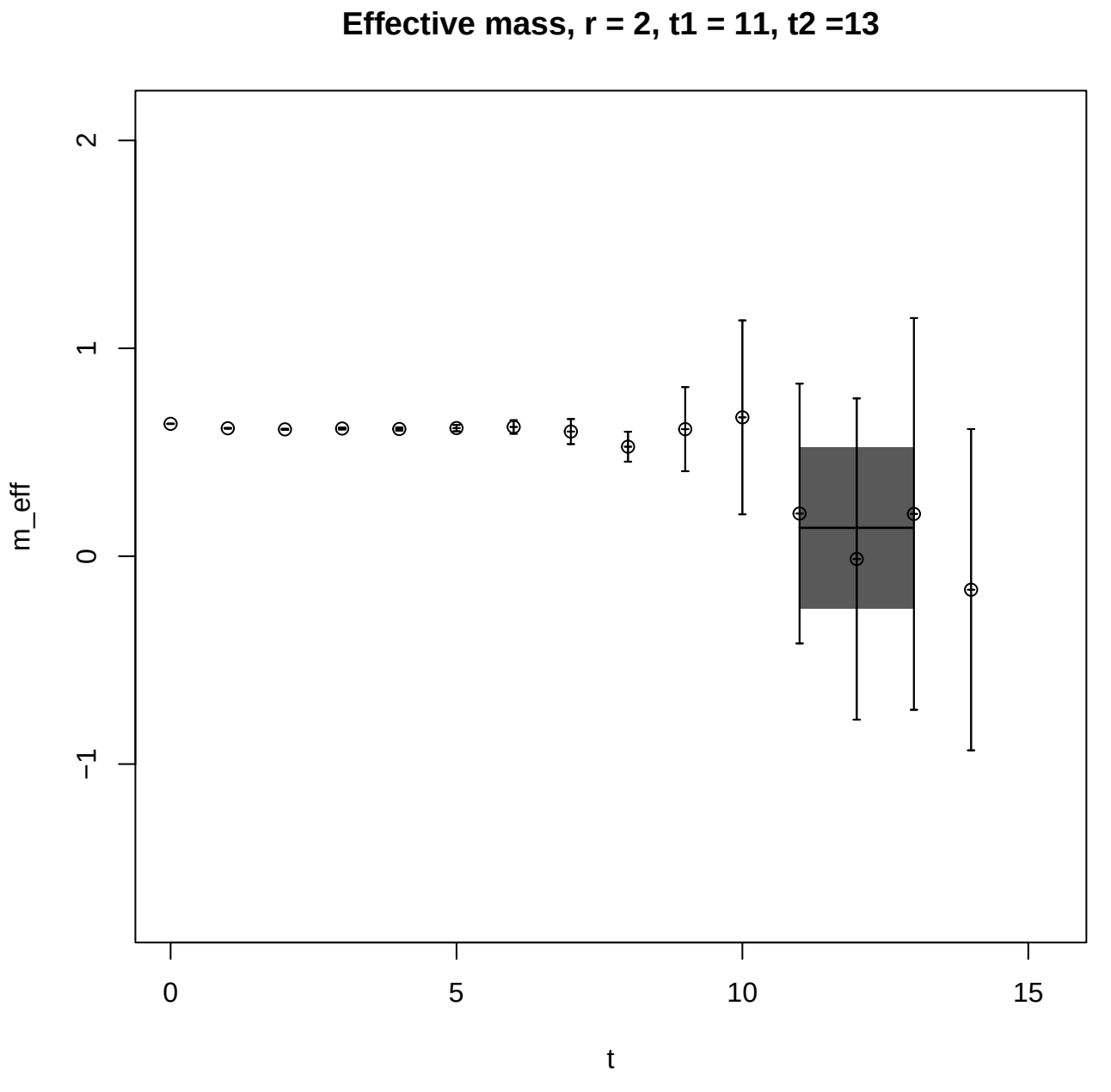

Effective mass, r = 2, t1 = 10, t2 =11
m_eff
2
1
0
−1
0
5
10
15
t

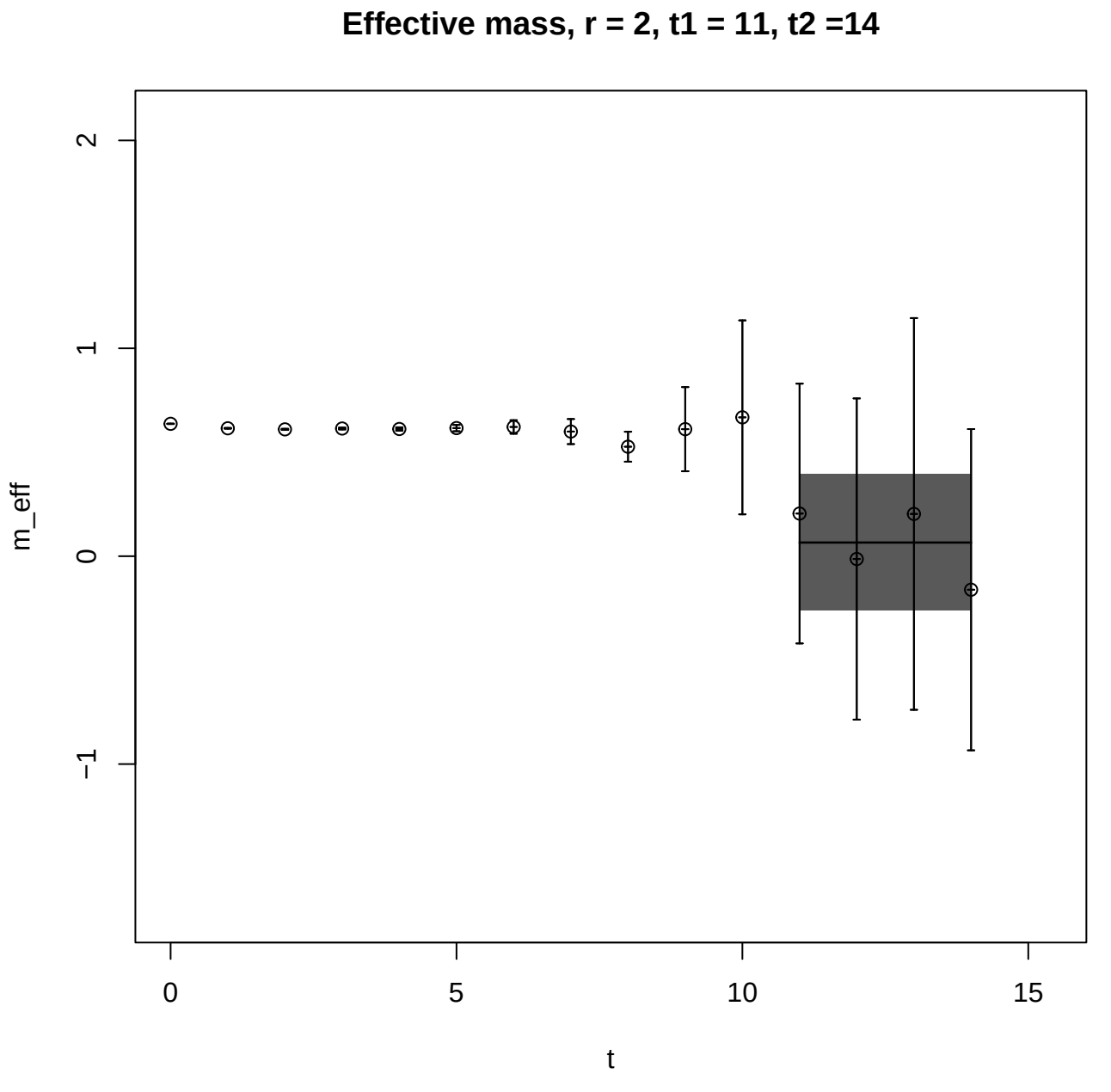

Effective mass, r = 2, t1 = 10, t2 =12
m_eff
2
1
0
−1
0
5
10
15
t

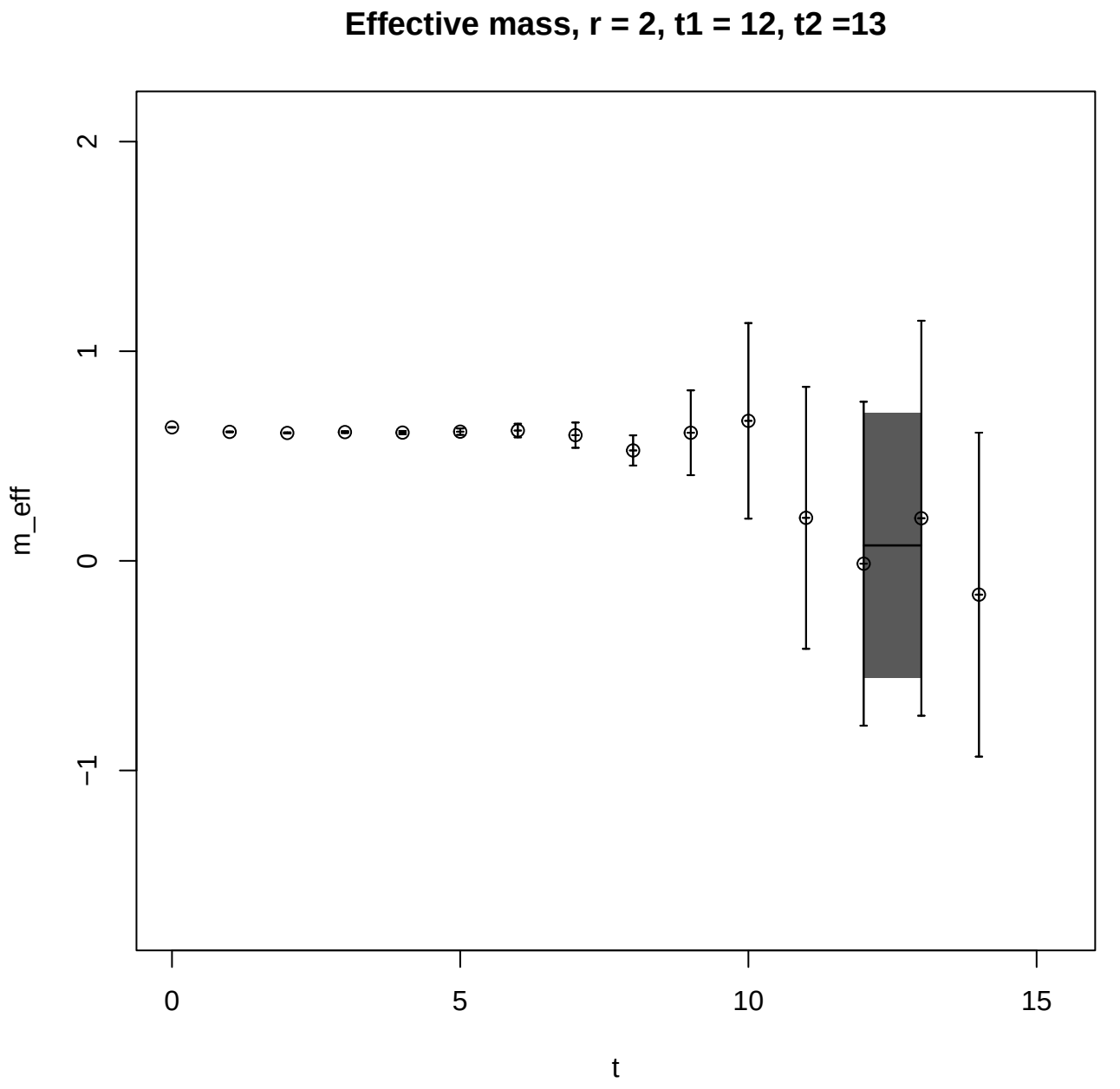

Effective mass, r = 2, t1 = 10, t2 =13
m_eff
2
1
0
−1
0
5
10
15
t

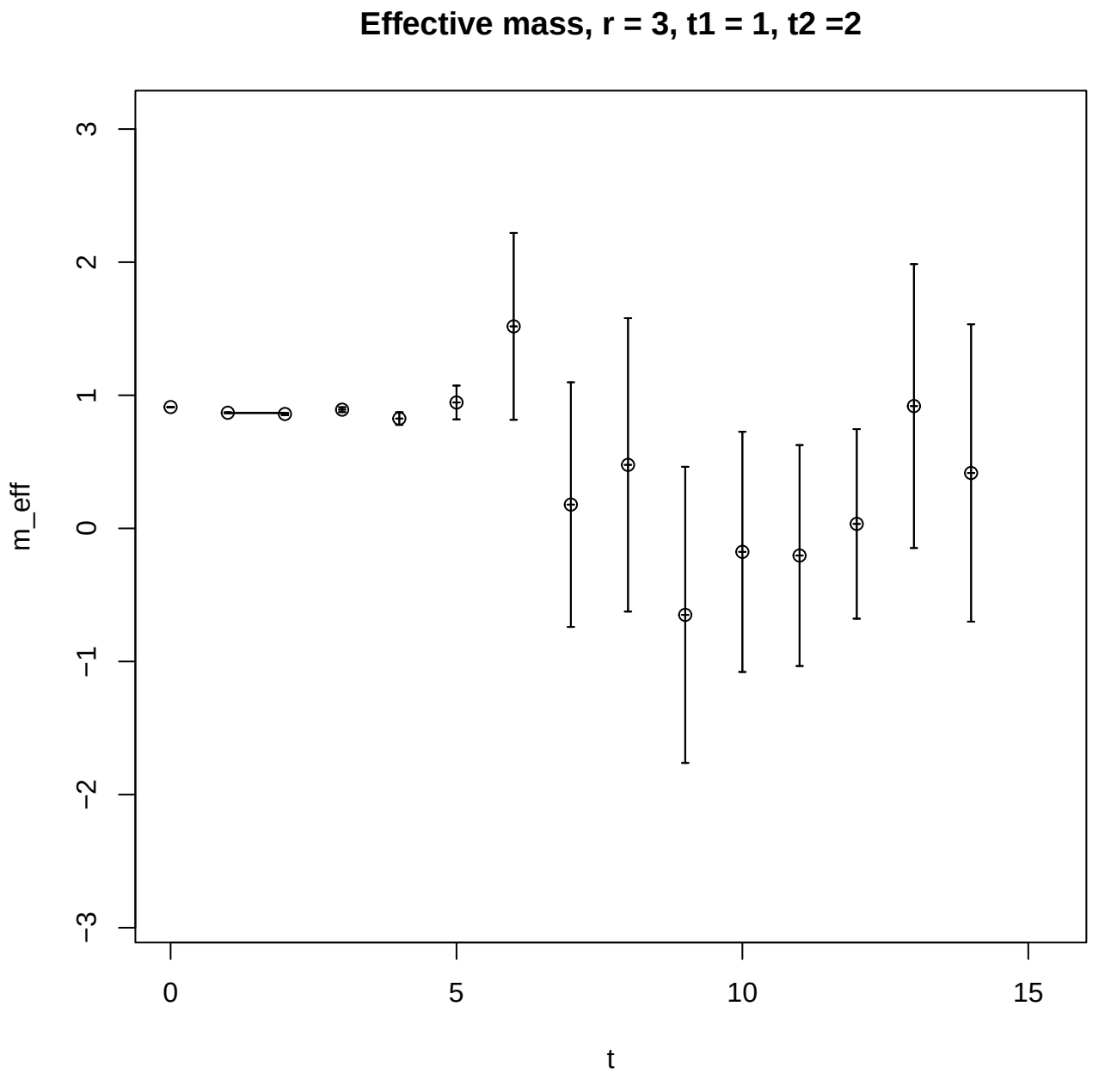

Effective mass, r = 2, t1 = 10, t2 =14
m_eff
2
1
0
−1
0
5
10
15
t

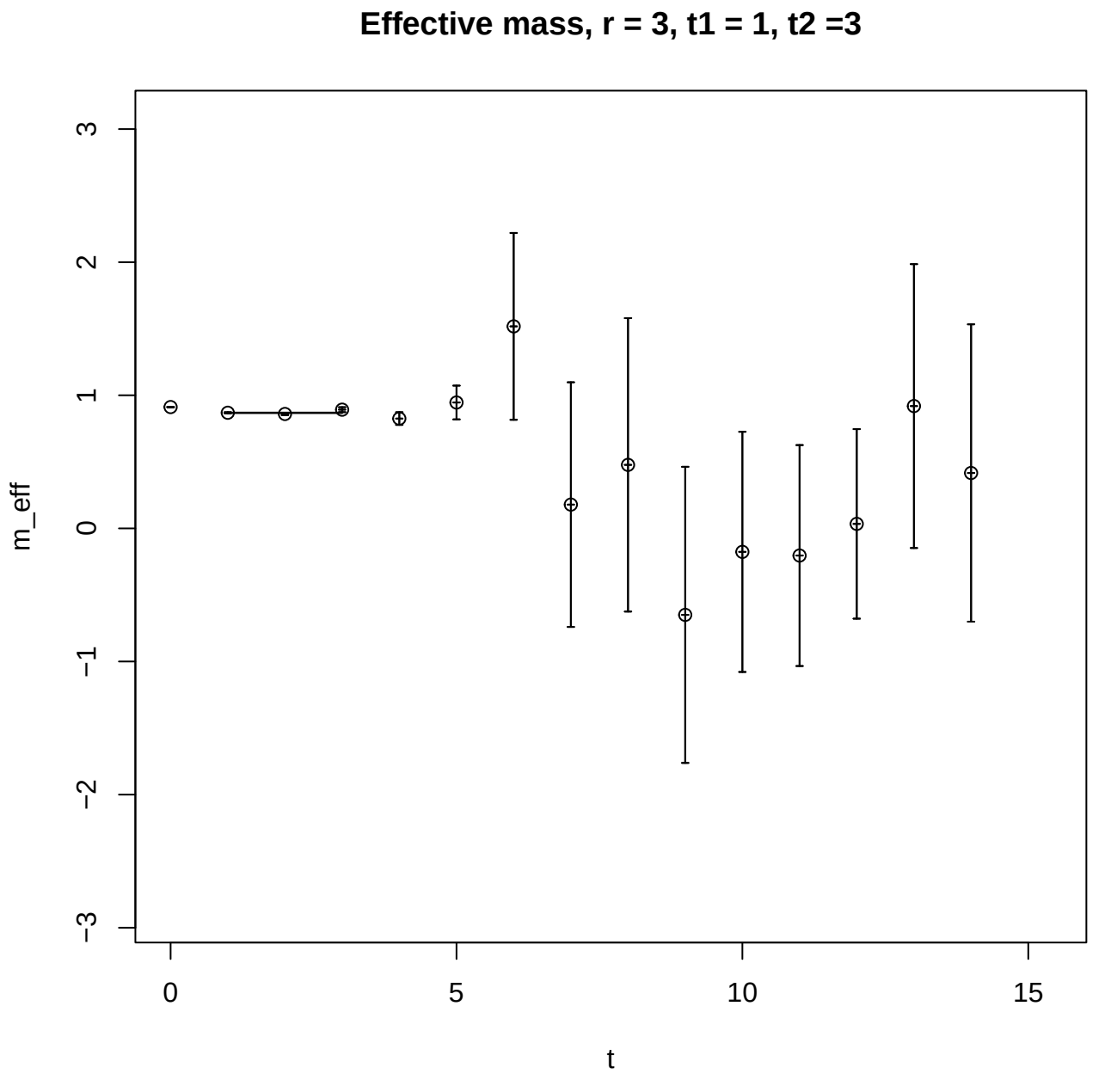

Effective mass, r = 2, t1 = 11, t2 =12
m_eff
2
1
0
−1
0
5
10
15
t

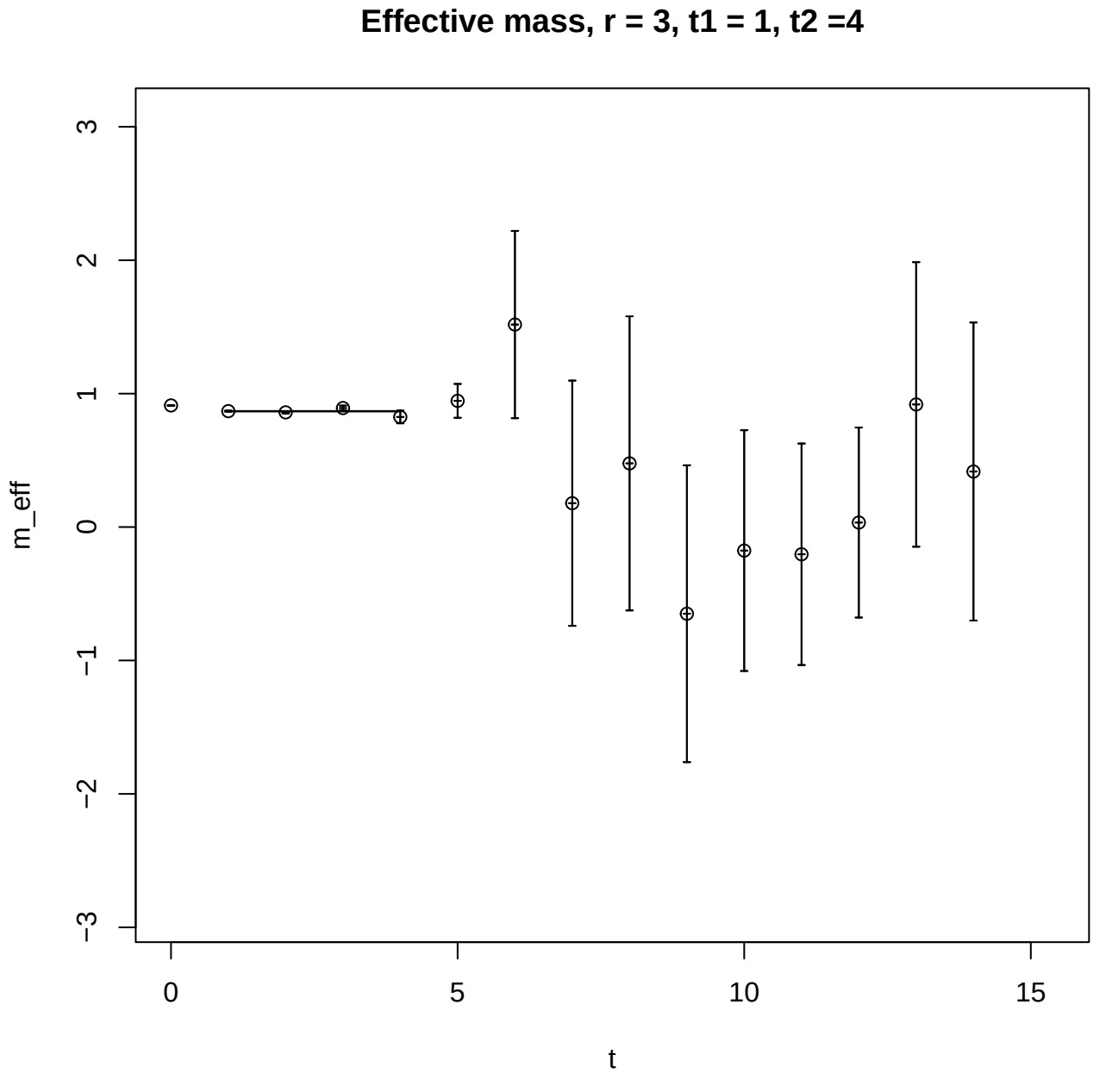

Effective mass, r = 2, t1 = 11, t2 =13
m_eff
2
1
0
−1
0
5
10
15
t

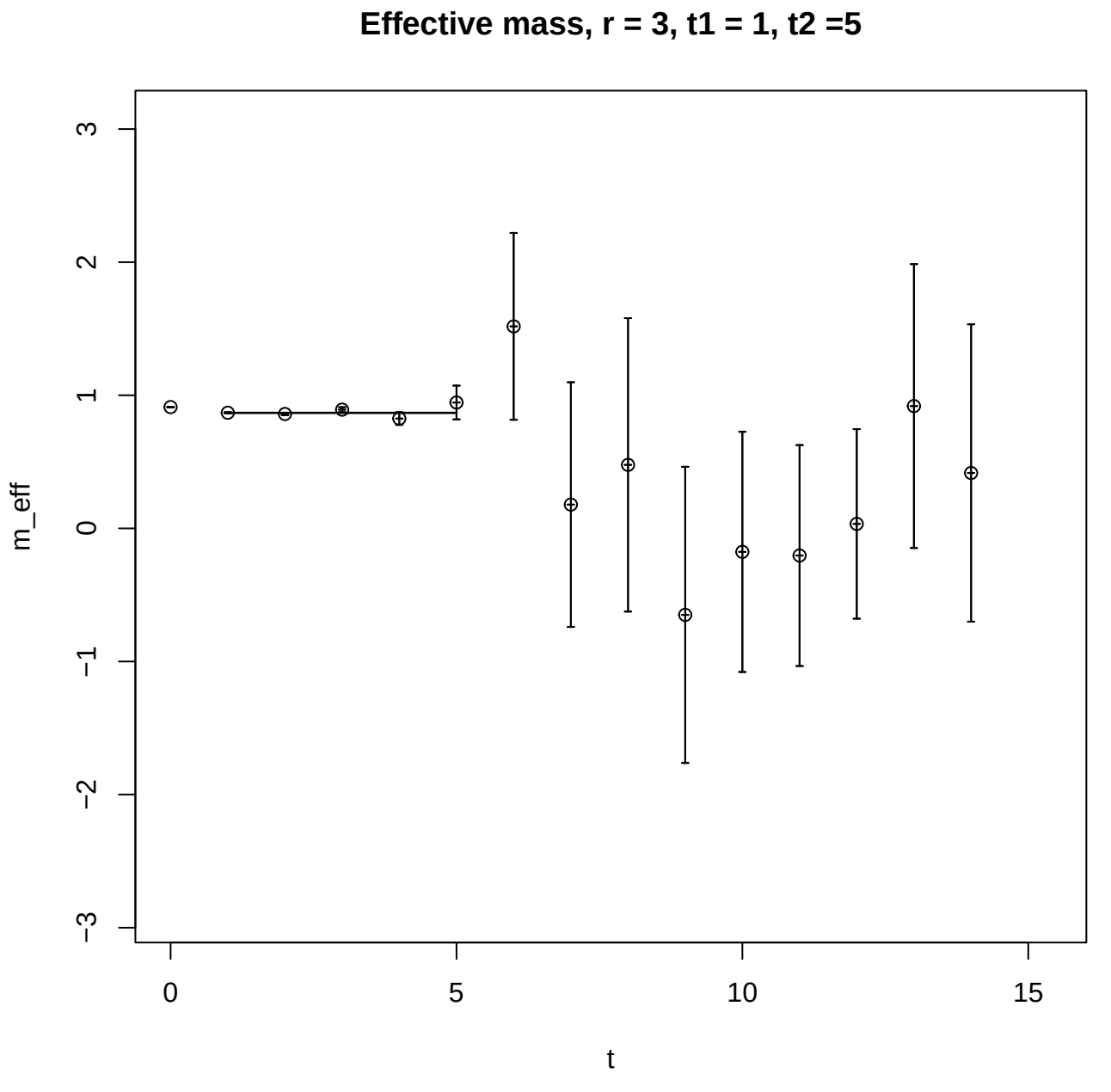

Effective mass, r = 2, t1 = 11, t2 =14
m_eff
2
1
0
−1
0
5
10
15
t

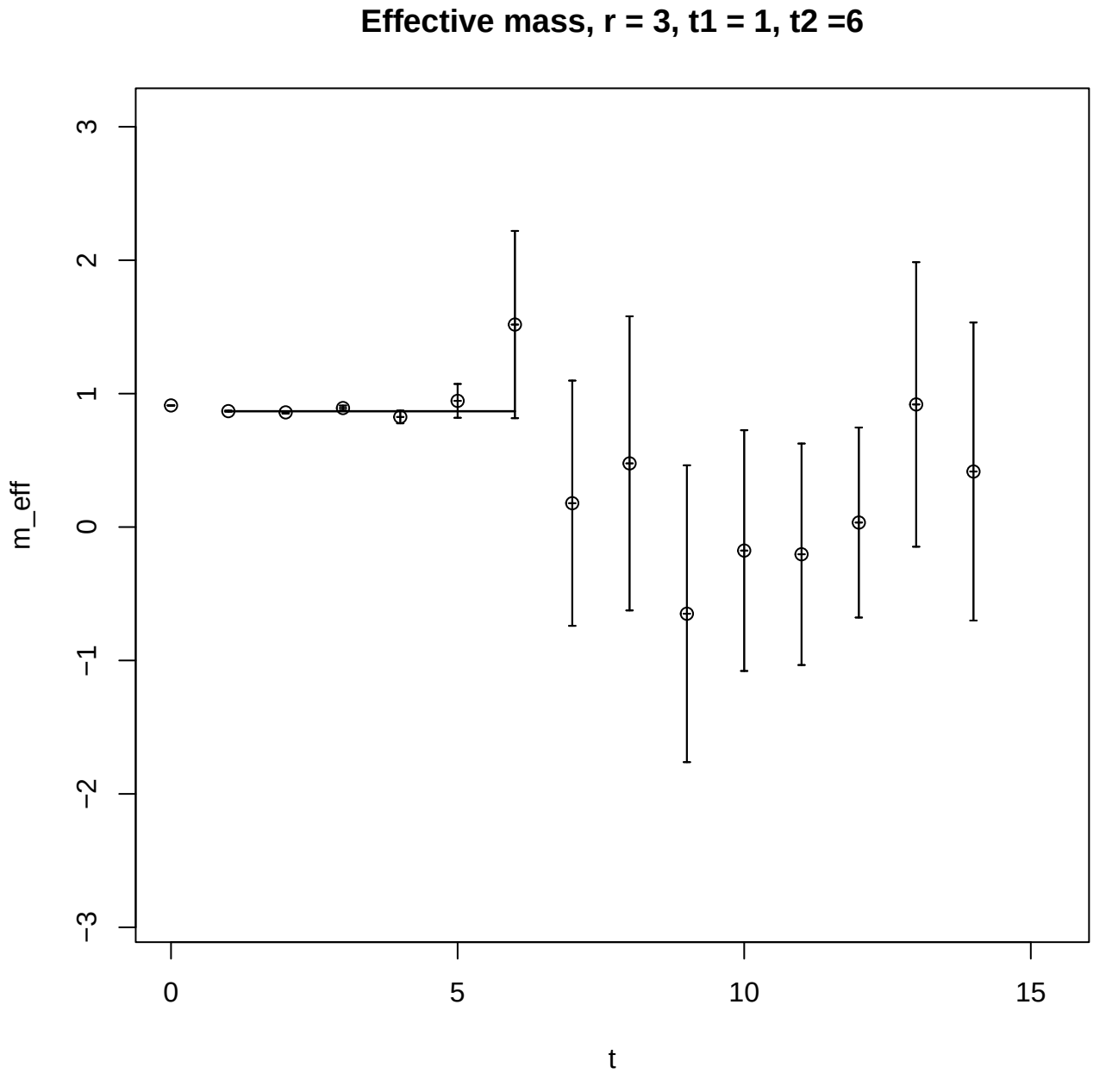

Effective mass, r = 2, t1 = 12, t2 =13
m_eff
2
1
0
−1
0
5
10
15
t

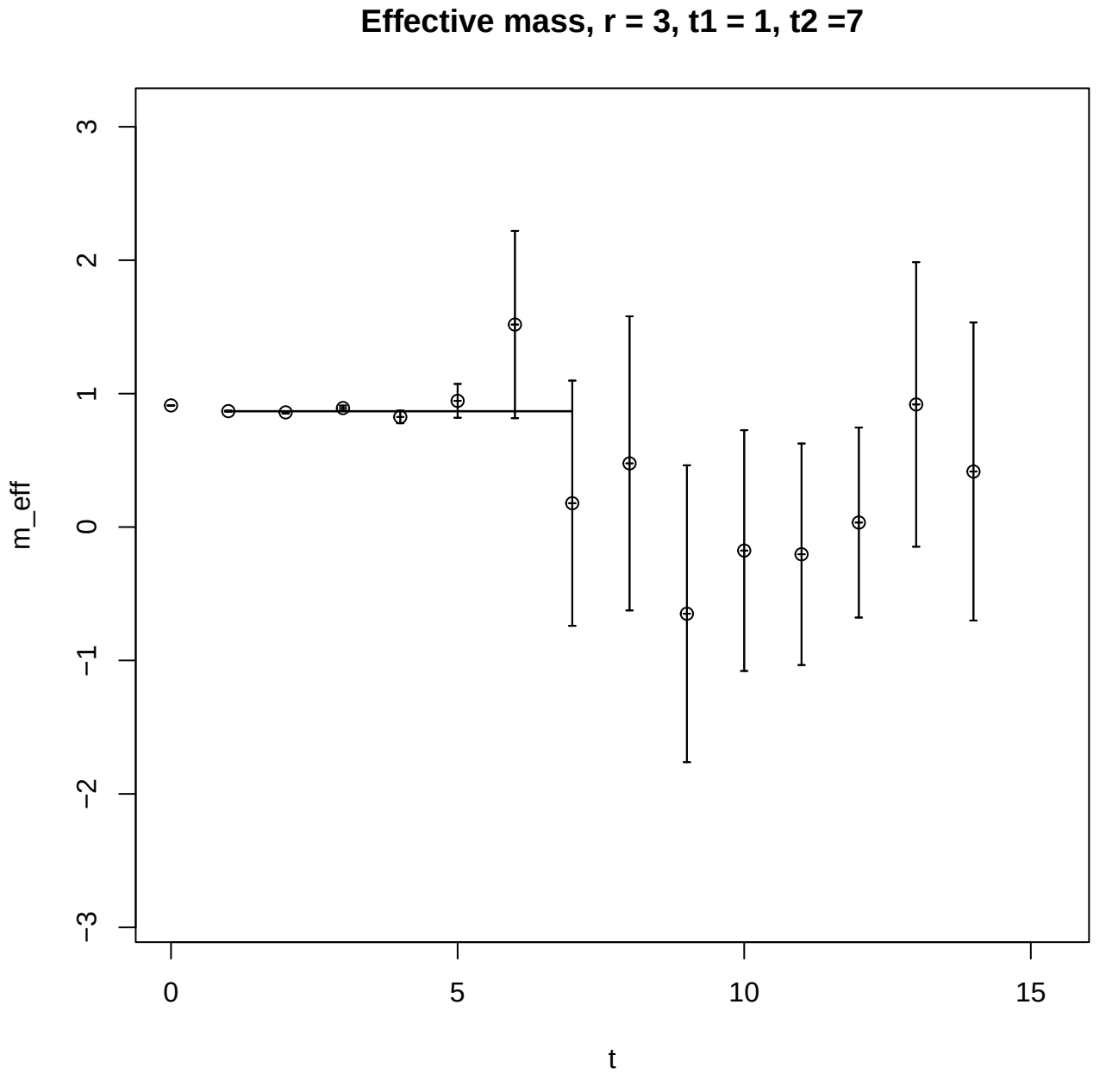

Effective mass, r = 3, t1 = 1, t2 =2
m_eff
t
3
2
1
0
−1
−2
−3
0
5
10
15

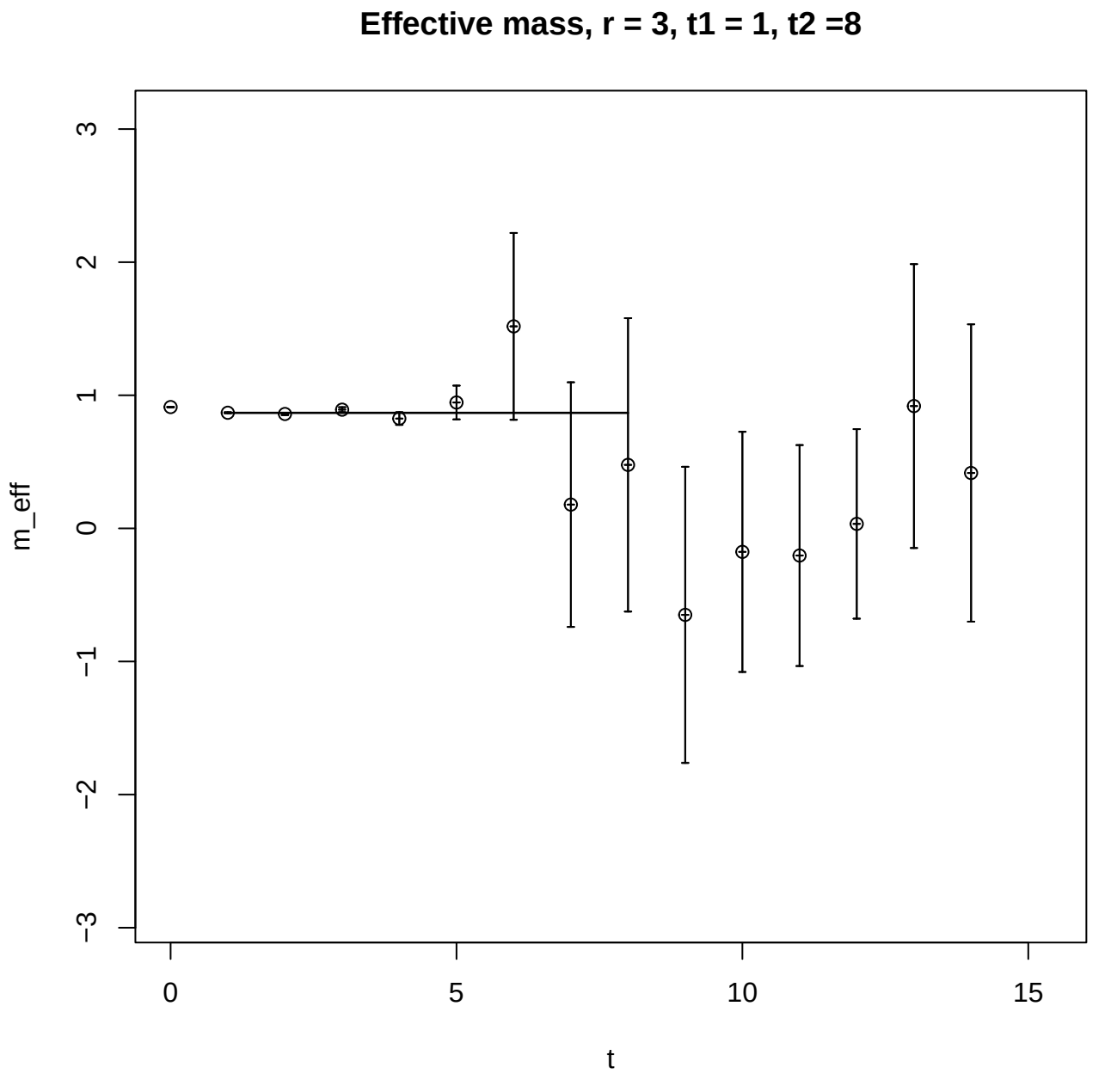

Effective mass, r = 3, t1 = 1, t2 =3
m_eff
t
3
2
1
0
−1
−2
−3
0
5
10
15

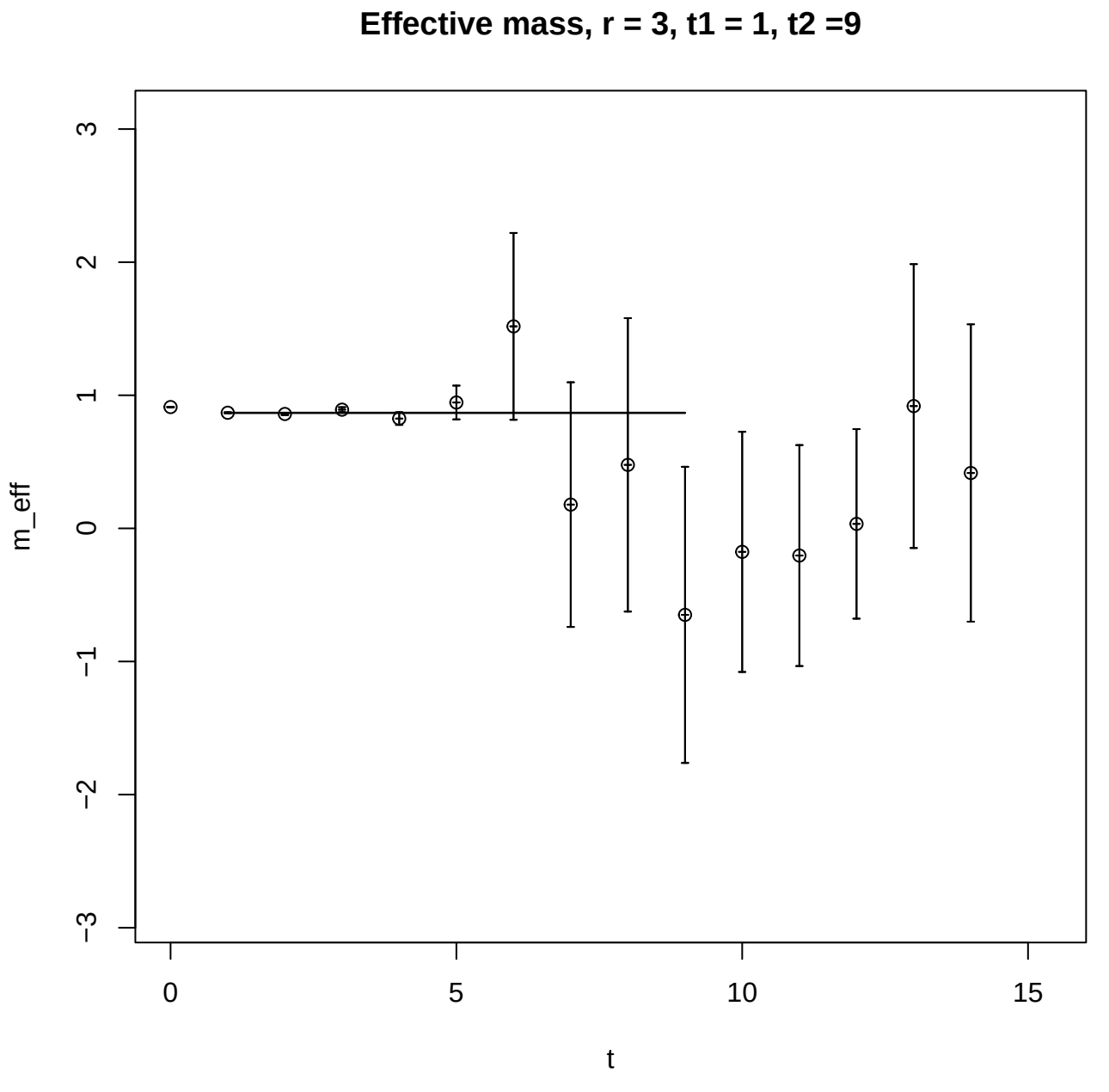

Effective mass, r = 3, t1 = 1, t2 =4
m_eff
t
3
2
1
0
−1
−2
−3
0
5
10
15

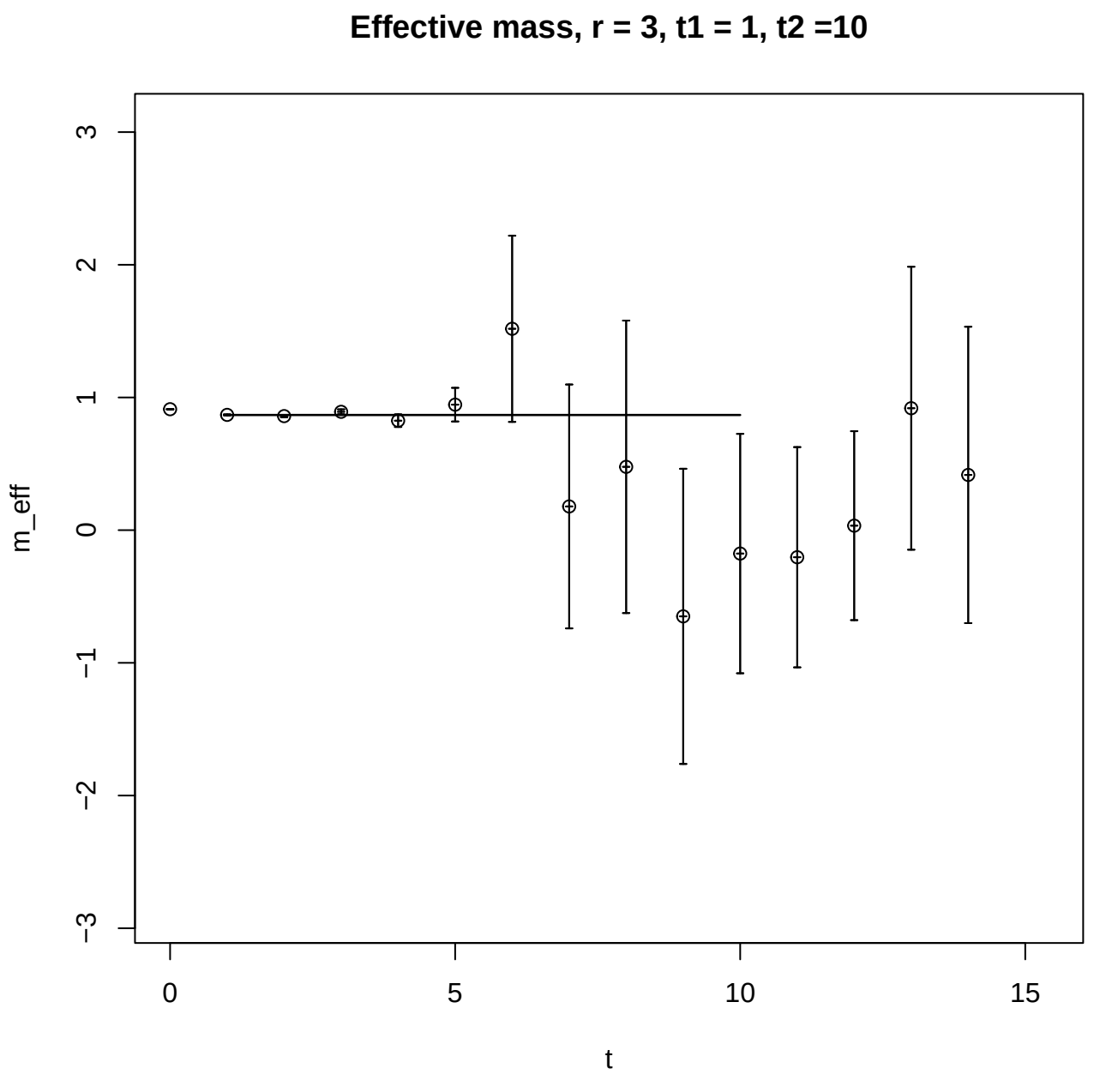

Effective mass, r = 3, t1 = 1, t2 =5
m_eff
t
3
2
1
0
−1
−2
−3
0
5
10
15

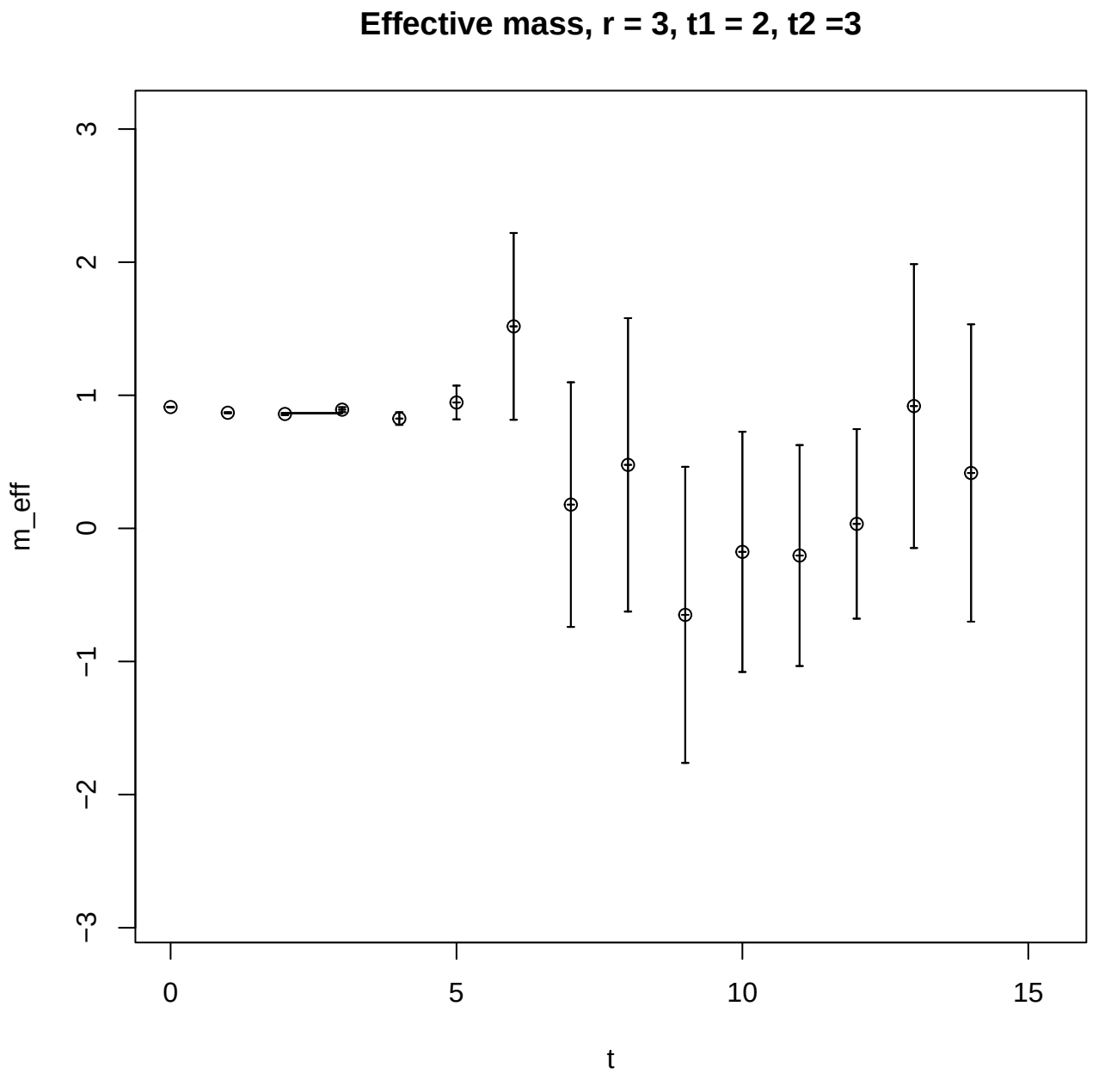

Effective mass, r = 3, t1 = 1, t2 =6
m_eff
t
3
2
1
0
−1
−2
−3
0
5
10
15

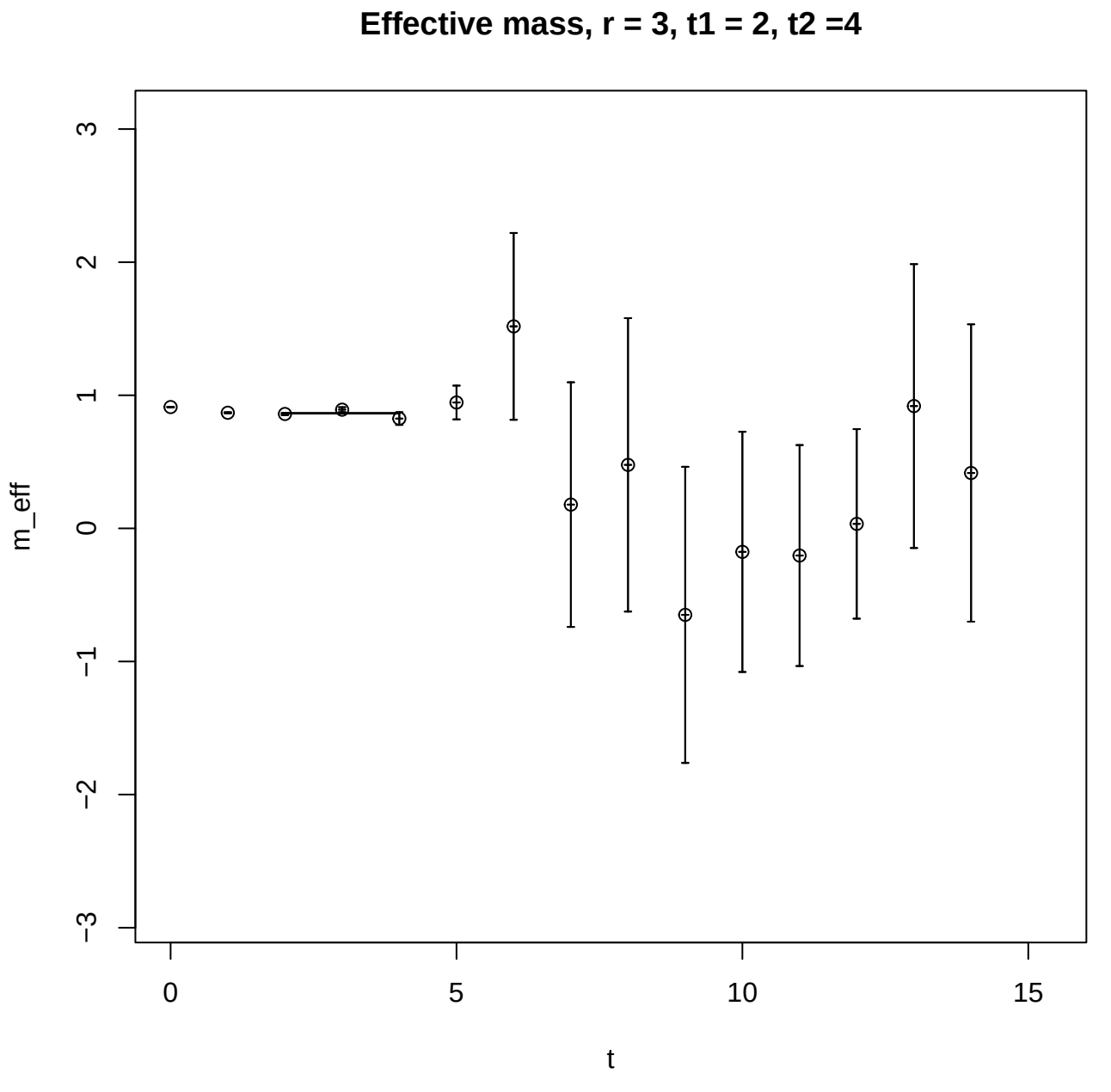

Effective mass, r = 3, t1 = 1, t2 =7
m_eff
t
3
2
1
0
−1
−2
−3
0
5
10
15

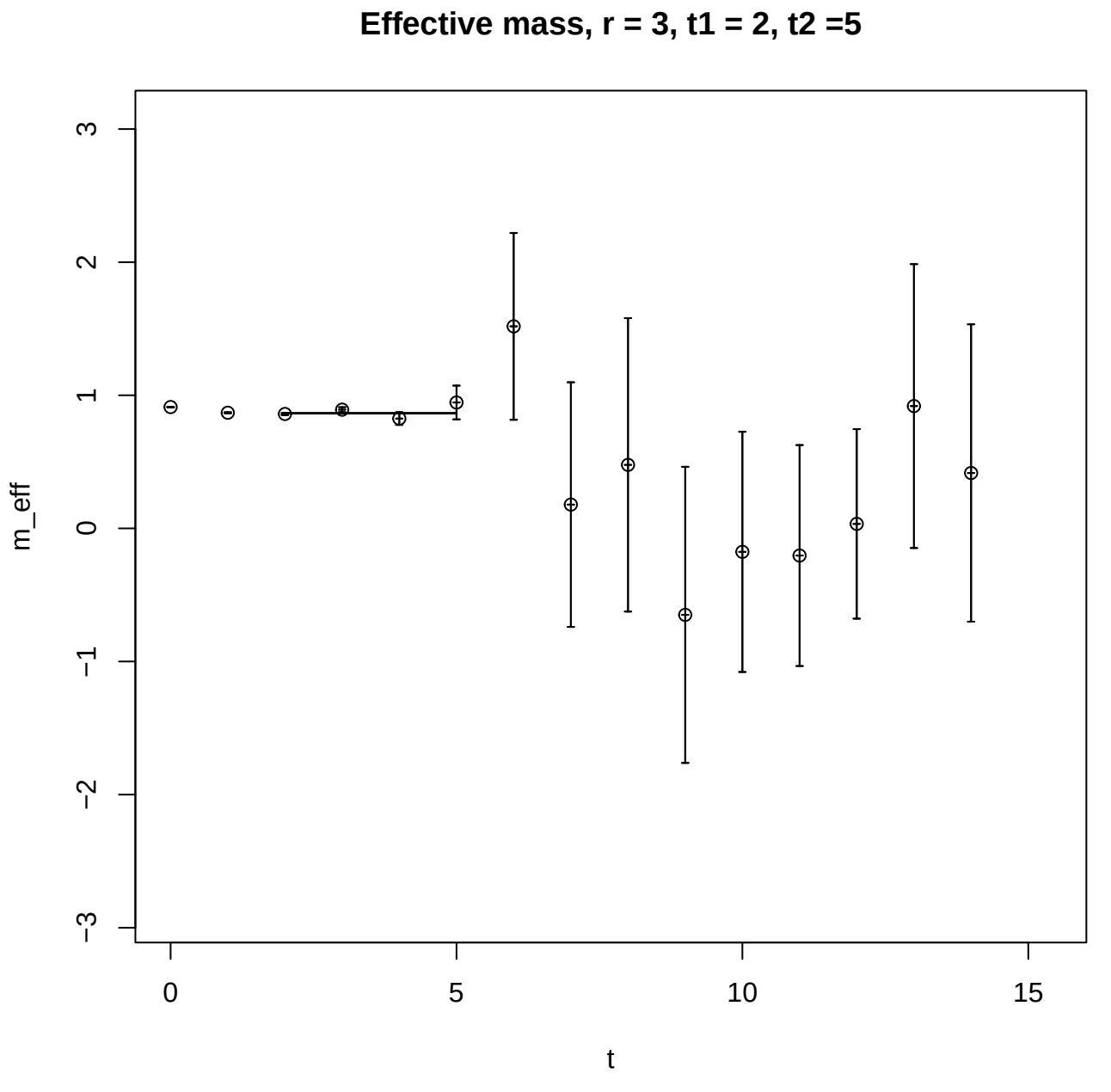

Effective mass, r = 3, t1 = 1, t2 =8
m_eff
t
3
2
1
0
−1
−2
−3
0
5
10
15

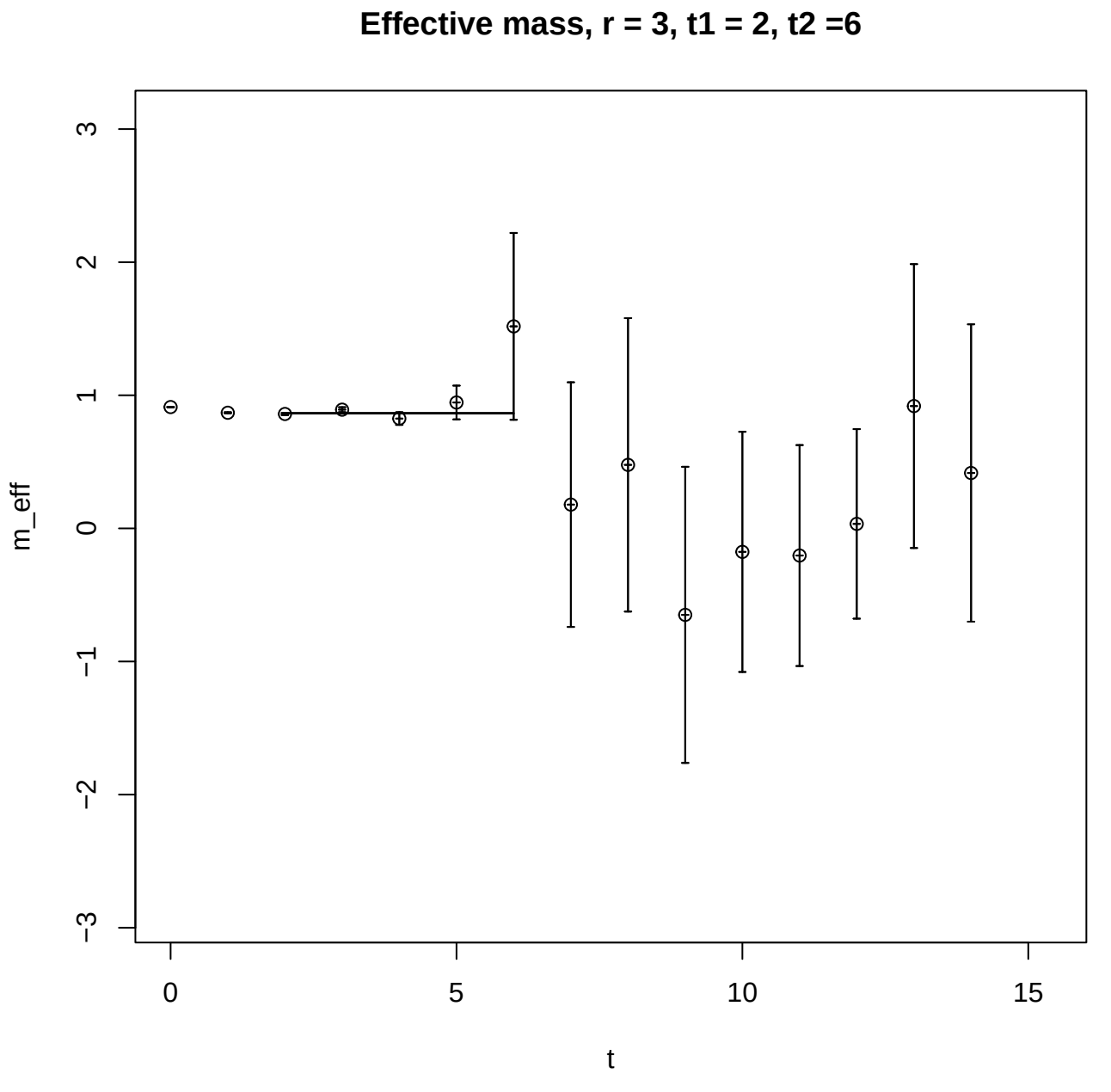

Effective mass, r = 3, t1 = 1, t2 =9
m_eff
t
3
2
1
0
−1
−2
−3
0
5
10
15

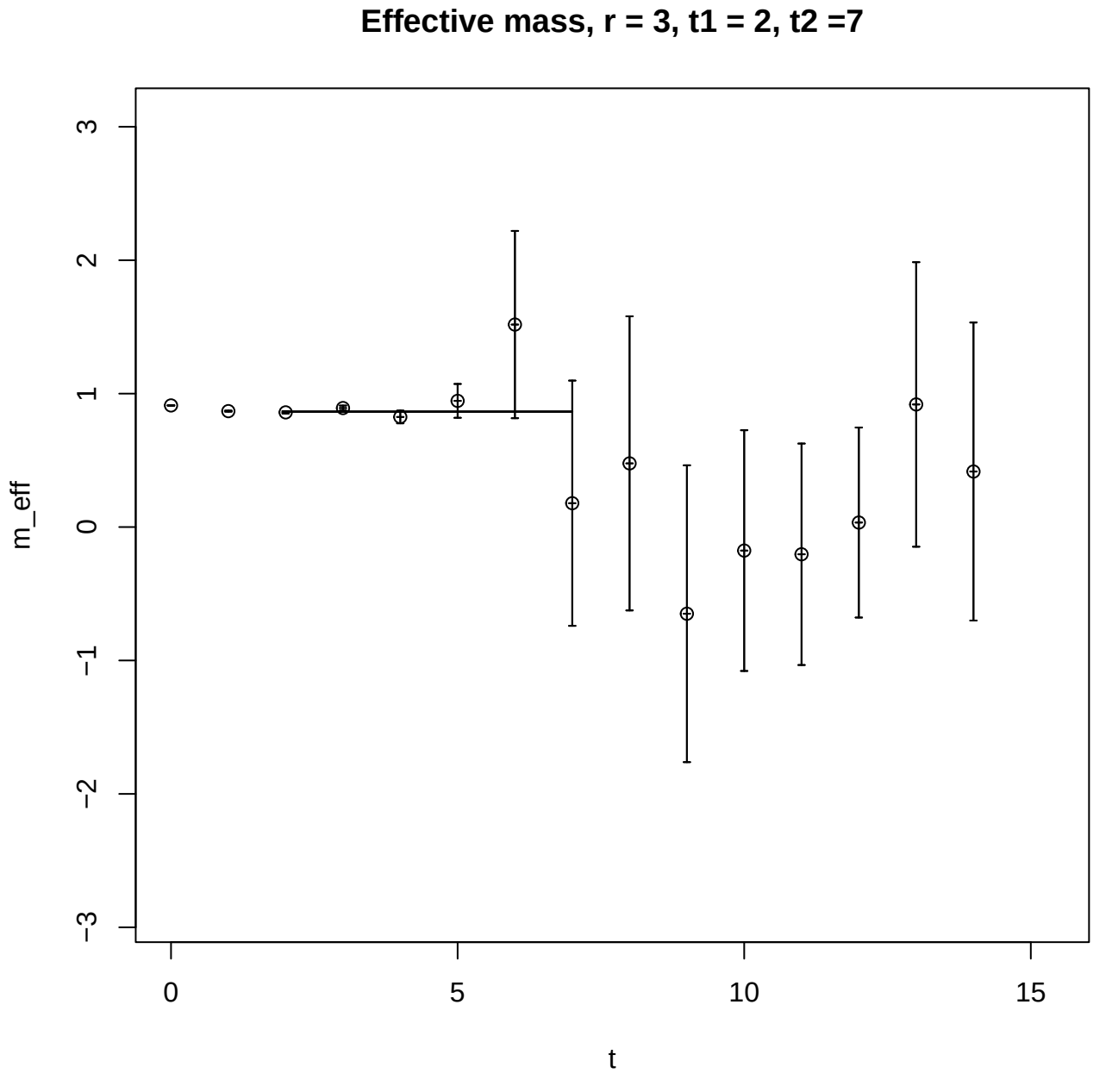

Effective mass, r = 3, t1 = 1, t2 =10
m_eff
t
3
2
1
0
−1
−2
−3
0
5
10
15

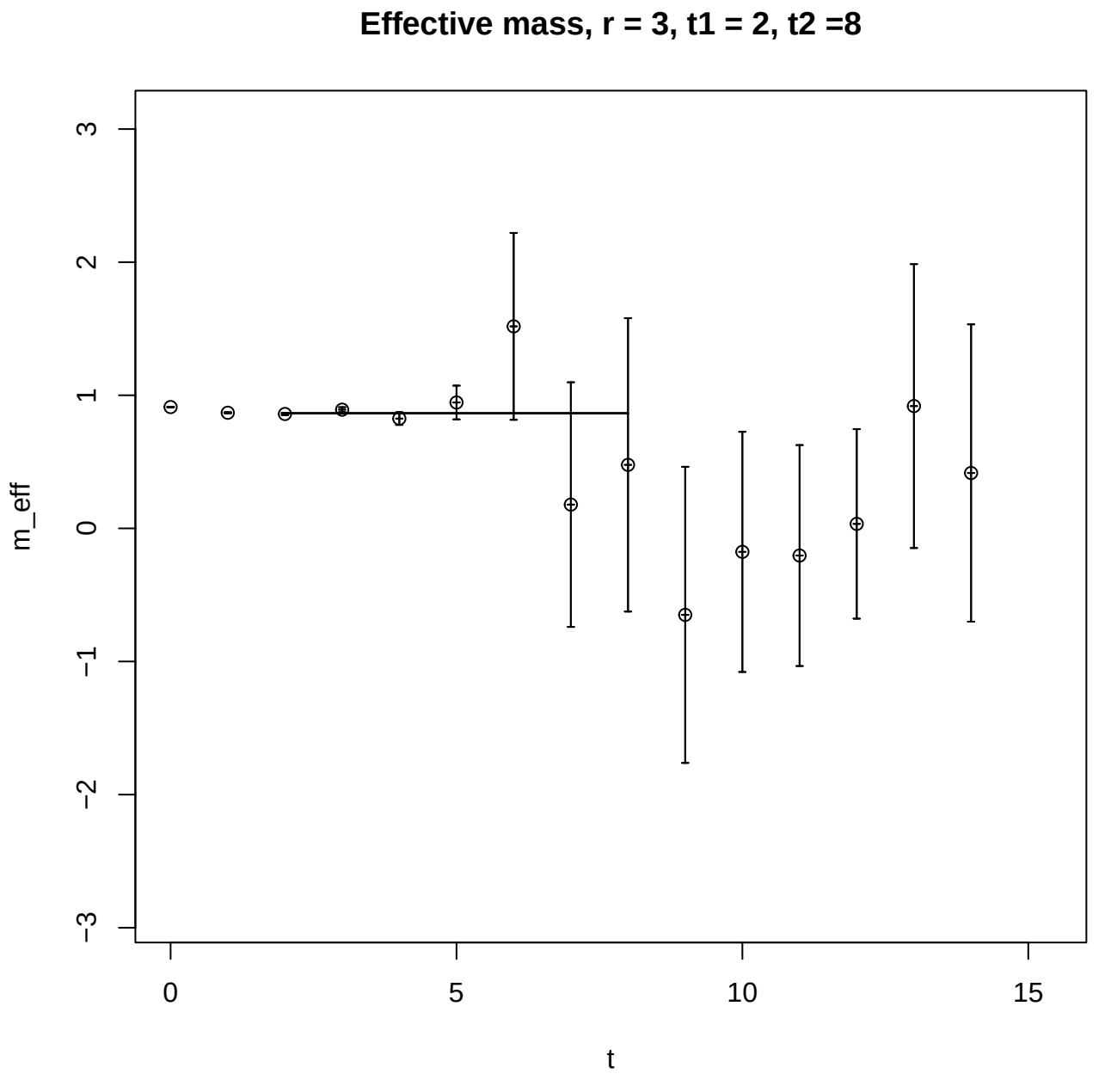

Effective mass, r = 3, t1 = 2, t2 =3
m_eff
t
3
2
1
0
−1
−2
−3
0
5
10
15

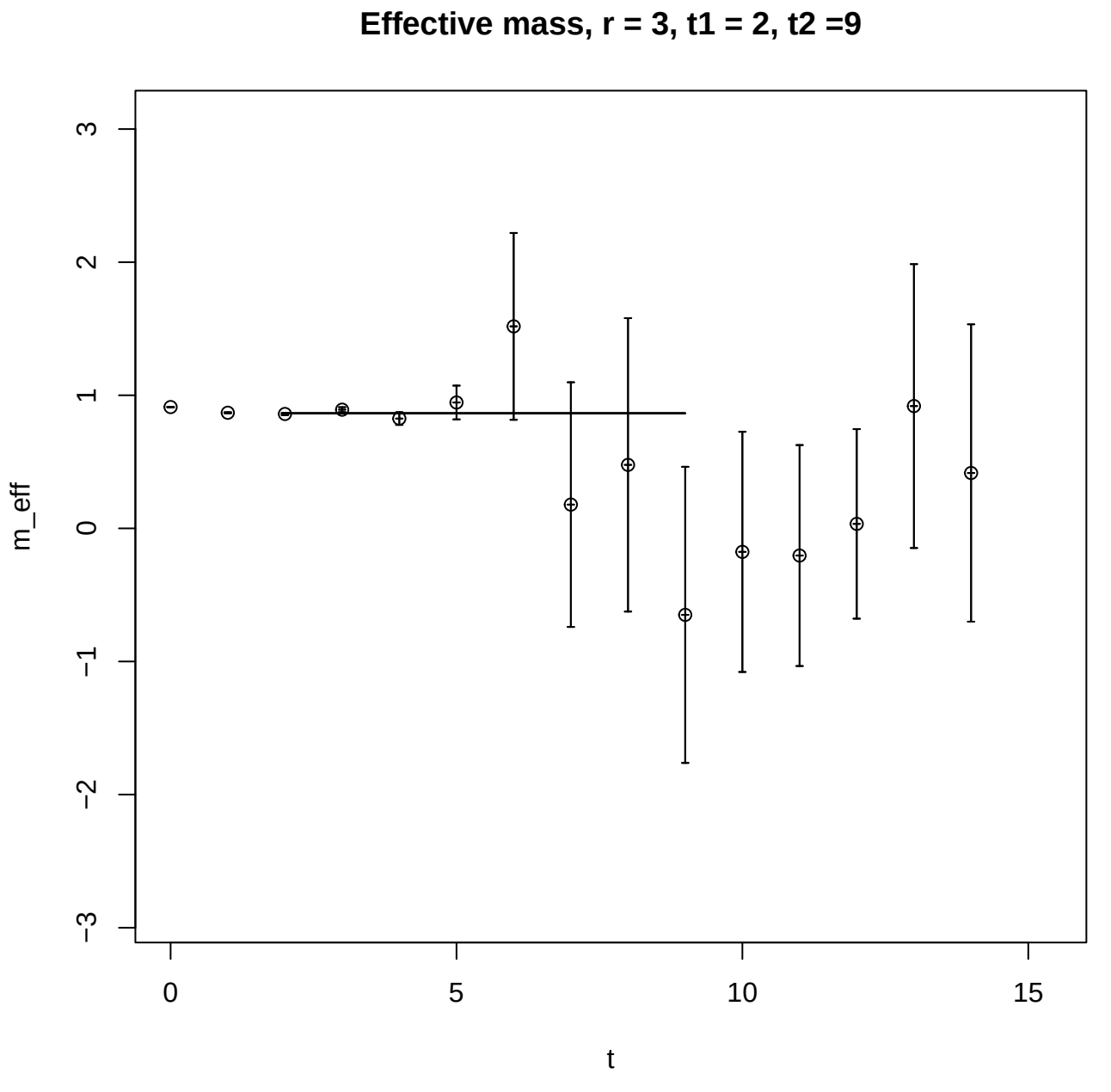

Effective mass, r = 3, t1 = 2, t2 =4
m_eff
t
3
2
1
0
−1
−2
−3
0
5
10
15

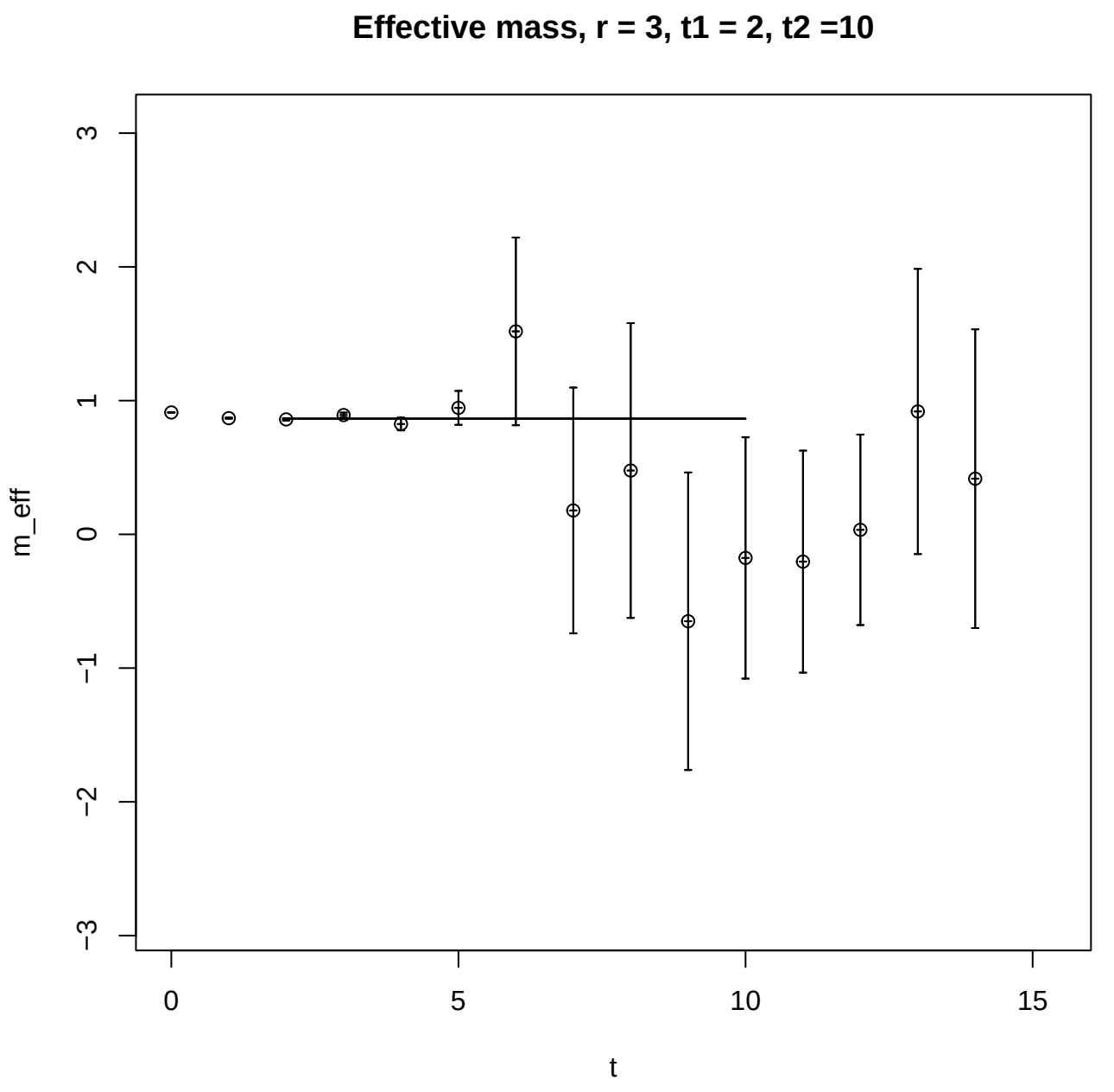

Effective mass, r = 3, t1 = 2, t2 =5
m_eff
t
3
2
1
0
−1
−2
−3
0
5
10
15

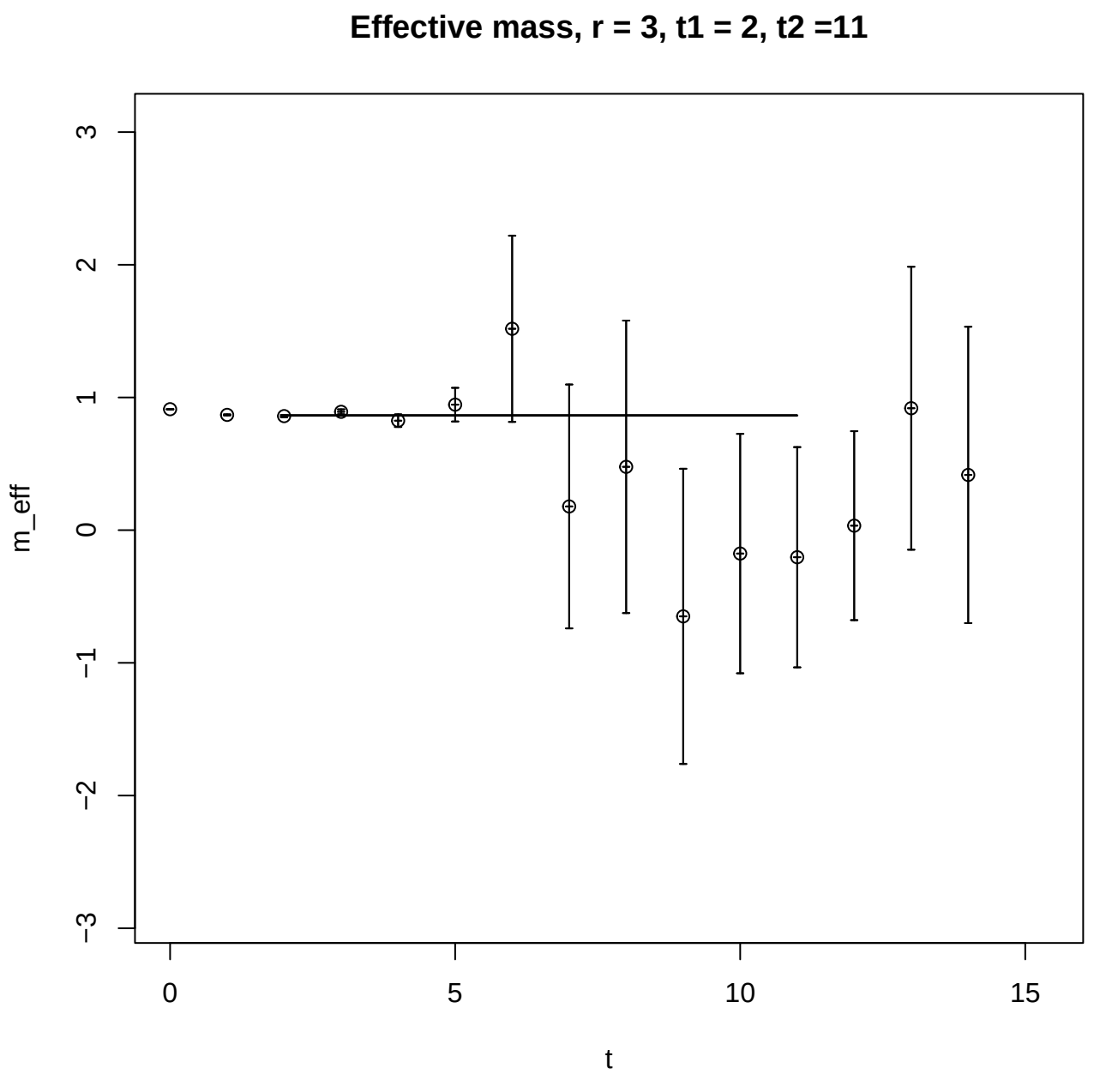

Effective mass, r = 3, t1 = 2, t2 =6
m_eff
t
3
2
1
0
−1
−2
−3
0
5
10
15

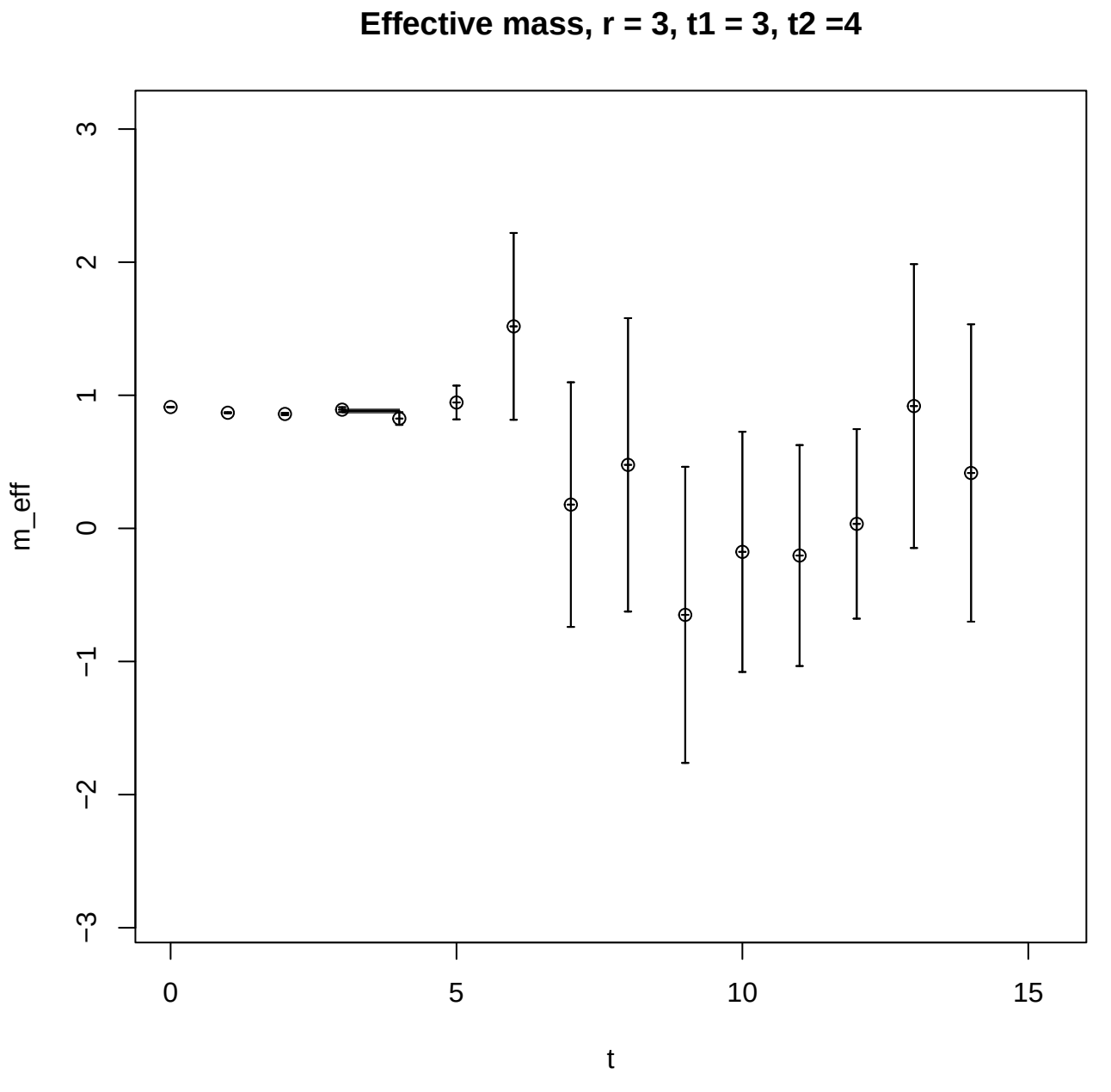

Effective mass, r = 3, t1 = 2, t2 =7
m_eff
t
3
2
1
0
−1
−2
−3
0
5
10
15

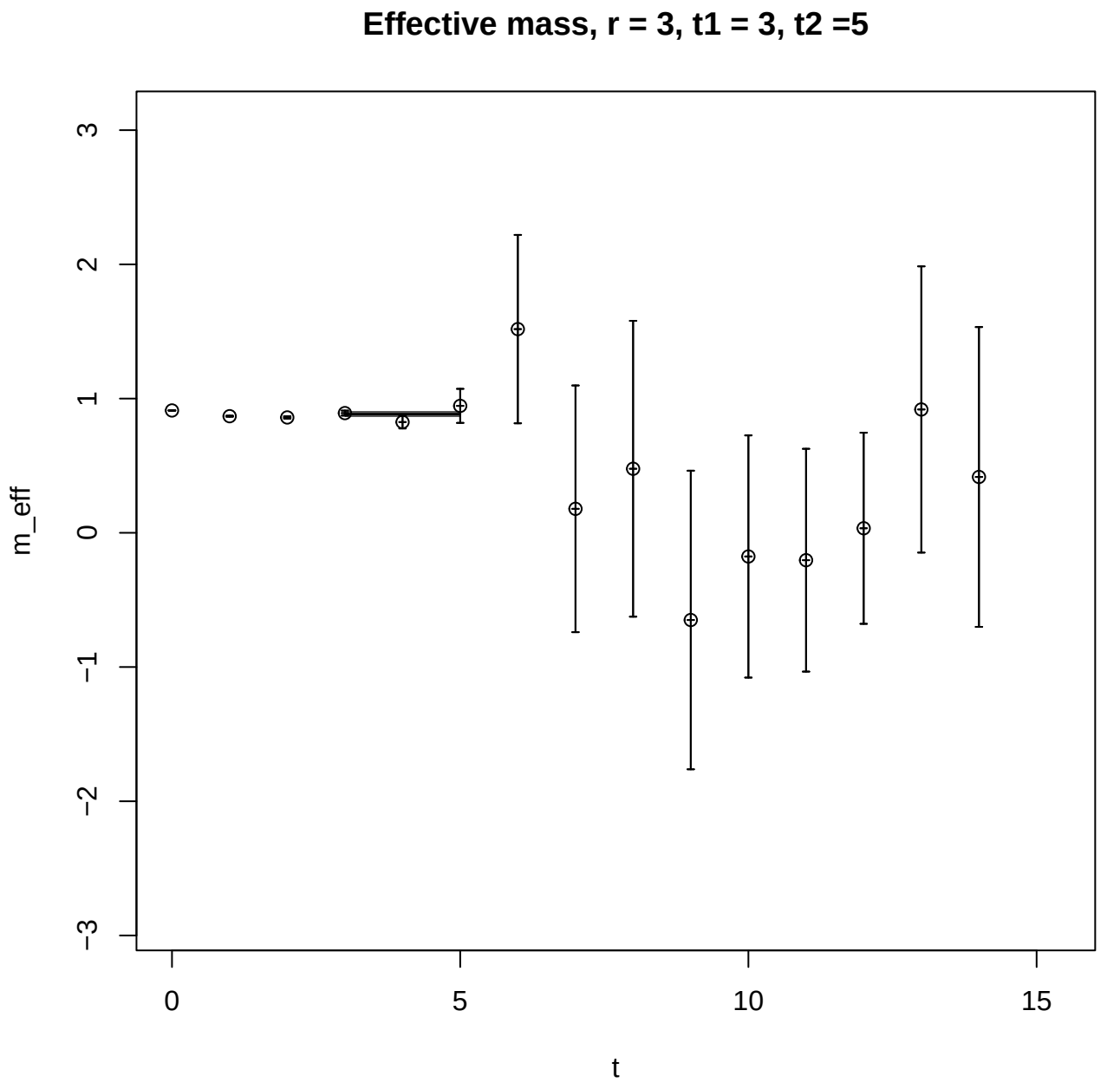

Effective mass, r = 3, t1 = 2, t2 =8
m_eff
t
3
2
1
0
−1
−2
−3
0
5
10
15

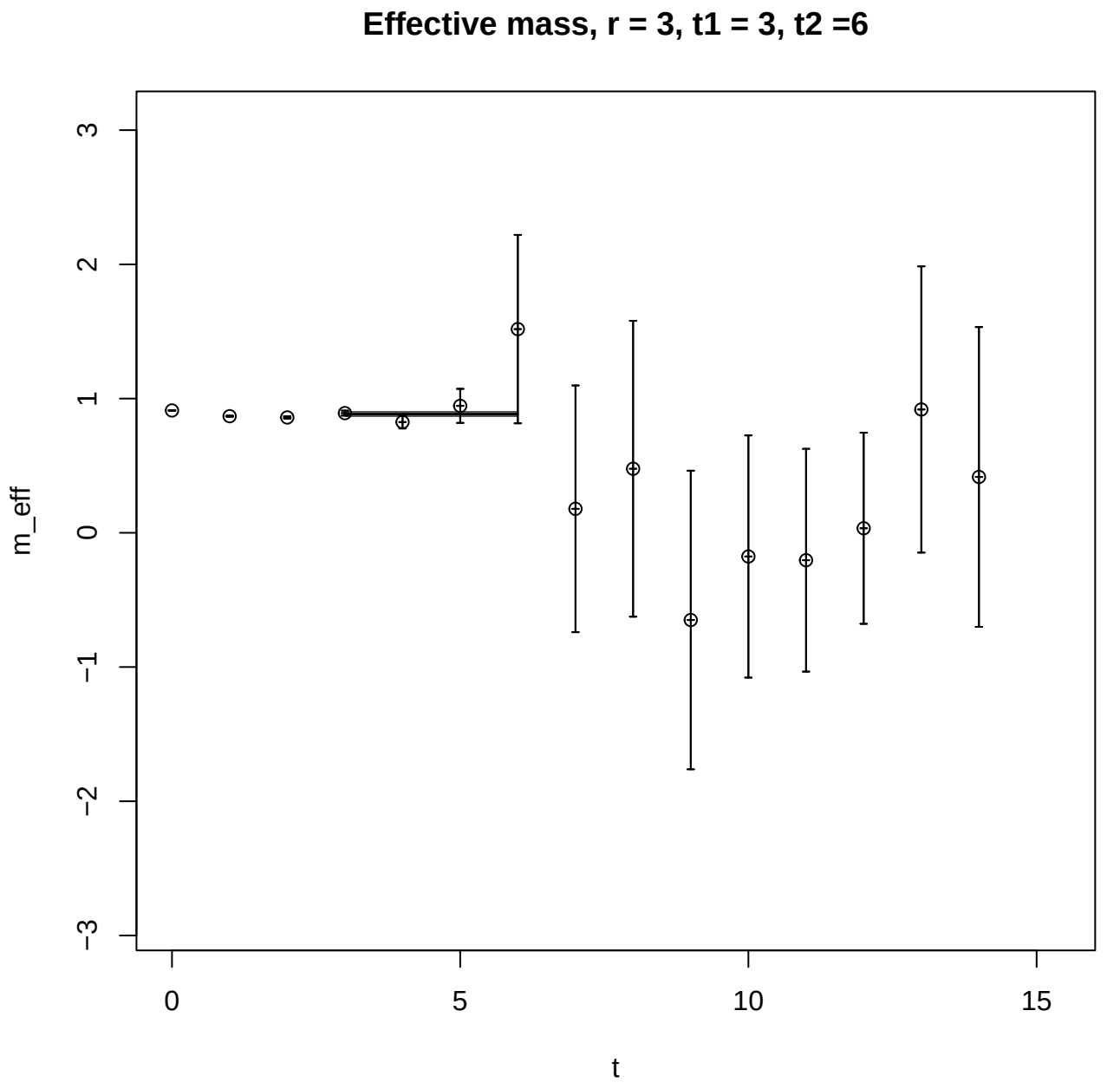

Effective mass, r = 3, t1 = 2, t2 =9
m_eff
t
3
2
1
0
−1
−2
−3
0
5
10
15


Effective mass, r = 3, t1 = 2, t2 =10
m_eff
t
3
2
1
0
−1
−2
−3
0
5
10
15


Effective mass, r = 3, t1 = 2, t2 =11
m_eff
t
3
2
1
0
−1
−2
−3
0
5
10
15


Effective mass, r = 3, t1 = 3, t2 =4
m_eff
t
3
2
1
0
−1
−2
−3
0
5
10
15


Effective mass, r = 3, t1 = 3, t2 =5
m_eff
t
3
2
1
0
−1
−2
−3
0
5
10
15


Effective mass, r = 3, t1 = 3, t2 =6
m_eff
t
3
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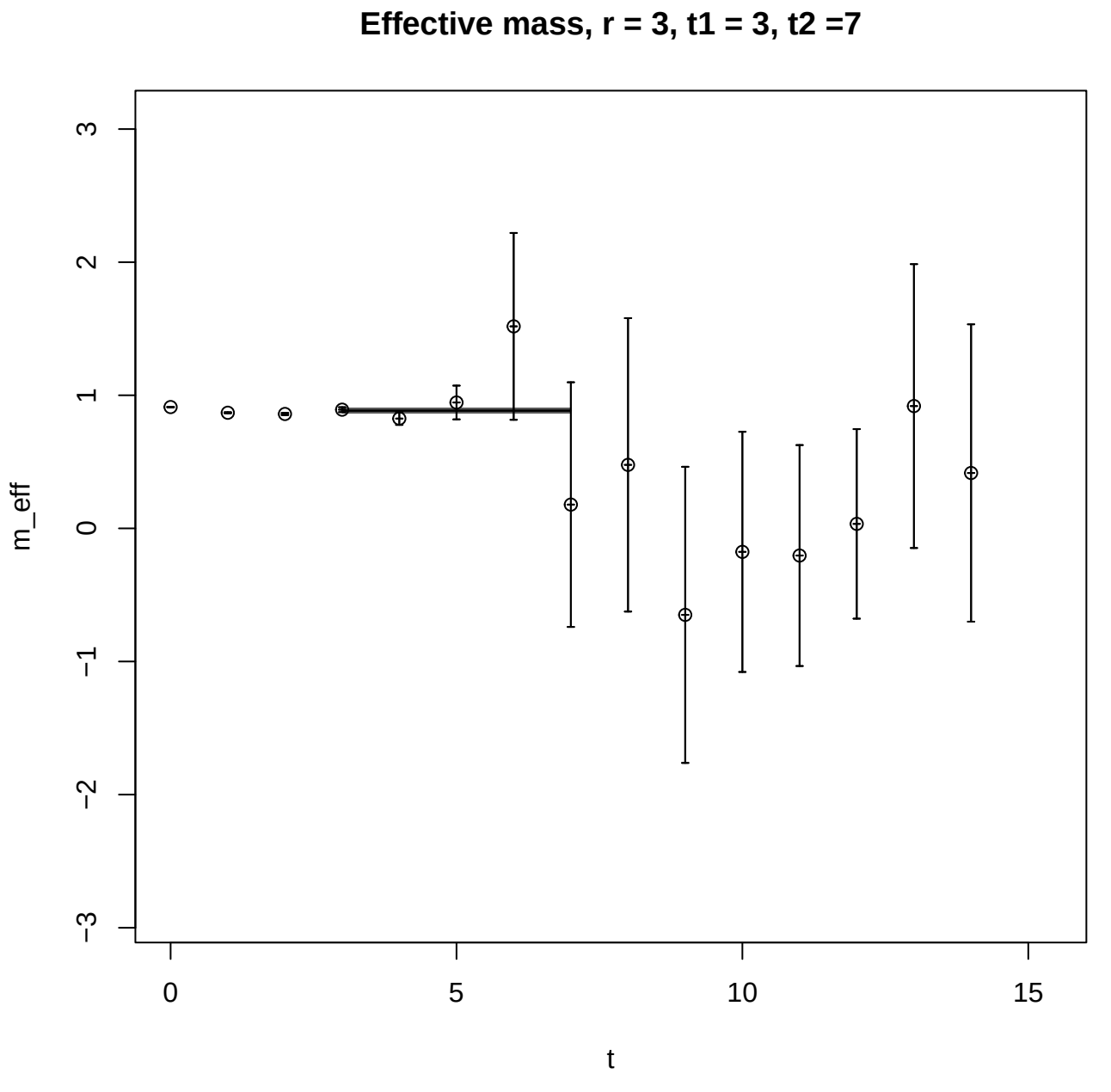
0
−1
−2
−3
0
5
10
15


Effective mass, r = 3, t1 = 3, t2 =7
m_eff
t
3
2
1
0
−1
−2
−3
0
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10
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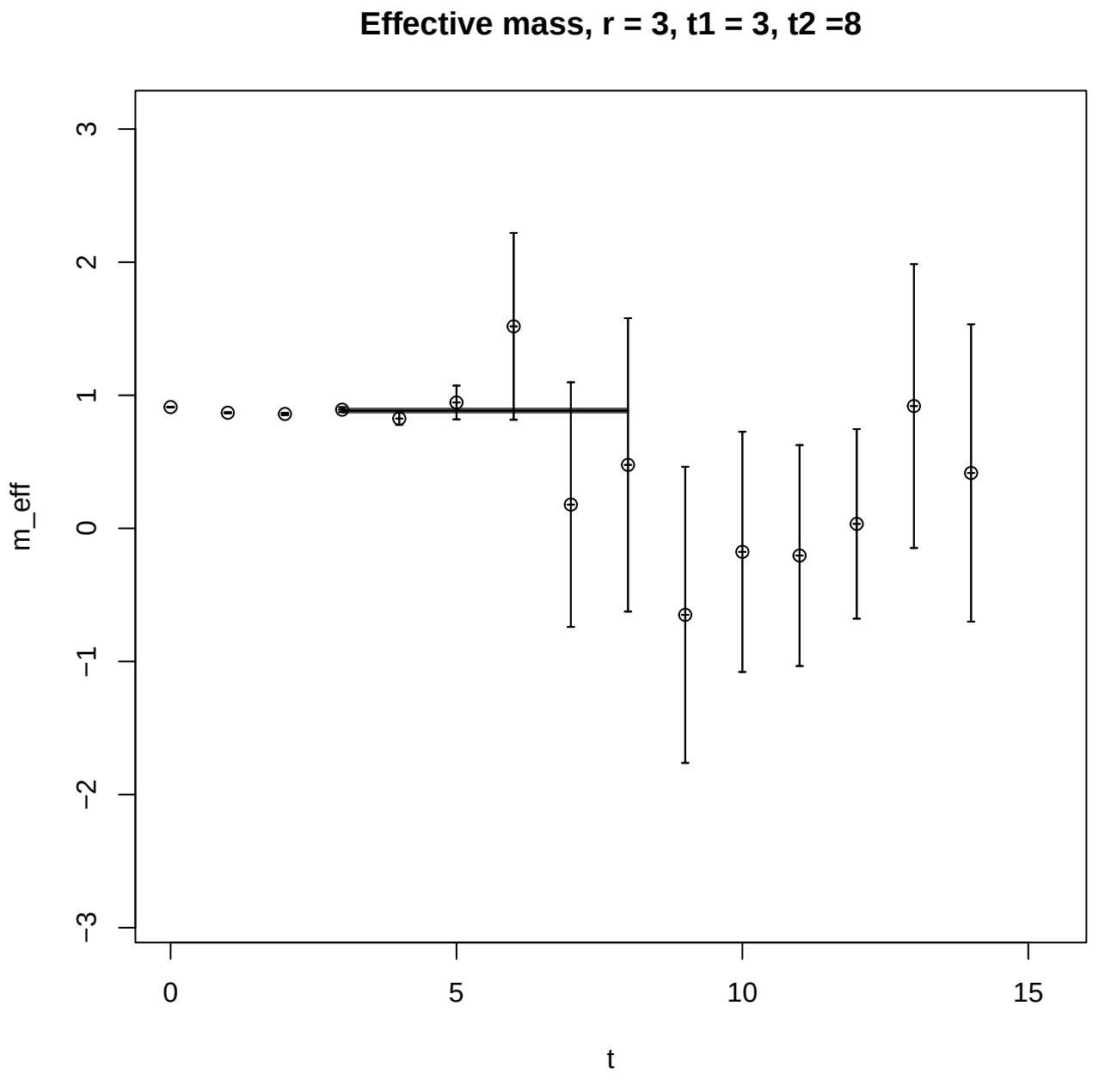
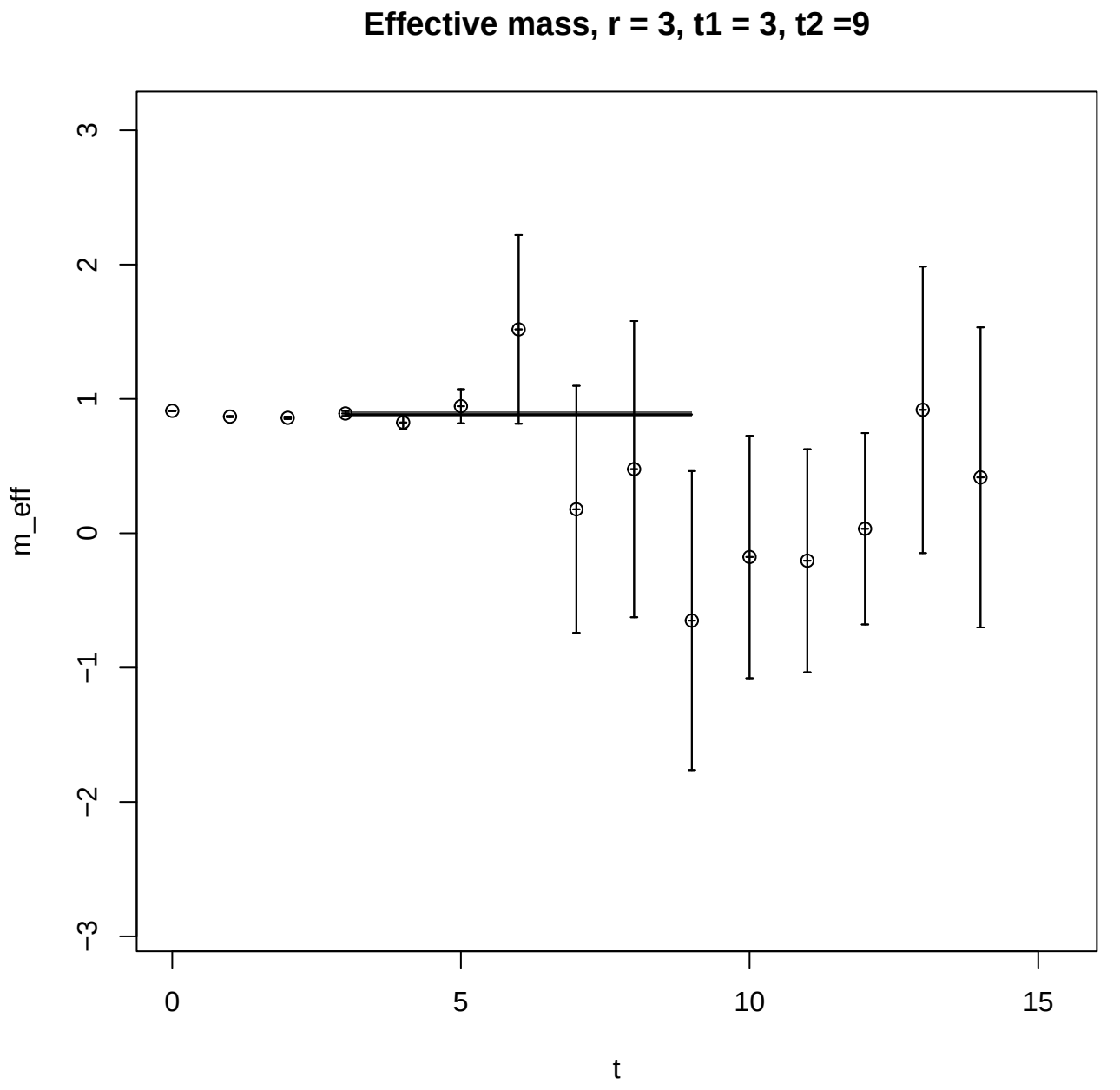
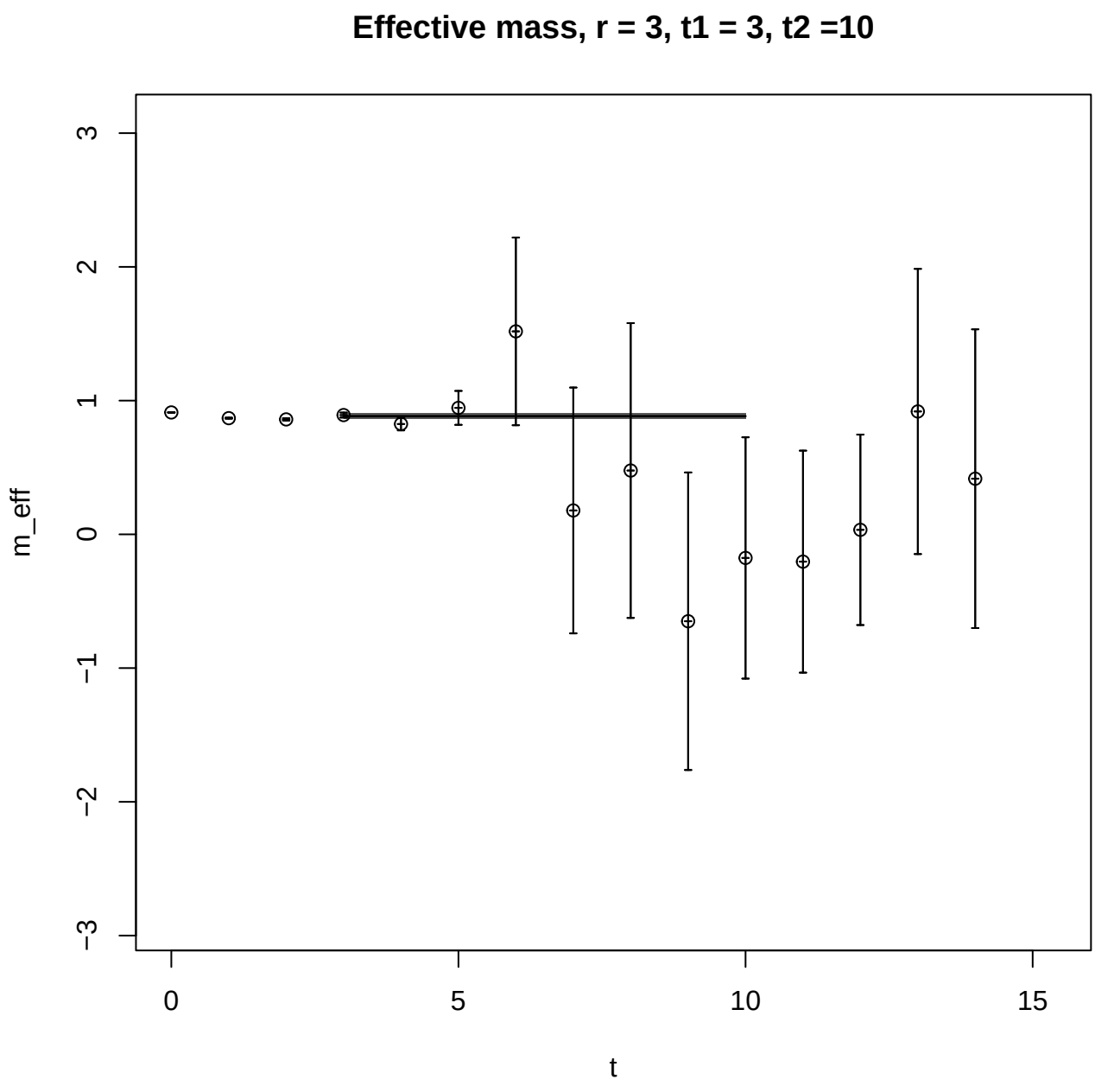
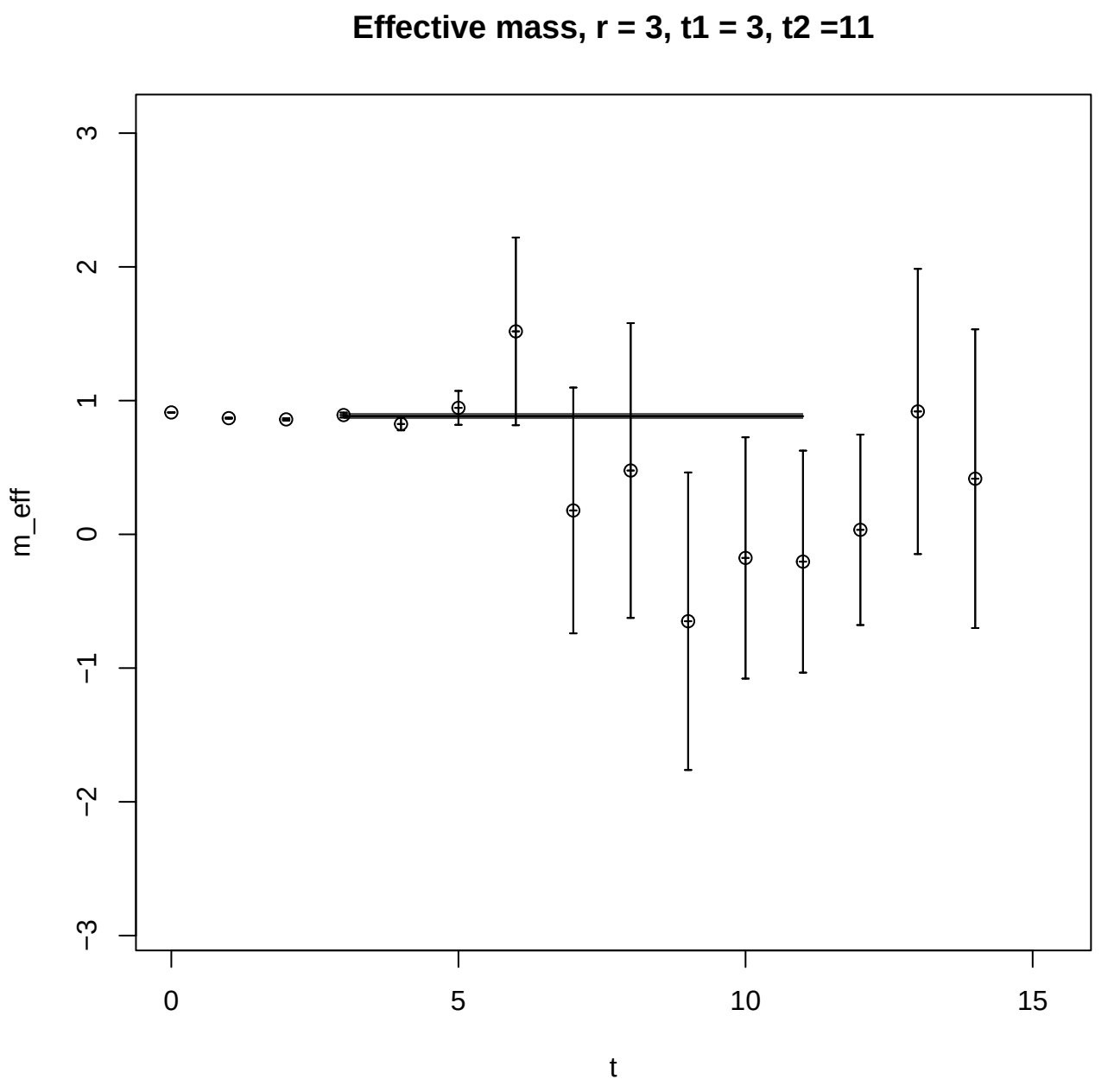
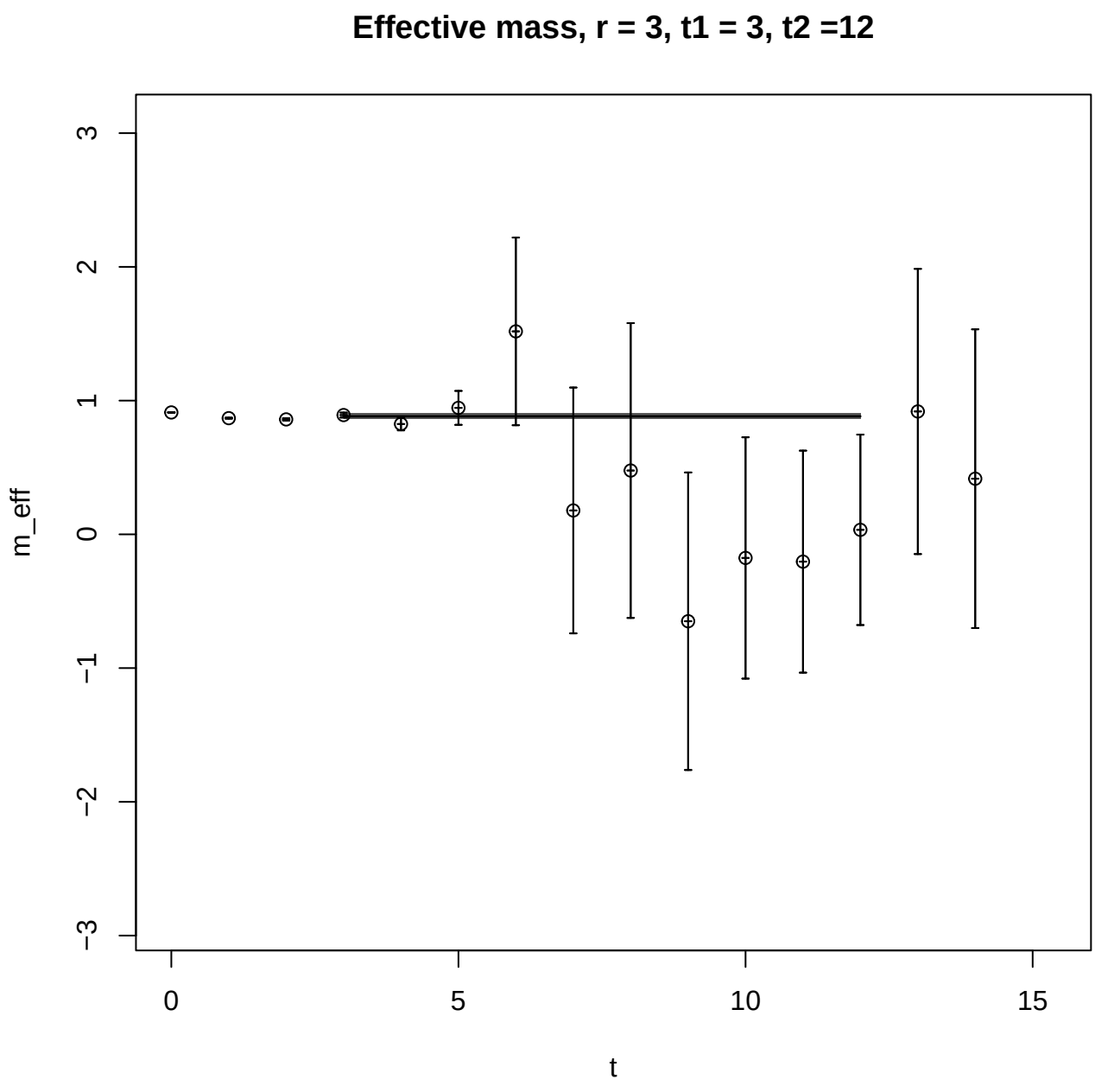
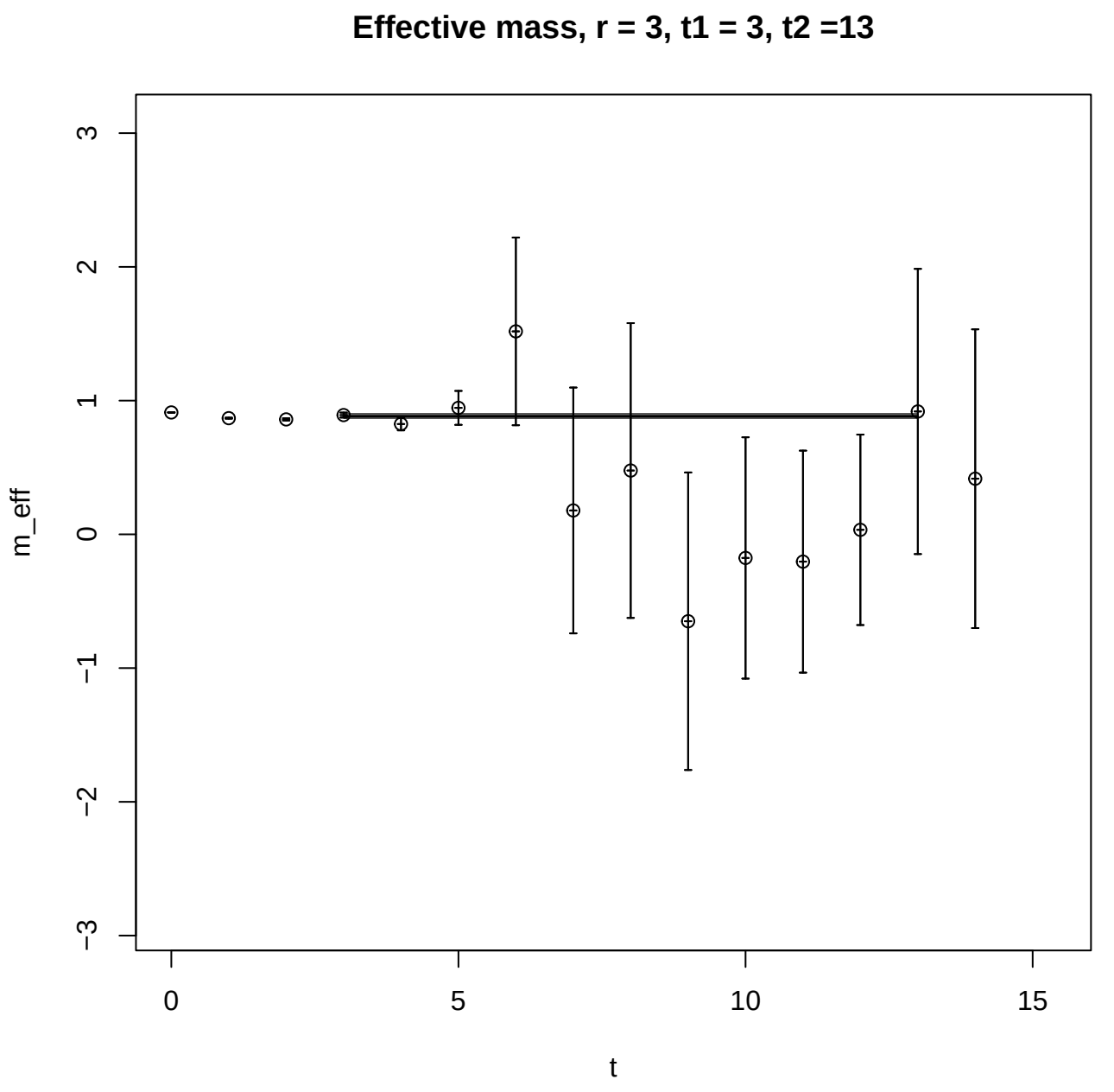

Effective mass, r = 3, t1 = 3, t2 =8
m_eff
t
3
2
1
0
−1
−2
−3
0
5
10
15

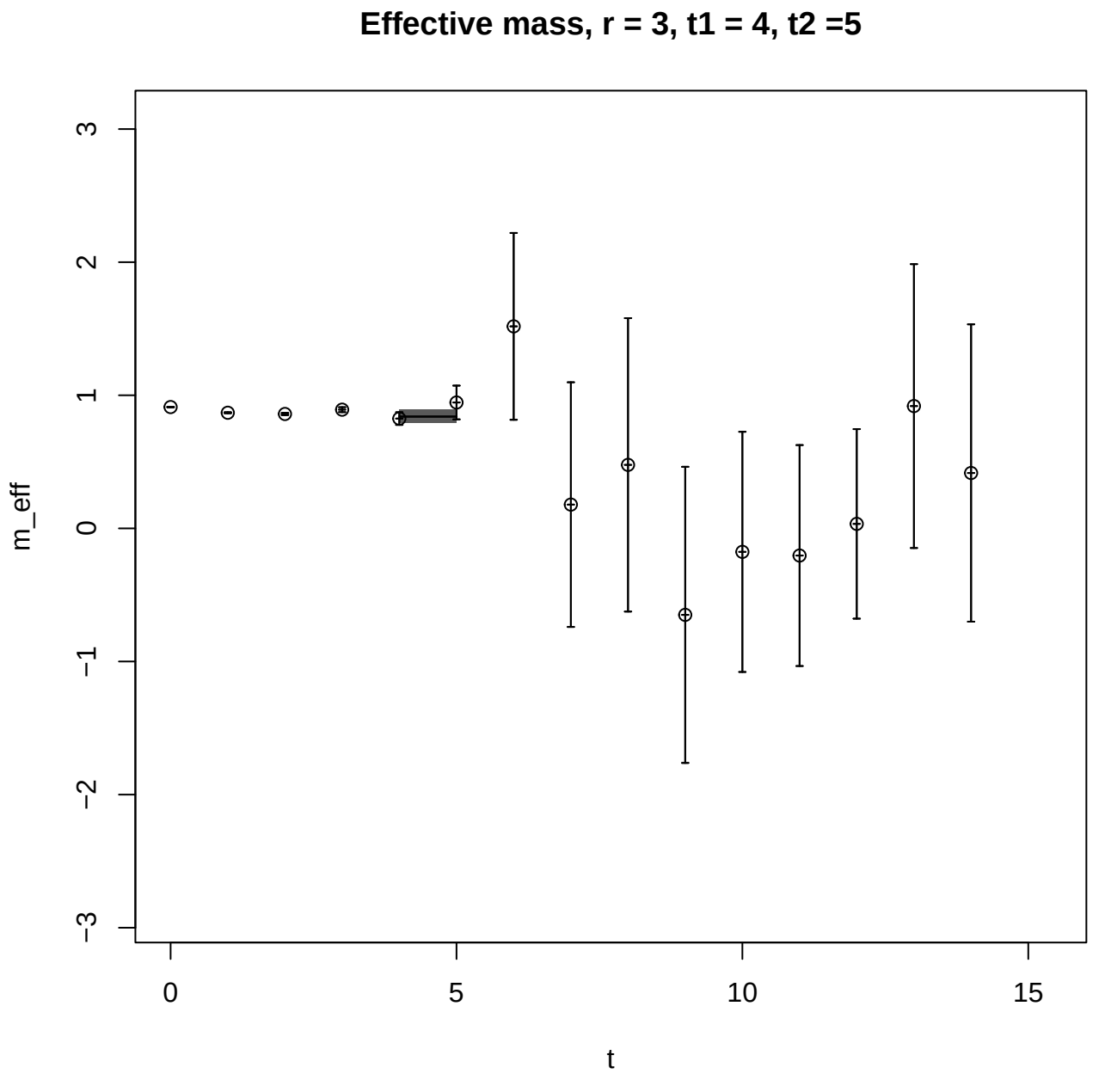

Effective mass, r = 3, t1 = 3, t2 =9
m_eff
t
3
2
1
0
−1
−2
−3
0
5
10
15

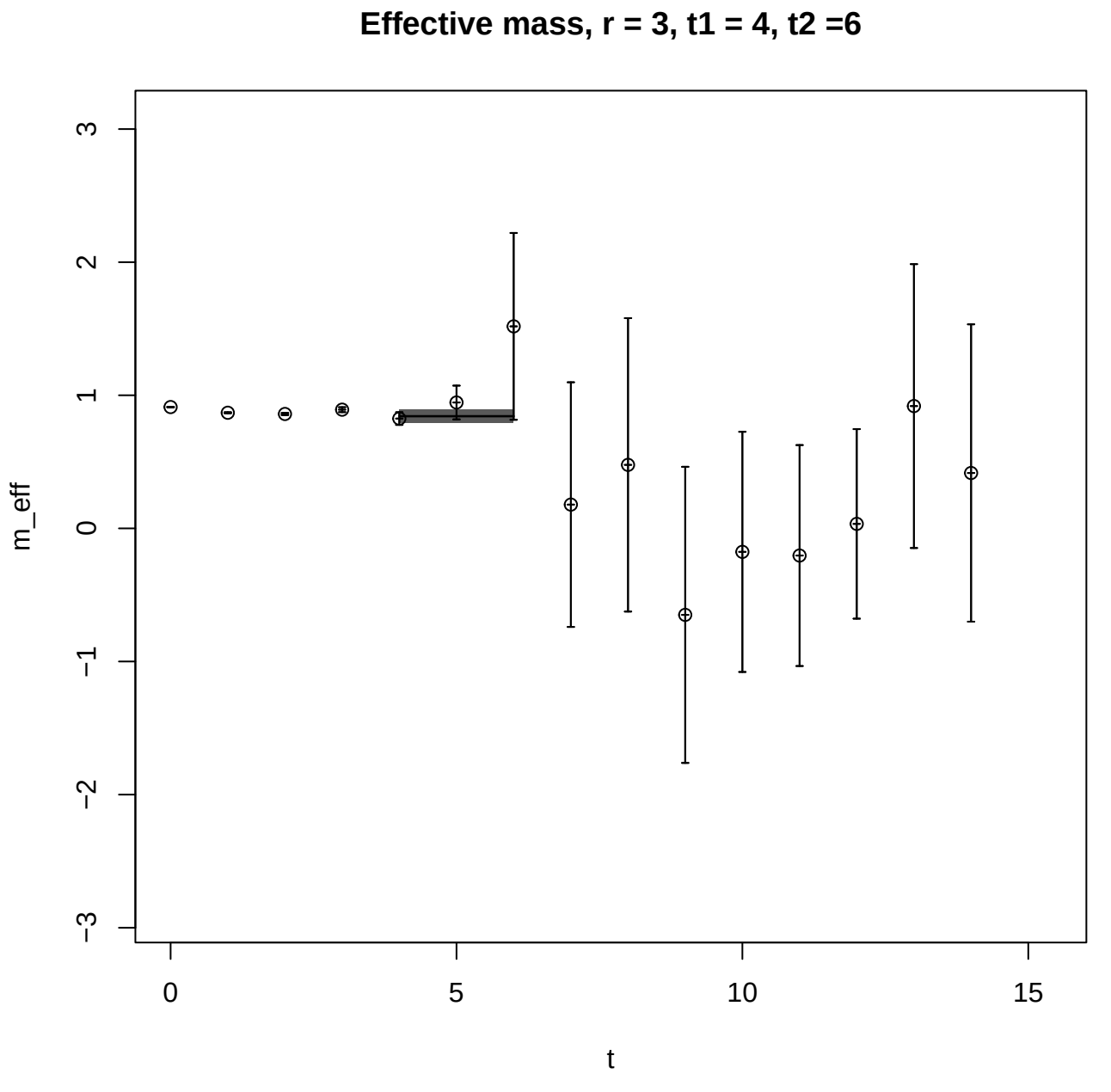

Effective mass, r = 3, t1 = 3, t2 =10
m_eff
t
3
2
1
0
−1
−2
−3
0
5
10
15

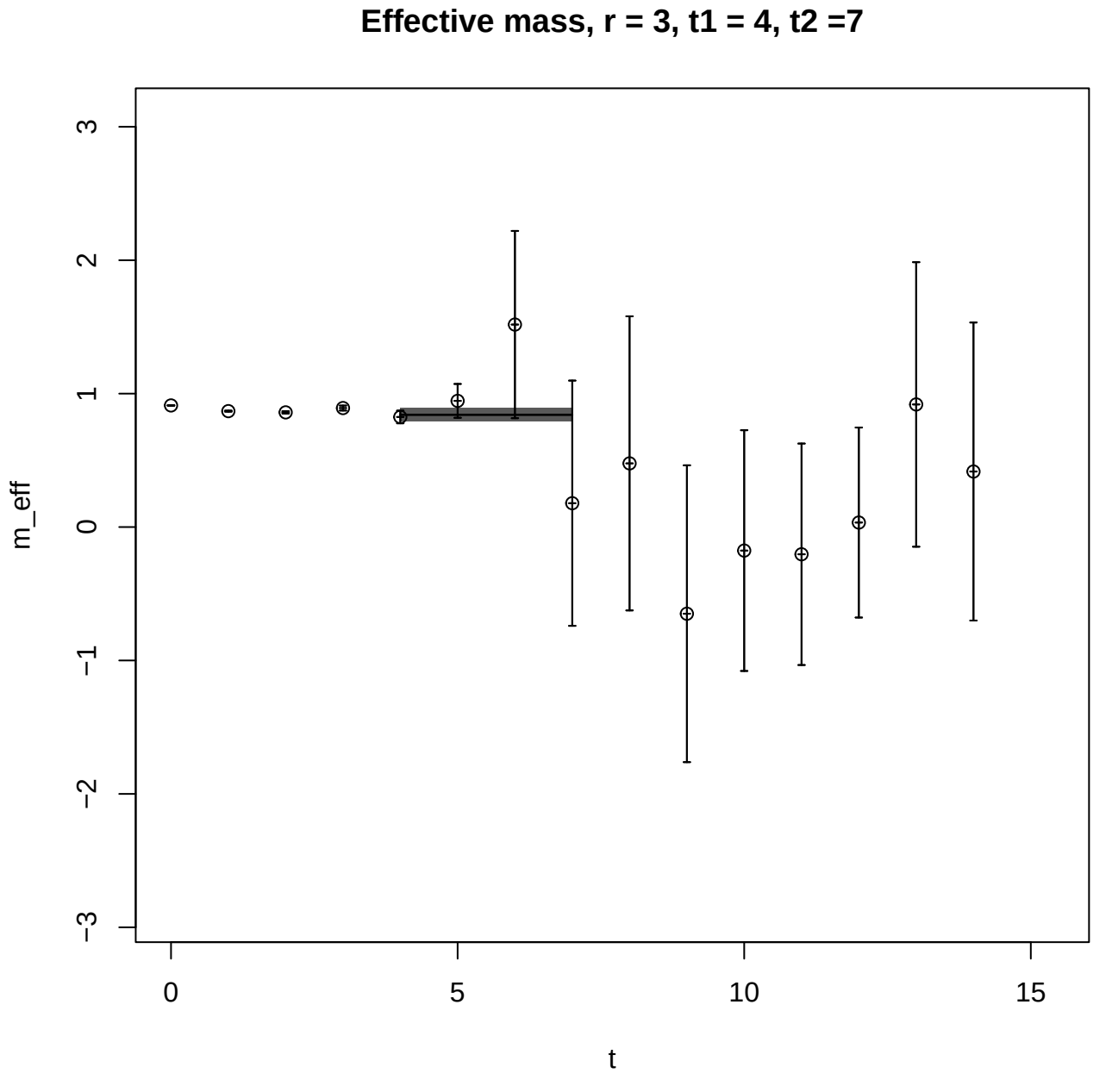

Effective mass, r = 3, t1 = 3, t2 =11
m_eff
t
3
2
1
0
−1
−2
−3
0
5
10
15

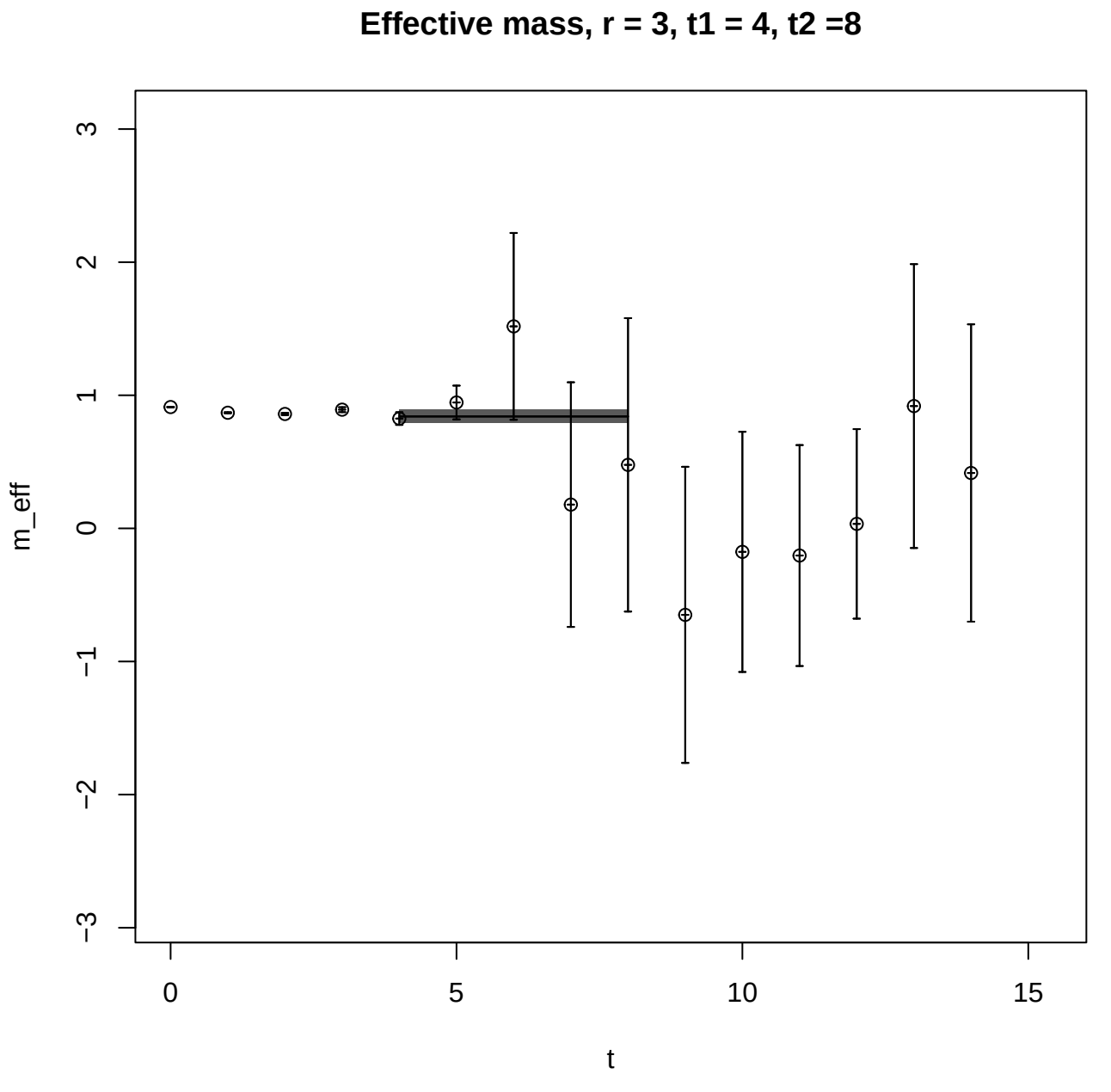

Effective mass, r = 3, t1 = 3, t2 =12
m_eff
t
3
2
1
0
−1
−2
−3
0
5
10
15

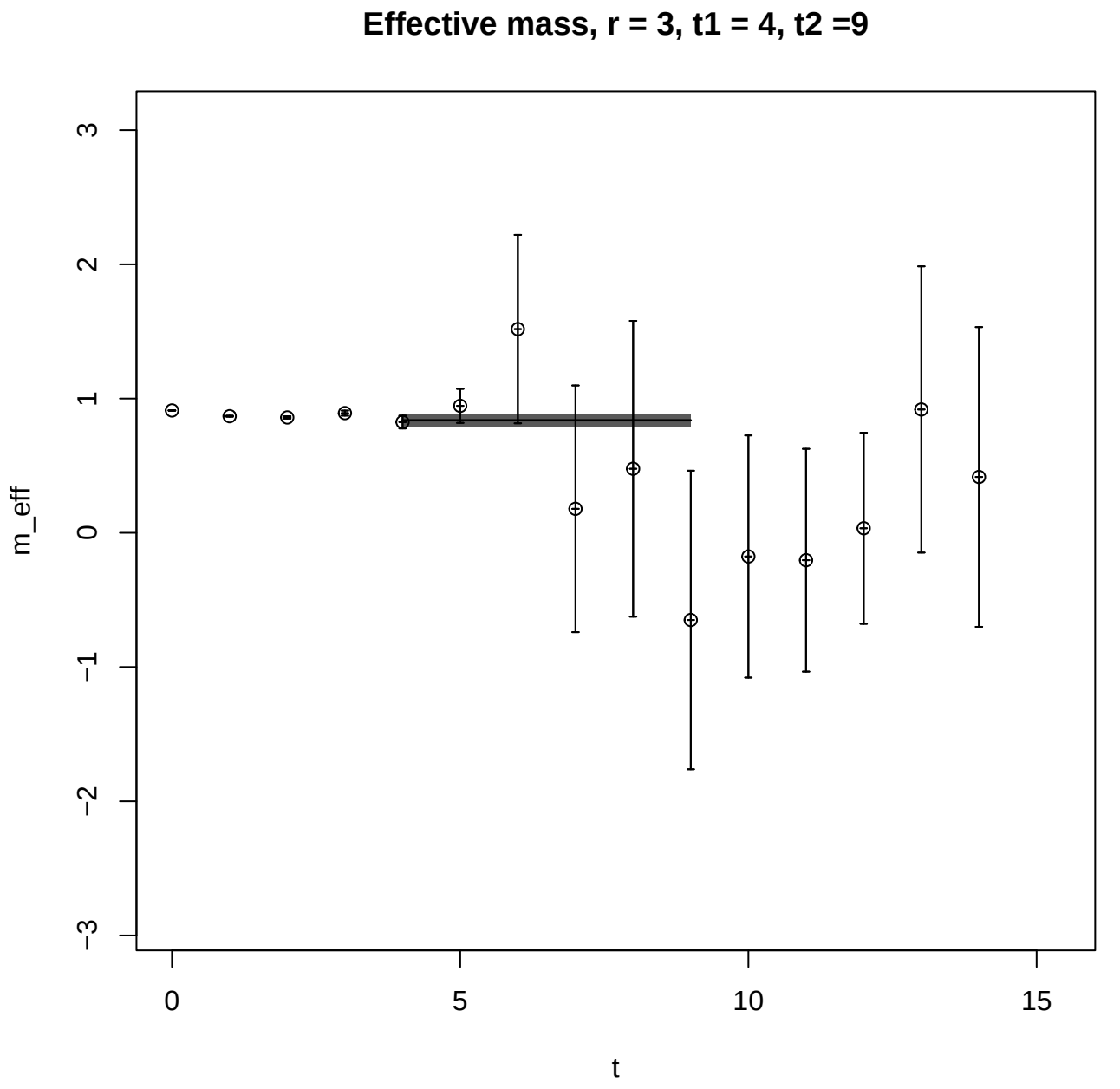

Effective mass, r = 3, t1 = 3, t2 =13
m_eff
t
3
2
1
0
−1
−2
−3
0
5
10
15

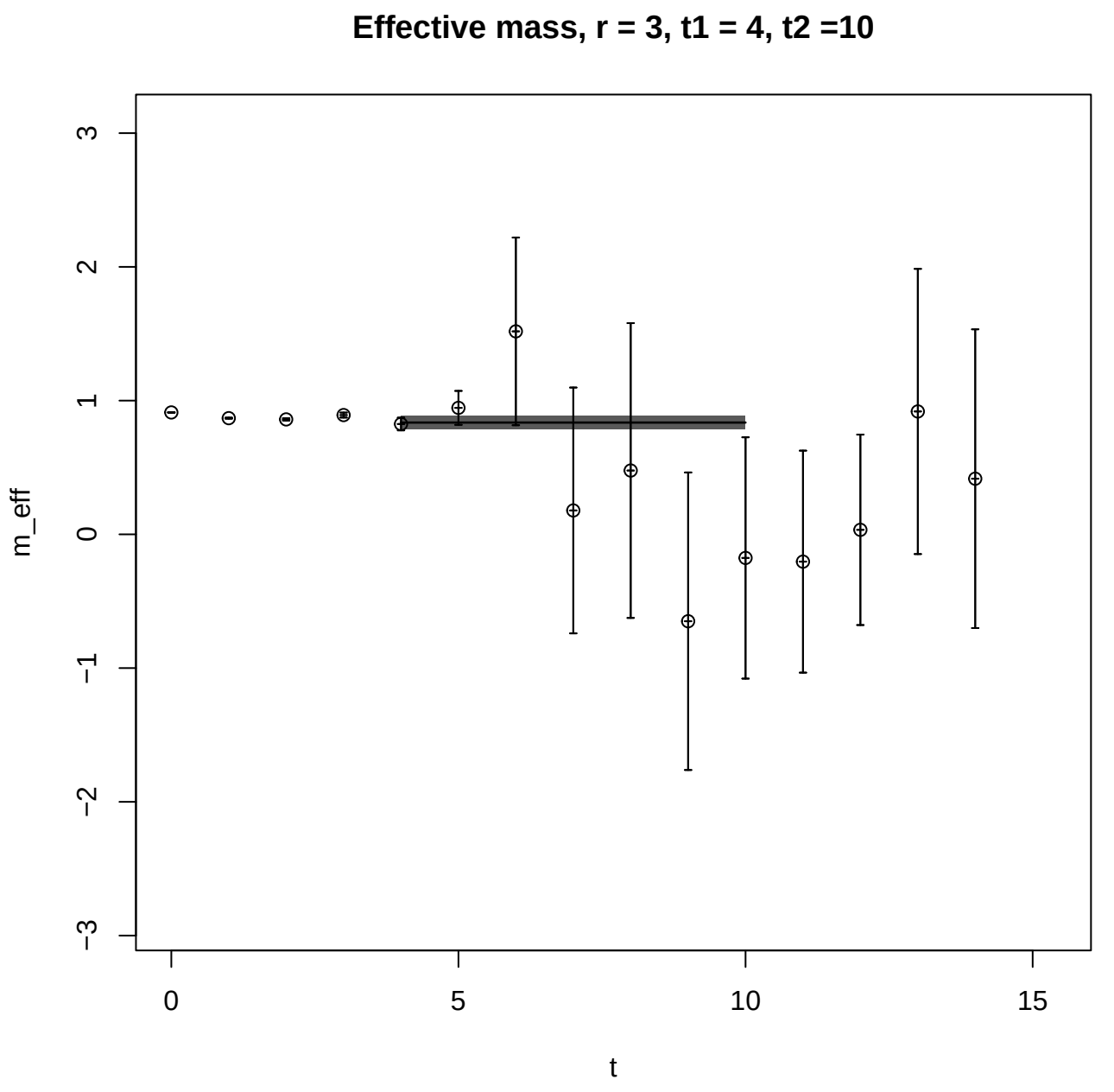

Effective mass, r = 3, t1 = 4, t2 =5
m_eff
t
3
2
1
0
−1
−2
−3
0
5
10
15

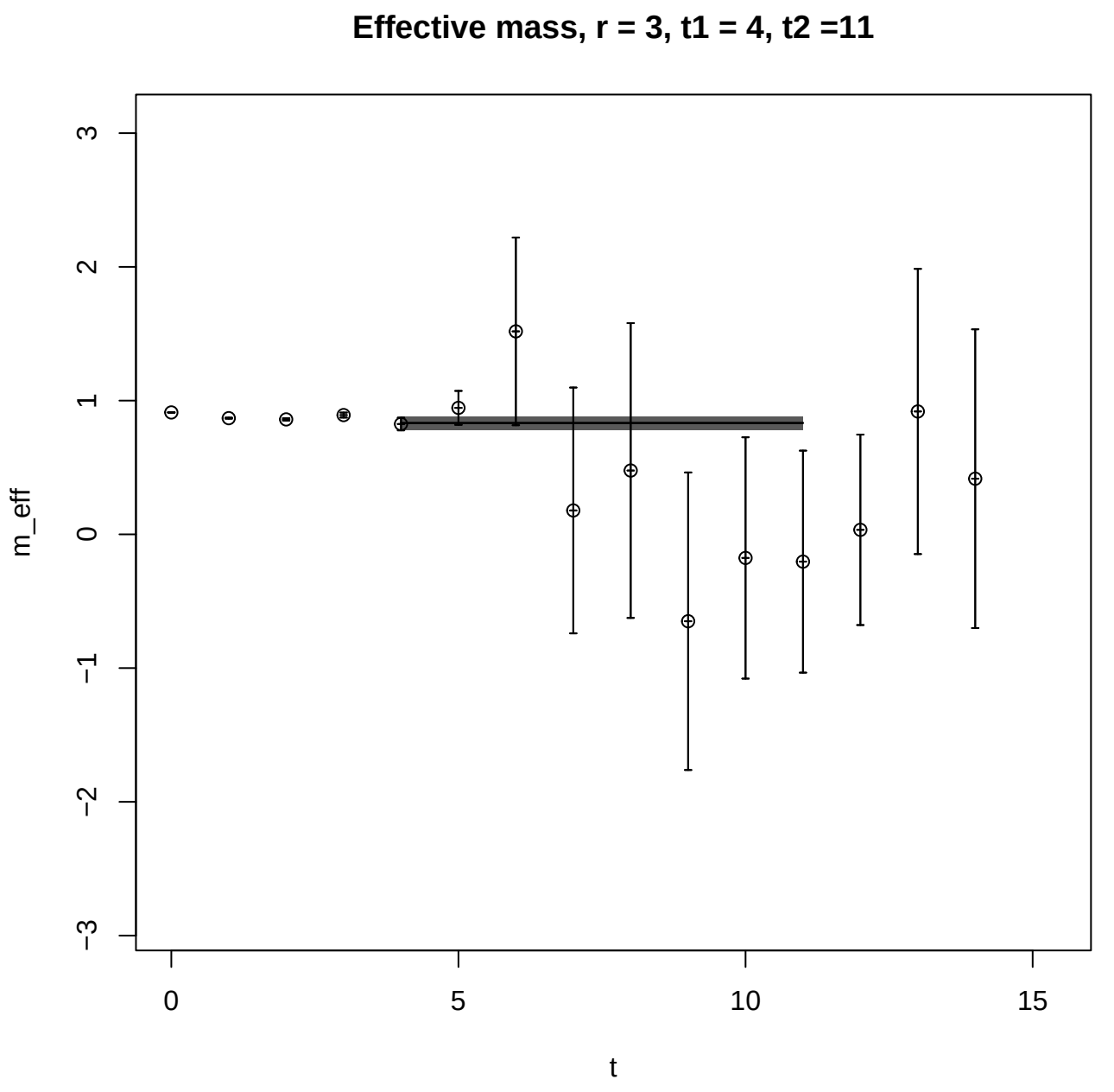

Effective mass, r = 3, t1 = 4, t2 =6
m_eff
t
3
2
1
0
−1
−2
−3
0
5
10
15

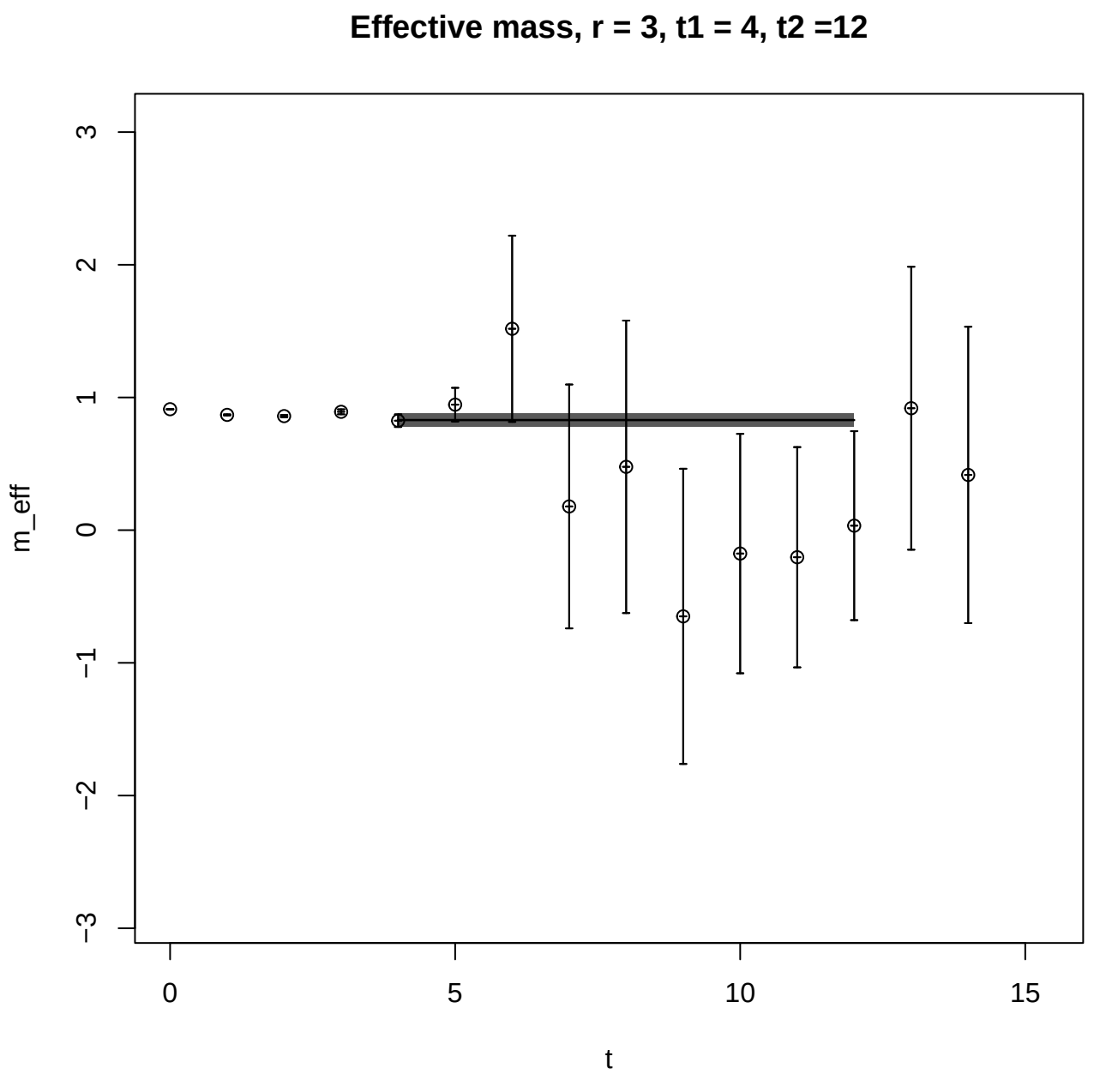

Effective mass, r = 3, t1 = 4, t2 =7
m_eff
t
3
2
1
0
−1
−2
−3
0
5
10
15

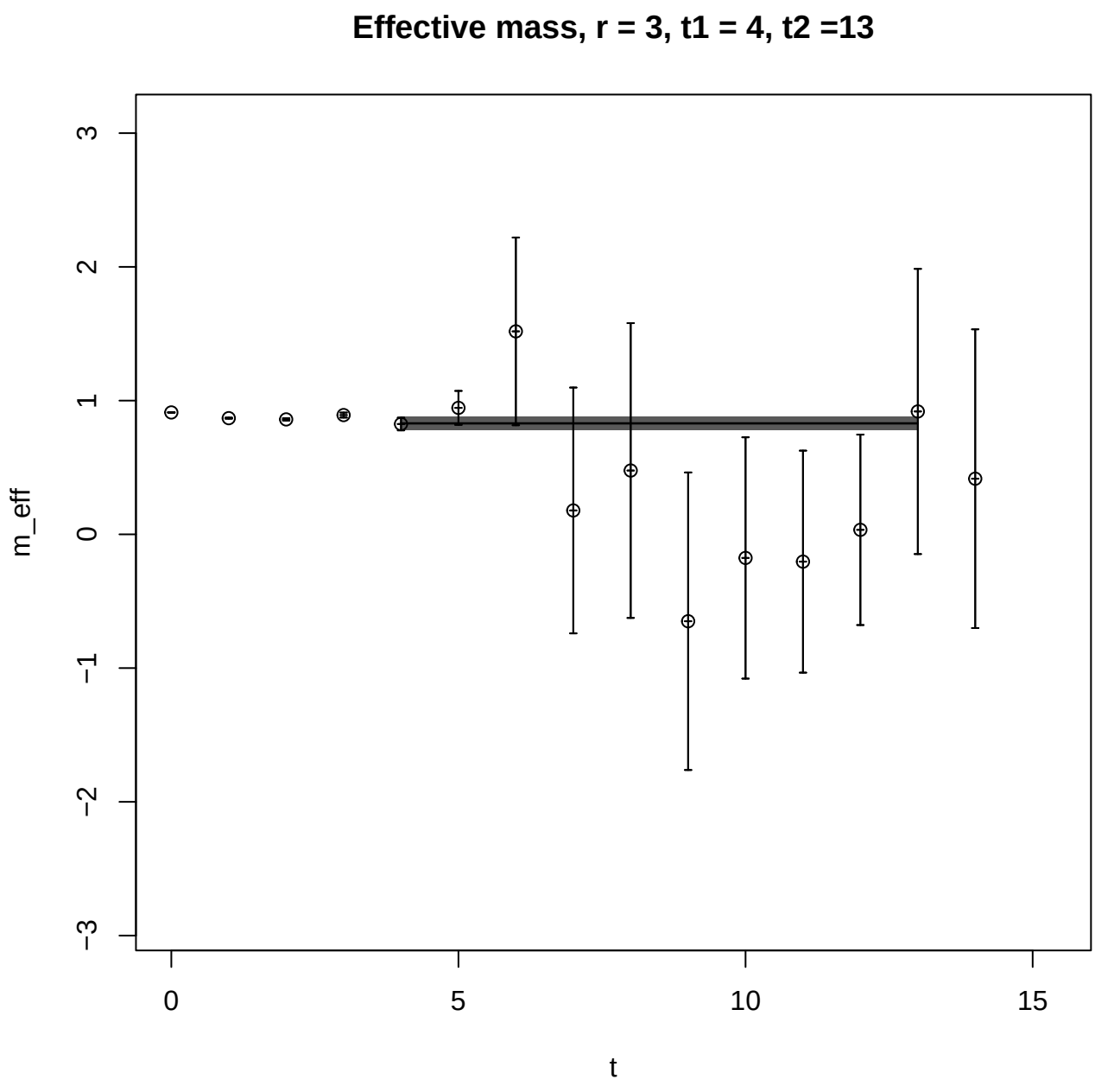

Effective mass, r = 3, t1 = 4, t2 =8
m_eff
t
3
2
1
0
−1
−2
−3
0
5
10
15

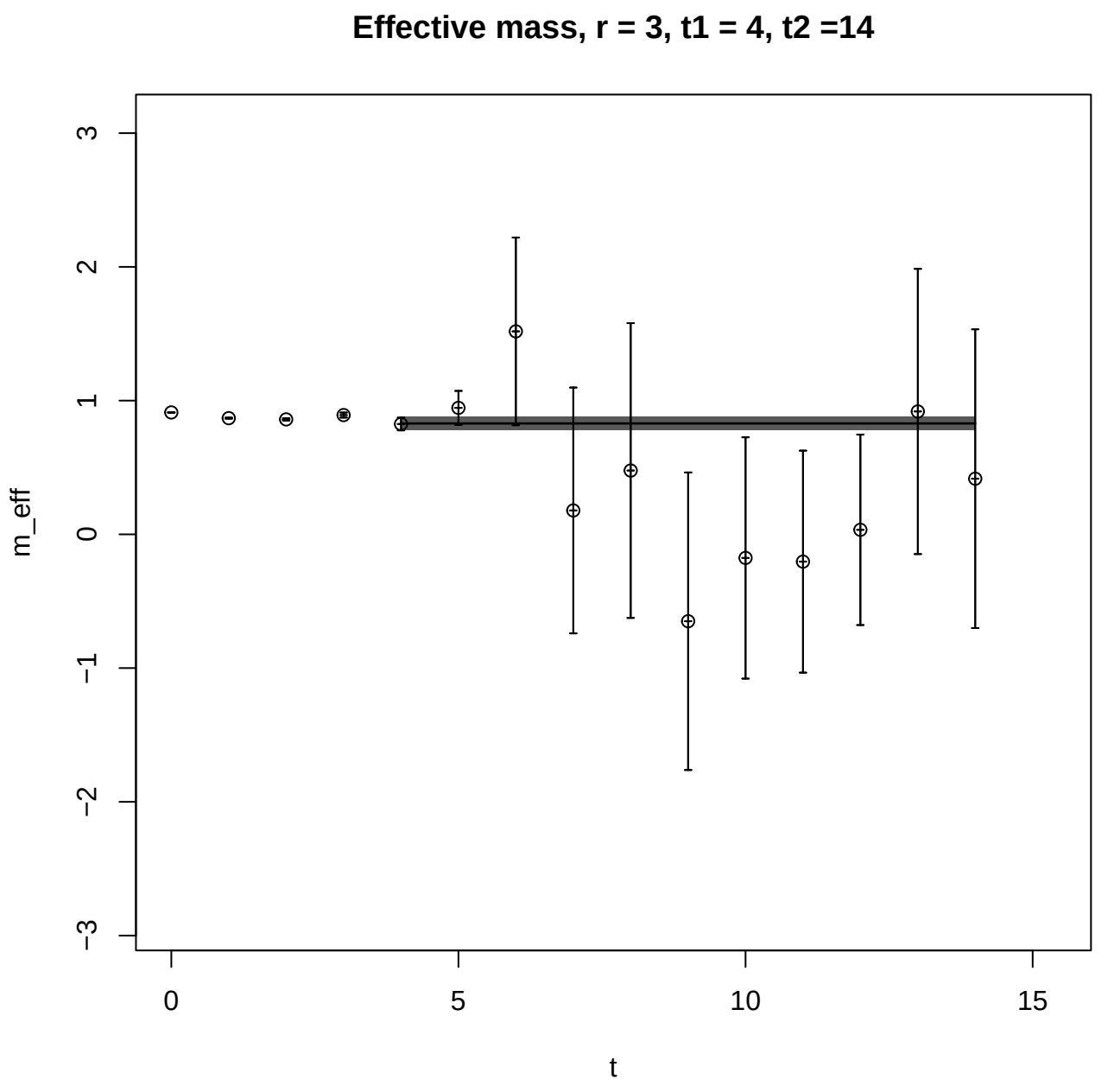

Effective mass, r = 3, t1 = 4, t2 =9
m_eff
t
3
2
1
0
−1
−2
−3
0
5
10
15

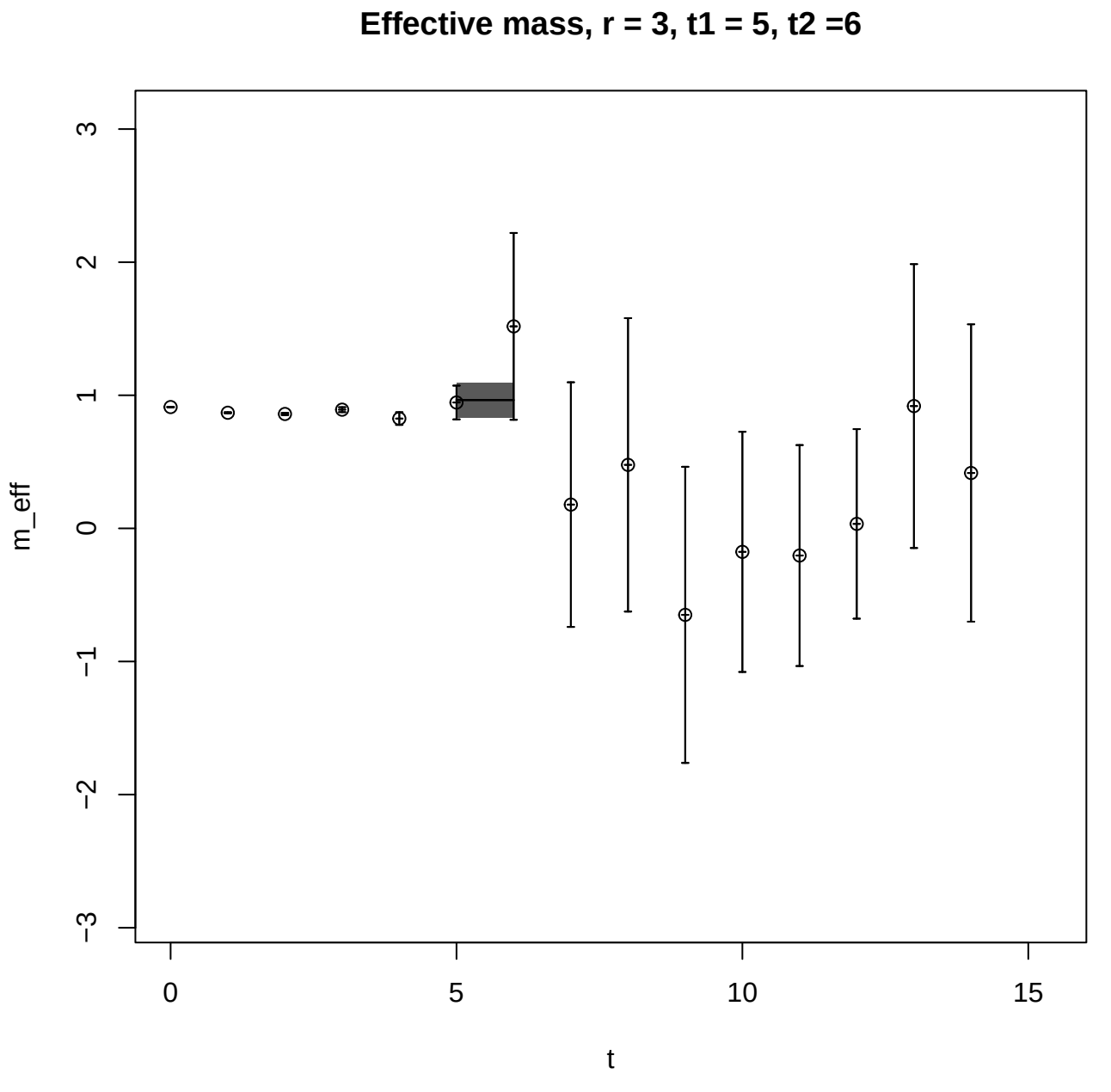

Effective mass, r = 3, t1 = 4, t2 =10
m_eff
t
3
2
1
0
−1
−2
−3
0
5
10
15


Effective mass, r = 3, t1 = 4, t2 =11
m_eff
t
3
2
1
0
−1
−2
−3
0
5
10
15


Effective mass, r = 3, t1 = 4, t2 =12
m_eff
t
3
2
1
0
−1
−2
−3
0
5
10
15


Effective mass, r = 3, t1 = 4, t2 =13
m_eff
t
3
2
1
0
−1
−2
−3
0
5
10
15


Effective mass, r = 3, t1 = 4, t2 =14
m_eff
t
3
2
1
0
−1
−2
−3
0
5
10
15


Effective mass, r = 3, t1 = 5, t2 =6
m_eff
t
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0
−1
−2
−3
0
5
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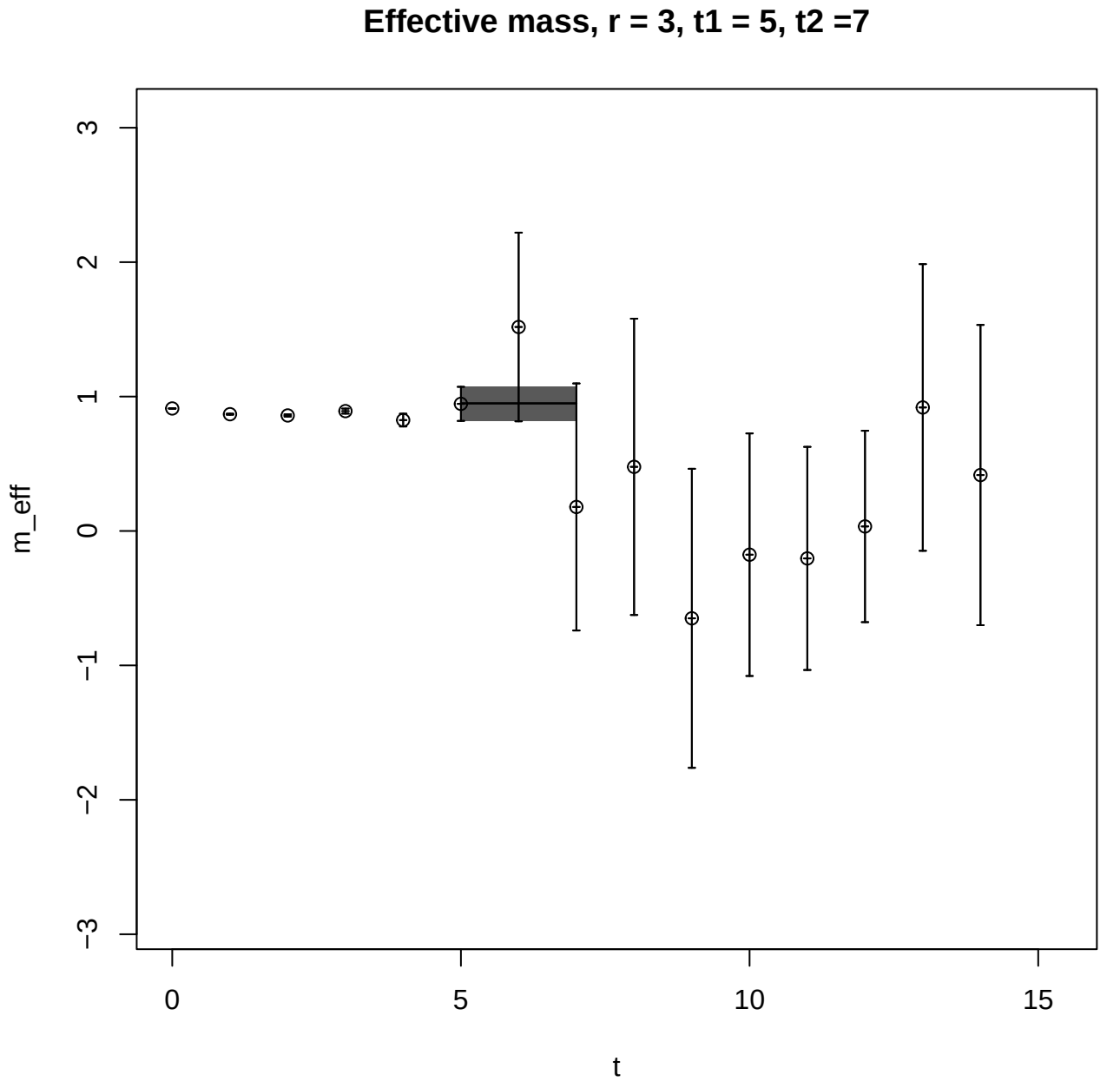
15


Effective mass, r = 3, t1 = 5, t2 =7
m_eff
t
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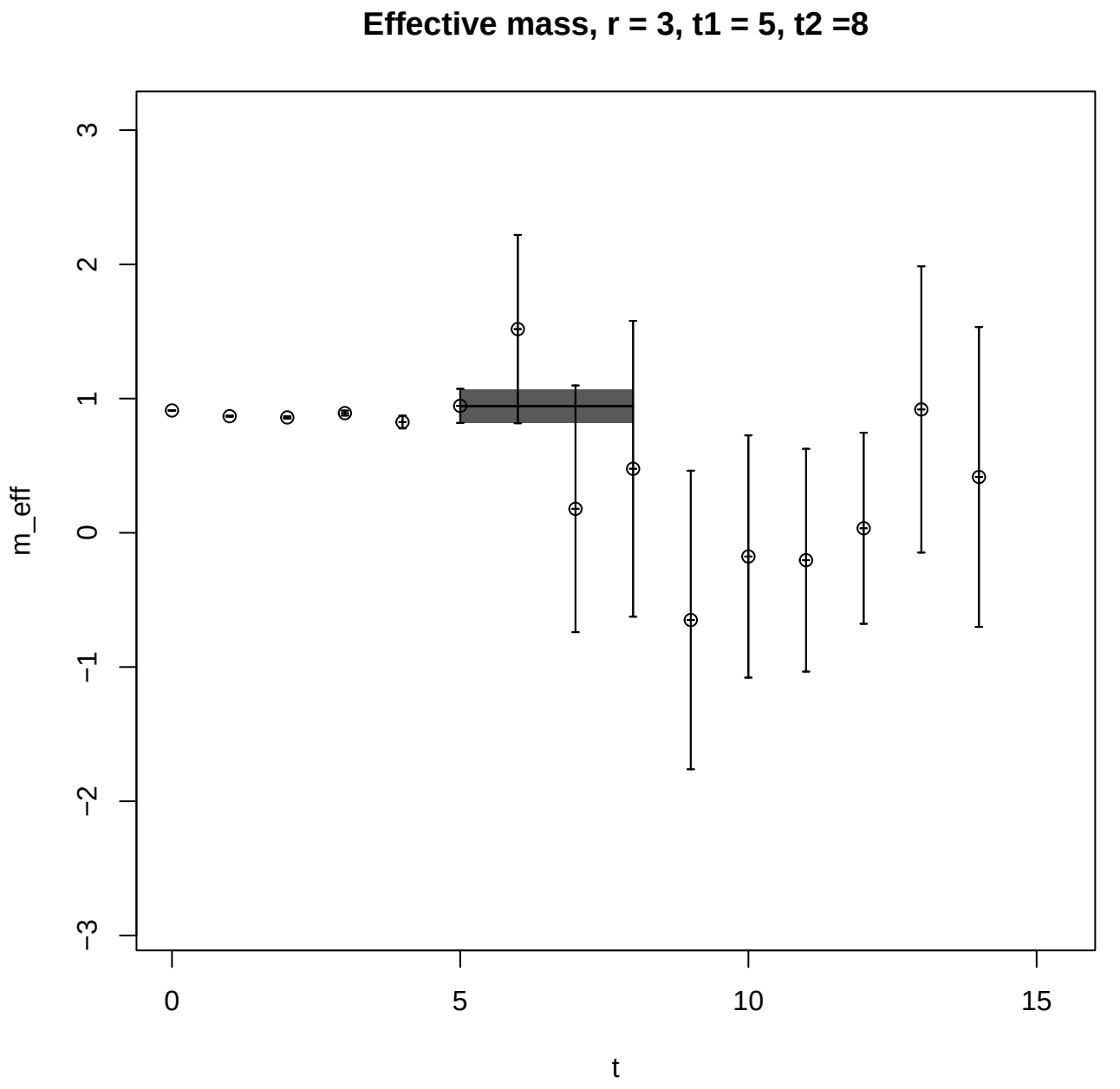
2
1
0
−1
−2
−3
0
5
10
15


Effective mass, r = 3, t1 = 5, t2 =8
m_eff
t
3
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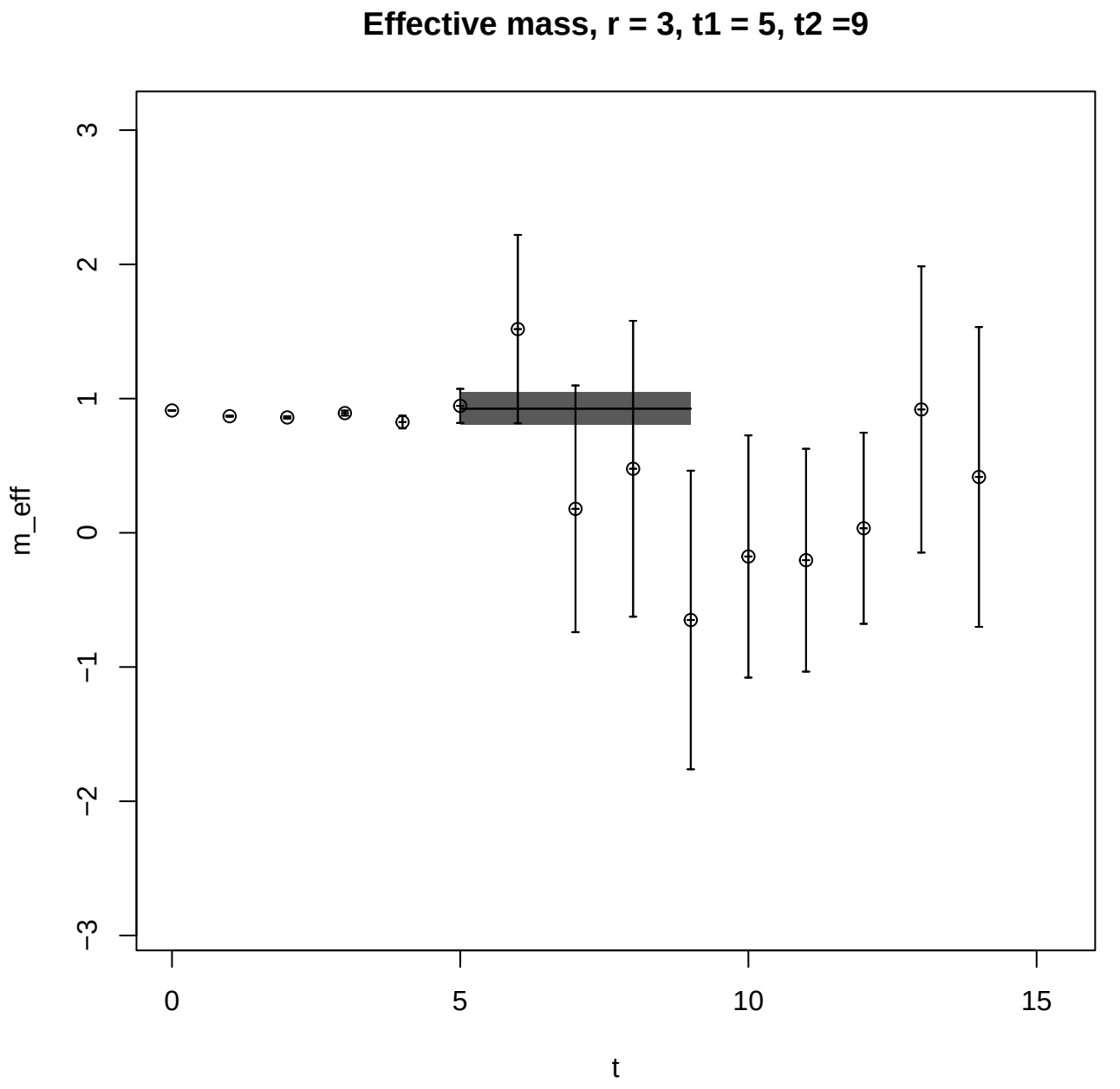
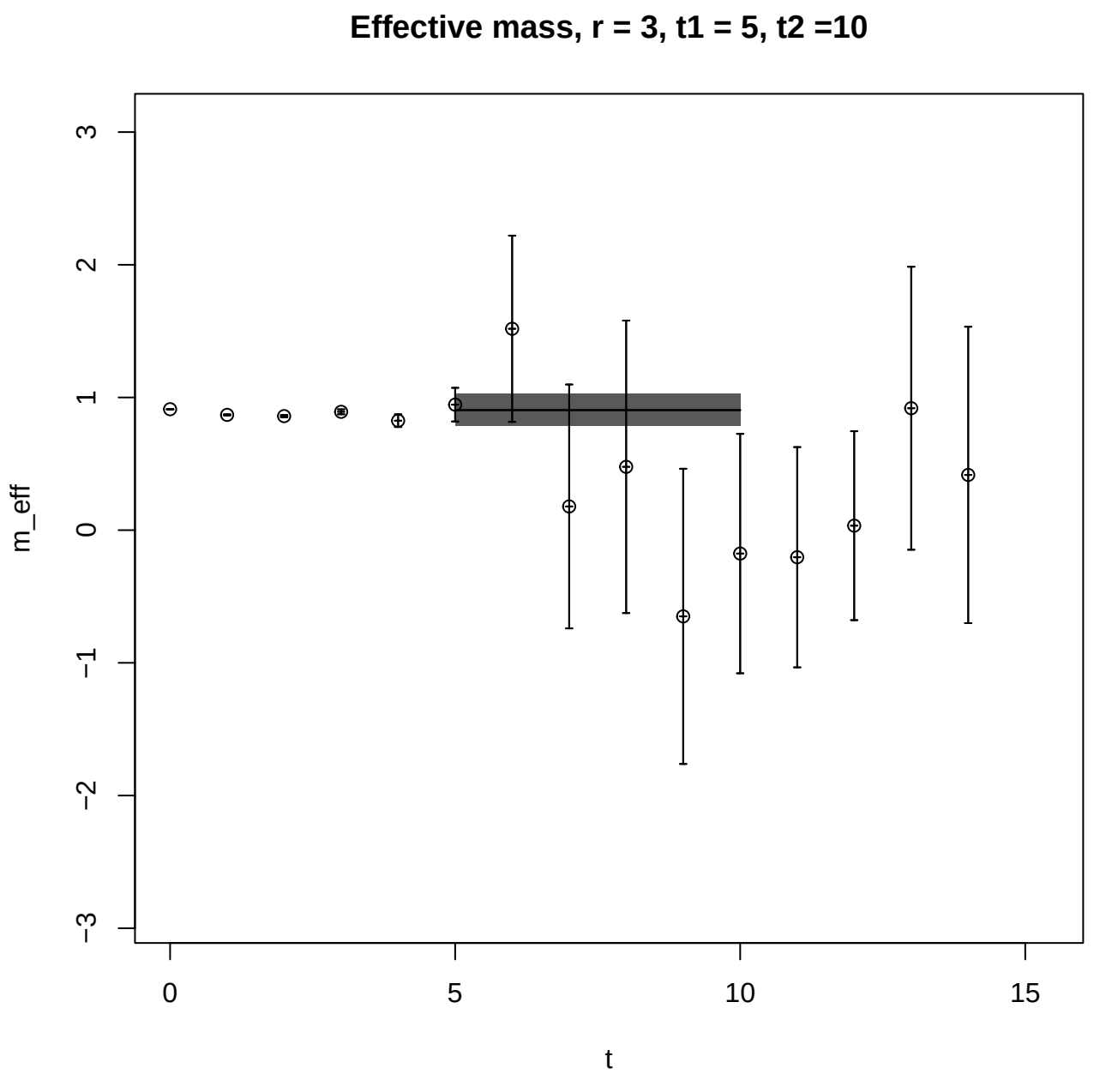
0
−1
−2
−3
0
5
10
15


Effective mass, r = 3, t1 = 5, t2 =9
m_eff
t
3
2
1
0
−1
−2
−3
0
5
10
15


Effective mass, r = 3, t1 = 5, t2 =10
m_eff
t
3
2
1
0
−1
−2
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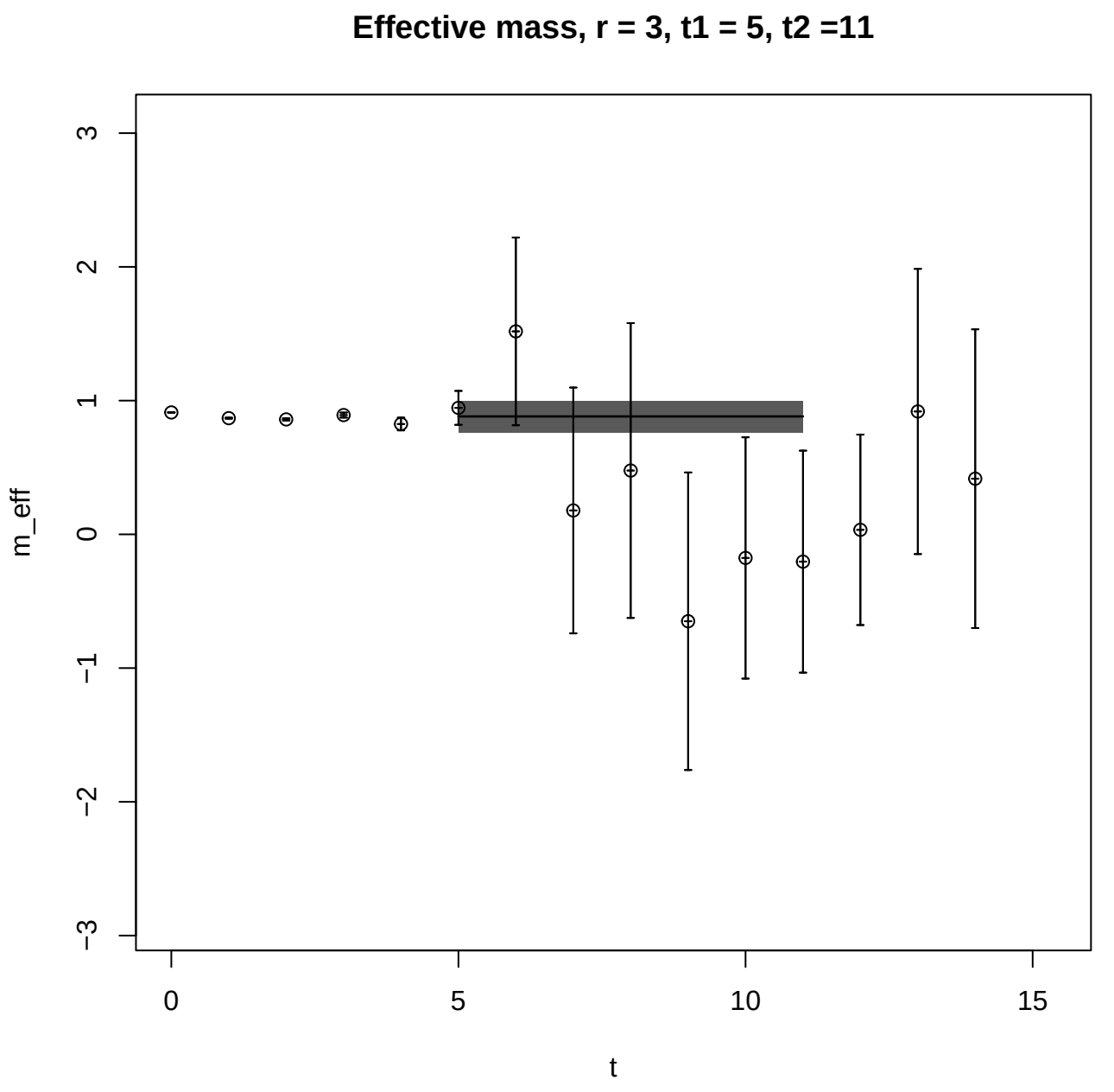
0
5
10
15


Effective mass, r = 3, t1 = 5, t2 =11
m_eff
t
3
2
1
0
−1
−2
−3
0
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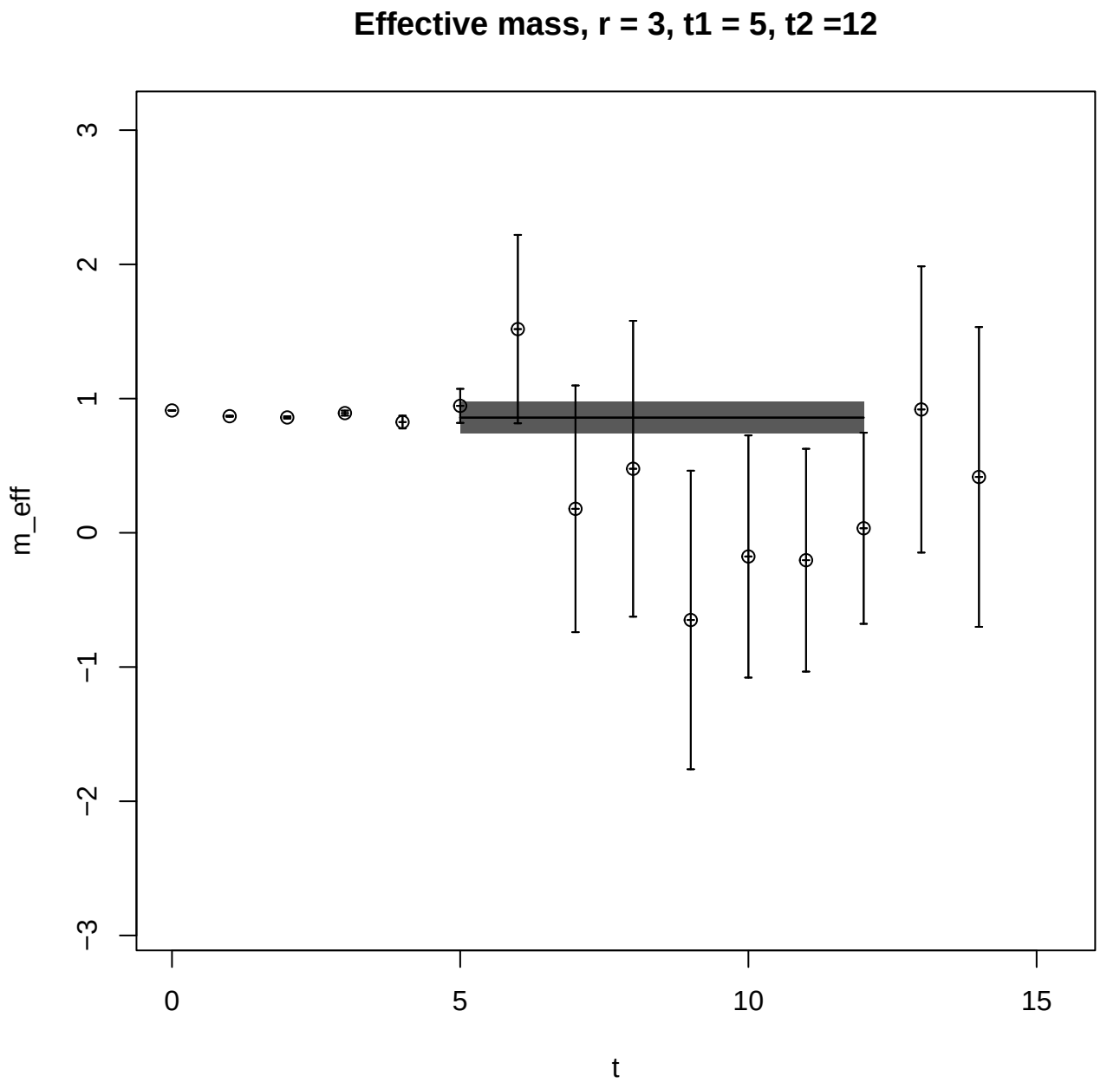
10
15


Effective mass, r = 3, t1 = 5, t2 =12
m_eff
t
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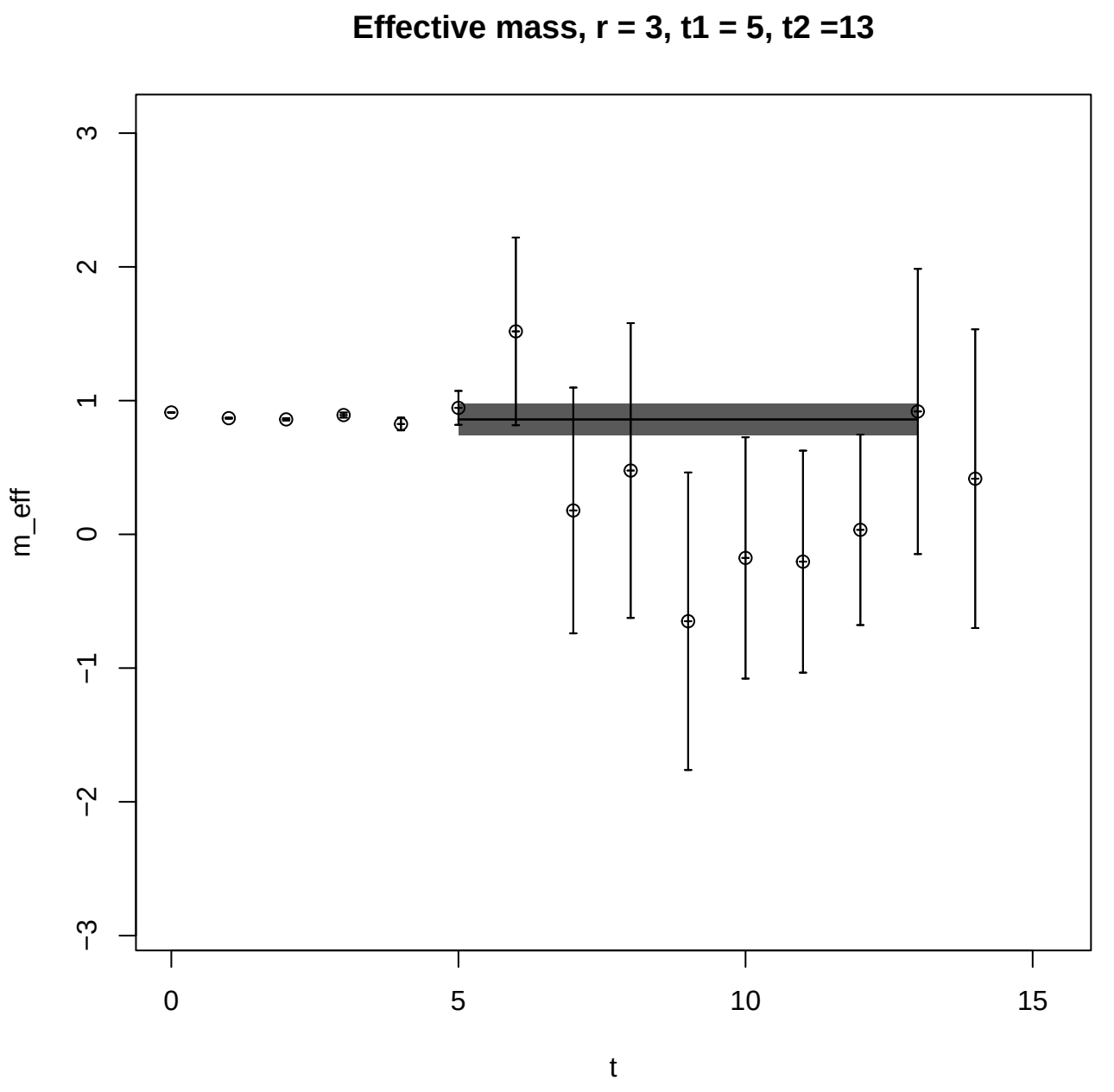
2
1
0
−1
−2
−3
0
5
10
15


Effective mass, r = 3, t1 = 5, t2 =13
m_eff
t
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2
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0
−1
−2
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10
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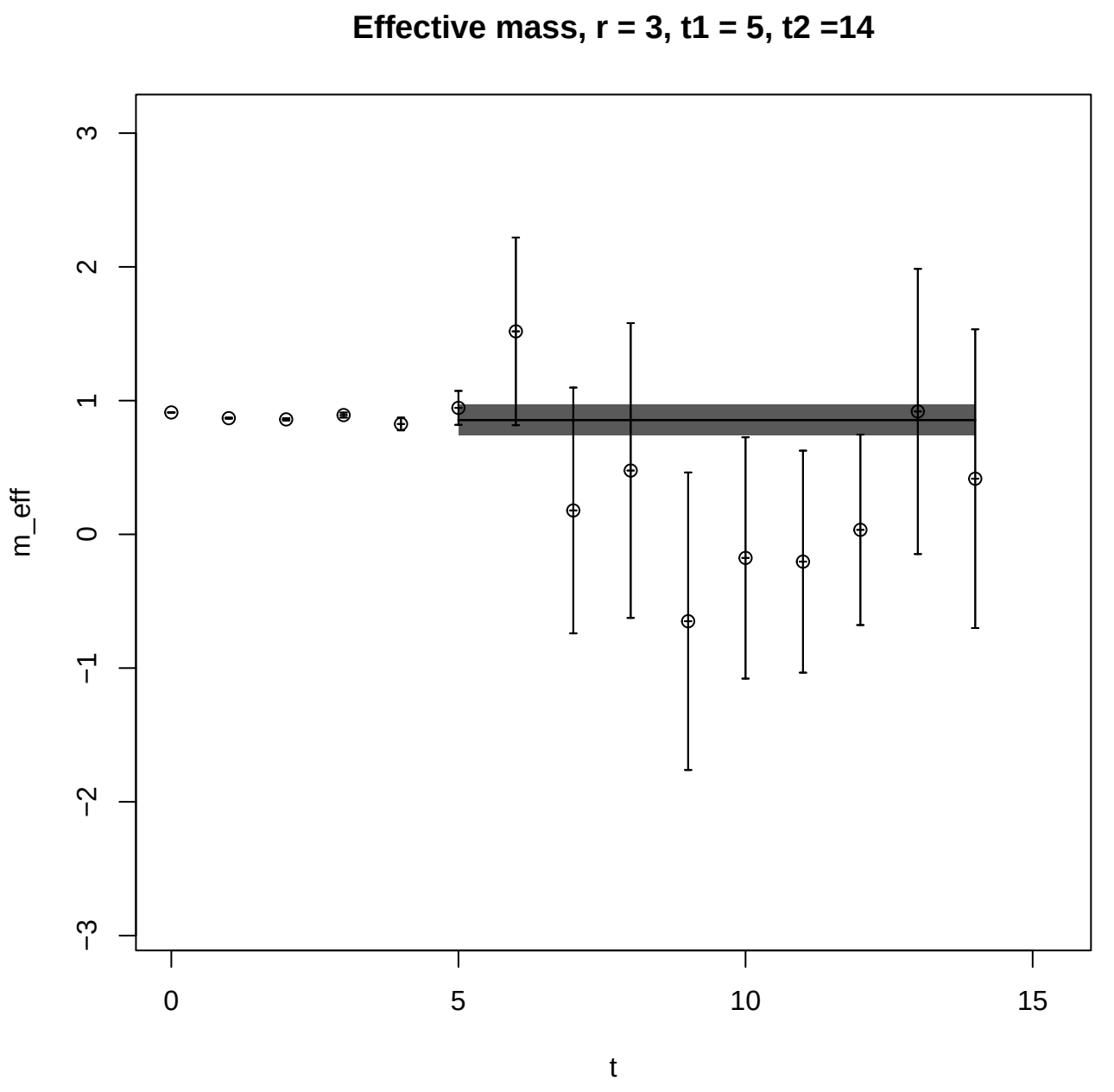
Effective mass, r = 3, t1 = 5, t2 =14
m_eff
t
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2
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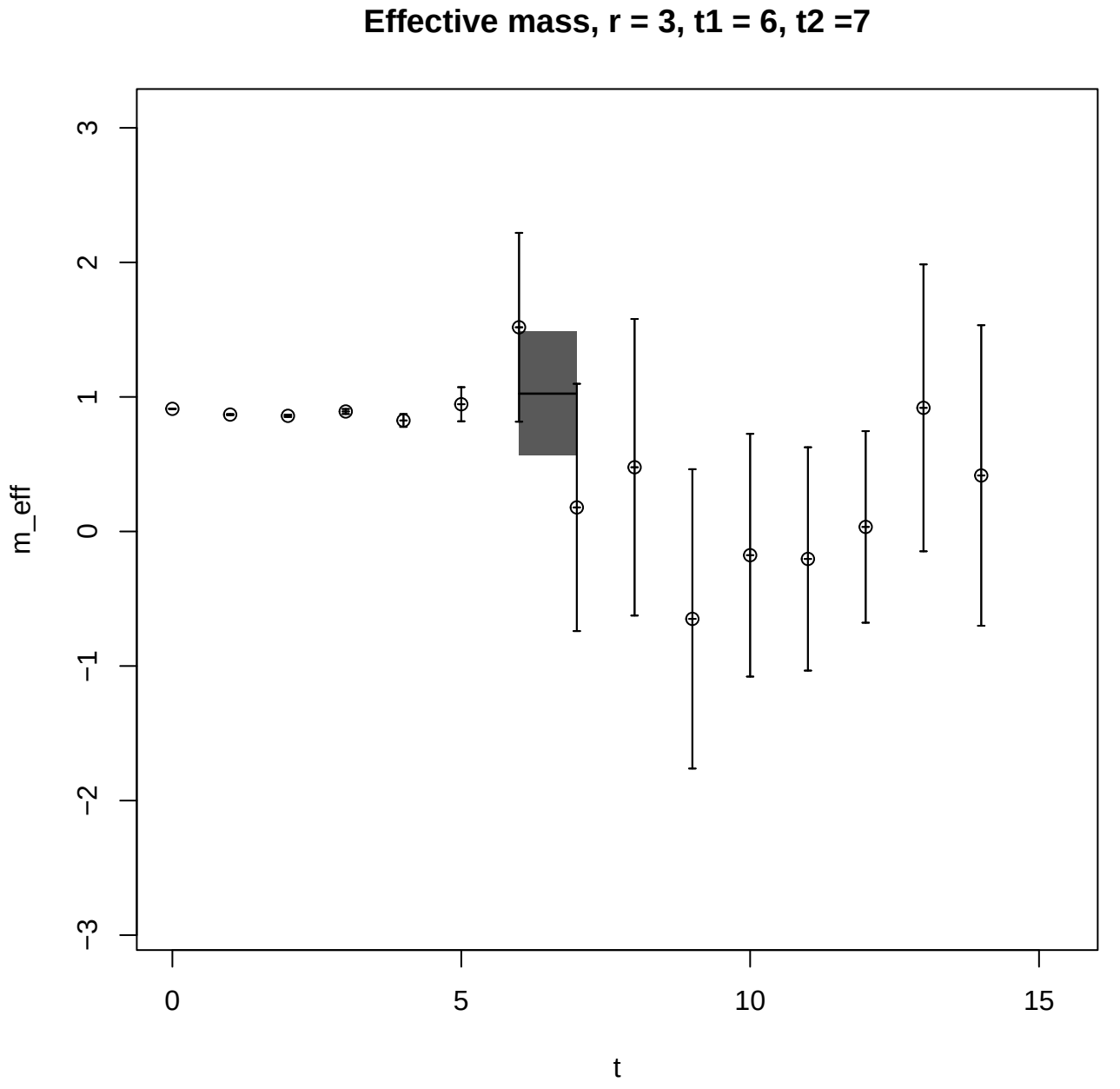
Effective mass, r = 3, t1 = 6, t2 =7
m_eff
t
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2
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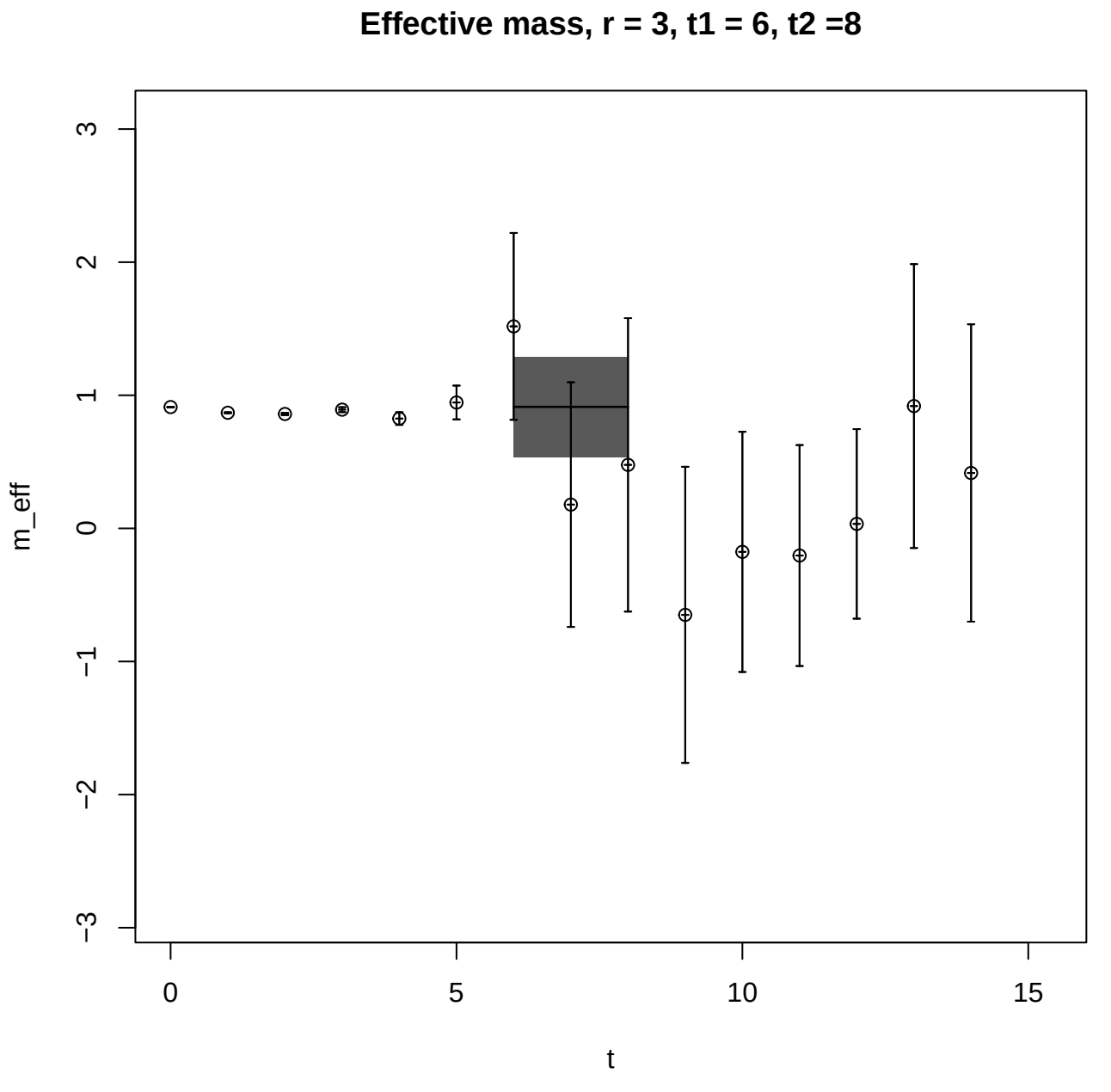
0
−1
−2
−3
0
5
10
15


Effective mass, r = 3, t1 = 6, t2 =8
m_eff
t
3
2
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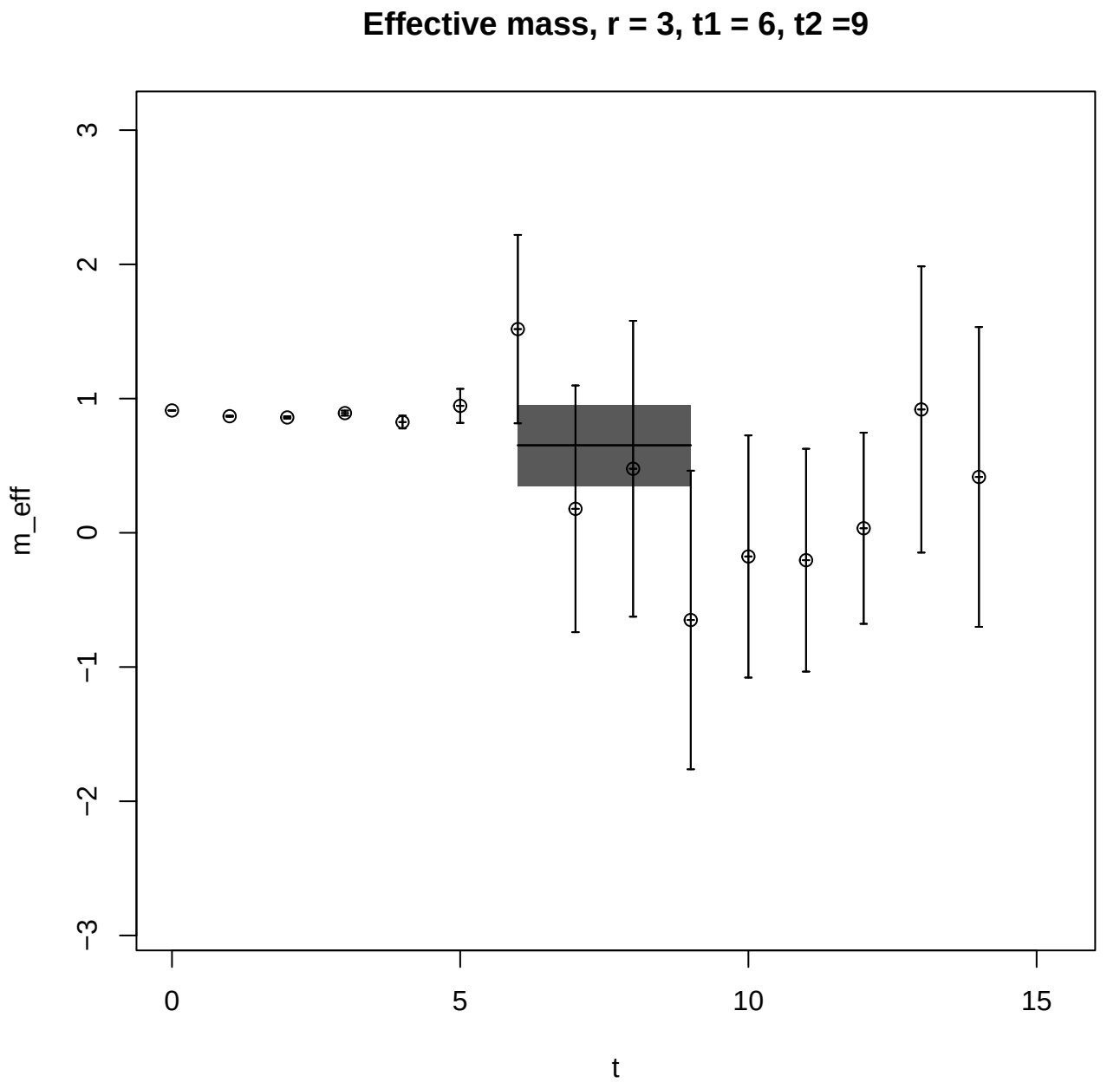
0
−1
−2
−3
0
5
10
15


Effective mass, r = 3, t1 = 6, t2 =9
m_eff
t
3
2
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0
−1
−2
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0
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10
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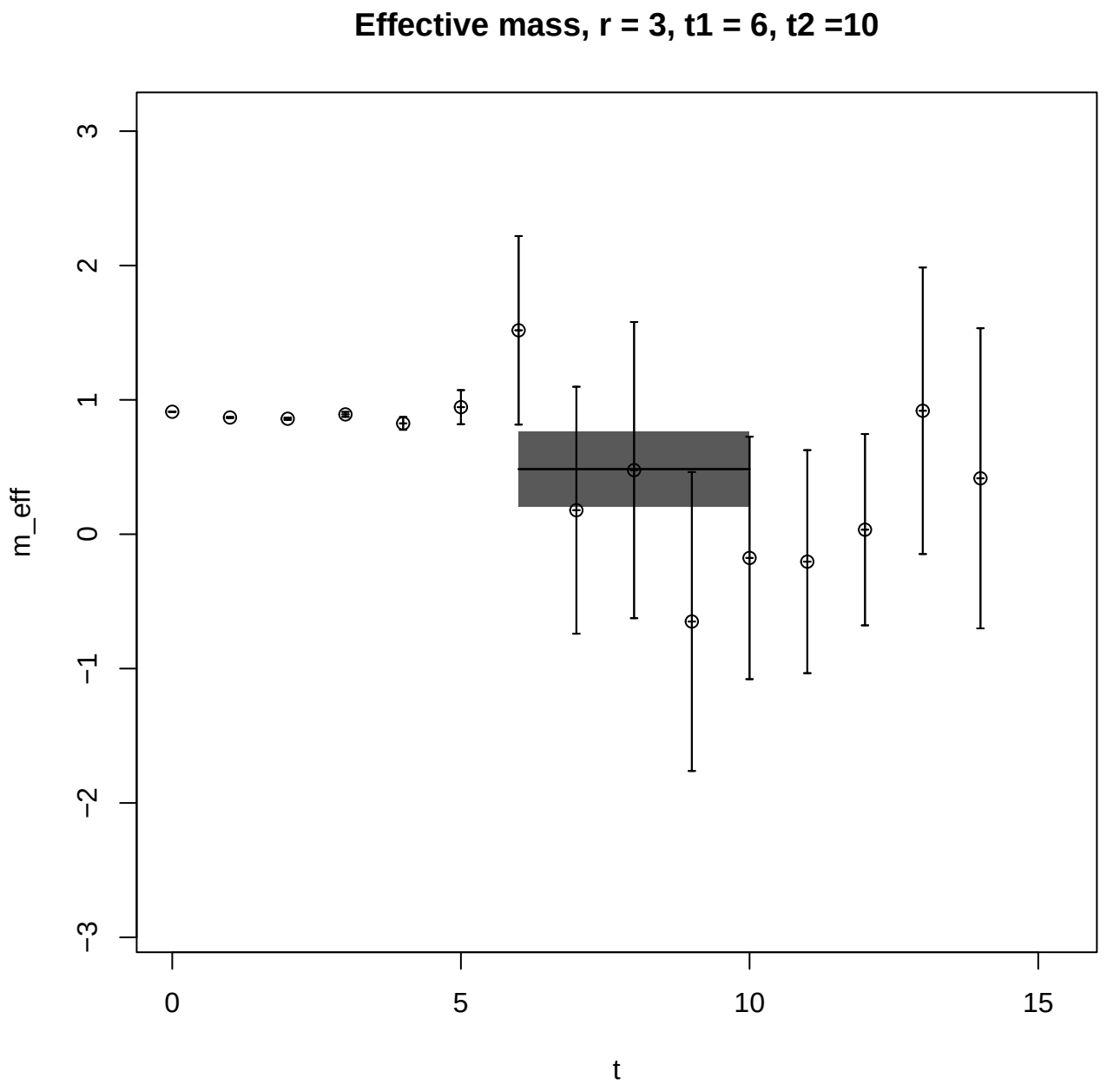
Effective mass, r = 3, t1 = 6, t2 =10
m_eff
t
3
2
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0
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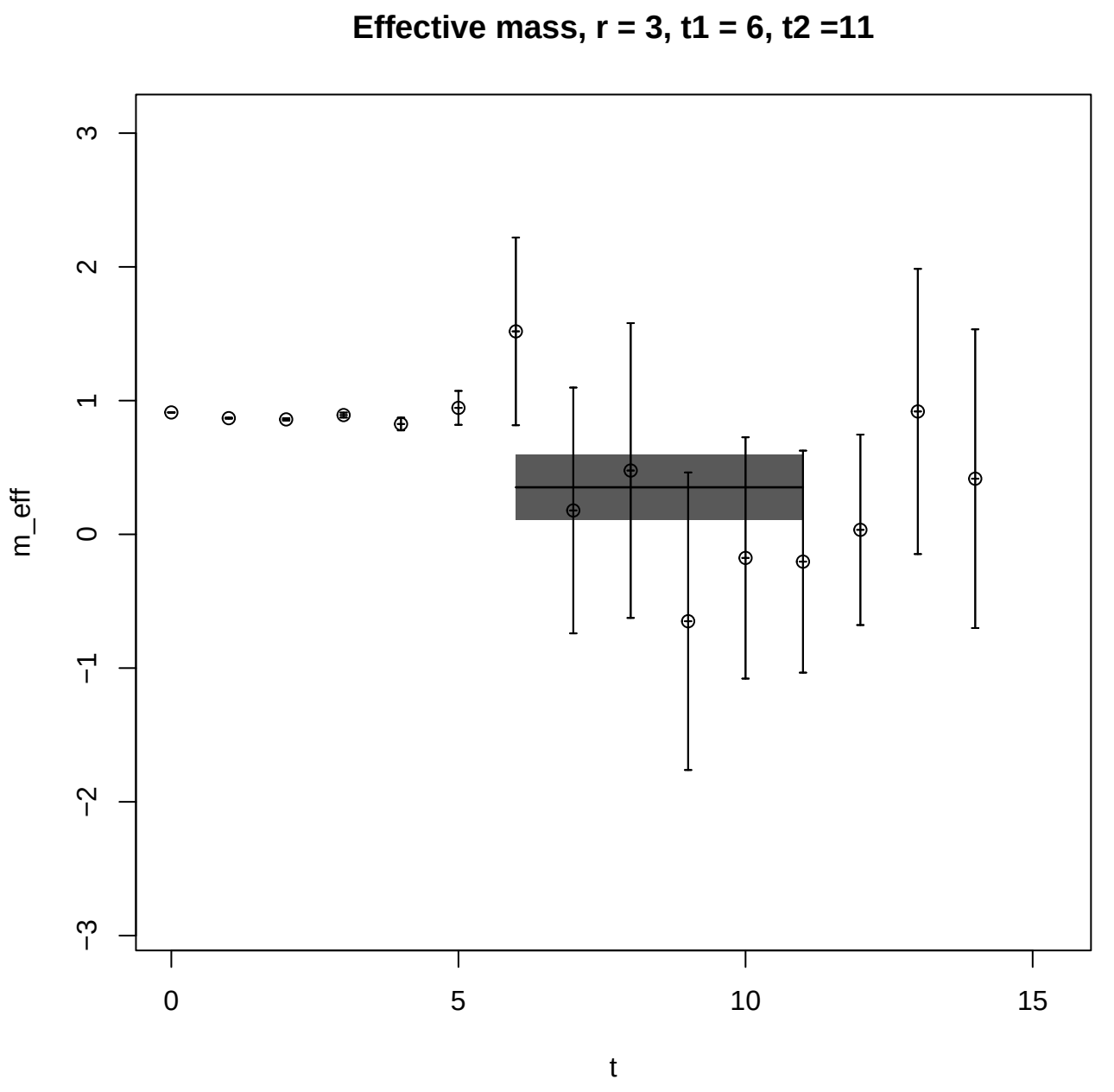
Effective mass, r = 3, t1 = 6, t2 =11
m_eff
t
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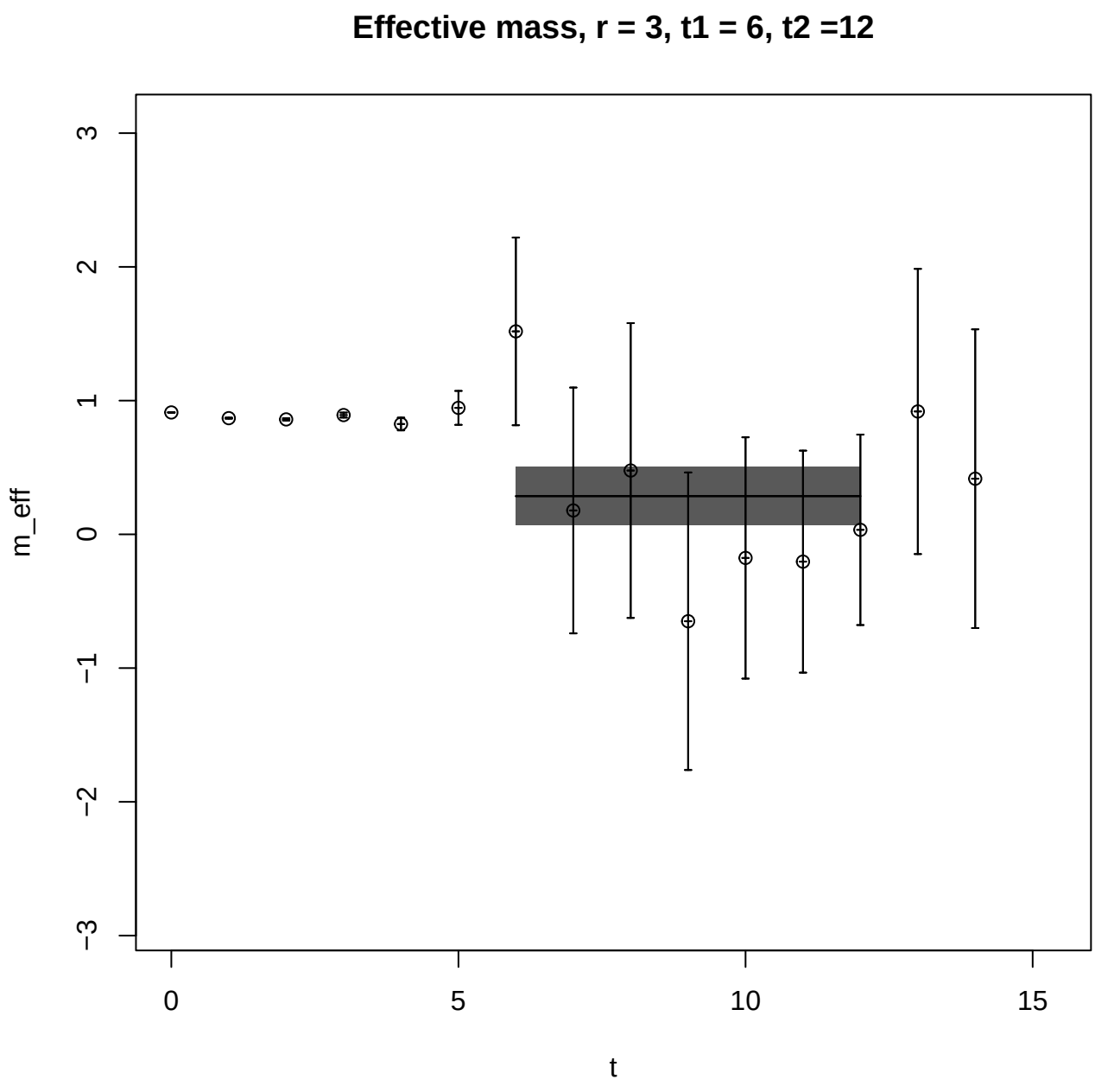
Effective mass, r = 3, t1 = 6, t2 =12
m_eff
t
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2
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0
−1
−2
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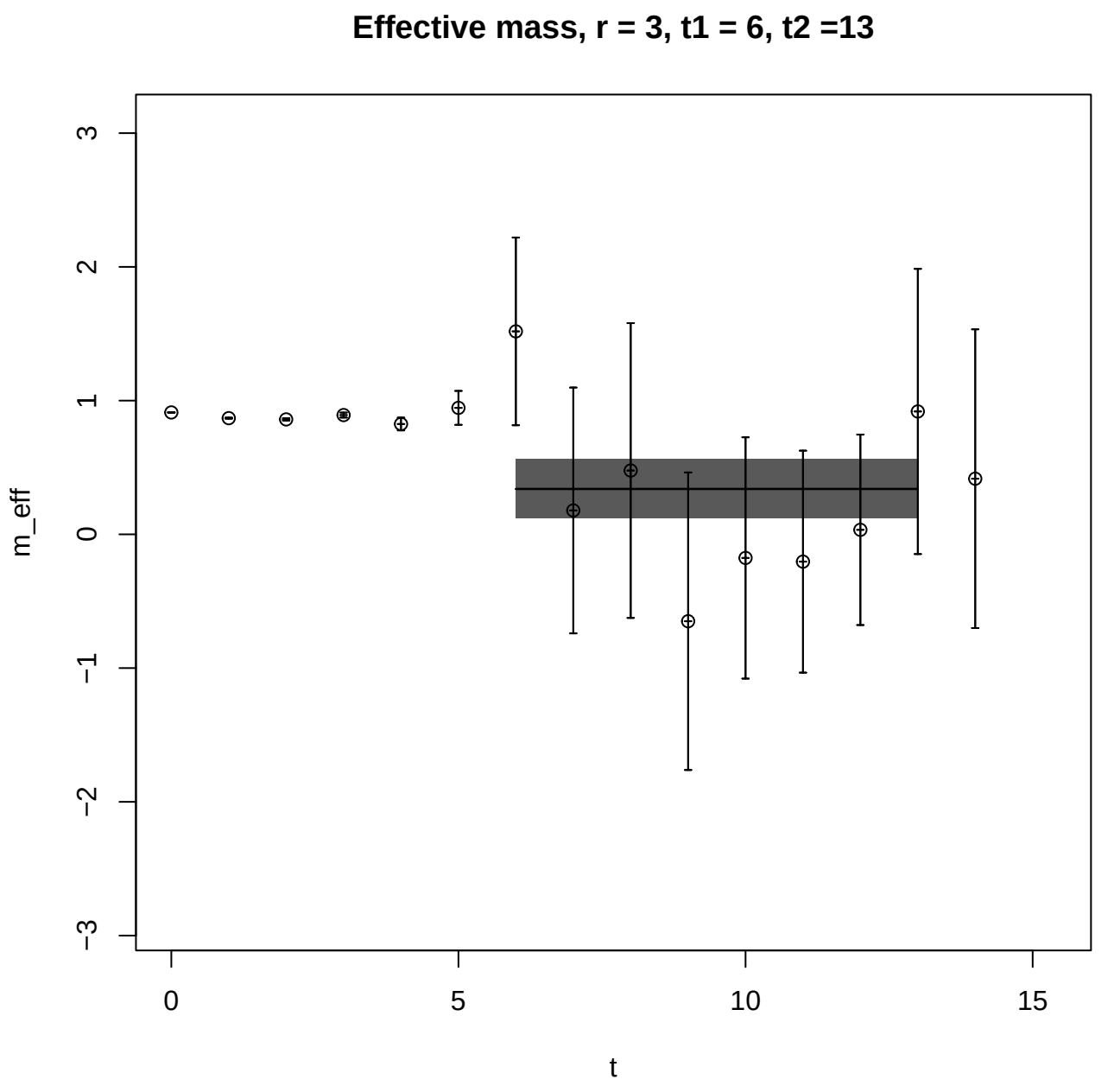
Effective mass, r = 3, t1 = 6, t2 =13
m_eff
t
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1
0
−1
−2
−3
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5
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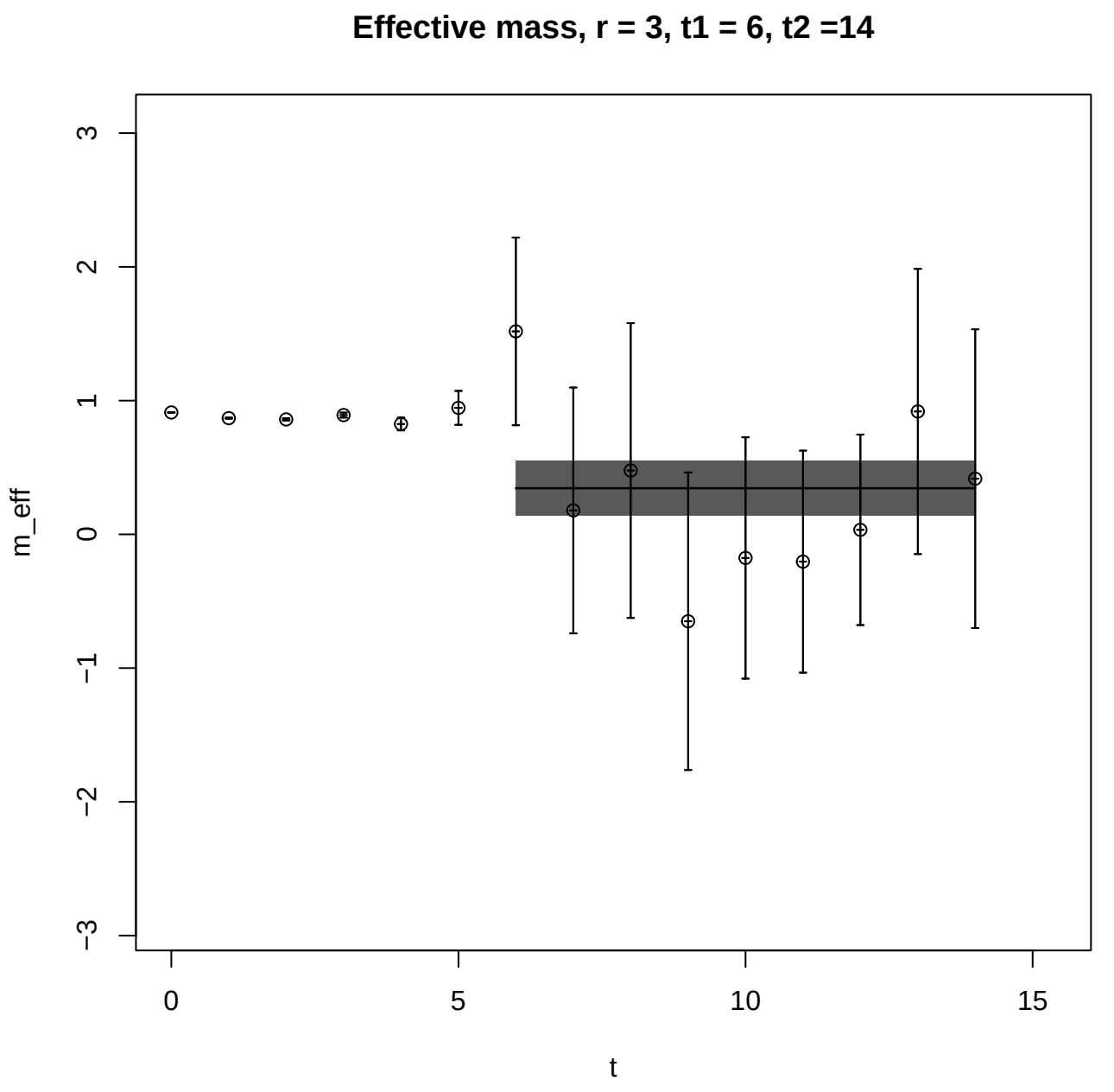
Effective mass, r = 3, t1 = 6, t2 =14
m_eff
t
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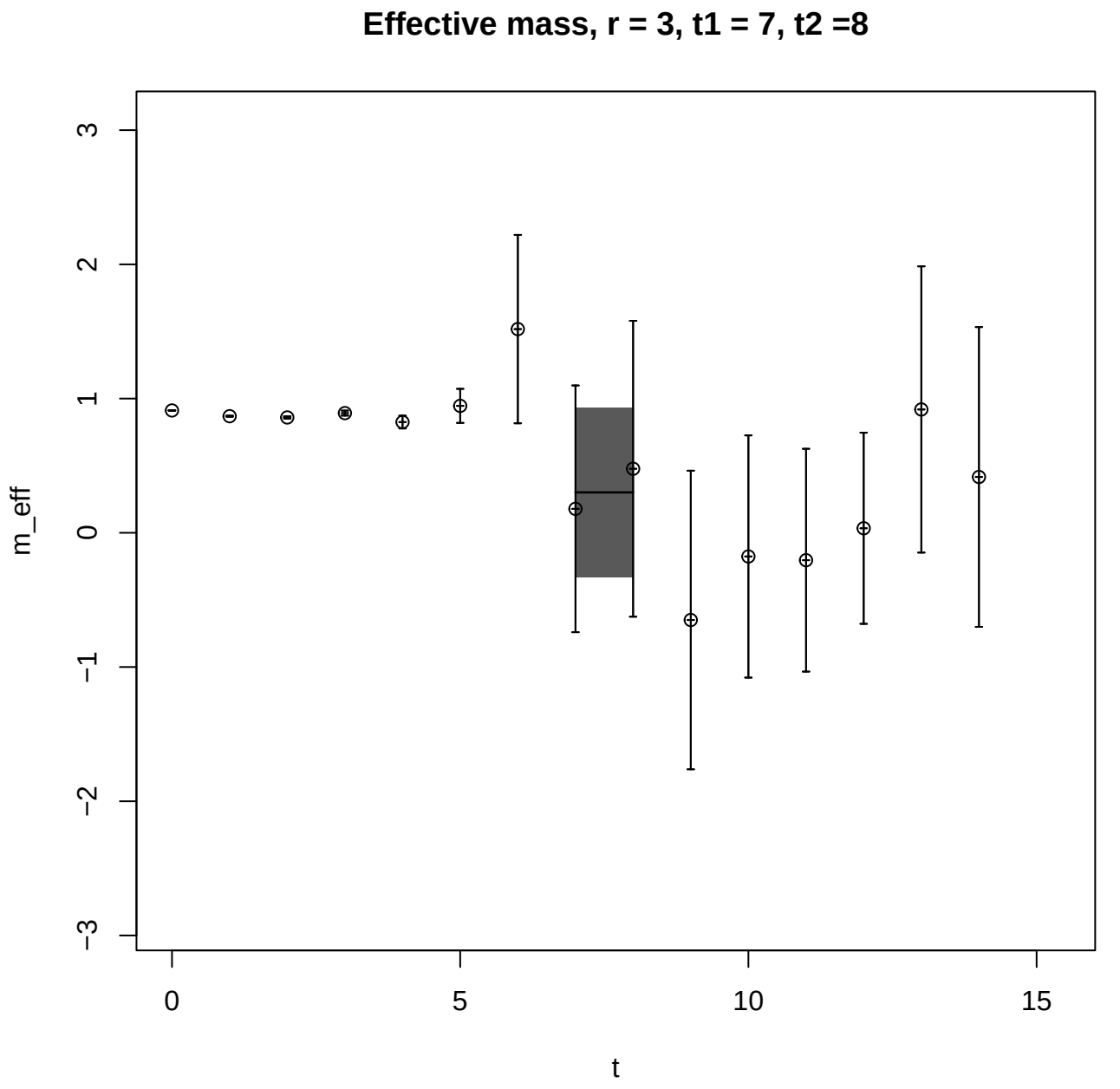
Effective mass, r = 3, t1 = 7, t2 =8
m_eff
t
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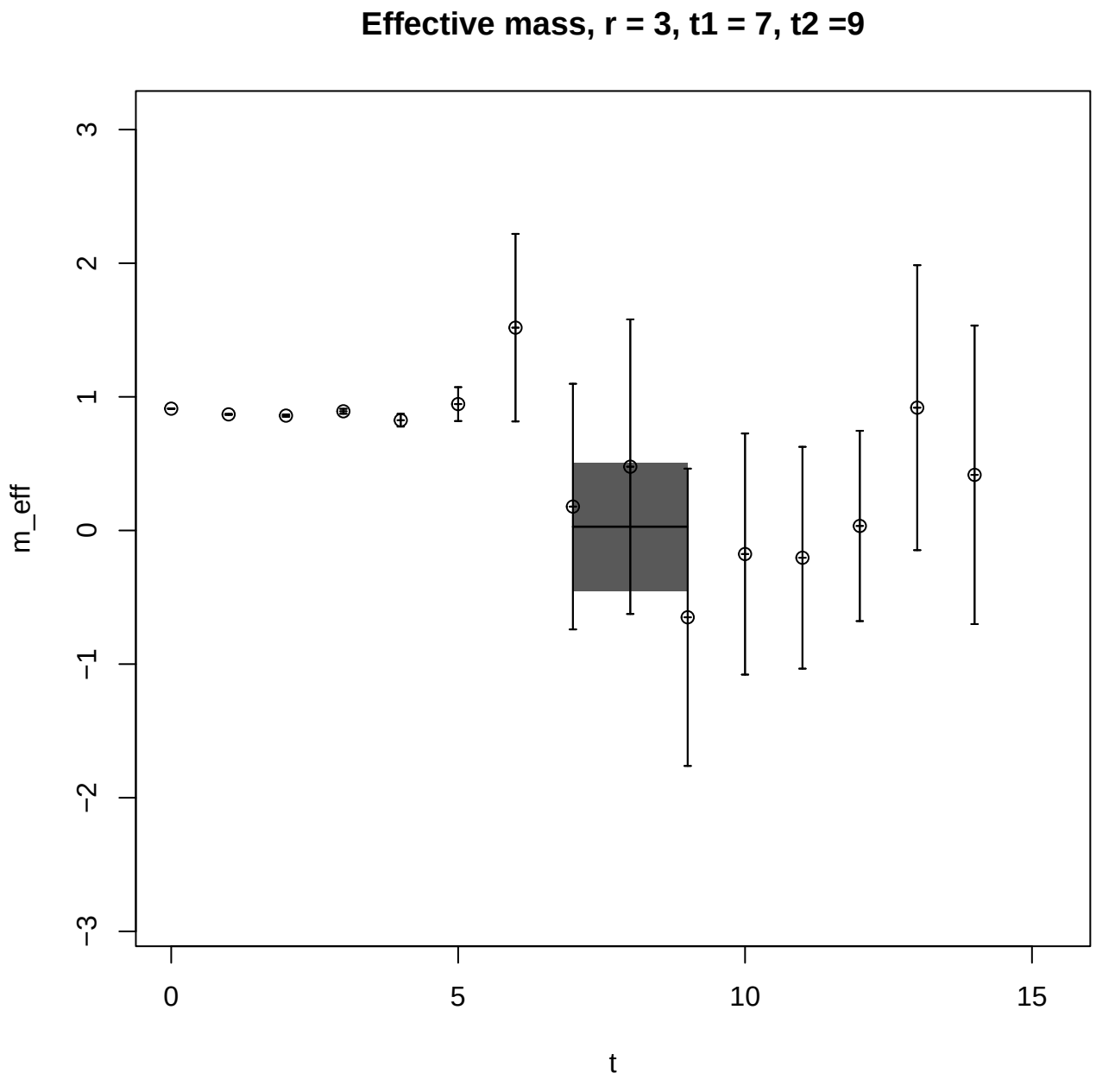
Effective mass, r = 3, t1 = 7, t2 =9
m_eff
t
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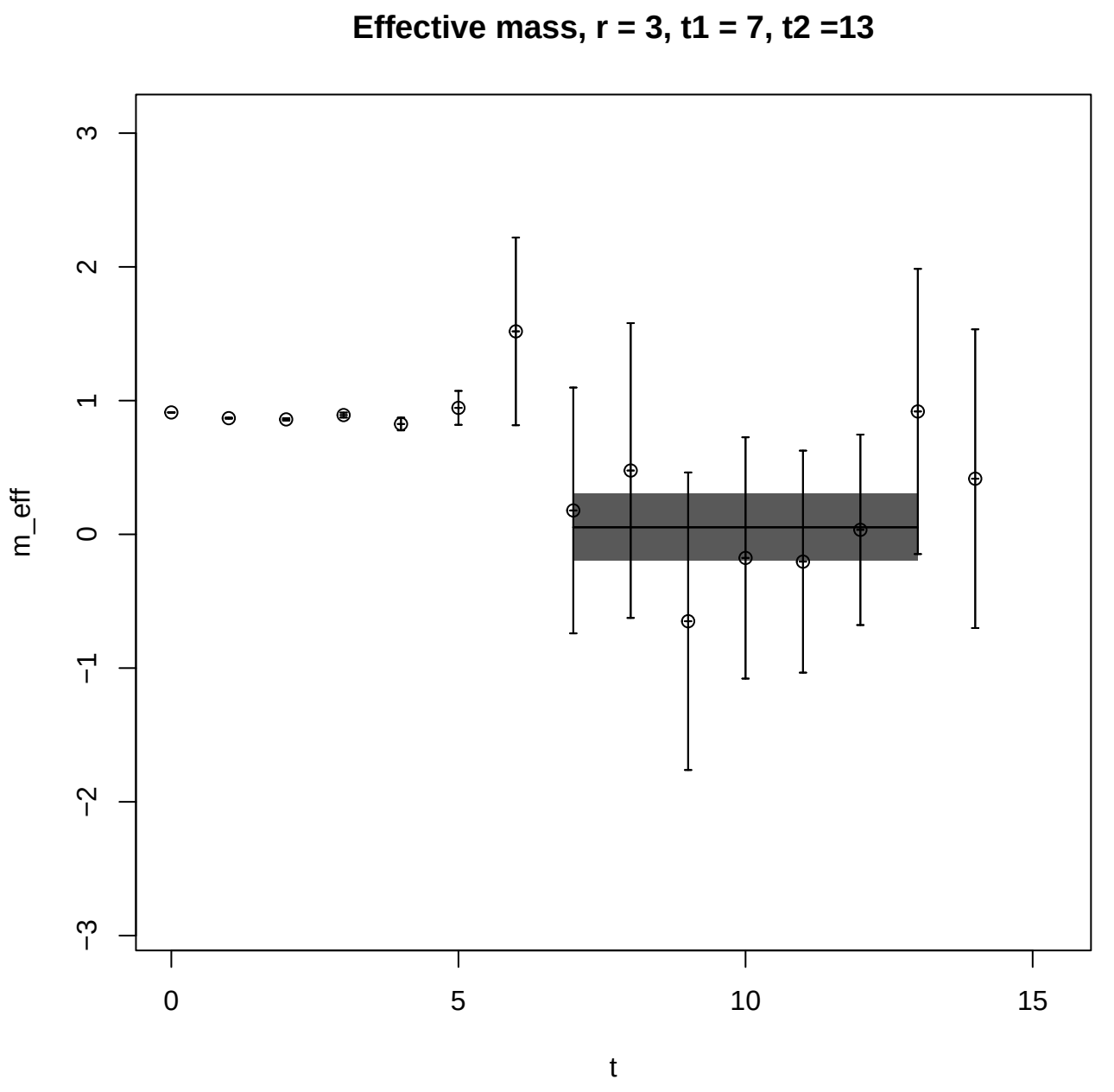
Effective mass, r = 3, t1 = 7, t2 =13
m_eff
t
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5
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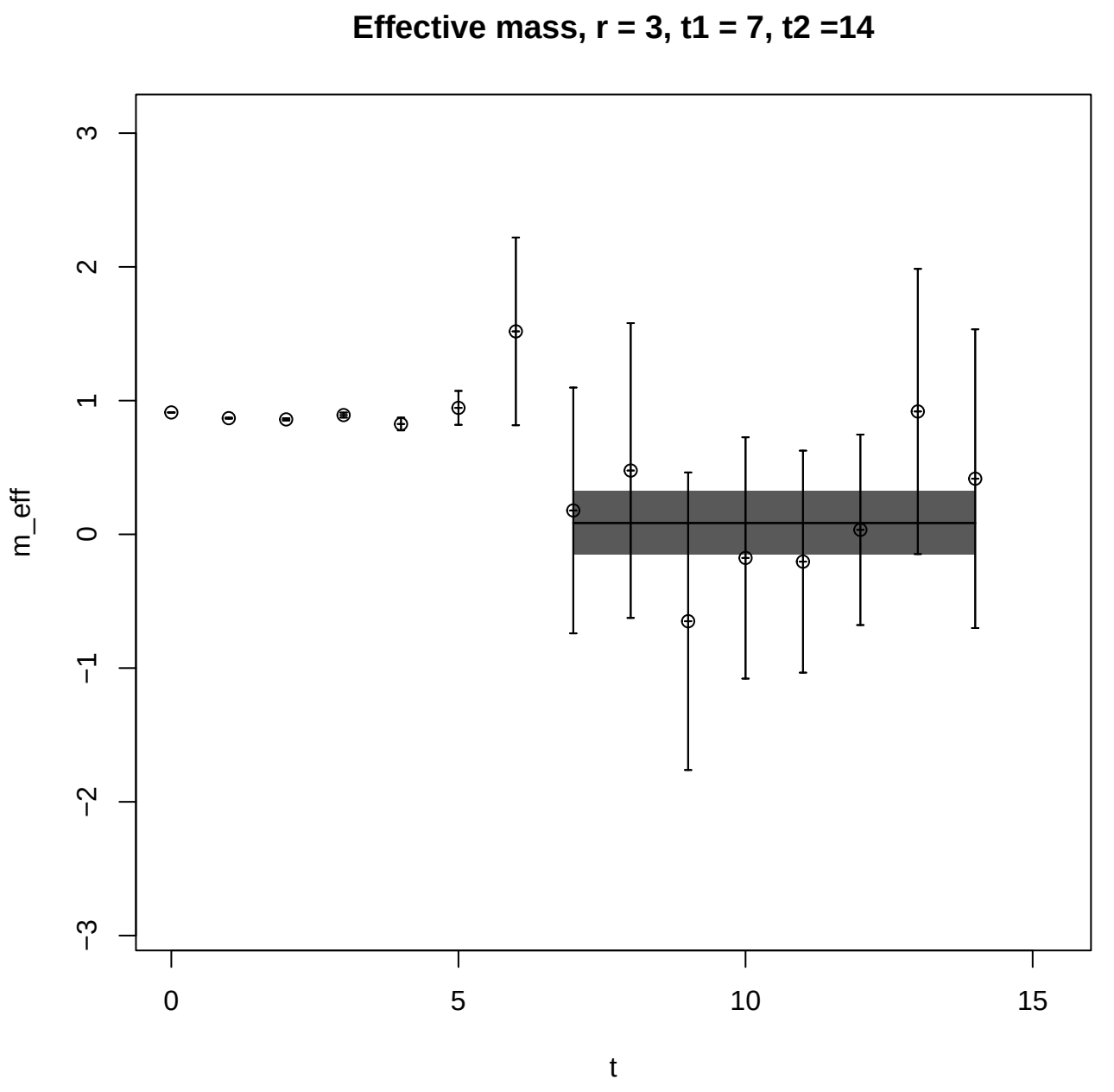
Effective mass, r = 3, t1 = 7, t2 =14
m_eff
t
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1
0
−1
−2
−3
0
5
10
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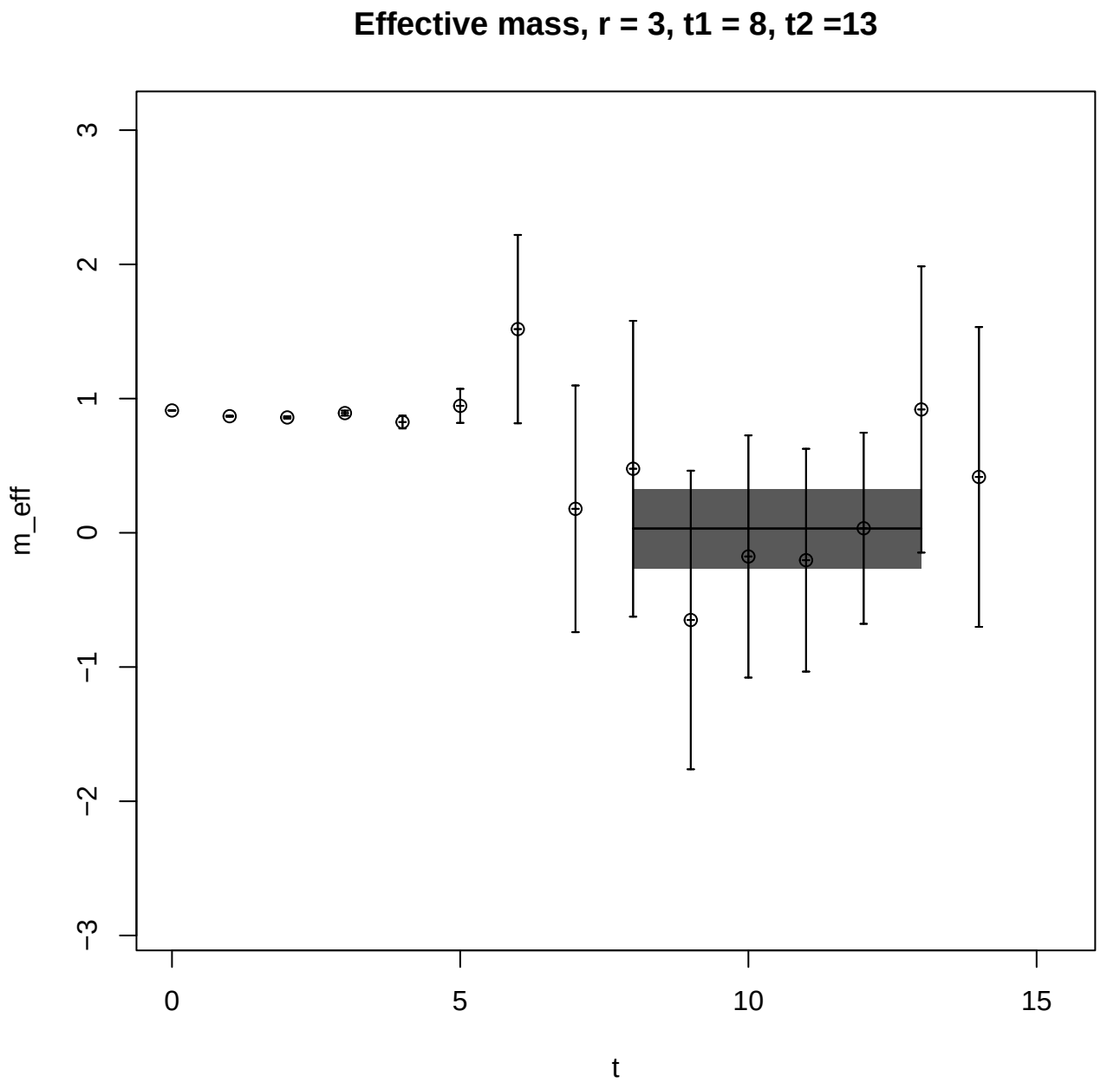
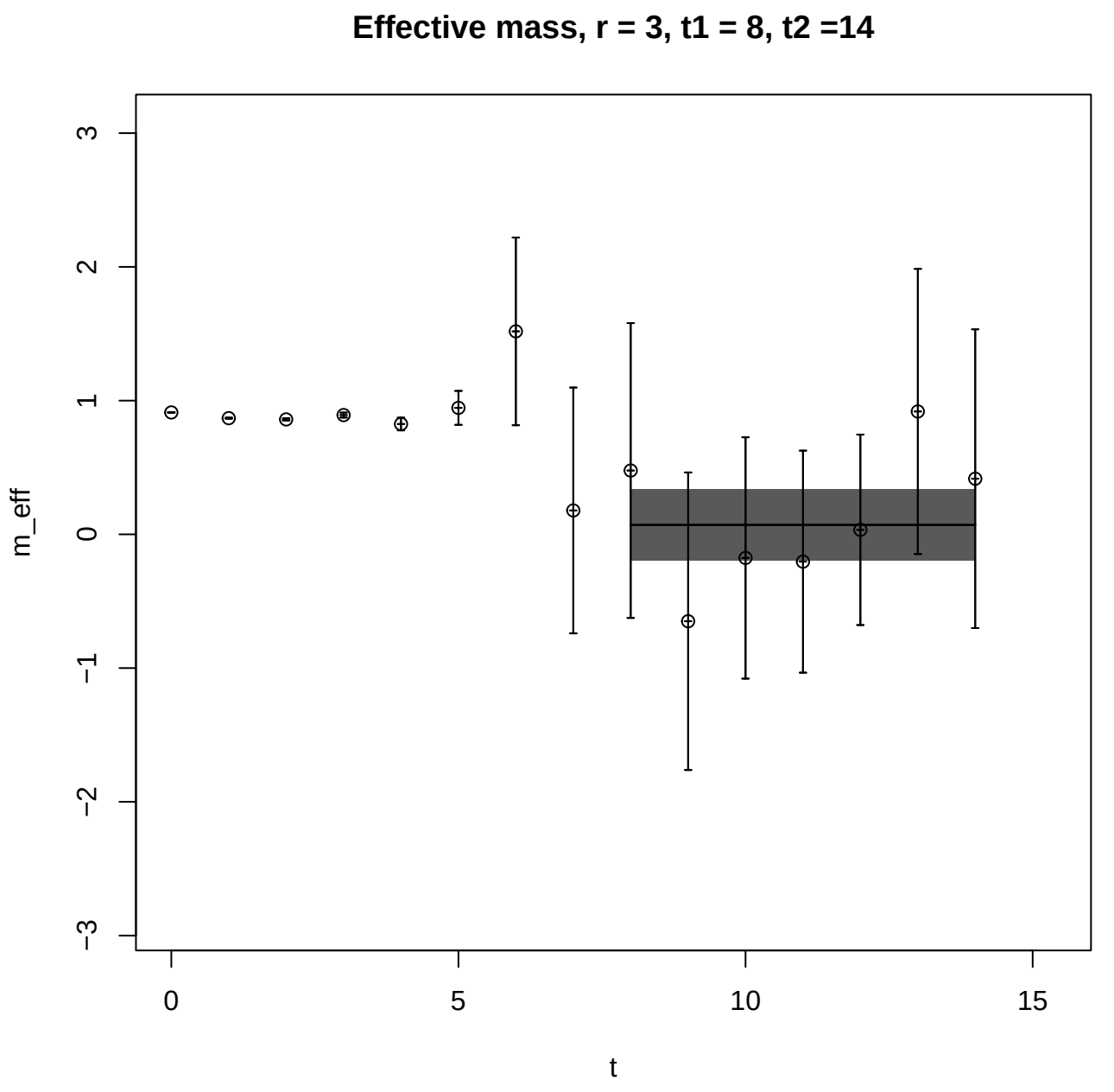
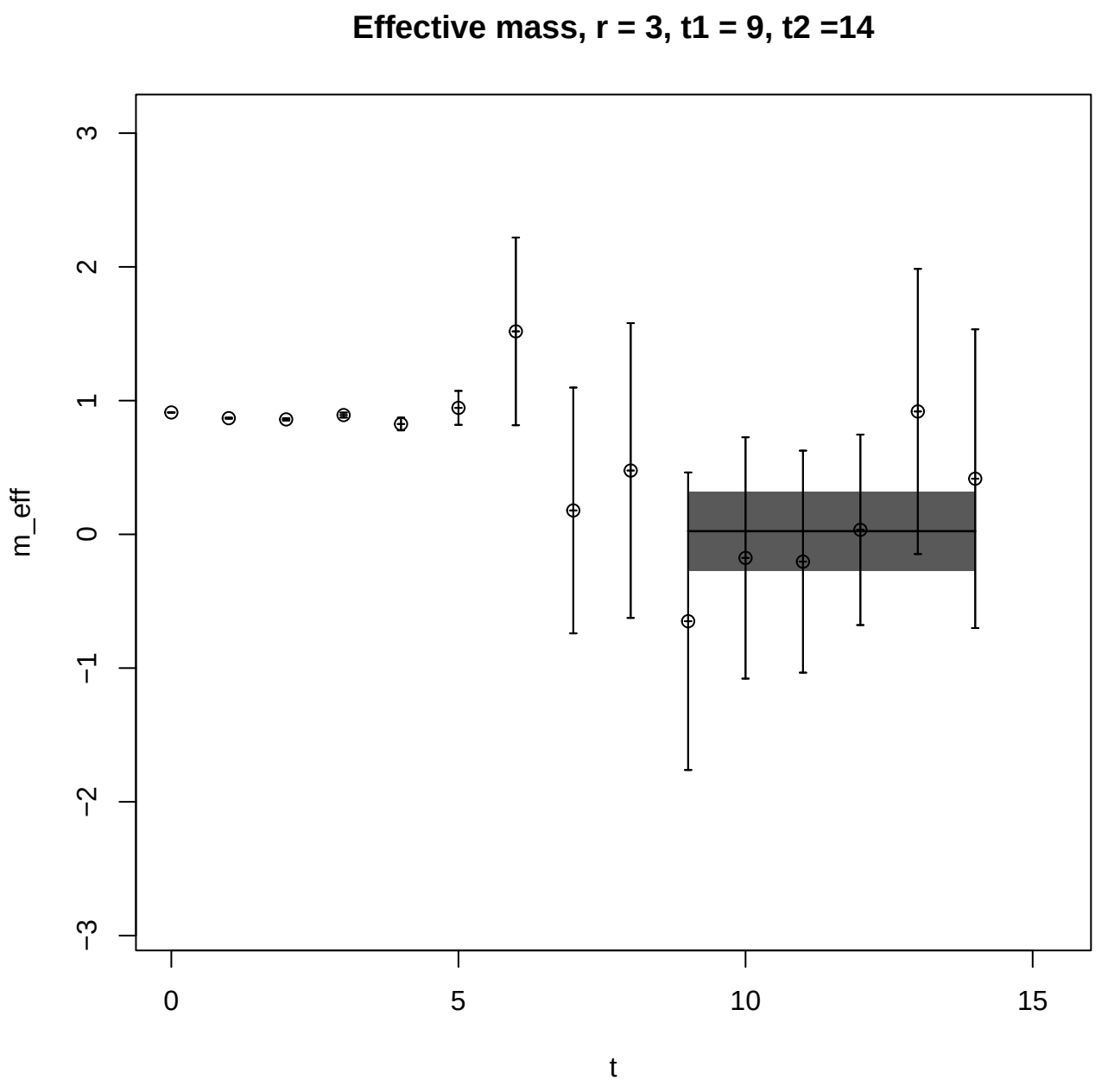
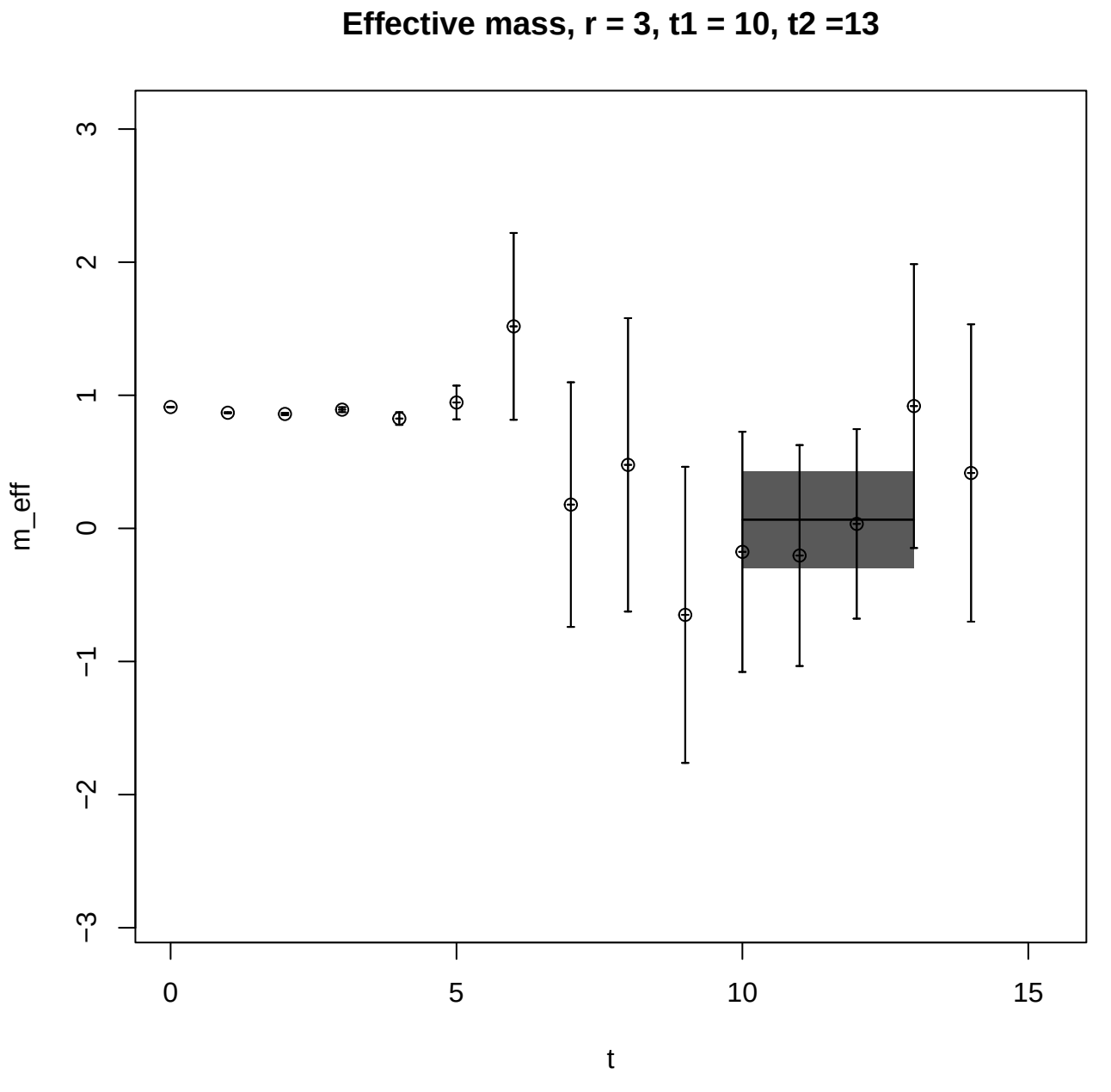
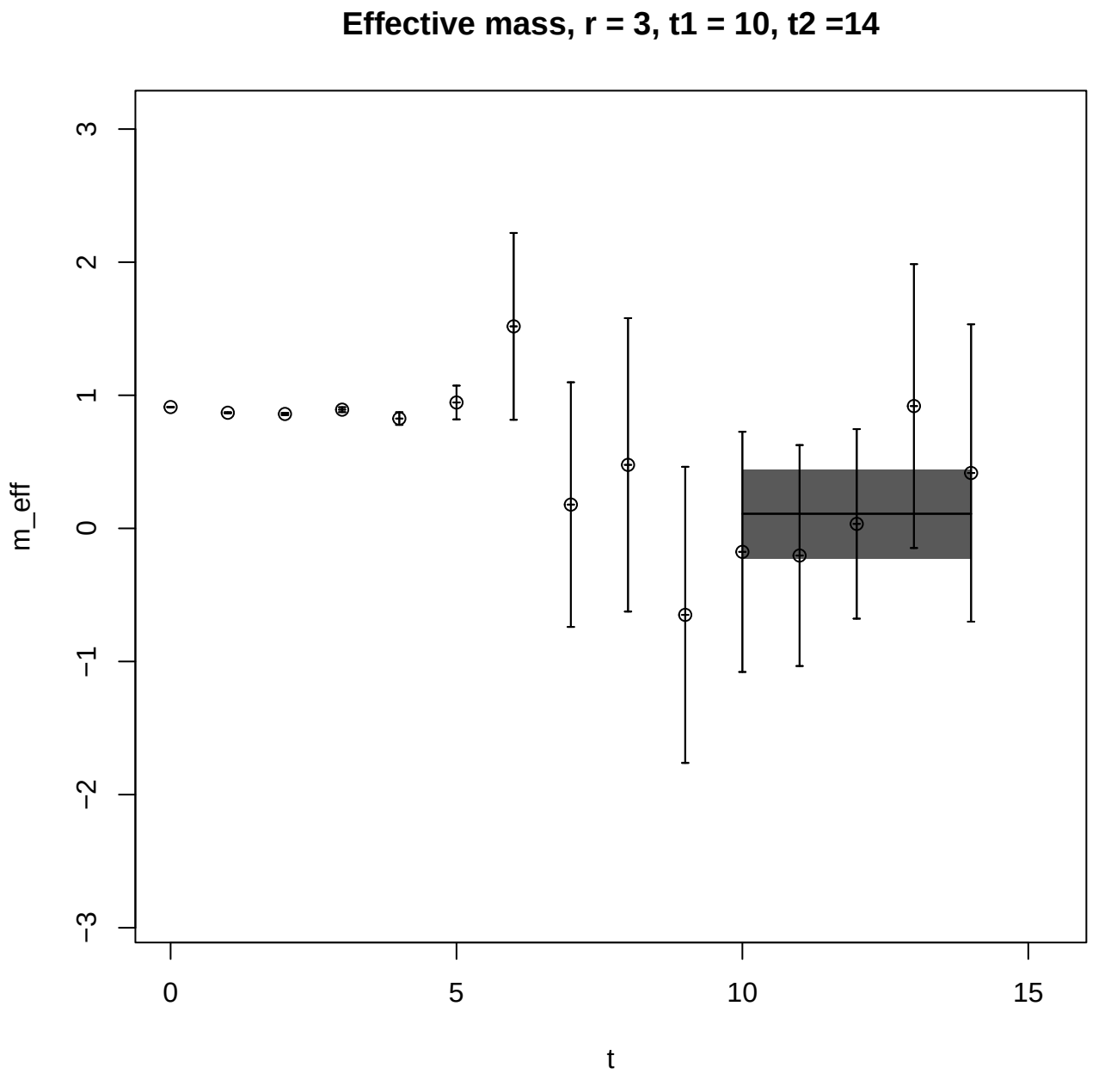
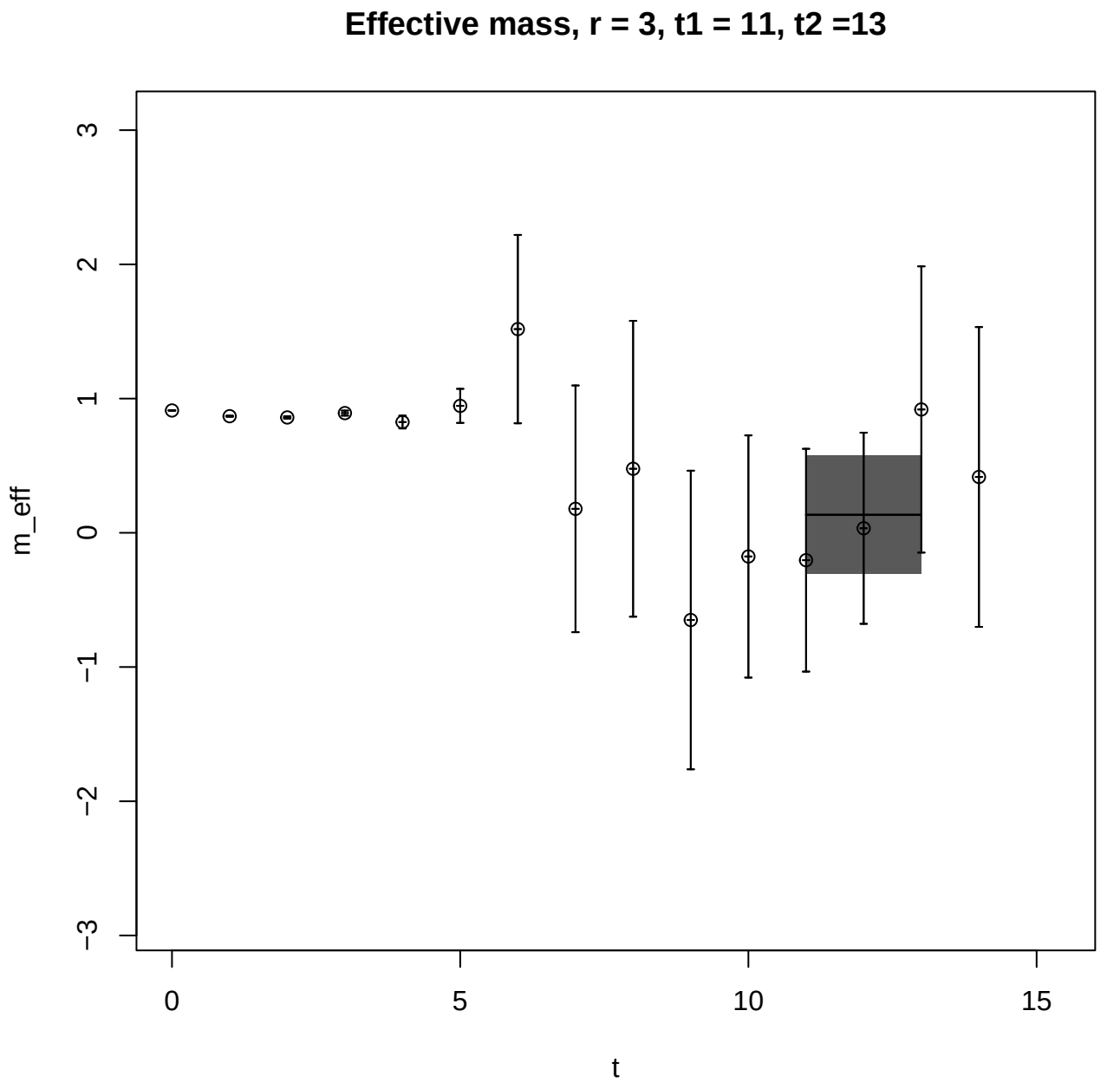

Effective mass, r = 3, t1 = 8, t2 =13
m_eff
t
3
2
1
0
−1
−2
−3
0
5
10
15

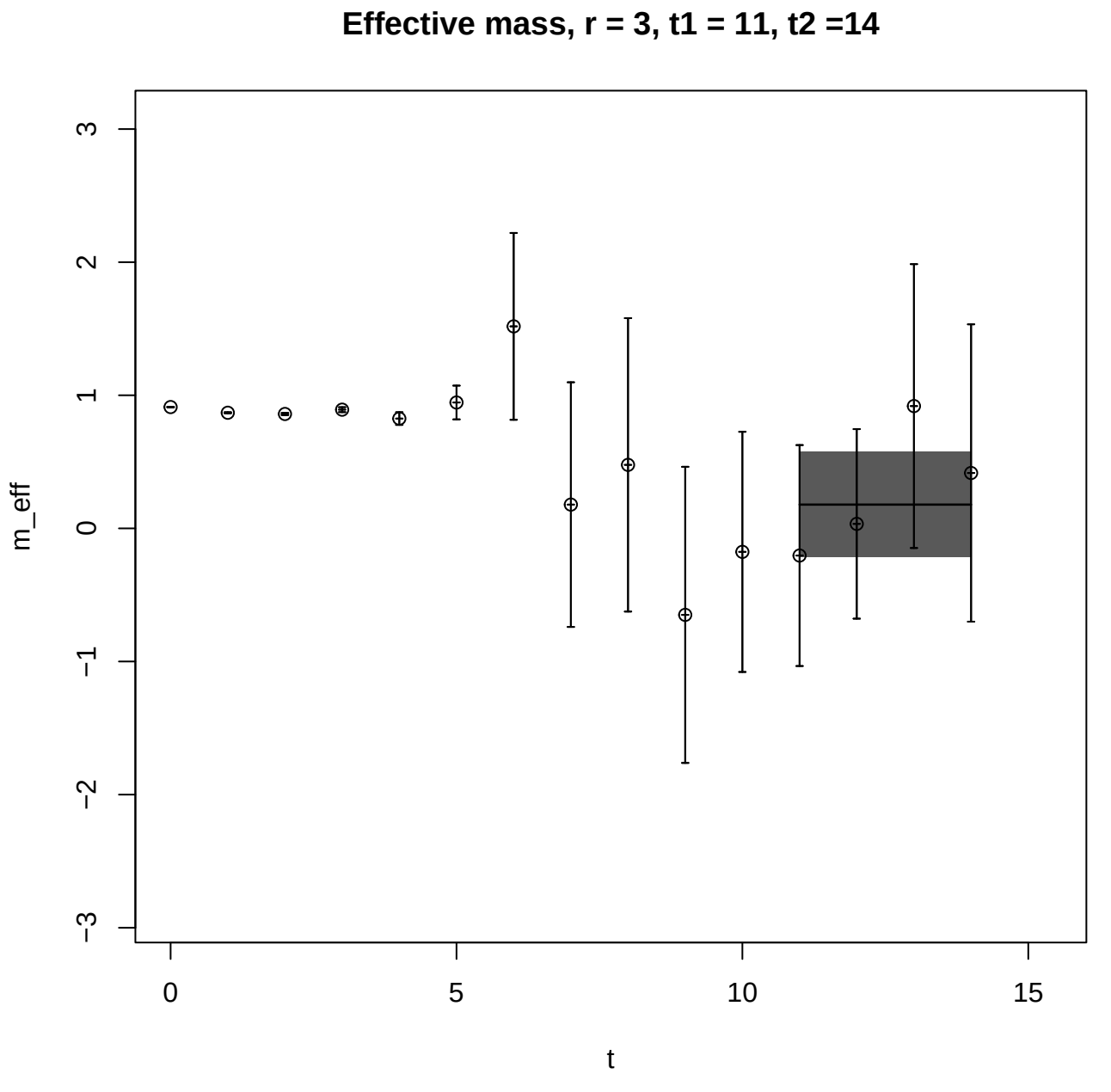

Effective mass, r = 3, t1 = 8, t2 =14
m_eff
t
3
2
1
0
−1
−2
−3
0
5
10
15

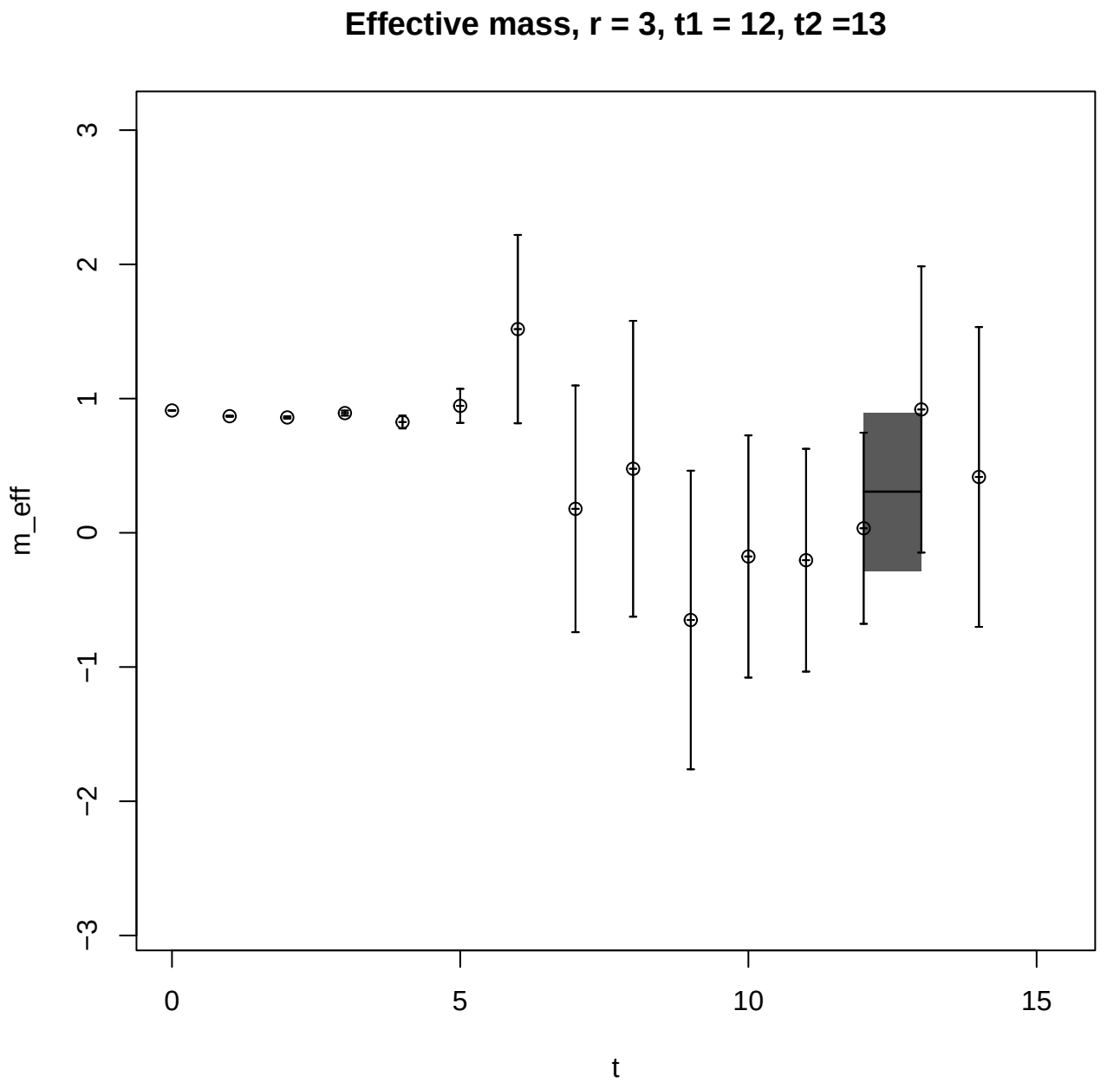

Effective mass, r = 3, t1 = 9, t2 =14
m_eff
t
3
2
1
0
−1
−2
−3
0
5
10
15

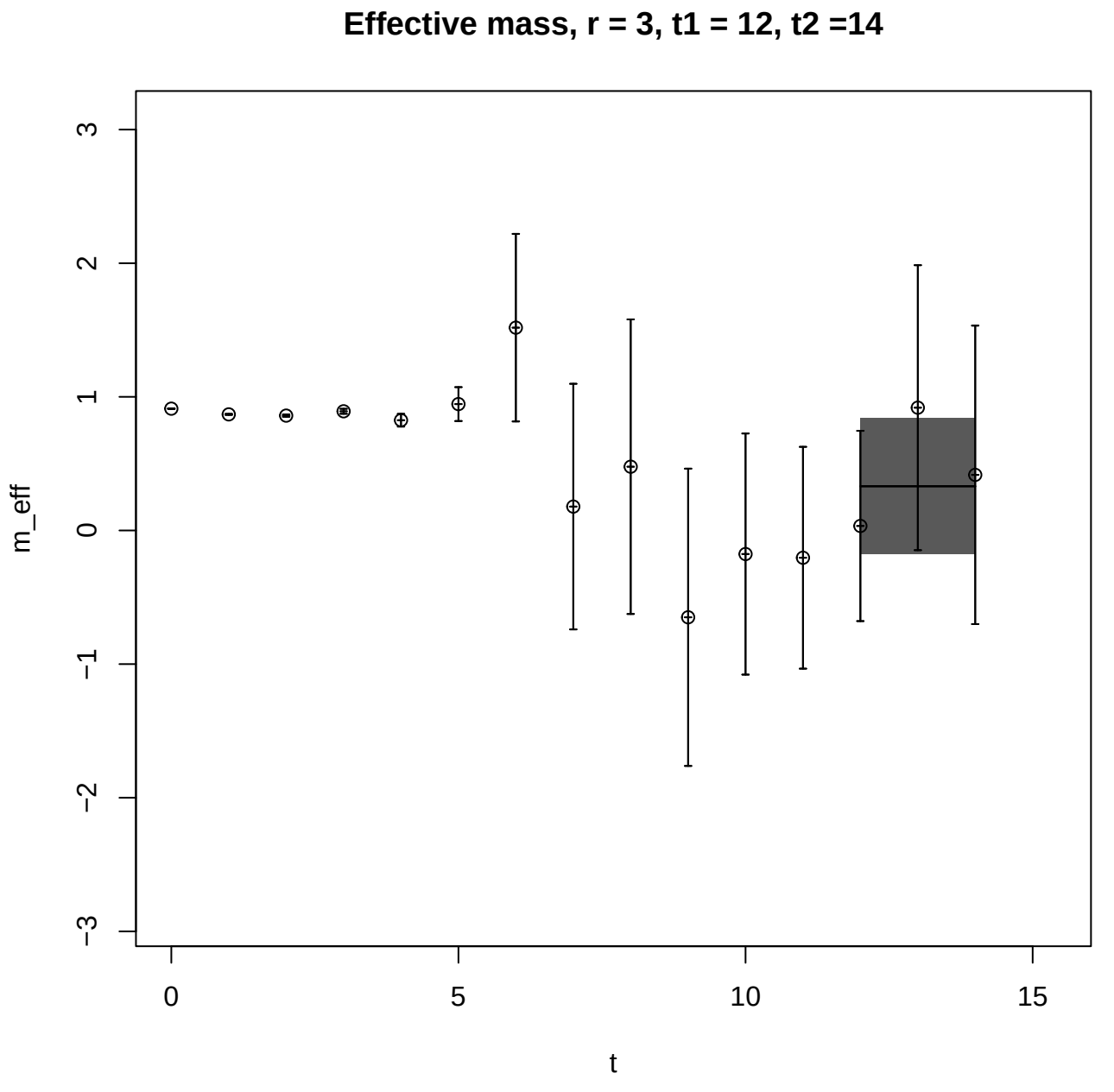

Effective mass, r = 3, t1 = 10, t2 =13
m_eff
t
3
2
1
0
−1
−2
−3
0
5
10
15

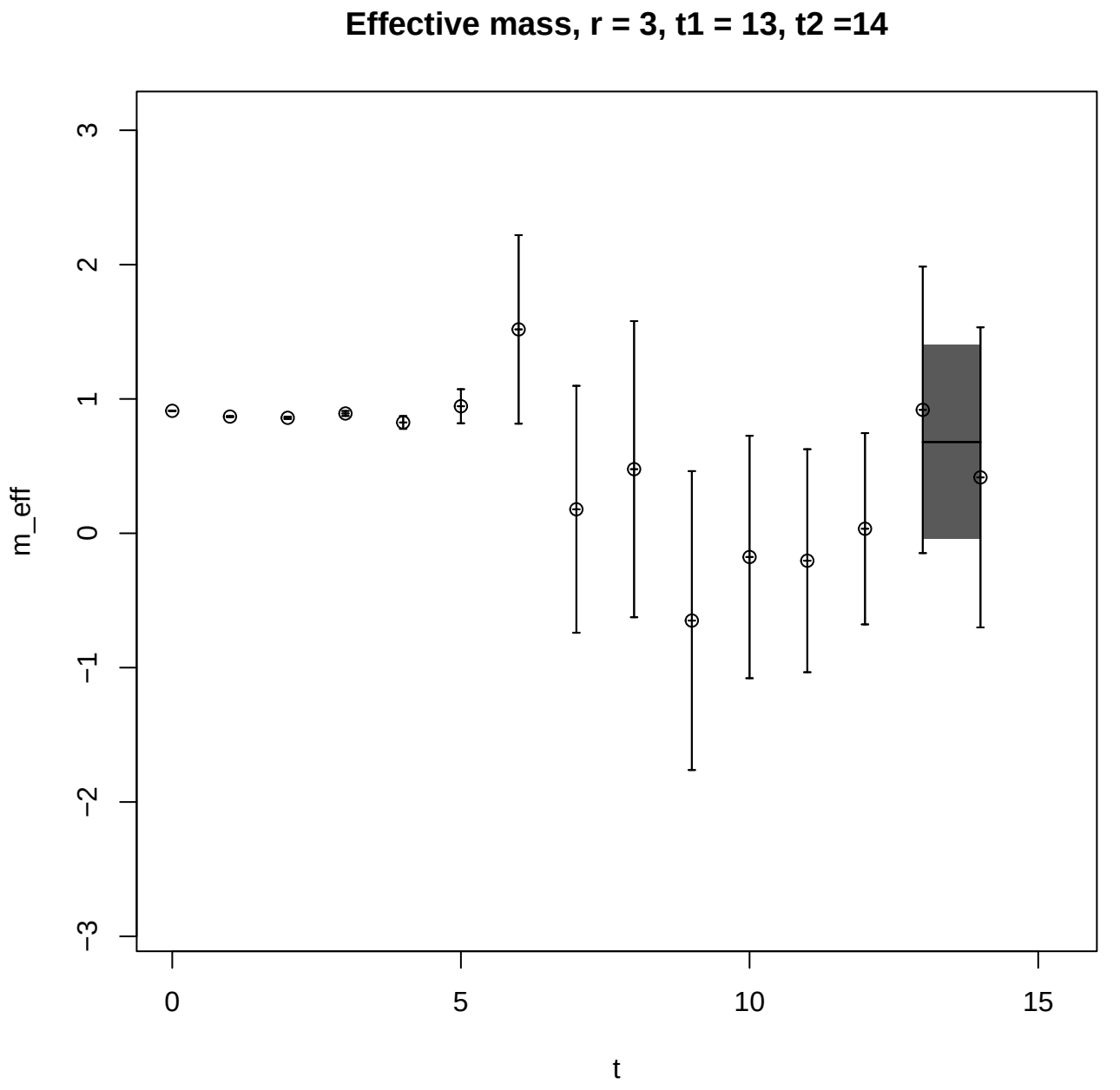

Effective mass, r = 3, t1 = 10, t2 =14
m_eff
t
3
2
1
0
−1
−2
−3
0
5
10
15

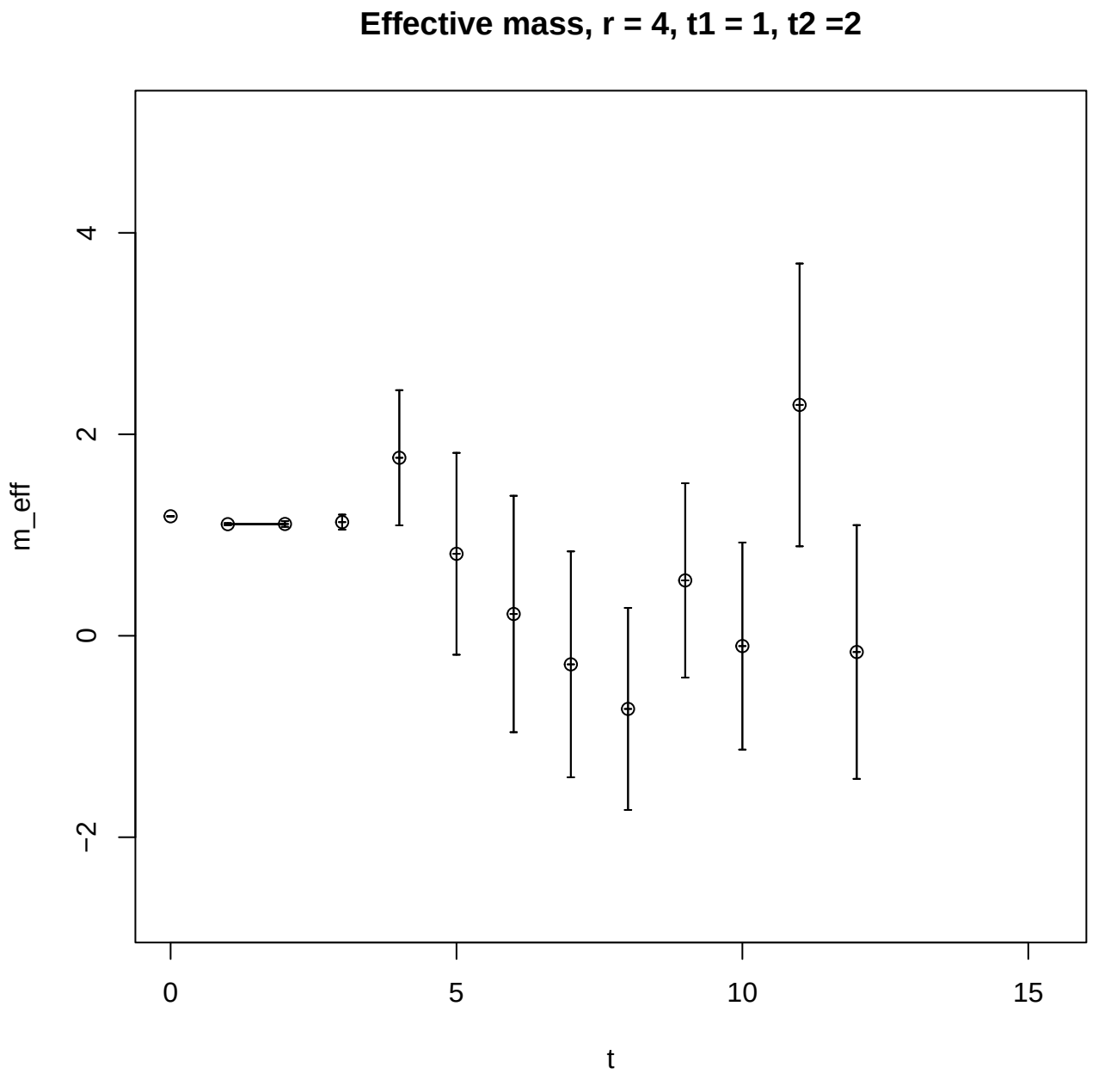

Effective mass, r = 3, t1 = 11, t2 =13
m_eff
t
3
2
1
0
−1
−2
−3
0
5
10
15

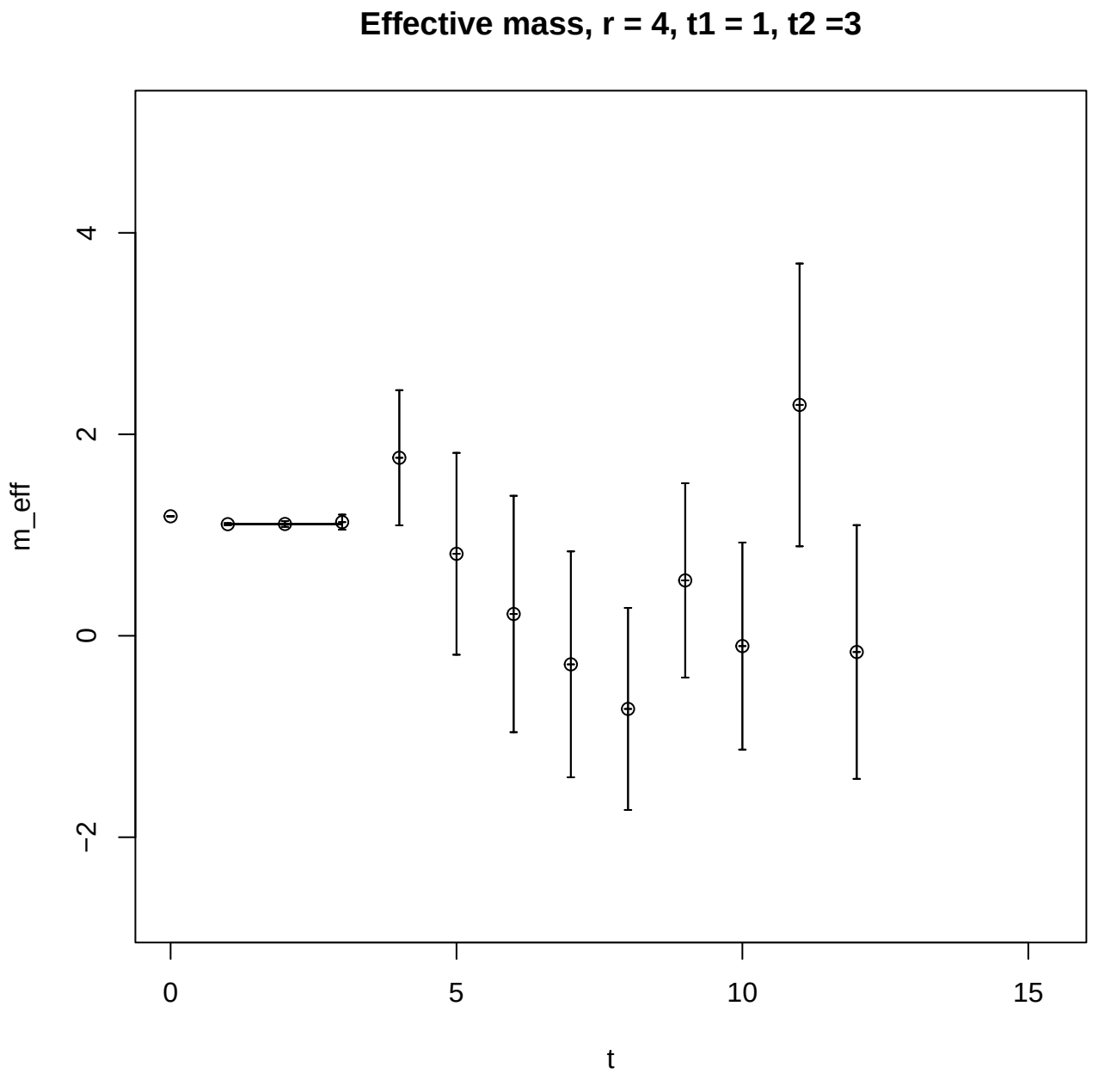

Effective mass, r = 3, t1 = 11, t2 =14
m_eff
t
3
2
1
0
−1
−2
−3
0
5
10
15

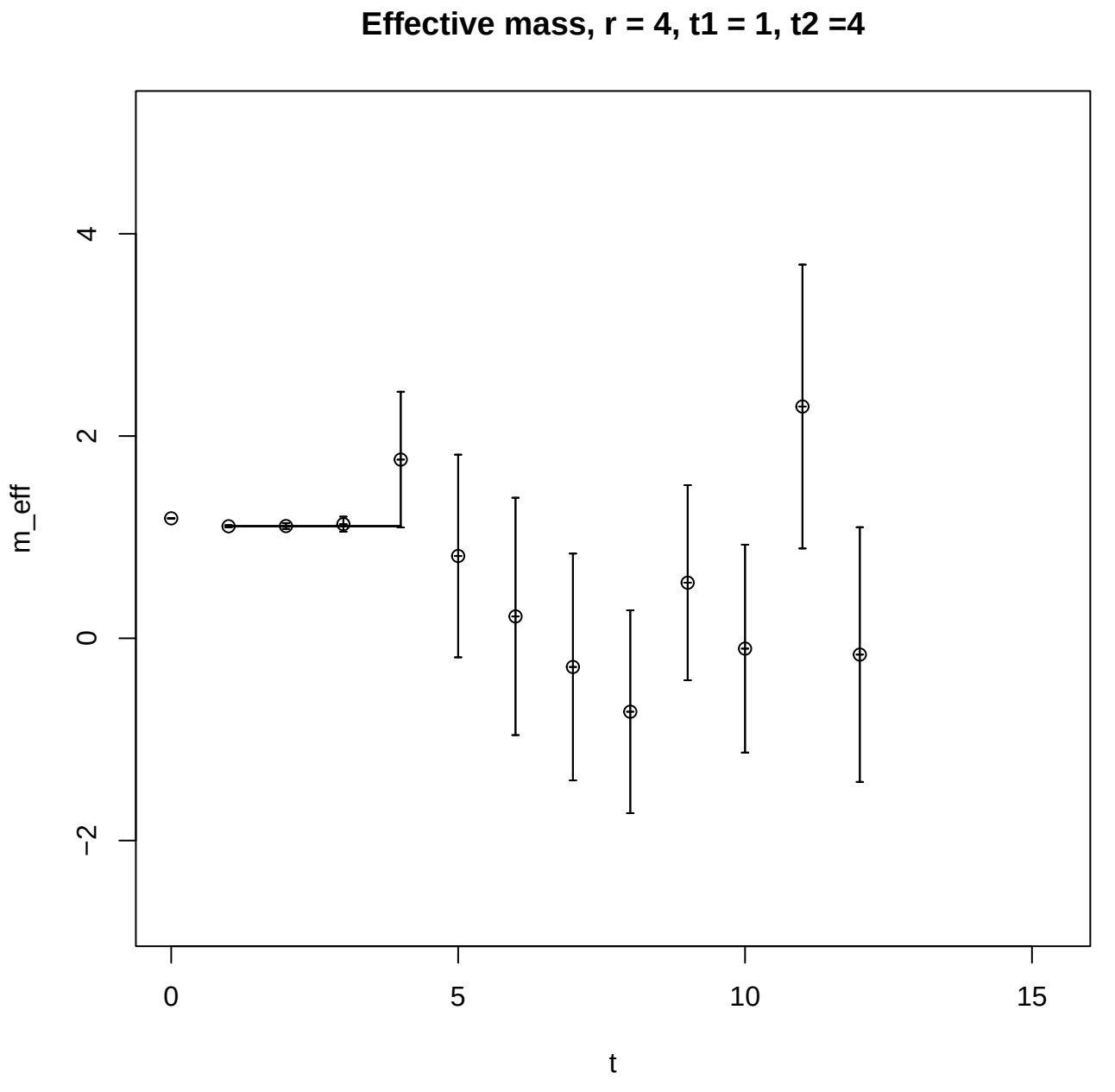

Effective mass, r = 3, t1 = 12, t2 =13
m_eff
t
3
2
1
0
−1
−2
−3
0
5
10
15

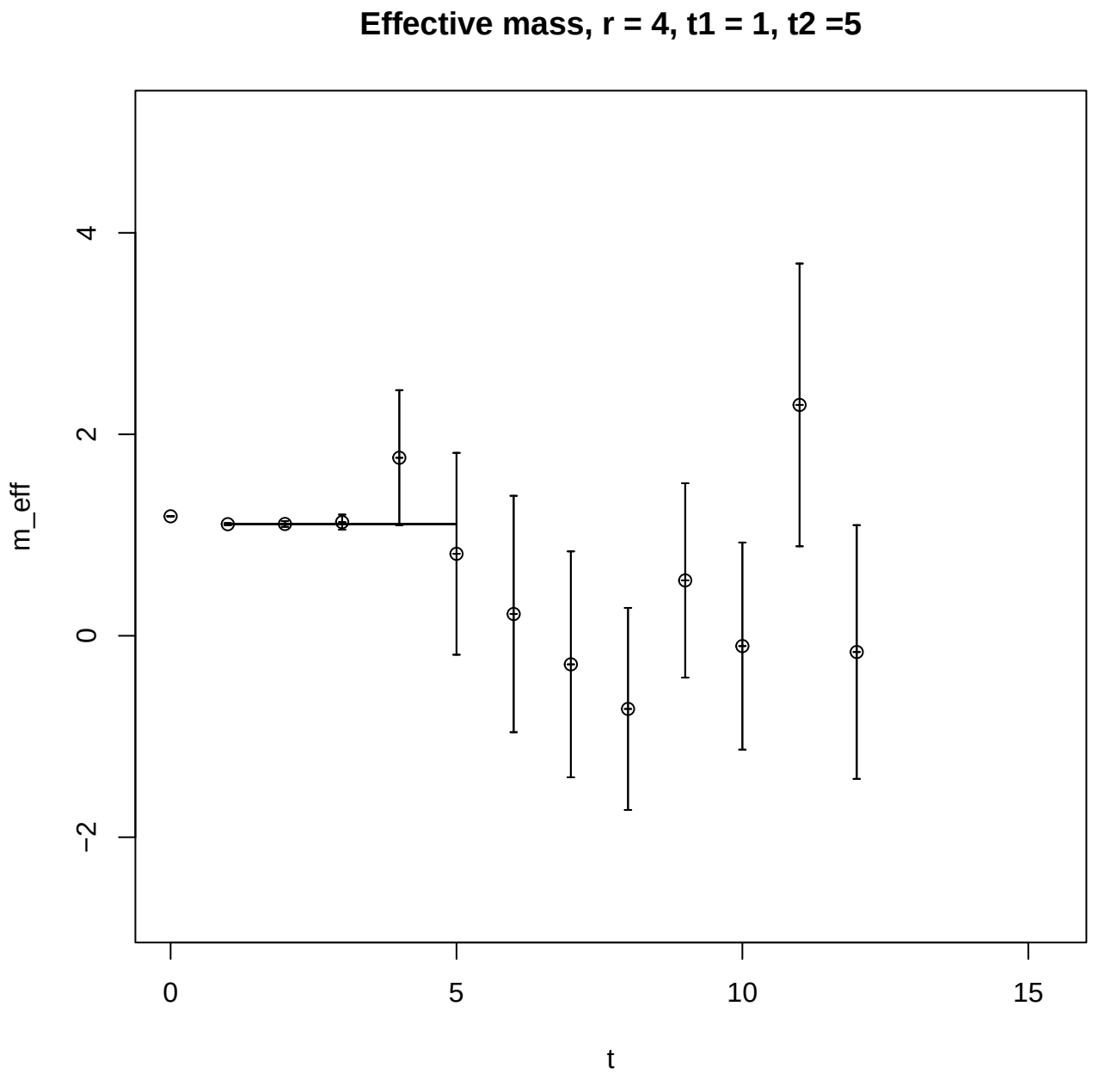

Effective mass, r = 3, t1 = 12, t2 =14
m_eff
t
3
2
1
0
−1
−2
−3
0
5
10
15

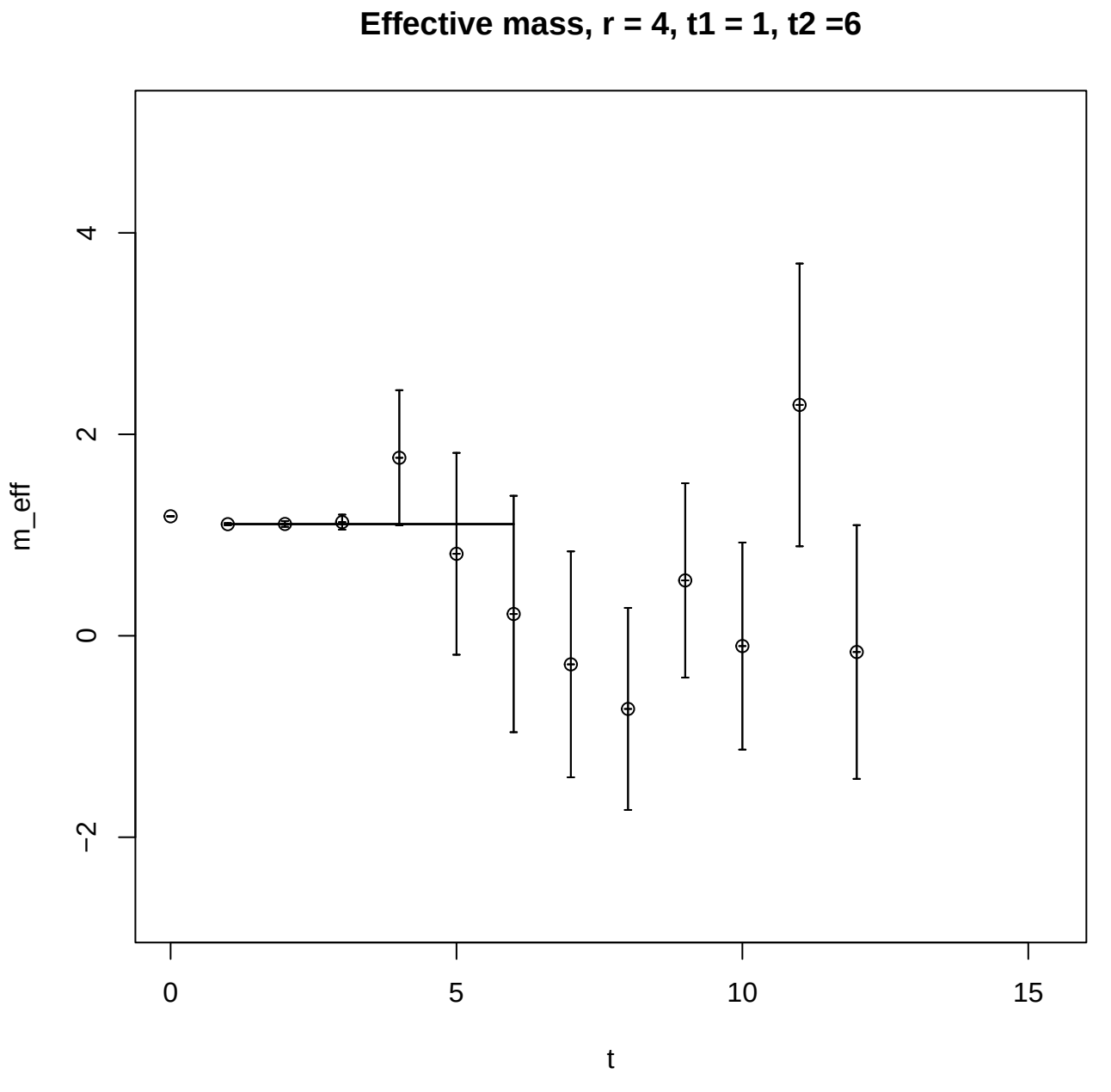

Effective mass, r = 3, t1 = 13, t2 =14
m_eff
t
3
2
1
0
−1
−2
−3
0
5
10
15

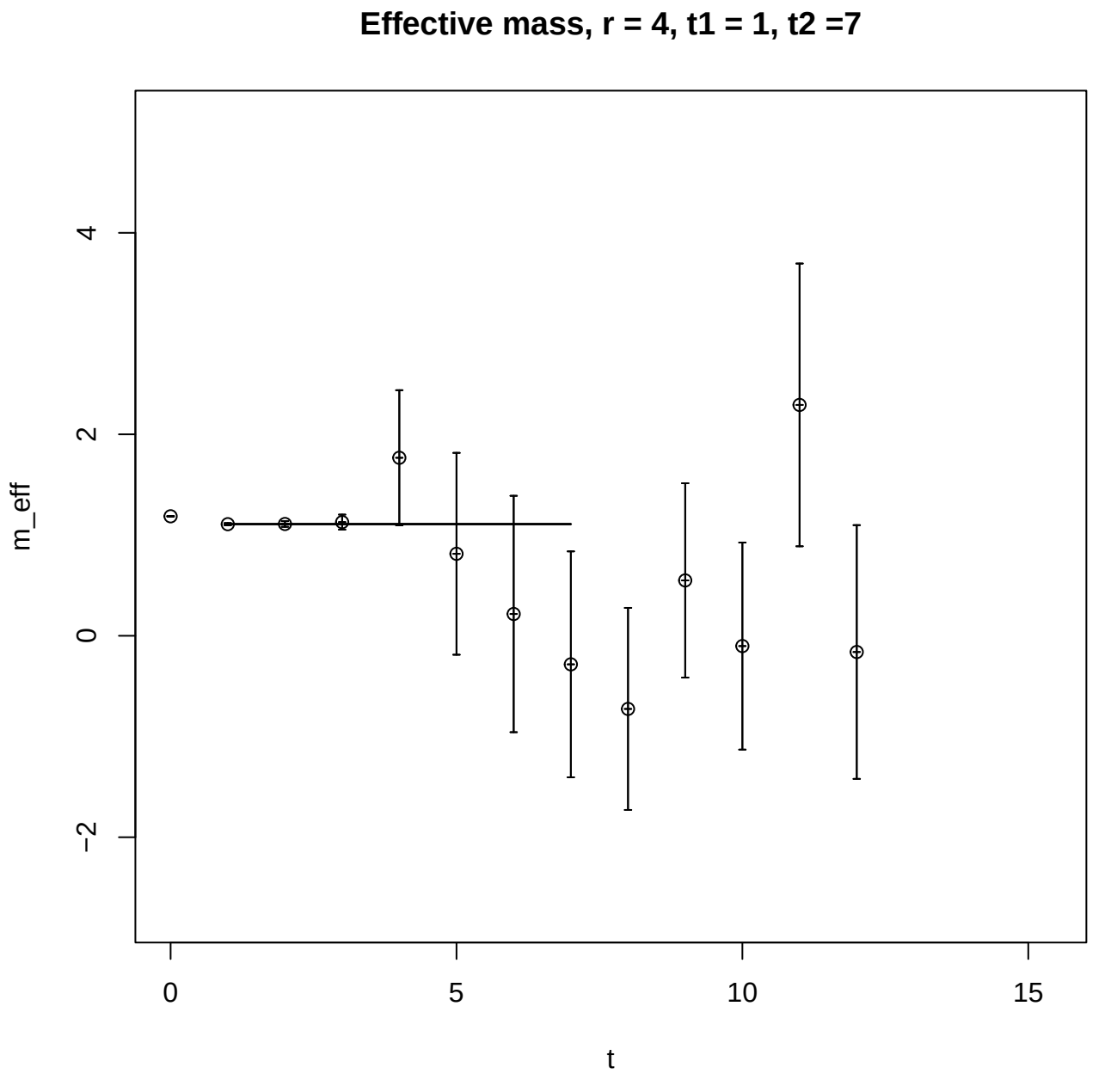

Effective mass, r = 4, t1 = 1, t2 =2
m_eff
4
2
0
−2
0
5
10
15
t

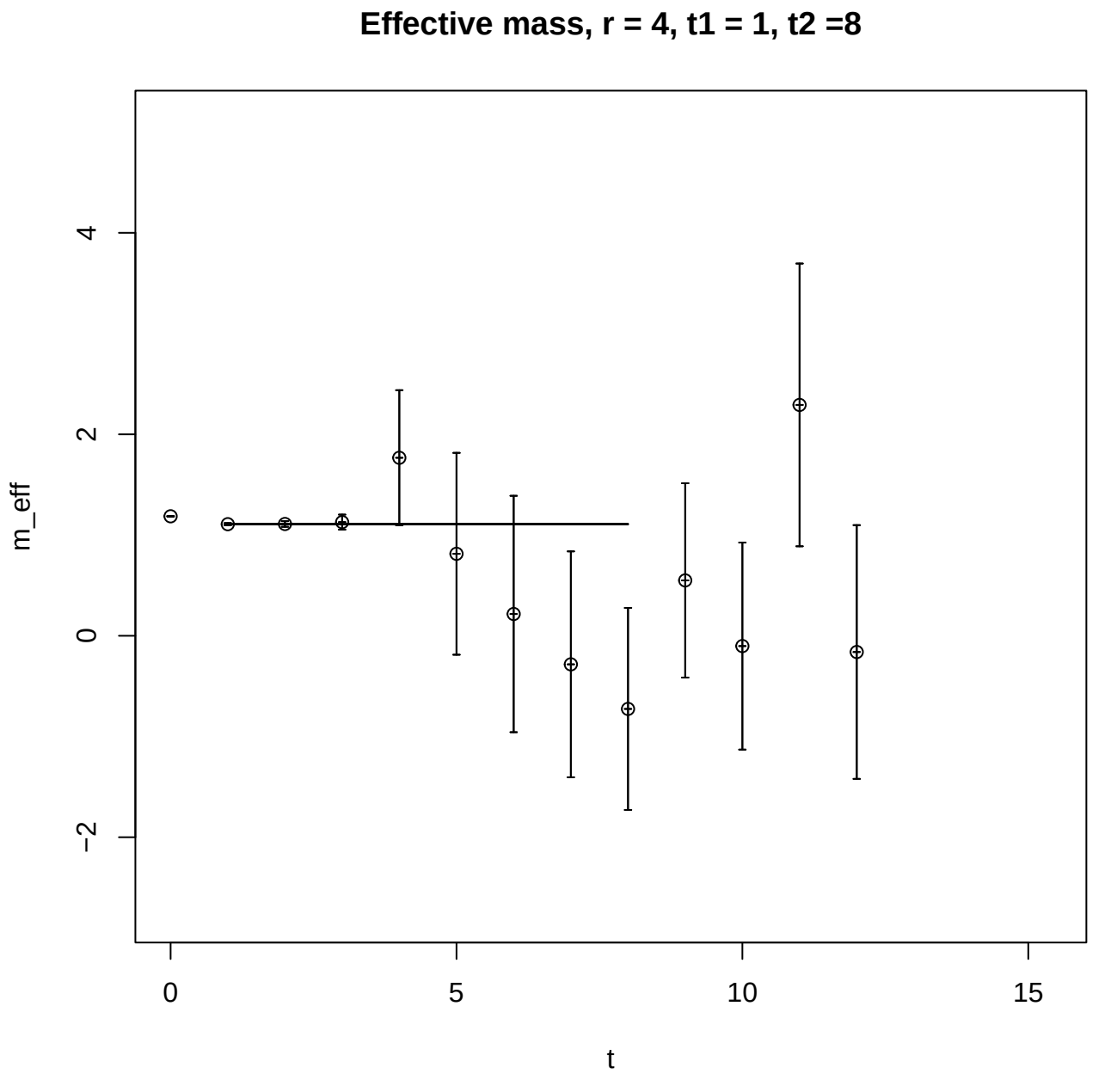

Effective mass, r = 4, t1 = 1, t2 =3
m_eff
4
2
0
−2
0
5
10
15
t

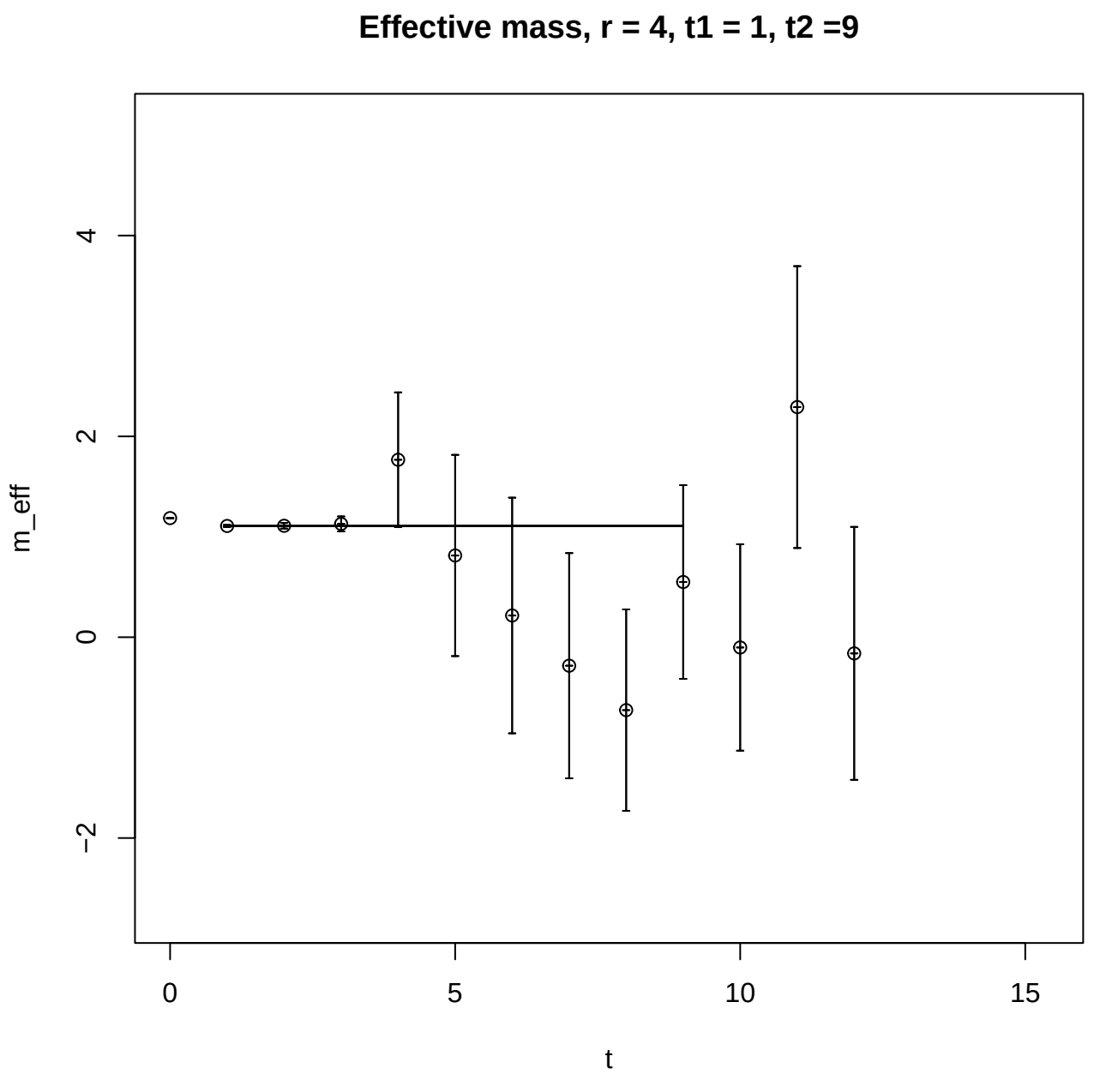

Effective mass, r = 4, t1 = 1, t2 =4
m_eff
4
2
0
−2
0
5
10
15
t

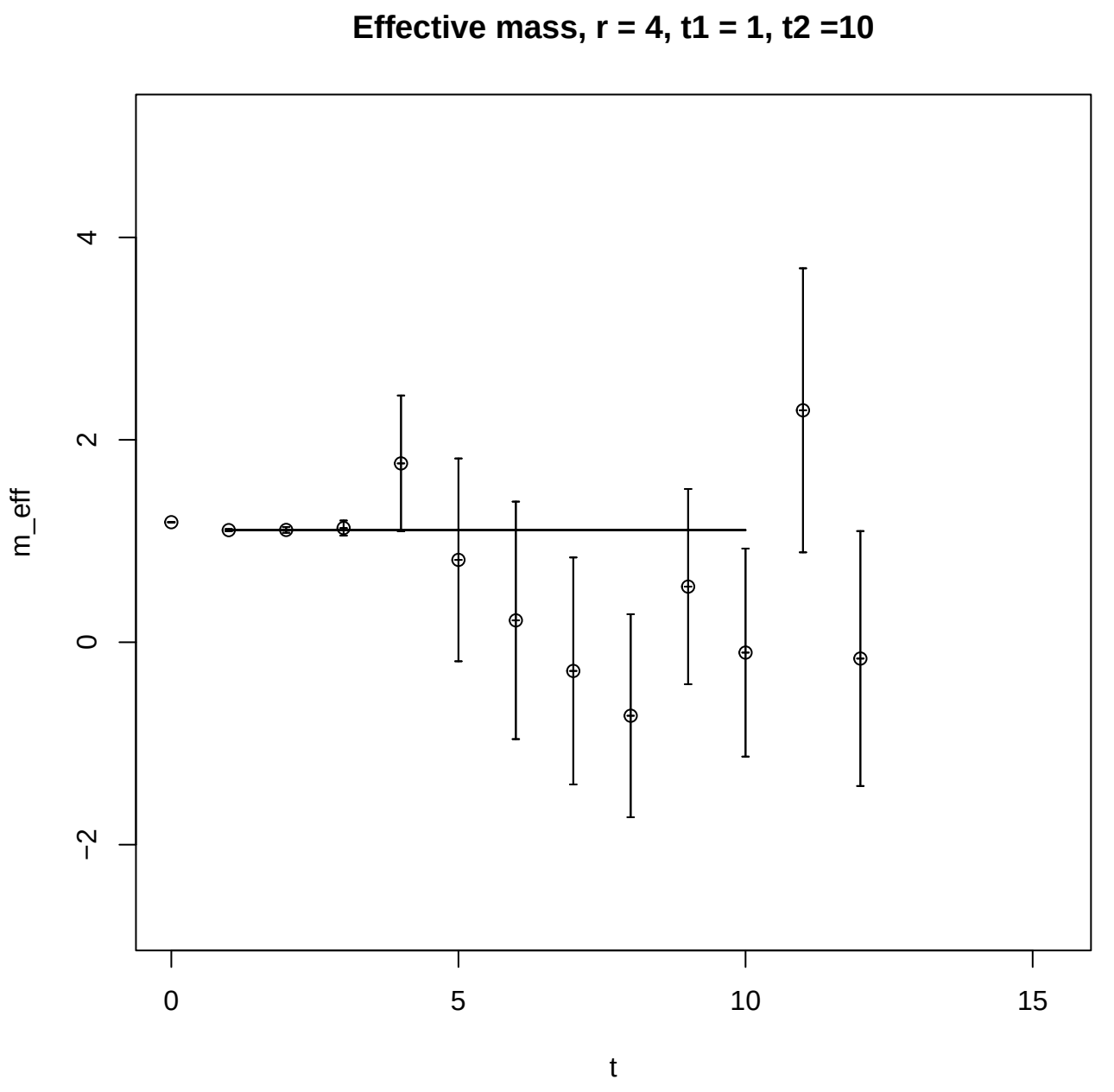

Effective mass, r = 4, t1 = 1, t2 =5
m_eff
4
2
0
−2
0
5
10
15
t

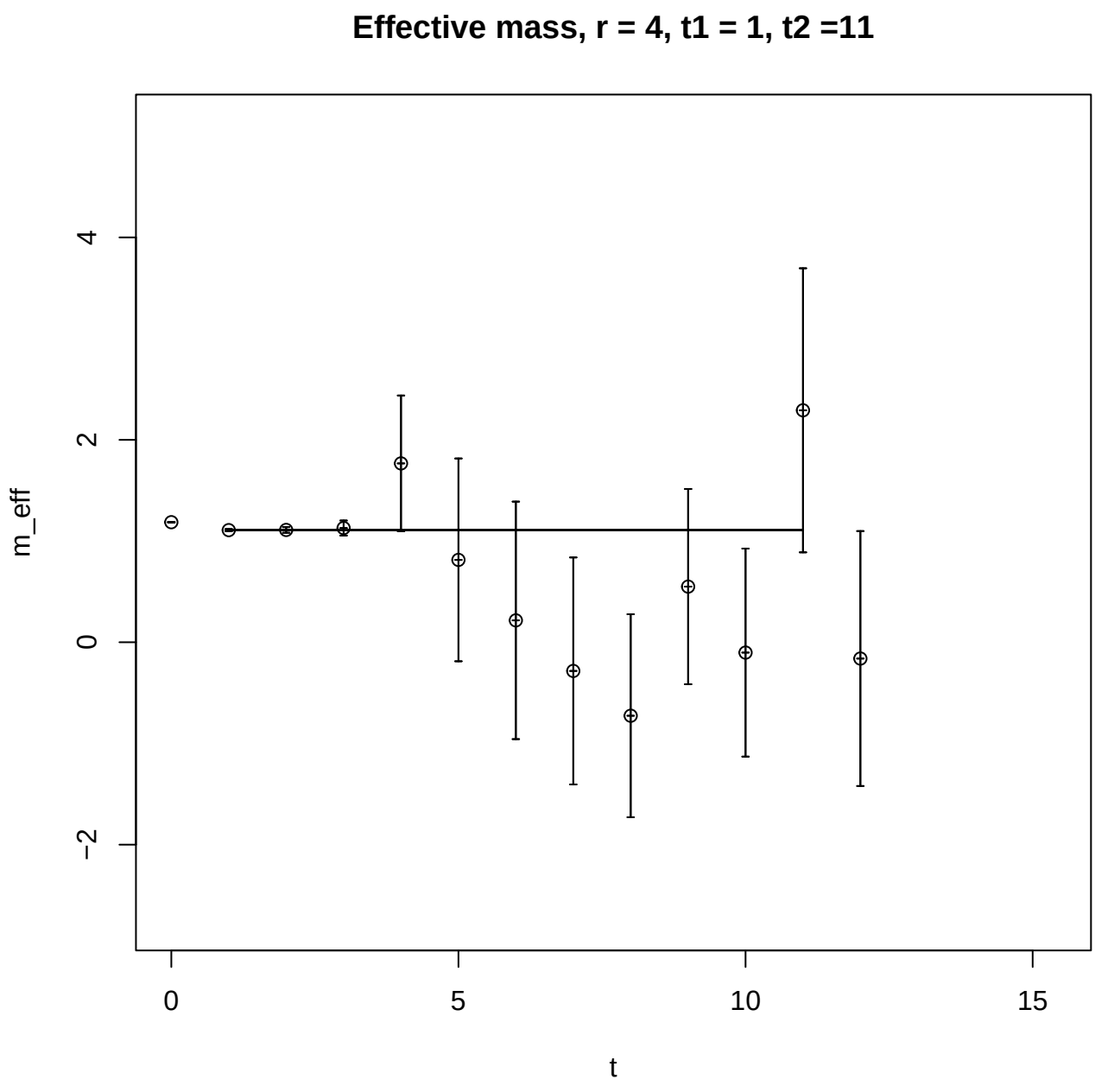

Effective mass, r = 4, t1 = 1, t2 =6
m_eff
4
2
0
−2
0
5
10
15
t

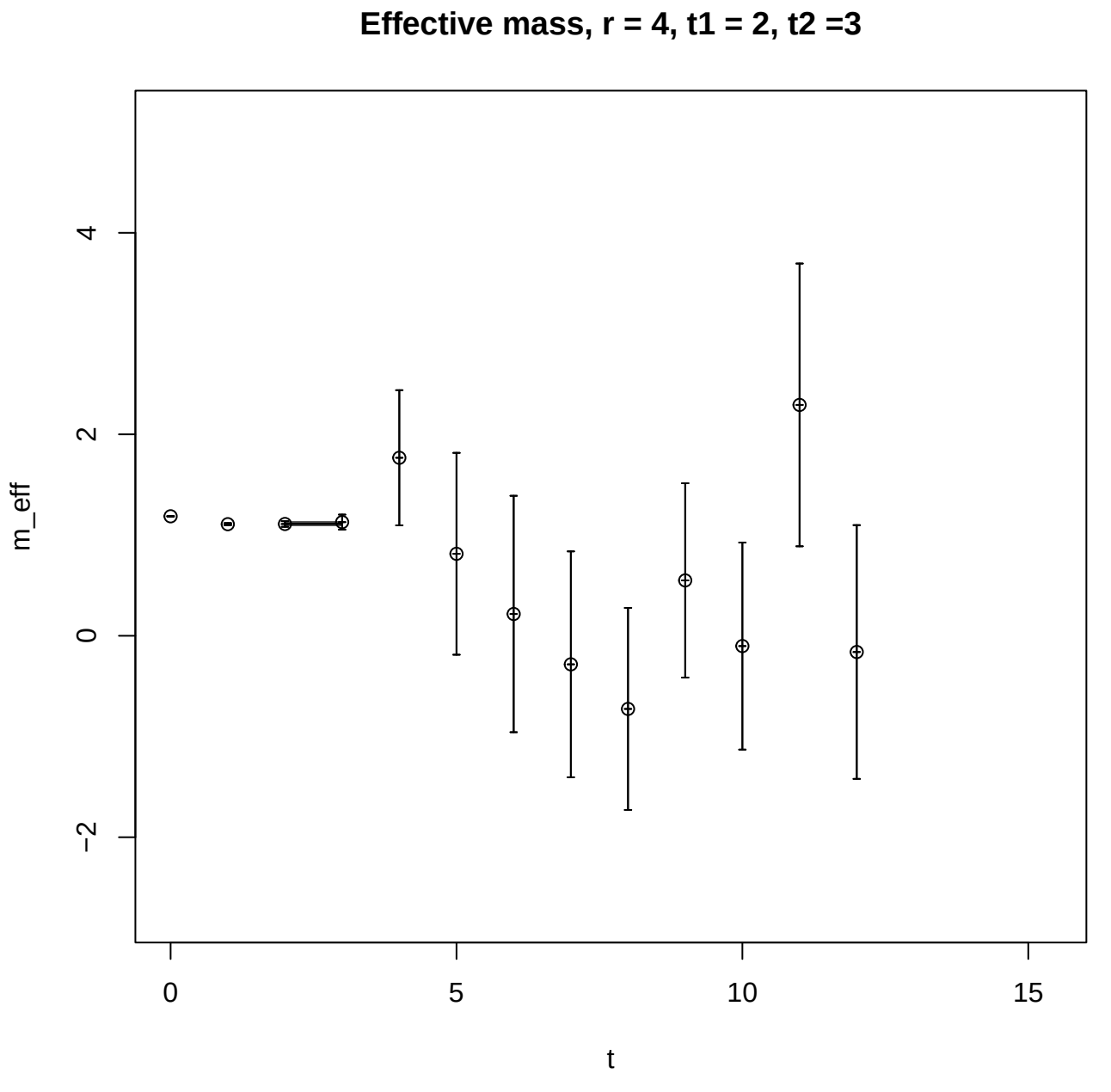

Effective mass, r = 4, t1 = 1, t2 =7
m_eff
4
2
0
−2
0
5
10
15
t

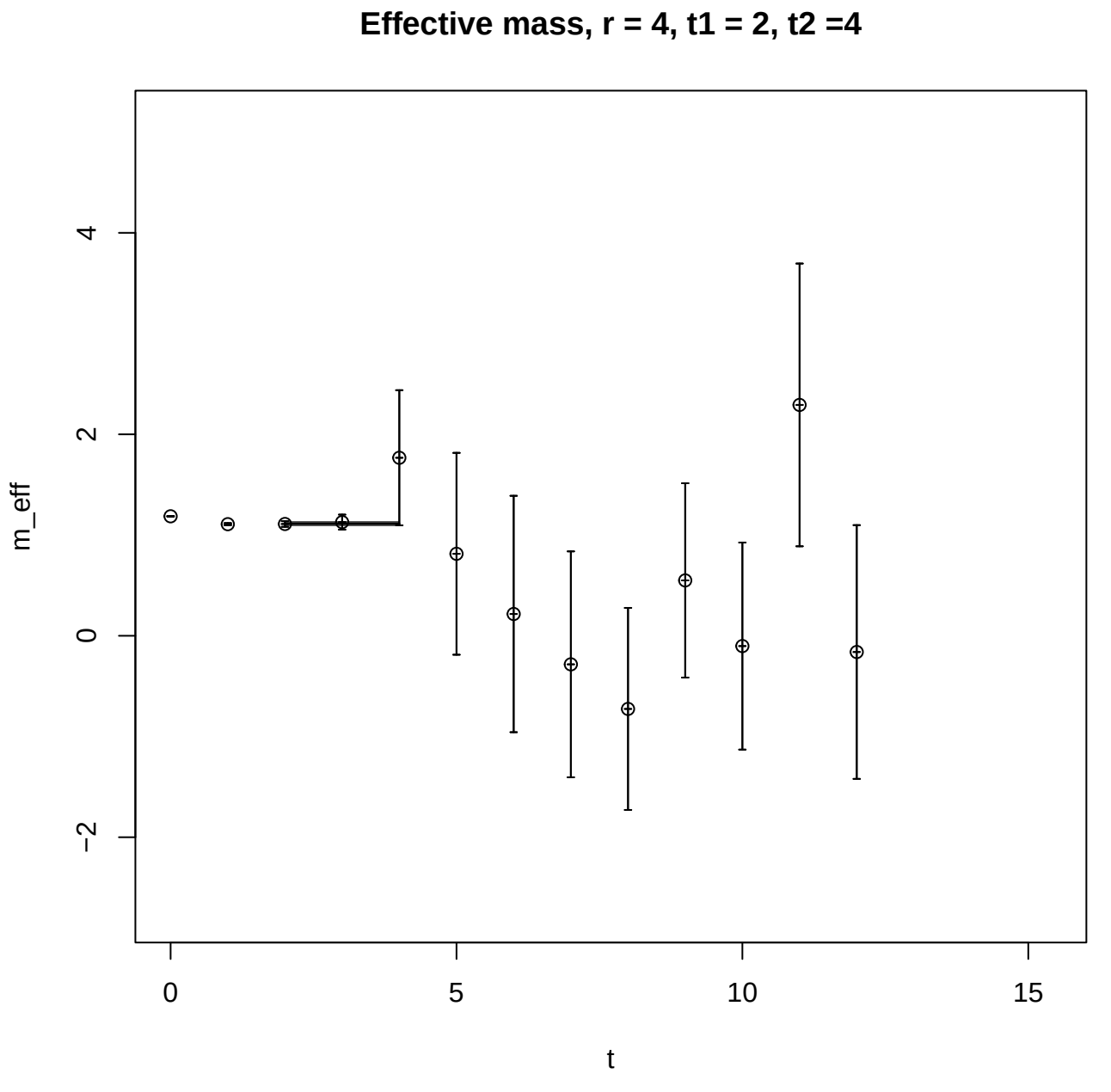

Effective mass, r = 4, t1 = 1, t2 =8
m_eff
4
2
0
−2
0
5
10
15
t

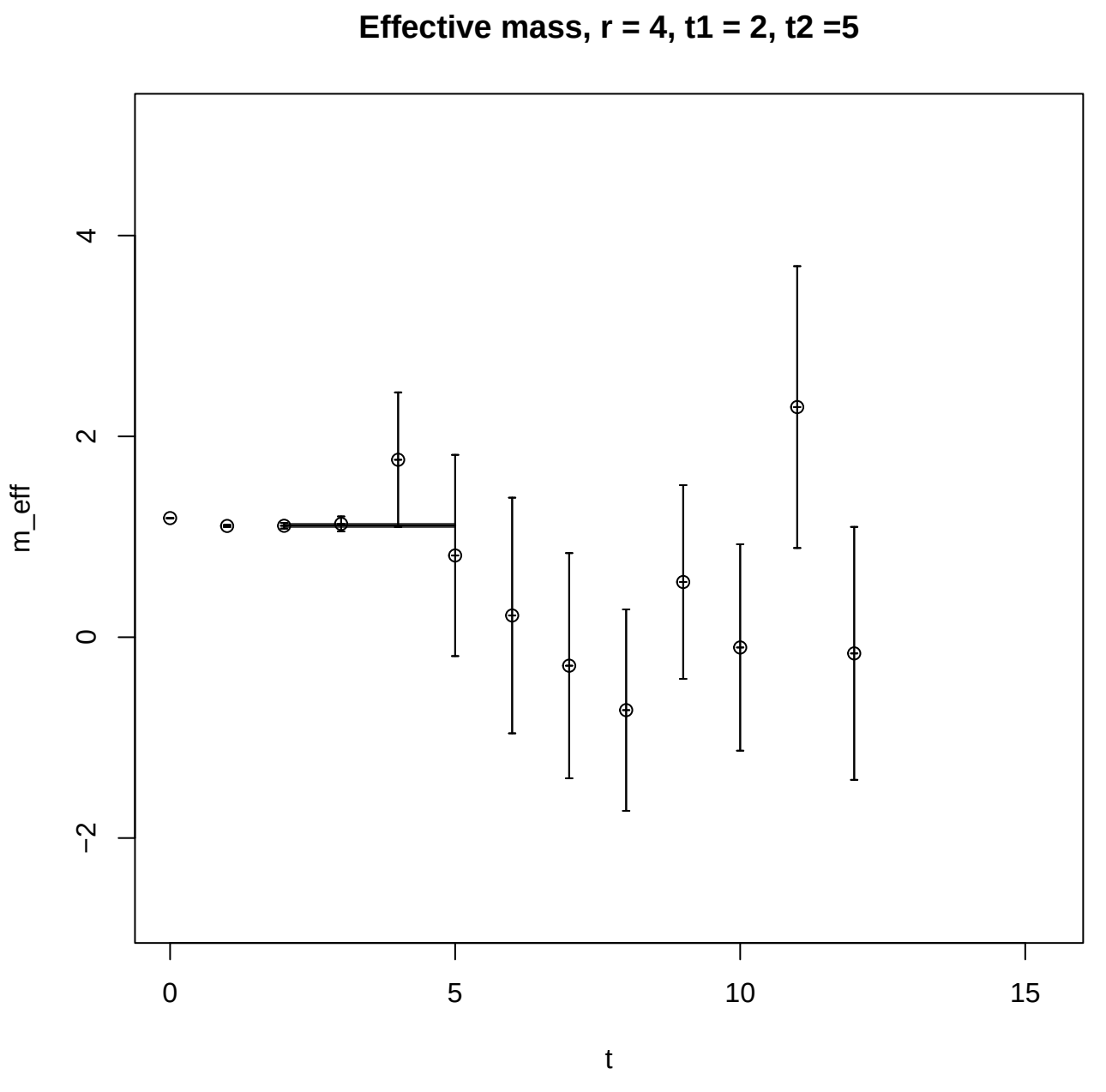

Effective mass, r = 4, t1 = 1, t2 =9
m_eff
4
2
0
−2
0
5
10
15
t

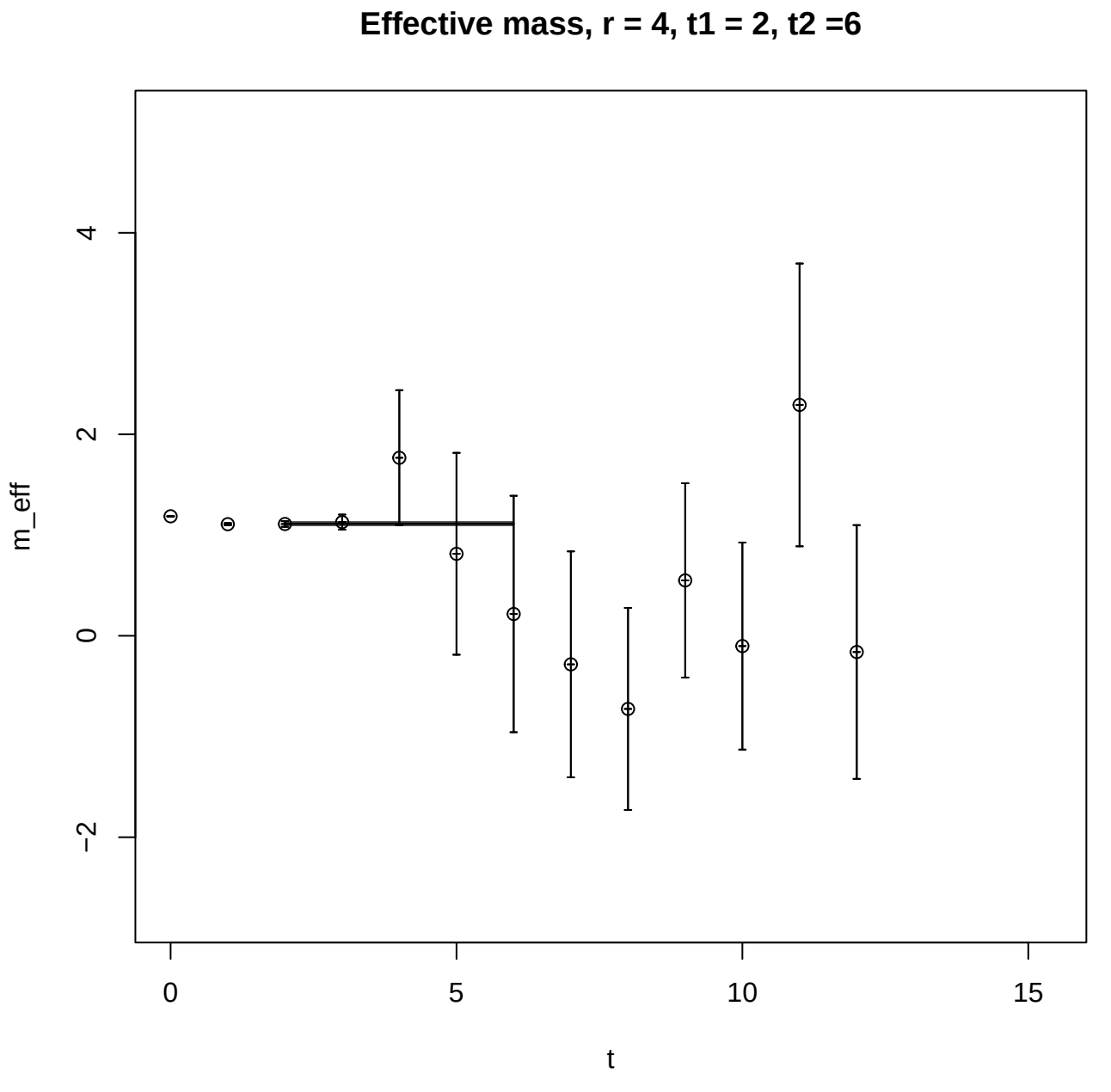

Effective mass, r = 4, t1 = 1, t2 =10
m_eff
4
2
0
−2
0
5
10
15
t

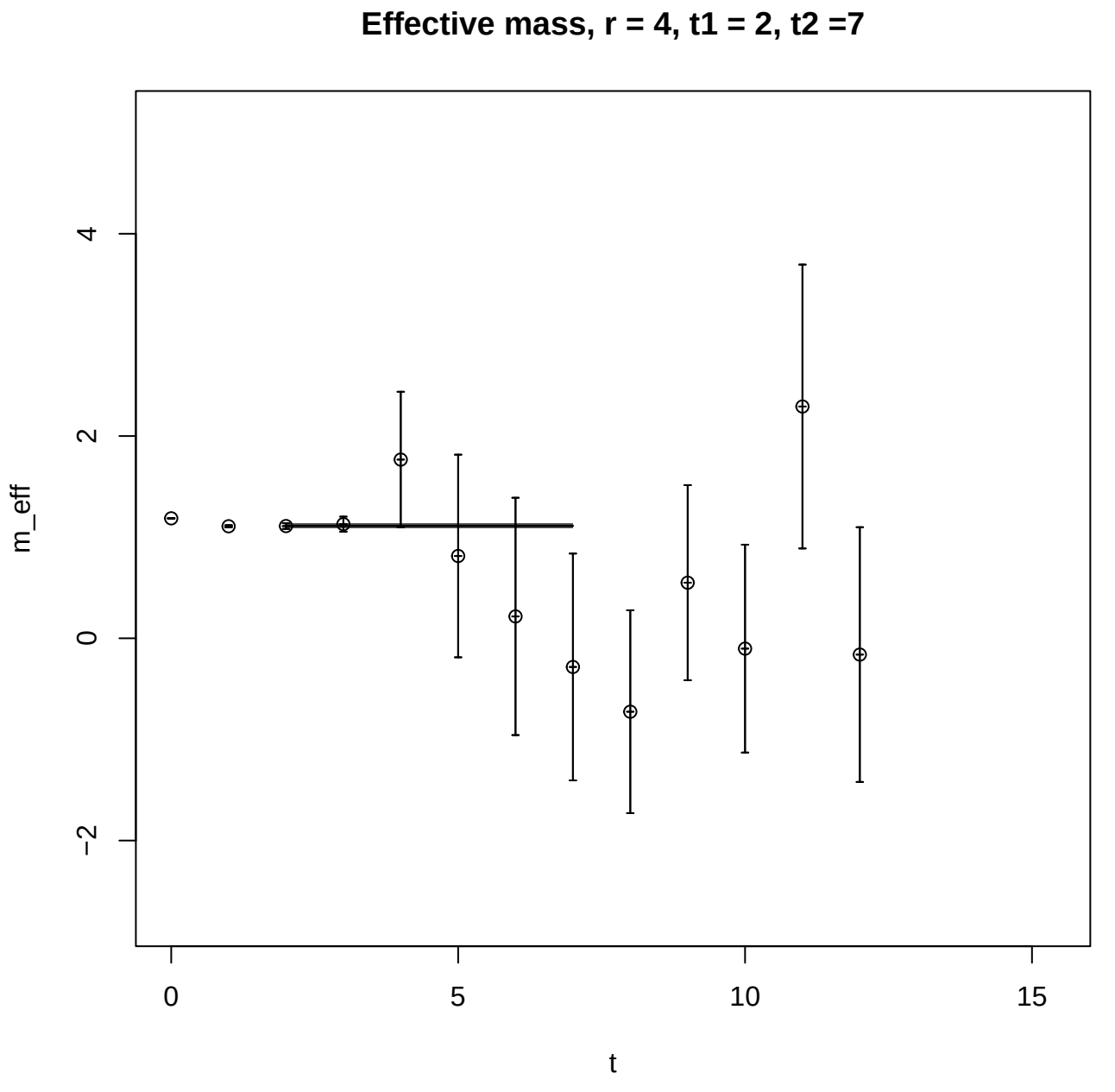

Effective mass, r = 4, t1 = 1, t2 =11
m_eff
4
2
0
−2
0
5
10
15
t

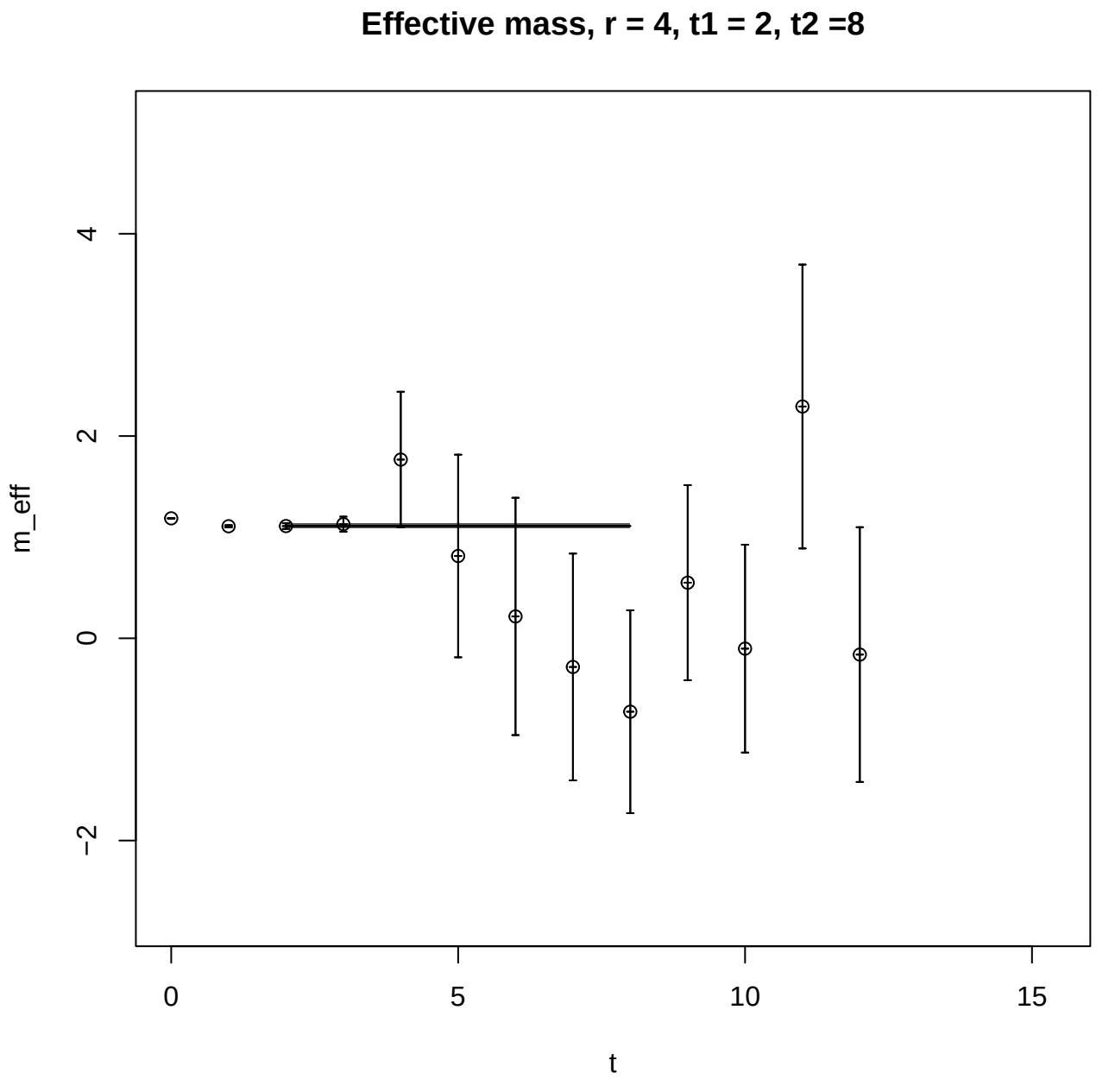

Effective mass, r = 4, t1 = 2, t2 =3
m_eff
4
2
0
−2
0
5
10
15
t

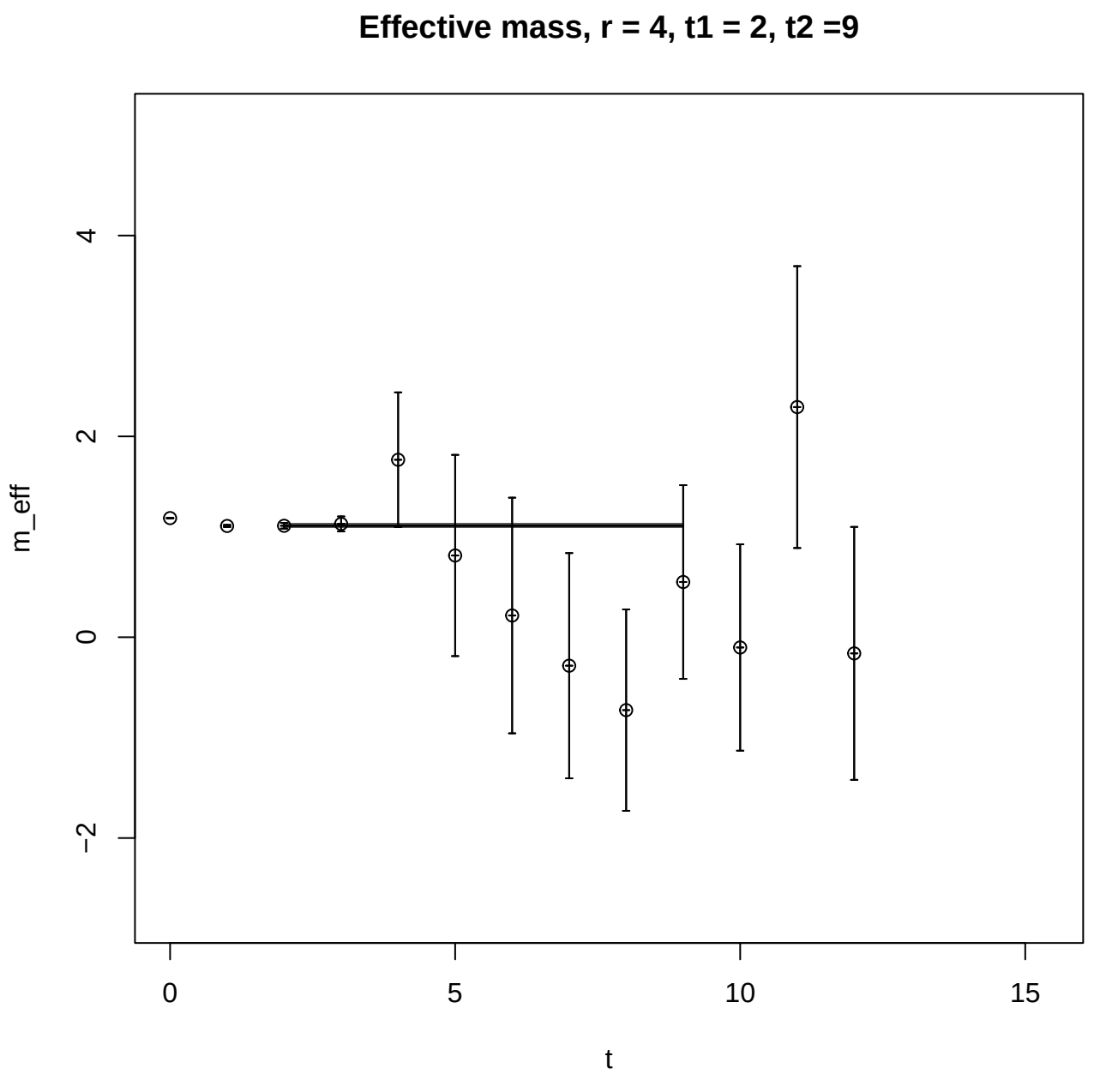

Effective mass, r = 4, t1 = 2, t2 =4
m_eff
4
2
0
−2
0
5
10
15
t

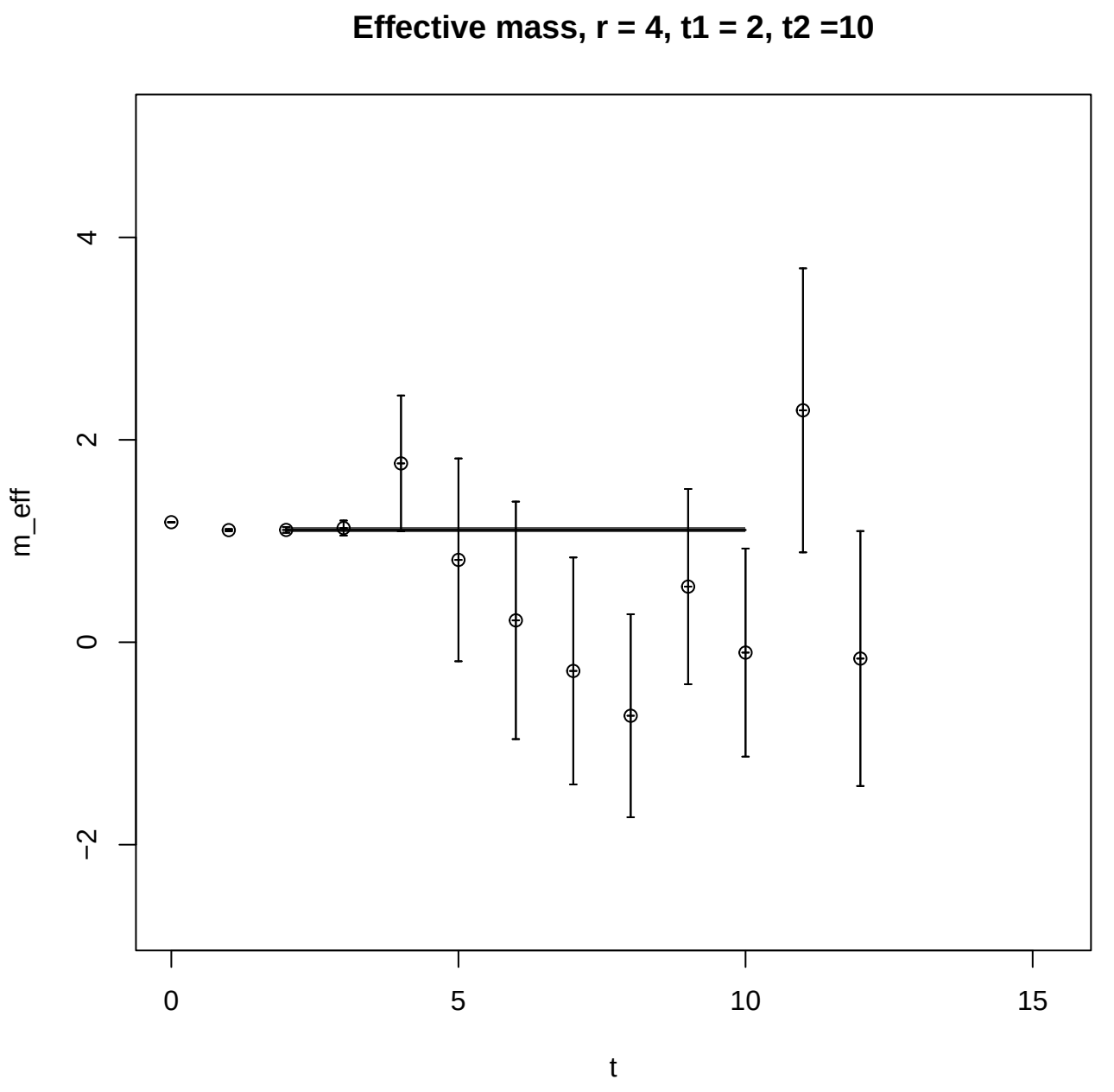

Effective mass, r = 4, t1 = 2, t2 =5
m_eff
4
2
0
−2
0
5
10
15
t

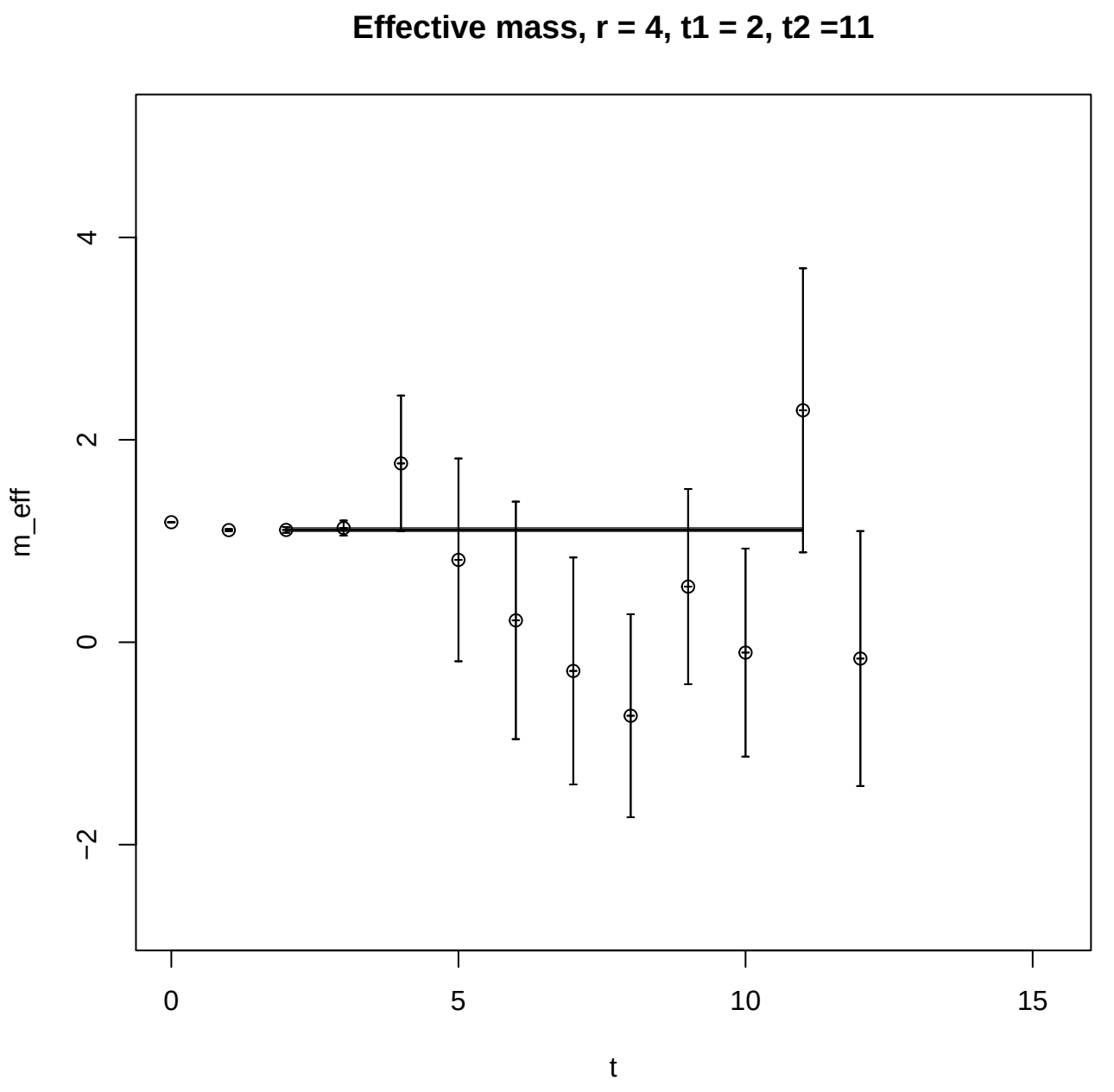

Effective mass, r = 4, t1 = 2, t2 =6
m_eff
4
2
0
−2
0
5
10
15
t

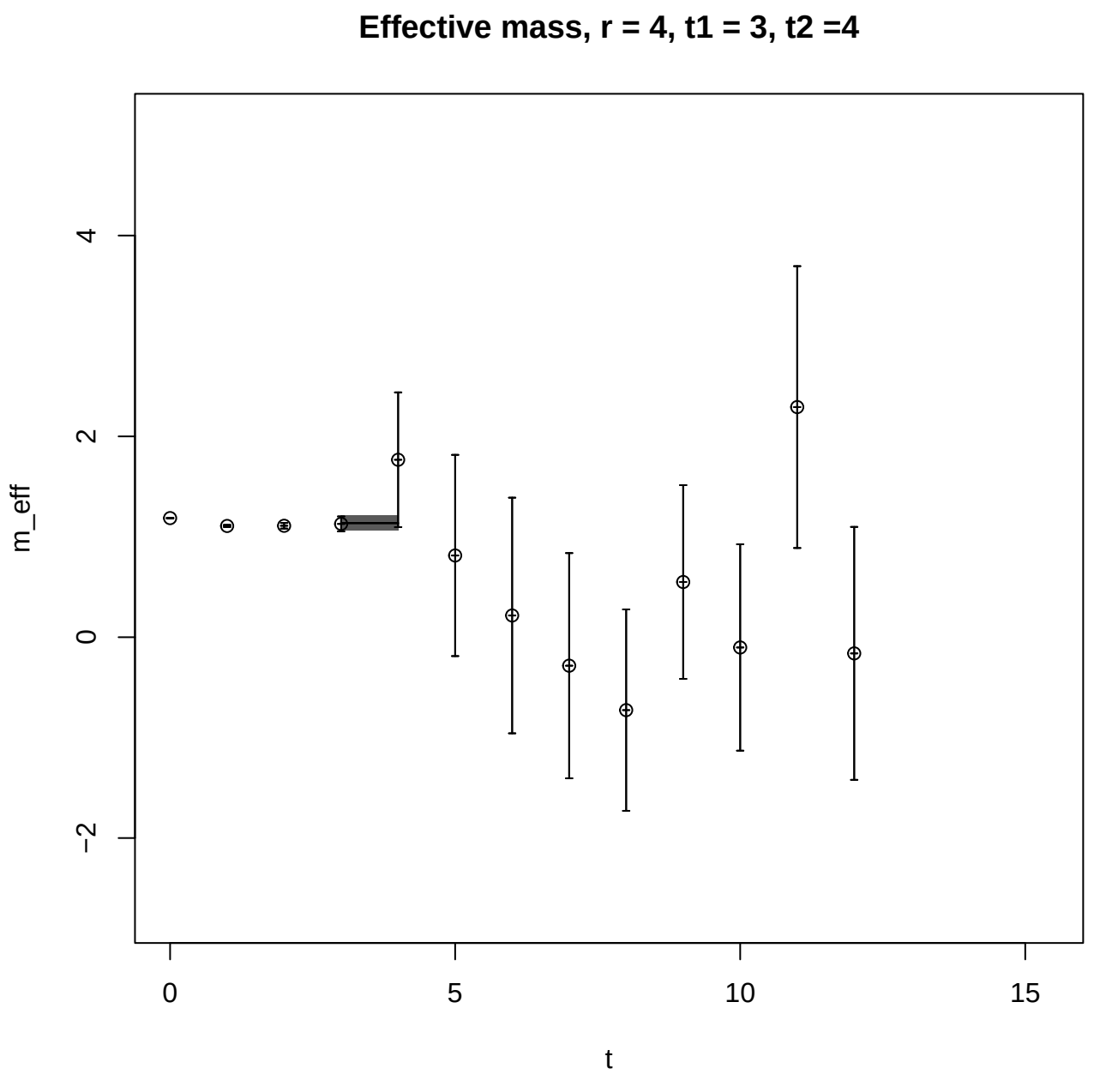

Effective mass, r = 4, t1 = 2, t2 =7
m_eff
4
2
0
−2
0
5
10
15
t

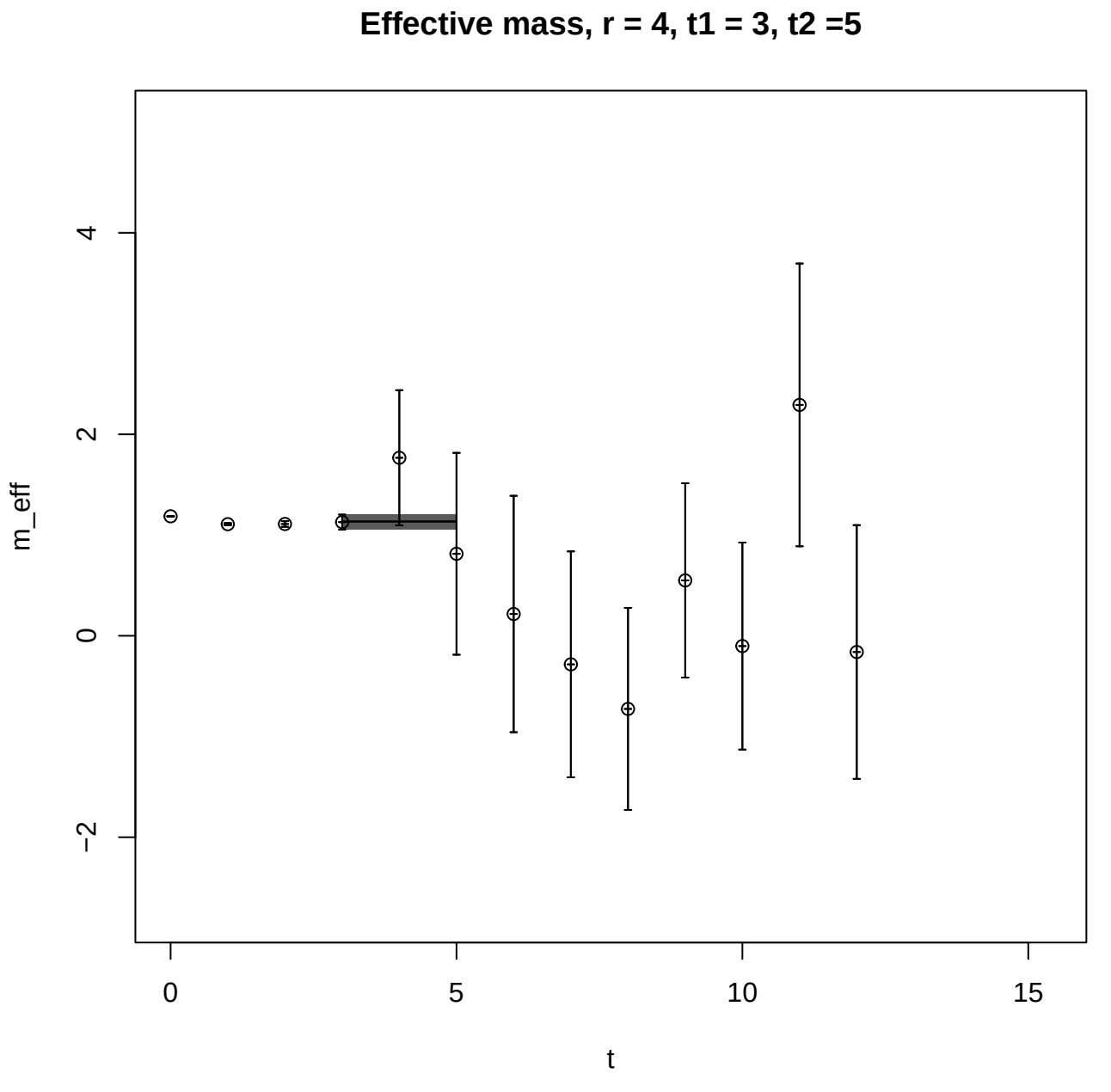

Effective mass, r = 4, t1 = 2, t2 =8
m_eff
4
2
0
−2
0
5
10
15
t

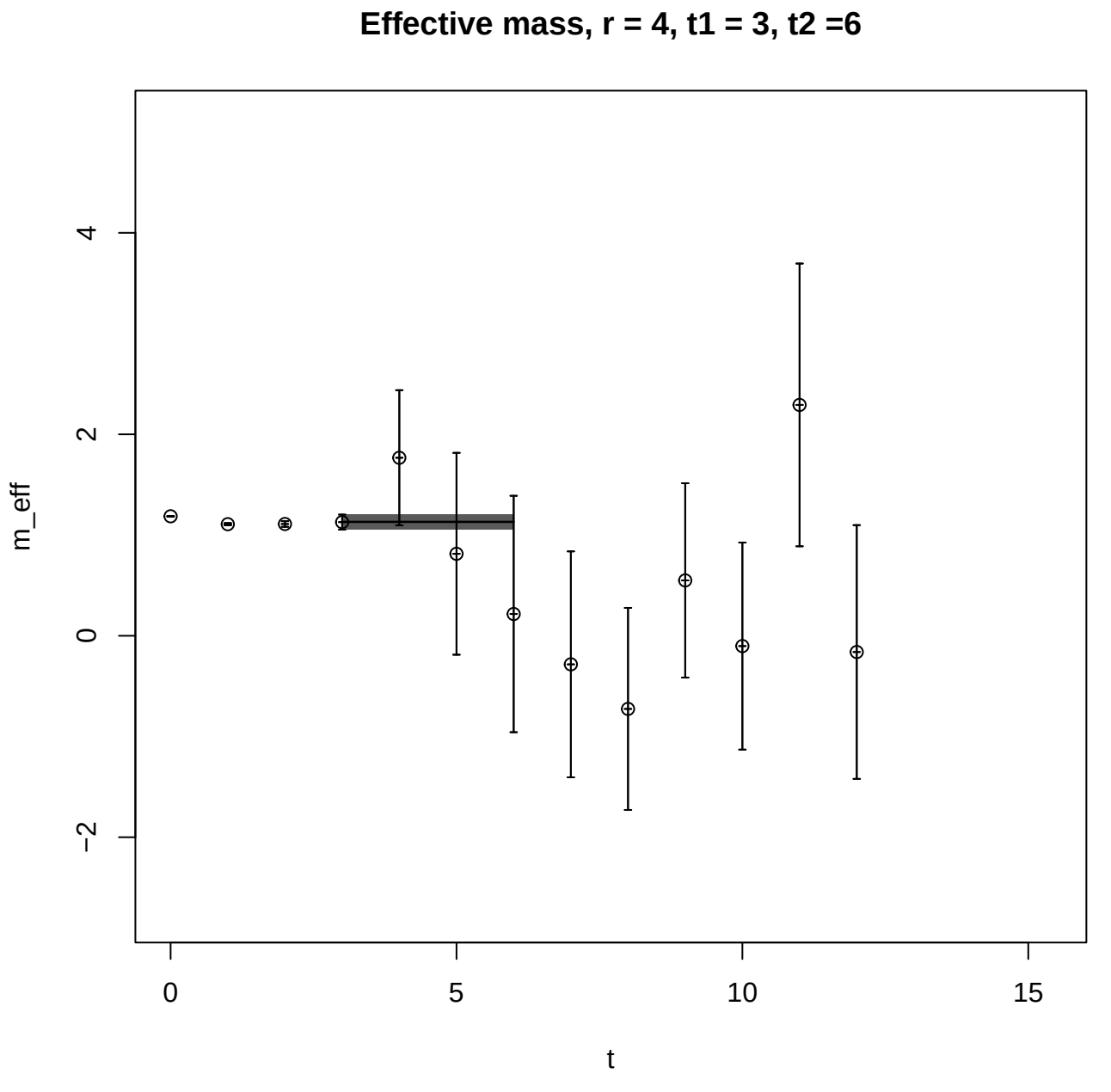

Effective mass, r = 4, t1 = 2, t2 =9
m_eff
4
2
0
−2
0
5
10
15
t

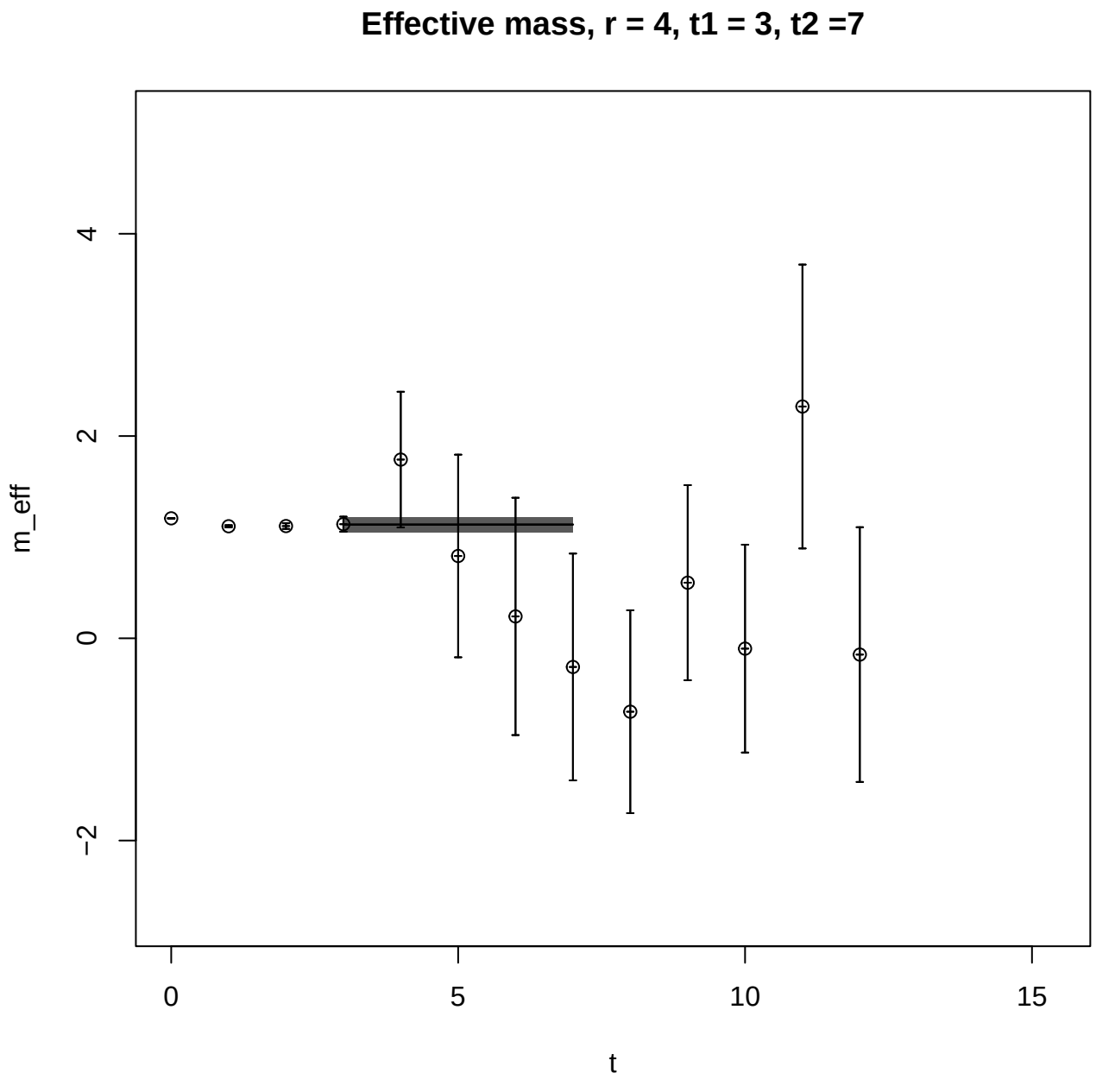

Effective mass, r = 4, t1 = 2, t2 =10
m_eff
4
2
0
−2
0
5
10
15
t

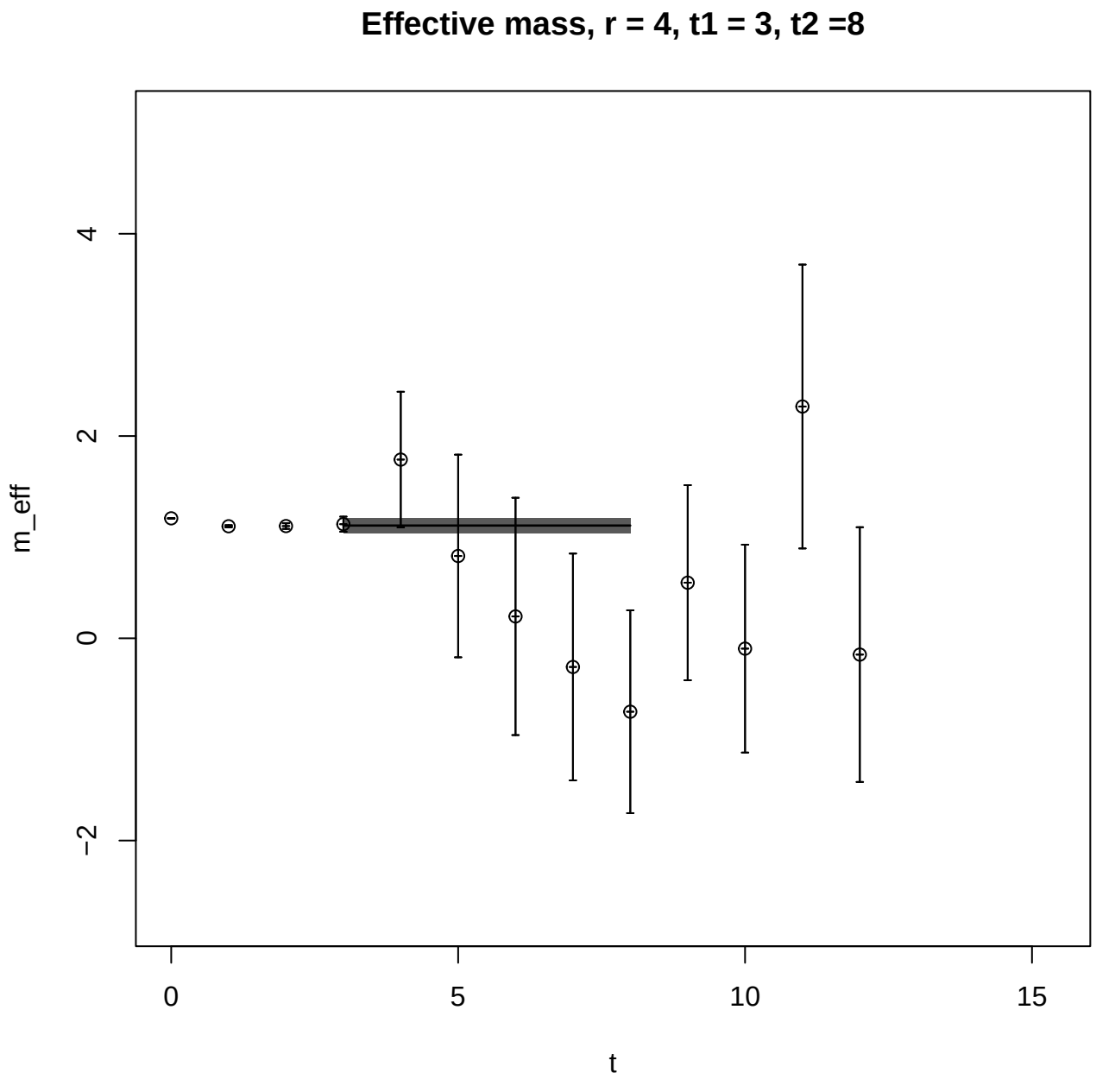

Effective mass, r = 4, t1 = 2, t2 =11
m_eff
4
2
0
−2
0
5
10
15
t

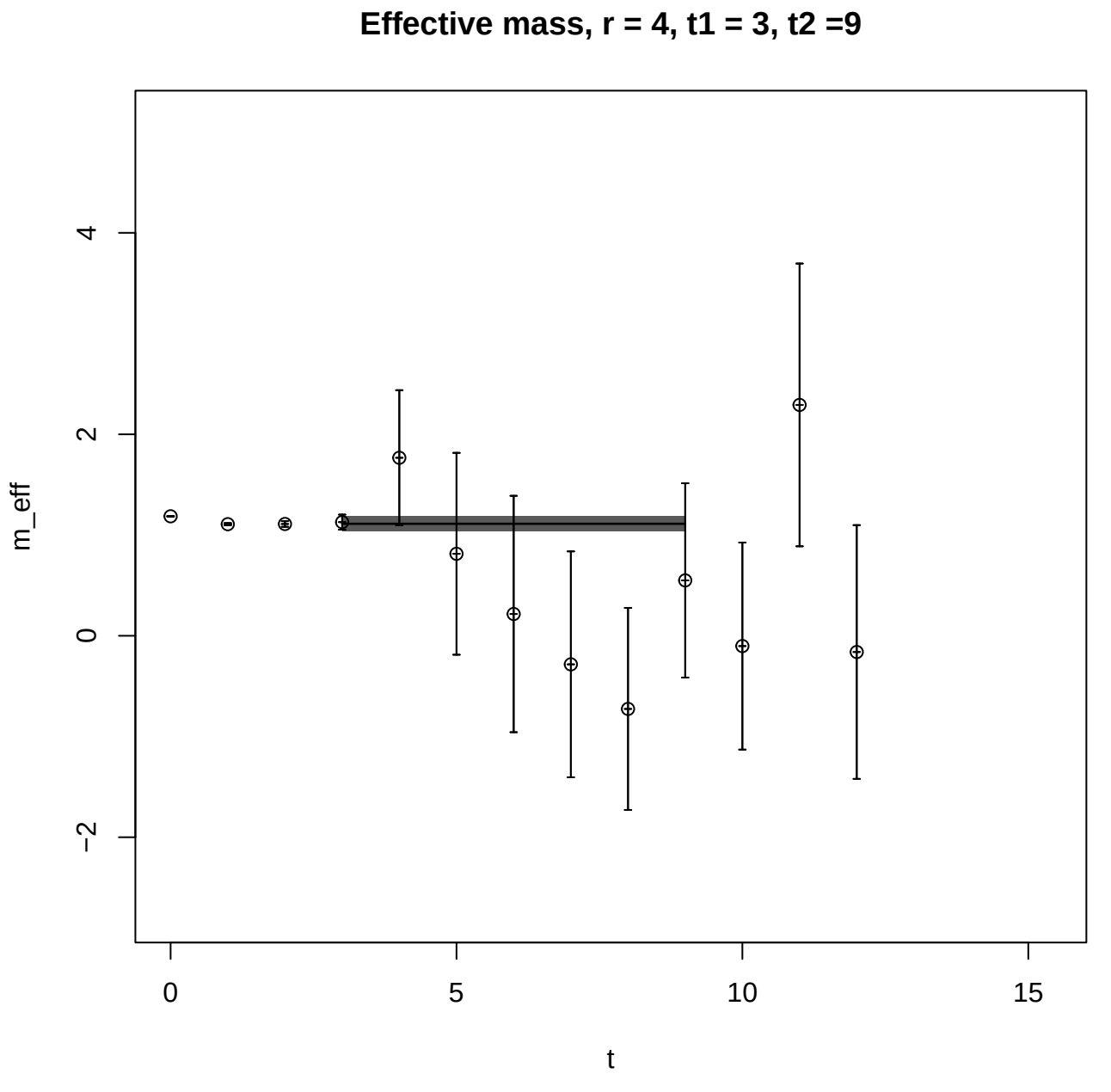

Effective mass, r = 4, t1 = 3, t2 =4
m_eff
4
2
0
−2
0
5
10
15
t

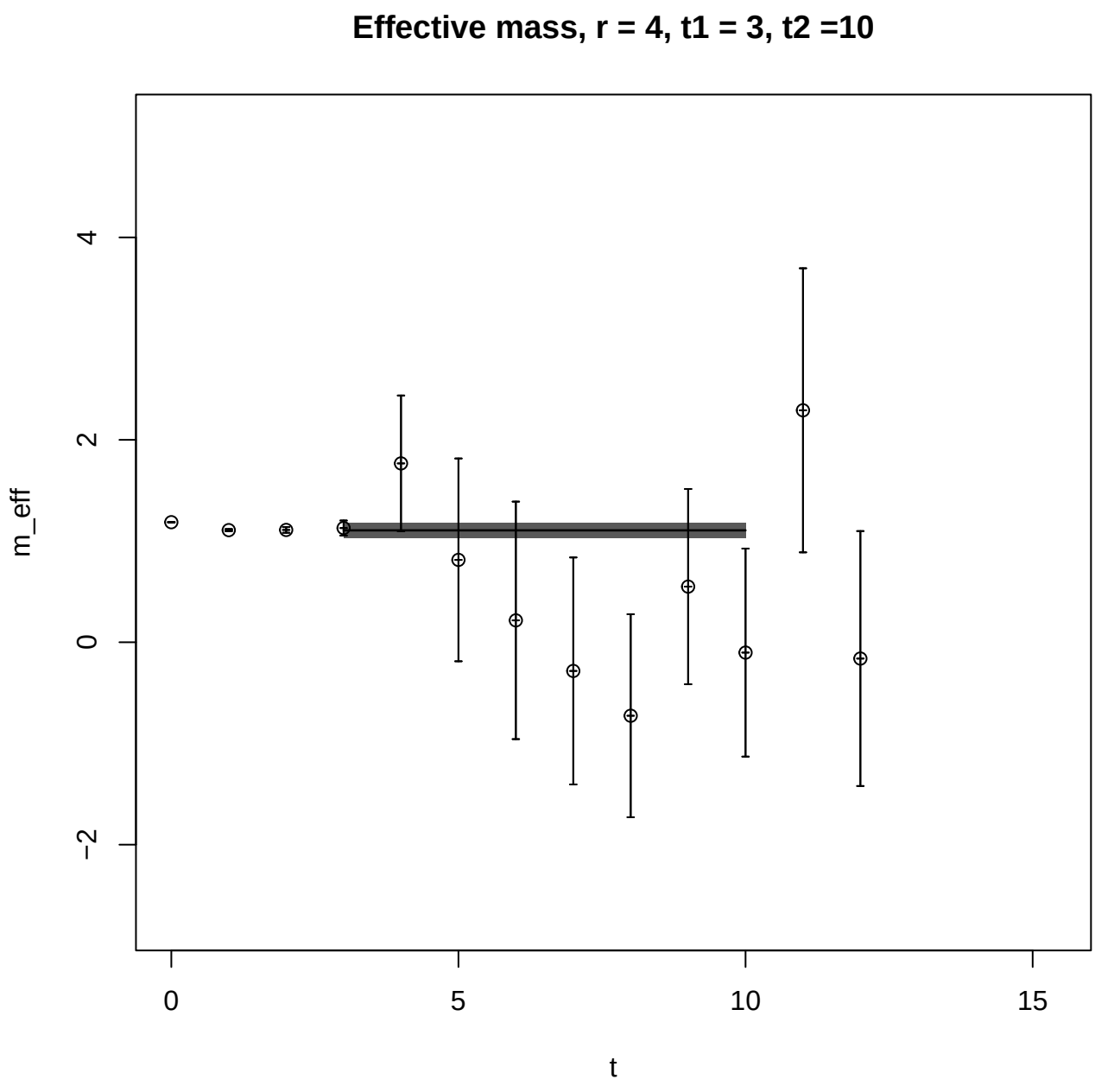

Effective mass, r = 4, t1 = 3, t2 =5
m_eff
4
2
0
−2
0
5
10
15
t

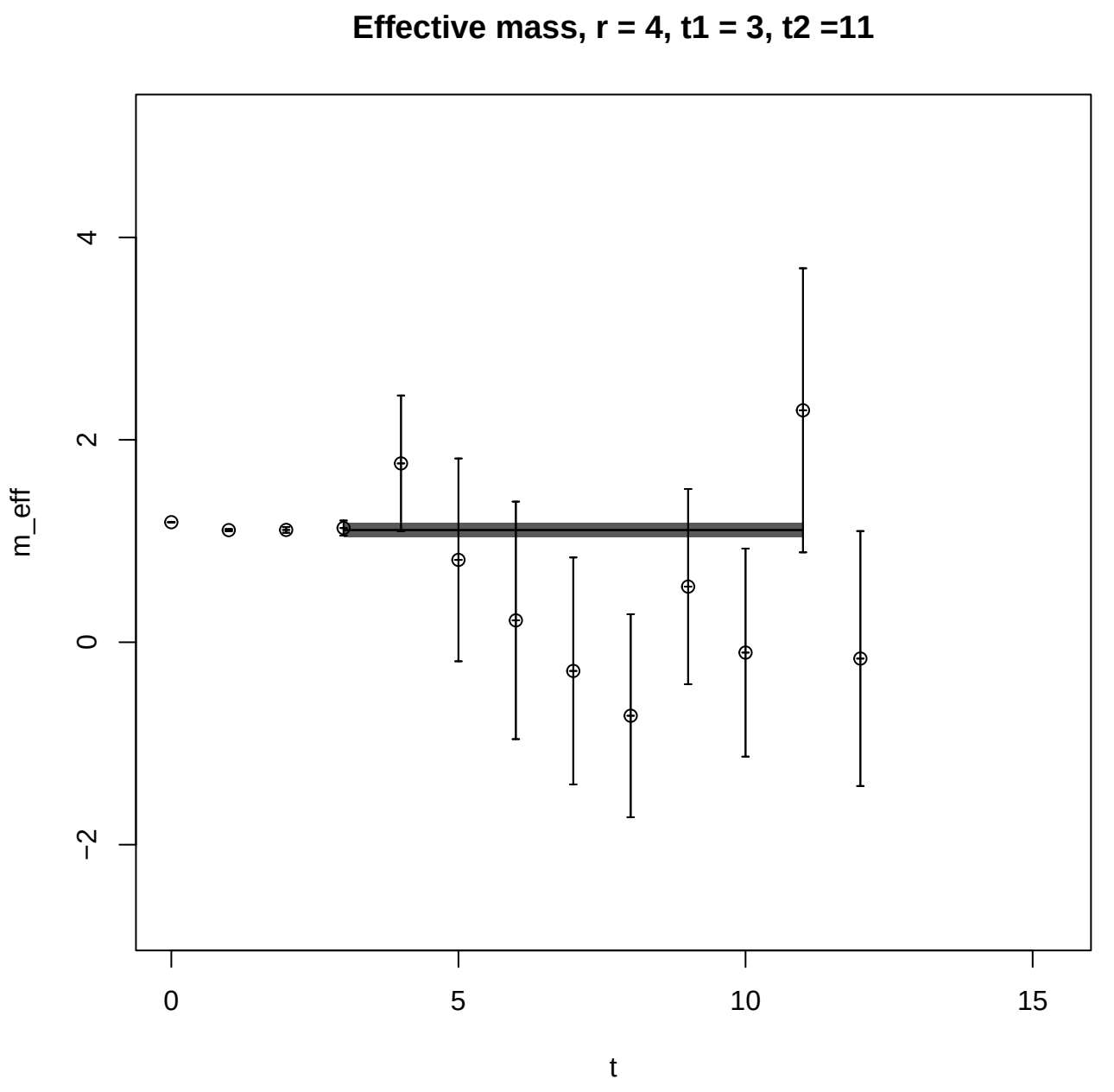

Effective mass, r = 4, t1 = 3, t2 =6
m_eff
4
2
0
−2
0
5
10
15
t

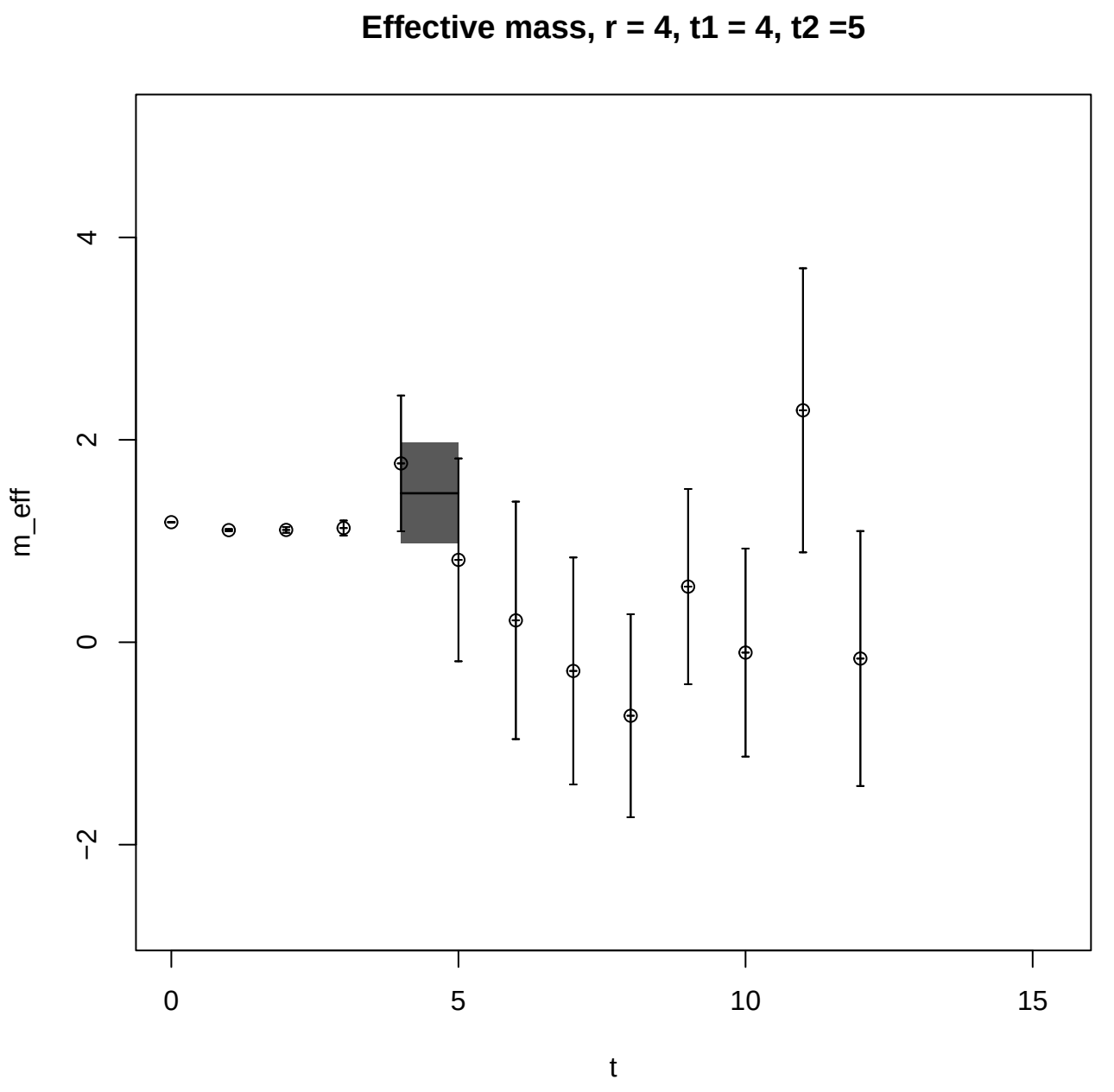

Effective mass, r = 4, t1 = 3, t2 =7
m_eff
4
2
0
−2
0
5
10
15
t

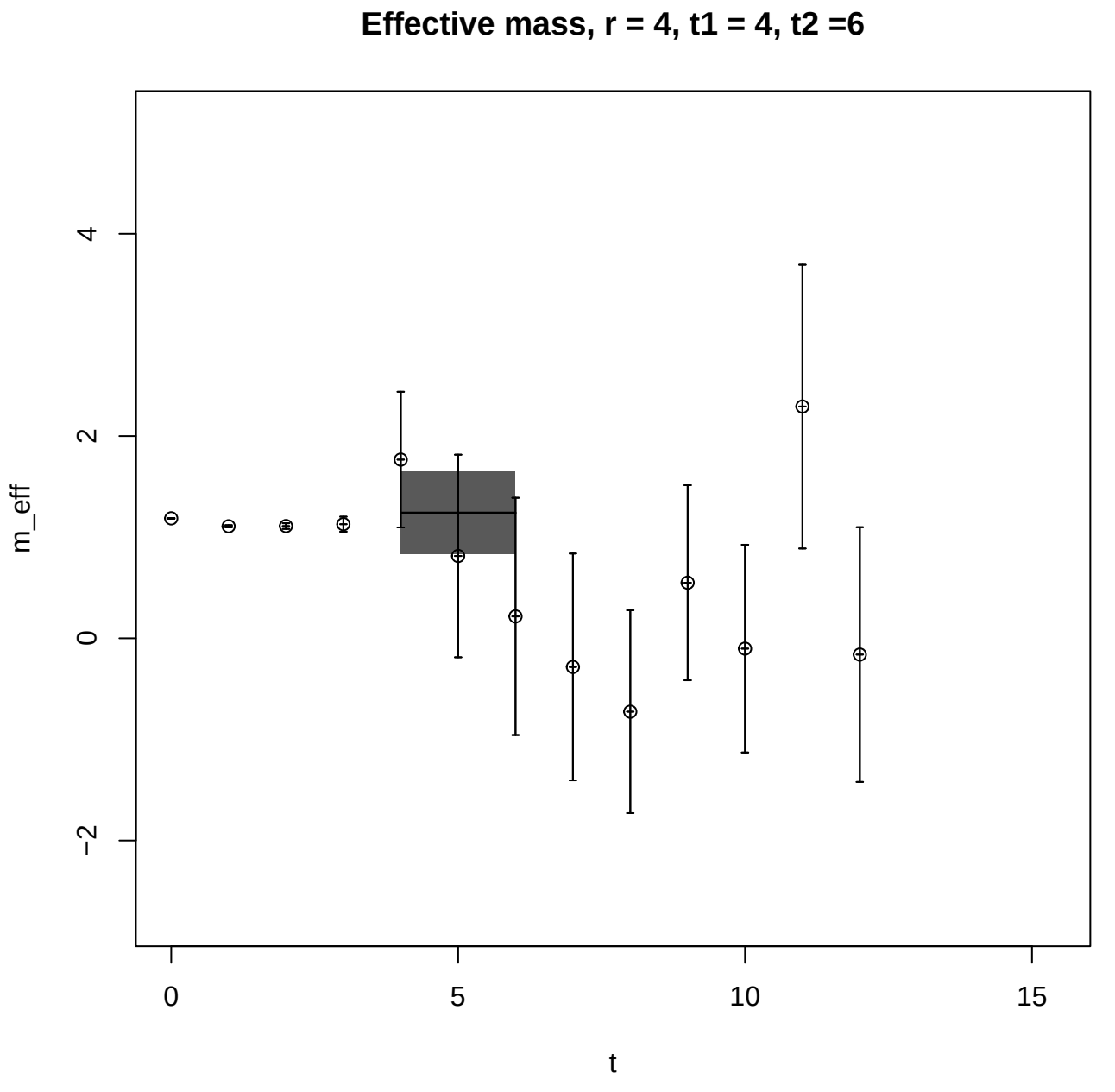

Effective mass, r = 4, t1 = 3, t2 =8
m_eff
4
2
0
−2
0
5
10
15
t

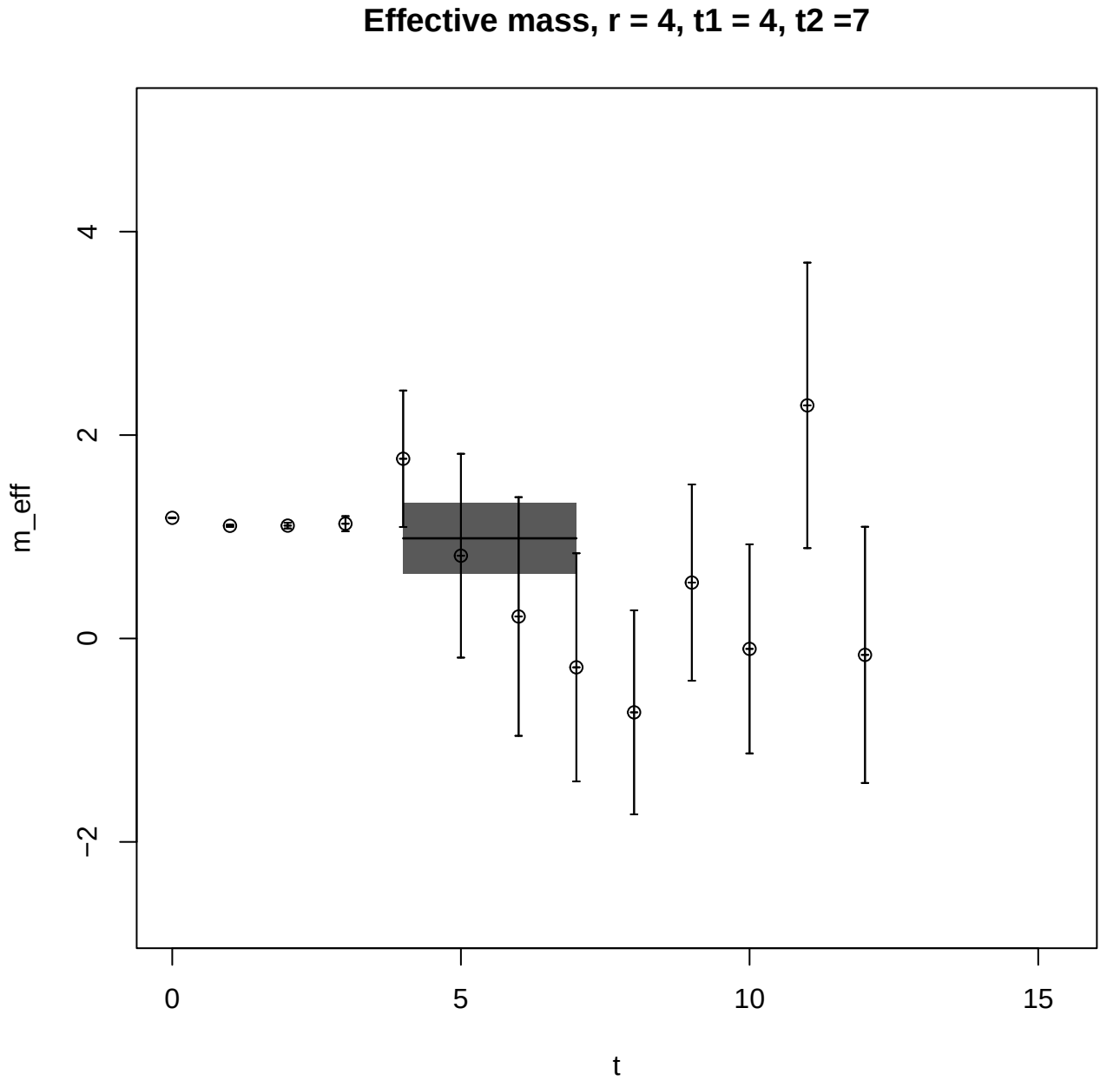

Effective mass, r = 4, t1 = 3, t2 =9
m_eff
4
2
0
−2
0
5
10
15
t

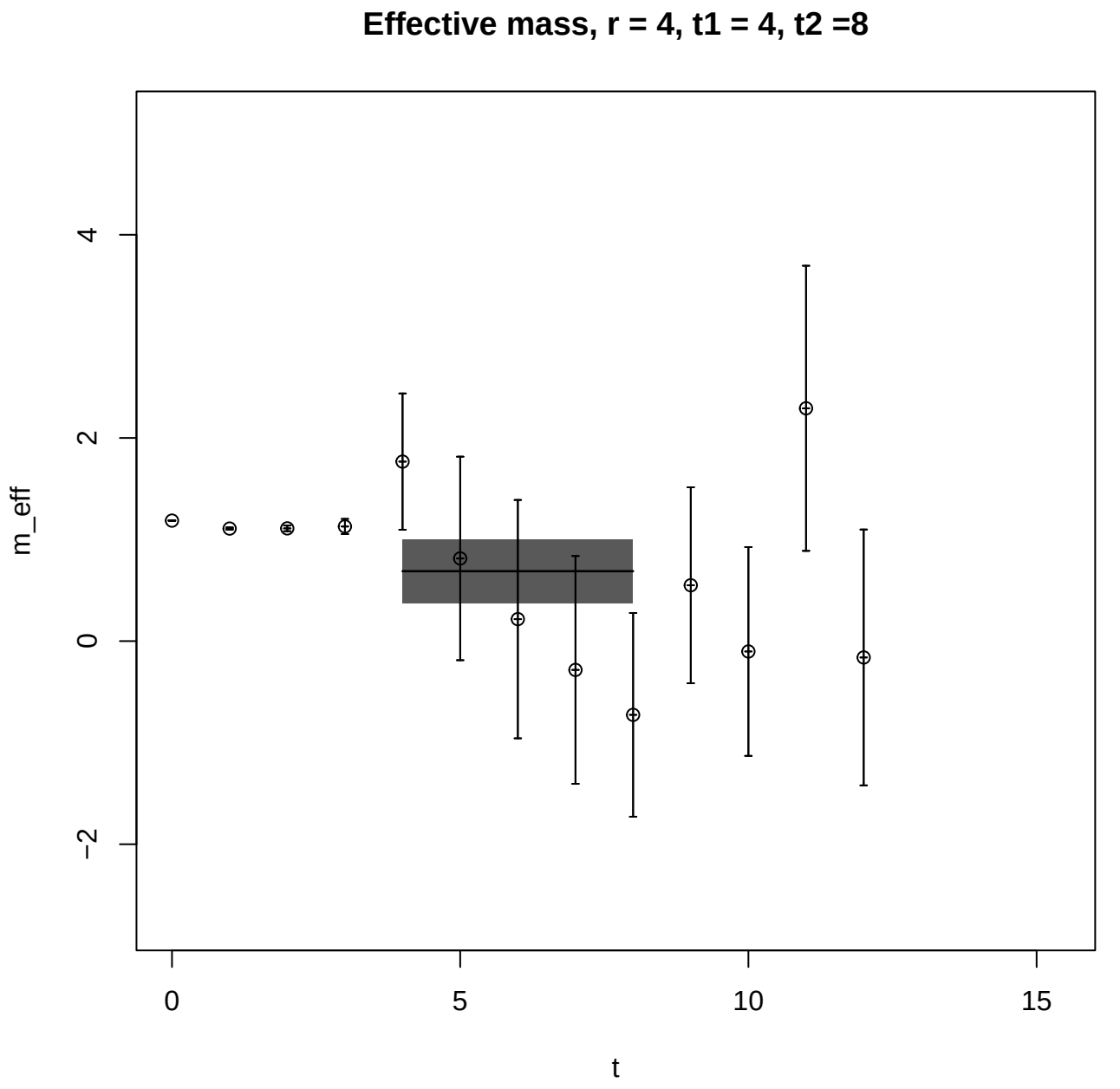

Effective mass, r = 4, t1 = 3, t2 =10
m_eff
4
2
0
−2
0
5
10
15
t

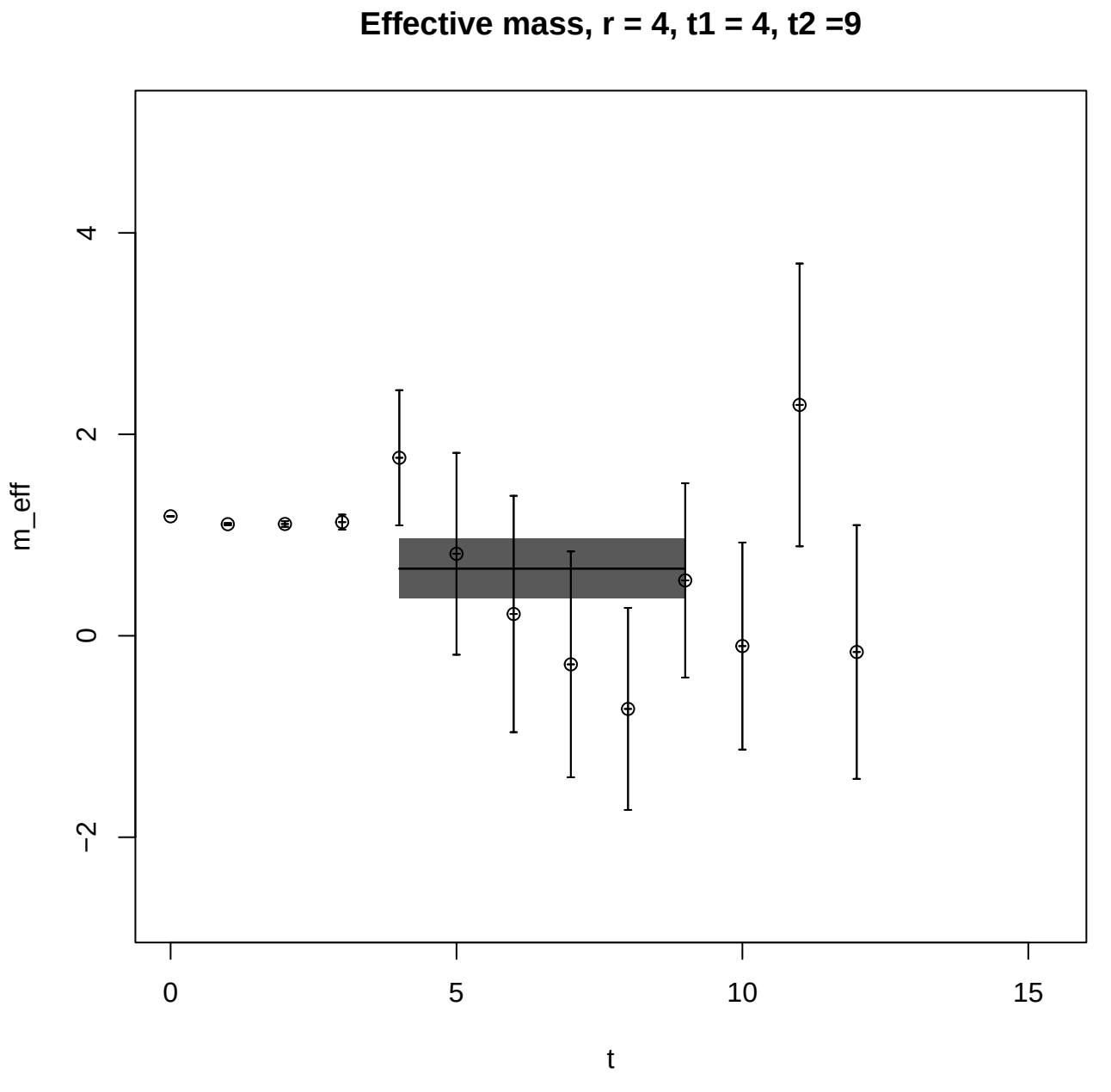

Effective mass, r = 4, t1 = 3, t2 =11
m_eff
4
2
0
−2
0
5
10
15
t

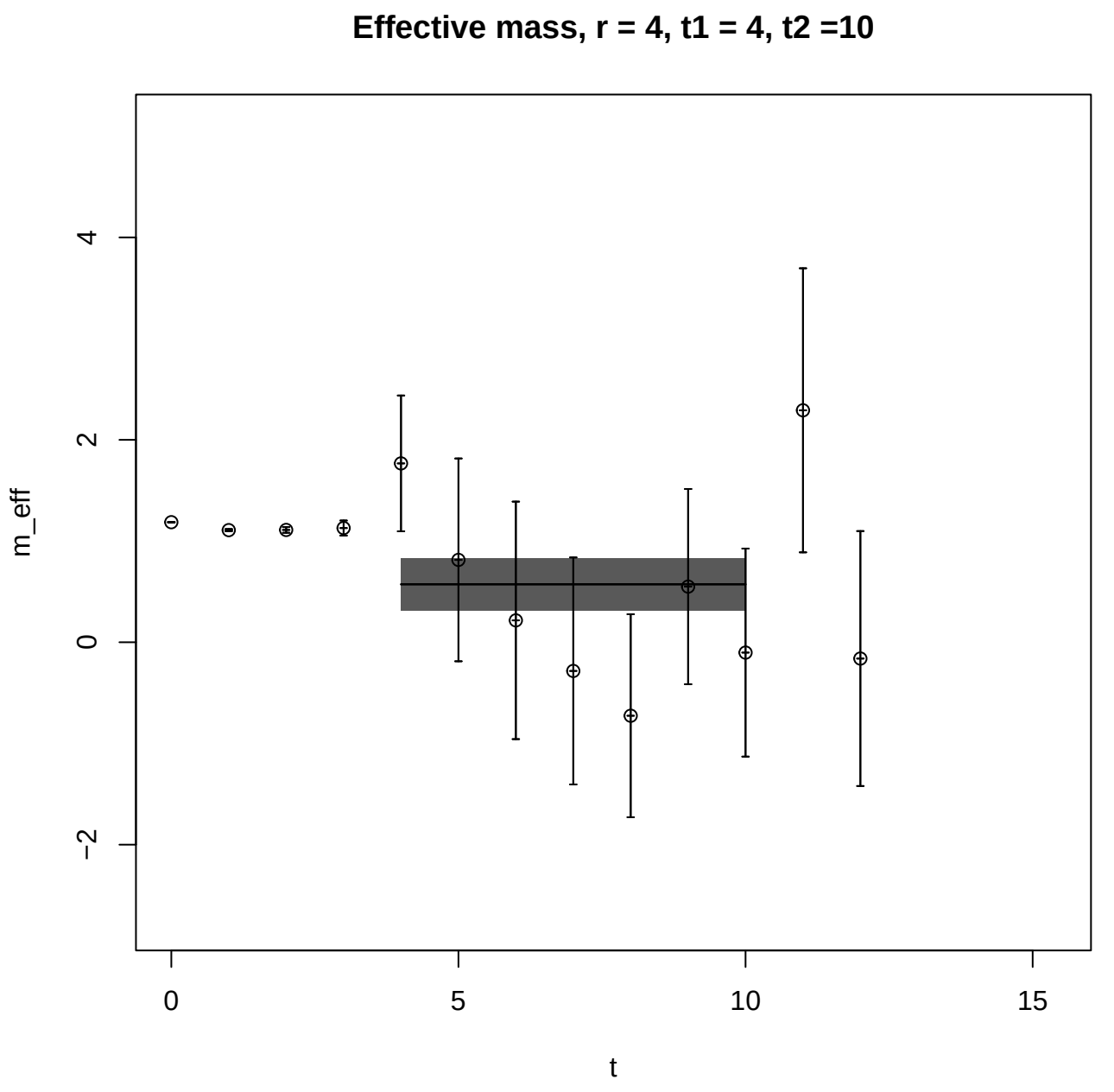

Effective mass, r = 4, t1 = 4, t2 =5
m_eff
4
2
0
−2
0
5
10
15
t

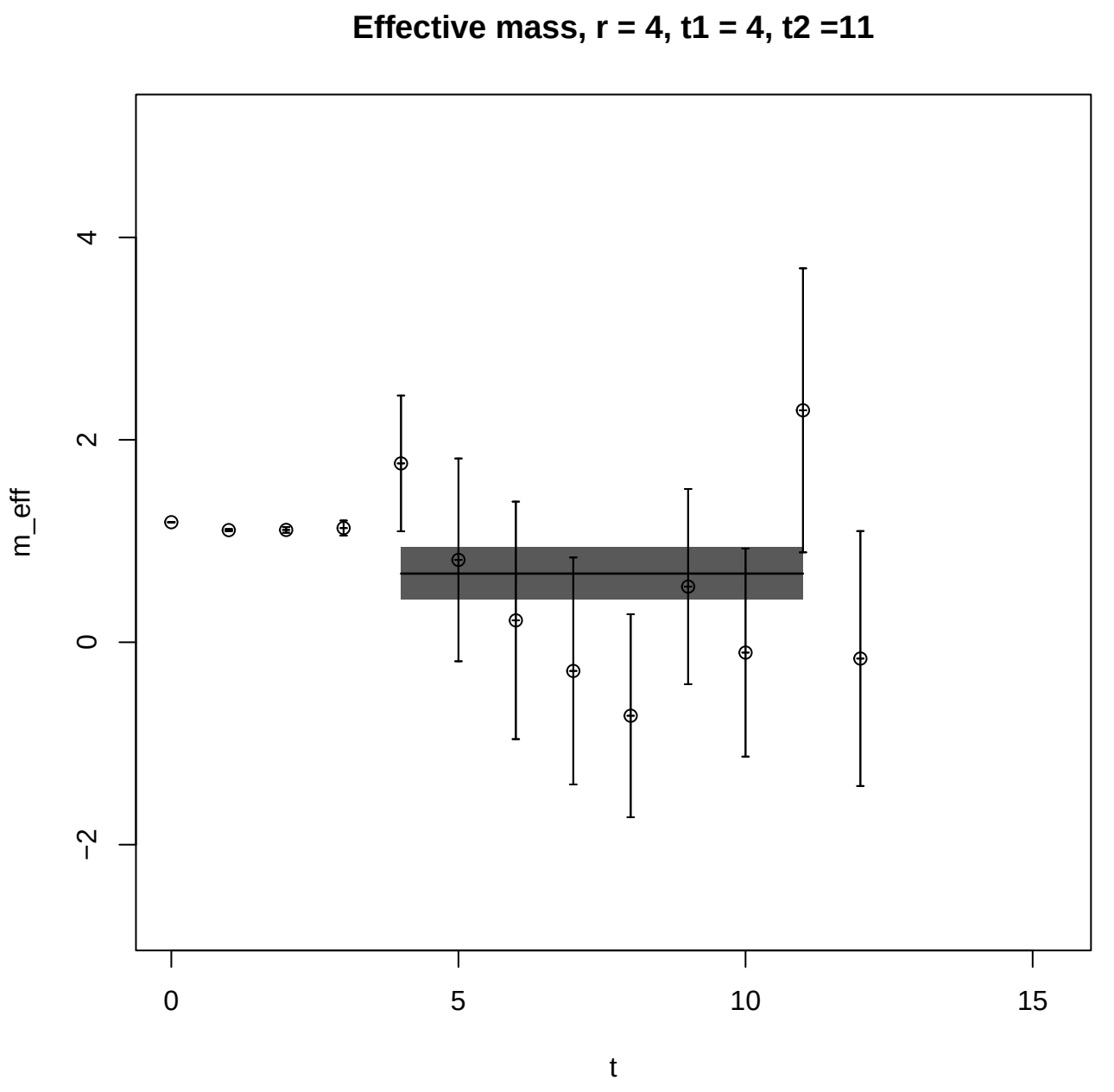

Effective mass, r = 4, t1 = 4, t2 =6
m_eff
4
2
0
−2
0
5
10
15
t

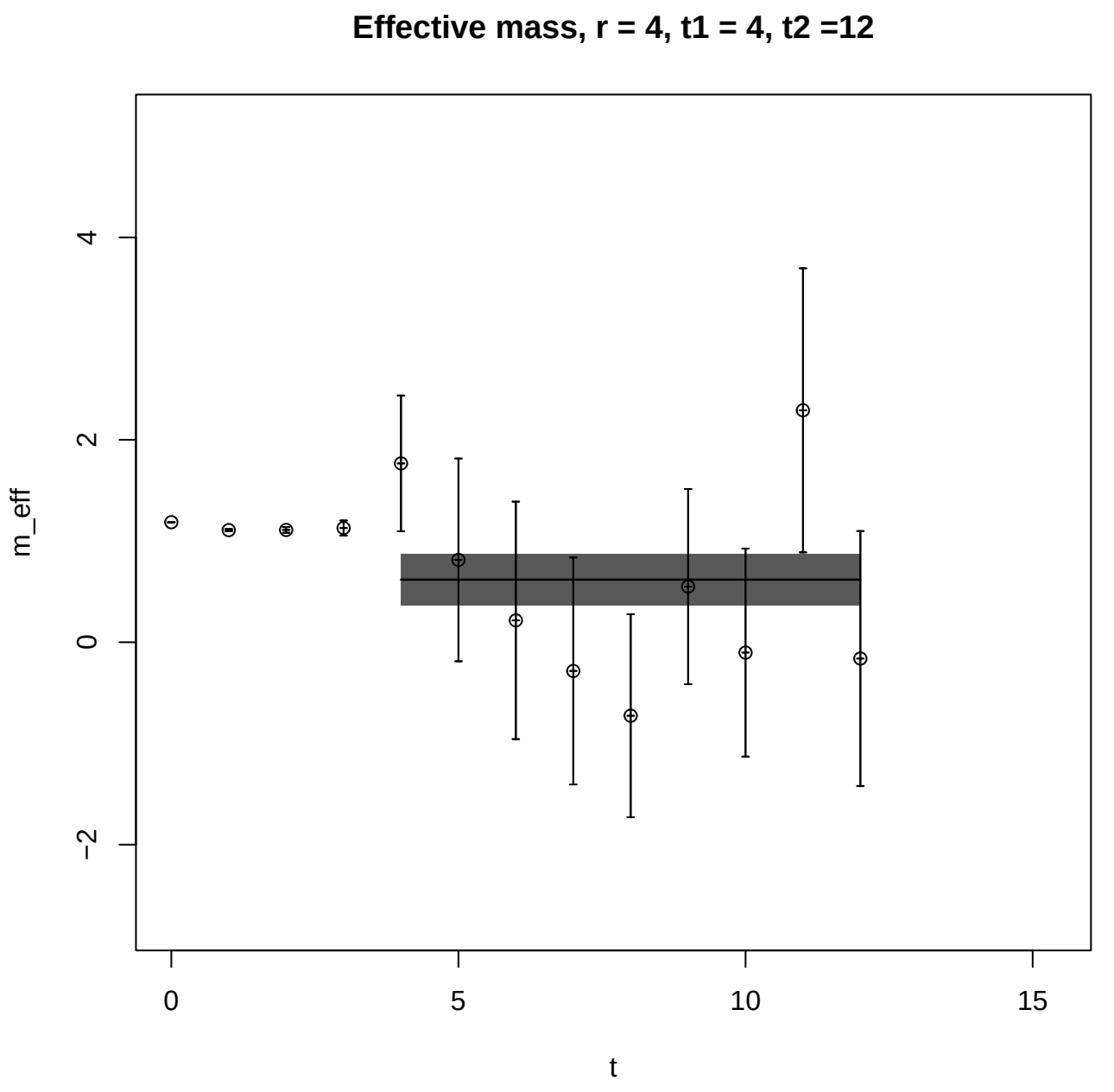

Effective mass, r = 4, t1 = 4, t2 =7
m_eff
4
2
0
−2
0
5
10
15
t

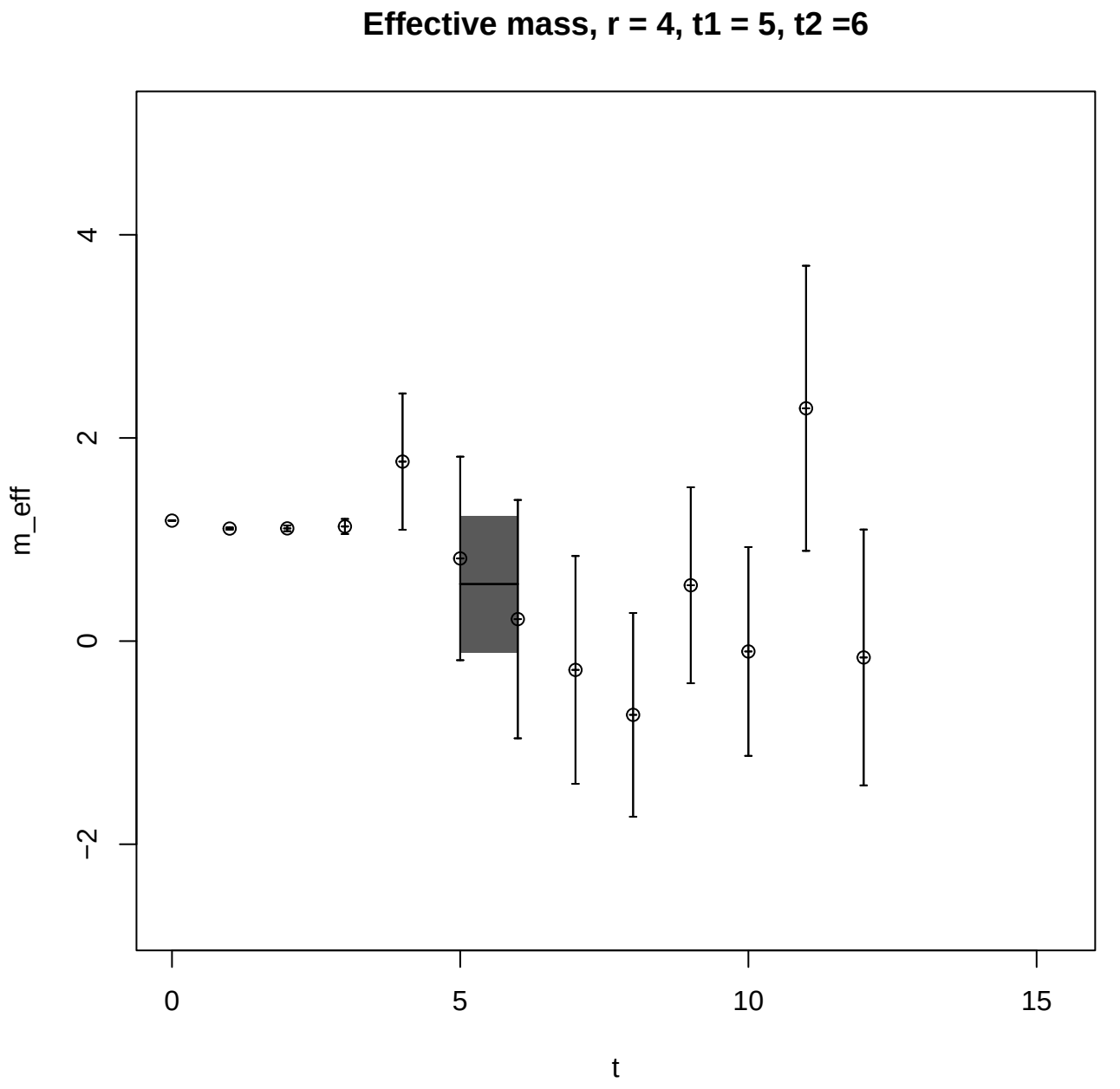

Effective mass, r = 4, t1 = 4, t2 =8
m_eff
4
2
0
−2
0
5
10
15
t

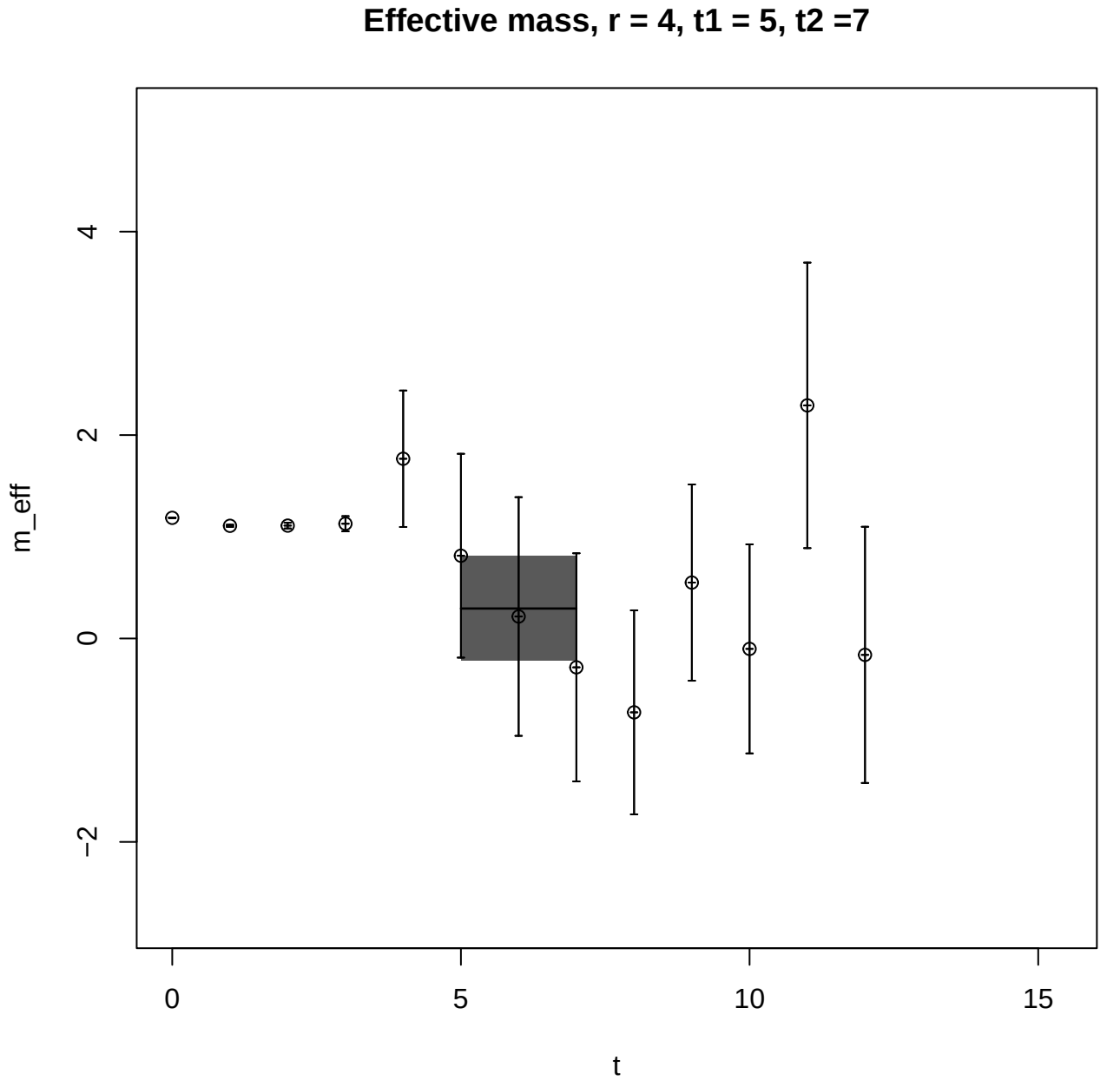

Effective mass, r = 4, t1 = 4, t2 =9
m_eff
4
2
0
−2
0
5
10
15
t

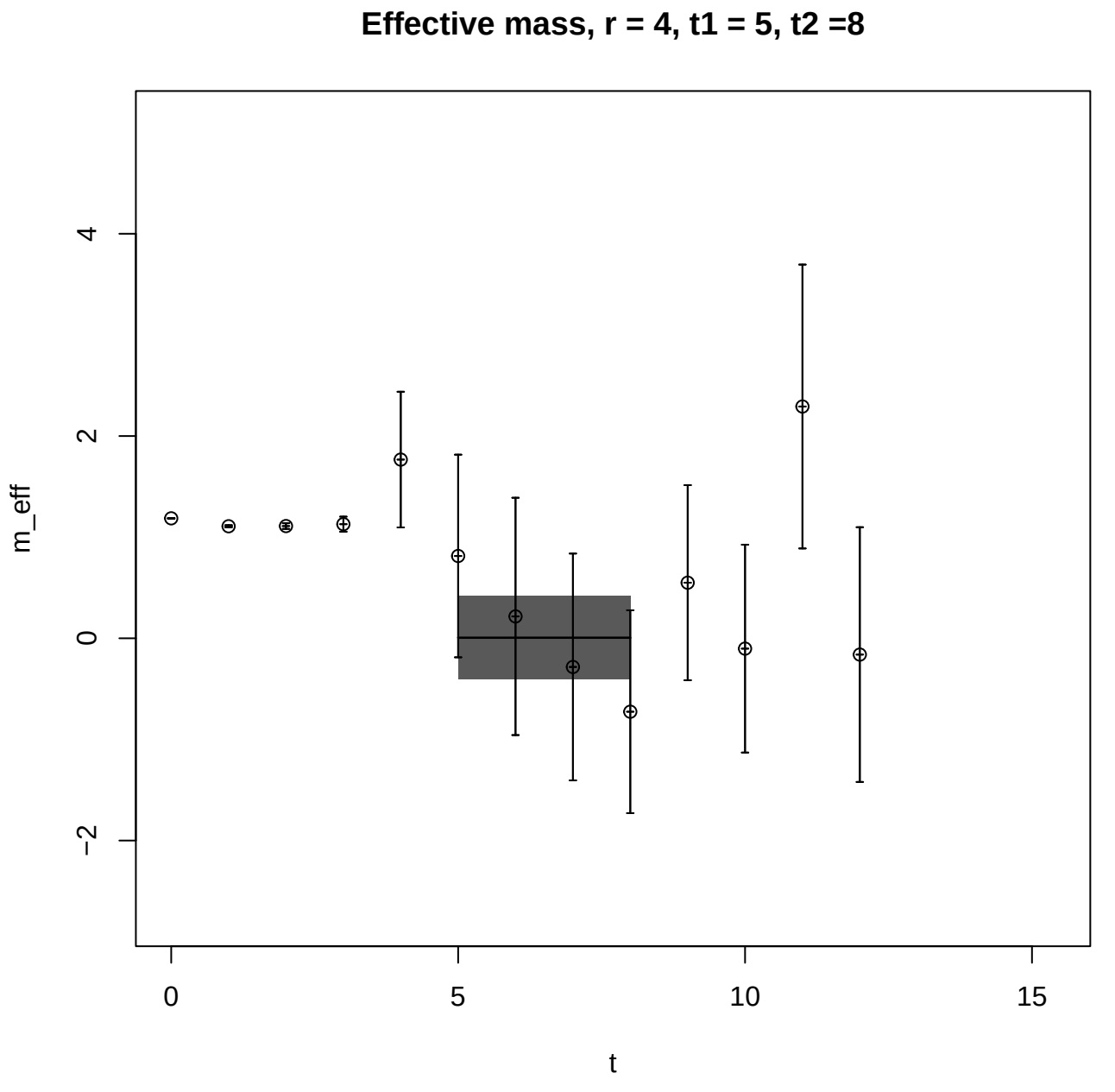

Effective mass, r = 4, t1 = 4, t2 =10
m_eff
4
2
0
−2
0
5
10
15
t

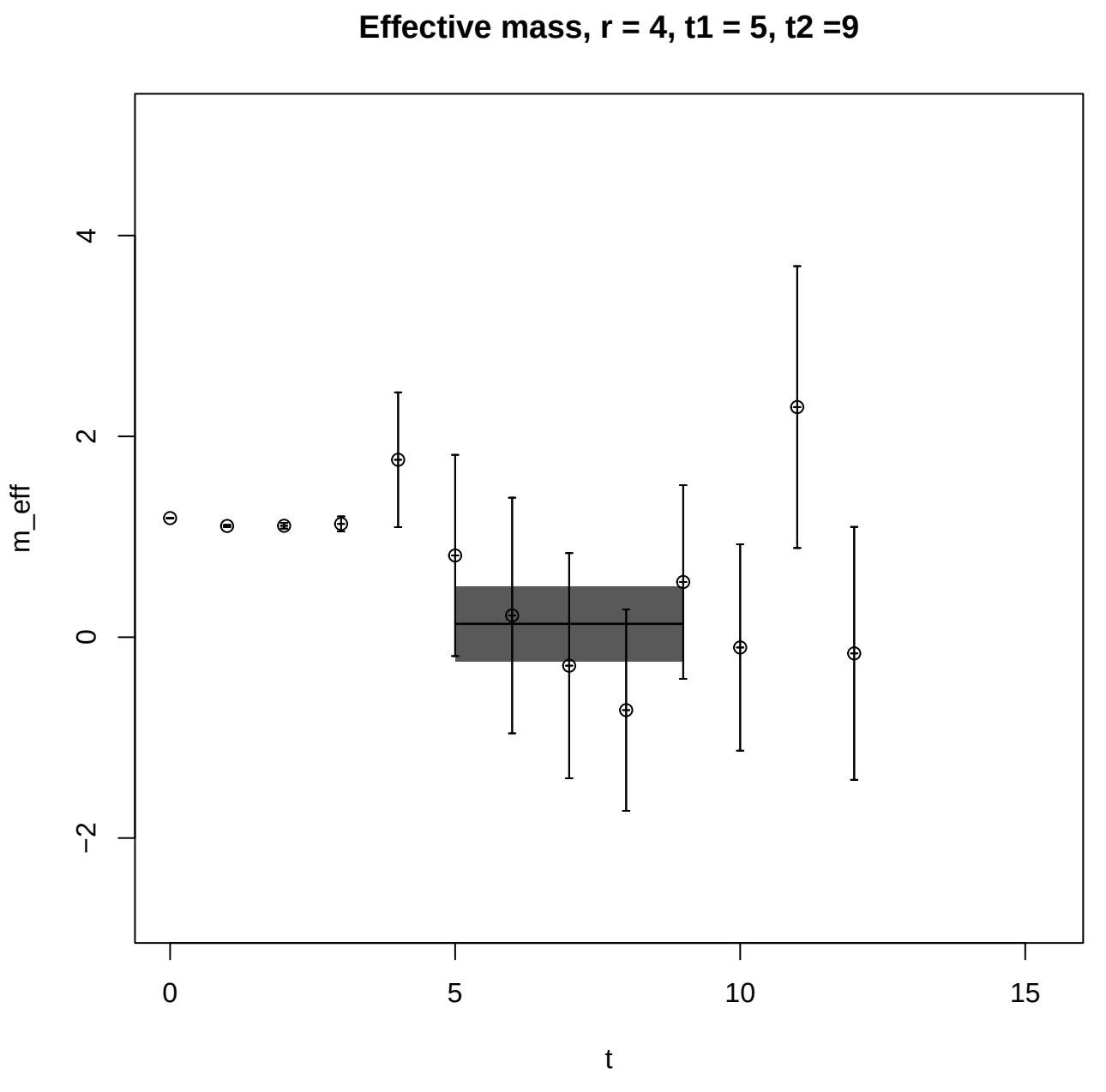

Effective mass, r = 4, t1 = 4, t2 =11
m_eff
4
2
0
−2
0
5
10
15
t

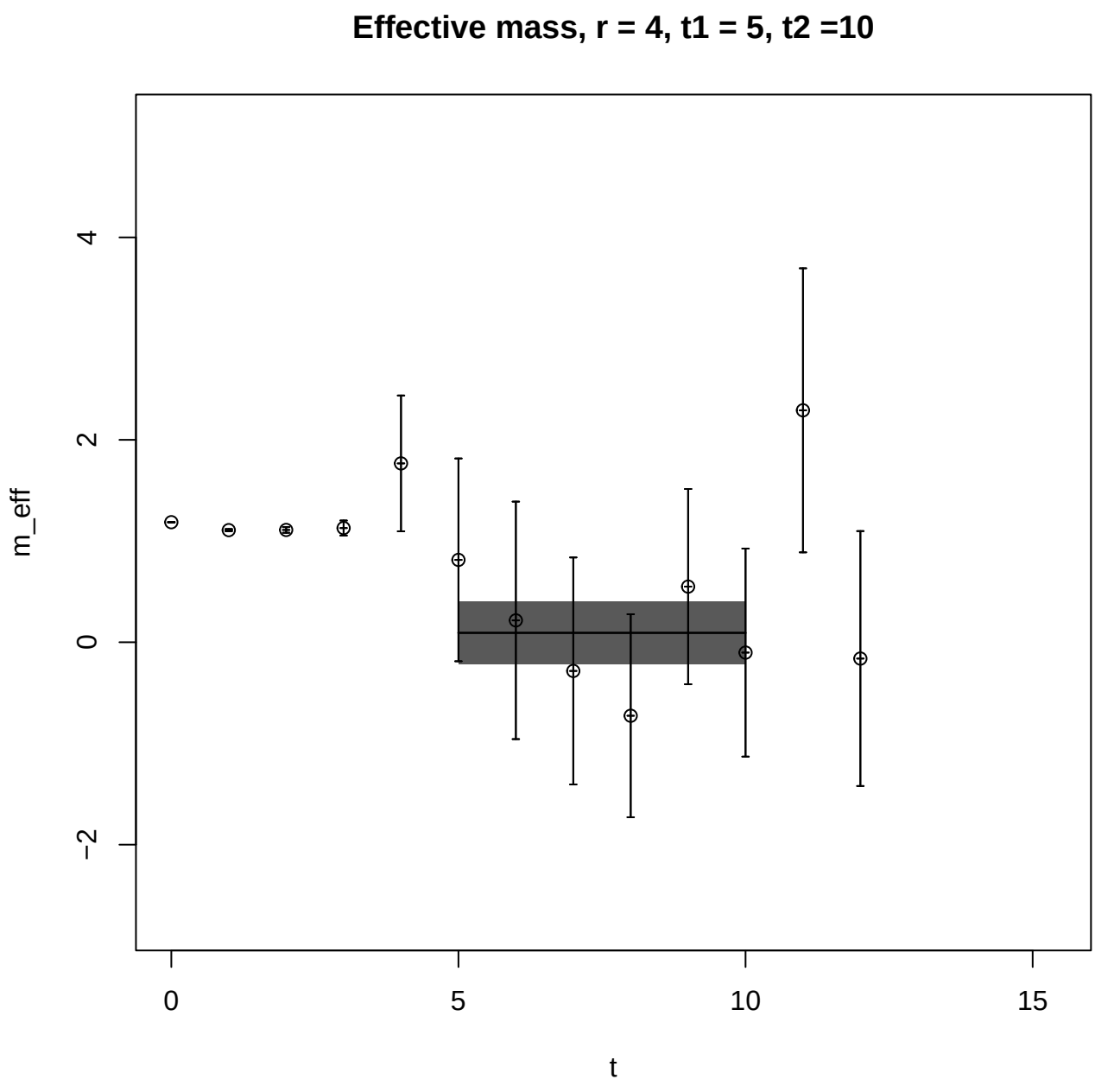

Effective mass, r = 4, t1 = 4, t2 =12
m_eff
4
2
0
−2
0
5
10
15
t

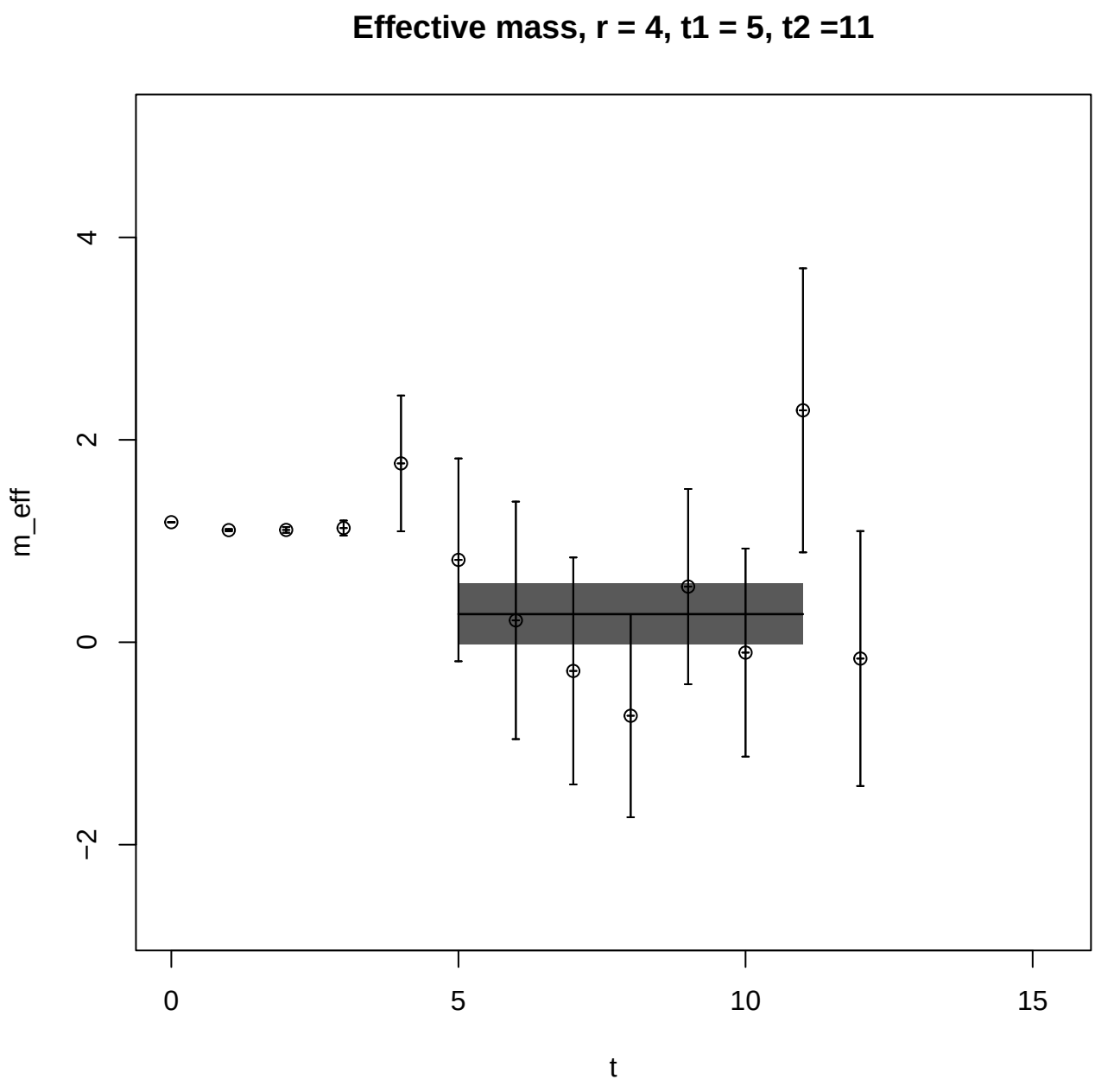

Effective mass, r = 4, t1 = 5, t2 =6
m_eff
4
2
0
−2
0
5
10
15
t

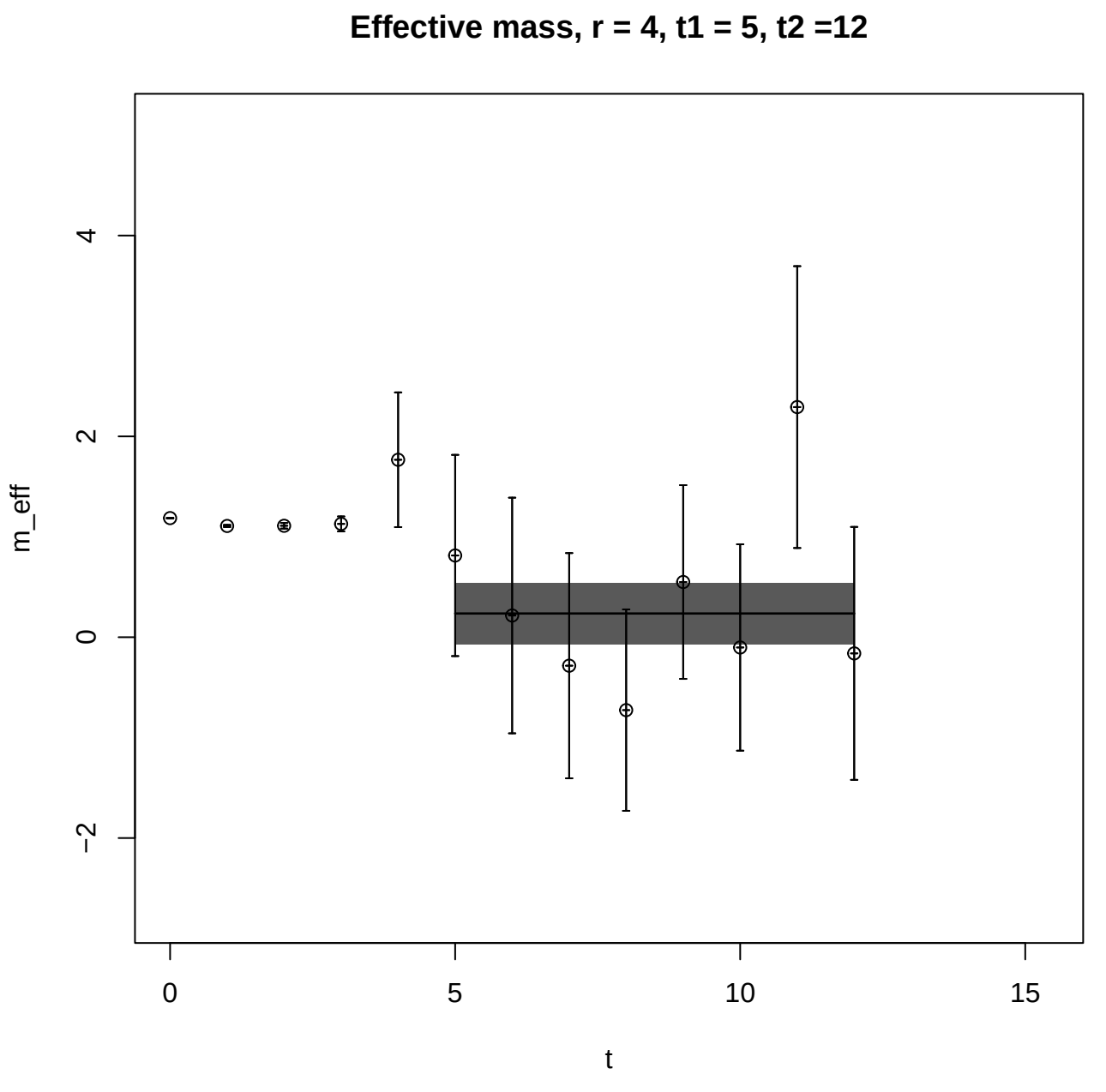

Effective mass, r = 4, t1 = 5, t2 =7
m_eff
4
2
0
−2
0
5
10
15
t

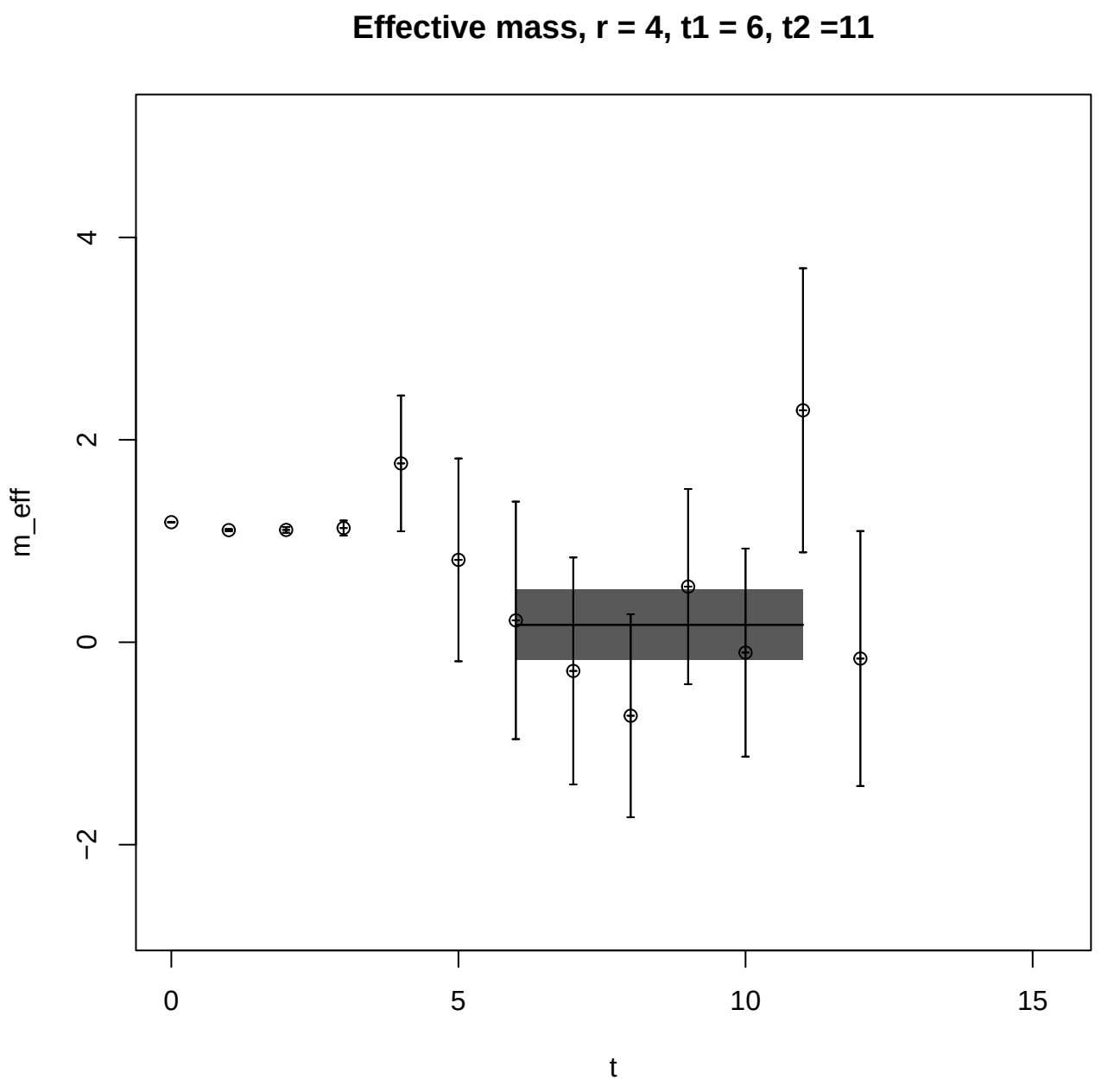

Effective mass, r = 4, t1 = 5, t2 =8
m_eff
4
2
0
−2
0
5
10
15
t

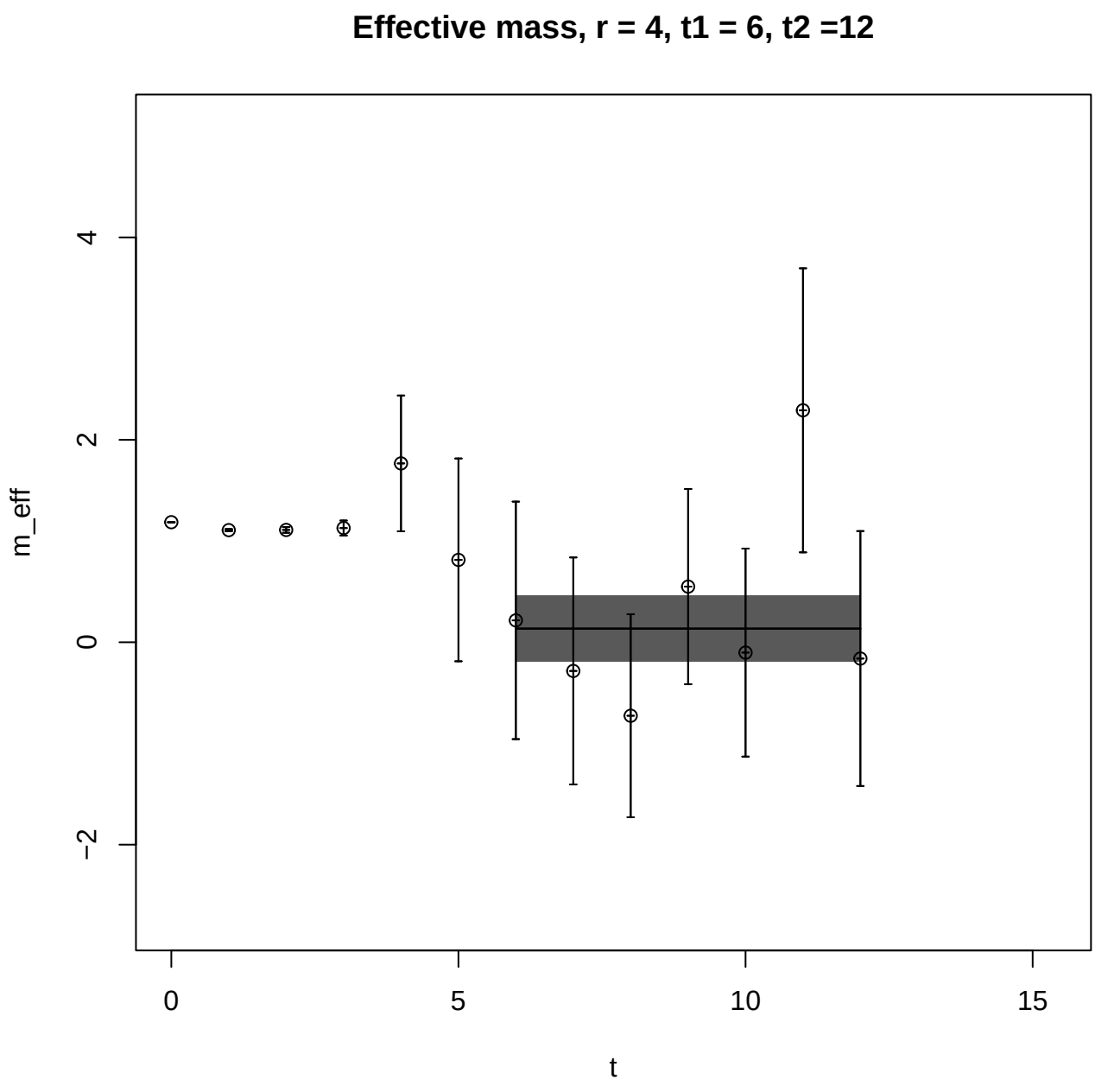

Effective mass, r = 4, t1 = 5, t2 =9
m_eff
4
2
0
−2
0
5
10
15
t

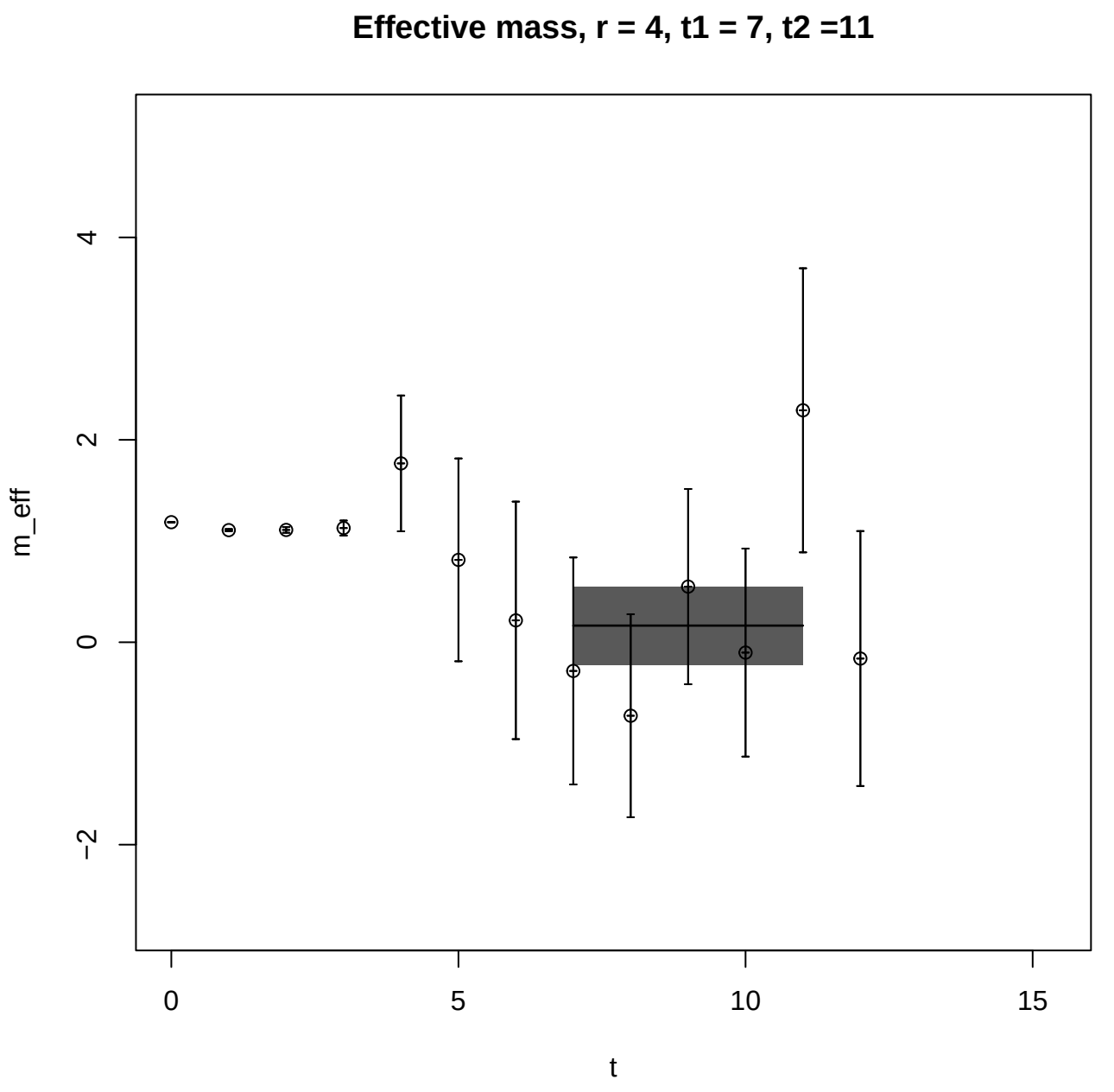

Effective mass, r = 4, t1 = 5, t2 =10
m_eff
4
2
0
−2
0
5
10
15
t

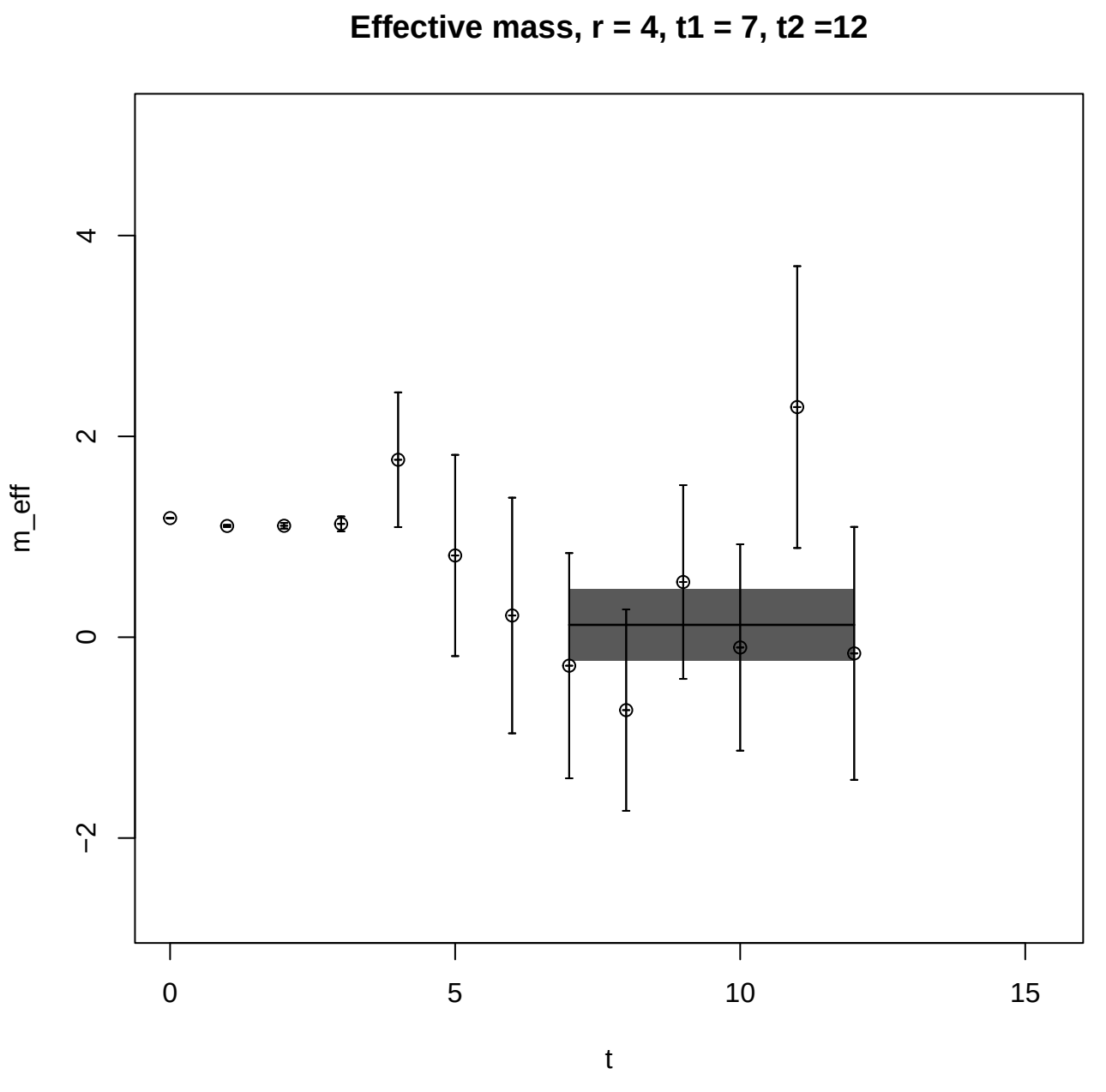

Effective mass, r = 4, t1 = 5, t2 =11
m_eff
4
2
0
−2
0
5
10
15
t

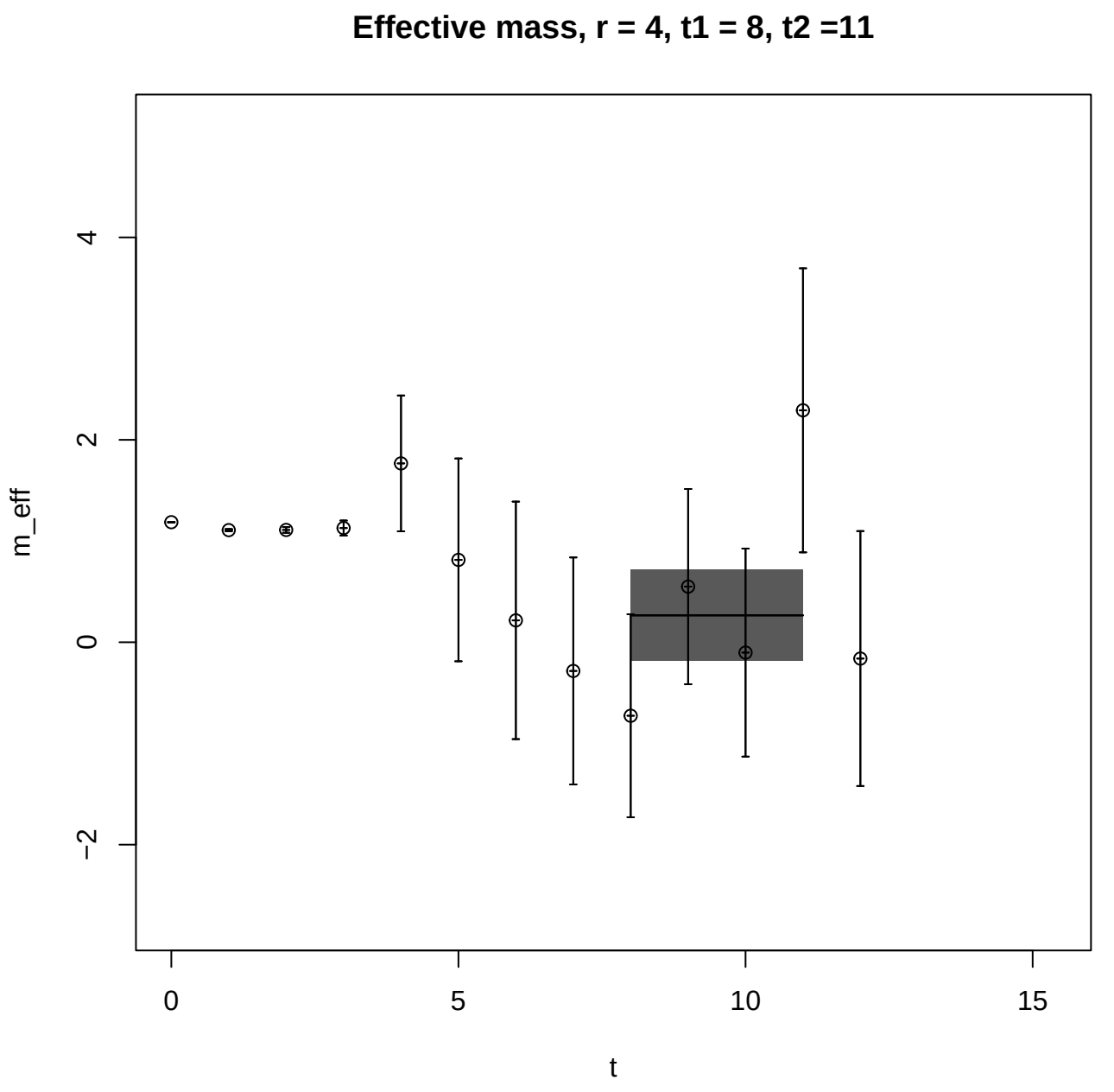

Effective mass, r = 4, t1 = 5, t2 =12
m_eff
4
2
0
−2
0
5
10
15
t

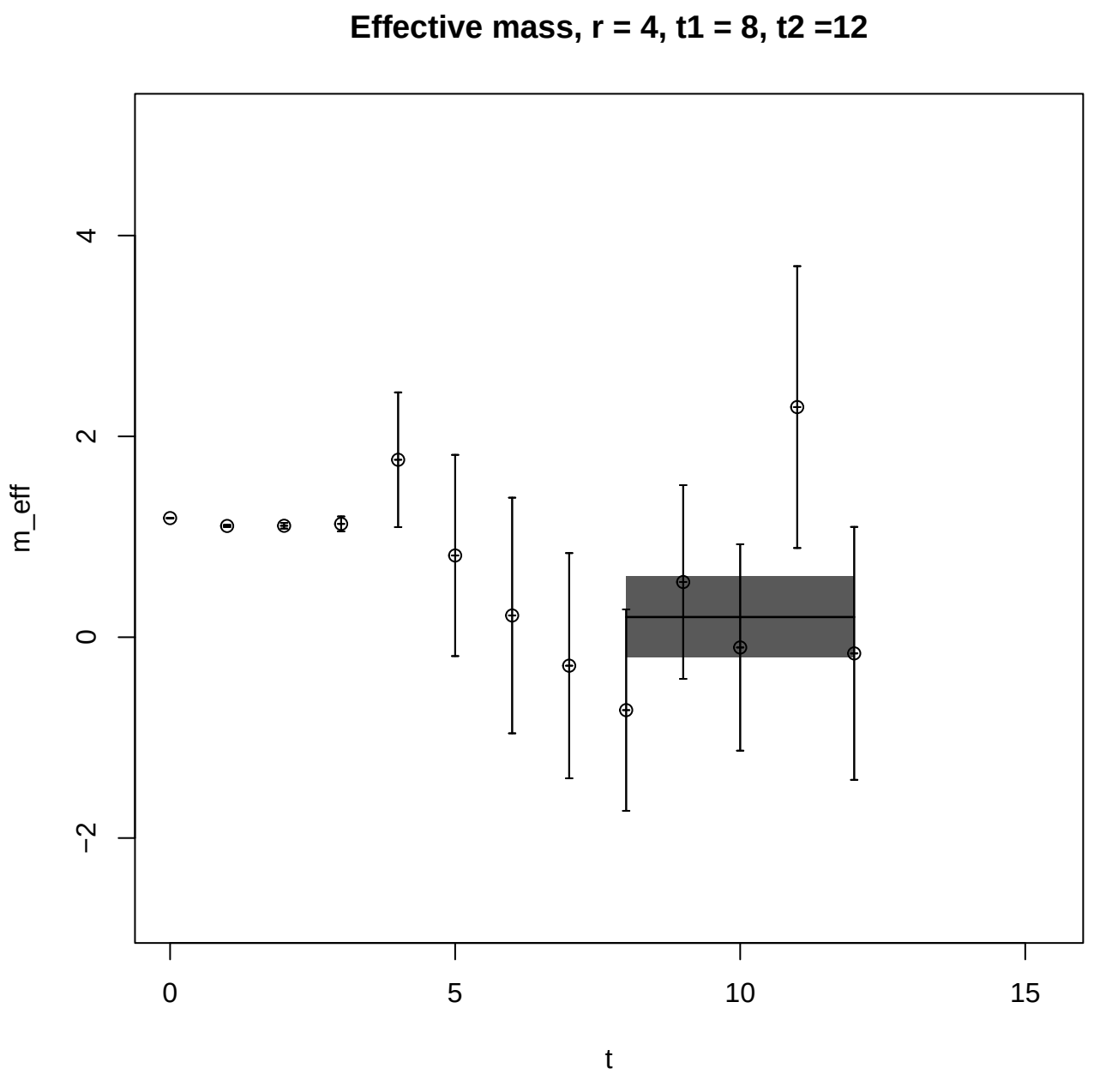

Effective mass, r = 4, t1 = 6, t2 =11
m_eff
4
2
0
−2
0
5
10
15
t

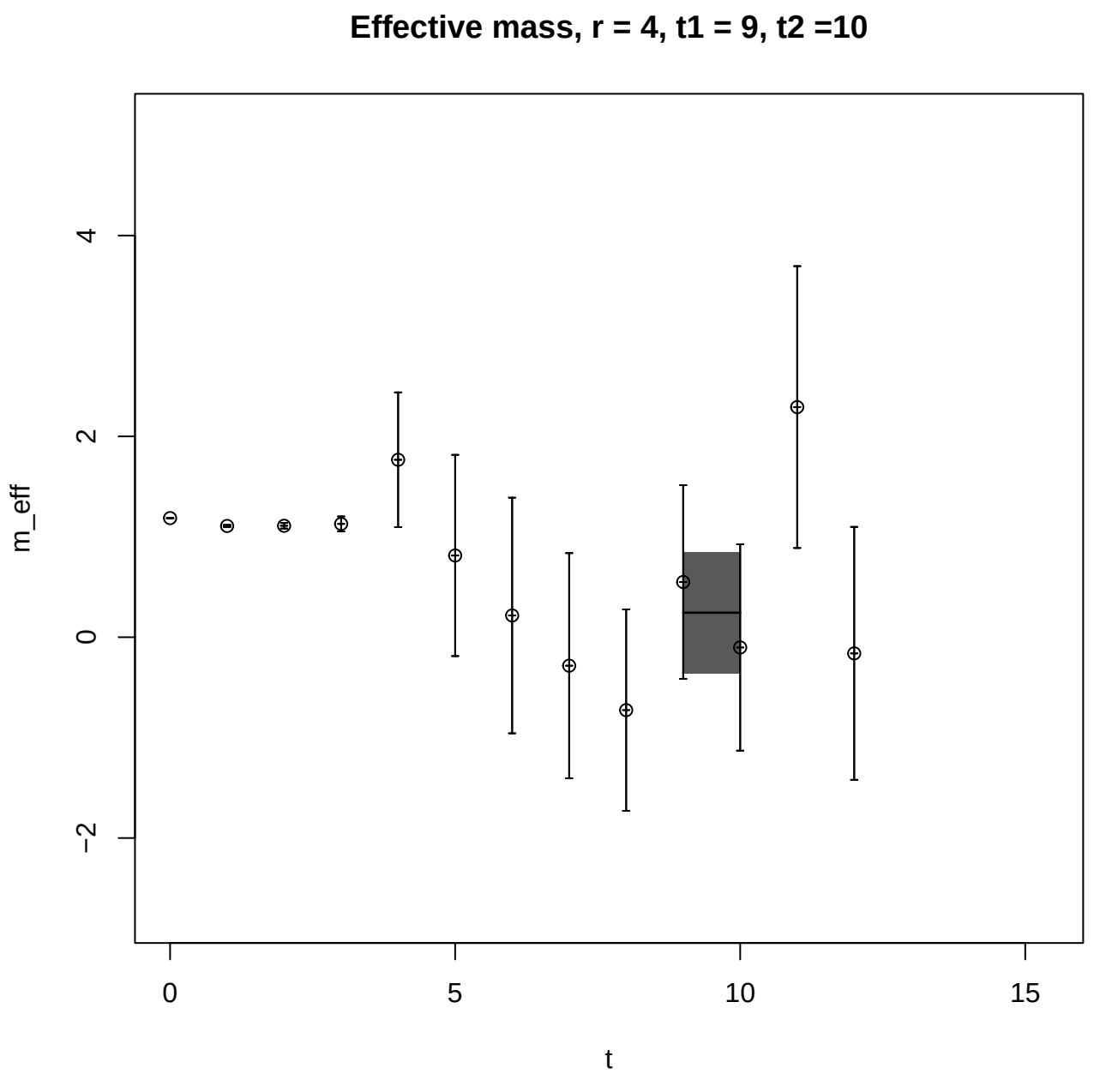

Effective mass, r = 4, t1 = 6, t2 =12
m_eff
4
2
0
−2
0
5
10
15
t

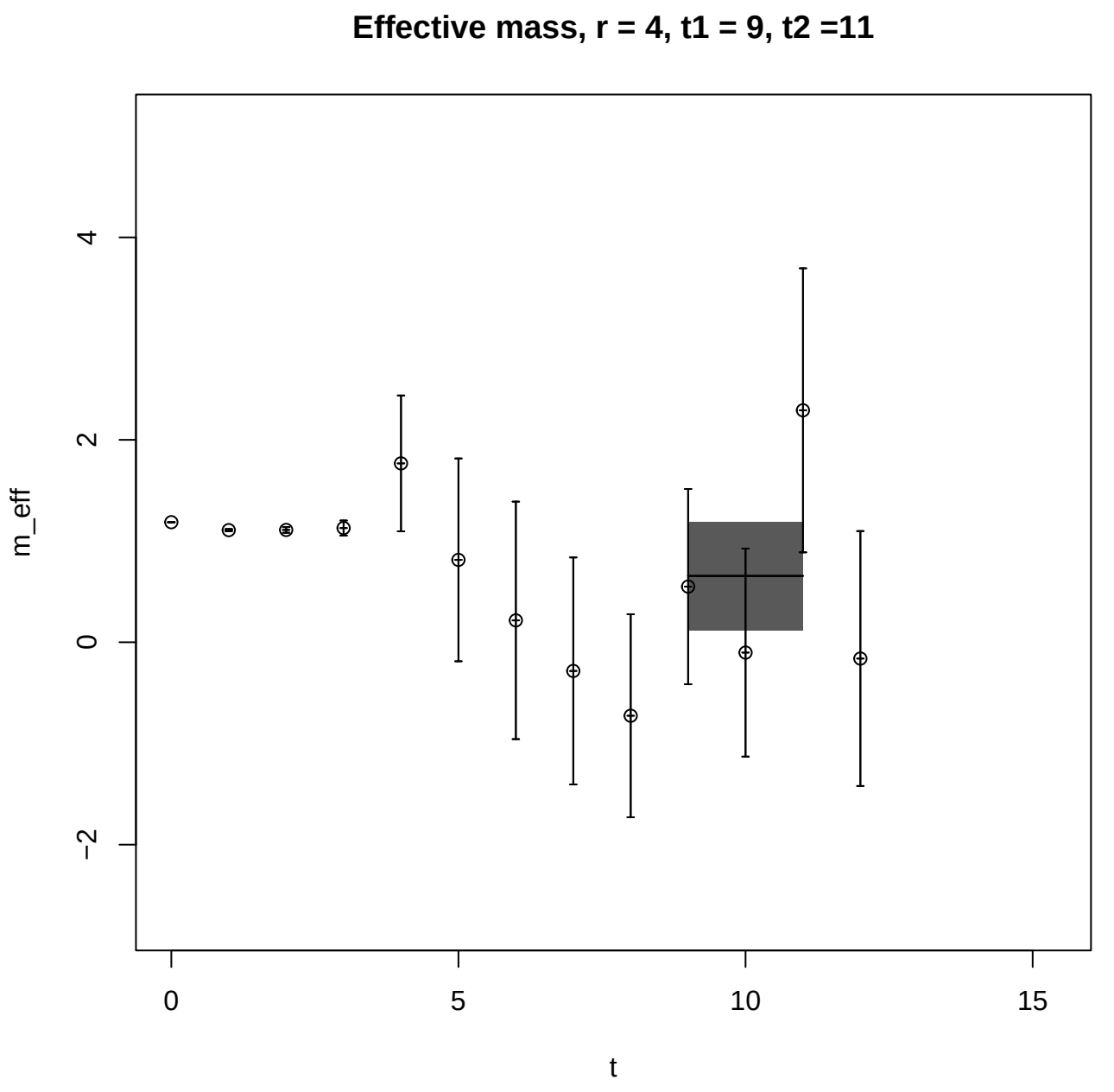

Effective mass, r = 4, t1 = 7, t2 =11
m_eff
4
2
0
−2
0
5
10
15
t

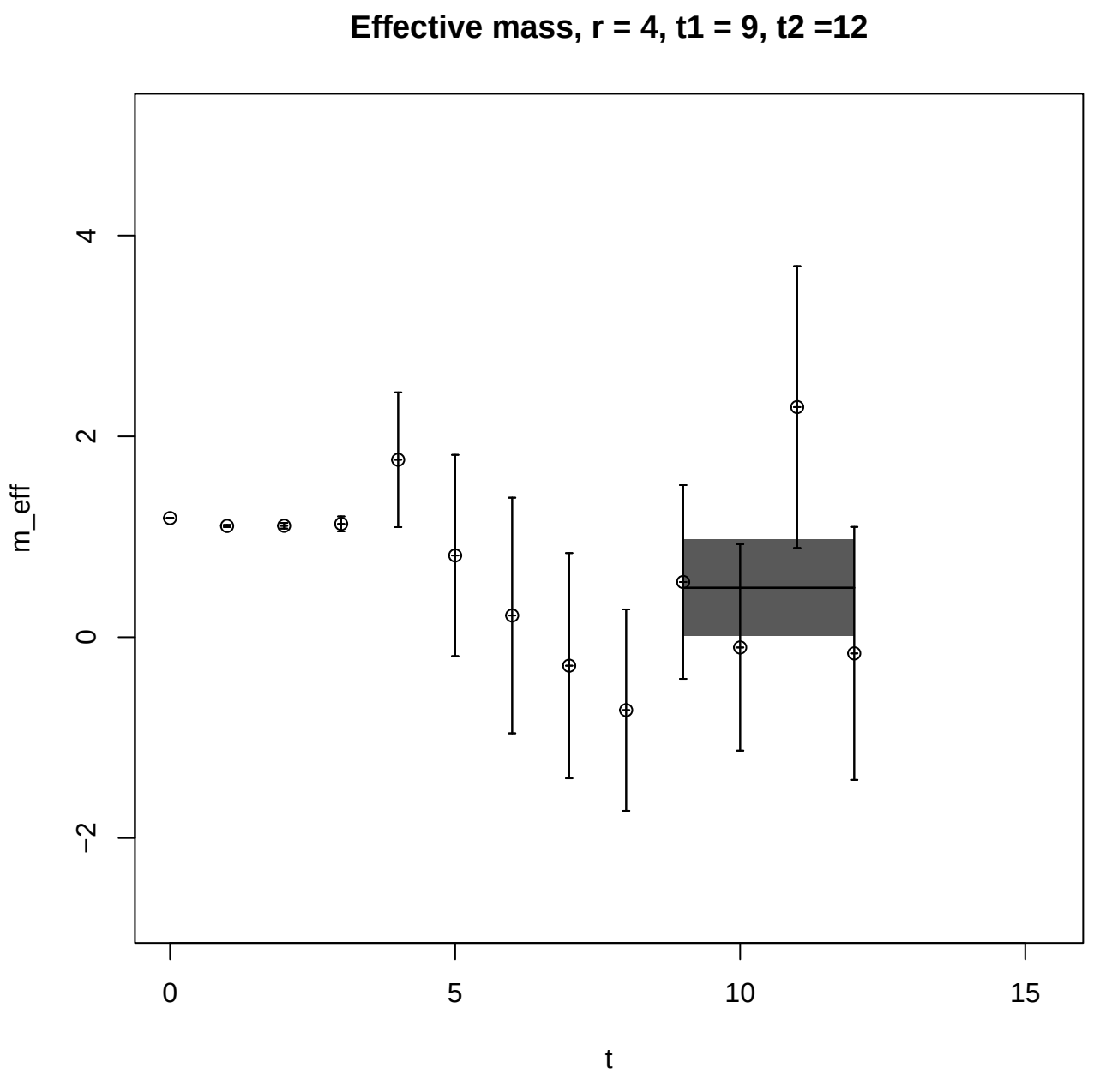

Effective mass, r = 4, t1 = 7, t2 =12
m_eff
4
2
0
−2
0
5
10
15
t

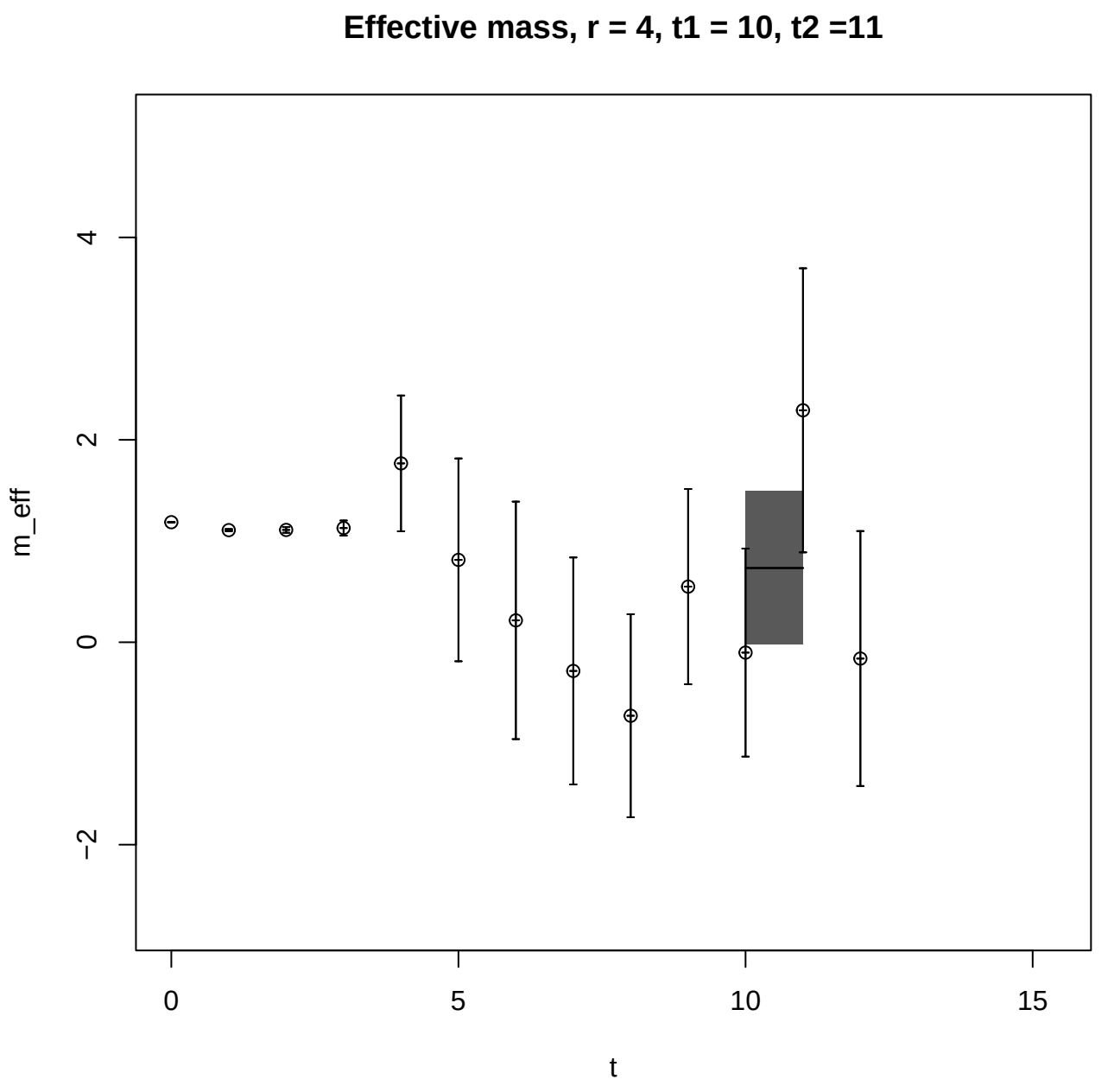

Effective mass, r = 4, t1 = 8, t2 =11
m_eff
4
2
0
−2
0
5
10
15
t

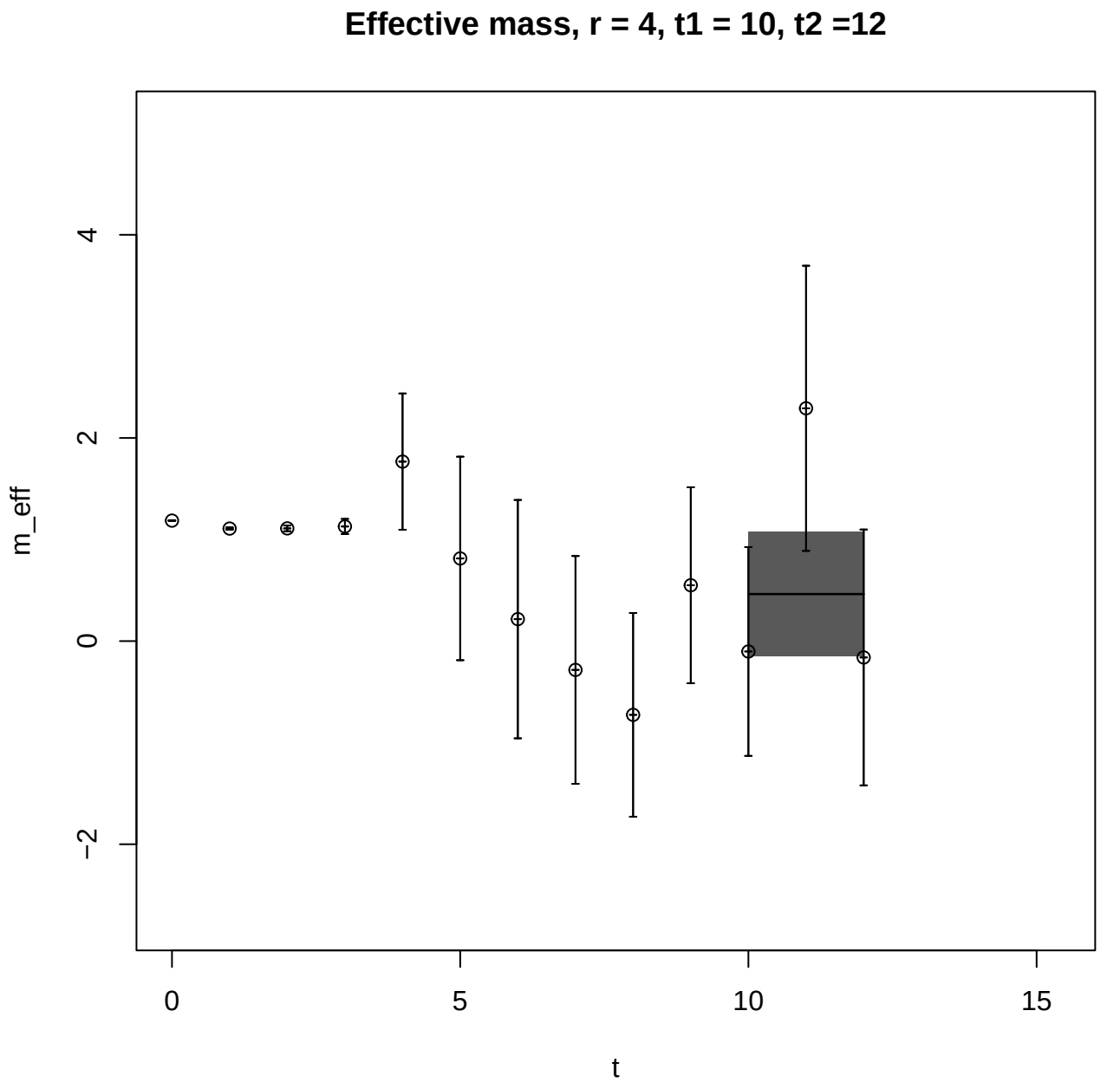

Effective mass, r = 4, t1 = 8, t2 =12
m_eff
4
2
0
−2
0
5
10
15
t

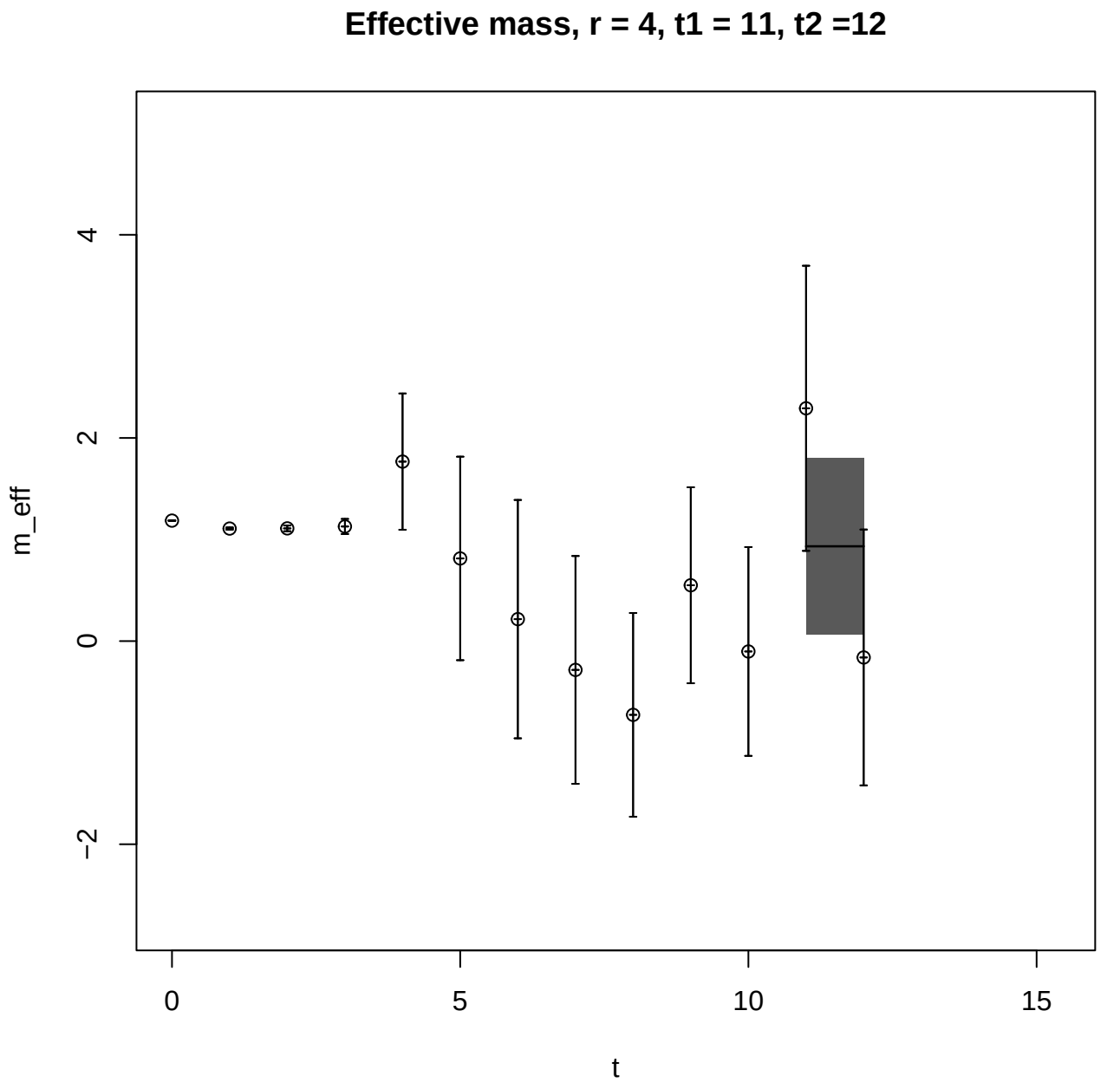

Effective mass, r = 4, t1 = 9, t2 =10
m_eff
4
2
0
−2
0
5
10
15
t

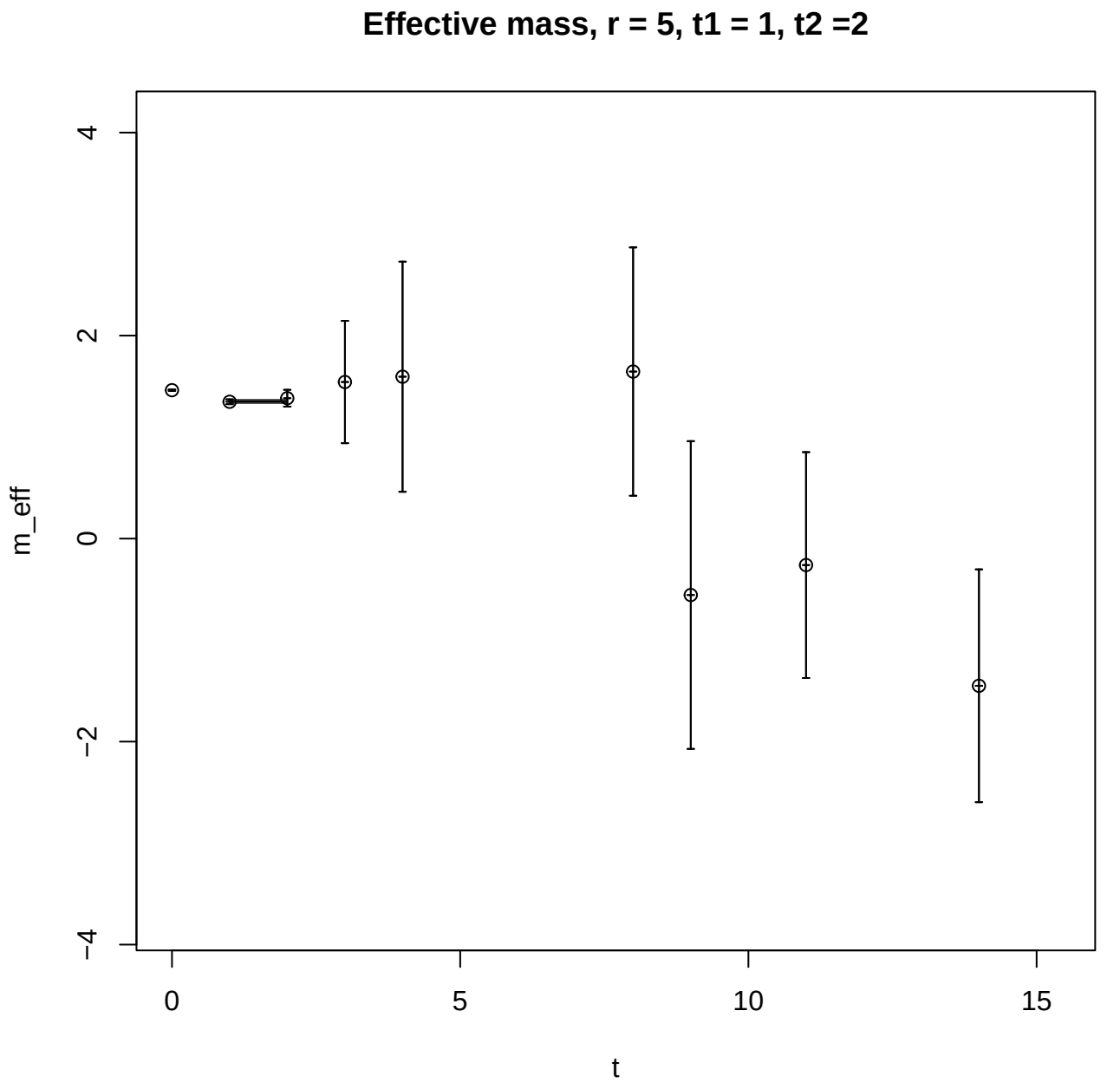

Effective mass, r = 4, t1 = 9, t2 =11
m_eff
4
2
0
−2
0
5
10
15
t

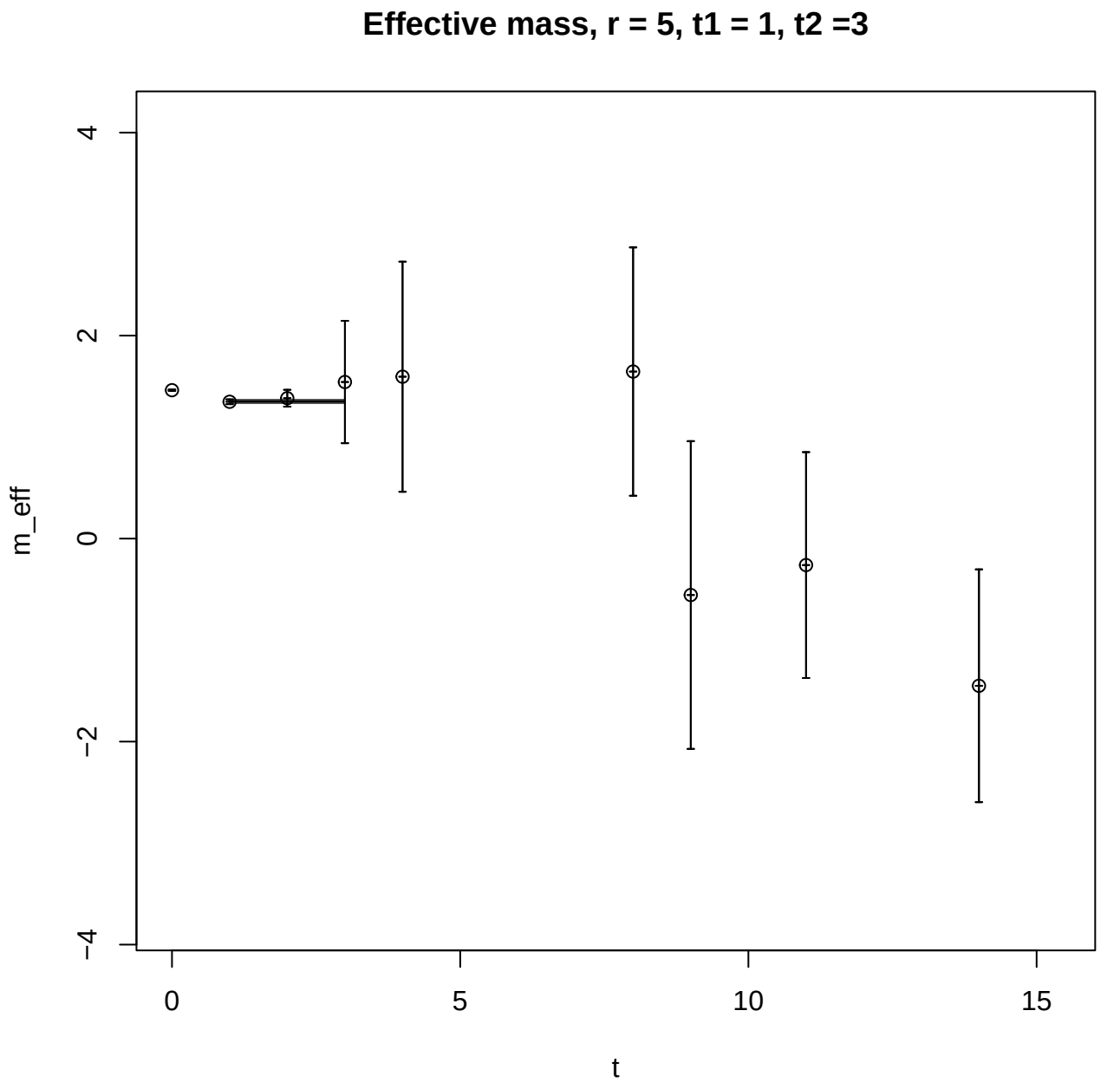

Effective mass, r = 4, t1 = 9, t2 =12
m_eff
4
2
0
−2
0
5
10
15
t

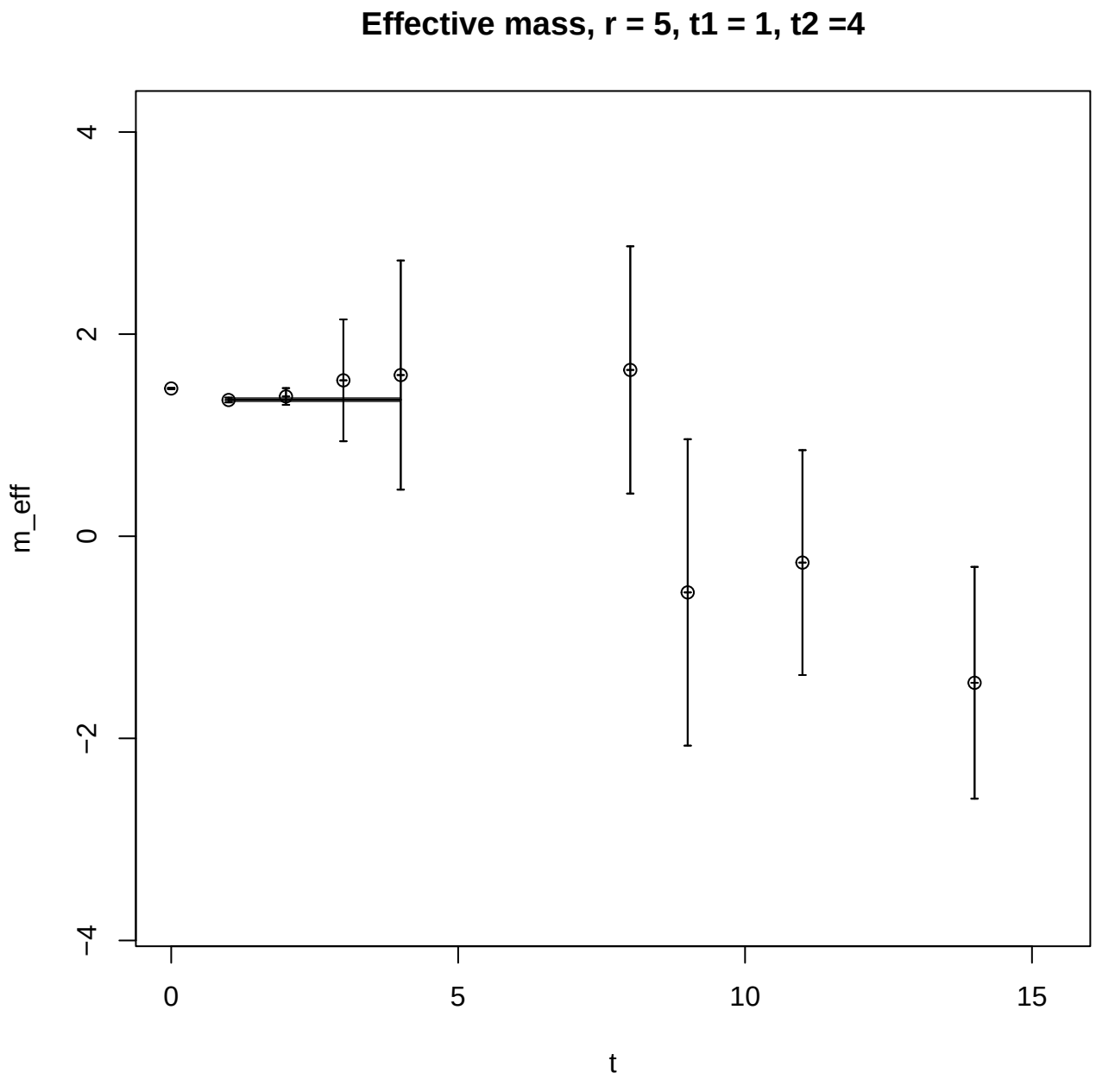

Effective mass, r = 4, t1 = 10, t2 =11
m_eff
4
2
0
−2
0
5
10
15
t

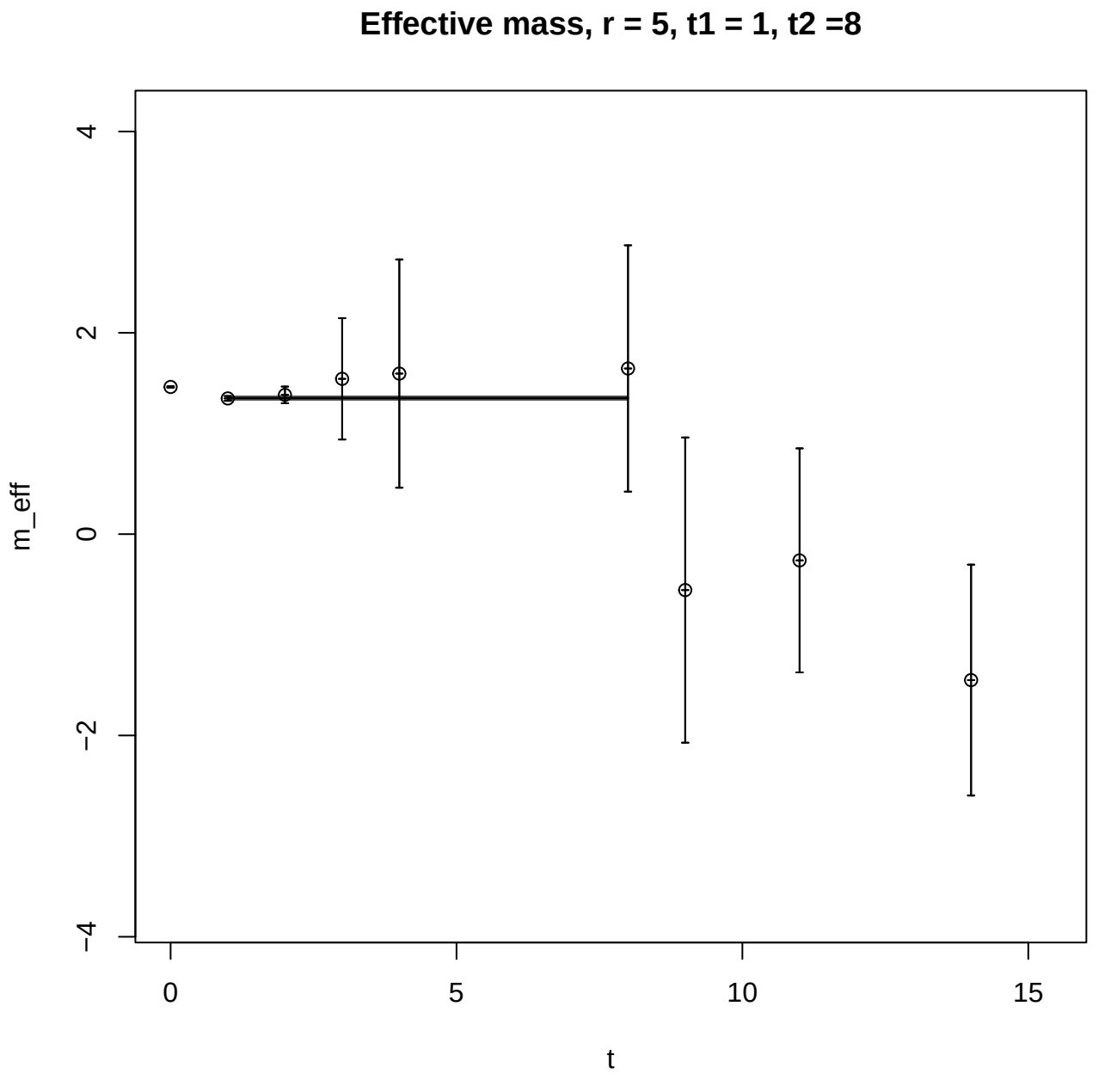

Effective mass, r = 4, t1 = 10, t2 =12
m_eff
4
2
0
−2
0
5
10
15
t

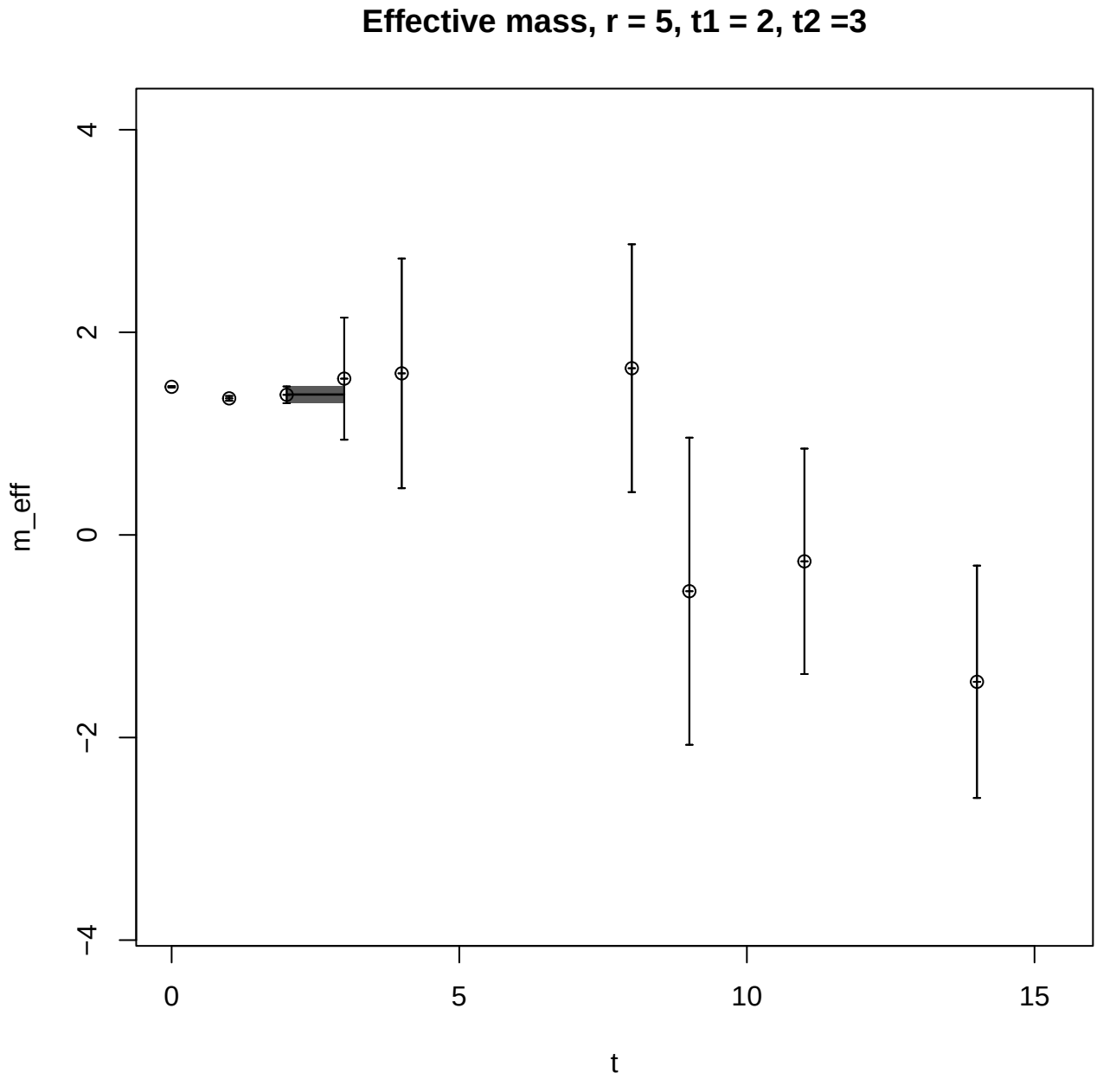

Effective mass, r = 4, t1 = 11, t2 =12
m_eff
4
2
0
−2
0
5
10
15
t

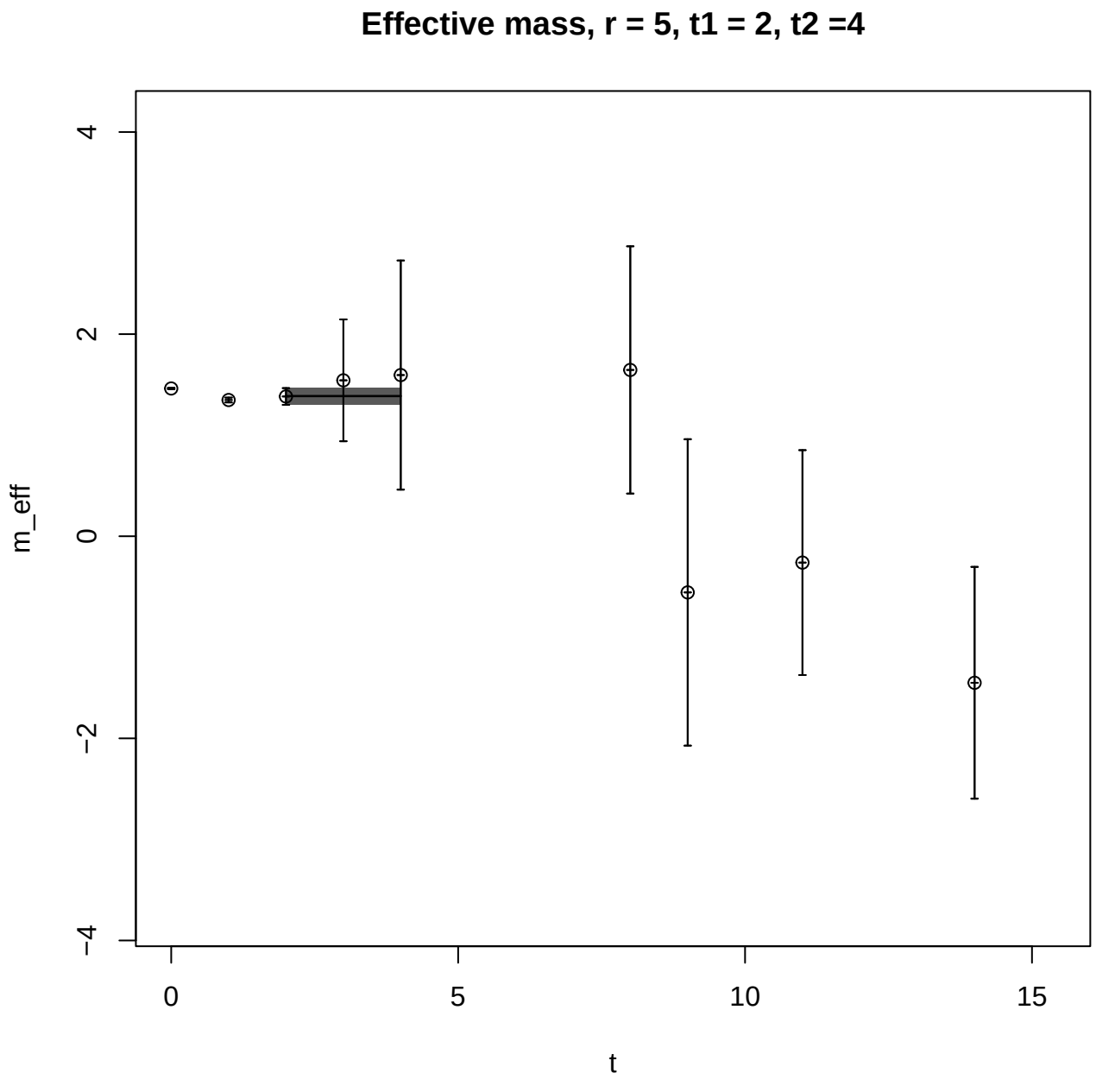

Effective mass, r = 5, t1 = 1, t2 =2
m_eff
t
4
2
0
−2
−4
0
5
10
15

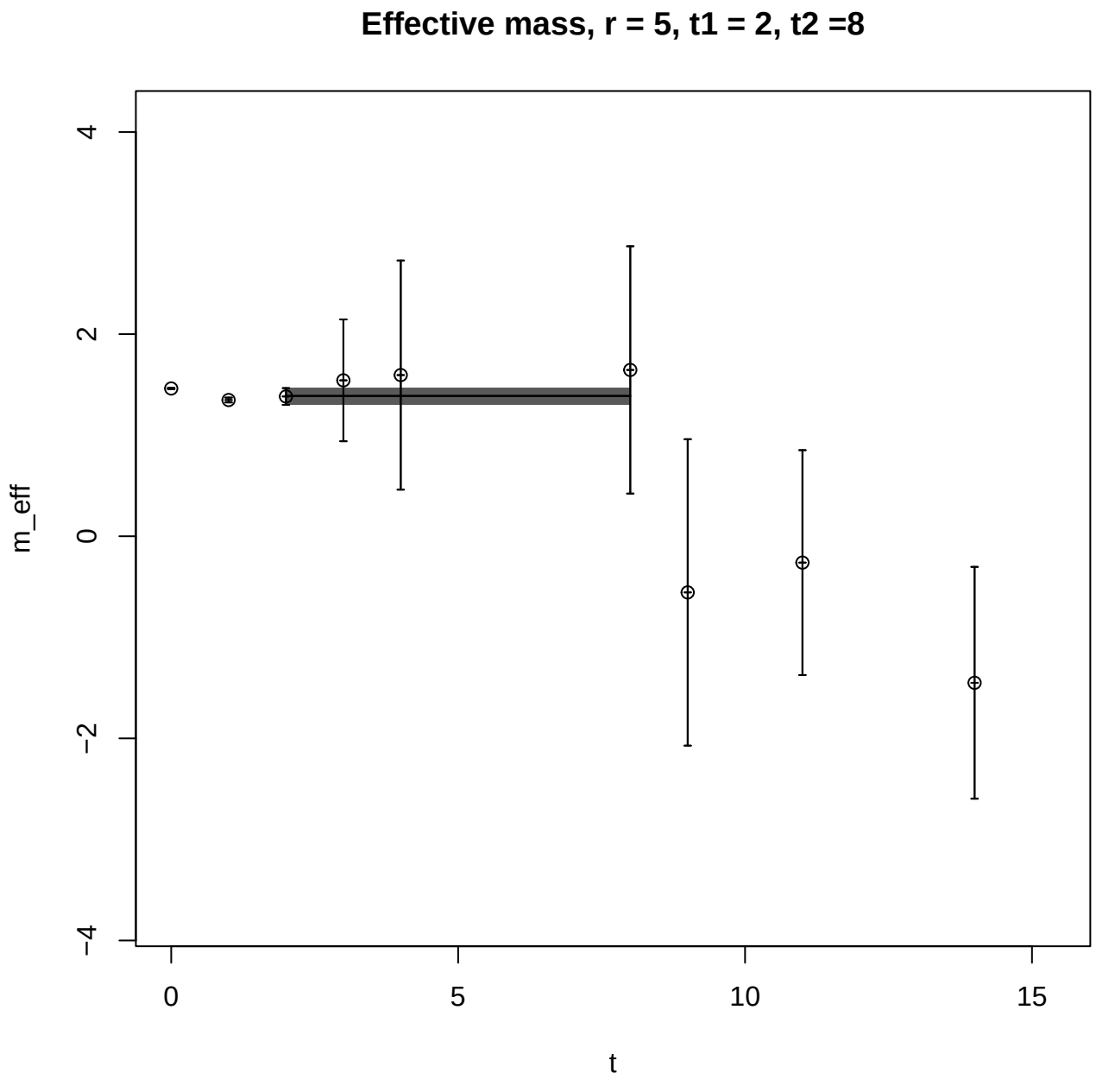

Effective mass, r = 5, t1 = 1, t2 =3
m_eff
t
4
2
0
−2
−4
0
5
10
15

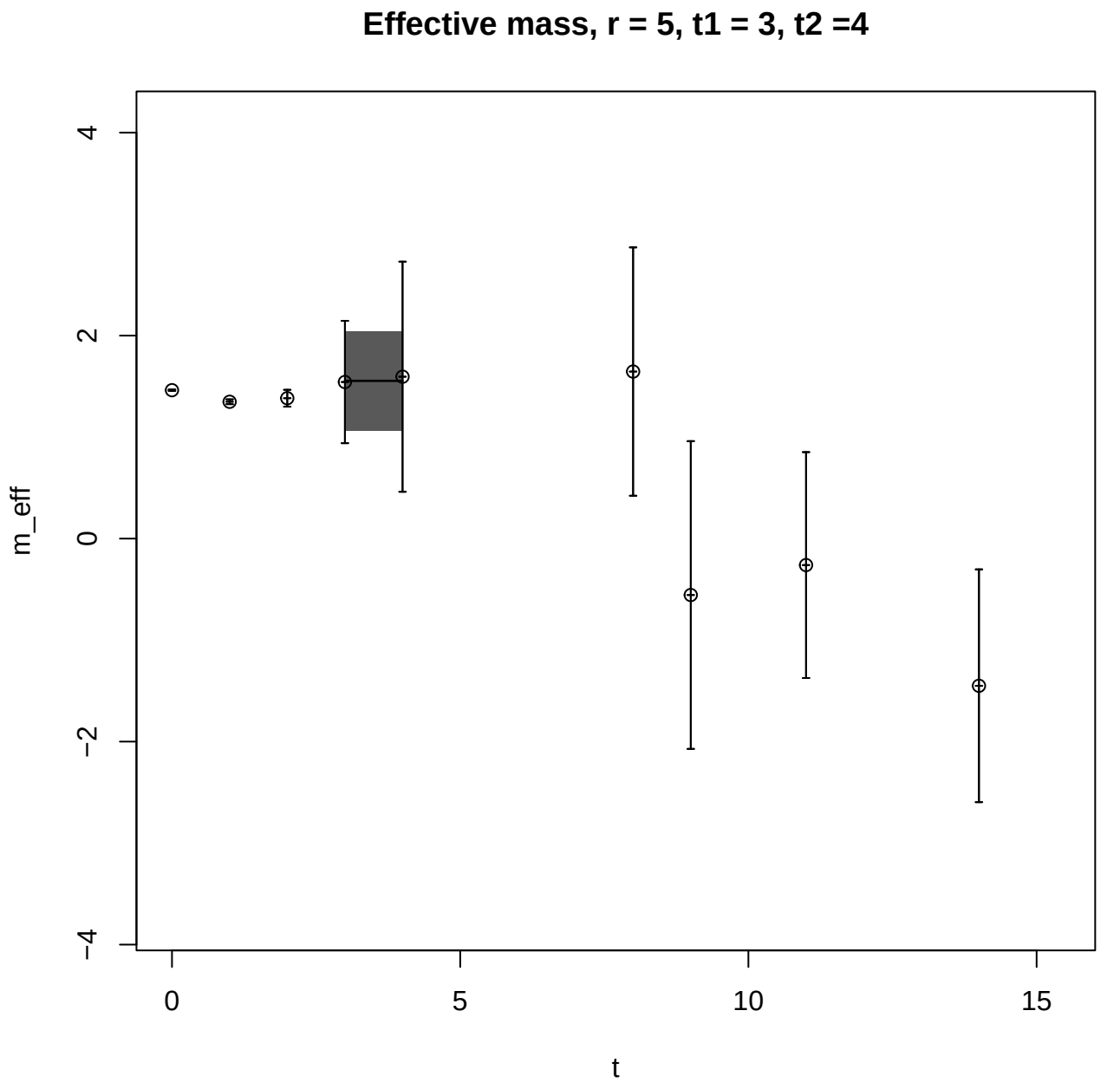

Effective mass, r = 5, t1 = 1, t2 =4
m_eff
t
4
2
0
−2
−4
0
5
10
15

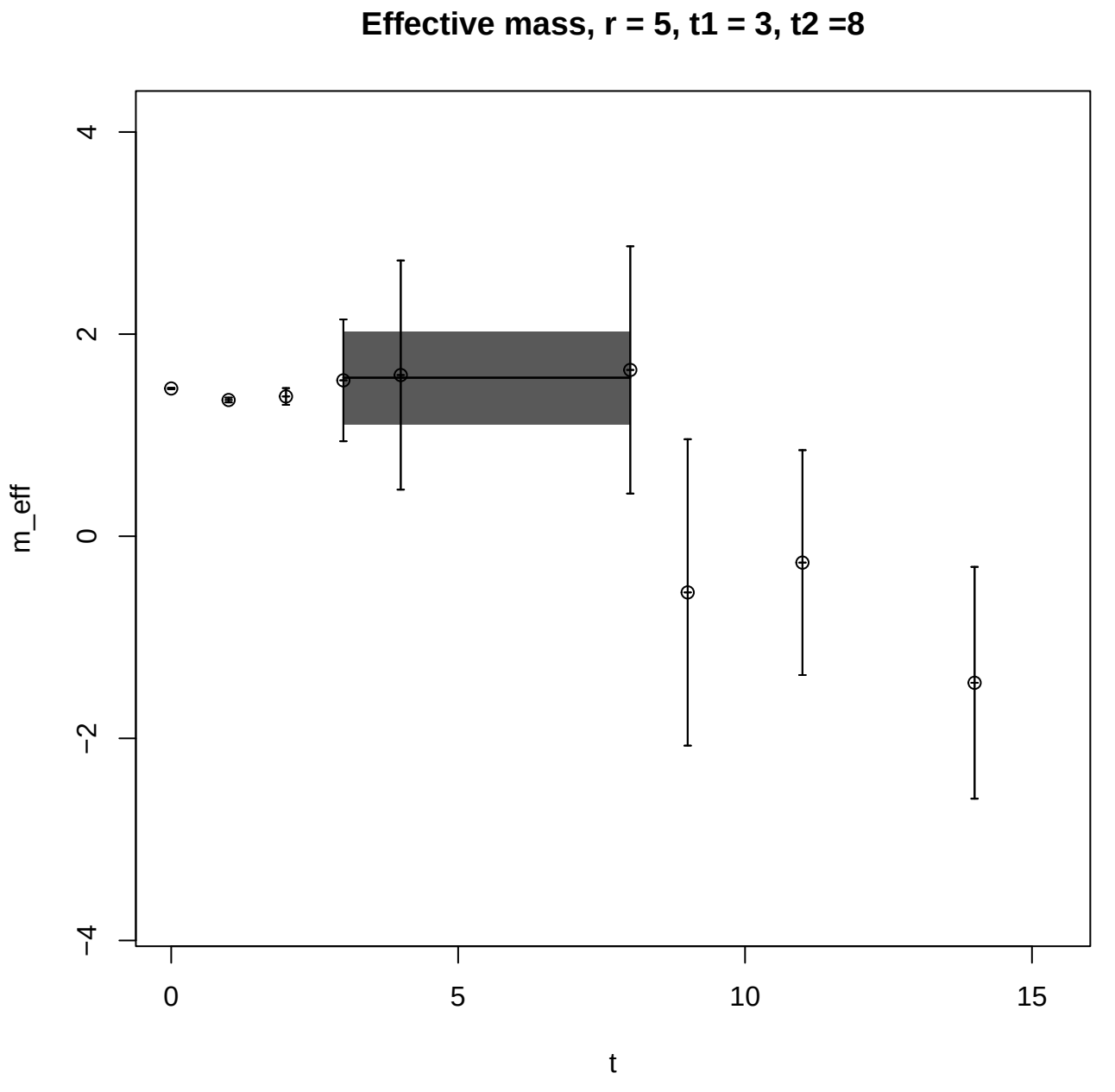

Effective mass, r = 5, t1 = 1, t2 =8
m_eff
t
4
2
0
−2
−4
0
5
10
15

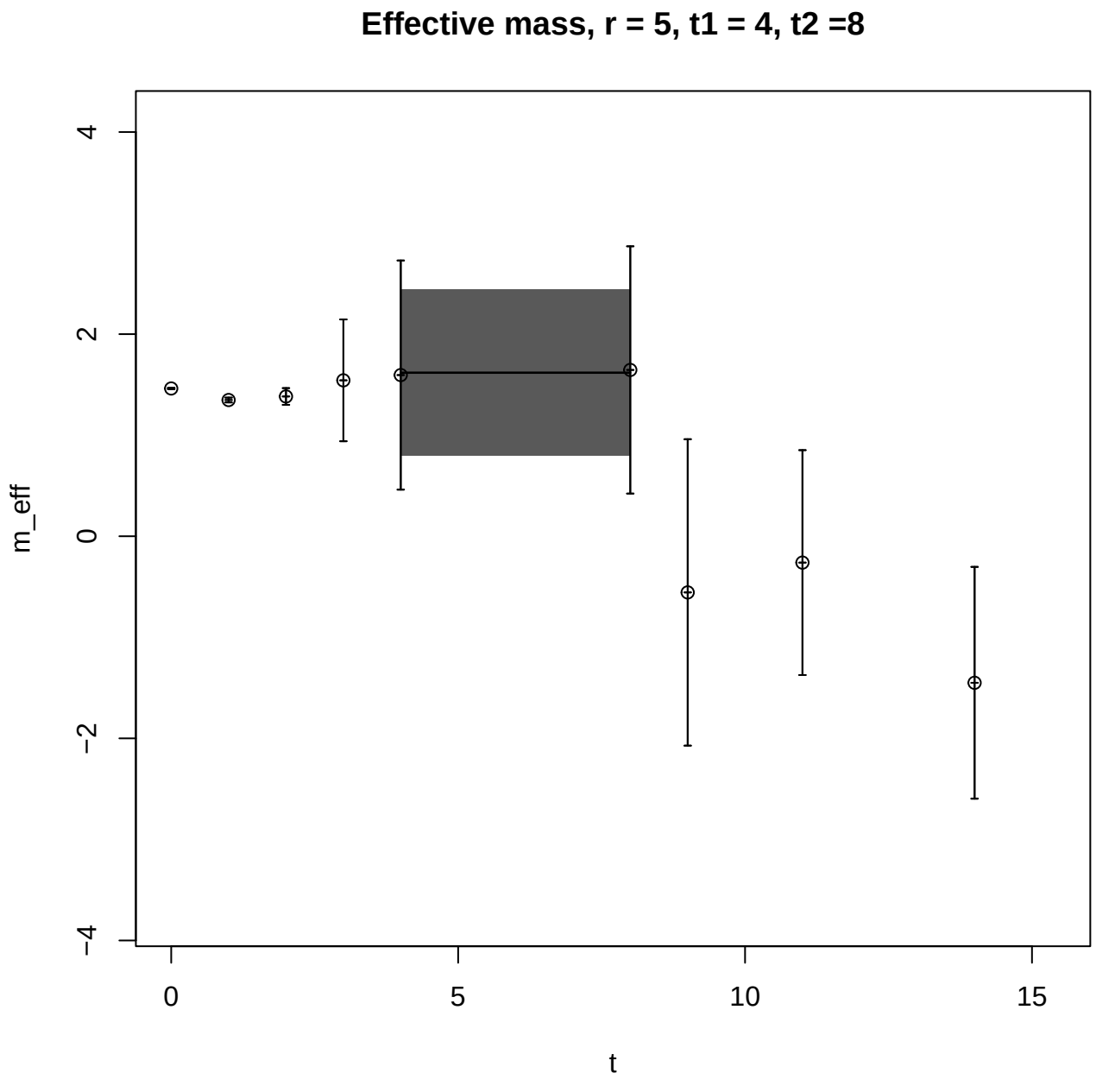

Effective mass, r = 5, t1 = 2, t2 =3
m_eff
t
4
2
0
−2
−4
0
5
10
15

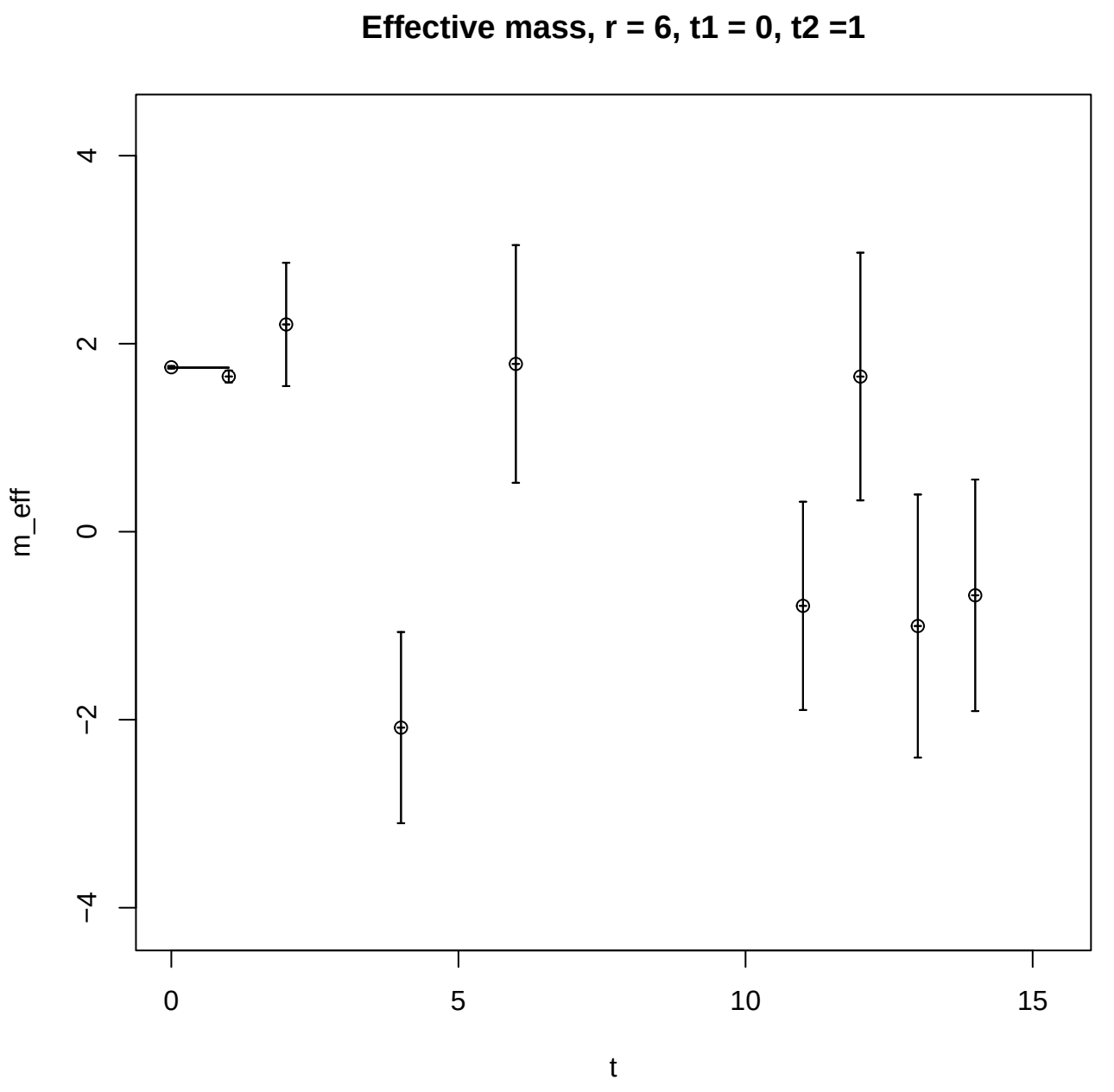

Effective mass, r = 5, t1 = 2, t2 =4
m_eff
t
4
2
0
−2
−4
0
5
10
15

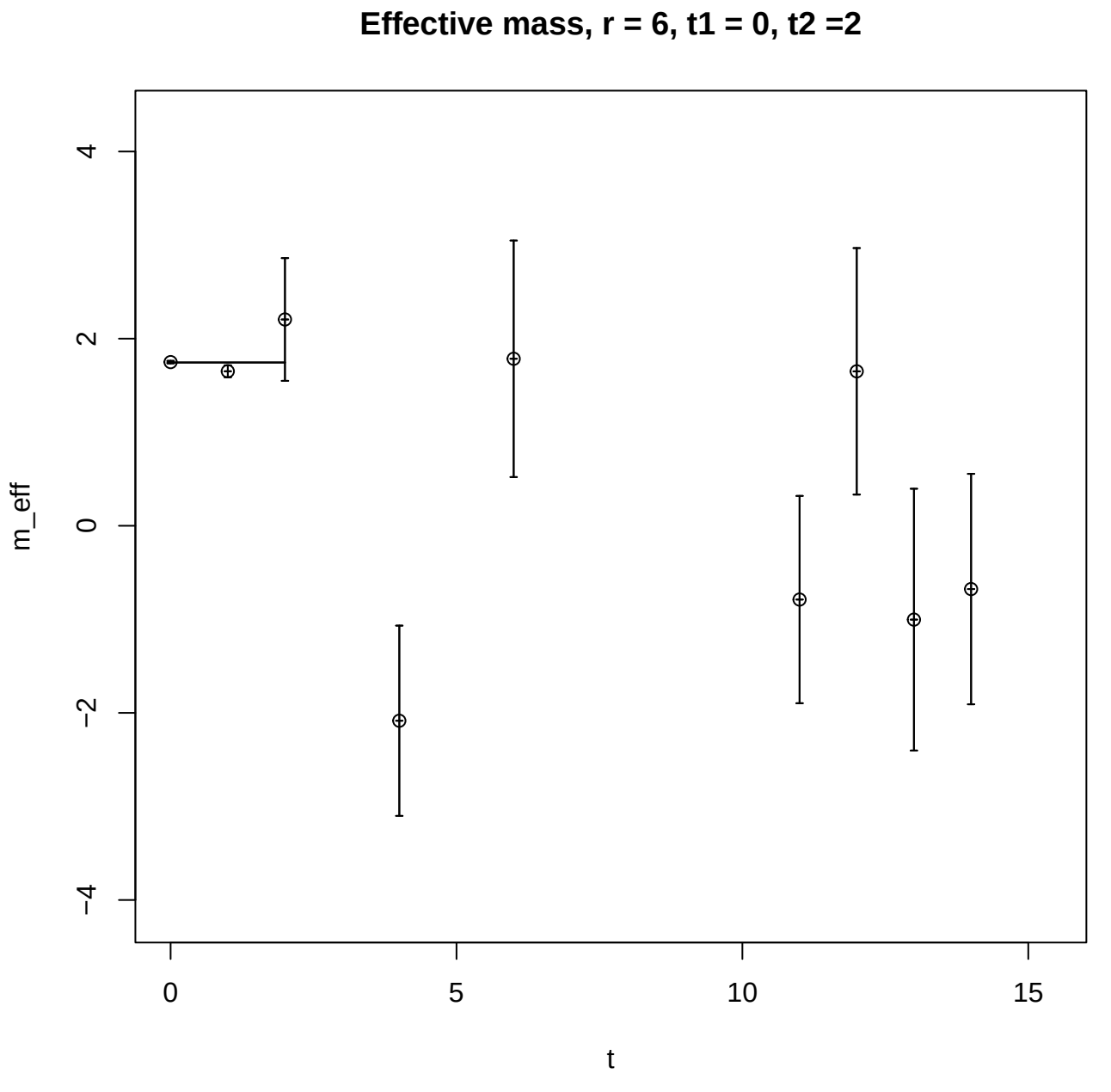

Effective mass, r = 5, t1 = 2, t2 =8
m_eff
t
4
2
0
−2
−4
0
5
10
15

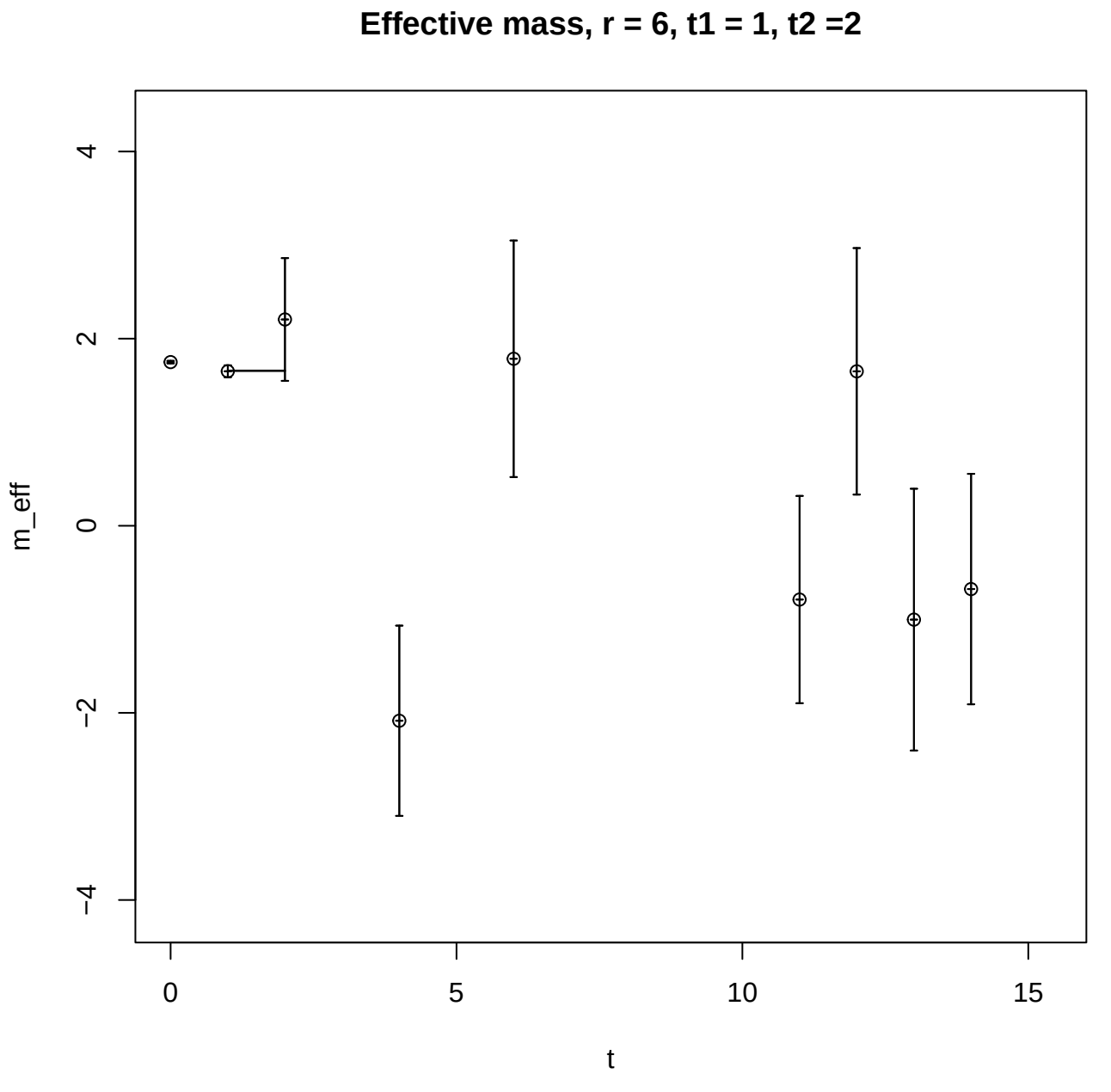

Effective mass, r = 5, t1 = 3, t2 =4
m_eff
t
4
2
0
−2
−4
0
5
10
15

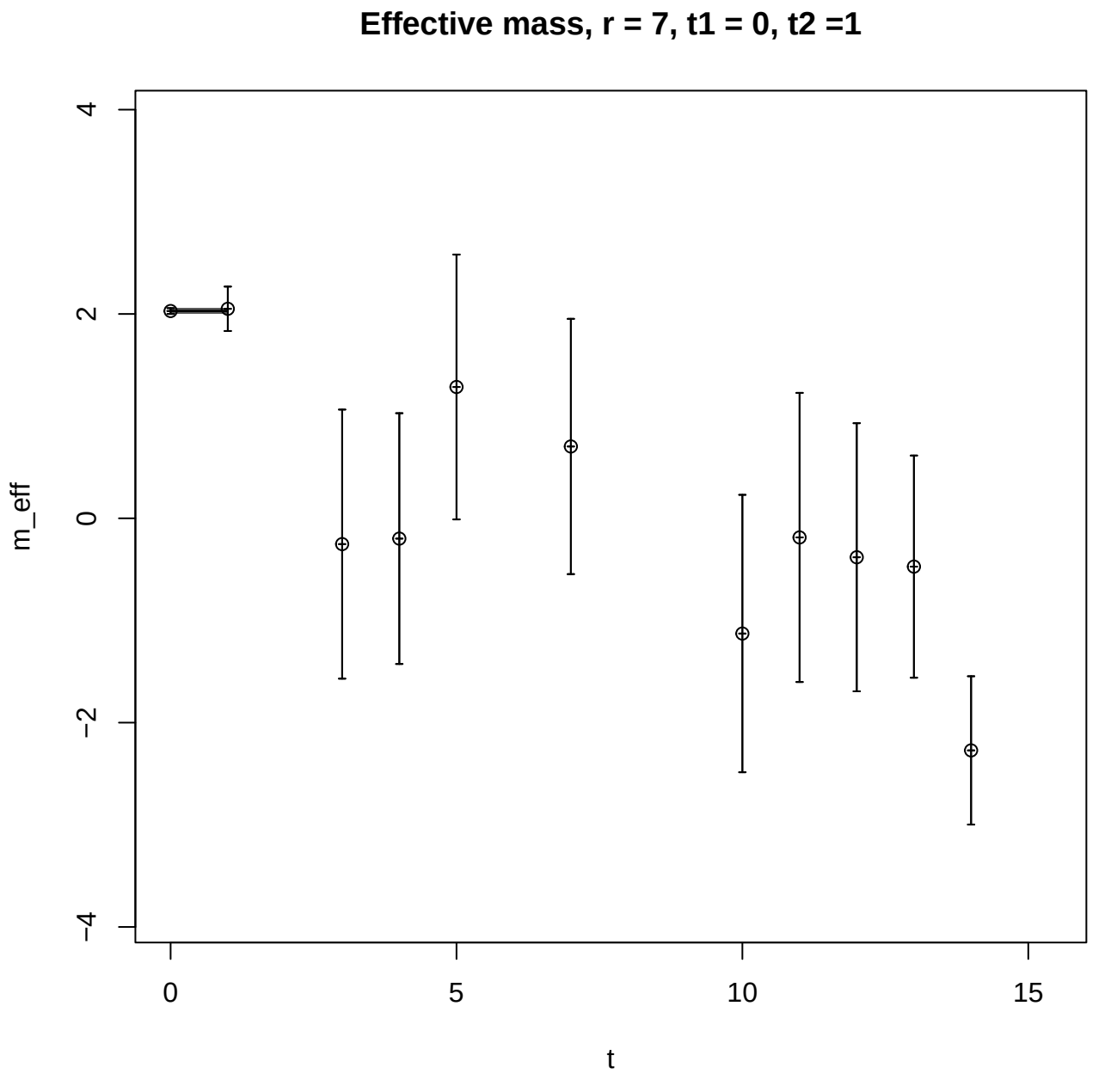

Effective mass, r = 5, t1 = 3, t2 =8
m_eff
t
4
2
0
−2
−4
0
5
10
15

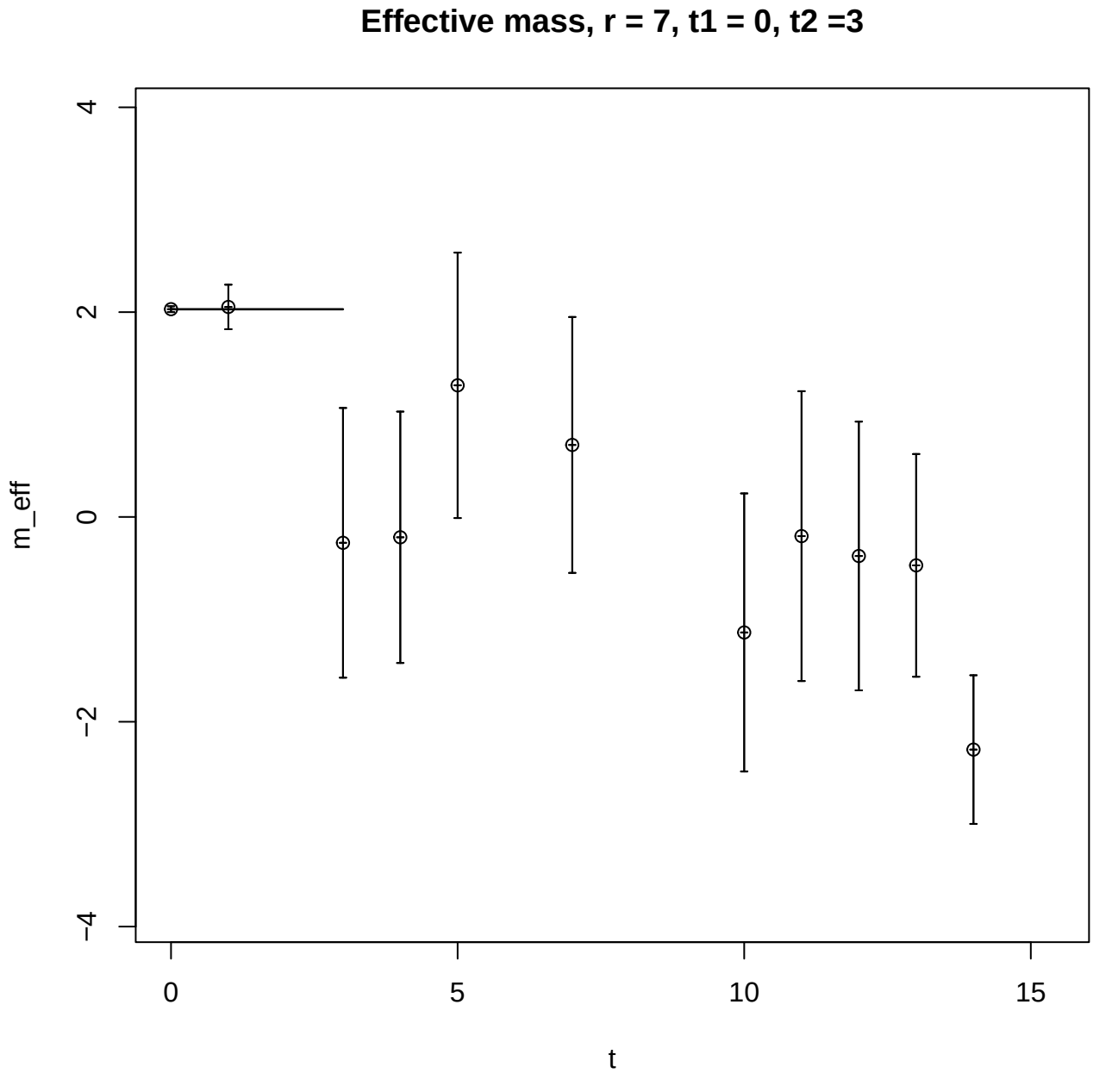

Effective mass, r = 5, t1 = 4, t2 =8
m_eff
t
4
2
0
−2
−4
0
5
10
15

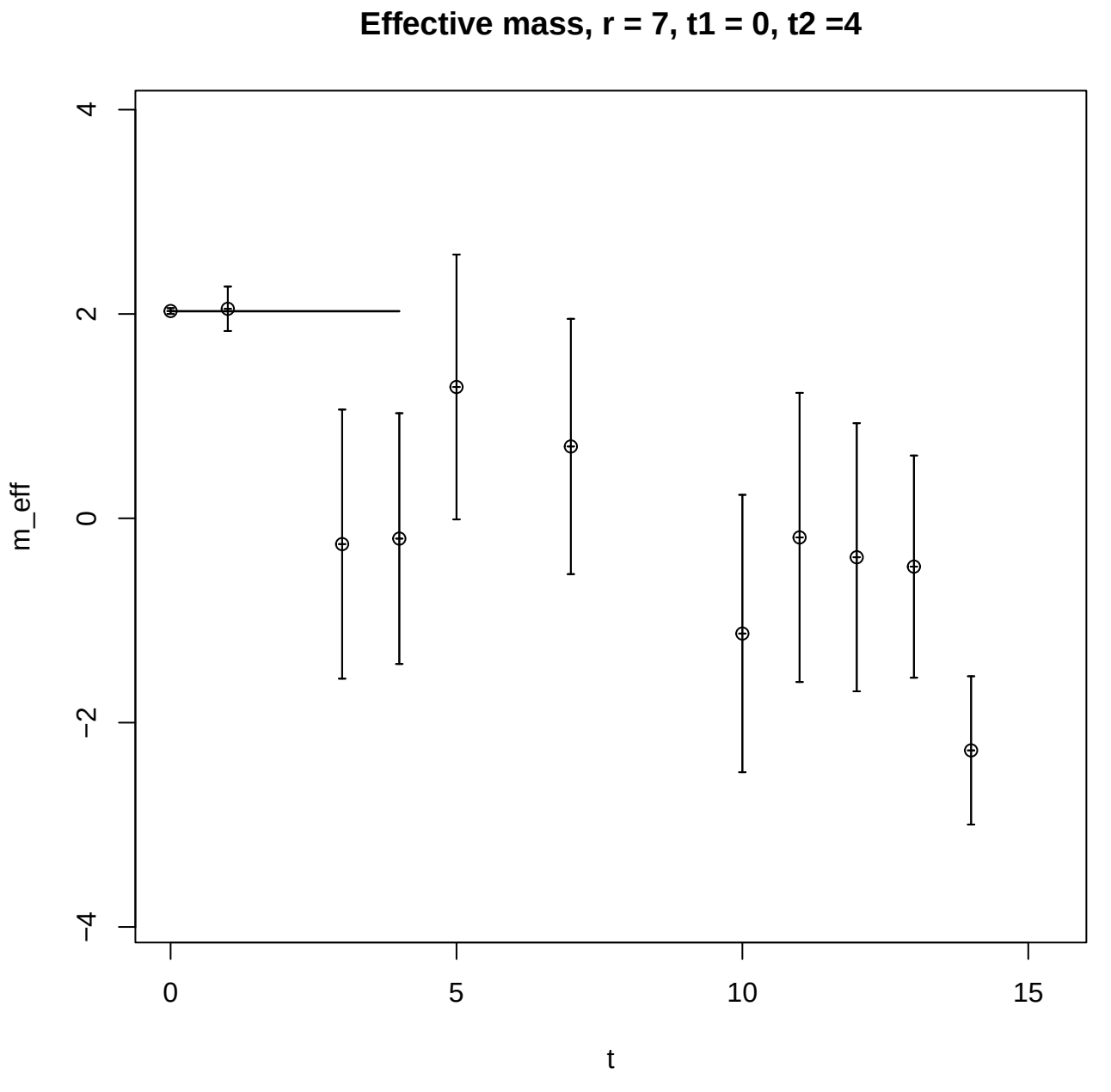

Effective mass, r = 6, t1 = 0, t2 =1
m_eff
t
4
2
0
−2
−4
0
5
10
15

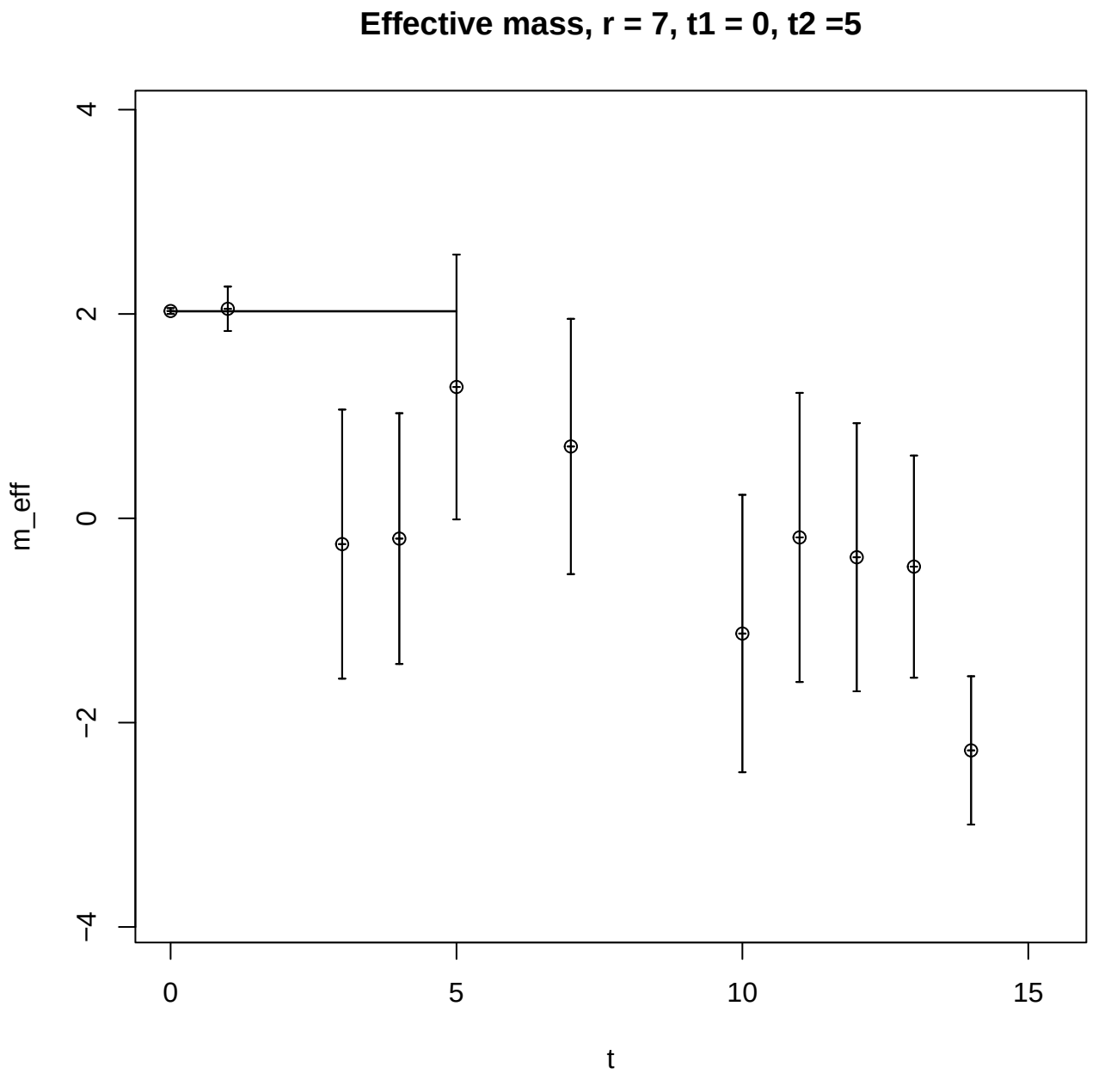

Effective mass, r = 6, t1 = 0, t2 =2
m_eff
t
4
2
0
−2
−4
0
5
10
15

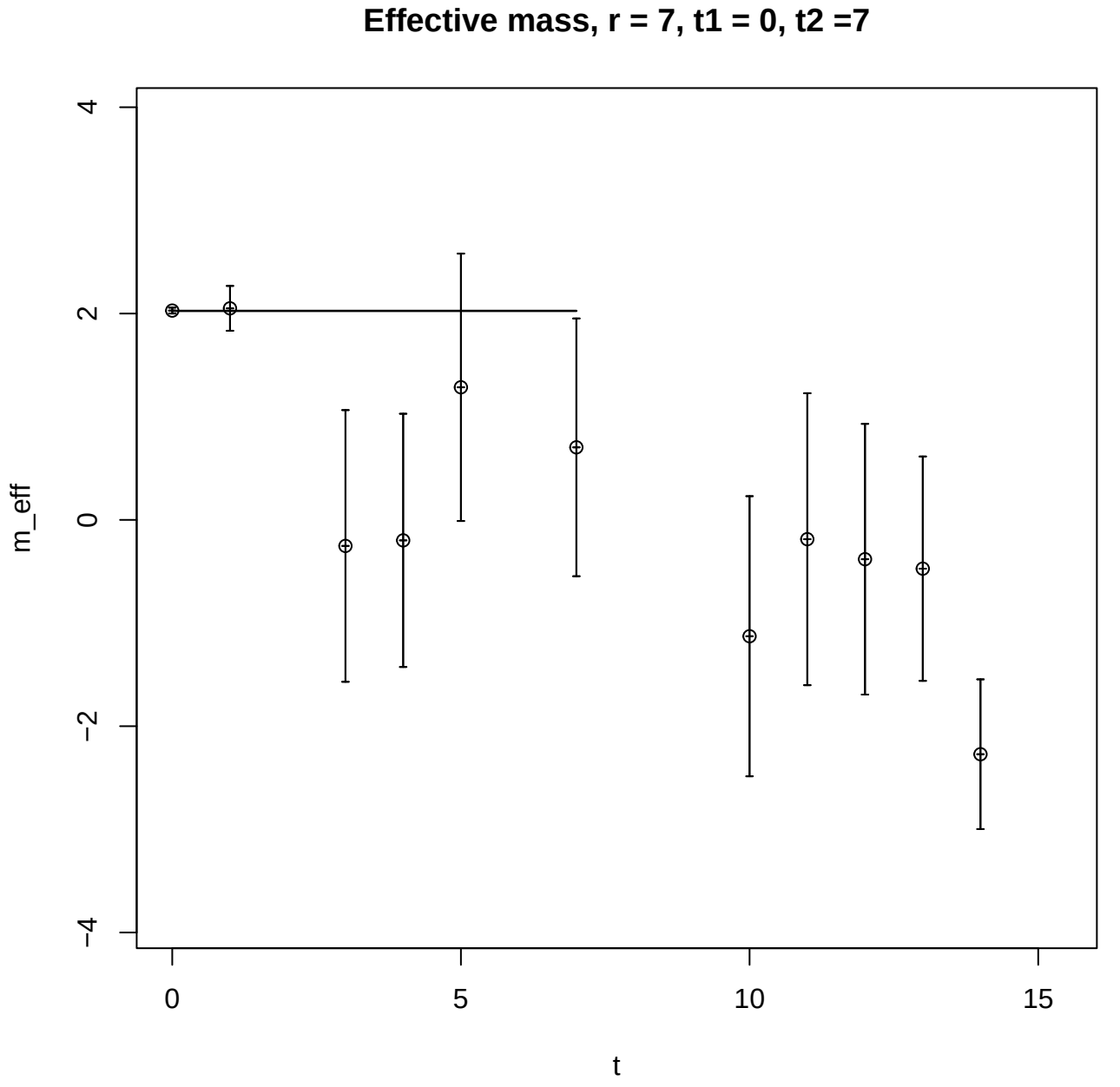

Effective mass, r = 6, t1 = 1, t2 =2
m_eff
t
4
2
0
−2
−4
0
5
10
15

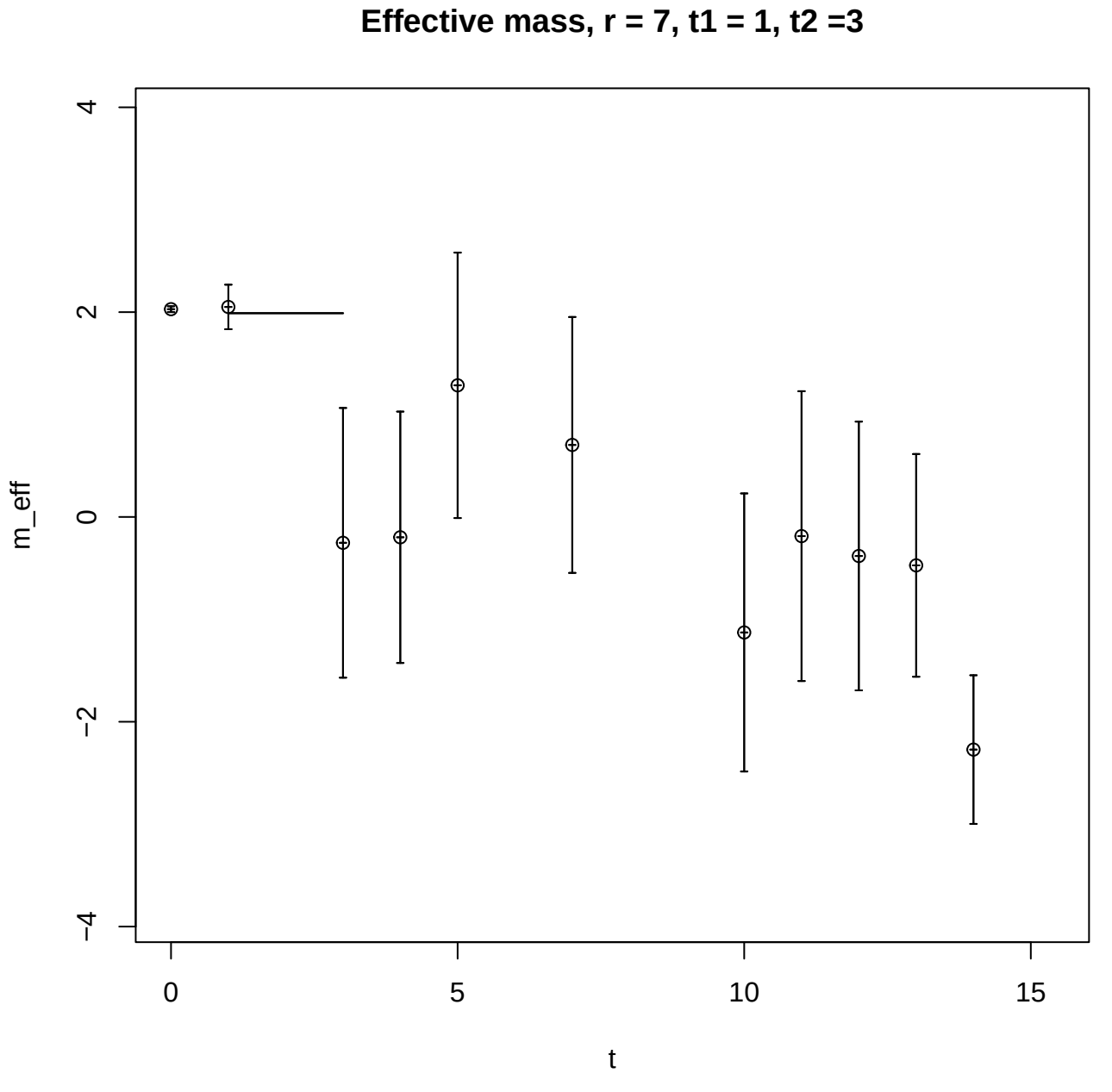

Effective mass, r = 7, t1 = 0, t2 =1
m_eff
t
4
2
0
−2
−4
0
5
10
15


Effective mass, r = 7, t1 = 0, t2 =3
m_eff
t
4
2
0
−2
−4
0
5
10
15


Effective mass, r = 7, t1 = 0, t2 =4
m_eff
t
4
2
0
−2
−4
0
5
10
15


Effective mass, r = 7, t1 = 0, t2 =5
m_eff
t
4
2
0
−2
−4
0
5
10
15


Effective mass, r = 7, t1 = 0, t2 =7
m_eff
t
4
2
0
−2
−4
0
5
10
15


Effective mass, r = 7, t1 = 1, t2 =3
m_eff
t
4
2
0
−2
−4
0
5
10
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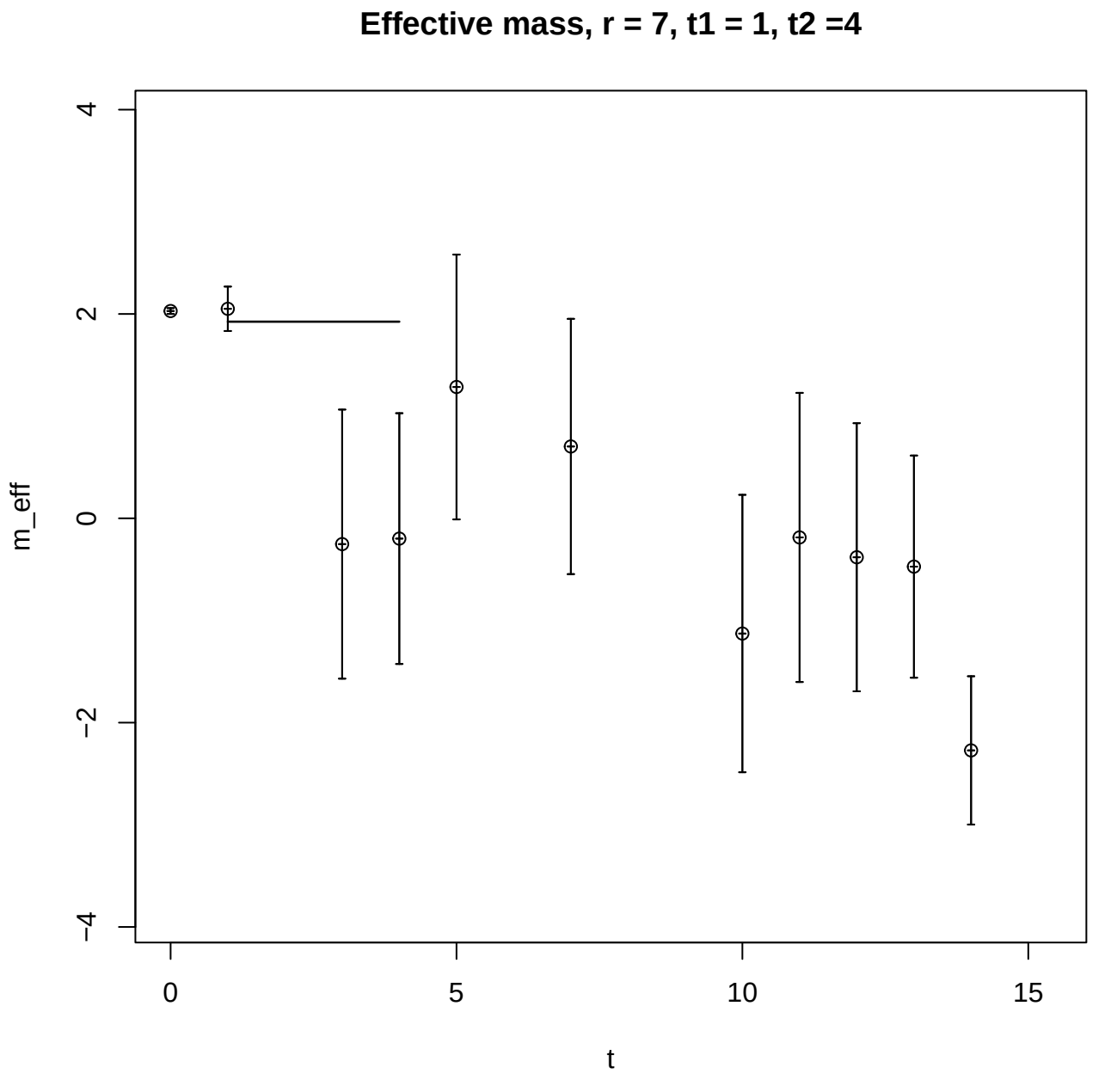
Effective mass, r = 7, t1 = 1, t2 =4
m_eff
t
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0
−2
−4
0
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10
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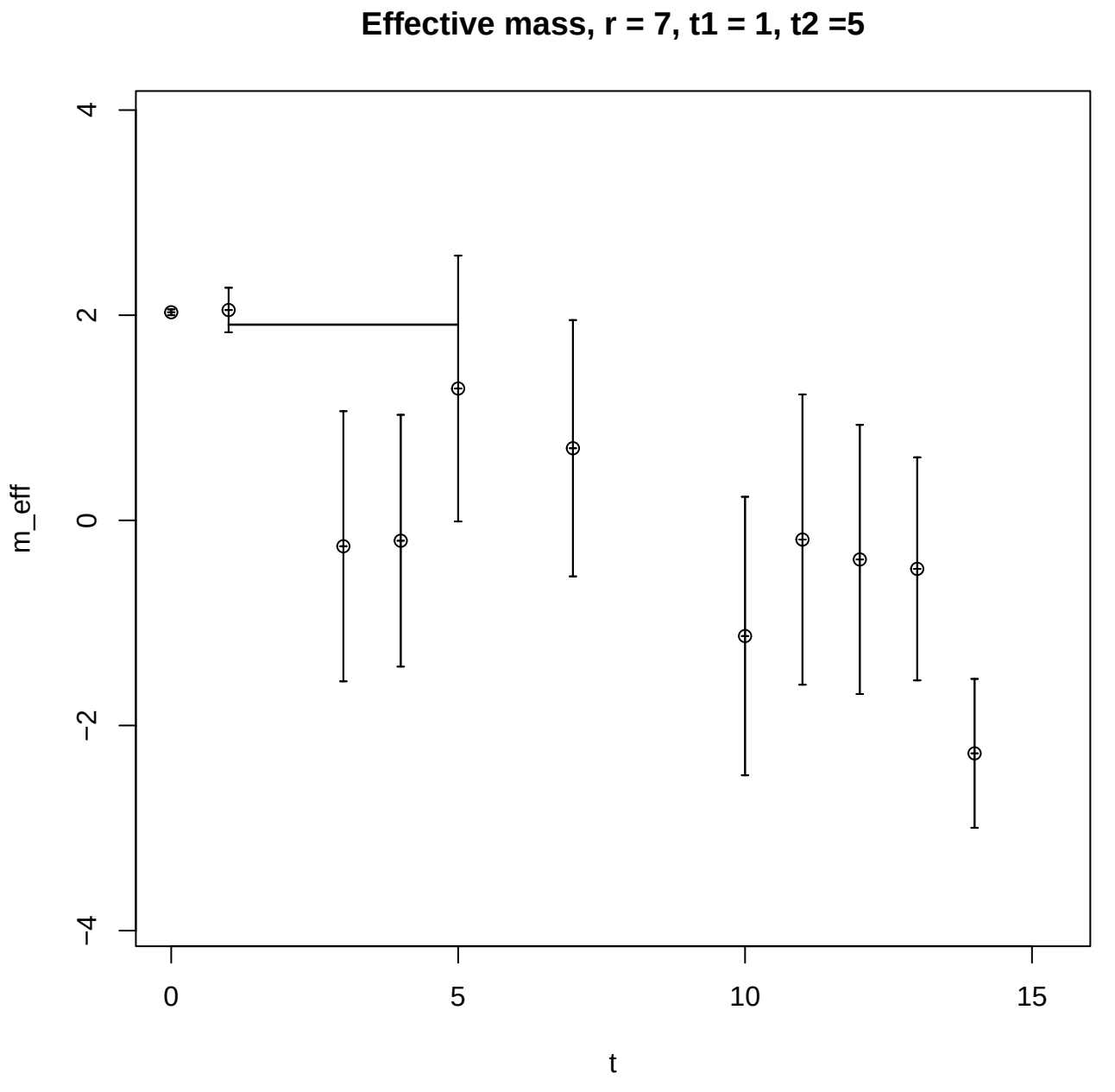
Effective mass, r = 7, t1 = 1, t2 =5
m_eff
t
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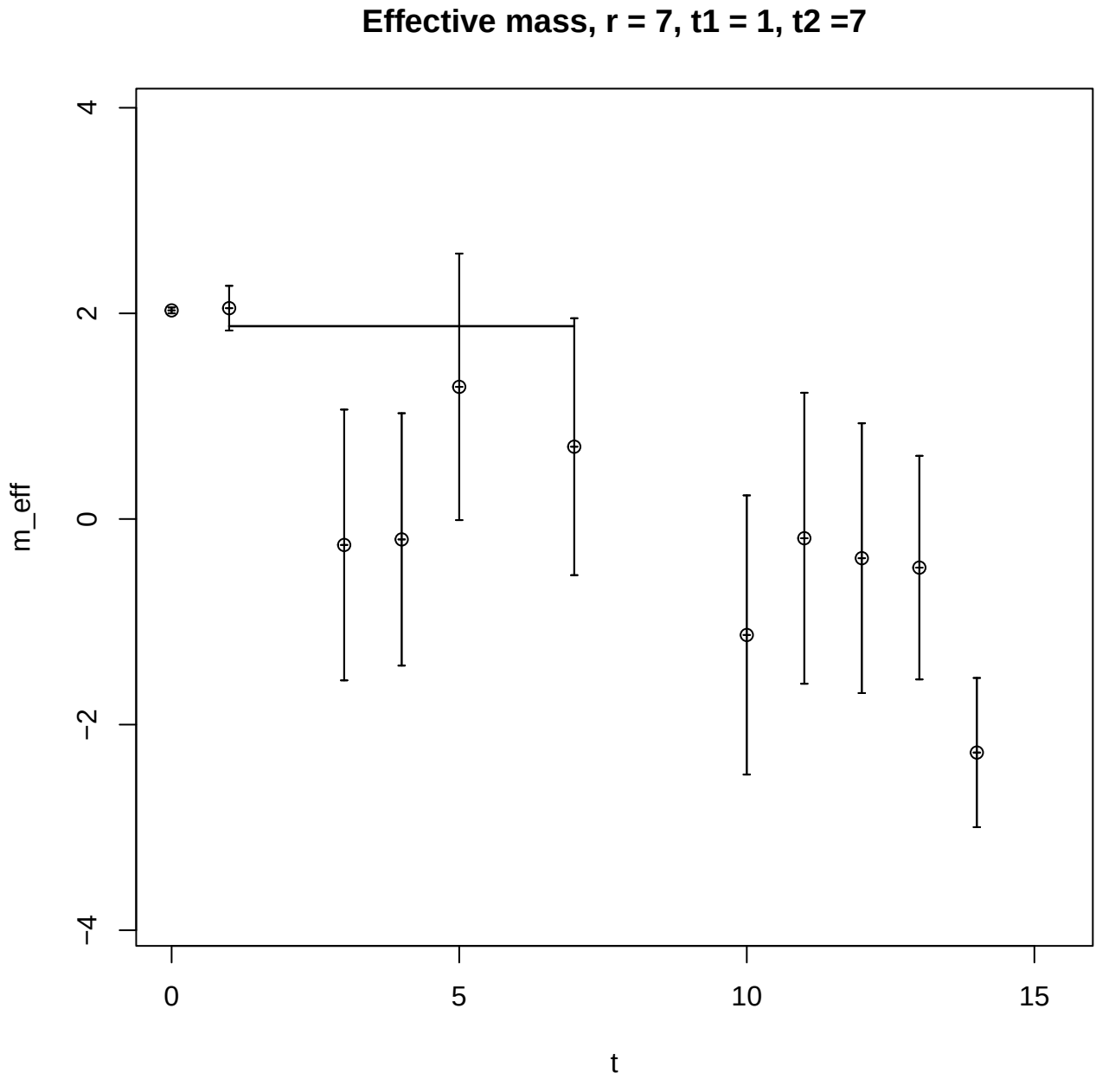
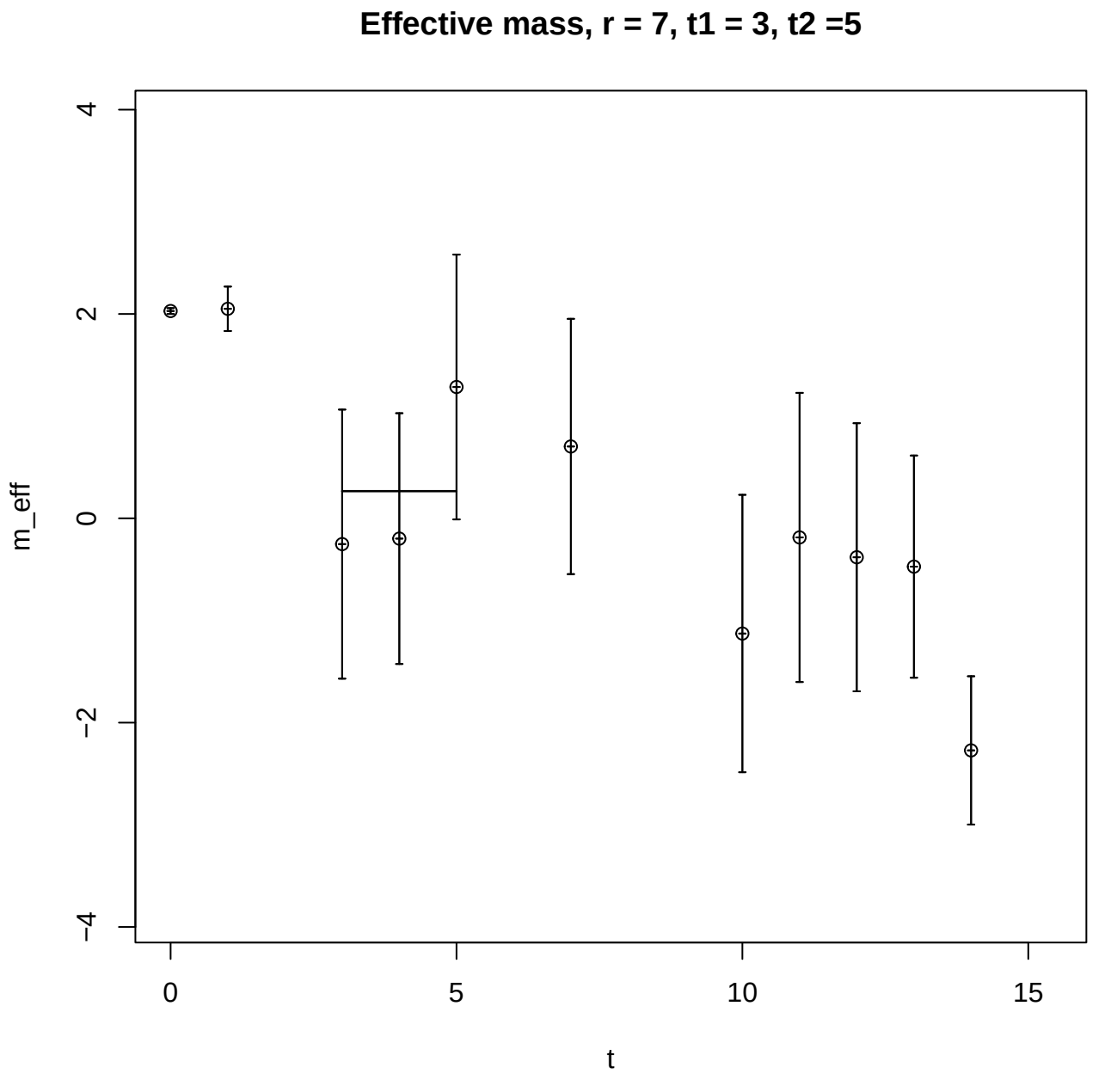
2
0
−2
−4
0
5
10
15


Effective mass, r = 7, t1 = 1, t2 =7
m_eff
t
4
2
0
−2
−4
0
5
10
15


Effective mass, r = 7, t1 = 3, t2 =5
m_eff
t
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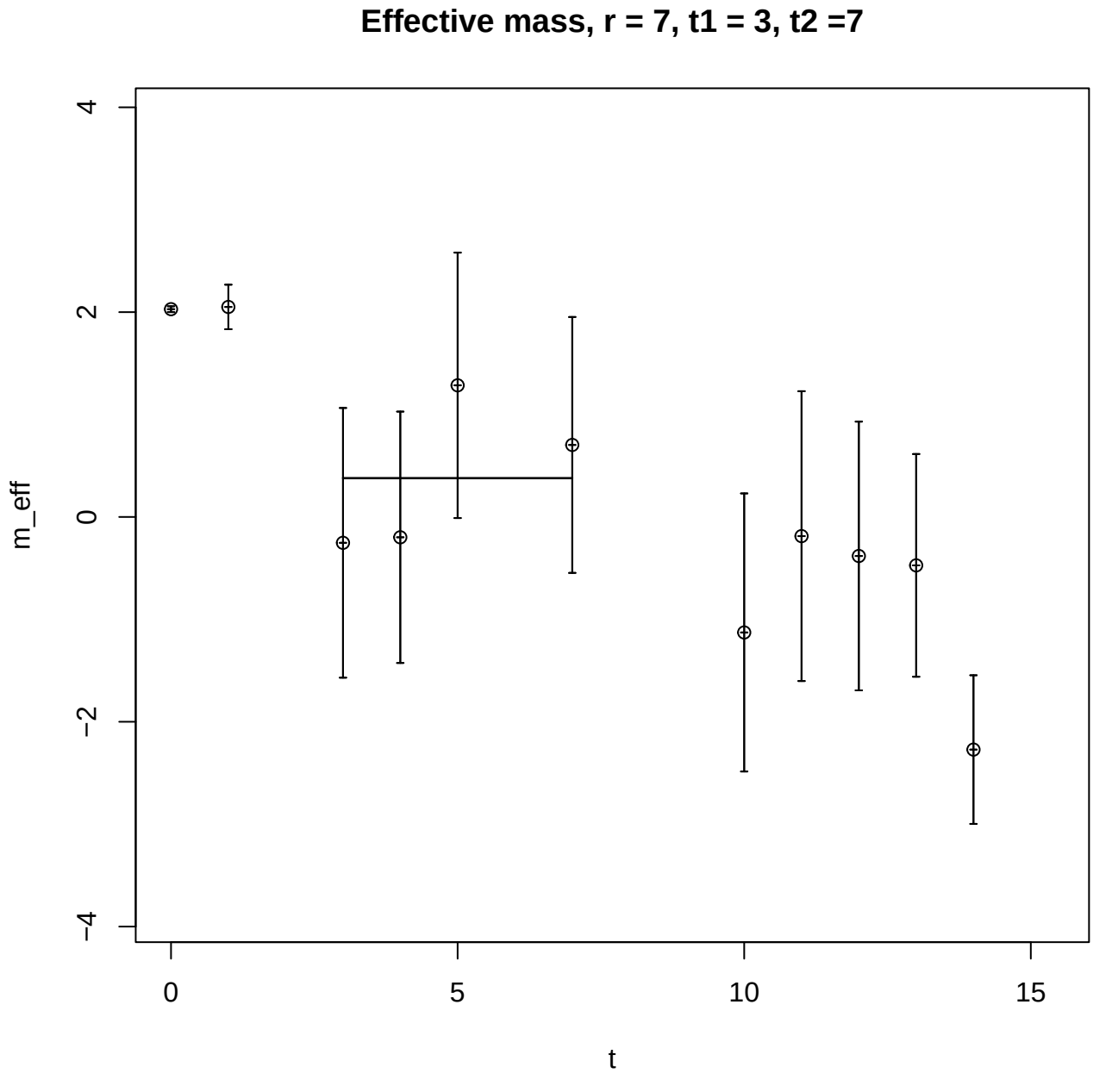
2
0
−2
−4
0
5
10
15


Effective mass, r = 7, t1 = 3, t2 =7
m_eff
t
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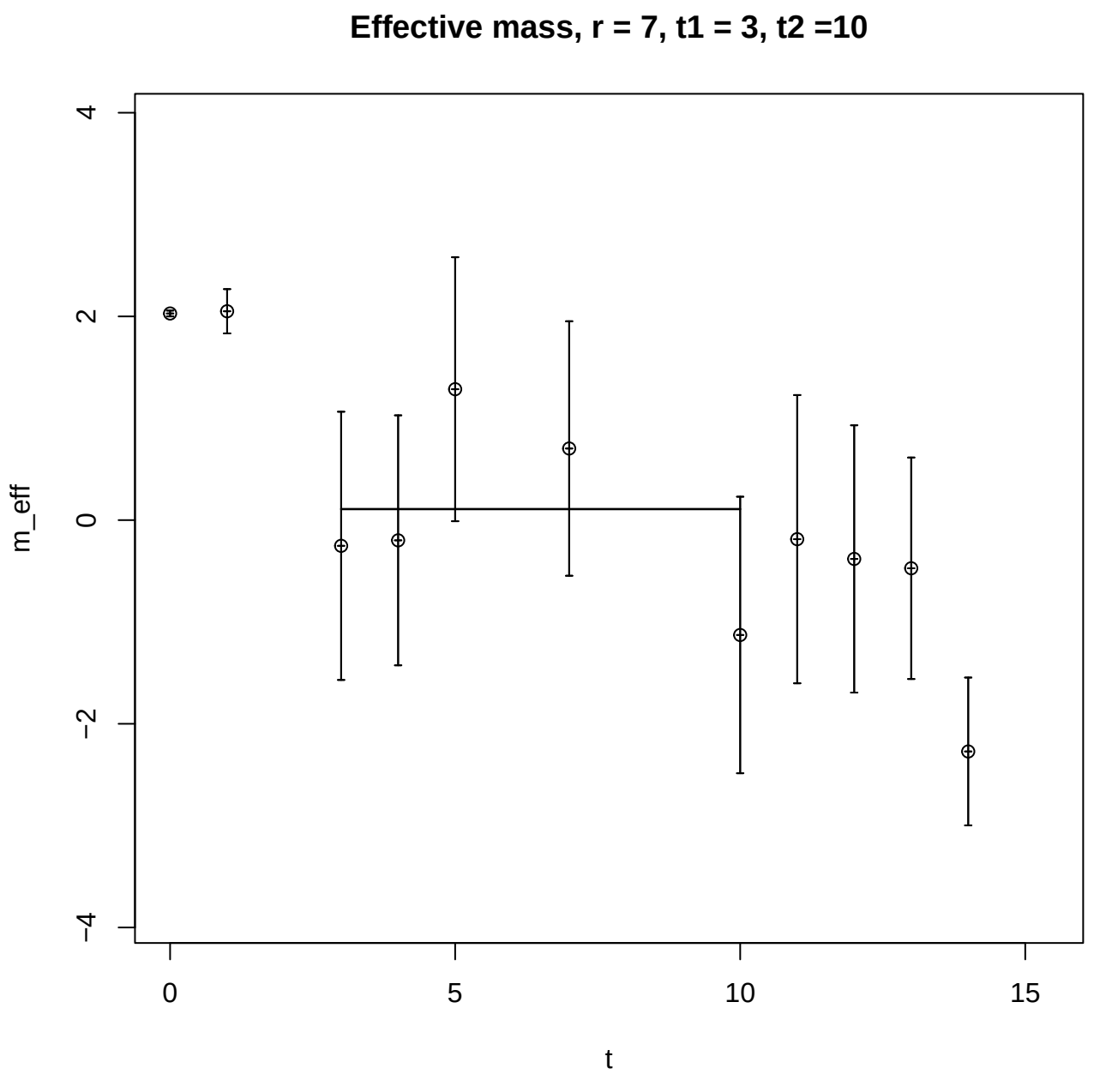
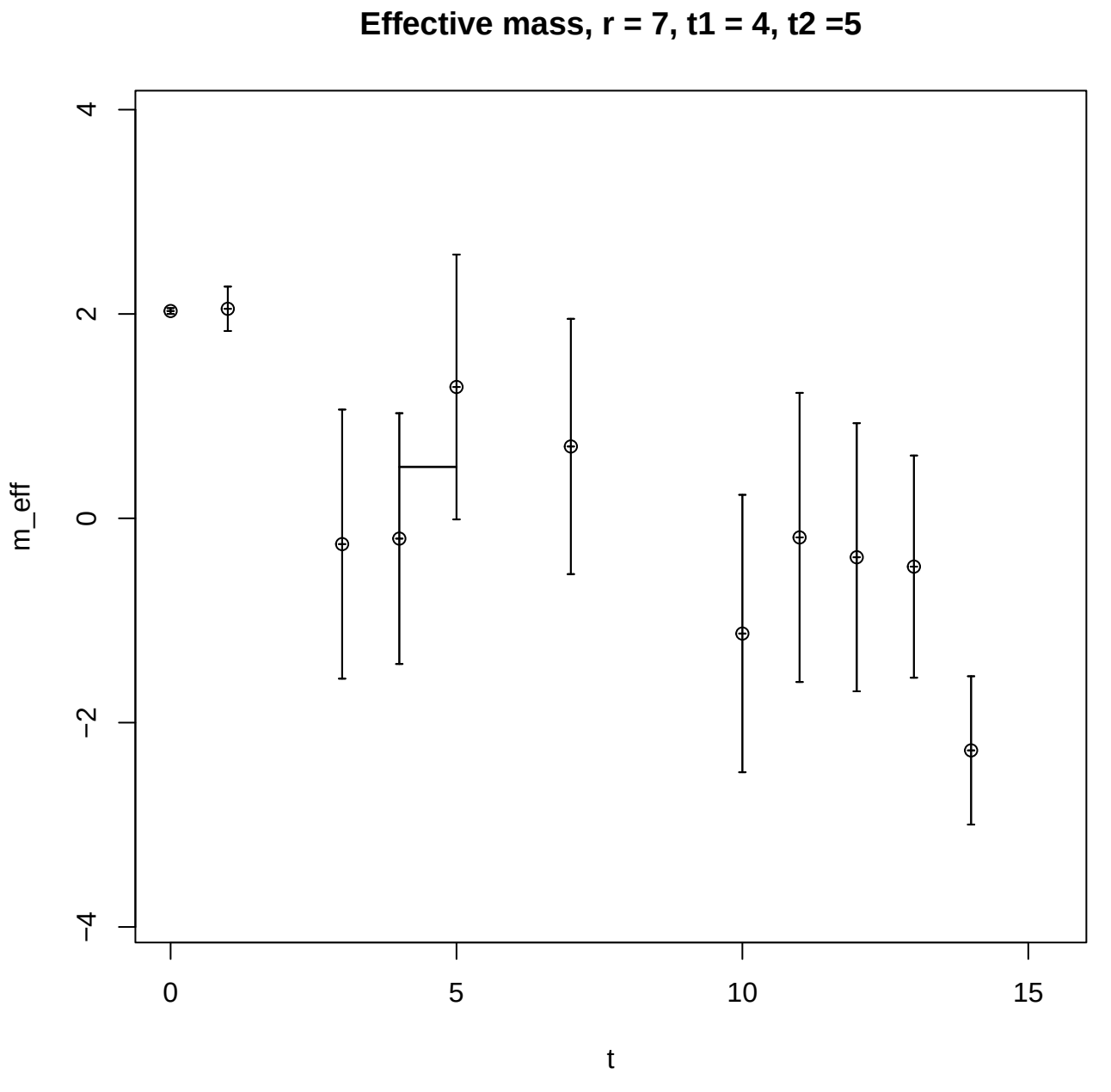
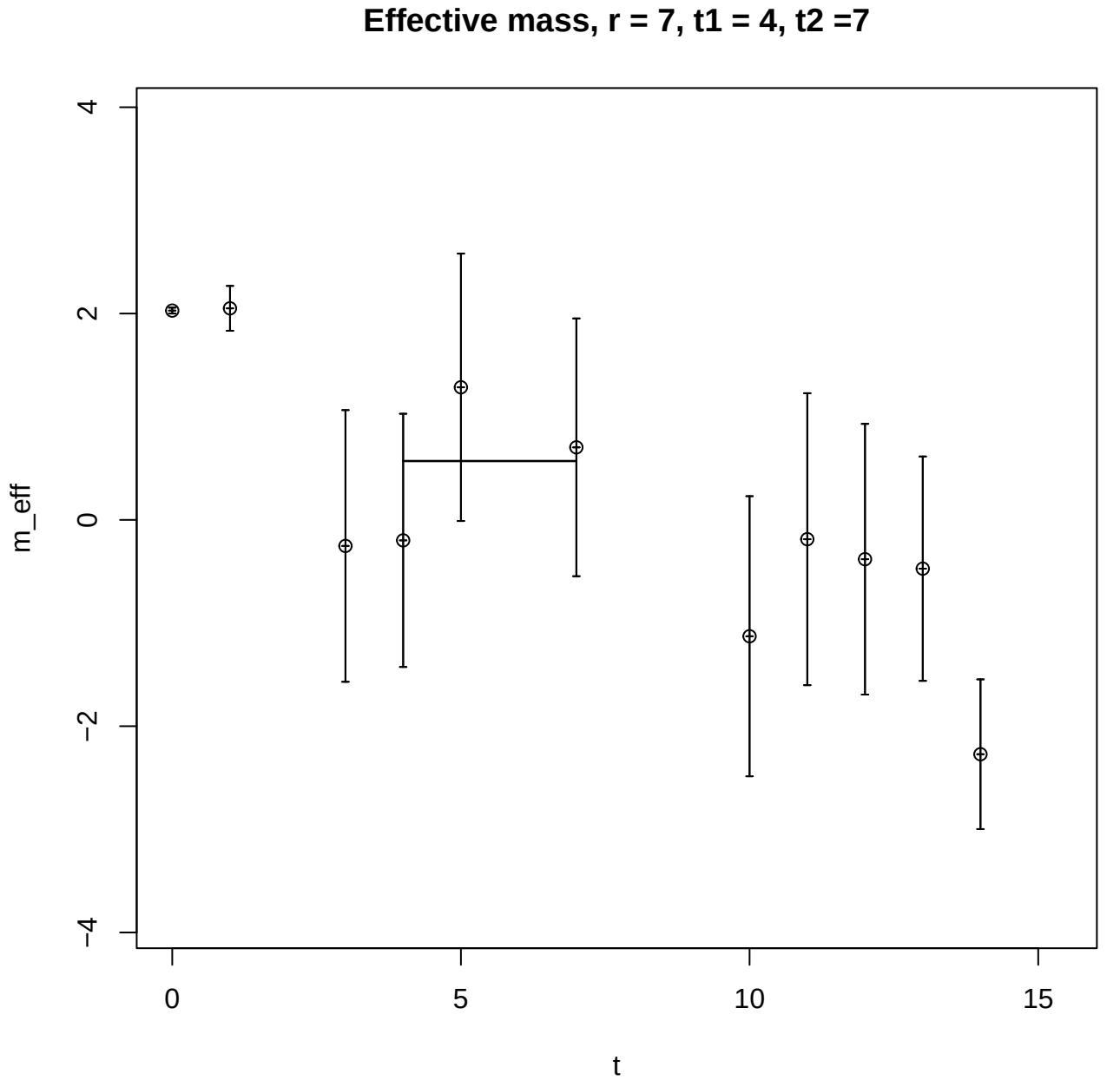
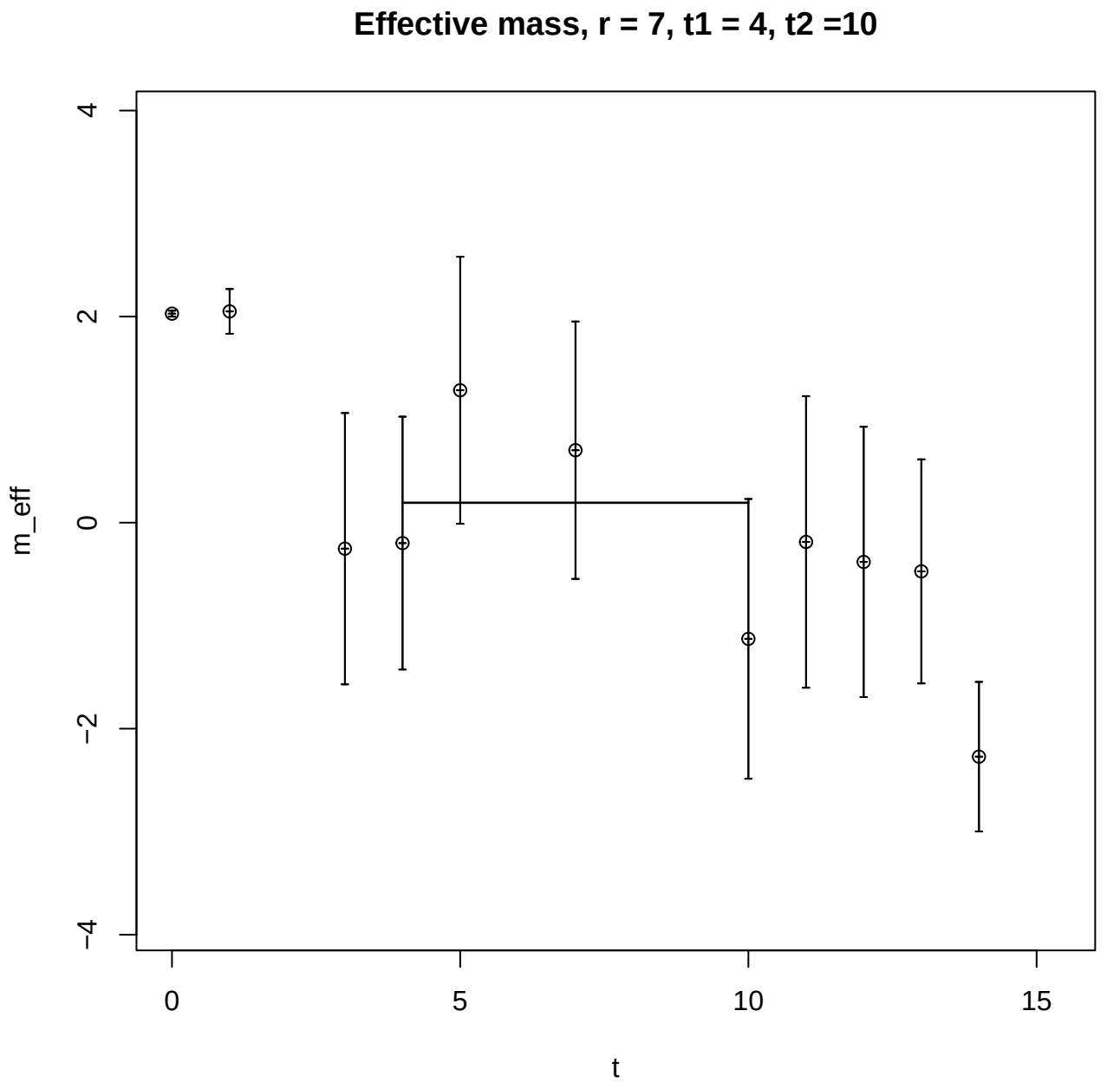
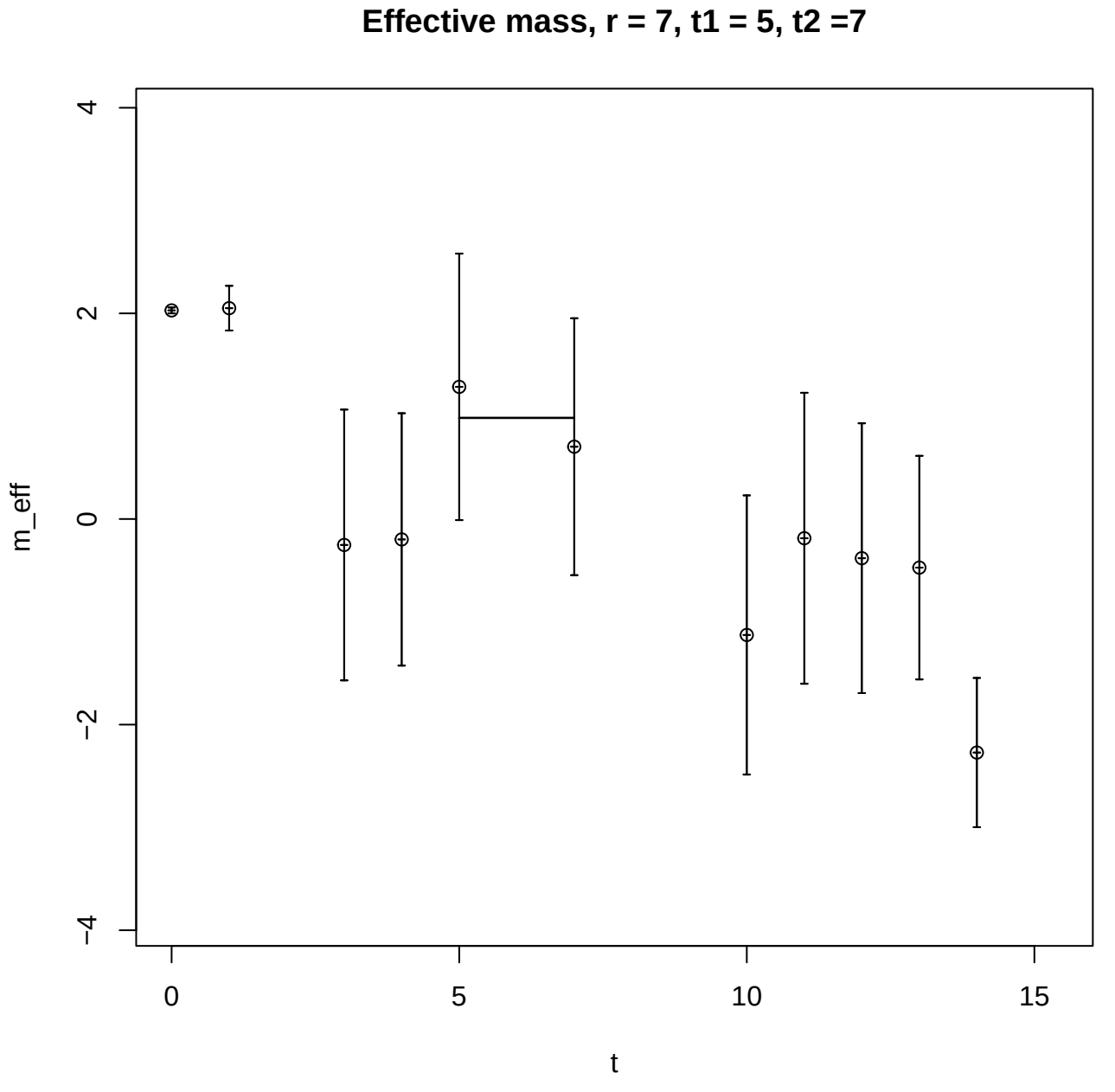
2
0
−2
−4
0
5
10
15

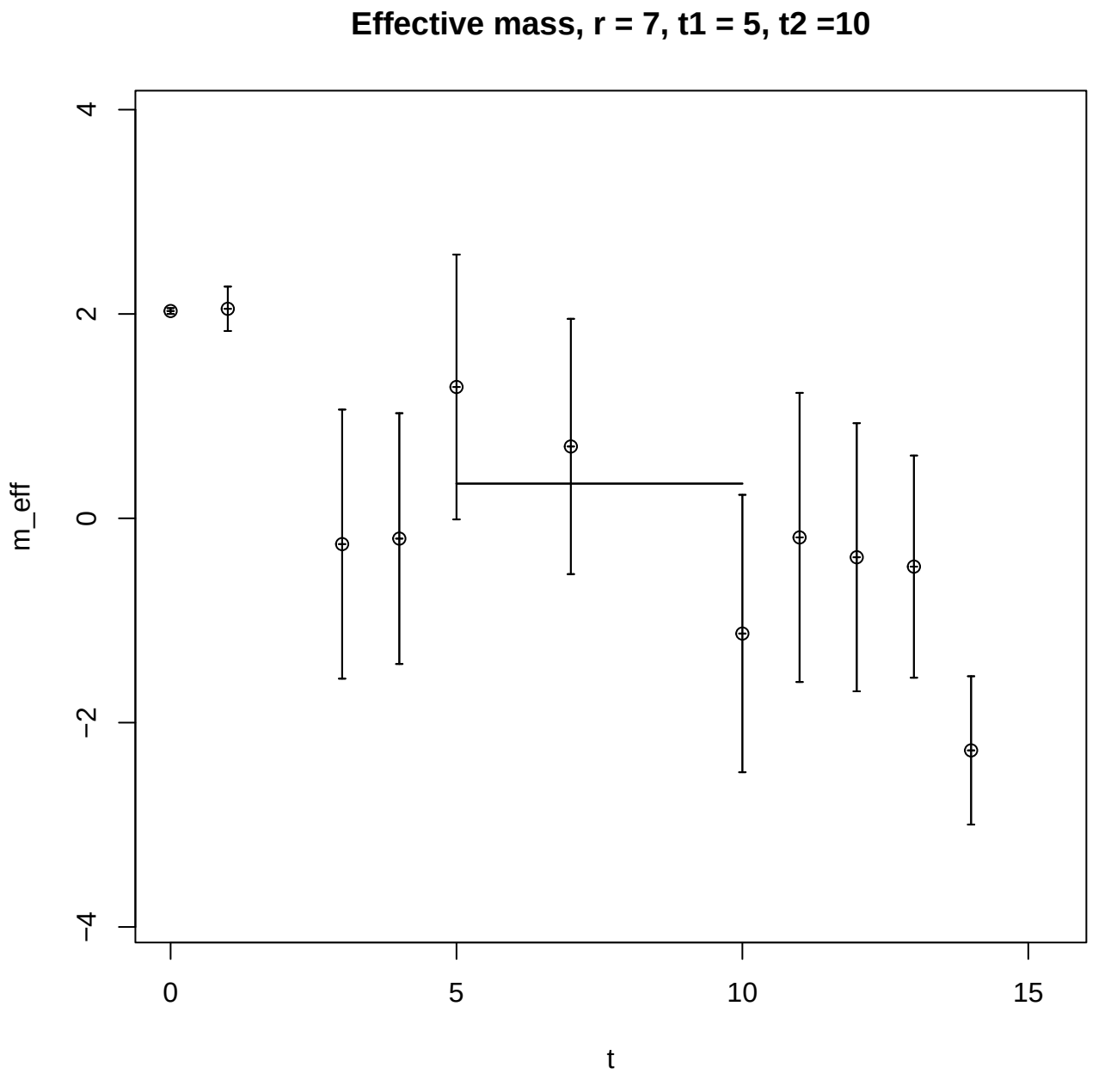

Effective mass, r = 7, t1 = 3, t2 =10
m_eff
t
4
2
0
−2
−4
0
5
10
15

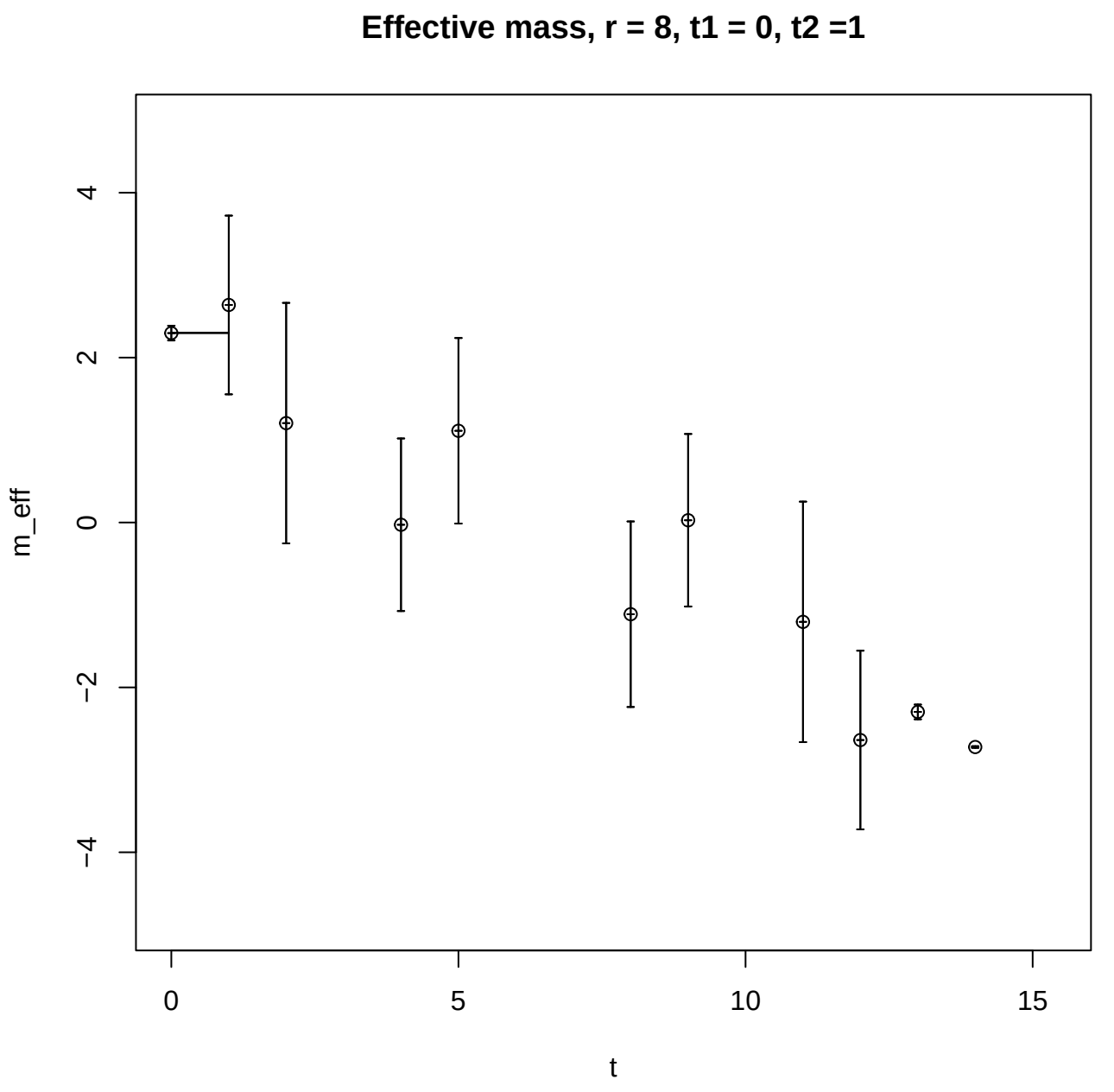

Effective mass, r = 7, t1 = 4, t2 =5
m_eff
t
4
2
0
−2
−4
0
5
10
15

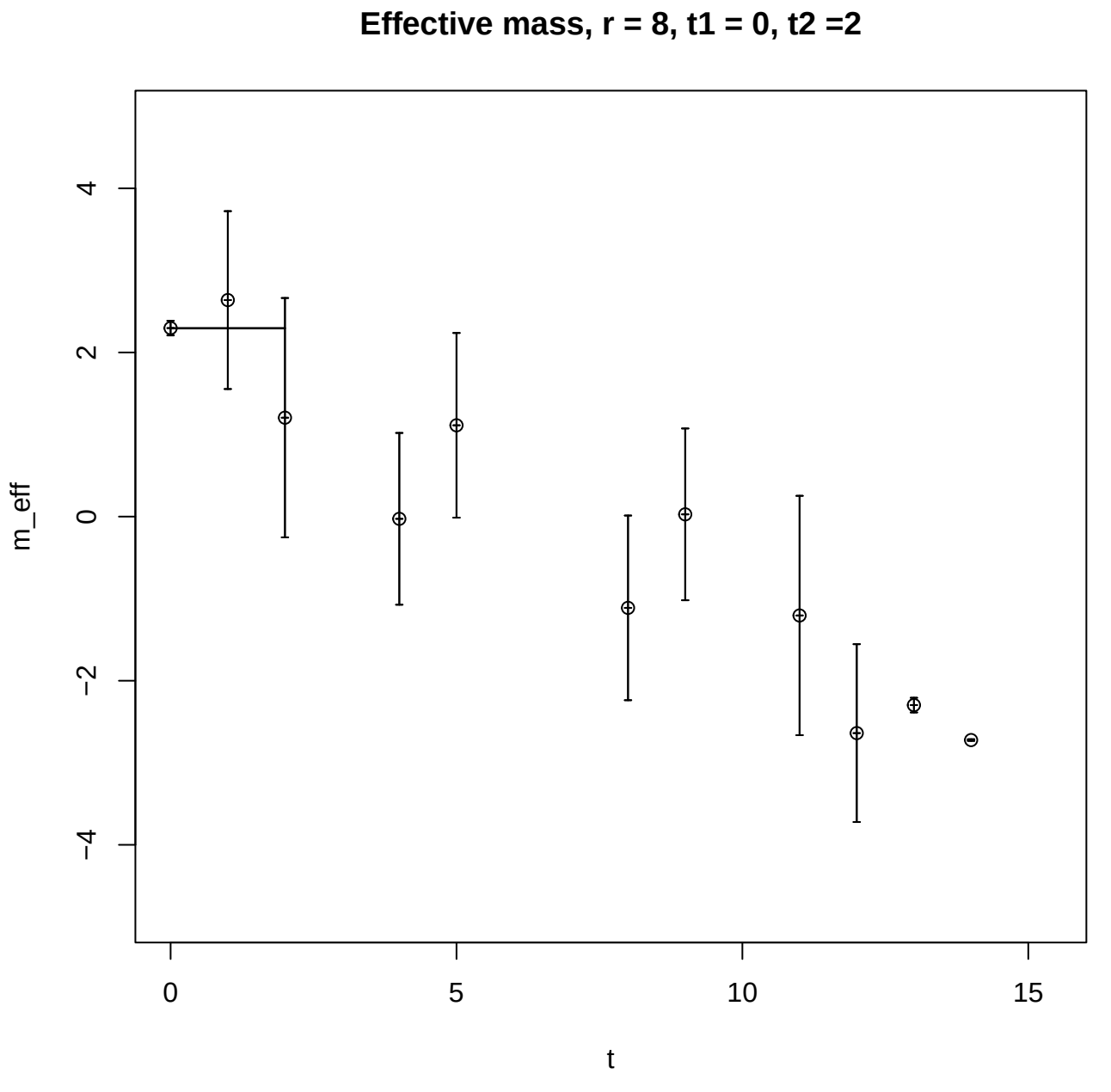

Effective mass, r = 7, t1 = 4, t2 =7
m_eff
t
4
2
0
−2
−4
0
5
10
15

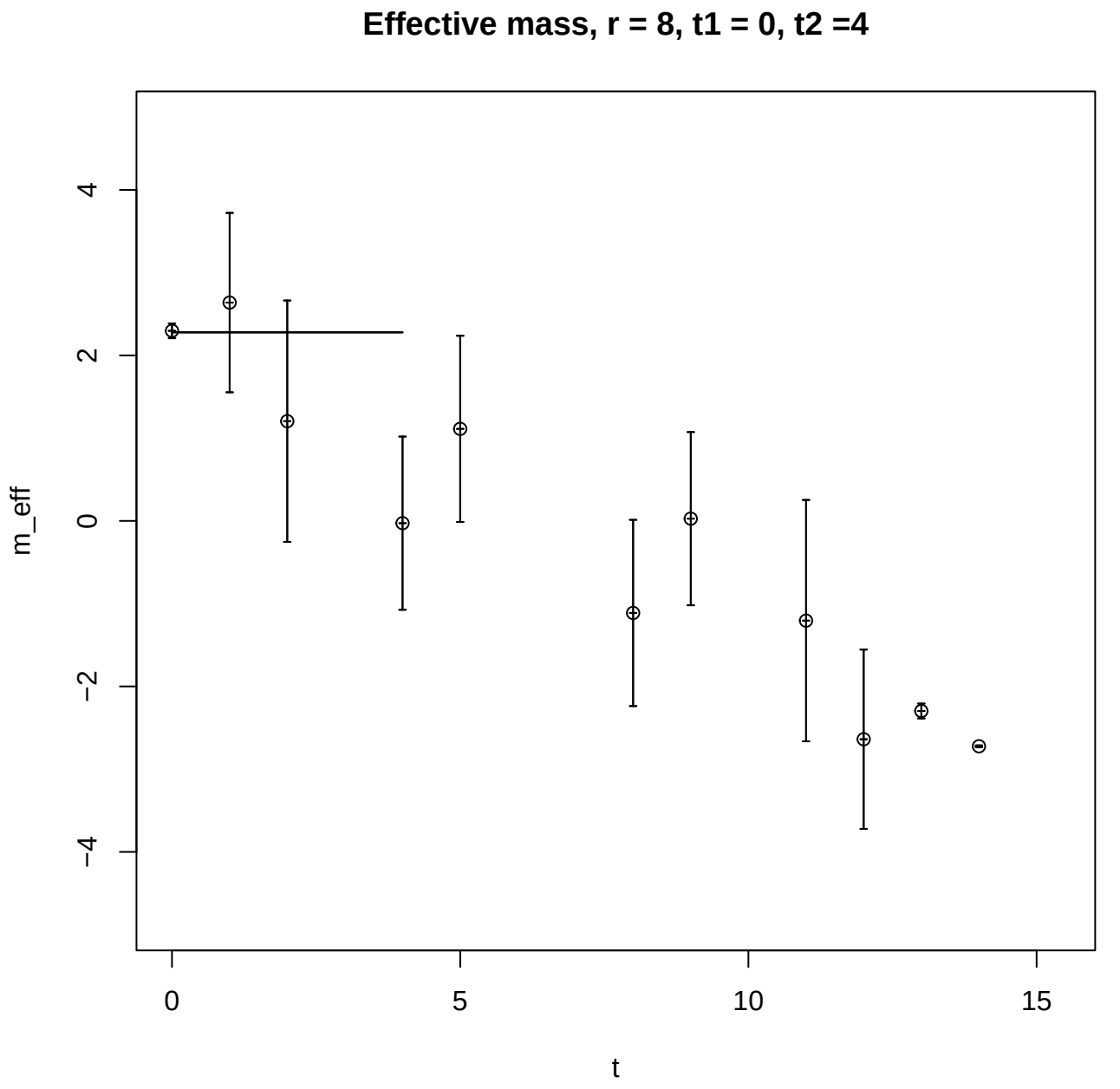

Effective mass, r = 7, t1 = 4, t2 =10
m_eff
t
4
2
0
−2
−4
0
5
10
15

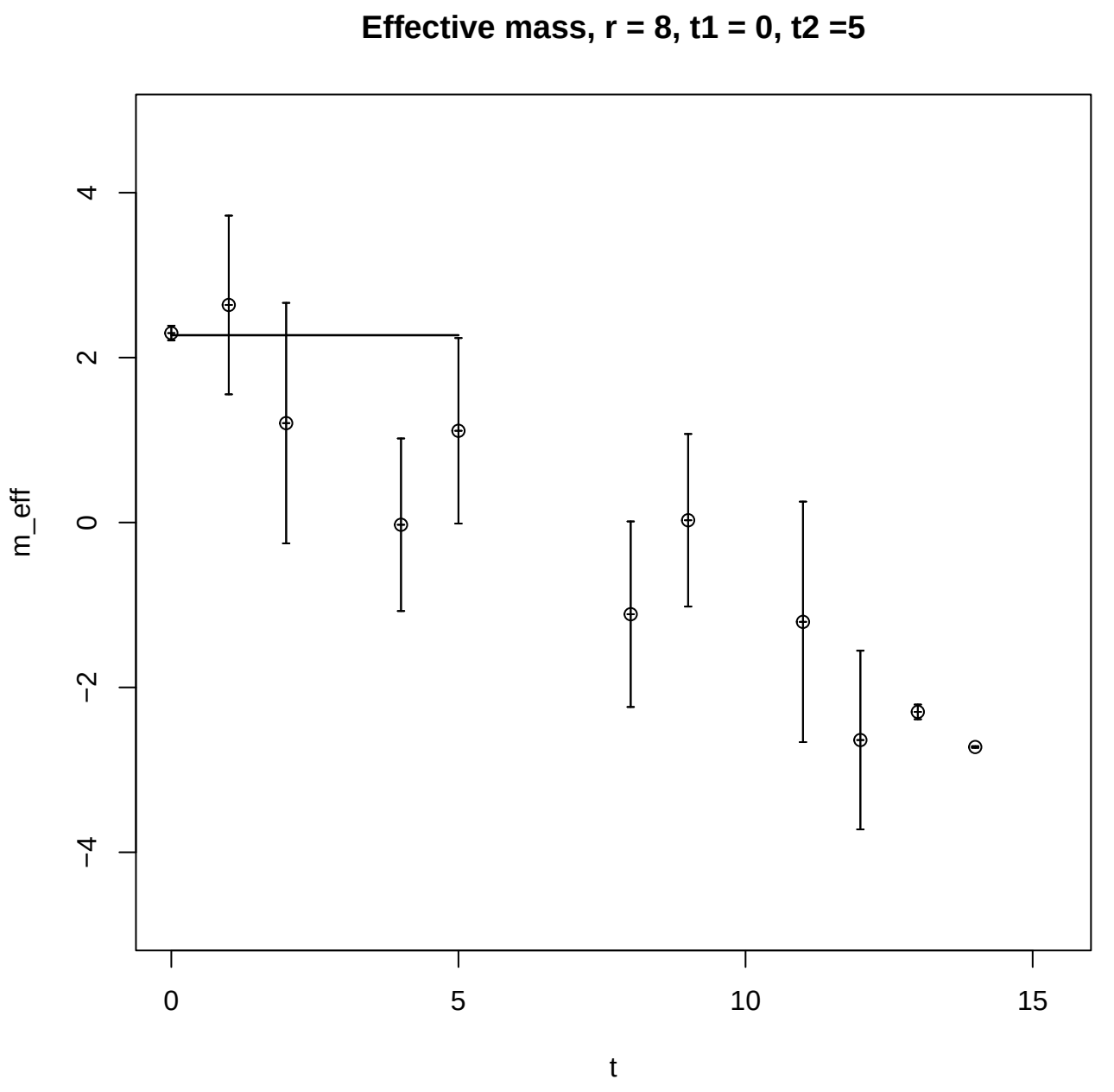

Effective mass, r = 7, t1 = 5, t2 =7
m_eff
t
4
2
0
−2
−4
0
5
10
15

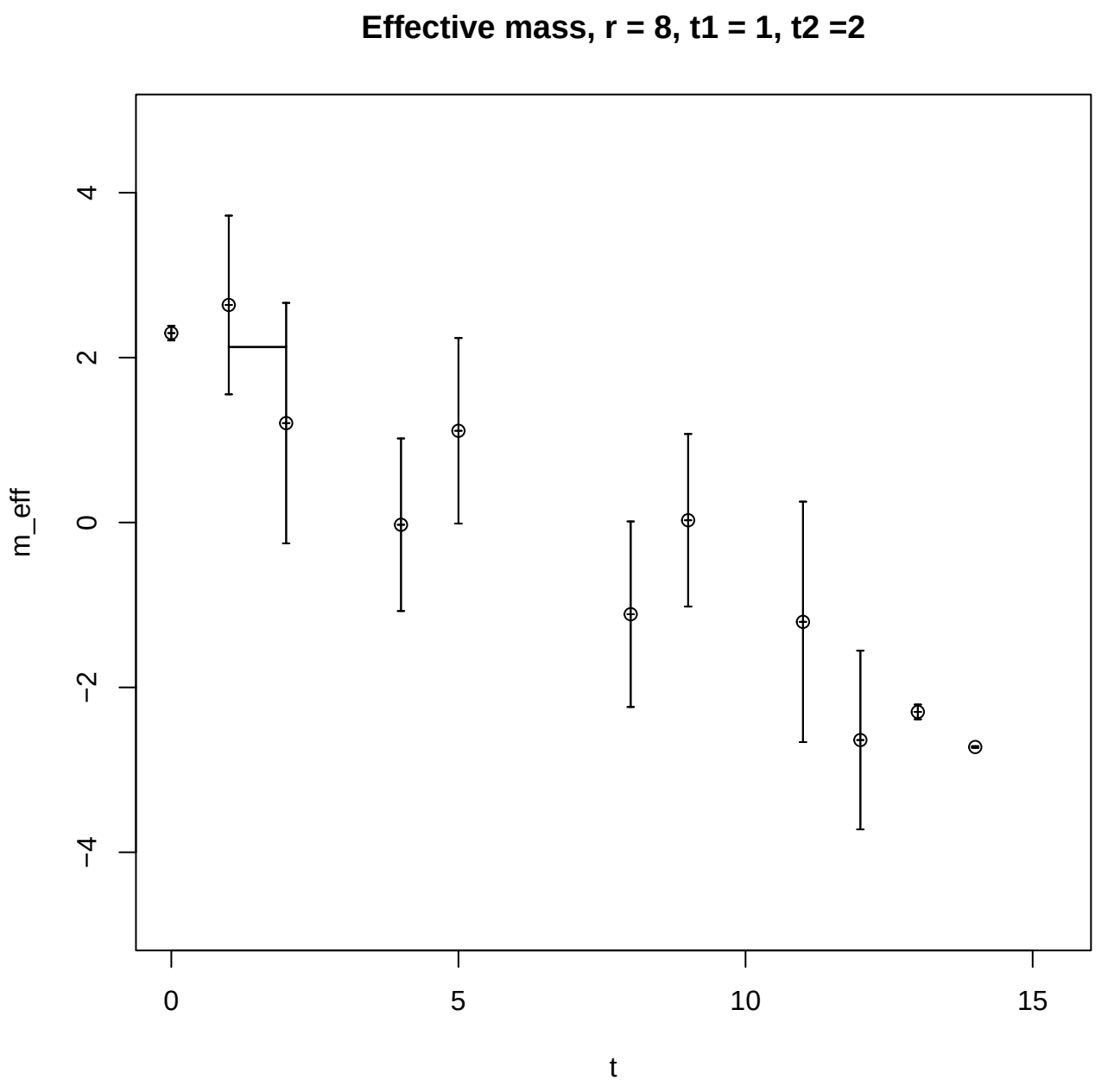

Effective mass, r = 7, t1 = 5, t2 =10
m_eff
t
4
2
0
−2
−4
0
5
10
15

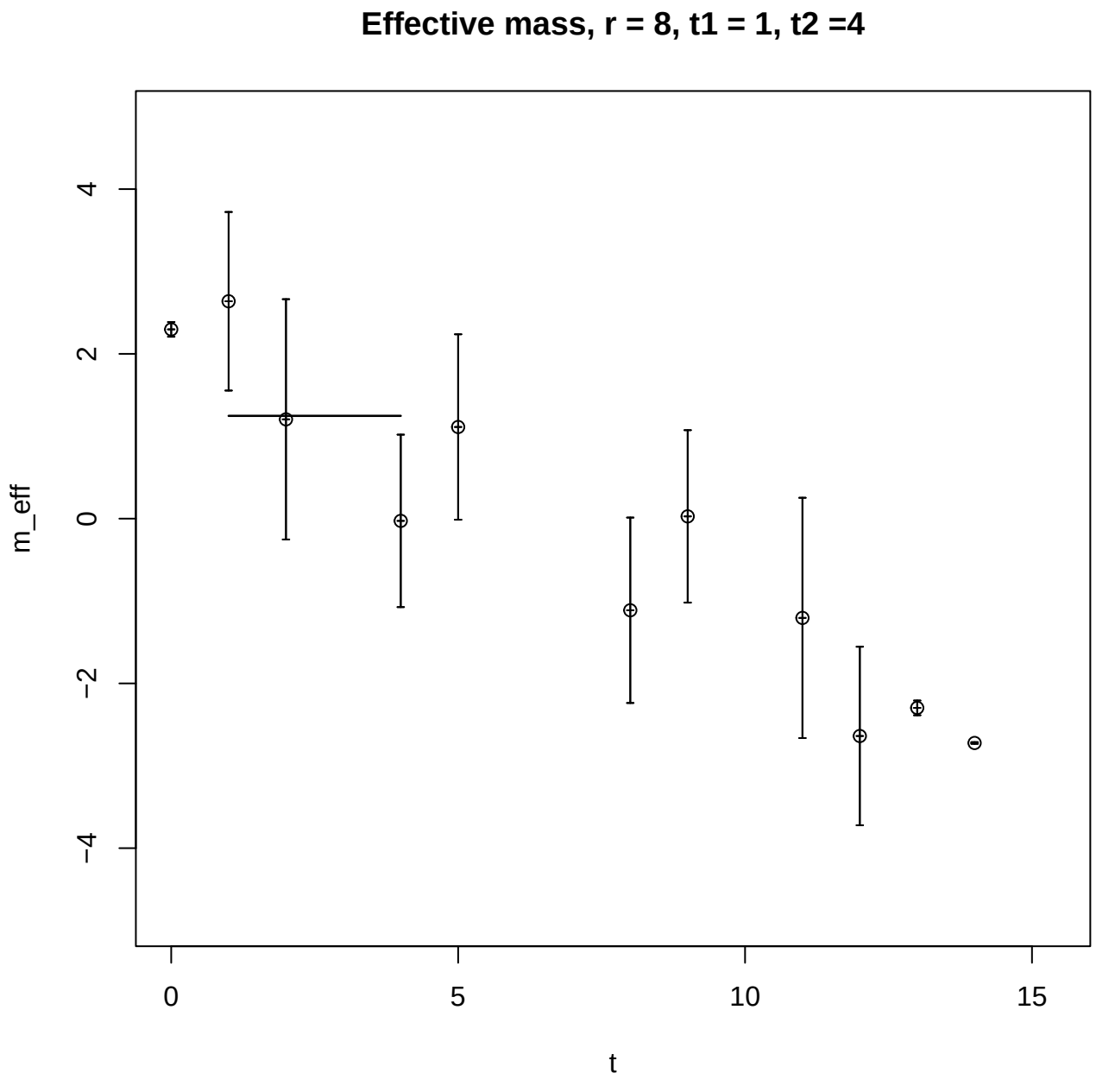

Effective mass, r = 8, t1 = 0, t2 =1
m_eff
4
2
0
−2
−4
0
5
10
15
t

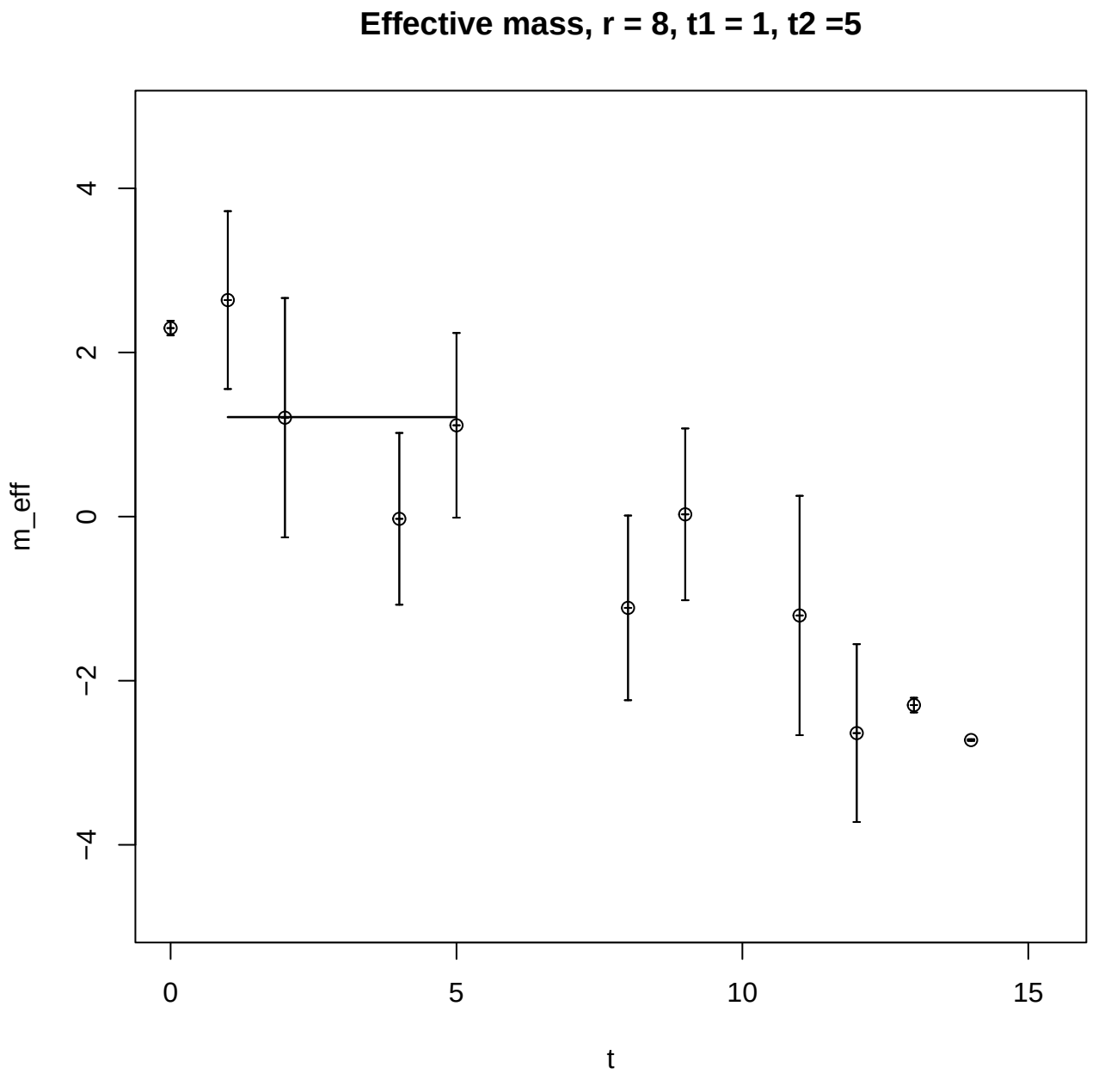

Effective mass, r = 8, t1 = 0, t2 =2
m_eff
4
2
0
−2
−4
0
5
10
15
t

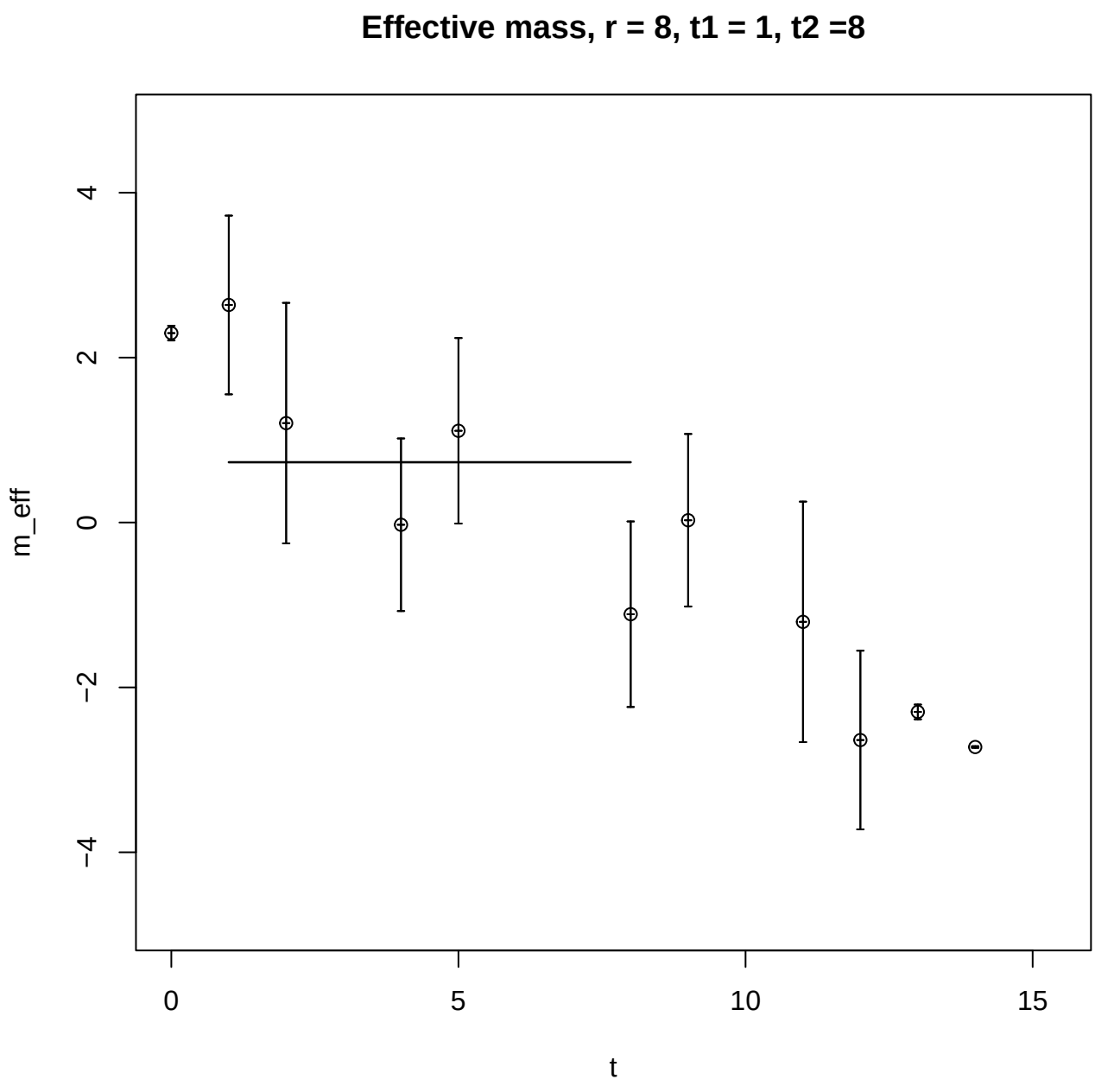

Effective mass, r = 8, t1 = 0, t2 =4
m_eff
4
2
0
−2
−4
0
5
10
15
t


Effective mass, r = 8, t1 = 0, t2 =5
m_eff
4
2
0
−2
−4
0
5
10
15
t


Effective mass, r = 8, t1 = 1, t2 =2
m_eff
4
2
0
−2
−4
0
5
10
15
t


Effective mass, r = 8, t1 = 1, t2 =4
m_eff
4
2
0
−2
−4
0
5
10
15
t


Effective mass, r = 8, t1 = 1, t2 =5
m_eff
4
2
0
−2
−4
0
5
10
15
t


Effective mass, r = 8, t1 = 1, t2 =8
m_eff
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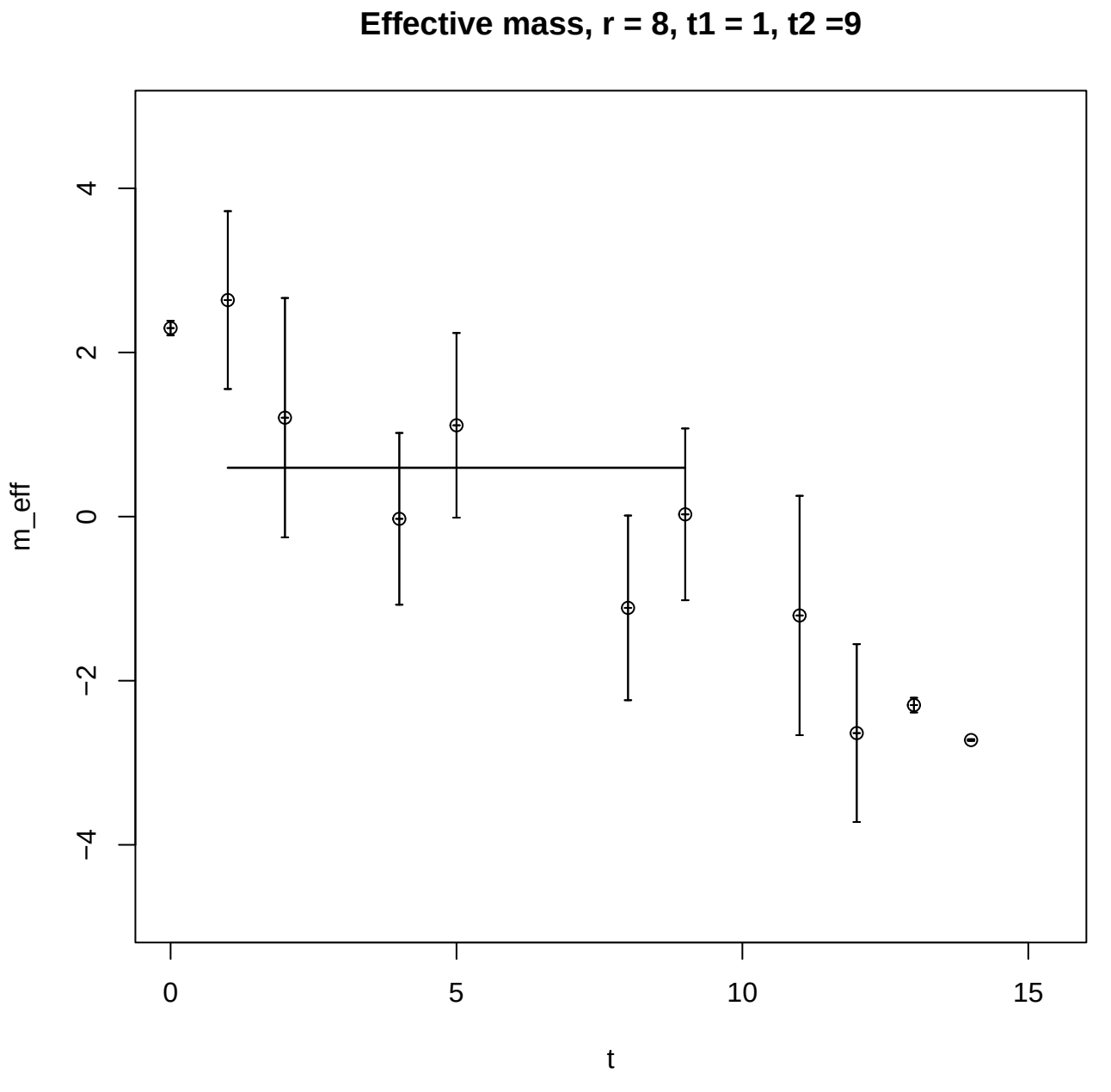
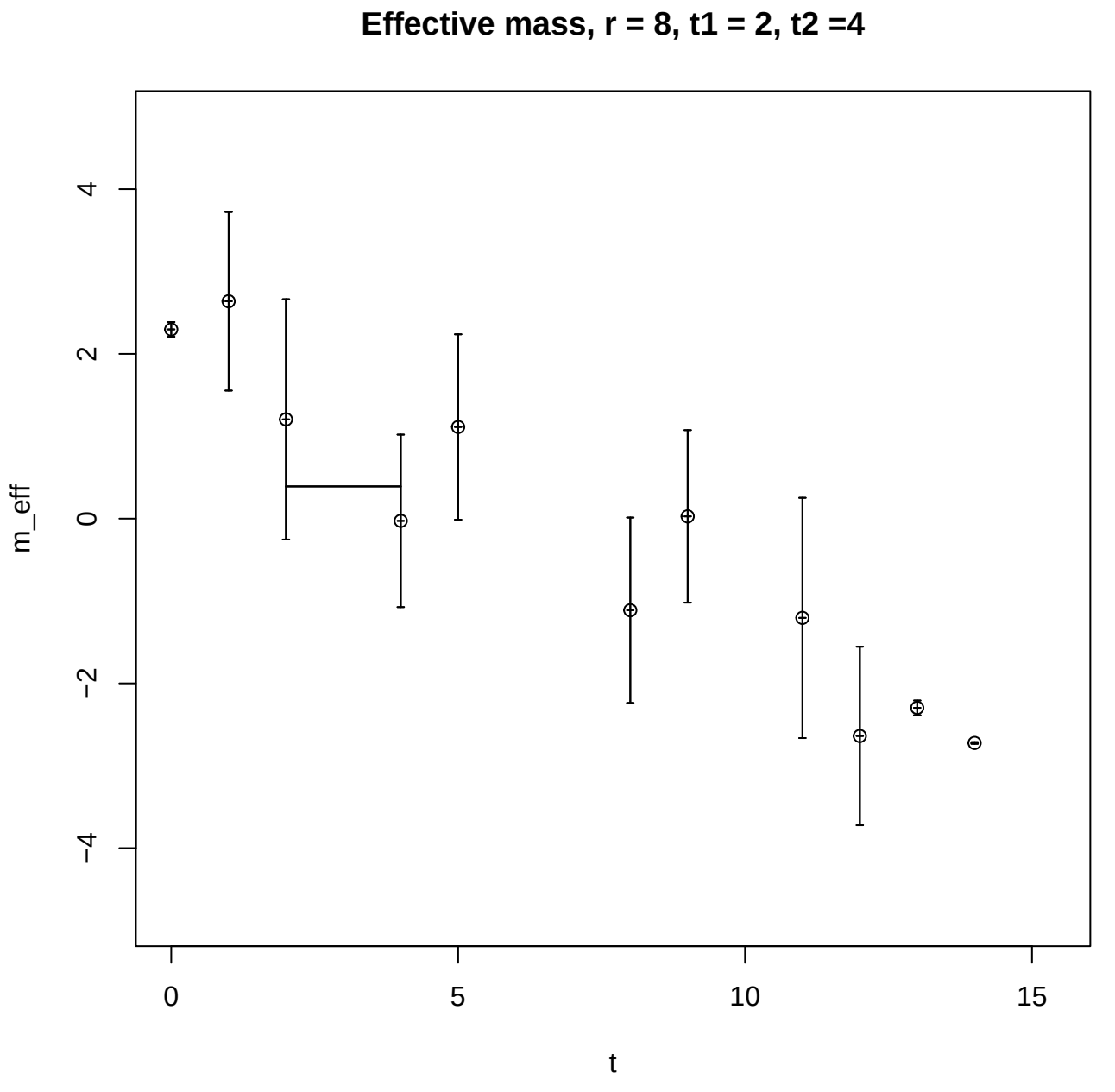
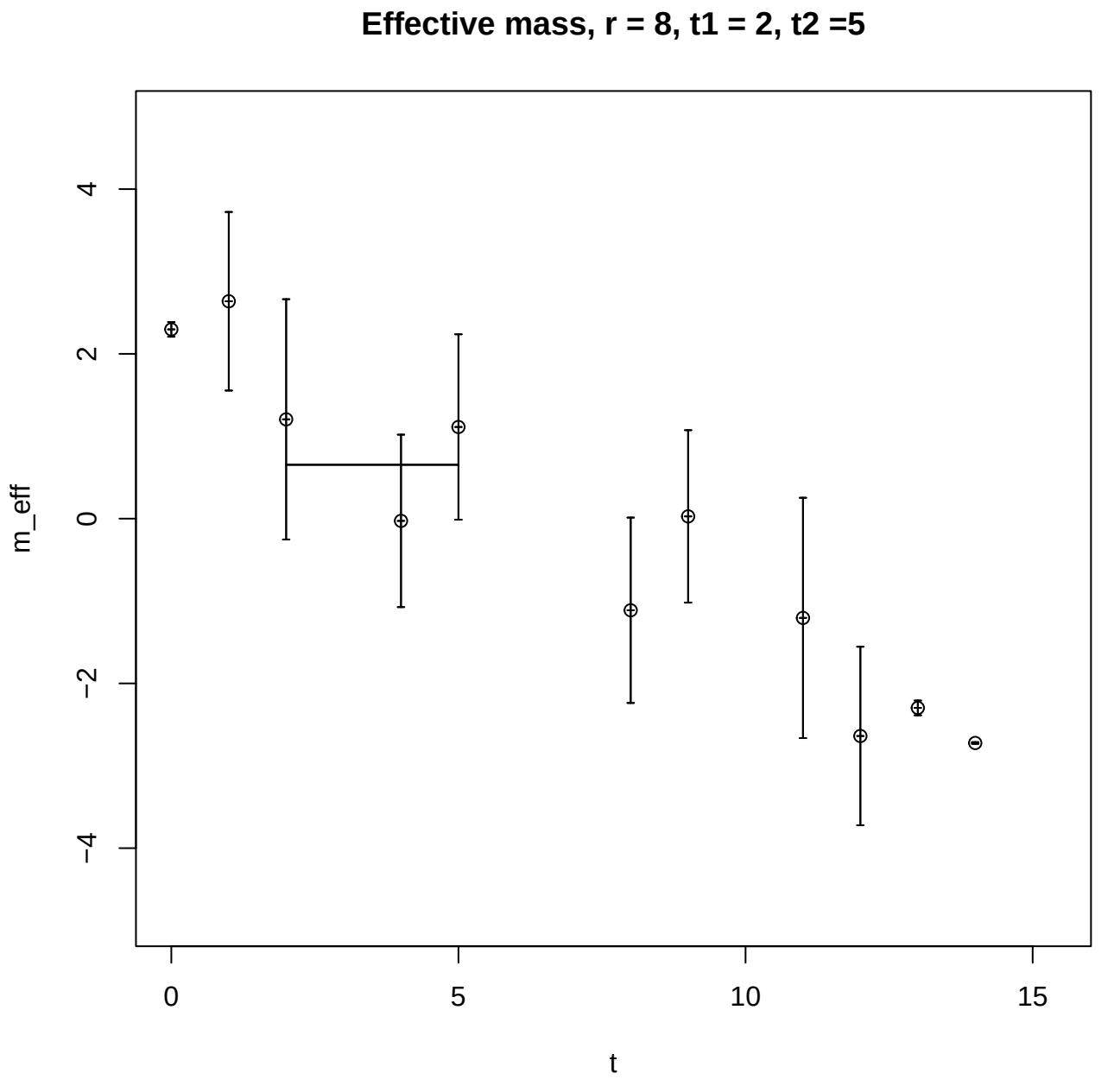
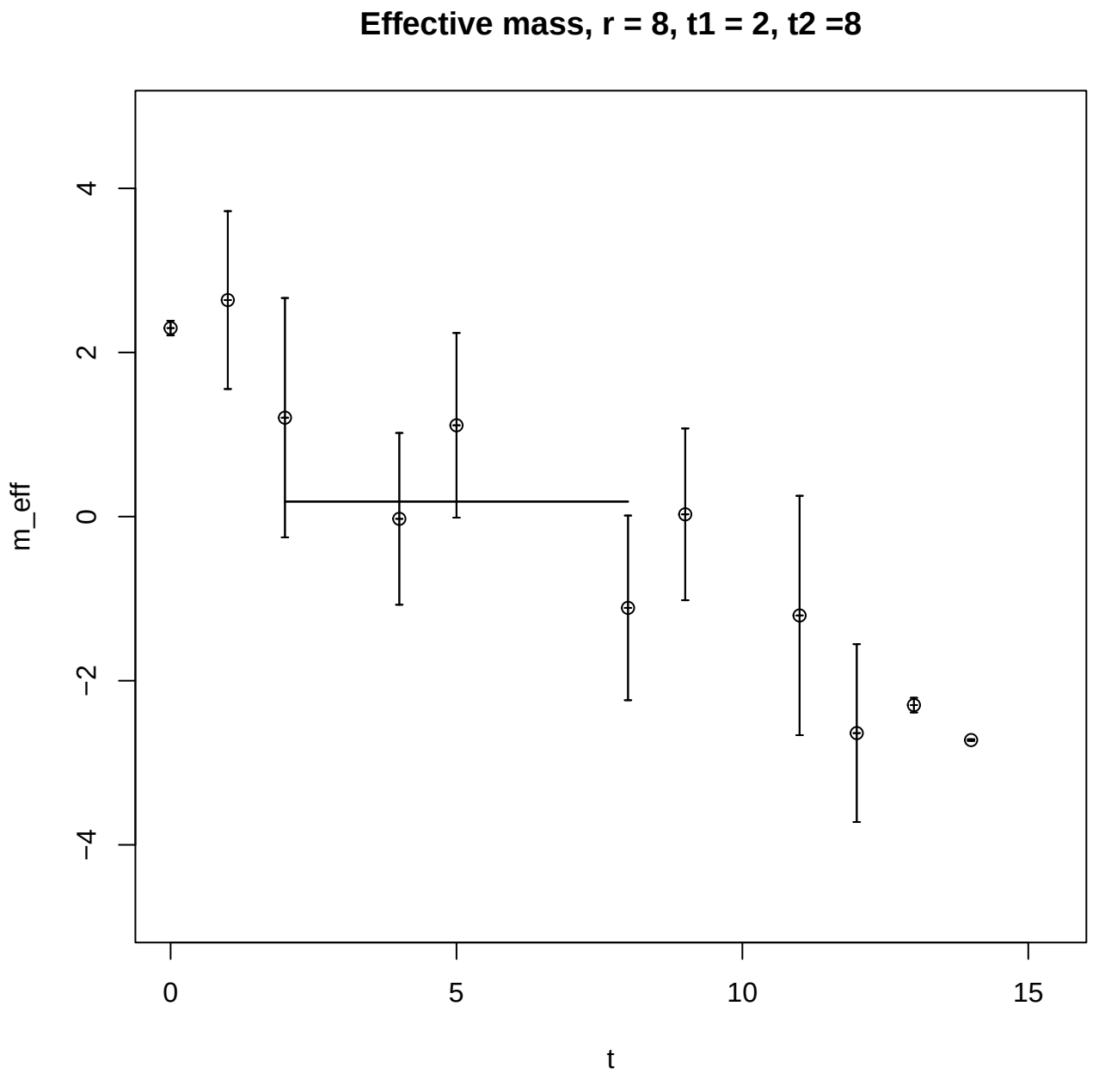
2
0
−2
−4
0
5
10
15
t


Effective mass, r = 8, t1 = 1, t2 =9
m_eff
4
2
0
−2
−4
0
5
10
15
t


Effective mass, r = 8, t1 = 2, t2 =4
m_eff
4
2
0
−2
−4
0
5
10
15
t


Effective mass, r = 8, t1 = 2, t2 =5
m_eff
4
2
0
−2
−4
0
5
10
15
t


Effective mass, r = 8, t1 = 2, t2 =8
m_eff
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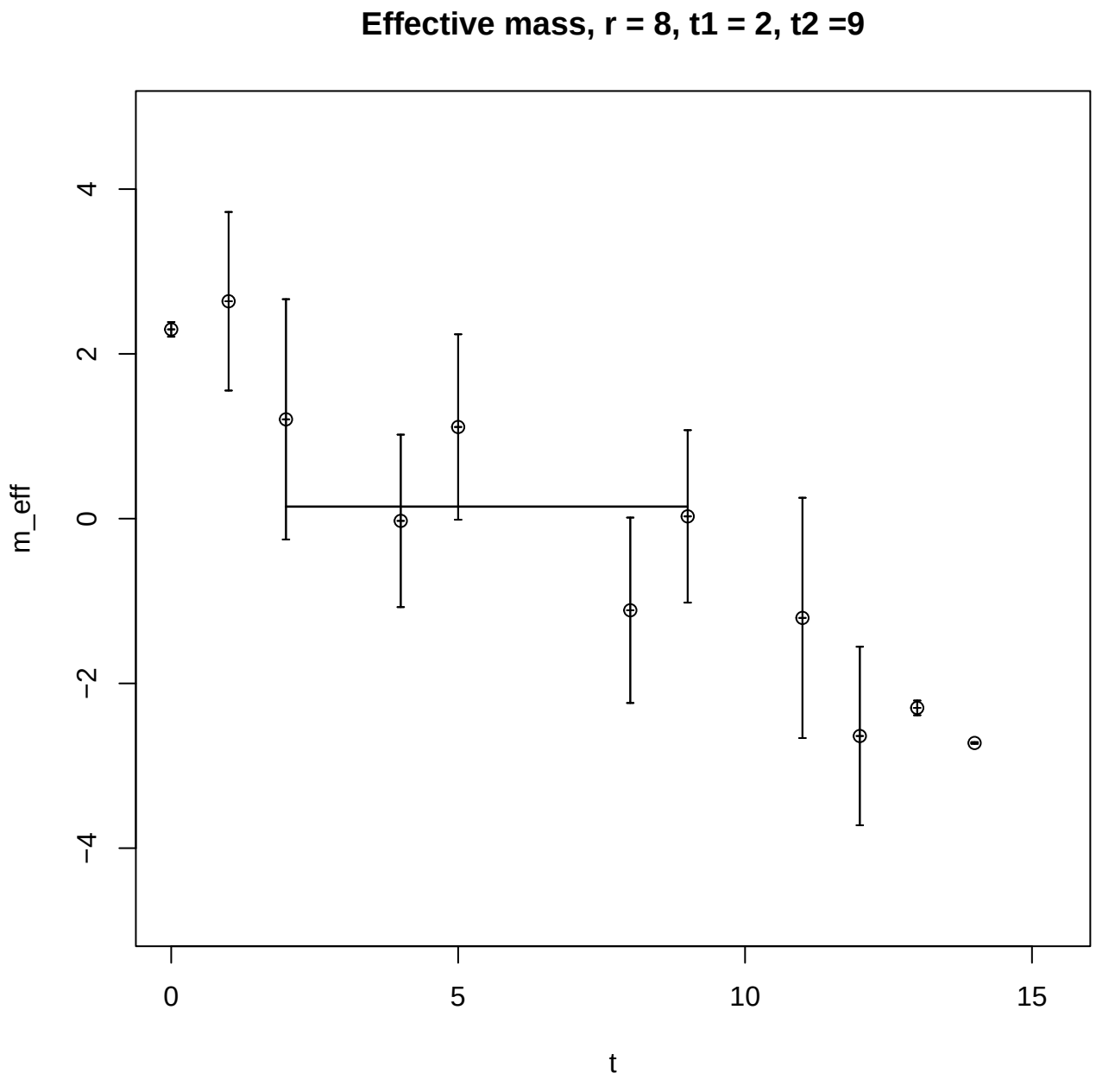
2
0
−2
−4
0
5
10
15
t


Effective mass, r = 8, t1 = 2, t2 =9
m_eff
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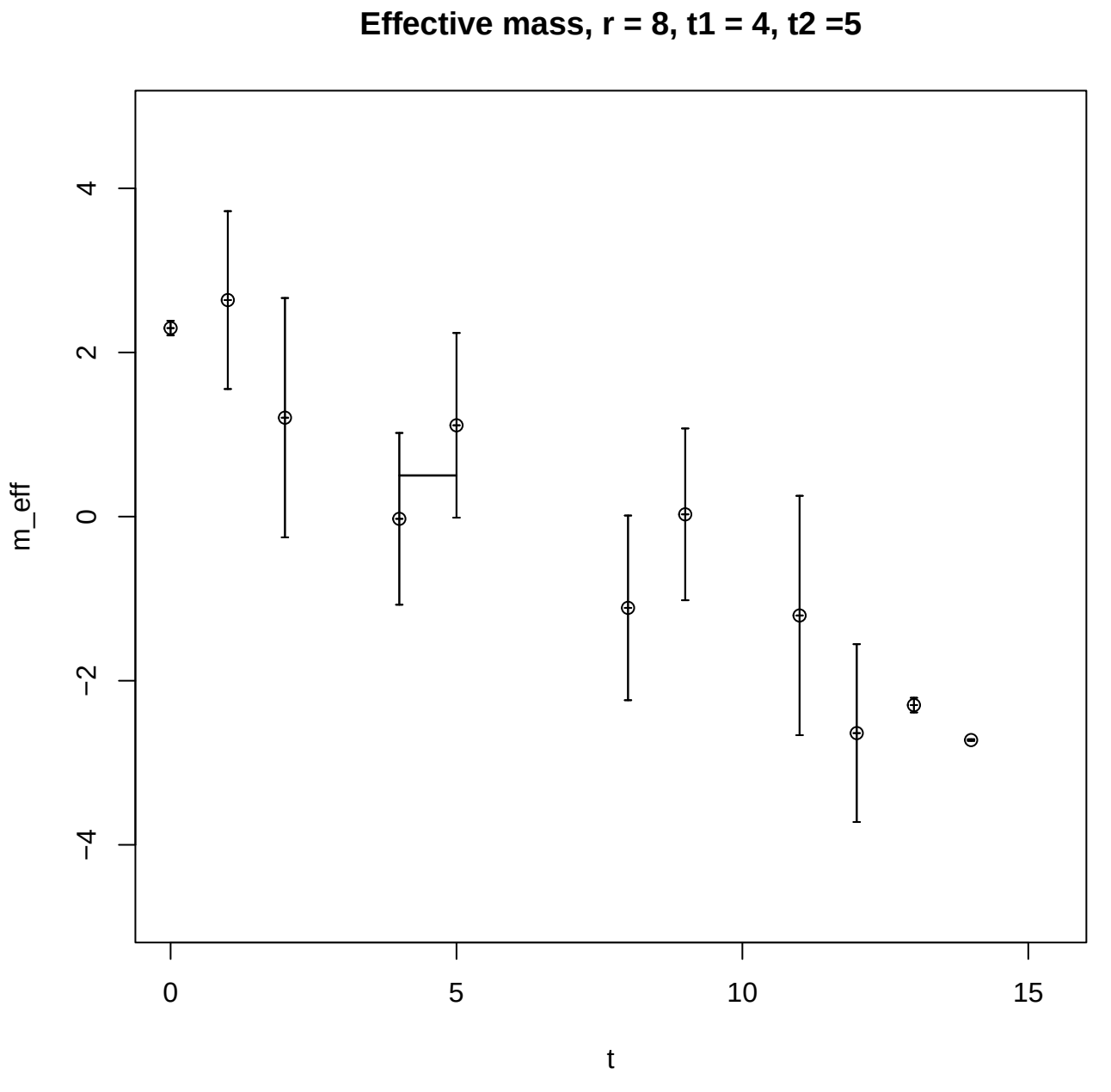
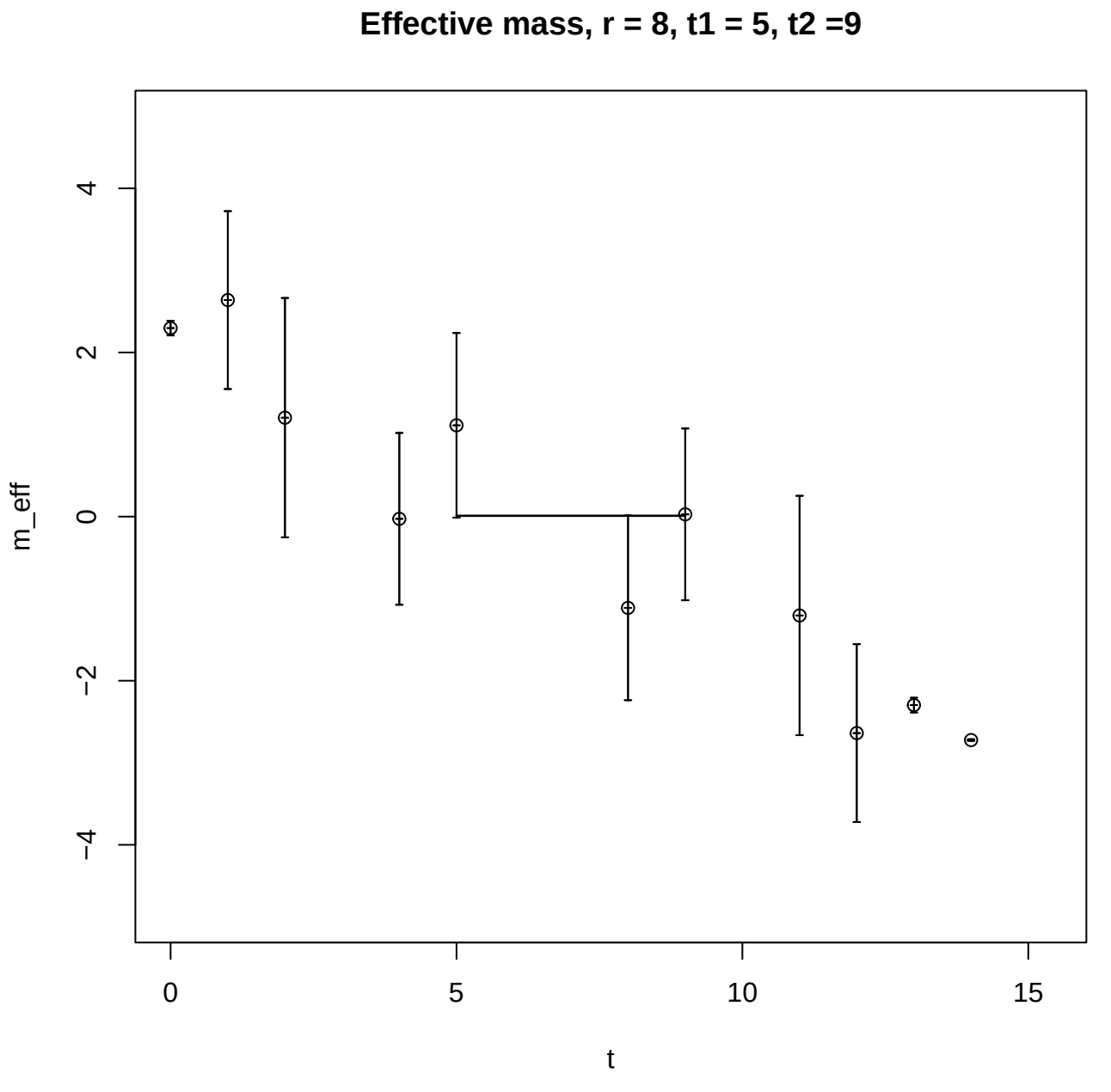
2
0
−2
−4
0
5
10
15
t


Effective mass, r = 8, t1 = 4, t2 =5
m_eff
4
2
0
−2
−4
0
5
10
15
t


Effective mass, r = 8, t1 = 5, t2 =9
m_eff
4
2
0
−2
−4
0
5
10
15
t